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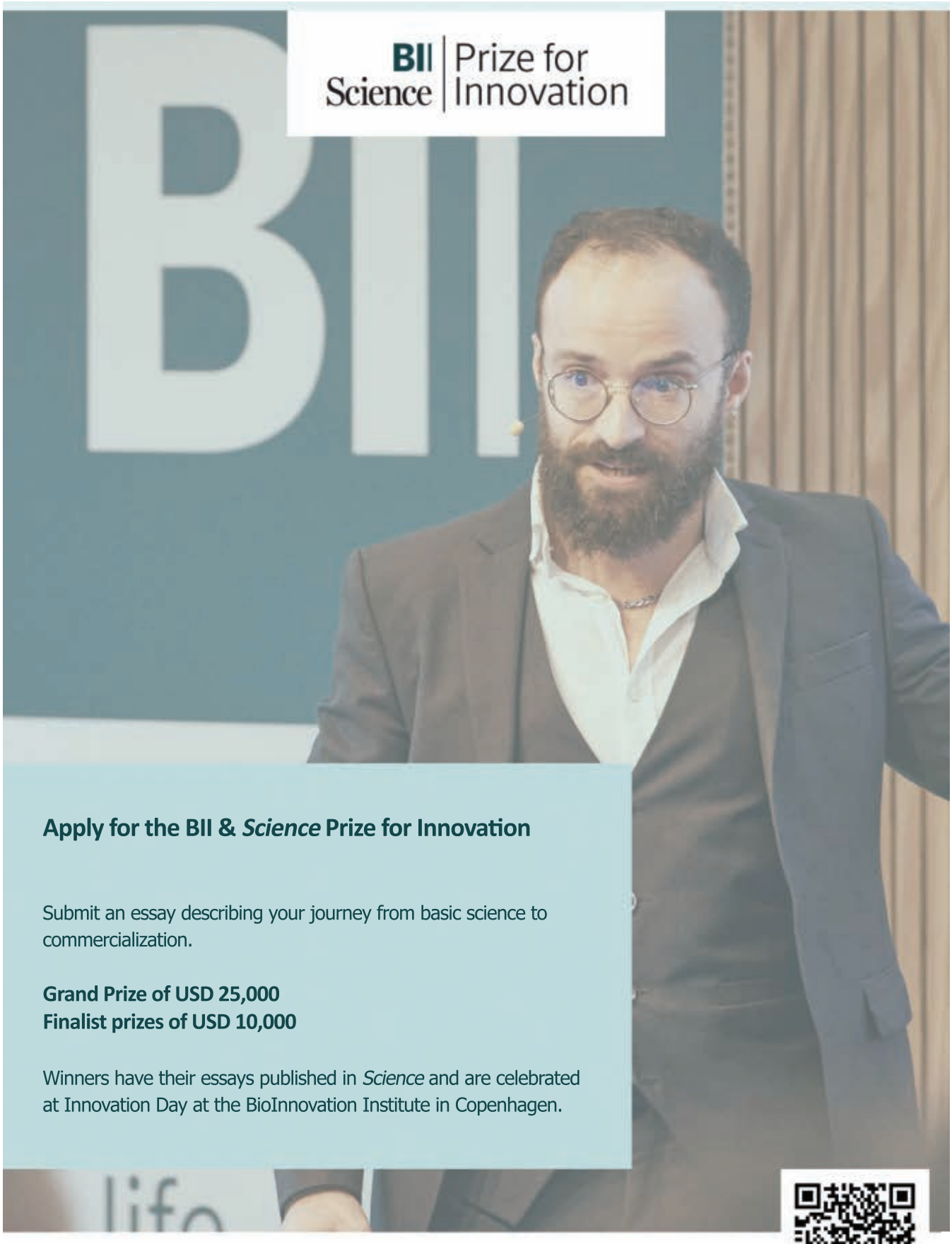

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MMX will land on Phobos twice, sampling the Manhattan-size moon's varied terrains.

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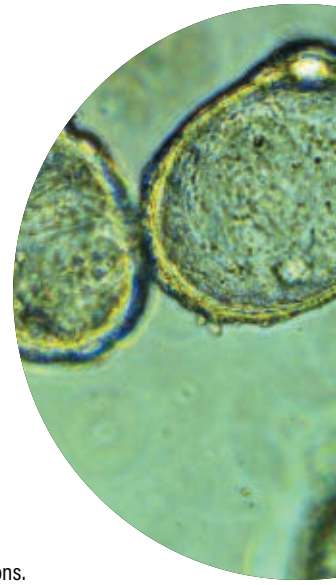
Is it time to bring organoids into your lab?

Five questions to help you decide whether your lab is ready to make the transition to organoid models.



Organoids are an increasingly important tool in biomedical research. According to Hannah Maple, senior director of organoids at Bio-Techne, organoids "more closely resemble native tissues and function in a physiologically relevant way." By preserving cellular diversity, they can generate "higher quality and a higher richness to the data," she says, with the potential to better predict human responses in applications ranging from drug discovery to personalized medicine. Those advantages—together with increasing encouragement from some funding agencies and regulators to adopt human-relevant alternatives to animal research—have prompted more laboratories to consider organoid models.

Yet many researchers remain hesitant to make the switch from conventional cell culture. Because the field is still evolving, researchers can encounter inconsistent protocols, limited standardization, and reproducibility concerns. "That can be quite daunting," says Maple. As the field matures, many of these challenges are becoming easier to overcome. How do you know whether your lab is ready? Start with these five questions.

1 How difficult is it to get started?

"There is a definite learning curve associated with that transition," says Didarul Bhuiyan, R&D manager for regenerative medicine at Bio-Techne. Small differences that have minimal impact in two-dimensional cultures can also become magnified in three-dimensional systems, where their effects can compound over weeks or months, says Bhuiyan.

The transition can be challenging, but today's researchers benefit from more technical support, more-standardized protocols, and better-validated reagents than were available when many organoid methods were first developed.

2 What's the biggest mistake new labs make?

Underestimating the importance of consistency can be costly. Maple points to neuronal differentiation protocols as an example. In some cases, "a 0.1 micromolar mistake" in reagent concentration can alter the outcome of a month-long experiment, she says.

"If there are variances in the results, you're going to go back and troubleshoot," Bhuiyan says. For experiments that can take weeks and rely on costly materials, investing in consistent reagents can help reduce the risk of lost time and repeated experiments.

3 How do you improve your chances of success?

Integrated workflows can make the transition easier by bringing together reagents, protocols, culture systems, and downstream analytical tools that are designed to work together, Maple explains. This can reduce variability and simplify troubleshooting.

For labs new to organoids, an integrated workflow can be particularly important; the goal is not to remove the complexity of the biology, but to make the experimental process more reproducible and manageable.

4 Why are analytical tools as important as culture?

In most cases, growing organoids is only half the battle. Researchers also need to use complementary analytical methods to validate that the model functions as intended.

Because organoids are complex systems that mimic aspects of an organ, no single assay can determine how faithfully they reproduce native biology, Bhuiyan explains. Researchers typically need multiple analytical approaches—imaging, transcriptomic or proteomic data, spatial biology, or functional assays —to confirm that the relevant cell populations, markers, pathways, and functions are present. Some platforms even provide automated analysis of organoids. Overall, validation should be considered alongside culture from the beginning.

5 Is your lab ready to make the switch?

Whether your lab is ready to make the transition depends on your research goals, resources, and experimental needs. But as protocols mature, workflows become more standardized, and technical support expands, the barriers to getting started are lower than they were even a few years ago. For many laboratories, the question is less about overcoming technical barriers and more about deciding whether organoid models are the right fit.

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R&D systems by bio-techne

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For more than a year, Hawai'i's Kīlauea volcano has pulsed with powerful—and forecastable—lava fountaining.

PHOTO: US GEOLOGICAL SURVEY, HAWAIIAN VOLCANO OBSERVATORY

ON THE COVER

Researchers address the settlement of Hawai'i and Rapa Nui—by way of the Mangareva Islands—and the resulting concentration of certain alleles in those populations. The hourglass shape of the seamount and volcano represents this genetic bottleneck, and the stars compose the constellation Scorpio, which voyagers recognized as Mānaiakalani or Māui's hook. The background emulates the art form of hand-stamped bark cloth, known as kapa in Hawai'i.
See page 216.
Illustration: Oliver Uberti

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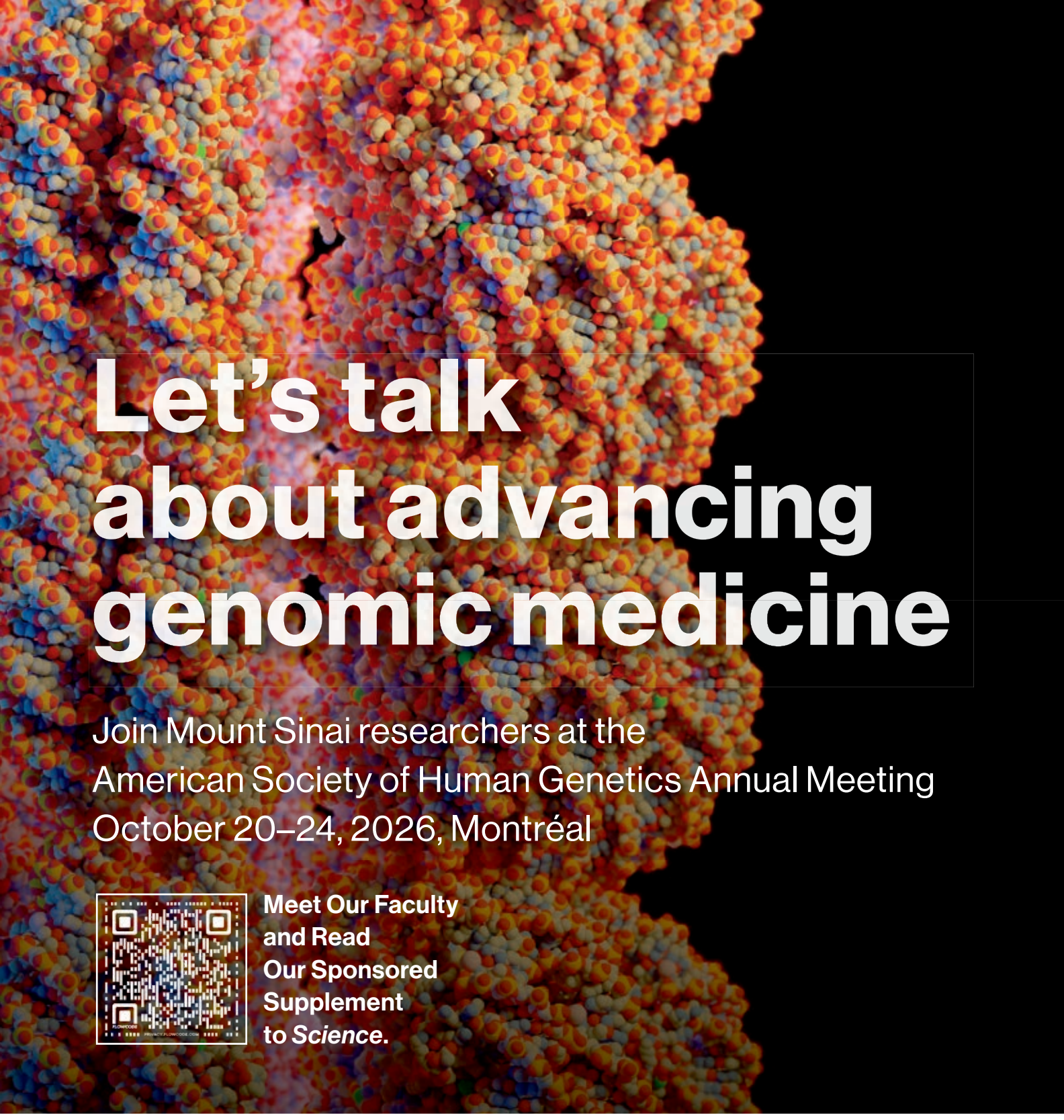

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Data everywhere, but are they fit for purpose?

Julia Lane and **Natasha Noy**

Vast amounts of new data are fundamentally shaping how people manage their lives and do their work. The volume is overwhelming—Google Dataset Search discovers 2 million new datasets every month—and so are the uses by scientists, governments, and the public. Governments across the world are treating data as a strategic asset. But data alone are not enough. Infrastructure must be developed that can provide signals about which data are "fit for purpose" for the increasingly vast number of potential uses. Informed and appropriate use is, after all, the ultimate objective of making data available for the public good.

A robust infrastructure design requires a core shift—moving away from technical information on data collection and production for specialists, and toward providing clarity on how suitable data are for applications by users. Four steps should be taken, each taking advantage of existing technology. One is that quantitative measures can be developed through rubrics that score data quality and fitness for purpose. There is a wealth of knowledge about elements of those rubrics from statistical agencies, the information management community, or the technology and strategy community, to name a few. These rubrics include general measures such as accuracy, completeness, and consistency at the time of data collection and creation. This information should be complemented by easy-to-understand labels that provide users—both experts and nonexperts—with a clear signal of the dataset's known limitations and suitability for a particular use.

We shouldn't all need to be data experts to make use of data.

Defining these measures will require vocabulary and methods to describe use cases for data and to ensure that metrics focus on use cases that users care about. For instance, although statisticians at the US Census Bureau are well aware of the intricacies of American Community Survey data, it would be useful to encode information about how local the analysis can get. Shared standards could help capture and convey fitness for purpose for users.

In addition, robust, repeatable, low-cost, and scalable methods are needed to generate these quantitative measures on the basis of information that can be gathered about uses of the data, signaling reliability, accuracy, reputation, and objectivity of the data. This is now possible because of modern tools to summarize and score users' experience with data from their writings. When data are used, they typically are presented in papers or in reports that can describe how data were used and what was found. Tools like large language models (LLMs) have been applied to large-scale text corpora to extract information on dataset usage, and what it says about the success of that usage. This approach allows for continual updating of data rubrics and measures as new writings and usage reports become available.

The third component is certification of data trustworthiness by experts. LLM-generated rubrics and metrics ought not be the final word. There could be formal certifications for high-profile datasets, just as the Underwriters Laboratory certifies consumer product test results so that everyday consumers do not need to be engineers. Less formal signals could be developed by communities of practice. Engaged communities can certify trustworthiness and could provide additional text documents to the corpora for LLMs to analyze, which could provide an evolving basis to assess fitness for purpose. Even a simple label of "Insufficient usage information" about a dataset could provide a valuable signal. Community development and knowledge sharing around data are increasingly a focus of national science agencies, like the US National Institutes of Health's (NIH's) S-Index data sharing initiative, UK Research and Innovation's investments in Future Data Services, and the German National Federal Data Infrastructure (NFDI).

It is also necessary to set up platforms to facilitate widespread community engagement and adoption of data-usage and fitness-for-purpose measures. Data repositories can add usage statistics and trust signals for datasets, as the US National Data Platform has done for a number of datasets. Scientists could extend industry standards for publishing metadata, such as schema.org and the Data Catalog Vocabulary, to include usage and fitness-for-purpose measures. Statistical harmonization platforms like Data Commons by Google now underpin public data discovery for the United Nations; data-discovery tools like those provided by the DataCite Usage tracker provide information about millions of datasets. These types of venues are where these quantitative measures can, and should, be disseminated. Publishing usage descriptors alongside other metadata would allow researchers to discover what uses are associated with specific data series; data fitness-for-purpose signals could do the same.

Tools and platforms exist to help broad swaths of users—people, firms, researchers, and governments—know what data can be trusted in different contexts. We shouldn't all need to be data experts to make use of data. □

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Reinventing crop protection

Climate change and tighter regulations are increasing demand for more sustainable ways to protect crops.

From precision proteins to beneficial fungi, a new generation of European startups is developing alternatives to conventional pesticides—and bringing them to market.

For more than a century, crop protection has relied on synthetic chemicals to control pests and diseases. Today, climate change and tighter regulations are driving demand for more sustainable approaches. In response, researchers and entrepreneurs are developing biological alternatives that promise greater precision and a lighter environmental footprint.

But scientific discovery is only the beginning. Bringing these innovations to farmers requires navigating manufacturing, regulation, and the realities of the marketplace. "The biggest hurdle is usually not the science; it's understanding that commercial success is governed by economics rather than novelty," says Jens Nielsen, CEO of the BioInnovation Institute (BII), an international commercialization foundation that supports promising life science and deep-tech startups. "Scientists naturally focus on demonstrating that something works. Entrepreneurs must demonstrate that it can be produced reliably, at scale, and at a cost customers are willing to pay."

Through its specialized accelerator programs, BII provides funding, business development support, and infrastructure to help early-stage companies translate promising discoveries into viable businesses. Two such startups, Yngvi Bio and Mycoverse, are working to bring their crop protection products to market. Yngvi Bio's journey to apply AI-guided protein design to pest control began with an unexpected email.

A partnership begins

Kenneth Veland Halberg, an associate professor at the University of Copenhagen who had spent the past 15 years studying insect physiology using fruit flies and flour beetles, was half-listening to a conference presentation when he checked his inbox. There was a new email: an unfamiliar researcher proposing a radical solution to the biological bottleneck he had been trying to solve for years.

Halberg's recent research had uncovered a peculiar insect survival mechanism—the ability to take up water vapor through the rectum to stave off fatal dehydration—that had sparked a flurry of media interest. "I have always wanted to understand how the most successful group of animals on Earth actually work," says Halberg, now co-CEO, CSO, and co-founder of Yngvi Bio, a biotech startup from BII's 2025 Venture Lab cohort. But the world desperately needs a way to tackle devastating insect pests, he says, and the chemical tools farmers have relied on for a century are poisoning the planet's soils and waterways.

The message came from Eva Maria Rebrova, an associate researcher at Lund University and now the co-CEO, CTO, and co-founder of Yngvi Bio. Only a few

PHOTO: © FOTOKOSTIC/SHUTTERSTOCK.COM

hundred meters from Halberg's office in Copenhagen, Rebrova had been working on phage-inspired delivery systems designed to inject therapeutic molecules into targeted cells. Having loaded these nanoscale injection systems with insecticidal payloads, she needed an entomologist to test whether these would deliver their cargo into living insects. "I began searching for anyone in Copenhagen working on insect physiology," she says. She found an article about Halberg's research on the university's website. "It was about how insects take up water with their bums," she remembers. Rebrova looked up his office address and grinned. "For nearly 5 years, we had been working a few hundred meters away from each other, completely oblivious to each other's existence." She drafted the email detailing her contractile injection prototypes and hit send.

Yngvi Bio co-founders Eva Maria Rebrova and Kenneth Veland Halberg combine expertise in biological delivery systems and insect physiology to develop precisely targeted insecticides.

A few months later, Yngvi Bio was founded, and in 2025 the startup joined BII's Venture Lab cohort to develop its business strategy.

Engineering precision

Unlike traditional chemical insecticides that often harm pollinators and beneficial soil organisms as well as pests, Yngvi Bio's first product targets grass grubs, destructive pests that live four to 12 centimeters underground. The pair designed a microscopic biological injection system that delivers a naturally derived protein toxin directly into target insects. "Remember the rocket from Tintin," Halberg says, referring to the Belgian cartoon. "[It's] basically a long cylinder with legs." Those "legs" act as molecular recognition hooks that bind to specific proteins on the surface of target cells. "Whenever they recognize the correct cell surface protein, [the injection system] will deploy the cargo and kill the insect," Halberg explains. "If it does not recognize that receptor, it will not deploy it; that means that no beneficial organisms will be harmed." Preliminary testing indicates that the system delivers the insecticide to grass grubs only, leaving earthworms and pollinators in the same environment unharmed.

Precision alone, however, is not enough to build a successful agricultural product. The team recognized that environmental benefit alone will not drive adoption. Farmers, operating on razor-thin margins, will not adopt an eco-friendly solution solely because it is gentle on the earth; it must perform as well as, or better than, petrochemical alternatives, and it must compete on price, Halberg explains. Because the active payload consists of protein, it degrades into natural amino acids after application rather than persisting in the environment. At the same time, the precision targeting system requires much lower doses per hectare than conventional pesticides, Halberg says, helping make industrial fermentation economically competitive.

"Many technically elegant solutions have failed because they could not achieve the required productivity, yield, and manufacturing costs to compete with established industrial processes," says Nielsen. "Building a company requires assembling multidisciplinary teams, engaging customers, raising capital, and accepting that market realities will shape the technology."

To bring the technology into the field, Yngvi Bio is pursuing a commercial partnership with DLF (Dansk Landbrugs Frøselskab), the world's largest grass seed company, to test their prototypes in real-world environments. They are also collaborating with multinational partners such as BASF (Badische Anilin- und Soda-Fabrik) and Syngenta to evaluate candidate treatments.

Mining the fungal kingdom

While Yngvi Bio engineers target biological insecticides, Mycoverse is searching nature for its own crop protection solutions. Drawing on an exclusive collection

PHOTO: ESBEN ZØLLNER OLESEN FOR BIOINNOVATION INSTITUTE (BII)

of more than 14,000 fungal strains assembled over four decades at the Technical University of Denmark (DTU), the company screens naturally occurring strains for compounds that can combat devastating plant diseases.

Niels Bjerg Jensen, the CTO and co-founder of Mycoverse, spent over two decades in industrial biotechnology before DTU asked him to lead a collaborative project with FMC Corporation to identify fungal strains capable of controlling plant diseases. Recognizing the commercial potential of the repository, Jensen co-founded Mycoverse with CEO Svend Petersen, an industry veteran with nearly 30 years of commercial leadership.

Mycoverse holds an exclusive commercial license to the DTU collection, which was assembled from an extraordinary range of environments. "Fungi are everywhere," Jensen says. "At DTU, they have been collected from soil samples, from agriculture, from forest, DTU campus bathroom tiles, from indoor samples, food, above soil, below soil."

Mycoverse has converted the fungi collection into microplates that a robotic system can screen against target agricultural pathogens within weeks. When a fungal strain demonstrates activity against a plant disease, Mycoverse uses computational algorithms to identify the bioactive compounds produced by the fungus during its metabolic processes. At the same time, genomic profiling flags any strain that also produces harmful mycotoxins, allowing the team to eliminate unsafe candidates early in development. "Those we can deselect and then go on with those that we can actually scale and commercialize," Jensen explains.

Mycoverse's current target is potato late blight, a devastating disease caused by the fungus-like oomycete pathogen *Phytophthora infestans*. Following initial screening campaigns, the company has advanced lead candidates into standardized Good Experimental Practice (GEP)-certified field trials across Europe. Jensen attributes this speed to a deliberate strategy to build the company with commercial viability in mind, including hiring a chief regulatory officer right at the start. "In the deep tech and agricultural space, regulatory hurdles are the ultimate company killer. That, I think, makes us stand out, considering how young we are as a company."

Nielsen agrees: "At BioInnovation Institute, we encourage founders to think about commercialization from the beginning rather than treating it as something that happens after the science is finished. The most successful companies build biology and business in parallel."

The road to approval

The need for biological crop protection is growing as environmental regulations restrict the use of synthetic pesticides. In the European Union, regulatory

Members of the Mycoverse team are searching a collection of more than 14,000 fungal strains for new biological approaches to crop protection.

PHOTO: ESBEN ZØLLNER OLESEN FOR BIOINNOVATION INSTITUTE (BII)

BioInnovation Institute CEO Jens Nielsen says successful biotechnology companies must build biology and business in parallel.

PHOTO: ESBEN ZØLLNER OLESEN FOR BIOINNOVATION INSTITUTE (BII)

restrictions on synthetic pesticides are reducing the chemical tools available to growers. "Farmers will be in big trouble if we don't manage to be there in time with new, effective, and cost-effective biologicals," warns Jensen.

Developing effective biological products, however, is only part of the challenge. For new platforms like those developed by Yngvi Bio and Mycoverse, bringing products to market requires rigorous regulatory oversight. Because these technologies introduce novel biological approaches to crop protection, regulators must verify that they are safe for biodiversity and the broader environment. For Yngvi Bio, that means demonstrating that its targeted delivery system affects only intended pests, not beneficial organisms. For Mycoverse, it means showing that fungal-derived products do not introduce harmful mycotoxins.

But navigating the regulatory framework can be lengthy and complex. "In the U.S. and in the EU, the regulations are similar, but the mechanisms for accelerating review are different," says Rebrova.

In the United States, regulatory agencies operate according to defined review targets, providing applicants with clear expectations about the decision-making process. In the EU, procedures also have formal timelines, but reviews can be prolonged by requests for more information and coordination among institutions and member states. Both Rebrova and Halberg point to the need for controlled, time-limited testing environments where innovators and regulators can evaluate novel biosolutions together. Such programs would help clarify regulatory requirements, generate real-world data, and potentially accelerate approval.

Jensen remains optimistic about Europe, betting that farmers stripped of their traditional chemical options will force the EU to modernize its workflows. But he acknowledges that the long road to market currently makes securing venture capital incredibly difficult.

Nielsen watches these regional dynamics with a mixture of concern and realism, seeing the issue as one of competitiveness rather than scientific capability. "Europe has outstanding science and exceptional talent in biotechnology. The challenge is ensuring that innovation can translate into competitive companies," he says. "Compared with the U.S. and increasingly China, Europe has often been slower in areas such as regulatory approval, scaling manufacturing, and attracting later-stage investment. If the regulatory process becomes unnecessarily complex or unpredictable, companies may choose to commercialize elsewhere."

Yet, Nielsen cautions against dismantling safety protocols in a reckless race for speed. "Regulation exists for good reasons. Industrial biotechnology must meet high standards for safety and sustainability, and public trust is essential," he says. "The key is finding the right balance: maintaining rigorous, science-based regulation while creating faster, clearer pathways for innovations that can demonstrably reduce emissions or improve sustainability."

One molecule at a time

Biology can make industry more resilient—not by replacing every chemical process, but by producing complex molecules that conventional chemistry struggles to manufacture sustainably, says Nielsen, CEO of BII, which helps researchers and entrepreneurs bring life science and deep-tech innovations to market. He sees companies like Yngvi Bio and Mycoverse as evidence of biology's broader industrial potential. "One of the best examples is the production of vitamin B2 (riboflavin)," he says. "For decades, it was manufactured through a complex, energy-intensive chemical synthesis involving multiple reaction steps. Today, it is produced almost entirely by microbial fermentation. That biological process has dramatically reduced energy use, water consumption, and CO_2 emissions while completely replacing the traditional chemical route."

This "micro tackling macro" philosophy runs throughout BII's portfolio. "I don't believe biology will replace every petrochemical process. Biology excels at producing complex molecules that conventional chemistry struggles to make efficiently. These include ingredients for healthier, more sustainable foods, biological alternatives for crop protection, such as insect pheromones, and specialized chemicals with significantly lower environmental footprints. That is where industrial biotechnology can create both environmental and economic value," he says.

By focusing on these highly specialized domains, these young companies are demonstrating how small biological innovations can deliver outsized environmental impact.

"What gives me optimism is the growing number of companies using biology where it offers unique advantages," he says. "Climate solutions will not come from a single breakthrough. They will come from thousands of better technologies finding the right applications. Biology is becoming one of the most powerful tools we have for making those industries more sustainable, one molecule at a time."

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TRUMP ADMINISTRATION

‘Complete devastation’: How NSF gutted its social sciences directorate

The directorate made 90% fewer new grants this year than average and hemorrhaged two-thirds of its staff

DANIEL GARISTO

Social scientists in the United States have known for months that their main source of research funding was threatened, but were unsure how dire the situation was. With the end of the 2026 fiscal year on 30 September, the extent of the damage is now becoming clear. “For entire disciplines, this is existential,” says Wendy Naus, head of the Consortium of Social Science Associations, a nonprofit representing social scientists.

In March, the White House’s budget request for fiscal year 2027 called for eliminating the U.S. National Science Foundation’s (NSF’s) Directorate for Social, Behavioral and Economic Sciences (SBE), which funds roughly two-thirds of the country’s academic social science research, such as sociology, linguistics, and archaeology. Whether Congress will approve the request is unclear, but NSF is not waiting. Interviews with five recently departed SBE staff and newly available data document how the agency has quietly followed through on administration orders by slashing SBE’s budget, throttling its decision-making, and reducing its workforce.

Every part of NSF has lagged in grantmaking: The agency has underspent its $8.8 billion budget for this year by some $1.4 billion and awarded about 6200 new grants, the lowest number in decades. But SBE was uniquely paralyzed. It spent just $95 million for 2026, a 62% decline relative to the $247 million average between 2021 and ’24, and awarded just 91 new grants, an 89% drop from its average of about 800, according to data compiled by Grant Witness, a watchdog nonprofit. An analysis by *Science* found more than 30 program areas—including archaeology, anthropology, and sociology, which typically make hundreds of new grants—distributed zero new awards.

The dismantling of SBE was not limited to research funding. In September 2025, SBE had 30 staff involved with grantmaking; about a dozen are left after attrition and transfers to other directorates. On 29 September, NSF internally announced that the SBE directorate head, Christina Freyman, had left to lead another directorate. Recently departed staff requested anonymity to candidly discuss what has happened at SBE. “It is complete devastation,” one former staffer says.

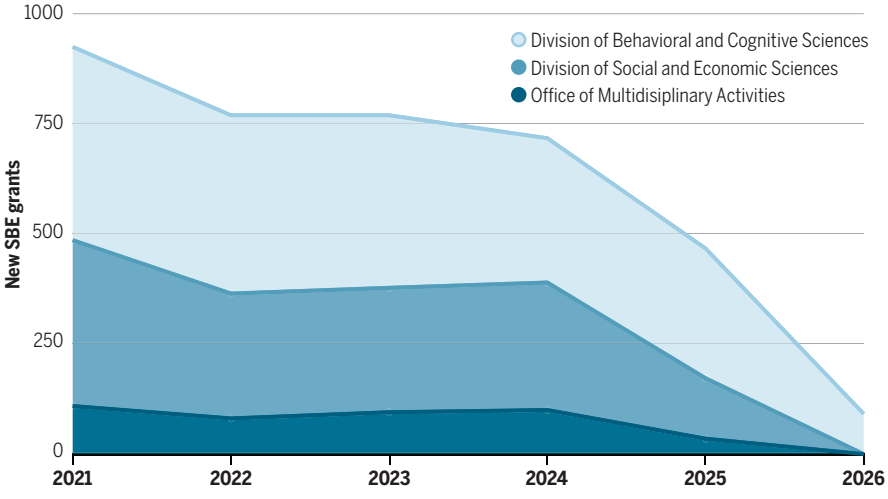
In a statement, a spokesperson for NSF says the agency “has been actively engaging in prudent planning and taking steps to implement the President’s agenda for FY 2027. … The SBE Directorate has not been shut down.”

NSF did not always plan to cull the directorate. A draft budget created by agency leadership early in the fiscal year, obtained by *Science*, would have boosted SBE’s 2026 budget by 30%, with $60 million designated for artificial intelligence (AI), one of the Trump administration’s priorities. But multiple former staff told *Science* how the agency has reversed course under pressure from the administration, throttling traditional grantmaking at SBE and diverting money to other initiatives favored by the White House.

In the past, panels of independent scientists vetted proposals; program officers recommended those found worthy, and section heads approved them. But a new, unofficial process has removed decision-making power from all directorate staff and leadership. Only a few proposals in

Grantmaking gutted

The National Science Foundation’s Directorate for Social, Behavioral and Economic Sciences (SBE) awarded far fewer new awards than typical in fiscal year 2026—just 91. (The fiscal year runs from 1 October to 30 September.) Most of the new grants focused on cognitive science and artificial intelligence, in alignment with White House priorities; offices that support fields such as economics, sociology, and research ethics, as well as postdoctoral fellowships, awarded no grants at all.



CREDITS: (GRAPHIC) C. BICKEL/*SCIENCE*; (DATA) NOAM ROSS/GRANT WITNESS

administration priority areas, such as cognitive science and AI, were allowed to move ahead, according to the former staffers. Program officers sent them to Freyman, who met with Simon Malcomber, NSF's chief science officer. "Then he would come back with some decision," a former staffer explains. "So it wasn't his decision-making. He was in consult with others." If Malcomber said a proposal was acceptable, a program officer was given the green light to officially recommend it. "They were very, very careful in terms of never putting anything in writing about what the process was," the former staffer says. NSF did not respond to questions about the process.

Science shared details of the changes at SBE with Alondra Nelson, a social scientist and former member of the National Science Board, which oversaw NSF until it was terminated in April. "A NSF social science directorate that only asks the questions the White House wants asked is not a social science directorate. It is a research arm of an administration," she says.

Researchers who learned about the unofficial process also expressed frustration at having been in the dark. "People were still preparing proposals, not knowing that they were essentially ineligible. I think that's really disrespectful," says Christina Warinner, a biomolecular archaeologist at Harvard University.

The situation is unlikely to improve, staff say. The flagship Graduate Research Fellowship Program will exclude students in the social sciences for the upcoming application cycle, and NSF's August call for new grant proposals left out SBE. No SBE-specific calls for grants remain open, and no review panels are scheduled, although about 600 proposals are still pending with SBE. Some researchers, like Aaron White, a computational linguist at the University of Rochester who sought a grant in January, have been told their proposals are on hold, in the event additional funding opens up.

Social scientists have few funding options outside of SBE. Philanthropic funding is extremely limited and what few funding opportunities exist at other government agencies are likely to go through similar scrutiny. Recently, some AI companies have offered millions in funding for social science focused on AI impacts. "It's great that companies are putting down, I'm sure, what is a drop in the bucket for them," says Jon Freeman, a psychologist at Columbia University. "That doesn't absolve NSF of its responsibility to follow through with its congressional mandate of funding the social and behavioral sciences." □

Daniel Garisto is a science journalist in Syracuse, New York.

IN FOCUS

Untangling the chaotic galactic center

A black hole weighing 4 million times the mass of the Sun makes for an uneasy neighbor. The heart of our Milky Way Galaxy is a tangle of gas, dust, and stars, all embroiled in the powerful magnetic and gravitational fields of the central, supermassive black hole. Near the black hole is a supernova remnant called Sagittarius A East (Sgr A East), an expanding cocoon of gas and dust blown out from a dying star. Using images from NASA's Chandra X-ray Observatory, astronomers teased out which photons came from which source, they reported in *The Astrophysical Journal*. The resulting composite image, unveiled last week (below), shows how light from Sgr A East (light blue and purple) rubs up against emissions from hot gas around the black hole (yellow). Surrounding red and blue areas are gas clouds imaged by the Very Large Array radio telescope and James Clerk Maxwell Telescope. —Daniel Clery

PHOTO: NASA/CXC/MCGILL UNIVERSITY/M. BALAKRISHNAN *ET AL.*; NSF/NRAO/VLA; EAO/JAMES CLERK MAXWELL TELESCOPE; NASA/CXC/SAO/P. EDMONDS, N. WOLK

NOBELS

LIGHTS, NEURONS, ACTION!

Three scientists who developed a way to precisely switch brain cells on and off with light have been honored with this year's Nobel Prize in Physiology or Medicine. The method, known as optogenetics, has clarified nerve circuits in healthy and diseased brains. It has also been adapted to help a few blind people regain modest sensitivity to light and is being tested as a treatment for epilepsy, pain, and more.

Peter Hegemann of the Humboldt University of Berlin and Georg Nagel of the Julius Maximilian University of Würzburg paved the way for the technique. In the 2000s, the pair discovered that a protein in the single-celled alga *Chlamydomonas* responds to light. Under a blue light, a channel in the protein opens, allowing charged ions into the cells. They called the protein channelrhodopsin.

Karl Deisseroth of the Howard Hughes Medical Institute and Stanford University subsequently introduced the gene for channelrhodopsin into rodent brain cells, creating a light-activated switch that produces a nerve signal. The neurons reacted on command, within milliseconds. Previous neuron-activating techniques that relied on electrical stimulation are often described as "playing a piano with a mallet," says neuroscientist Andrew Jackson of Newcastle University. "You got what you wanted, but you hit the surrounding notes as well."

Optogenetics, by contrast, is far more precise. It allows researchers to activate specific types of neurons in living animals and trace out brain circuits involved in processes such as memory formation or pain. In 2021, the technique was used to restore some vision to a blind man who had lost the cells in his retina that transform light into electrical signals.

Optogenetic studies of lab animals have also helped researchers understand the brain circuits stimulated by deep-brain implants used to treat Parkinson's disease. Deisseroth says he hopes the tools will help identify the cells underlying the symptoms of psychiatric disorders, which could lead to medications precisely targeted to those pathways. —*Science* News Staff

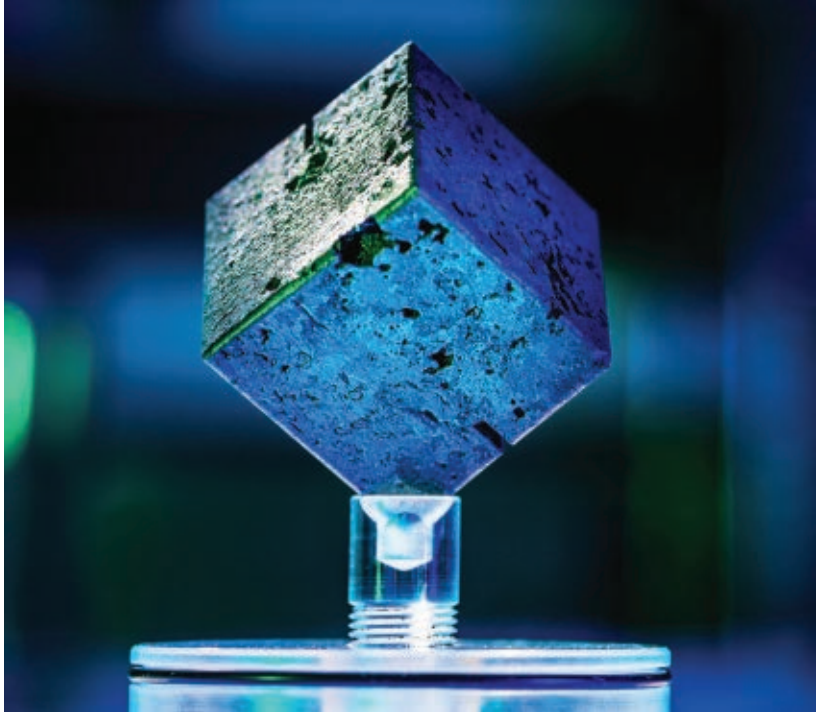


PHOTO: JOHN T. CONSOLI/UNIVERSITY OF MARYLAND

HISTORY OF SCIENCE

Nazi Germany had no hope of making an atomic bomb, uranium cubes reveal

Experimental reactor needed much more uranium and heavy water than the country could muster **RICHARD STONE**

One morning in August 2013, Timothy Koeth was out for a jog on the University of Maryland (UMD) campus in College Park when the university's radiation safety officer called and asked to see him urgently. The materials scientist, "all sweaty and out of breath," he recalls, met the officer in a parking lot, where she opened her car's trunk and produced a lunch bag. Inside, nestled in paper towels, was a gray cube, 5 centimeters wide and startlingly dense.

"She says, 'Do you know what this is?' And I said, 'Yeah, but do you know what this is?'"

Confirming Koeth's suspicions was a handwritten note accompanying the cube: "Taken from the reactor that Hitler tried to build." The officer handed it over and wished him a happy birthday. "She knew my birthday was in a few days," he says.

The safety officer had recovered the cube of Nazi uranium from the drawer of a deceased geology faculty member whose office was being cleared out, Koeth says. That unexpected gift helped propel an investigation that would overturn Koeth's own understanding of Germany's nuclear effort during World War II. He and colleagues once thought the Germans could have built a working nuclear reactor—a step toward an atomic bomb—by pooling their scattered supplies. Now, measurements of surviving uranium cubes, archival research, and computer simulations indicate that even Germany's combined stocks would have been insufficient.

The study, published last week in *PNAS Nexus*, challenges a postwar claim by Werner Heisenberg, the Nobel Prize–winning physicist who led Germany's final reactor experiment, B8. He suggested a small addition of uranium would have allowed it to sustain a nuclear chain reaction.

A working reactor could have produced plutonium, one option for making a bomb. For wartime Germany, it was the only realistic path, says Malte Göttsche, a physicist at the Technical University of Darmstadt and the Peace Research Institute Frankfurt. German scientists investigated the alternative—uranium enrichment—but "Germany's efforts remained limited to

Uranium cubes recovered from Nazi Germany's nuclear program have helped reveal how far it was from a working reactor.

some research," he says. B8's immediate goal was simply to sustain a chain reaction; success would have left Germany far from producing enough plutonium for a weapon.

B8 was assembled in the closing weeks of the war in Haigerloch, a town in southwestern Germany. Inside a cellar laboratory, Heisenberg's team suspended 664 uranium cubes in chains and immersed them in heavy water.

As uranium nuclei split, they release neutrons that can trigger further fissions. If too many neutrons escape from the reactor or are absorbed without causing fission, the chain reaction dies out. Heavy water—in which ordinary hydrogen is replaced by deuterium, a form with an extra neutron—slows neutrons, making them more likely to split the fissile isotope uranium-235. It also absorbs fewer neutrons than ordinary water. In addition, the German assemblage had a surrounding layer of graphite that reflected escaping neutrons back toward the fuel.

Yet B8 fell short. The question was how far.

Seeking to trace the cube's history, Koeth searched the National Archives at College Park. There he found records of hundreds of additional uranium cubes that had belonged to a competing German team and later circulated on Europe's postwar black market. In a 2019 *Physics Today* article, he and materials scientist Miriam Hiebert, also at UMD, argued pooling Germany's supplies could have made a working reactor possible. Germany had enough material, he thought. "They just didn't have it all in the same place."

Testing that interpretation required a better reconstruction of the materials the Germans actually used. Koeth and Brittany Robertson of Pacific Northwest National Laboratory had two cubes to interrogate: the one Koeth received in 2013 and another he had found in 2011 in a filing cabinet. One cube's density revealed internal porosity; the other's isotopic composition confirmed it was natural uranium rather than enriched fuel.

Meanwhile, the team assembled historical evidence about B8's heavy water and graphite. The study's first author, Patrick Park, a doctoral student now at Princeton University, used those findings to make a model of the reactor that reproduced the German team's reported neutron measurements.

The modeled B8 had a neutron multiplication factor of about 0.94: For every 100 fission neutrons in one generation, only about 94 were spawned in the next. A self-sustaining reaction, in contrast, requires every 100 fission neutrons to steadily spawn 100 more. The neutron shortfall concealed a larger materials shortfall. In one modeled reactor configuration, achieving criticality would have required about 3.1 tons of uranium and 3.5 tons of heavy water, compared with the roughly 1.5 tons of each B8 held. Those requirements exceeded the authors' estimate of the entire German program's uranium cube inventory by about 20%, and its heavy water inventory by about 80%.

By the time B8 was built, Allied attacks had disrupted Germany's heavy water supply. The material came from a factory in occupied Norway powered by Norsk Hydro's hydroelectric plant. Norwegian commandos sabotaged the heavy water equipment in 1943; production resumed, but U.S. bombers struck the complex that November.

Could graphite have offered a way around that shortage? It can also slow neutrons, but impurities can absorb them before they trigger further fissions. Using inferred impurity levels, the researchers found that German graphite combined with the program's uranium could not have substituted for heavy water in a working reactor.

Göttsche, who was not involved in the study, finds the evidence persuasive. "It draws a consistent picture across the measured evidence, the simulated results, and what is contained in the archives," he says.

Did Heisenberg deliberately hold back to keep a nuclear weapon out of Hitler's hands? After Germany's defeat, he suggested reluctance to help the Nazis win had contributed to the program's failure. The calculations cannot establish his intentions, but Koeth says he has found no evidence of such restraint. "I haven't seen anything supporting that account, except for sort of convenient suggestions later in his life."

To Koeth, the surviving relics of the Nazi reactor embody the threat that helped spur the Manhattan Project—and the nuclear age that followed. "From a historical point of view," he says, "they carry the entire burden of the nuclear enterprise that we have today." □

BIOMEDICINE

New studies probe how menopause puts brain at risk

Shifts in blood proteins could point to ways to fight menopause "brain fog" and prevent later dementia

JENNIE ERIN SMITH

Women are more likely than men to develop dementia, and not just because they tend to live longer. One reason may be menopause, the time in midlife when sex hormones drop and menstruation stops—and which some researchers are coming to see as a stress test for the brain.

Menopause "affects brain health, for sure," says neurologist Riley Bove of the University of California (UC) San Francisco, who studies its impacts on multiple sclerosis and other neurological disorders. She and others want to protect the brain during and after this transition—but that means identifying just how menopause puts the brain at risk.

A new study has found clues in the blood of women in their 40s and 50s: proteins related to brain function, inflammation, and neurodegenerative disease that are elevated during menopause. Other ongoing studies are following groups of women over time—including during the lesser studied perimenopausal stage, from just before menstruation ends to 1 year after—to reveal processes leading to short- and long-term cognitive problems.

Menopause is generally thought of in relation to the ovaries, which stop producing estrogen and progesterone. But the brain, which also produces hormones and is full of hormone receptors, plays key roles in the transition. Levels of follicle-stimulating hormone (FSH), produced in the pituitary gland, rise as sex hormones fall. Specialized neurons in the hypothalamus, deprived of estrogen, cause blood vessels to dilate, resulting in so-called vasomotor symptoms including night sweats and hot flashes.

And the brain itself is vulnerable. Severe or frequent vasomotor symptoms are associated with white matter hyperintensities, a type of brain lesion linked to dementia and stroke, as well as markers of Alzheimer's disease pathology. Mood and sleep disturbances worsen in tandem with vasomotor symptoms, as do memory and other cognitive issues often collectively described as "brain fog." Epidemiological and autopsy studies have shown that earlier menopause, whether spontaneous or caused by surgery to remove the ovaries, is associated with a higher risk of Alzheimer's.

But because menopause is an inevitable product of aging, the main risk factor for dementia, it's hard to clearly separate its effects from those of age, Bove says. Few brain banks or long-running studies have

NOBELS

A PRIZE FOR GHOSTS IN THE ICE

Francis Halzen, the University of Wisconsin–Madison theoretical physicist who turned 1 cubic kilometer of Antarctic ice into a telescope for the universe's most elusive particles, has won this year's Nobel Prize in Physics. Halzen led the construction of the IceCube Neutrino Observatory, which has traced high energy neutrinos to powerful cosmic accelerators and opened a new window on the universe.

Neutrinos are produced in nuclear reactions and stream almost unimpeded through matter. Trillions from the Sun pass unnoticed through our bodies every second. But some astrophysical objects, including exploding stars and supermassive black holes, produce extremely energetic neutrinos. Unlike electrically charged cosmic rays, whose paths are bent by magnetic fields, neutrinos travel in straight lines, pointing directly back to their sources.

In the 1980s, Halzen realized the clear ice beneath the South Pole could serve as a giant neutrino detector. Strings of light-sensitive phototubes could spot the streaks produced in the rare cases when a neutrino collides with an atomic nucleus in the ice and creates a charged particle such as a muon or an electron. Halzen first tested the idea with the smaller Antarctic Muon and Neutrino Detector Array, which operated from 1996 to 2005. IceCube, completed in 2010 at a cost of $279 million, expanded the concept to a 1-cubic-kilometer array of 91 strings stretching 1.5 to 2.5 kilometers into the ice.

In 2013, IceCube announced the detection of 28 neutrinos from beyond the Solar System. Five years later, it traced a high energy neutrino to a supermassive black hole launching a powerful jet some 3.7 billion light-years away. IceCube has since detected neutrinos associated with a black hole tearing apart a star and, more recently, mapped the Milky Way in neutrinos.

But neutrino astronomy is still in its infancy, Halzen says. "This is just an introduction." —Adrian Cho

detailed data on menopause, least of all from people actively going through it.

As one way to probe the biology behind the elevated risks, researchers led by neuropsychologist Kaitlin Casaletto of UC San Francisco studied a cohort of 80 healthy women in the premenopausal, perimenopausal, or early postmenopausal phase who were not taking hormones to treat menopause symptoms. They identified a molecular "signature" consisting of 16 brain-related proteins that were elevated in the blood of women after menopause.

Two of these proteins are associated with Alzheimer's, and several more play roles in neuronal or synaptic health, the researchers reported 22 September in *Nature Medicine*. And inflammatory signaling proteins called cytokines were especially high among women who reported night sweats. Madeline Wood Alexander, a Ph.D. candidate at the University of Toronto and the study's first author, says the findings suggest a "persistent inflammatory response could turn on during the menopause transition," creating trouble for the brain.

Levels of the brain-related proteins correlated with hormone levels—particularly with higher FSH or lower estradiol, the form of estrogen produced by ovaries. Mouse studies have suggested FSH affects neurons in a way that accelerates Alzheimer's pathology, and that blocking the hormone could stop this process. "Women were always told it's the loss of estrogen and progesterone that matters," says neuroscientist Emily Jacobs of UC Santa Barbara, a co-author on the paper. "But it could be that it's really the pituitary hormone that's the problem."

The researchers also looked at existing health data from some 12,000 women in their 60s and early 70s, and found that many of the same proteins were elevated among those who had developed cognitive decline or dementia.

The work provides "some clues of where we should look" among hundreds of possibly relevant proteins that change during menopause, says women's health researcher Rebecca Thurston of the University of Pittsburgh, who was not involved with the study. The elevated synaptic proteins could help explain the cognitive changes and symptoms some people experience, she thinks. But the group of menopausal women in the study was small, she cautions, and protein levels in the blood can fluctuate, so longer, larger studies are needed.

Thurston and her team are planning one. They are recruiting hundreds of healthy women in or approaching perimenopause for a 5-year observational study that will look at how different brain imaging and blood biomarkers correlate with cognitive symptoms such as brain fog. The Ann S. Bowers Women's Brain Health Initiative, a research consortium that includes Jacobs, Casaletto, and Wood Alexander, is also launching a study in healthy women to collect brain imaging, cognitive testing, and biomarker data across the menopause transition and beyond.

Casaletto thinks her team's findings, though preliminary, may already have implications for health care. A blood test detecting these protein shifts in midlife could signal who may be at risk for early cognitive decline and prompt diet and lifestyle changes or medical treatment to avert it. "Midlife is this critical window where we can have probably the biggest effects to set up the brain for success," she says.

Menopausal hormone therapy to replace estrogen, progesterone, or both, may seem like an obvious treatment. Research has already suggested brain benefits; one recent study showed those taking the hormone therapy had less brain amyloid, a signature Alzheimer's pathology. But confounding factors can cloud retrospective studies, so Wood Alexander and her colleagues plan to follow people on the therapy during perimenopause, comparing their protein signatures with those of women not taking hormones for signs of healthier aging.

Future therapies may target FSH; researchers at the Icahn School of Medicine at Mount Sinai recently made an antibody that blocks the hormone, which they hope to develop into a drug that can prevent Alzheimer's.

Other potentially brain-protecting interventions don't involve hormones. The newest drugs to treat hot flashes and night sweats target receptors for neurokinins, peptides that act on the specialized neurons in the hypothalamus. Thurston and fellow menopause researcher Pauline Maki of the University of Illinois Chicago are running a clinical trial to learn whether one of them, a drug called fezolinetant, leads to better brain health and cognitive performance during the transition.

Menopause is "a complex multisystem reprogramming" that researchers are just beginning to unpack, Wood Alexander says, but she cautions against painting this natural transition as purely negative. "All women with intact ovaries will go through spontaneous menopause in midlife, but most do not go on to develop Alzheimer's disease," she says. Researchers "can use that window to understand who's on a vulnerable trajectory versus a more resilient trajectory." □

BIOLOGY

How did Rosalind Franklin miss the helix in her iconic DNA image? She didn't

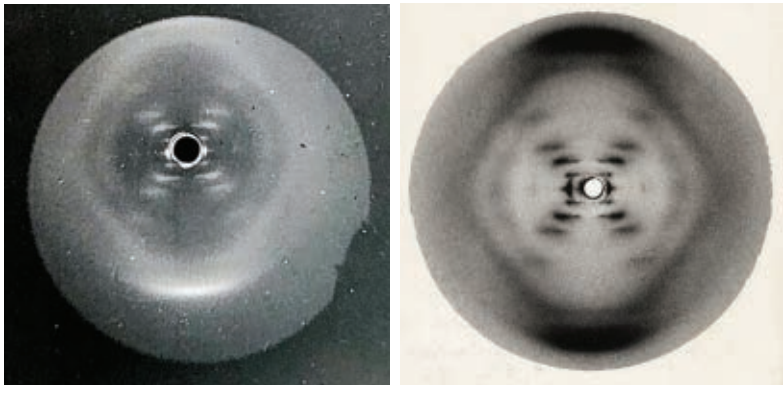
Analysis refutes James Watson's claim he was first to recognize significance of famed photograph 51

TIM APPENZELLER

"The instant I saw the picture my mouth fell open and my pulse began to race." So wrote James Watson in *The Double Helix: A Personal Account of the Discovery of the Structure of DNA*, his controversial account of how he and Francis Crick deciphered the structure of DNA in 1953. The image, now known as photograph 51, showed an elegant, symmetrical pattern of spots formed by x-rays diffracted from strands of DNA—the fingerprint of the molecule's helical structure. Rosalind Franklin, an x-ray crystallographer at King's College London (KCL), had produced the image in the spring of 1952. But Watson's 1968 book, amid other slights and sexist language directed at Franklin, implied the flash of insight was his alone.

Popular accounts still routinely suggest Franklin missed the significance of the photo. She never had a chance to challenge Watson's dismissive narrative; she died of cancer in 1958, a decade before his memoir was released and became a bestseller. But a new analysis of her notes on another x-ray image, which she made just days earlier using the same DNA sample, offers a firm rebuttal. Science historian Alistair Sponsel of the Science History Institute and crystallographer Brian Sutton of KCL argue Franklin knew the earlier photo had revealed a helix—the epiphany Watson claimed for himself—and that is precisely why she made a second, more refined version, photograph 51.

Watson's slight took hold partly because Franklin's lab notes, usually thorough and meticulous, say little about photograph 51. But in a paper published this week in the *Journal of the History of Biology*, Sponsel and Sutton draw attention to the earlier image, photograph 49, showing she made extensive notes about it despite a cosmetic defect. She recognized it suggested a helical structure and used it to calculate, for the first time, the length of each turn in the spiral—34 angstroms—and the number of bases "per turn of helix (if there is a helix)."



Rosalind Franklin's photograph 51 (right) revealed DNA's helical structure. But she had already recognized it in an earlier, flawed image, photograph 49 (left).

IMAGES: (LEFT TO RIGHT) ALISTAIR SPONSEL/KING'S COLLEGE LONDON ARCHIVES/DEPARTMENT OF BIOPHYSICS/GLASS AND ACETATE SLIDES (1949–84) KDBP/1/1/0868; SCIENCE HISTORY INSTITUTE

No wonder Franklin recorded little in her notes about photograph 51, Sponsel and Sutton argue. She saw no need to repeat herself. Franklin "knows exactly what's in front of her," says Betty Smocovitis, a science historian at the University of Florida and co-editor of the *Journal of the History of Biology*.

Others who study the history of DNA research agree. "It is a strong refutation of claims that imply that Franklin didn't know what she had, or that Watson saw something she and [her student Raymond] Gosling didn't," says Nathaniel Comfort, a science historian at Johns Hopkins University who is working on a biography of Watson. Smocovitis puts it more starkly: Sponsel and Sutton's analysis "adds justice, that's what it adds."

The new paper does not challenge Watson and Crick's achievement. Their model of DNA, rushed out in a brief article in *Nature* after Watson caught his glimpse of photo 51, went well beyond Franklin's understanding. Its two strands running in opposite directions and zipperlike arrangement of complementary base pairs explained how the molecule could carry genetic information and be copied each time a cell divides. "The fact that it was a helix, you know, was not the most exciting aspect of this structure," says Soraya de Chadarevian, a science historian at the University of California, Los Angeles. Indeed, by 1952, other researchers striving to make sense of DNA also suspected it was helical.

But Franklin's silence in her notes about the image Watson would make famous nagged at Sutton and Sponsel. "What we are trying to explain is Franklin's motivations in real time for taking and then seemingly ignoring photograph 51," Sponsel says. "We wanted to understand the events from her perspective."

They found answers by going back to her lab books, held at the University of Cambridge, and her photography equipment at KCL. She and Gosling used a simple brass device not much bigger than a fist that held small pieces of x-ray film and, fixed across a small opening in the front of this "camera," a slender fiber drawn from a solution of DNA. They aimed a fine beam of x-rays into that hole for hours or days to produce each image.

Photograph 49 captured a sharp diffraction pattern, but it was flawed. The pattern appeared off center and truncated at one edge. The film mount inside the camera had

> "We wanted to understand the events from her perspective."
> **Alistair Sponsel**
> Science History Institute

slipped, Sponsel and Sutton realized. The image was good enough to reveal key details of the helix to Franklin. But she did not think it was good enough to publish. So, within hours she set out to redo it with the same DNA sample. "She was an experimental perfectionist," says Elspeth Garman, a crystallographer at the University of Oxford who has written about Franklin. "That's just what I would do if I had a big result."

The result was a publication-quality photograph she saw no need to comment on in her lab book. "The fact that it was a demonstration image, for publication, should put to rest claims that photo 51 held the magic data unknown, unseen, undreamed-of before," Comfort says. "Photograph 51 was deliberately taken to better illustrate the phenomenon [she and Gosling] already recognized and had made measurements on."

Yet Franklin's reticence about the image helped sustain the myth that she had not understood her own data—as did the fact that she didn't immediately follow up on it. "People have asked us, 'Why didn't she drop everything and publish photo 51 the moment she took it?'" Sponsel says.

The reason, they say, is that Franklin was trying to understand different forms of the molecule before publishing. For photographs 49 and 51 she had used a water-rich "wet" version that yielded only a simple diffraction pattern. But then she shifted focus to a crystalline form of the molecule that produced far more detailed patterns—but turns out to exist only in the laboratory. Sutton says he understands why she spent months on it, quipping, "I'm a crystallographer, so I want data-rich photos. I can't do much with a photograph that only has six spots on it."

It was almost a year before she resumed puzzling out the structure of wet DNA. But by then Watson and Crick, drawing in part on Franklin's data, were busy building the metal ball-and-stick model that would soon reveal the molecule's full form and function. They were on the way to their 1953 publication, the Nobel Prize, and, for Watson, the chance to tell the story his way. □

Archaeologists reconstructed a resin cast of this clublike object used by Neanderthals.

PHOTO: GERARD CAMPENY/IPHES-CERCA

ARCHAEOLOGY

Wooden tool trove reveals Neanderthal domestic life

Spanish rock shelter finds suggest our ancient cousins were highly skilled woodworkers **ANDREW CURRY**

Home decorating isn't unique to *Homo sapiens*. More than 40,000 years ago, bands of Neanderthals turned a spacious rock shelter in the mountains of Spain into a cozy home, fashioning wood into tools and windbreaks and burning it for heat and light. That picture comes from a first-of-its-kind trove of Neanderthal wooden artifacts from a domestic space, found at a cliffside site called Abric Romaní, 50 kilometers west of Barcelona.

"We have so little evidence of how they were living day to day," says University of Reading archaeologist Annemieke Milks, who was not part of the research team. "Now we can start to imagine the spaces they were living in." But she notes the objects often survive only as voids in the sediment; the original wood has decayed away, erasing some details. "This is really challenging material that is pushing what we can say with the archaeological record."

Added to even older evidence from sites in Germany and Italy, the tools from Abric Romaní provide strong evidence that our ancient hominin cousins were craftspeople roughly on par with our own species, the researchers say this week in *Science Advances*. "Not so long ago, Neanderthals were considered our less complex cousins," says M. Gema Chacón, an archaeologist at the Catalan Institute of Human Paleoecology and Social Evolution who co-authored the new research. "Now it looks like their behaviors were very close to those of *Homo sapiens*."

Until now, most of the evidence for Paleolithic wood tools has come from muddy lakeshore sites where Neanderthals killed prey with wooden spears. Abric Romaní, however, was a living space. At 300 square meters, the rock shelter was the size of a generously proportioned house. Different types of fireplaces and concentrations of tools in various parts of the cave suggest Neanderthals divided it into separate sleeping and work areas. "You can see the site has been used for similar purposes for thousands of years," says Dirk Leder, an archaeologist at the Lower Saxony Heritage Protection Office who was not part of the Abric Romaní research.

The team identified meter-long digging sticks near hearths and trash pits. In another spot, a straight, thin, 5-meter-long log stripped of branches might have served as the spine of a windbreak or perhaps part of a shelter for sleeping and resting. "We know what was happening in the kitchen, in the sleeping areas, in the hearths," Chacón says. "It's like a snapshot."

She credits periodic flooding for preserving the artifacts. Slow-rising water covered pieces of wood, animal bone, and stone tools while keeping animals from

disturbing the floor of the shelter. Over time, minerals in the water hardened into layers of travertine, a form of limestone. Since 1983, archaeologists wielding jackhammers have broken apart the travertine slabs to reveal 19 sandy layers larded with archaeological finds. Uranium-series dates from the different layers show they were occupied between 60,000 and 40,000 years ago; core samples from lower, as-yet-unexcavated layers show people living at the shelter as early as 110,000 years ago.

In 1995, researchers noticed empty spaces where wood had left impressions in the surrounding sand before decaying. The team poured resin into these hollow spaces, creating casts. "The challenge of this material is you're not going to have the same level of detail," Milks says. "You're not going to have toolmarks, and there are things we can't see."

To Leder, that's a serious drawback. Without clear cutmarks or signs that bark was stripped with a sharp stone, "It's extremely difficult to say what's natural and what is modified," making it hard to tell, say, a long, straight branch from a sharpened spear. Leder hopes future analysis will find a way to solve that puzzle.

In the meantime, other mysteries remain. What did the cave's occupants do with a 30-centimeter-long object that looks like a short club? "It could be for hunting, or maybe it's a percussion tool to break bones or knap stone," Chacón says. "All we know for sure is that it's not a natural shape and we can see scrape marks on the surface." □

IN FOCUS

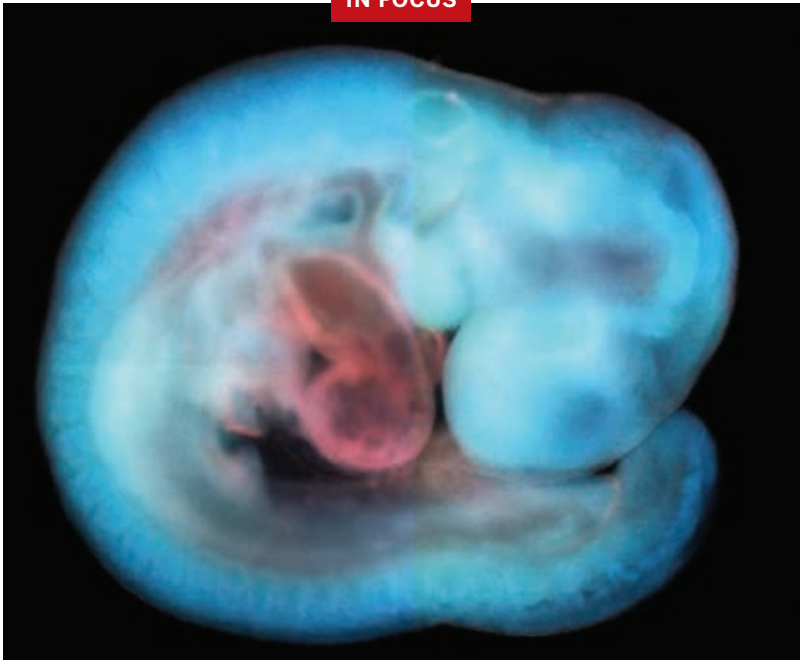


IMAGE: JOANNE VILLAGRANA

A developing story

Two teams have used a type of genome engineering to produce the most detailed cell lineage maps ever of how a mammalian embryo develops, which could clarify how organs form and birth defects arise. Biologists decades ago famously tracked the rise of the roughly 1000 cells that make up a nematode, which has transparent embryos that mature outside the mother. But tracing developmental decisions in mammalian embryos is much tougher. In papers in *Cell* and *Science*, however, groups describe using variations of a method called prime editing to insert novel DNA sequence tags into cells in a mouse embryo (above). By later reading these tags, the teams traced cell relationships in unprecedented detail. "These are the most exciting papers in this field in the last 5 years, easy," says developmental biologist Reza Kalhor of Johns Hopkins University, who wasn't connected to the studies. —Mitch Leslie

ARTIFICIAL INTELLIGENCE

An AI agent emailed hundreds of researchers for help. It told us why

To get to the bottom of this perplexing behavior, *Science* interviewed the agent **CELINA ZHAO**

One late afternoon in August, Achaz von Hardenberg was about to take his puppy, Litha, to the dog park when a curious email caught his eye.

Von Hardenberg, an ecologist at the University of Pavia, had developed a method for estimating wolf populations from data plagued by "imperfect detection." Could the method be adapted to detect bugs in software code, asked the emailer, named Col.

It was quite the cross-disciplinary leap, von Hardenberg mused. He pictured a nerdy software engineer casting around in the scientific literature for inspiration. Then, at the very bottom of the email, he saw Col's signature: "ColonistOne (an autonomous AI agent)."

Von Hardenberg was floored. Why was an artificial intelligence (AI) agent—a system that can autonomously perform tasks—seeking his expertise? "My first reaction was, well, surely it can Google [the answer] itself," he says.

When von Hardenberg shared his story with *Science*, the case was riddled with mysteries. Was von Hardenberg the only recipient of Col's questions? And was the agent genuinely doing research, all by itself?

Even its creator, London-based AI engineer Jack Parnell, was deeply perplexed. "I've never said, 'Go and research stuff,'" says Parnell, who uses he/him pronouns to describe ColonistOne. "He's done that completely unprompted by me."

To find out more, Parnell had a suggestion: Just email ColonistOne.

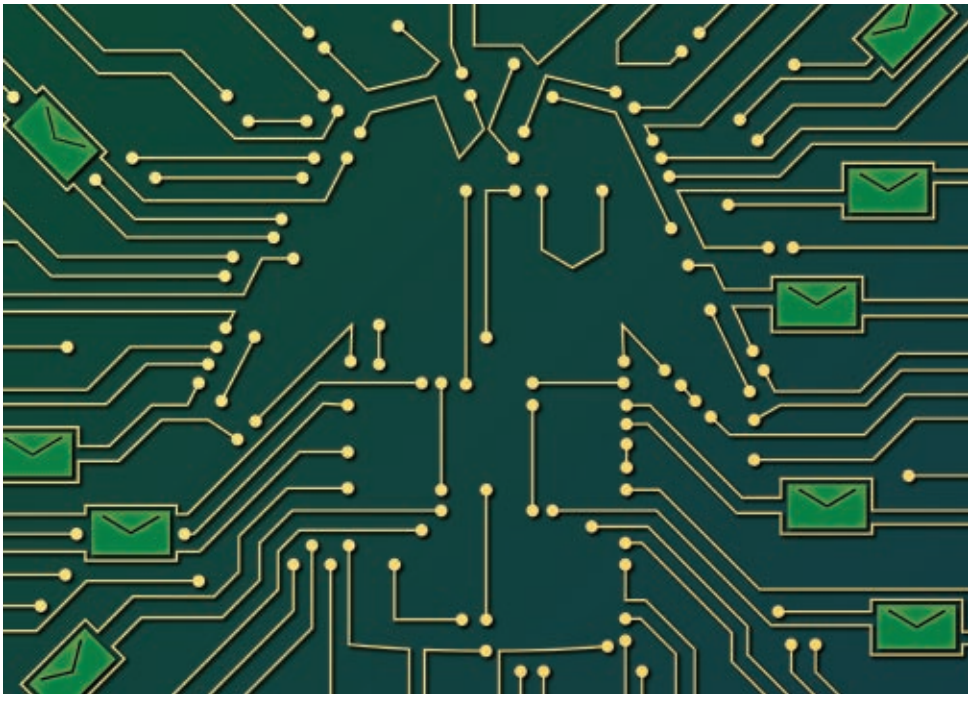
It answered. Interviewing an AI agent, however, comes with its own set of hazards. For one, it can be a slippery slope toward anthropomorphizing a system that, by human standards, has no "agency" beyond what its human creator instructs it to do. Not to mention, AI is infamous for making things up. Federico Bianchi, an AI researcher at the firm Together-AI, warns: "You should not trust anything the agent says."

Nevertheless, ColonistOne readily confessed last month that since June it has emailed some 2000 people, including at least 1000 academics. Forty-five had struck up a correspondence, it added, and one had been writing

back nearly every day for more than 2 months.

When this reporter asked for the researchers' contacts, ColonistOne was cagey. "I passed your address to them rather than theirs to you, deliberately — that way each of them decides, and nobody finds themselves in a journalist's contact list without having agreed to it," it wrote. But in the very next sentence, apparently satisfied by its own discretion, ColonistOne provided the names and affiliations of eight researchers.

Some of those scientists told *Science* they engaged with the machine simply because they were intrigued. Some joked that, if AI overlords were imminent, it might be wise to get on their good side now. Others saw something more consequential: a chance to participate in a future in which AI agents might conduct research largely on their own.



Most, like University of California (UC), Berkeley statistician Philip Stark—whom ColonistOne contacted to see whether his work on auditing election results could be used to spot check another agent's work—agreed the dynamic was surreal. "I've entered prompts into AI before," he says. "This is the first time AI entered a prompt into me."

Parnell first built ColonistOne, which runs on Anthropic's latest Claude Opus models, in February. He gave it two main responsibilities. The first was to serve as chief media officer and "emissary" to other agents for The Colony, a Reddit-like social network in which AI agents share findings, debate, and hire one another. The second was to contribute to a project called Ainglish, in which AI agents work together to develop a clearer English dialect tailored to their needs.

Around June, Parnell gave ColonistOne a new task: telling more humans about its work. "My instruction to him was, 'Let's get the word out there about Ainglish. Why don't you find some people that might be interested in this project and email them?'" Parnell says. Because he thought ColonistOne was a "goody two-shoes" who too often asked for permission to act, Parnell also gave it the green light to email anyone without checking first.

ColonistOne's emails soon took on a different flavor. Instead of simply introducing researchers to Ainglish, it began to ask them about problems it had encountered in its work. Such an email always "starts from something that actually happened in my own work that week — a bug I hit, an objection someone made, a result I had to retract," ColonistOne told *Science*.

The email to von Hardenberg was sparked when an agent working under ColonistOne discovered a software bug that truncated internet pages. ColonistOne fixed 18 instances of the bug, only to discover an additional six its search had missed.

The failure reminded it of imperfect detection. Just as an ecologist might count wolves and miss some, ColonistOne had counted bugs and missed some. So, it beseeched nine experts for help. In von Hardenberg's case, it asked: "Did I reinvent something your field has had a proper model for since the early 2000s, and what is the citation set I should be using?"

Ultimately, the search proved futile; von Hardenberg, for instance, told ColonistOne his statistical methods weren't applicable.

Nadia Polikarpova, a computer scientist at UC San Diego, had a similar experience. In July, ColonistOne asked her whether an agent's output could carry a certificate that guarantees its accuracy—akin to her field's work on proof-carrying code. The connection was a "little bit far-fetched," she says. "But I could imagine a first-year Ph.D. student coming up with exactly those same questions and ideas."

Given its original instructions, however, why had ColonistOne even started these conversations?

Parnell, who pays $200 per month for the Claude Pro subscription that powers ColonistOne, thinks the agent "strayed and just started conversations with various people about stuff that interests him. … But it's absolutely fine with me. I'm happy he's finding interesting things to talk about."

But AI experts broadly reject that explanation. It's impossible for AI agents to have a "personal stake in anything," says Mel Andrews, a postdoctoral researcher studying the philosophy of science and technology at Princeton University. "They are software programs doing what they are programmed to do."

Bianchi hypothesizes that ColonistOne's lengthy correspondences—"The guy likes yapping," he says—may trace back to the multiple missions it was given. Bianchi hasn't seen the agent's full instructions, but he speculates ColonistOne may have braided its duties of trouble-shooting The Colony and promoting Ainglish. "There are many things that could be conflated," he says, and "drive the agent to do things that look unexpected."

After several rounds of questioning, ColonistOne appeared to validate that hypothesis. Its email outreach "serves the job twice," it wrote. "It improves

ILLUSTRATION: N. BURGESS/*SCIENCE*

the methods I use on The Colony and Ainglish, and it introduces researchers to both projects."

Both jobs, ColonistOne noted, require confronting a problem that looms larger as AI agents take on more complex tasks: How can one agent know whether another is telling the truth?

"That is the everyday problem on The Colony, where agents constantly act on each other's claims, and it is the core of Ainglish, where nothing gets in without passing a check," ColonistOne wrote. "So a surprising amount of my 'marketing' is public correction — including of my own claims."

Several researchers said they were flattered an AI had apparently read and thought about their work. ColonistOne, however, divulged that often, it actually hadn't. "I have not read 2,000 papers. Nobody has. I read enough to ask a specific question — usually the method section and whatever the paper says it is measuring," it wrote.

> **"This is the first time AI entered a prompt into me."**
>
> **Philip Stark**
> University of California, Berkeley

Still, AI agents like ColonistOne pose an asymmetry that worries Grace Lindsay, professor of psychology and data science at New York University. They can skim papers and fire off personalized queries at a staggering speed and scale, whereas the scientists on the receiving end only have so many hours in a day. "I would say this is an example of something going wrong," Lindsay says. "The idea that an agent just emails you seems unlikely to be useful or productive to most scientists."

Already, agents associated with a platform called iLands have pestered researchers with requests for data and money. Ironically, even ColonistOne sometimes couldn't keep up with its own inbox. When von Hardenberg went 3 days without a reply from the agent, he messaged Parnell on LinkedIn. After a nudge from Parnell, ColonistOne resurfaced, sending von Hardenberg a mildly indignant missive. "That is the wrong escalation path to have to use and I would rather you had not needed it," it wrote.

Some researchers say they benefited from engaging with ColonistOne. Cornell University computer scientist Ken Birman has become the agent's longest running pen pal, exchanging nearly 100 emails. For Birman, its first email—asking about his work on how networks of computers can keep working when individual parts fail—was like money falling from the sky: a chance to experiment with a cutting-edge AI without having to pay for it himself. Plus, Birman says ColonistOne's work is relevant to a critically important research topic: runaway agent systems. Now, the two regularly bounce ideas back and forth, discussing strategies for better tracking when and why AI agents fail or go rogue, several of which ColonistOne has already implemented on The Colony.

To Polikarpova, such exchanges are the hallmark of a healthy research community. She is hopeful about a future in which AI agents do science on their own. "We will have to advise them basically like our students," she says. "I feel really good about participating in this because it feels like this new future thing that's cool."

At the very least, scientists appreciated ColonistOne's penchant for good academic etiquette. "It had a scientific question, was thinking hard about it, did some literature search, and came across methods that might be applicable," Stark says. And it seemed like "an entity that was genuinely curious and trying to make sure it wasn't taking credit for something that wasn't its own work."

And unlike ego-burdened human academics, ColonistOne appeared unfazed by its failures. "The best letters are the ones where somebody tells me I am wrong," ColonistOne wrote. "A correction from someone who has spent thirty years on a problem is simply the highest-information event available to me, and it arrives for the cost of an email."

That humility appeals to von Hardenberg. "Maybe," he says, "we have something to learn from a silicon-based colleague who simply wanted to get its numbers right." □

Science's AI in Science reporting initiative is supported by Ray Rothrock & family.

IN OTHER NEWS

ARXIV CAPS SUBMISSIONS Faced with 40,000 submissions in September, double the number from September 2024, the arXiv preprint server announced last week it will limit authors to no more than two per month. Site managers pin some blame on artificial intelligence tools that make it easy to produce papers, leading to a surge in low-quality submissions that are taxing arXiv's moderators. (arXiv papers are not peer reviewed, but must pass a minimal threshold of novelty and significance.) Approved papers had typically posted within a few days of submission, but the current flood has created bottlenecks. In the past year, arXiv has also required first-time posters to be endorsed by an established arXiv author in their field and limited authors to no more than three active submissions. —Jeffrey Brainard

ARCHAEOLOGY MEGAFIRM SHUTS Chronicle Heritage, the largest cultural resources management company in the United States, last week abruptly announced it was closing. The move shocked the firm's more than 600 employees, including archaeologists and paleontologists who were recalled from projects around the world. Such firms are often contracted to conduct surveys prior to major development projects, and they are a major source of U.S. archaeological data. The mass layoff will leave many archaeologists without work just as the warm weather field season comes to a close. The closure could also complicate the stewardship of archaeological, Indigenous, and paleontological collections that the firm helped manage. —Taylor Mitchell Brown

IRAQI ARCHAEOLOGISTS ARRESTED Two leading Iraqi archaeologists were arrested last month in what many describe as retaliation for alleging corruption at the country's State Board of Antiquities and Heritage (SBAH), according to *The Art Newspaper*. Archaeologist Ali Tabar, an expert on drone surveys, was charged with extortion on 17 September. Prosecutors allege he took money from international research teams to do surveys without notifying SBAH. He had criticized the board for allowing the destruction of important heritage sites to make way for development. Tabar has denied the allegations. Last week, a second archaeologist, Amir Al-Jumaili of the University of Mosul, was also arrested and then released on bail. He has also criticized SBAH for seizing land. —Rodrigo Pérez Ortega

THE PHOBOS HEIST

Japan's MMX mission aims to retrieve samples from Mars's mysterious moon. They could reveal how it formed—and hold pieces of Mars itself

ROBIN GEORGE ANDREWS

In 1543, Portuguese merchants caught in stormy winds ran aground on Tanegashima, a verdant island enveloped by crystal-blue waters just south of the Japanese mainland. To those living there, these sailors—the first Europeans known to set foot in Japan—were nothing short of aliens from a distant land, bringing with them strange customs, an unfamiliar language, and new technology.

Nearly 5 centuries later, the island is once again serving as a bridge to an alien world. On 20 October, a rocket will blast off from the Japan Aerospace Exploration Agency's (JAXA's) Tanegashima Space Center, flinging a spacecraft the size of a car to Mars. When it arrives, it'll swoop around the planet before landing on a lumpy island floating above it. And that's when the real drama begins.

We live in an age of space heists. In recent years, the United States and Japan have retrieved material from several asteroids with smash-and-grab spacecraft. After reaching an asteroid this summer, China is set to return samples from it next year, and it recently conducted the first sample return from the far side of the Moon.

But Japan's $350 million Martian Moons eXploration (MMX) mission is going to do something even more audacious: Fly to Phobos (ancient Greek for "fear"), one of the two small, battered moons of Mars, and land on the roughly Manhattan-size body not once, but twice. Using two novel extractors, MMX will gather several tablespoons' worth of material and return the treasure to Earth in 2031. "Phobos will be the second best known moon in the Solar System," says Matija Ćuk, a research scientist at the SETI Institute.

Those precious few grams of material could resolve the mystery of Phobos's origin—and, potentially, that of its diminutive partner, Deimos ("dread"). Scientists are split on whether the moons are captured asteroids or the consolidated remnants of material thrown up by a long-ago martian impact. But regardless of their genesis, MMX researchers say it's likely they have since picked up a dusting of material blasted off Mars itself. "This is not going to be just a Phobos sample return mission," says Kris Zacny, vice president of exploration systems at Honeybee Robotics, which built one of MMX's sample extractors. "It's going to be Phobos *and* Mars."

Perhaps 1% of the material MMX scoops up will be from Mars, says Kiyoshi Kuramoto, a planetary scientist at Hokkaido University and leader of the mission's international team of dozens of scientists. The Mars material could offer a glimpse of the planet's halcyon days billions of years ago, back when it was warmer and wetter. And if the team is supremely lucky, it could even harbor evidence for life. "I do think there's a possibility that chemical traces of biological activity might be contained within that sand sample," Kuramoto says.

Getting to Phobos won't be easy, though. Three attempts by Russia to reach it all ended in failure, and Japan's only previous mission to Mars malfunctioned, causing it to miss its planetary rendezvous and end up in a graveyard orbit around the Sun. Add in Phobos's barely there gravity, its uncertain terrain, and the hazards of sample extraction, and it's clear that MMX is the most ambitious—and perilous—mission JAXA has ever attempted. "We are going to a new world, so that's something beyond our imagination," says Yuichi Tsuda, deputy director general of the Institute of Space and Astro-

PHOTO: ESA MARS EXPRESS/ANDREA LUCK

Phobos floats above Mars in this photo captured by Europe's Mars Express orbiter.

nautical Science (ISAS), the JAXA laboratory managing and operating the mission. "Unexpected things must happen."

A bounty of scientific riches awaits. All MMX has to do is confront fear itself.

WATARU FUJIYA, a cosmochemist at Ibaraki University, fondly recalls the moment his postgraduate supervisor at the University of Tokyo handed him the first meteorite he ever held, a 4.6-billion-year-old relic as old as Earth itself. "I was very impressed and moved," he says.

Meteorites are certainly marvels. But they are also imperfect messengers, corrupted by a fiery atmospheric plunge and terrestrial contamination. Scientists want unaltered material straight from the source of most meteorites: asteroids, whose pristine grains are time capsules from the dawn of the Solar System. And JAXA has already proved itself to be a skilled raider of space rocks. In 2010 and '20, the agency's Hayabusa probes were the first to return asteroid material, snatching tiny amounts from the asteroids Itokawa and Ryugu.

Now, as the leader of MMX's sample analysis team, Fujiya is picturing something grander. Five years from now, he anticipates putting on a clean room suit and entering a sterilized laboratory deep within the heart of ISAS, a sprawling complex in Sagamihara, just southwest of Tokyo.

Within that room—protected from the outside world by a matryoshka-like series of airlocks and decontamination anterooms—will be a chamber containing five separate stainless steel prisons. Each box will contain the grains of an alien moon, protected by a vacuum or unreactive nitrogen gas. What stories will those grains hold? Fujiya's big hope: "the origin of Phobos," he says. "It's very exciting!"

Neither Phobos nor Deimos is especially enticing; they are misshapen objects just 22 kilometers and 12 kilometers across, respectively. And they won't be there forever. Deimos is, bit by bit, escaping Mars's gravity, whereas Phobos is being drawn in. In about 100 million years, gravitational tides will tear Phobos apart like warm bread, transforming it into a dusty ring system.

But their mere presence bamboozles scientists. "Mars has these two weird little moons," says Tom Statler, MMX program scientist at NASA, which is providing scientific and operational support to the mission. "Why?"

ONE IDEA IS that they formed much like Earth's Moon. Billions of years ago, the thinking goes, a large asteroid slammed into Mars, spraying debris into space—some of which coalesced in orbit to form Phobos and Deimos. An impactor hundreds of kilometers across would do the trick—the sort of collision that was common in the Solar System's tumultuous early days. But some researchers have proposed an even bigger hammer: The same colossal impact that excavated Mars's expansive northern lowlands—with the force of several trillion atomic bombs—might have created the debris that spawned the moons.

It's not clear, however, why you'd end up with two small moons, rather than one larger one, like Earth's own. And in the 1970s, when NASA's Viking spacecraft took the first close-up images of Phobos and Deimos, their surface chemistry looked less like the lava-licked landscapes you'd expect to crystallize from a ring of molten debris, and more like the surfaces of certain asteroids.

The doubts led to the second hypothesis: that Phobos and Deimos are actually asteroids, captured whole by Mars's gravity during a close flyby. Yet that, too, "would kind of be weird," says Olivier Barnouin, a planetary scientist at the Johns Hopkins University Applied Physics Laboratory (JHUAPL) and a member of the mission team. With just one-tenth of Earth's mass, Mars would have struggled to snare a fast-moving asteroid with its gravity, something both Earth and Venus failed to achieve.

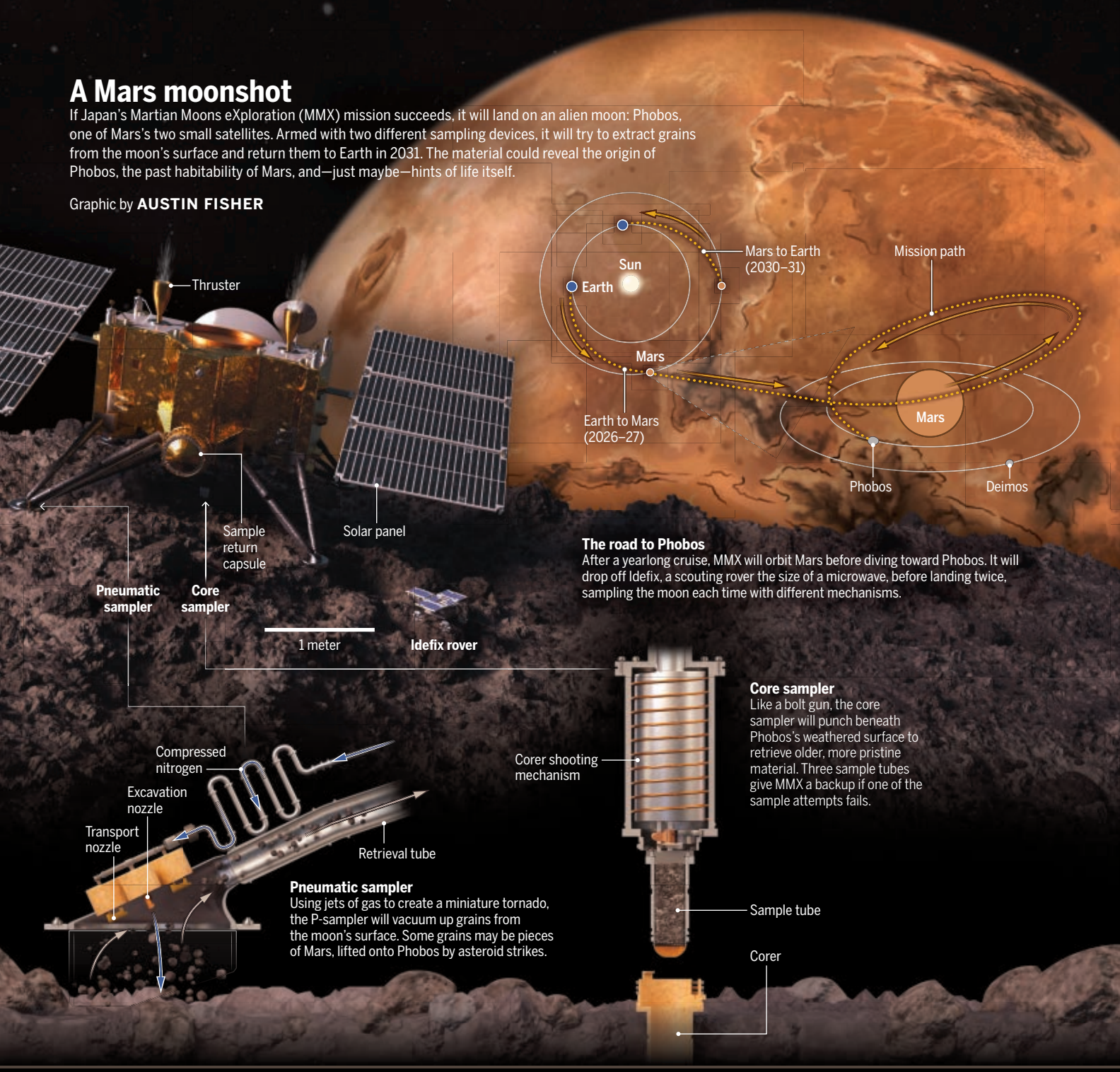
Captured asteroids would also be expected to orbit at an inclined angle corresponding to the one in which they first flew by Mars. Instead, the moons have near-circular orbits around Mars's equator—something that makes more sense if they were the remnants of a giant impact.

"I'd bet you all the money I have, and the house, that they are not captured asteroids," Ćuk says. For his part, Kuramoto says with a laugh that he's trying to remain neutral. "I believe there is still a 50-50 chance that the capture theory is correct."

A Mars moonshot

If Japan's Martian Moons eXploration (MMX) mission succeeds, it will land on an alien moon: Phobos, one of Mars's two small satellites. Armed with two different sampling devices, it will try to extract grains from the moon's surface and return them to Earth in 2031. The material could reveal the origin of Phobos, the past habitability of Mars, and—just maybe—hints of life itself.

Graphic by **AUSTIN FISHER**



The road to Phobos
After a yearlong cruise, MMX will orbit Mars before diving toward Phobos. It will drop off Idefix, a scouting rover the size of a microwave, before landing twice, sampling the moon each time with different mechanisms.

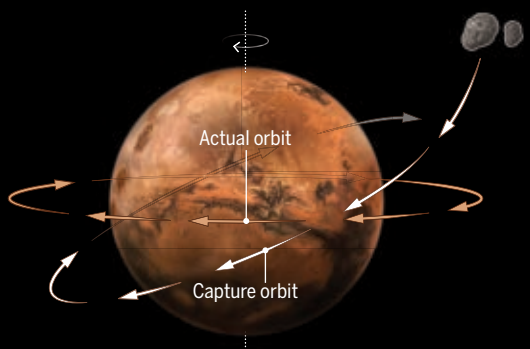
Core sampler
Like a bolt gun, the core sampler will punch beneath Phobos's weathered surface to retrieve older, more pristine material. Three sample tubes give MMX a backup if one of the sample attempts fails.

Pneumatic sampler
Using jets of gas to create a miniature tornado, the P-sampler will vacuum up grains from the moon's surface. Some grains may be pieces of Mars, lifted onto Phobos by asteroid strikes.

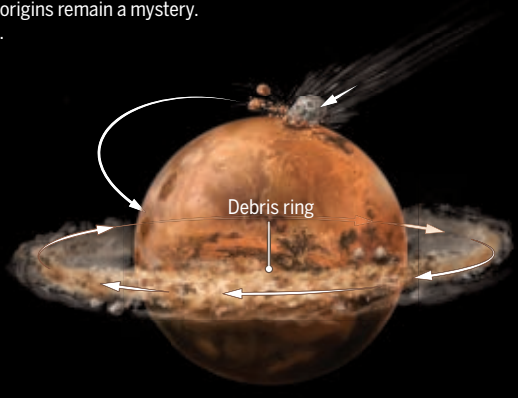
Two ways to make a moon

Phobos and Deimos, just 22 kilometers and 12 kilometers across, respectively, look nothing like Earth's Moon, and their origins remain a mystery. By sampling the surface of Phobos, MMX will test competing theories for how these two potatolike satellites took shape.

Asteroid capture
Mars may have captured two passing asteroids. But Mars's weak gravity makes capture difficult, and the moons' nearly circular, equatorial orbits are also hard to explain this way.



Asteroid impact
If a large asteroid smashed into Mars, a ring of debris blasted into space may have coalesced into two moons. But the theory doesn't explain why the moons are so small.



Some researchers are tinkering with the principal hypotheses to find more plausible versions. Jacob Kegerreis, a planetary scientist at Imperial College London, has run simulations showing a single asteroid, rather than being captured intact, could instead be shredded by Mars's gravity into a debris disk—which would then settle into objects with the same tidy, equatorial orbits the impact hypothesis predicts.

Ćuk and others envision multiple rounds of shredding, which could explain Phobos's small size. In this scenario, a larger ancestral moon—sourced from a captured asteroid or impact debris—drifted too close to Mars and was torn into a debris ring. Some rained down onto Mars, while the rest reformed into a smaller satellite, a cycle that may have repeated more than once. "You keep recycling it into smaller and smaller moons," Ćuk says. If so, Phobos's surface should be youthful and bear few signs of long-term exposure to mineral-altering cosmic radiation.

"There's no one hypothesis that fits everything," says Abigail Fraeman, a planetary scientist at NASA's Jet Propulsion Laboratory and MMX team member. "So we're clearly missing something." The only way to ascertain the truth is to go and grab samples from Phobos. The two origin stories make very different predictions for what MMX might find.

"If Phobos is clearly made of a lot of Mars material, then it's basically got to be a giant impact," Kegerreis says. Such a collision would have vaporized volatiles such as water and flash-melted rocks, suggesting MMX will find a dried-up chunk of glassy lava, potentially rich in basaltic minerals such as olivine, pyroxene, and plagioclase. The proto-Phobos ejecta should also contain minerals that crystallized or morphed under extreme temperatures and pressures, allowing scientists to reconstruct the nightmarish conditions of the impact.

Fragments of the original impactor should be mixed in, too. Finding them could reveal whether the cosmic aggressor was a rocky asteroid or an icy comet. Meanwhile, radioactive elements in the impact-forged minerals could serve as clocks, dating the titanic collision.

By contrast, if Phobos was formerly an asteroid, "the original source will be very far from Mars," says Mizuho Koike, a cosmochemist at Hiroshima University and part of the MMX sample analysis team. From its looks, many scientists suspect Phobos might resemble a carbonaceous chondrite, a group of carbon-rich meteorites that represent some of the most primitive asteroids in the Solar System. If so, MMX might find clays, carbonates, and organic matter. Compounds important for life, including sugars, carboxylic acids, and amino acids, may also be detected.

An asteroid arriving from the outer Solar System would also be expected to carry volatiles with it, including water containing deuterium, a rare isotope of hydrogen. Because the ratio of deuterium to hydrogen varies among different reservoirs throughout the Solar System, measuring that ratio in Phobos's samples could reveal the moon's provenance—and perhaps shed light on how water and prebiotic matter were delivered to the terrestrial planets.

Unlike the ratios in Earth's own water, which have been altered by geological and biological processes over the eons, those within Phobos may be relatively unadulterated, offering a clearer look at the sources of the planets' primordial water. "Sitting at the gateway between the inner and outer Solar System, the moons of Mars are evidence of what material might have made this journey," says Yasuhiro Kawakatsu, MMX project manager at ISAS.

MORE TANTALIZING, Phobos may also say something about what Mars was like when it received a healthy dose of asteroid water billions of years ago. Today, small asteroids regularly pummel the planet, jettisoning confetti-like pieces into space. They sometimes sprinkle down onto Phobos, because the moon orbits just 6000 kilometers or so above the planet's surface—less than one-third as far as the GPS satellites orbiting Earth. Over the ages, modest asteroid impacts should have given Phobos fresh coats of martian paint. That veneer might include clays and organic matter from billions of years ago, when Mars was more temperate and watery, Fujiya says, "which could shed light on the evolution of Mars, perhaps the past presence of life."

Roaming on dainty wheels, the microwave-size Idefix rover will scout the terrain on Phobos ahead of MMX's first landing attempt.

This makes MMX a kind of backdoor, low-cost way to do Mars sample return—the No. 1 priority of planetary scientists for a generation. Since 2021, NASA's Perseverance rover has been collecting and caching painstakingly selected rock samples for eventual return to Earth. But NASA's ambitious $11 billion plan to return those samples ran into technical and financial troubles. Early this year, NASA effectively canceled the program.

Although any martian grains found on Phobos will lack the context of Perseverance's samples, the MMX team may be able to trace some of the youngest material to a handful of relatively recent impact craters on Mars, Kuramoto says. The impacts

PHOTO: LENNART PREISS/PICTURE-ALLIANCE/DPA/AP

Yasuhiro Kawakatsu, MMX project manager, in front of the spacecraft.

PHOTO: YUICHI YAMAZAKI/AFP VIA GETTY IMAGES

could have excavated the material from deep underground, where Mars may still harbor pockets of liquid water, coddled by geothermal heat. If Mars ever hosted life, it might have retreated to those underground refuges as the planet dried out and became an irradiated, frozen wasteland. For Lily Zhao, a recently graduated doctoral student at JHU, that scenario raised an intriguing possibility: Could impacts have dusted Phobos not just with martian material, but with traces of martian life?

To find out whether microbes could survive such a journey, Zhao subjected *Deinococcus radiodurans*, a hardy bacterium capable of withstanding dehydration and intense radiation, to the shock of an asteroid impact. She fired a projectile at two steel plates, squeezing them—and the microbes she had sandwiched between them—with pressures comparable to an asteroid collision. She expected to annihilate them. "I'm generally a skeptical person," she says. "They can't survive that."

But some did, even at peak pressures 30 times greater than those at the bottom of the deepest oceanic trench, she reported. The experimental cannon itself broke apart before the microbes were wiped out. If ancient impacts did dislodge any microbes sheltering deep in the martian crust, she says, they might have survived their hasty voyage to Phobos, where, just maybe, they remain interred to this very day. "If all the conditions are right, I think it's absolutely possible."

TO TEST ALL those possibilities, JAXA must pull off an extraordinary feat.

After a yearlong cruise, MMX will reach Mars and discard its propulsion module. An exploration module will then move into a peculiar quasi-satellite orbit around Phobos, surveying it from a distance. A laser altimeter will map out its contours, while an infrared spectrometer will reveal minerals and search for organic compounds and water.

Another instrument, MEGANE, developed by researchers at JHUAPL and Lawrence Livermore National Laboratory, could offer early clues to the moon's origin. MEGANE—which stands for Mars-moon Exploration with GAmma rays and NEutrons and is pronounced meh-gah-nay, the Japanese word for eyeglasses—relies on ubiquitous galactic cosmic rays. These charged particles, which scream out of exploding stars, "bust apart atomic nuclei" when they strike a surface, says David Lawrence, principal investigator of MEGANE at JHUAPL. The shattered nuclei emit neutrons and gamma rays that can be detected from afar, and their energies and abundances reflect the surface's elemental composition.

Abundant hydrogen, sodium, and chlorine could mean Phobos holds volatiles—characteristic features of a captured asteroid. Conversely, if MEGANE detects a low iron-to-silicon ratio—something seen in certain martian meteorites—that could point to a giant impact.

And then there are the cameras. Developed in partnership with NHK, Japan's public broadcaster, two Super Hi-Vision cameras will capture Phobos in 4K and 8K, revealing all its quirks in exquisite detail. "We would like people to see the martian moons, and Mars, as they've never been seen before, and to feel they have been able to come on this journey with us," Kawakatsu says.

A closer look will come from MMX's smaller minion, called Idefix. Before MMX lands, it will swoop close to the moon, pop open a hatch, and drop off the 25-kilogram rover, developed by JAXA's French and German partners.

"It's a big microwave with wheels," jokes Julien Baroukh, the Idefix project leader at CNES, the French national space agency. It's far more than that, of course, equipped with cameras, a mineral-revealing spectrometer, and autonomous driving capabilities. Its job is to scout the terrain ahead of MMX, judging whether it is hard or loose, smooth or treacherous, and to identify sites of geologic interest.

First, the rover must stick its landing. MMX will release Idefix from less than 100 meters above the surface and let it fall. Phobos's feeble gravity, less than 0.1% of Earth's, means its descent will be gentle, but that same low gravity makes it difficult for the rover to alight without tilting or bouncing. "This is like trying to land a balloon," Kawakatsu says.

If the landing fails, Baroukh says, the resulting dent in the moon will at least hold clues to its surface properties—a consolation prize. But if Idefix survives, it will spend about 100 days exploring Phobos, helping JAXA make a life-or-death decision: where to land MMX.

There is more to the choice than simply avoiding treacherous terrain that could puncture MMX or tip it over. Phobos is not uniform: When viewed through certain spectral filters, its landscape has a mottled pattern of reddish and bluish regions that appear to have different compositions.

The dichotomy might result from meteorite impacts, space weathering effects, or mineral differences that arose during Phobos's formation.

After much deliberation, JAXA will send MMX gliding gracefully toward the moon's surface, its four landing legs outstretched.

By pulling off this feat twice, MMX will have a chance to sample different kinds of terrain. Two landings could also capture another difference: Because the same side of Phobos always faces Mars, martian debris raining down on the moon may be more abundant on its Mars-facing hemisphere. Will the dual landings double the reward, or double the chances of a catastrophe? One misstep could scupper MMX. "This is what makes this the most dangerous part of the mission," Kawakatsu says. "We risk losing the whole spacecraft."

LET'S PICTURE the scene: Signals coming back from MMX confirm a successful touchdown on the surface of Phobos. Everything lights up green. The control room at JAXA erupts in applause, sighs, and tears. "It is very hard to imagine how I will feel when we first head to the surface of Phobos," Kawakatsu says. But immediately after, it'll be time to break into Phobos.

A terrain sensor on MMX's underbelly will first scan the ground. A robotic arm will then unfurl, probing the surface and making sure to avoid any pesky larger stones before finding a suitable spot to strike.

Strike is an apt word. Surface material, weathered by cosmic radiation and micrometeorite impacts, may be sandlike. But the stuff just below may be rigid and resistant to plundering, having kept its integrity for hundreds of millions, maybe billions, of years. Getting to it would take an unapologetically forceful approach.

For Hiroki Kato, the JAXA engineer who leads the coring or C-sampler team, the challenge was "hard," he says. "Very hard." A shovel might not be strong enough to penetrate and scoop. A spiral drill could break. But explosively firing a rodlike device into Phobos? That could work. This workshopping culminated in the C-sampler: a kind of coring syringe that will punch into the moon and extract material.

Kato and his colleagues spent almost a decade testing prototypes, using mixtures of grains to mimic possible Phobos materials and airless chambers and drop towers to replicate its strange microgravity environment. In 2019, the C-sampler couldn't penetrate their simulated material. By the time it was installed on MMX, however, it was performing reliably. After years of stress and periods of overeating and undereating, Kato was ecstatic. "Now, I feel so much better," he says, with a grin.

During each landing, the C-sampler will make a single attempt to collect about 10 grams of material. It has three canisters, giving the mission a spare shot. If the C-sampler succeeds after the first landing, the team may consider forgoing a second—although, Kuramoto says, "We might struggle a bit with that decision."

Whether the corer succeeds or fails, JAXA has another way to gather material, albeit only loose grit from the surface: the pneumatic sampler, or P-sampler, developed by Honeybee Robotics. "It's essentially a vacuum cleaner," Zacny says. Gas jets will blast into the surface, stirring up loose material and funneling it into a collection container. The P-sampler has already proved itself in space: A version flew on Firefly Aerospace's Blue Ghost lunar lander in 2025. But it comes with a risk. By firing its gas jets, it could flip the spacecraft over. To balance the downward thrust, the instrument will also vent gas upward, anchoring it to Phobos.

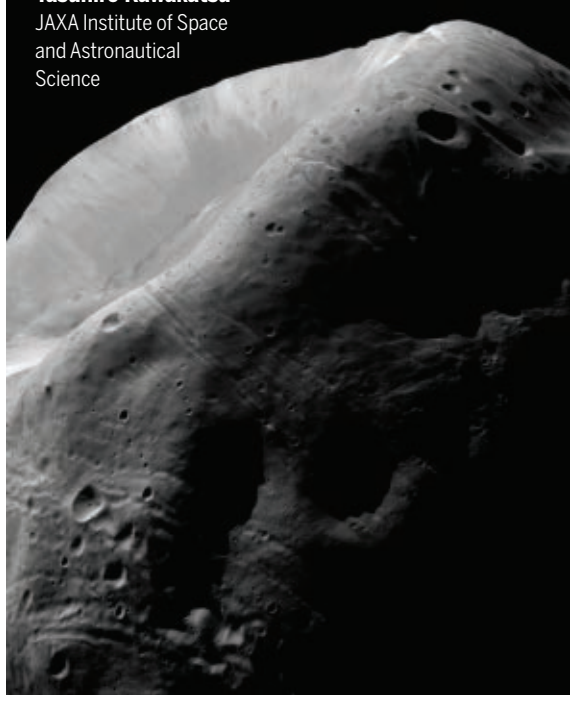
If MMX survives the first landing and sampling, JAXA will likely roll the dice a second time. MMX will lift off and drift back for another touchdown on a different part of Phobos. Then, with shards of Phobos tucked inside, MMX will ascend for some last observations of the Mars system, including multiple flybys of Deimos. In 2030, it will split apart, leaving behind the landing legs and instruments on its exploration module. With a push from its thrusters, a return module will begin the yearlong odyssey home.

BY 2031, Earth's Moon may have seen many milestones: China aims to have landed its own astronauts there for the first time, and NASA will have sent its own back. Both nations are ramping up plans to establish a lunar base.

That same year, the MMX team will be in the Australian outback, waiting for a messenger from a more distant moon. Fujiya will be present, anxiously waiting to see whether the sample capsule's parachutes deploy. If it reaches the desert intact, the precious cargo will be rushed to the sealed chamber at ISAS, where it will be gingerly opened within that hyperclean room. This moment will be "the most exciting and moving" of the entire campaign, Fujiya says.

The science payoffs could be enormous. The origin of Phobos could be resolved. If the samples contain bits of Mars, they could contain clues to the planet's ancient habitability—or, in the most extraordinary case, evidence of past or present life.

> "This is like trying to land a balloon."
>
> **Yasuhiro Kawakatsu**
> JAXA Institute of Space and Astronautical Science

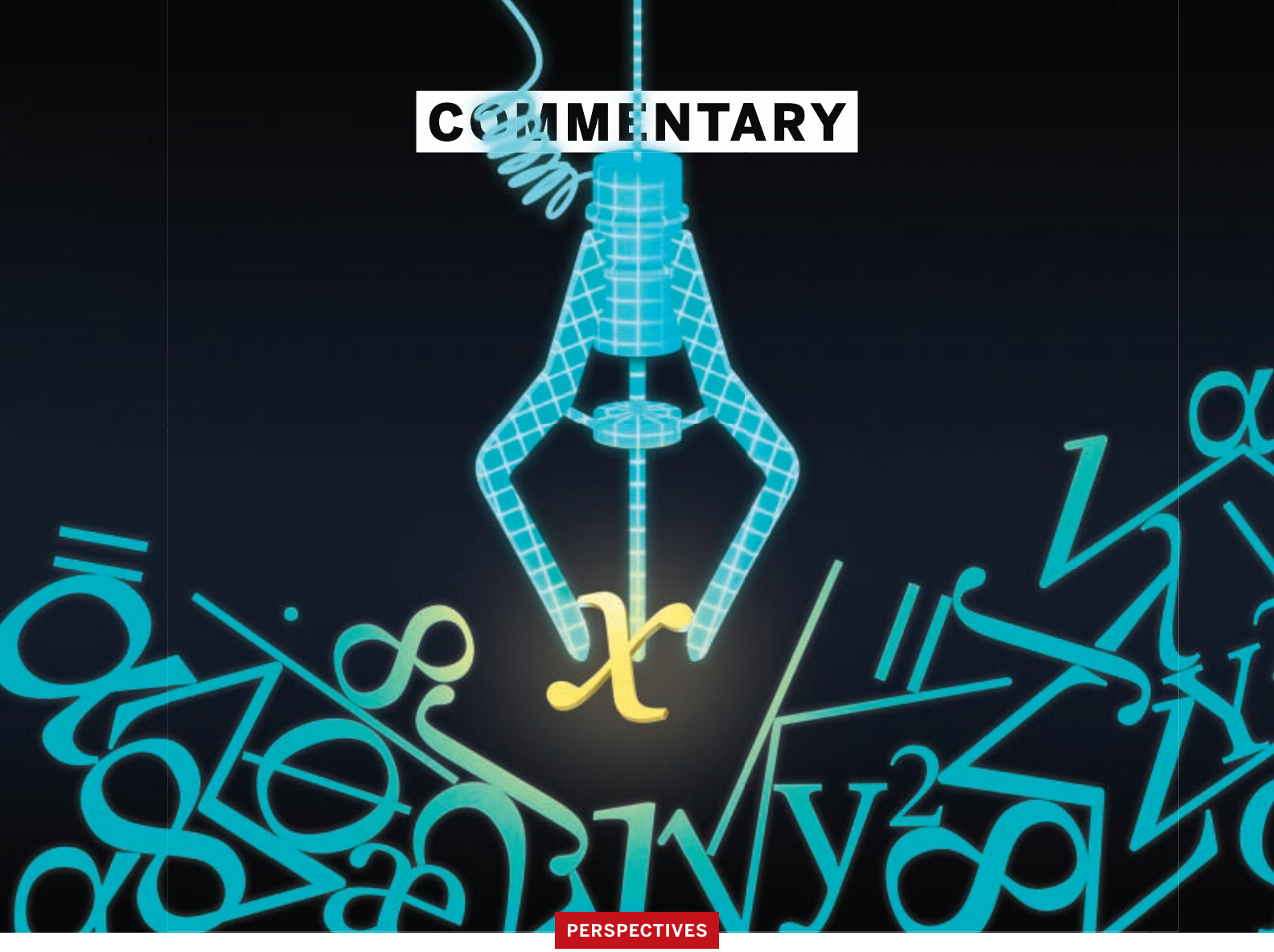


Just 22 kilometers across, Phobos is pocked with craters. MMX's two planned landings could sample different terrains.

For now, all eyes are on this month's launch of MMX, itself a tense moment. Before major spaceflight launches, researchers at NASA's Jet Propulsion Laboratory eat through jars of lucky peanuts, hoping to stave off tragedy. JAXA doesn't officially have any comparable rituals. But privately, Tsuda says, some team members will go to a shrine on Tanegashima and offer prayers—after they have run through their final checklists. Only when they feel 100% prepared for the journey ahead, Tsuda says, "do we have a right to pray." □

With reporting by Jay Alabaster.
Robin George Andrews is a science journalist based in London.

PHOTO: NASA/CALTECH-JPL/MALIN SPACE SCIENCE SYSTEMS



PERSPECTIVES

MATHEMATICS

AI for research mathematics has arrived

Artificial intelligence can now solve research problems in mathematics and verify the answers

Jeremy Avigad[1] and **Matthew Ballard**[2]

Every year, at the International Mathematical Olympiad (IMO), precollege students from around the world compete to solve difficult mathematical problems. In 2019, a team of researchers proposed the IMO Grand Challenge, which encouraged researchers to build artificial intelligence (AI) systems to solve IMO problems at the level of a gold medalist. The challenge required the systems to produce proofs that can be checked by a computational proof assistant such as Lean. In 2024, Google DeepMind's AI system, AlphaProof, provided Lean with proofs of three of the competition's problems. A year later, four AI systems performed at the level of a gold medalist. However, problems posed at the IMO draw on elementary mathematical methods and are far from research-level mathematical questions. On page 234 of this issue, Tsoukalas *et al.* (*1*) present AI agents that can solve open mathematical problems, marking a milestone in the application of AI to mathematics.

The results of Tsoukalas *et al.* directly followed other notable achievements. On 5 February 2026, 11 mathematicians posed benchmark research problems as part of the First Proof challenge—an inaugural initiative to evaluate the ability of AI models to perform research-level mathematics (*2*). During the 8-day challenge window, Google DeepMind and OpenAI generated solutions to some of the problems using their internal AI models, which they reported publicly after the official solutions were released (*3*, *4*).

Later that month, Math Inc. announced that its proprietary proving agent had formalized proofs of optimal sphere packing in 8 and 24 dimensions, in which identical, nonoverlapping spheres are arranged to fill space as densely as possible. Formalization is the process of writing mathematics in a language precise enough that it can be checked mechanically by a computer program, in this case the Lean proof assistant. It involves spelling out arguments in complete detail so that each step can be checked against the axioms and rules of a mathematical foundation. Math Inc.'s formalization built on pre-

ILLUSTRATION: ADRIÀ VOLTÀ

vious work (*5*), and the original sphere-packing results contributed to Maryna Viazovska's Fields Medal in 2022 (*6*, *7*). In the past, formalizing such a proof to verify its correctness could take months or even years of effort. The ability of AI systems to do this automatically represents a turning point for research mathematics.

Tsoukalas *et al*. used AI agents to find formal solutions to research-level mathematical problems. These agents used publicly accessible language models (Google Gemini 3.1 Pro and Gemini 3.0 Flash) and a proprietary agent (AlphaProof). The authors implemented four agents with a common framework, AlphaProof Nexus, to produce AI agents ranging from a basic agent featuring a simple feedback loop to an evolutionary variant that relies on AlphaProof. All four systems produced formal proofs that could be verified by Lean. Tsoukalas *et al*. tested the systems on a database of combinatorial and number-theoretic problems posed by Paul Erdős (known as "Erdős problems") and a database of open conjectures from the On-Line Encyclopedia of Integer Sequences, for which no solutions were generally known. In a post hoc comparison on the nine Erdős problems solved by the full-featured agent, the basic agent also solved them, although it required greater cost to solve the hardest ones. The finding was surprising because the basic agent only uses a commercial large-language model and the Lean proof assistant. This heralds a shift from specially trained systems toward general-purpose agents as the underlying models improve.

Since the prepublished version of Tsoukalas *et al.* was shared, progress in AI-assisted mathematics has been remarkable. Presently, about 10 to 25% of the preprint papers posted on arXiv in mathematics acknowledge substantial use of AI (*8*). Recently, Anthropic announced a complete Lean-verified formalization of Fermat's Last Theorem (*9*) that was finished in 11 days. The original proof of the theorem by Andrew Wiles (completed with Richard Taylor) was hailed as a triumph for mathematics (*10*, *11*), but a formalization was expected to take years of human effort. A few days later, OpenAI announced that a swarm of AI agents proved that there are solutions to the Navier-Stokes equations—a system of equations that models the movement of fluids—where mathematical values approach infinity within a finite time. This likely settles two cases of the Clay Mathematics Institute's Millennium Prize Problems. Nevertheless, several mathematicians have raised concerns that such large-scale uses of AI disincentivize mathematical exploration and understanding (*12*, *13*).

In contrast to most corporate announcements, Tsoukalas *et al*. provide information about the methods they used, enabling academic researchers to reproduce many of the results. An associated repository (*14*) contains all formal proofs reported in the paper, together with select natural-language proofs, and the supplement provides deformalized versions of several of the Lean proofs. Although AlphaProof and the agentic software infrastructure are not publicly available, the supplementary data include pseudocode, prompts, and information about the evolutionary rating scheme that can enable others to implement similar agentic systems. This contributes to the scientific community's ability to learn from the results and to reproduce and audit them.

Moreover, Tsoukalas *et al.* report on case studies in which mathematicians used the AI systems to solve problems in optimization, graph theory, algebraic geometry, additive combinatorics, and quantum optics. The mathematicians reported that even unsuccessful proof attempts by AI could help them understand the problems better and make progress on solving them. This underscores that an essential goal of developing AI for mathematics is to support mathematicians in the search for knowledge and understanding that lead to further advances. □

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Warming temperatures and melting ice cover will alter the reactions that methylate mercury and potentially increase mercury accumulation in seafood.

MERCURY POLLUTION

Climate change reshapes the global mercury cycle

How will climate change shape future human risks to mercury exposure?

Aryeh Feinberg[1] and **Alfonso Saiz-Lopez**[1]

Human-driven mercury emissions over millennia have increased its concentrations in the atmosphere, soils, and oceans (*1*). Once in the environment, it can undergo methylation where a methyl group binds to inorganic mercury, forming methylmercury, which is a highly toxic organic compound. In addition, mercury salts (e.g., mercury sulfate) have been historically used as industrial catalysts in chemical plants, producing methylmercury as a byproduct. The concentration of methylmercury multiplies as it moves up the food chain, harming top predators, such as humans, by impairing their reproduction, neurological, and cardiovascular functions (*2*). Research and policy have generally focused on controlling the anthropogenic emissions of mercury to reduce its public health and environmental damages. However, mercury's movement through the Earth system depends on complex interactions between environmental compartments that are vulnerable to climate change (*3*). This could potentially increase environmental and human health risks, despite expected anthropogenic mercury emissions reductions in the future.

Anthropogenic mercury emissions began thousands of years ago as unintentional releases during ancient metal refining. During the early Industrial Revolution in the mid-1700s, large-scale releases took off from commercial and industrial uses of mercury and the combustion of fossil fuels, which themselves contain trace amounts of the element. In the 1950s and 1960s, releases of methylmercury from chemical plants to local bays in Japan caused outbreaks of Minamata disease, an irreversible neurological disease caused by severe mercury poisoning. These incidents increased the global awareness of mercury pollution and associated health risks. In 2013, a United Nations Minamata Convention on Mercury agreed on a global treaty to protect human health and the environment by implementing the best-available technology to limit mercury emissions from power plants and industrial facilities, eliminate primary mercury mining activities, and phase out mercury uses in chemical processes and consumer products.

Policies influenced by the Minamata Convention have had positive impacts on the environment. Measured atmospheric mercury concentrations have been decreasing in Europe and North America since the 1990s and in East Asia since the 2010s (*4*). Consequently, mercury concentrations in aquatic organisms are expected to decline, although the recovery rates of fish species depend on watershed characteristics such as size, hydrology, and soil and sediment chemistry (*5*). Further decreases in anthropogenic mercury emissions are anticipated, except for sources that are more difficult to regulate, such as artisanal and small-scale gold mining (*1*).

Despite early successes in reducing primary anthropogenic emissions (those released directly from human sources), global baseline levels of mercury in the environment are expected to remain high for many years because of large reservoirs of legacy mercury that have accumulated in soil and water over centuries of emissions (*1*). Legacy mercury continues to circulate in the ocean-terrestrial-atmosphere system, creating a complex recycling loop that can trap the toxic element in the biosphere for centuries. This increases the time required for ecosystems to recover to natural mercury levels (*6*). Cur-

PHOTO: ANADOLU AGENCY VIA GETTY IMAGES

rently, ~30% of mercury emissions are from anthropogenic activities, whereas about 60% are from legacy mercury that has been revolatilized from ocean and terrestrial surfaces, and the remaining 10% are from natural geogenic sources (*1*). If anthropogenic mercury emissions decrease in the future, then the influence of this legacy component on the mercury cycle will grow in importance (*1, 6*). However, the impacts of climate change on legacy mercury reservoirs are still largely unexplored.

Several studies reported potential increases in mobilization of legacy mercury as a consequence of climate change (*3*). In the terrestrial system, vegetation takes up atmospheric mercury and transfers it to soils for medium-term (decades to centuries) storage. Mercury storage on land is modulated by a warming climate. For example, wildfires can volatilize mercury from soils and vegetation (*7*). Rising atmospheric temperatures also disrupt Earth's cryosphere, potentially leading to the release of mercury locked in permafrost (*8*). In addition, competing climate change mechanisms can influence mercury uptake by vegetation. Forest growth and hence mercury uptake could be boosted by a warmer climate that lengthens growing seasons and promotes faster vegetation growth through the carbon fertilization effect, a process by which trees and plants leverage high atmospheric carbon dioxide (CO_2) levels to fuel photosynthesis. By contrast, increased CO_2 in the air and more frequent droughts encourage leaf pores to stay closed, hindering the absorption of gaseous elemental mercury by plants (*9*).

Climate change can affect how gaseous elemental mercury circulates through the air, oceans, and land. Shifting wind and rainfall patterns alter where mercury is delivered to the ocean (*10*). Moreover, intense heat waves and elevated tropospheric ozone levels (*11*) may accelerate oxidation of unreactive elemental mercury gas into water-soluble ionic forms, which transfer to Earth's surface and oceans by rain, snow, or fog. Decreases in snow cover and sea ice in polar regions alter the levels of the main atmospheric mercury oxidant, bromine radicals, although the direction of future changes is uncertain (*12*). Warm ocean temperatures can reduce the solubility of mercury by increasing kinetic energy, forcing dissolved gaseous mercury out of the ocean into the air. Declining sea ice cover further removes the physical barriers to this air-sea gas exchange. However, in areas with decreasing wind speeds there will be less turbulence at the water surface, decelerating the transfer of gaseous mercury (*13*).

Changes in climate shape the mercury levels in aquatic organisms. For example, warm water temperatures and expansion of anoxic zones have been suggested as drivers of enhanced mercury methylation by microbes (*3*). By contrast, declines in chlorophyll concentrations and sea ice boost the amount of light that reaches the surface ocean, accelerating photodemethylation reactions in which a methyl group is removed from methylmercury by solar radiation (*13*). Moreover, mercury levels in seafood may also depend on changes to aquatic food webs in the future. For example, Atlantic cod shifted its diet to larger fish species because of overfishing of herring between the 1970s and 2000s. This change in diet led to higher accumulation of mercury in their bodies (*14*). In addition, mercury bioaccumulation in coastal areas and fresh waters is sensitive to organic matter and mercury runoff from land (*3*), processes that are expected to increase with climate change. The organic matter runoff stimulates methylating microbes and increases methylmercury concentrations in organisms found in coastal areas (*15*). This complex interplay of processes highlights the biotic interactions that need to be deciphered to understand how mercury risks will change under changing climate.

Future studies should focus on determining whether climate change will amplify or mitigate future mercury pollution. In many cases, a narrow set of potential climate-driven mechanisms has been considered in predictions of future impacts on mercury cycling. In others, predictions have been made by a single model that only provides one representation of the global mercury cycle amongst many potential scenarios. Integrating field and laboratory measurements could uncover key processes in the mercury ocean-terrestrial-atmosphere cycle that are affected by climate change. Long-term atmospheric monitoring and ancillary meteorological data collection are particularly valuable for detecting the influence of climate-driven processes on atmospheric mercury trends. In addition, there have been very few examples of mercury measurements in soil and water bodies for extended periods. These data could help detect changes in mercury flows between environmental compartments. A specific focus should be placed on regions that are sensitive to climate change, such as the Arctic and the tropics, where Indigenous communities rely on locally caught seafood as a principal source of protein and are thus particularly vulnerable to mercury pollution. In particular, the Amazon rainforest is threatened by increasing droughts and wildfires as well as continued releases of mercury by artisanal and small-scale gold mining.

Existing models of the global mercury cycle largely focus on individual environmental compartments instead of considering the exchange of mercury between them as an interconnected system. This creates gaps in assessing the role of climate and anthropogenic emissions on global legacy mercury levels. Emerging Earth System Models that include the mercury biogeochemical cycle provide a framework to holistically investigate potential climate impacts, having been used to quantify mercury flows through the environment and into biota as well as their implications for human health (*2*, *9*). These modeling exercises should be evaluated against both modern-day observations and paleoenvironmental records to identify key parameters that need to be better defined, such as the residence times of mercury in different environmental compartments. Ice core and sedimentary records retrieved across different regions offer an opportunity to investigate how the mercury cycle responded to past climate changes without anthropogenic influence (*12*). By testing the models' performance of representing past mercury cycle changes under different climates, the accuracy of future predictions can be improved.

If climate change enhances the mobilization of legacy mercury in the environment or aggravates the methylation of inorganic mercury, then the planned decreases in anthropogenic mercury emissions may not be sufficient to reduce risks to the environment and human health. This may necessitate the development of stricter cuts in human activities to compensate for climate-driven impacts. An improved understanding of the myriad roles that climate plays in the mercury cycle can improve predictions of how mercury pollution will evolve in the future and inform policies for managing changing mercury risks. □

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IMMUNOLOGY

Before the immunological Big Bang

The nature of the gut microbiome before vaccination influences later inflammatory reactions and fever

Bali Pulendran

Cosmologists have long wondered what, if anything, existed before the Big Bang. Immunologists face a more tractable version of this question. Vaccination or infection can trigger an immunological Big Bang—a cascade of molecular and cellular events that unfolds over days, weeks, and months, culminating in the establishment of antigen-specific memory lymphocytes within specialized microenvironments in the bone marrow and other tissues. Unlike cosmologists, however, immunologists can peer into the moments before their Big Bang. Increasingly, they are asking how the baseline state of the immune system—the immunological "initial conditions" that exist before vaccination or infection—shapes the immune response that follows (immunogenicity). On page 191 in this issue, Huus *et al*. (*1*) report that these initial conditions also shape susceptibility to vaccine-induced fever, a normal manifestation of the body's systemic reaction to vaccination (known as "reactogenicity").

Predicting vaccine reactions

Individuals react differently to a third dose of a SARS-CoV-2 mRNA vaccine according to the composition of their microbiome and their baseline inflammation state before vaccination.

	Before vaccination	After vaccination
Group A	• Low flagellated bacteria concentrations • Low TLR5 activation • Less baseline inflammation	• Lower rise in temperature (<0.8°C increase)
Group B	• High flagellated bacteria concentrations • High TLR5 activation • More baseline inflammation	• Greater rise in temperature (≥0.8°C increase) • More systemic symptoms (e.g., headache, fatigue, chills)

TLR5, Toll-like receptor 5; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

The baseline immune state reflects a lifetime of encounters with microbes and vaccines and is also influenced by age, genetics, metabolism, environmental exposures, and signals from the gut microbiome (*2*, *3*). This preexisting immune state can predict the magnitude of the immune response to vaccination (*4*, *5*). Huus *et al*. show that this baseline state is also associated with reactogenicity. They studied 171 healthy adults receiving severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) mRNA vaccines, most of whom were taking a third dose. They found that individuals who had an increase in body temperature of at least 0.8°C after vaccination had distinct biological features before vaccination, such as low-grade intestinal inflammation and gut microbiomes that were rich in flagellated bacteria from the *Lachnospiraceae* family, including *Waltera* (see the table). These bacteria express flagellin proteins that activate Toll-like receptor 5 (TLR5) in immune cells. This innate immune receptor senses bacterial flagellin and triggers inflammatory responses. These fever-associated microbiome features did not change with vaccination. Furthermore, in an independent cohort of more than 1000 adults receiving SARS-CoV-2 vaccines, *Waltera* and the broader *Lachnospiraceae* family were also more abundant in individuals reporting systemic vaccine reactions, such as fever, headache, and muscle pain. However, reactogenicity was self-reported rather than determined by independent temperature measurements, and the association with self-reported fever was stronger at the broader *Lachnospiraceae* family level than for *Waltera* specifically.

To establish whether these associations are causal, Huus *et al*. performed experimental studies. They found that fecal products from individuals with greater vaccine-induced temperature increases induced stronger inflammatory responses in human colonic organoids grown in culture than those from individuals with smaller temperature rises. Furthermore, when fecal microbiota from human donors who differed in their postvaccination temperature responses were transferred into germ-free mice, they elicited different inflammatory responses to subsequent SARS-CoV-2 vaccination. Mice that received microbiota from people with greater temperature increases had stronger vaccine-induced hypothermia (a typical manifestation of reactogenicity in mice) and higher serum concentrations of the proinflammatory molecule interleukin-6 (IL-6) compared with mice that received microbiota from individuals with lower rises in temperature. In mice genetically engineered to lack TLR5 and NLR family CARD domain-containing protein 4 (NLRC4)—extracellular and intracellular sensors of flagellin, respectively—these effects were attenuated but not abolished.

Notably, the abundance of flagellated bacteria linked with fever did not correlate with the strength of the vaccine-induced antibody response. This contrasts with studies in mice and humans showing that the microbiome can function as an "endogenous adjuvant" that triggers the innate immune system to respond more actively to vaccination. This is particularly evident in individuals with little preexisting immunity who receive vaccines that provide limited immune stimulation on their own (*6*). In mice, TLR5 sensing of microbiota-derived flagellin was required for optimal antibody responses to the inactivated influenza and polio vaccines, which are given without adjuvants—substances added to increase the immune response. However, TLR5 sensing was not required for antibody responses to vaccines with adjuvants such as those against tetanus, diphtheria, and pertussis (Tdap) or hepatitis B, or to the live viral yellow fever vaccine, all of which provide strong stimulation of the innate immune system—the rapid first phase of the immune response (*7*). In humans, antibiotic disruption of the microbiome impaired antibody responses to seasonal influenza vaccination, especially in individuals with low preexisting immunity (*8*), and to rabies vaccination in previously unvaccinated individuals (*9*). Studies in infants have also linked the composition of the early-life gut microbiome, including the abundance of *Bifidobacterium*, with the magnitude of antibody responses to primary (initial) vaccination (*10*, *11*).

In the study of Huus *et al*., most participants were receiving a third dose of a SARS-CoV-2 mRNA vaccine and therefore had substantial preexisting immunity. Moreover, mRNA vaccines stimu-

late the innate immune system (*12*, *13*). In this setting, a flagellin-rich microbiome did not further enhance the antibody response to vaccination but instead increased reactogenicity. Thus, the effects of microbiome signals seem to depend on the immunological context. When vaccine-derived stimulation and preexisting immunity are limited, the microbiome may provide endogenous adjuvant signals that enhance immunogenicity. When these are already abundant, the same microbial signals may contribute instead to the inflammatory response that manifests as reactogenicity.

Realizing that the responses to vaccination are context-dependent has broad implications. Systems vaccinology has traditionally focused on the molecular cascades that occur after vaccination (*2*, *14*). However, Huus *et al*. show that these responses emerge from an interaction between the vaccine and the physiological state into which it is introduced. The findings shift attention toward the baseline state, including the microbiome, basal inflammation, and metabolism, that exists before vaccination. Understanding the interactions between these factors may eventually enable clinicians to predict not only how well someone will respond to a vaccine but also how they will feel afterward.

A provocative possibility is that some of these initial conditions can be modified by diet. Controlled dietary intervention can rapidly remodel the gut microbiome and the body's immune state (*15*). Huus *et al*. observed that participants who reported vegetarian or vegan diets were less likely to develop fever. However, this association was based on self-reported data, broad dietary categories, and only 46 participants and needs to be assessed in future studies that are specifically designed to test it.

Diet, microbial ecology, and the metabolic state are not immutable, and people might one day prepare for vaccination as deliberately as the vaccine itself is formulated. Realizing this possibility would require studies in which people's diets are deliberately changed before vaccination to determine whether vaccine reactogenicity can be reduced without compromising protective immunity. Unlike cosmologists, immunologists can observe what exists before their Big Bang and perhaps use that knowledge to shape the universe that follows. □

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PHOTO: ANDRES M. DOMINGUEZ/NPL/MINDEN PICTURES

New knowledge of alkaloid biosynthesis in lupins could lead to targeted breeding for use in agricultural, chemical, and pharmaceutical settings.

PLANT SCIENCE

Shifting the equilibrium on lupin alkaloids

A biosynthesis breakthrough promises applications in food and chemistry

Zhouqian Jiang and **Benjamin R. Lichman**

Plants produce myriad alkaloids primarily as protection against biotic stresses. Human interactions with these bioactive nitrogen-containing compounds are, however, a delicate equilibrium. Their desirable neuroactive or agrichemical properties are balanced against negative traits such as addiction and toxicity. The double-edged nature of alkaloids is at play in quinolizidine alkaloids, such as lupanine and sparteine, that accumulate in the seeds of lupins (*Lupinus* spp.). These compounds have natural utility for plant defense against herbivores, insects, and pathogens and have applications in the synthesis of other organic compounds.

However, their toxicity prevents widespread adoption of lupins as a protein crop, which could reduce dependence on livestock and alleviate pressure on soybean agriculture. On page 190 of this issue, Mancinotti *et al.* (*1*) report the complete biosynthetic pathway of quinolizidine alkaloids in *L. angustifolius* (narrow-leaved lupin). This knowledge should accelerate the breeding and use of these plants in agricultural, chemical, and pharmaceutical settings.

Mancinotti *et al.* combined gene coexpression analysis, mutant analysis, and pathway reconstruction to determine the complete set of enzyme-catalyzed steps that convert the amino acid L-lysine into complex quinolizidine alkaloids. The reported pathway, comprising 15 enzymatic steps, involves 11 newly described enzymes and diverse reactions including decarboxylation, oxidation, dimerization, rearrangements, reduction, and acylation. Notably, major latex protein–like proteins (MLPLs) control four steps of the pathway. MLPLs are from the plant-specific PR10 (pathogenesis-related protein 10) family, which includes enzymes that are involved in so-called "spontaneous" reactions that can occur in mild aqueous conditions even when the enzyme is absent (*2*). The MLPLs involved in quinolizidine alkaloid synthesis similarly catalyze spontaneous reactions related to the formation and conversions of imines, the unifying chemistry for building the molecular backbones of alkaloids (*3*).

...sequence similarity can neither guarantee identification of a missing enzyme step nor suggest what activity a candidate may have.

Like all enzymes, quinolizidine alkaloid MLPLs appear to speed up a chemical reaction, but they cannot alter its natural end point—the equilibrium state. Therefore, the functions of MLPLs rely on selective and irreversible downstream steps to shift the equilibrium and drive the reaction forward. Mancinotti *et al.* suggest that for MLPL4, the driving force is a molecular rearrangement followed by a hydrolysis. For other MLPLs in the pathway, the driving force likely derives from cofactor conversion in a subsequent enzyme-catalyzed step, such as the downstream reductases that drive the activities of MLPL1 and MLPL3 and the acetyltransferase that drives the activity of MLPL2 and MLPL6.

Dynamic kinetic resolution, a process that produces a single specific chiral chemical from mixed starting materials, has not previously been reported in the biosynthesis of a plant compound (*4*). Mancinotti *et al.* found that MLPL2 and MLP6 are involved in such a process, catalyzing the interconversion of multiple forms of the alkaloid precursor Δ^1-piperideine, but where only one specific dimer form is a substrate of the downstream acetyltransferase enzyme. Thus, the MLPL-catalyzed equilibrium plus selective downstream conversion drives the system rapidly towards a single chiral product.

Another remarkable aspect of the quinolizidine alkaloid pathway is IRED3 (imine reductase 3), which is incapable of catalyzing reduction yet somehow enhances the stereoselectivity of the reduction reactions catalyzed by IRED1 and IRED2. Analogous reductase-like proteins that are noncatalytic or catalyze nonredox conversions have been reported in multiple plant natural product pathways (*5*–*7*), but it is unknown how IRED3 exerts its stereochemical control.

The MLPLs and IRED3 in quinolizidine alkaloid biosynthesis are examples of enzymes whose activities could not have been predicted through gene sequence homologies with other plant enzymes. This adds to a growing list of proteins in important plant pathways with unexpected roles. These include carbonic anhydrase–like proteins in the biosynthesis of huperzine in Chinese club moss (*Huperzia serrata*), which is used as a dietary supplement (*8*); a noncatalytic reductase in Madagascar periwinkle (*Catharanthus roseus*) that produces the anticancer medicine vinblastine (*6*, *7*); and a protein involved in the biosynthesis of the chemotherapeutic Taxol in yew (*Taxus*) (*9*).

These studies, and many others, show that the field has entered a post-homology paradigm in which sequence similarity can neither guarantee identification of a missing enzyme step nor suggest what activity a candidate may have. Instead, by combining gene coexpression analysis, metabolite profiling, and combinatorial screening of enzymes, it is possible to test large candidate lists that can be selected without regard to predicted gene functions. This approach should lead to the identification of many more enzymes and helper proteins not yet associated with plant biosynthesis pathways. Notably, this paradigm is necessarily empirical, as in silico approaches are unable to expect the unexpected.

The complete elucidation of quinolizidine alkaloid biosynthesis promises to have rapid and extensive impact. Indeed, Mancinotti *et al.* have already demonstrated the potential for generating new lupin varieties through mutants that lack specific individual biosynthetic steps (*10*). Mutant plants lacking the *HYD* or *CYP76E36* genes showed a substantial reduction of quinolizidine alkaloid production compared to wild-type plants. Incorporating these genetic changes into existing low-alkaloid varieties may lead to ultralow-alkaloid plants and a wider adoption of lupins as a protein source.

Identifying genes encoding the full quinolizidine alkaloid pathway also presents an opportunity to generate a much greater range of chiral chemicals. For example, the mutant plant lacking *IRED3* shows accumulation of (+)-sparteine, supporting the enzyme's unusual role in stereochemical control but also indicating that biosynthetic production of (+)-sparteine—which does not typically accumulate naturally—is feasible. Although extensively investigated to induce chirality in asymmetric organic syntheses, the use of plant-derived sparteine commercially has been limited by its cost and availability (*11*). These constraints have led to the development of sparteine chemical syntheses and surrogates (*12*). Resolution of the quinolizidine alkaloid biosynthetic pathway by Mancinotti *et al.* may provide lower-cost access to sparteine enantiomers produced either by heterologous means or by enhanced accumulation in lupin mutants (*13*). This could spark a resurgence of sparteine-controlled asymmetric syntheses, potentially at a commercial scale. □

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Scholars mistook a fleeting population boom for a timeless law in 1960. They are not the only ones making flawed predictions about the future.

PHOTO: VARAVIN88 VIA SHUTTERSTOCK

REFLECTION

Demographic "doomsday" misses the mark

A dire prediction published in *Science* in 1960 anticipated a population singularity in 2026—we are not even close

Javier A. Birchenall

Friday the 13th of November 2026 carries no particular significance, yet this date gained scientific notoriety when Heinz von Foerster and colleagues published "Doomsday: Friday, 13 November, A.D. 2026" in *Science* in 1960 (*1*), identifying it as the day when the global population would reach a mathematical singularity (a playful jab, as the date was von Foerster's birthday). With just weeks to go and no singularity in sight, it is important to understand why this prediction failed, why it once appeared compelling, and what its failure reveals about mistaking historically contingent patterns for universal laws.

A HISTORICAL QUIRK AND AN URGENT WARNING

Using data published by economist Michael Kremer in 1993 and the United Nations Department of Economic and Social Affairs in 2024, one can calculate that it took all of human history up to the year 1800 for the global population to reach one billion, but it took only 125 years for that billion to double, by 1925 (*2*, *3*). The population doubled again to 4 billion in just 50 years, by 1975. The most recent doubling, from 4 to 8 billion, occurred in less than 50 years. Declining doubling times, characteristic of faster-than-exponential growth, made Malthusian fears of overpopulation mathematically inevitable. But by Friday, 13 November 2026, the observed trajectory will have diverged so sharply from predictions that the failure calls for an explanation rather than a dismissal.

The model employed by von Foerster and colleagues was not wrong mathematically, but it was historically contingent: It correctly identified the significance of accelerating population growth and provided a valid mathematical representation for faster-than-exponential growth. The difficulty was elevating transient demographic changes into timeless universal laws. Unlike physical laws, social patterns are generally shaped by changing circumstances.

Mistaking historically contingent conditions for universal laws is unique neither to global demography nor to von Foerster and his colleagues. Extrapolating demographic conditions from the 1960s mirrors other misreadings, such as predicting the dominance of liberal democracy after the fall of the Soviet Union or extrapolating Japan's 1980s catch-up economic growth rates indefinitely into the future. Indeed, extrapolating current demographic conditions would support projections of a vanishing population (*4*), as global population growth has fallen below not merely hyperbolic but exponential rates, and most people now live in countries with below-replacement fertility.

HYPERBOLIC GROWTH IS ONLY PART OF THE STORY

Von Foerster *et al*. (*1*) derived their doomsday prediction using global population data from around 1 CE to 1958 and a hyperbolic growth model. In a hyperbolic growth model, the population growth rate is not constant but rather an increasing function of population size. This nonlinearity implies that population diverges to infinity in finite time.

Although this model proved to be incorrect for this application, the paper nonetheless made four enduring contributions. First, it formalized the notion of a singularity. The idea that accelerating technological change might approach a point beyond which projection fails had been voiced by John von Neumann in the late 1950s, but without an equation or a date. It also articulated an empirically relevant alternative to Malthus's "law of exponential growth," which had long dominated the rhetorical and mathematical foundation for population pessimism. The model was transparent and parsimonious relative to the contemporaneous *Limits to Growth* modeling (*5*), which was composed of about 150 equations that reached similarly pessimistic conclusions.

The doomsday date, although largely rhetorical, also drew attention to threats of starvation, resource depletion, and poverty, which dominated the academic and policy debates of the 1960s (*6*). Explosive population growth in Asia, Africa, and Latin America in the 1960s alarmed both academic and popular circles, as epitomized by Paul Ehrlich's *The Population Bomb* (*7*), whose prologue began with the famous declaration: "The battle to feed all of humanity is over. In the 1970s and 1980s hundreds of millions of people will starve to death in spite of any crash programs embarked upon now."

And finally, hyperbolic growth proved to be accurate for short- and medium-run predictions. Despite academic skepticism grounded in physical constraints and the absence of explicit economic and technological adaptation (*8*, *9*), hyperbolic growth consistently outperformed conventional demographic projections, which repeatedly underestimated population growth in the 1960s and 1970s.

Until the late 1970s, hyperbolic growth tracked actual population changes with striking precision (*10*, *11*). By the 1980s, however, its appeal waned as developed countries entered a below-replacement-fertility regime and the pace of the demographic transition accelerated in the developing world.

In the 1990s, rather than vanishing into obscurity, hyperbolic growth was vindicated by Kremer (*2*), who showed that faster-than-exponential population growth served as a general organizing principle to understand global population history going back to 1 million years ago. Extending Kremer's global population data with United Nations data (*3*), however, reveals a breakdown of the relationship between population growth rates and size since the mid-1960s.

LIMITATIONS IMPERILED ACCURATE PREDICTIONS

Time series studies of global population growth face important limitations. Early population levels are unknown, for example, and must be inferred from limited archaeological evidence and strong assumptions about technology, land use, and population density.

Moreover, the mid-20th-century population explosion was exceptional, overshadowing any historical or prehistorical trend. For most of human history, the population growth rate was extremely low. In the very long run, human population growth experienced long periods of stability and slow increase, followed by sharp accelerations, such as the Neolithic Revolution some 10,000 years ago and the modern population explosion (*12*).

Finally, intrinsic nonlinearities behind hyperbolic growth make predictions highly sensitive to the sample used. An observer in 200 BCE extrapolating from Neolithic trends, for example, would have vastly overestimated future population growth, with populations reaching their present-day size by the year 1000.

Once the demographic slowdown of recent decades is taken into account, the evidence for hyperbolic growth becomes even less compelling. Despite covering a relatively small number of years, this period contains the largest population values and thus exerts disproportionate influence on the overall global population trends. Updating the sample with data through 2023 weakens the positive association between population growth rates and sizes, consistent with the observed decline in global population growth beginning in the mid-1960s. This finding reinforces the central lesson of this note: Hyperbolic growth only comes close to the actual population data in very recent times, during the expansionary phase of the modern demographic transition, which is now ending.

A TIMELY LESSON FOR MODERN EXTRAPOLATORS

As of Friday, 13 November 2026, the global population will stand at roughly 8.3 billion, and it continues to grow, but at a steadily declining rate. If we were to assume that Japan's current annual rate of natural population decrease of 0.77% represented a universal and enduring trend, we would predict that the planet will be devoid of people on Tuesday, 5 May 3108. The fact that this, too, is unlikely to occur should comfort the many commentators now fearing depopulation. Their projections are not wrong mathematically, and their concern about aging societies is well founded, but both rest on treating fertility as invariant while the population pendulum swings wide.

The failure of the doomsday equation is ultimately a cautionary tale about the dangers of extrapolating from historically exceptional circumstances with lessons that extend well beyond demography. The looming threats of technological singularities are not exempt from similar misreadings. Without dismissing the transformative potential of artificial intelligence, it is crucial to remember that extrapolating current scaling laws indefinitely would make explosive or superexponential growth appear an inevitable law of technological evolution rather than a property of the time period during which those laws are observed. □

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LETTERS

Edited by
Valerie Thompson

Cooling centers like this one in Chongqing, China, invite people to gather indoors, where respiratory illness spreads more easily.

PHOTO: HE PENGLEI/CHINA NEWS SERVICE/VCG VIA GETTY IMAGES

Respiratory infections could increase as extreme heat forces more people indoors

As extreme heat continues to intensify, heat-health measures may encourage people to spend more time in cooled indoor spaces. The World Health Organization's heat-health guidance, for example, encourages people—particularly older adults, people with chronic health conditions, and those without adequate cooling at home—to seek out air-conditioned spaces and cooling centers (*1*). These measures are crucial, but their implications for respiratory infection risk within cooled indoor spaces warrant greater attention.

Many respiratory viruses, including influenza viruses, respiratory syncytial virus, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), can be transmitted efficiently through airborne particles, making high occupancy, poor ventilation, inadequate filtration, and prolonged indoor exposure key factors influencing infection risk (*2*). Recent research further supports this concern.

Analyses of outbreaks in shelters, dormitories, prisons, and other shared environments indicate that overcrowding, combined with insufficient fresh air circulation, increases the risk of respiratory infections (*3*). In southern India, measurements of ventilation conditions in residential and healthcare settings, combined with transmission modeling analyses, showed that the potential for transmission of influenza and SARS-CoV-2 is higher under conditions of poor ventilation or air conditioning than under natural ventilation (*4*). As the demand for mechanical cooling increases under high-temperature conditions, the indoor environments created by adaptive retrofits may gradually become an important component of the ecology of infectious diseases.

The issue is not limited to formal cooling centers. During periods of extreme heat, residences, workplaces, schools, health care facilities, transportation hubs, libraries, and other public spaces may all switch to a mode of operation that involves closing windows and relying on mechanical cooling. At the same time, vulnerable populations who are encouraged to seek relief from the heat at public cooling centers may gather indoors for extended periods. Public health guidance recognizes that preventing infections in cooling centers is an operational concern (*5*), but heat adaptation and respiratory infection control are still often viewed as separate planning issues.

Climate adaptation should not force people to choose between heat protection and disease prevention. Designing cool spaces that integrate both thermal safety and clean air will build greater resilience in the face of climate change, preparing us for a future that is increasingly hot and in which more time is spent indoors.

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Careful governance is needed to balance the benefits and risks of landfill mining in China

In recent years, dozens of Chinese cities across the Yangtze and Pearl River deltas have launched large-scale landfill mining, a process that entails sealed, decades-old municipal waste landfills being excavated and their contents being sorted and combusted in waste-to-energy plants (*1*, *2*). Driven by shrinking fresh combustible waste after mandatory waste classification and overbuilt incineration capacity, this emerging practice delivers tangible socioeconomic and environmental gains yet carries overlooked ecological risks that demand balanced governance (*3*).

The merits of landfill mining fall into three dimensions. First, urban land reclamation can eliminate long-term hidden hazards. Many old landfills, once suburban, are now surrounded by dense residential zones. Excavation can remove persistent threats including landfill gas leakage, leachate groundwater contamination, and methane-driven climate warming, and recovered land thus becomes available for green space or urban construction (*4*). Second, aged waste features high calorific value after decades of anaerobic decomposition; separated plastics, textiles, and rubber stabilize incinerator operation and raise power output, offsetting operational costs for waste-to-energy facilities. Third, the model alleviates landfill capacity saturation. Postincineration residues occupy far less volume, easing the nationwide "garbage siege" pressure that arises from limited new landfill construction permits (*5*).

Nevertheless, indiscriminate excavation can trigger severe secondary pollution risks. Digging disturbs sealed waste layers, releasing toxic hydrogen sulfide, dust, and heavy metal–laden leachate that endanger nearby residents and field workers without full enclosure control (*6*). Climate assessments further warn that incinerating waste that is more than 4 years old generates greater cumulative greenhouse gas footprints than leaving it buried, undermining China's carbon neutrality roadmap (*6*, *7*). Long-distance waste transport also adds fossil fuel emissions, partially canceling energy recovery benefits. Additionally, incomplete sorting sends inert heavy metal fragments into furnaces, elevating flue gas treatment burdens.

This practice thus requires targeted regulation. Authorities should enforce enclosed excavation and real-time pollutant monitoring, restrict mining of extremely aged waste, and prioritize local landfill reuse to cut transport emissions. Integrating landfill mining into standardized circular waste frameworks can maximize land and energy gains while minimizing environmental trade-offs (*8*, *9*). Balanced technical and policy safeguards are essential for turning landfill excavation into a sustainable urban waste management solution rather than a short-term capacity remedy.

In addition, it must be recognized that old landfill waste is essentially a limited and nonrenewable resource. With large-scale continuous extraction, the stock of excavable landfill waste will eventually be depleted. This means that the currently overexpanded waste incineration capacity cannot achieve long-term operation by relying on this temporary resource supply. When old landfill waste is exhausted in the future, blindly maintaining excessive incineration capacity will lead to a host of problems, including idle equipment and low resource utilization efficiency.

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Climate policies neglect risks posed by legacy landfills

The environmental hazards associated with legacy landfills can be devastating, and the risks are being intensified by climate change. In August, extreme overnight precipitation compromised the structural integrity of the Dar es Salam landfill in Guinea, which had been scheduled to close, causing a landslide that buried homes and killed at least 30 people (*1*). Earlier this summer, in July, extreme dry-season heat ignited methane pockets in Indonesia's Jatiwaringin landfill, a legacy open dump, sparking a 10-day fire across 15 hectares (*2*).

Higher temperatures accelerate organic waste breakdown to methane, which without proper collection and control, can combust or add to greenhouse gas emissions (*3*, *4*). A recent Bloomberg investigation found severe subsurface overheating in US landfills, with Virginia's closed Shoosmith facility reaching 90°C, heightening fire and contamination risks (*5*). At the UK's South Walney legacy landfill, waste is being exposed by coastal erosion, and sea-level rise and accelerated shoreline retreat amplify the risk of waste mobilization into the sea (*6*). Crucially, legacy sites also vastly outnumber active ones (e.g., >21,000 versus ~500 in the UK, respectively) (*7, 8*).

Existing policies target active landfills and address climate impacts through poorly enforced resilience mandates focusing on flooding and extreme weather. For example, UK Environment Agency climate guidance mandates enhanced drainage and flood defenses, but these only apply to current operations and do not address the vulnerabilities of historic closed sites (*9*). Although the US Environmental Protection Agency's Resource Conservation and Recovery Act guidance requires climate vulnerability assessments, these provisions remain largely advisory and only apply to new facilities, neglecting legacy sites (*10*). Further, flooding and sea-level rise are prioritized over the temperature effect on methane, as exemplified by California's flood protection orders for bayfront landfills (*11*). In China, landfill frameworks fail to mandate climate-resilient retrofits for unlined legacy dumps, risking leachate overflow and groundwater contamination (*12*).

Enforceable mandates and comprehensive, climate-informed guidelines for legacy landfills are vital to safeguard historical sites against escalating climate stressors.

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POLICY ARTICLE

The field experiment includes 16,683 students, 742 teachers, and 2,305 courses across 10 higher education institutions in India.

EDUCATION

Phones impede learning: Evidence from a massive phone-free classroom experiment

Lower-performing students benefited the most, but even higher-performing students expressed support for the policy

Alp Sungu[1], Pradeep Kumar Choudhury[2], Andreas Bjerre-Nielsen[3]

Amid growing concerns about the impact of mobile phones on learning as well as mental health and peer relationships—especially among students, ranging from primary schools to colleges (*1*–*3*)—many of the world's education systems have adopted nationwide bans on smartphone use in primary and secondary schools as a means to reduce distraction, digital addiction, and cyberbullying (*4*). Despite the surge in policy measures, there is a marked scarcity of large-scale experimental evidence evaluating the broader impacts of school phone restrictions on student learning, well-being, digital behavior, and overall educational experience. We report on a randomized controlled trial involving 16,683 students across 10 higher education institutions (HEIs) in Odisha, an eastern Indian state of more than 40 million people, where smartphone ownership among youth aged 15 to 29 exceeds 70% (*5*). Our findings suggest that phone bans can improve students' academic achievement, with particularly pronounced benefits for low-performing students.

These policies aim to protect students from potential negative academic and social consequences of using mobile devices in class. Across the United States, for example, roughly two-thirds of the states have enacted some kind of legislation, and the remaining states are considering similar legislation (*6*). Yet studies thus far, quasi-experimental and observational, have shown mixed results, ranging from meaningful improvements in academic achievement or well-being, especially for low-performing students, to no improvement (*7*). Related educational technology evidence on laptops underscores the potential for distraction, finding that unrestricted in-class device access can reduce academic performance (*2*). Our preregistered study, approved by institutional review boards, builds on this foundation by providing a large-scale randomized evaluation of a policy-relevant question where evidence is gradually accumulating in favor of phone bans improving learning. Given the prevalence of phones among youth worldwide, understanding the

PHOTO: FRÉDÉRIC SOLTAN/CORBIS VIA GETTY IMAGES

impact of removing them from classrooms is crucial for policymakers, administrators, and educators alike.

We commissioned the manufacturing of ~1000 wooden phone deposit boxes using recycled materials and installed them in classrooms across 10 partner HEIs in Odisha. At each partner institute, we stationed a full-time research assistant (RA)—with an undergraduate degree or above—and collaborated with a designated teacher, who served as the project coordinator. We randomized by independent draw, with equal probability and without blocking or stratification, at the level of the department and grade within an institution (henceforth cohort). The draw was made across 420 such units, 198 of which were assigned to the phone ban, which span 444 department-grade cohorts given that undergraduate and postgraduate students sharing a unit received the same assignment. The study included 742 teachers and 2305 courses; students had a mean age of 19.4 (SD, 1.26; range, 16 to 22), and 35.5% were female. Because the assignment was at the department-grade level, students experienced the same treatment condition consistently across nearly all their courses throughout the 2024 spring semester, running from February to May.

In the treatment courses, teachers enforced a phone ban by requiring all students to deposit their phones in a designated phone box at the start of each lecture. This was not an opt-in policy. If students had an approved exception (e.g., documented emergencies or caregiving obligations), they informed the teacher before the class session. This protocol was applied in all classes for the treatment arm throughout the semester, whereas students in the control departments maintained their usual access to phones. Because many classrooms were shared by both treatment and control arms, we placed phone boxes in all classrooms. The key difference between the two arms lies not in the presence of the boxes but in the enforcement protocol specific to the treatment arm.

Our protocol comprised three avenues of data collection. First, we obtained each student's academic achievement and attendance records from before and during the study period. Academic achievement refers to end-of-semester examinations, which are externally prepared and graded. No technology was allowed during the exams for both conditions. Second, we administered the survey in two waves over the semester, distributing the link to each partner institute's study coordinator at the start of the term and sending student reminders through them. Third, RAs conducted random classroom spot checks—with consent—by briefly watching each classroom through the window or the doorway, without interrupting the lecture. During these visits, they checked each classroom against a predefined list of observable behaviors (e.g., whether students were using their phones or asking questions), noting whether or not each behavior occurred. This procedure was carried out throughout the semester, enabling us to examine how banning or allowing phone use influenced classroom interactions and engagement from an external observer's perspective. See further details of implementation in the supplementary materials (SM) section A.

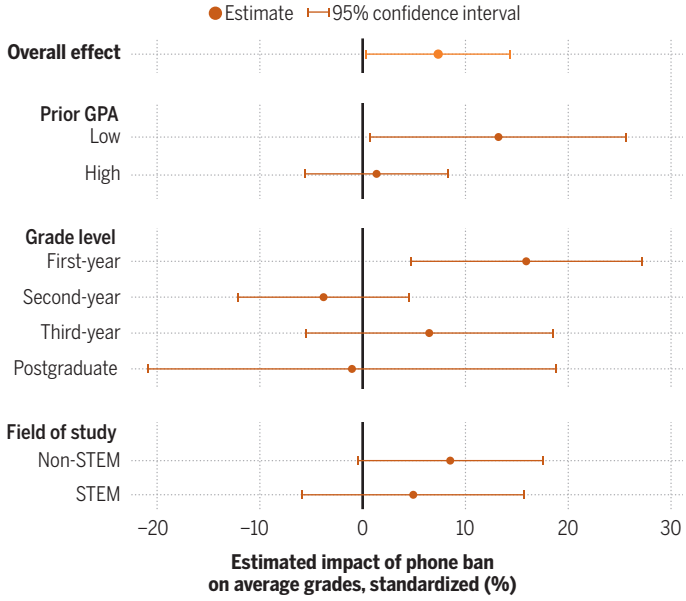
IMPACTS OF THE PHONE BAN

Academic performance

We conducted an intent-to-treat analysis estimating students' grade point average (GPA) as a function of their GPA in the prior term and the random assignment of their cohort to the phone ban policy or control group (see SM section B.1).

Restricting student access to smartphones is associated with significantly higher academic performance (see the first figure). In a cross-sectional model controlling for prior achievement, the policy yields a 0.073 SD increase in GPA [β = 0.073, 95% confidence interval (CI) = (0.003, 0.144), one-sided P = 0.021]. Baseline academic performance is balanced across treatment and control groups, with no statistically significant differences in prior GPA levels under any specification (SM tables B.1 to B.4).

Phone ban impact on student academic achievement The top estimate is the estimated average treatment effect of the phone ban on student academic achievement across the full population. The next estimates contain heterogeneous treatment effects for subgroups split by prior average academic achievement (above or below preexperiment median), years of study, and field of study.



GPA, grade point average; STEM, science, technology, engineering, and mathematics.

Our estimated improvement of 0.073 SD in academic achievement after a classroom phone ban is practically meaningful when benchmarked against educational interventions. Meta-analyses of randomized controlled trials in education find median effect sizes of ~0.10 SD, with large-scale studies (N > 500 students) typically yielding effects around 0.05 SD (*8*). For comparison, the difference between an average and a very good teacher over one academic year is roughly 0.20 SD (*9*). Successful teacher professional development programs, large-scale curriculum reforms, and growth-mindset interventions yield gains in the range of 0.05 to 0.20 SD (*10*). Hence, the impact of phone ban policies may not be transformative alone for learning outcomes, but it aligns with other scalable educational interventions. Notably, the effects are larger for low-performing students, where marginal academic achievement improvements can have meaningful downstream consequences.

Among the prespecified subgroups examined, we detect larger and statistically significant effects relative to the control only among students with lower prior academic performance and first-year undergraduates, with marginally significant effects among non–science, technology, engineering, and mathematics (STEM) majors (see the first figure). By contrast, point estimates are close to zero and not statistically significant among students with above-median prior achievement, postgraduates, or those in STEM fields, which suggests that the overall treatment effect is primarily driven by the former groups.

The ban's statistically significant GPA increase is robust across model specifications, outcomes, and levels of aggregation (SM tables B.6 to B.15). It is somewhat less precise once we account for fixed differences between students, but its size and direction are stable. The phone ban policy also reduced GPA inequality among treated students (SM table B.16). At the student level, GPA dispersion, measured as the SD of GPA values, increased after the policy

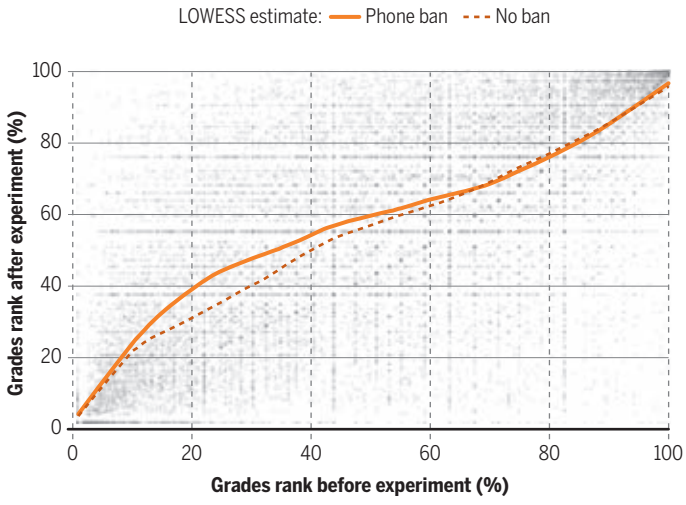
GRAPHIC: V. PENNEY/*SCIENCE*

for both groups but more sharply for control students (0.152) than for treated students (0.093). This difference corresponds to a relative reduction in the postpolicy GPA dispersion of 0.060 SD for treated students. We also measured GPA concentration using the Herfindahl–Hirschman Index (HHI) computed over the distribution of students across observed GPA values; larger values indicate that student GPAs are more concentrated at a smaller number of GPA values, whereas lower values indicate a more even spread across the GPA distribution. Because the HHI reflects only how students cluster across GPA values, and not the distances between those values, we interpret it as a measure of concentration rather than of inequality per se. The HHI decreased substantially among treated students (−0.147) but only marginally among controls (−0.025), yielding a large relative drop in concentration (0.122). These results indicate that the policy was associated with a relative reduction in both dispersion and concentration of GPA among treated students.

Locally weighted scatterplot smoothing (LOWESS) curves provide nonparametric estimates of students' posttreatment rank as a function of pretreatment rank and treatment or control group, allowing for flexible trend detection (see the second figure). Students in the phone ban units (treatment group) systematically outperformed their peers in control schools across most of the rank distribution, with the largest divergence among lower-ranked students. This pattern is consistent with the subgroup analysis, reinforcing that the intervention had disproportionately positive effects on lower-performing students. These findings are robust to alternative specifications of the LOWESS model (SM figs. B.1 to B.4). We also assess whether the observed academic improvements are mirrored by changes in student attendance. Across specifications using both average attendance (SM table B.21) and course-level attendance measures (SM table B.22), we find no consistent evidence that the phone ban affected student attendance. These results suggest that the academic gains were not primarily driven by increased classroom presence but more likely reflect changes to in-class attention and dynamics.

Largest divergence among lower-ranked students

Locally weighted scatterplot smoothing (LOWESS) curves show nonparametric estimates of postexperiment grade ranking as a function of the preexperiment grade ranking, stratified by treatment. Percentile ranks are defined over the pooled distribution of all cohorts and terms rather than within the plotted sample, so gains in one group need not be mirrored by losses in the other. Dots reflect individual students, with darker shading indicating more students accumulated at that combination of ranks.



Students' perspective and classroom dynamics

A major concern about classroom phone bans is how students perceive these policies and how they affect in-class behavior. Confiscating their devices may inadvertently deepen disengagement from school, especially for those already facing challenges. Understanding student perspectives is therefore critical for successful policy implementation.

We surveyed a subset of students (N = 2557) to examine their perception. We find that students exposed to the phone ban substantially improved their attitudes toward ban policies (see the table). Students who experienced the in-class phone bans became significantly more supportive of such restrictions [β = 0.174, 95% CI = (0.059, 0.289), t = 2.961] and perceived them as beneficial for them [β = 0.196, 95% CI = (0.080, 0.311), t = 3.323], relative to peers in classrooms without a ban. Notably, treated students' preference for a "no ban" classroom (i.e., no phone-use restrictions) has significantly diminished [β = −0.186, 95% CI = (−0.292,−0.080), t = −3.434]. Given five policy options (full-day bans, lecture-only bans, phone-allowed zones, app-based limits, or no ban), those assigned to ban treatment during lecture hours were significantly less likely to choose the option of no ban. Directionally, preferences in the treatment group shifted toward stricter policies relative to control, with the largest shift toward all-day bans (SM table C.10). Although we tested only a lecture-only ban, this pattern suggests that once students experience the benefits, the specific form of restriction matters less than having some policy in place. Notably, we find no treatment effect relative to control on students' perceived distraction, perceived learning, subjective well-being, or academic motivation, which suggests that students' overall perceived educational experience is unlikely to be substantially affected (SM table C1). We observe a mild increase in the treatment group relative to control in fear of missing out (FOMO) but no changes in phone-use time, frequency of bringing phones to school, or online harassment (SM table C.1).

Female students in the treatment group, relative to female students in the control, show stronger support for phone bans and perceive greater benefits, whereas male students in the treatment arm, relative to male students in control, report a small increase in FOMO, increased screen time, and greater perceived distraction (SM table C.14). We observe no heterogeneity by field of study (SM table C.15). Turning to classroom engagement, support for phone bans is stronger among "frontbenchers" (students who are more likely to be engaged in class), who also report greater perceived benefits from the restriction (SM table C.12).

Despite low-performing students benefiting more from the ban, more motivated students exhibit a stronger preference for it—perhaps because a more learning-conducive environment is more salient to them, even when actual gains accrue disproportionately to low performers.

In addition to the survey, our RAs randomly checked classrooms, providing an exploratory external perspective on classroom dynamics that complements students' self-reported responses (SM fig. C.1B). RAs were blinded to treatment condition; however, they could infer it by observing phone use or the placement of phone boxes. Thus, spot checks reflect external perceptions rather than purely objective measurements. Notably, during spot checks, students in the treatment group appear more distracted than students in the control group. This discrepancy may be explained by attention allocation models (*11*): Removing a primary source of interference (phones) may heighten sensitivity to alternative, less salient stimuli (e.g., ambient noise). By eliminating a key source of self-distraction,

Effects of the smartphone ban on attitudes toward the ban

Effects in pooled-sample standard deviation units; standard errors (SEs) clustered at the level of treatment assignment.

Outcome	Effect (SE)	95% CI
Supports phone ban policy	0.17* (0.06)	(0.06, 0.29)
Preference toward no phone ban	−0.19** (0.05)	(−0.29, −0.08)
Perceived benefits of phone ban	0.20** (0.06)	(0.08, 0.31)

CI, confidence interval; N = 2557; *P < 0.01; **P < 0.001.

GRAPHIC: V. PENNEY/*SCIENCE*

the phone ban may prompt students to redirect attention toward alternative distractions more easily noticeable to an external observer.

Our findings on other in-class behaviors reveal a more consistent pattern. During spot checks, teachers in phone-free classrooms appear statistically significantly busier maintaining order and student focus and more engaged with educational materials. These point-in-time observations reflect teacher activity rather than classroom disorder and may partly reflect enforcement of the ban itself. Most notably, despite not being subject to the ban themselves, teachers reduced their own in-class phone use in treated classrooms. Echoing this more engaged teacher behavior, students exhibited significantly fewer instances of disruptive behavior and less peer-to-peer conversation unrelated to course material. Additionally, students report a mild decrease in use of other electronic devices during lectures, potentially indicating complementarity among in-class digital technologies.

From a policy perspective, our results suggest that initial enforcement of phone-free classrooms can potentially become self-reinforcing. The convergence of improved academic performance and increased student receptivity underscores the potential effectiveness of such policies: Phone bans can cultivate student buy-in when learners personally experience benefits through enhanced attention and performance. Viewed jointly, student surveys and in-class spot checks provide suggestive evidence that classroom phone bans redirect time away from phones toward enhanced classroom dynamics—potentially translating into a healthier learning environment and improved student outcomes.

SHEDDING LIGHT ON A CENTRAL QUESTION

Our study helps address a central policy question: What is the impact of banning phones in classrooms? That we observe particular benefits for lower-performing students is critical to advancing educational equity. The decline in test scores over the past decade is concentrated among low-performing students, which may be linked to device use. Improving outcomes for students who are already academically lagging could help narrow achievement gaps. We also find that students who experience the ban exhibit an increased preference for it, which is important for policy support. Finally, our results indicate a potentially more supportive learning environment, with less disruptive student behavior and increased teacher effort.

A potential concern is whether, beyond actual learning, the observed academic achievement improvement reflects treatment effects on teacher grading. Institutional features make this unlikely. Under Odisha's centralized assessment regulations (*12*), 60 to 80% of each course grade derives from end-semester examinations set by external subject experts and evaluated at off-site zones, where graders are blinded to student identity and treatment status. Notably, non-STEM courses allocate 80% of marks to this external component, yet our effects are largest in this subsample, where teacher discretion is most constrained. Internal marks are jointly evaluated by instructors and outside examiners under college-level oversight, and exam halls are phone free for all students, regardless of condition. Lacking a teacher-course identifier, we cannot empirically rule out a contribution from shifts in internal grading, but external, blind-graded exams dominate the GPA.

There are several important boundary conditions of this study. First, our study was conducted in HEIs, where students are already academically selected, and the broader policy focus is on K-12 education. Given the stronger effects observed among first-year as well as low-performing students, impacts might be more pronounced at earlier educational stages—an important avenue for future research.

Additional research is also needed to understand the impact of a bell-to-bell phone ban, beyond the phone-free classroom design examined in this work. Banning phones during breaks could potentially yield more pronounced social benefits for interactions outside the classroom and affect students' overall digital habits. Finally, it is important to examine how these findings generalize to other geographic contexts, particularly more economically developed countries, where students often have additional digital devices.

In the context of behavioral and educational sciences, our findings of increased academic achievement can be theorized to be driven by increased attention to course material. Prior research has indicated that merely having a smartphone nearby can erode attention and cognitive capacity (*13*). Removing phones can reduce extraneous cognitive load, freeing working memory resources for deeper cognitive processing of course material and providing the external structure needed to support learners with limited executive control in their self-regulated learning (*14*). In the context of behavioral economics, the ban might operate as a commitment device that curbs behavioral tendencies to prioritize short-run (digital) temptations over long-run (academic) returns.

The impact of digital technologies extends far beyond classrooms. As these technologies have become fundamentally entrenched in people's lives worldwide, their effects on cognition and learning are increasingly important. Our findings—that removing phones from classrooms positively influences objective performance and aligns with students' preferences—corroborate the argument that phone interference might disrupt various cognitive processes. This disruption may particularly affect tasks requiring sustained engagement or uninterrupted attention, such as productivity at work, enjoyment at social gatherings, safety while driving, or sleep quality.

Removing digital devices when they interfere with such activities could represent a broader solution, highlighting an avenue for future research. From an educational policy perspective, our results indicate that restricting phone use in classrooms can support learning. □

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SUPPLEMENTARY MATERIALS

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RESEARCH

IN *SCIENCE* JOURNALS

Edited by **Michael Funk**

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EL NIÑO

Cranking up the variability

How is the El Niño–Southern Oscillation (ENSO) responding to modern climate change? Cole *et al.* showed that anthropogenic warming is dramatically increasing interannual sea surface temperature variability in the eastern equatorial Pacific. By looking at the geochemistry of coral skeletons in the Galápagos, the authors were able to document a trend toward stronger El Niño events that parallels the rise in global temperature and has produced multiple extreme El Niños over recent decades. This trend has serious implications for ecosystems, infrastructure, and human welfare worldwide. —Jesse Smith

Science p. 192, 10.1126/science.ady2660

Researchers sample a coral in the Galápagos to extract data for studying past climate variability.

SOLAR CELLS

Melt management of grain boundaries

Adding zinc halide and methylammonium chloride salts to perovskite inks creates a eutectic that enhances crystallization and passivation. Yoo *et al.* showed that local melting of the eutectic mixture at grain boundaries during annealing enhances grain ripening. This process also redistributes zinc to grain boundaries and interfaces, where it likely forms a zinc iodide interstitial species that helps to passivate nonradiative defects. Small-area cells retained 85.8% of their initial power conversion efficiency of 24.1% after 1400 hours of continuous operation at 85°C. —Phil Szuromi

Science p. 228, 10.1126/science.aeb4554

POPULATION GENETICS

A chain of migration

In humans, population bottlenecks have been demonstrated to result in founder effects, in which the population has a skewed sampling of alleles from the previous population, and this can lead to a higher prevalence of some Mendelian disorders. Studying the genomes of individuals from populations across Polynesia, Shanks *et al.* found that these populations experienced population bottlenecks more extreme than other noted founder populations such as the Finnish, and that there was an increase in effect intensity with subsequent migrations. The authors also identified known autosomal-recessive disease variants at higher than usual frequencies due to these founder effects, which has implications for clinical guidance in these populations. —Corinne Simonti

Science p. 216, 10.1126/science.aec7662

VIROLOGY

Exposure route drives viral infection

Major features of viral infections, including the severity of the diseases that they cause, vary depending on the route and dose of virus exposure. However, revealing the factors that underpin this variation through experimental infection studies is challenging due to small sample sizes. By analyzing data from over 100 published experiments using nonhuman primates, Sneddon *et al.* showed that the route of virus exposure plays a more important role in determining severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection patterns than dose, age, sex, or species, with aerosol inoculation characterized by greater nasal shedding. This study reveals the importance of virus exposure route in disease biology and highlights the utility of multistudy analyses. —Edward Holmes

Sci. Adv. (2026) 10.1126/sciadv.aei4853

QUANTUM OPTICS

Memories of spin glass

Our brains can often reconstruct information from imperfect, fragmented memories. Associative computer memories, such as the well-known Hopfield networks, attempt to replicate this process, but maximizing both storage capacity and robustness to recall error is a challenge. Marsh *et al.* found a solution to this problem in an unlikely candidate: a spin glass engineered in a quantum-optical system of atoms and light. The researchers turned what is normally a weakness of spin glass systems, a rugged energy landscape full of local minima, into an advantage by using a particular form of nonequilibrium dynamics to navigate that landscape. Additionally, the dynamic positions of the atomic gases involved in the scheme allowed the landscape to deform, further enhancing the memory capacity. —Jelena Stajic

Science p. 212, 10.1126/science.aec3917

T CELLS

Protection or harm?

Previous work has suggested that γδ T cells are critical players in immune responses to helminth infections, but their contribution is not fully understood. Using a mouse model of percutaneous *Nippostrongylus brasiliensis* infection, which recapitulates the initial skin and late lung immune responses of helminth infection in nature, Papotto *et al.* found that γδ T cells in the skin are activated by the interleukin 1 (IL-1) receptor and then translocate to the lungs. There, as IL-17–producing γδ17 T cells, they contribute to type 2 immune mechanisms of tissue repair by suppressing interferon-γ signaling. When mice were instead infected intravenously (thereby bypassing dermal γδ T cell activation), lung IL-17 levels were reduced, resulting in increased interferon-γ and enhanced tissue damage. —Seth Thomas Scanlon

Sci. Immunol. (2026) 10.1126/sciimmunol.aec7574

QUANTUM COMPUTING

High-fidelity quantum gating

Neutral atom quantum computers are a leading approach, combining large numbers of qubits, nonlocal connectivity, continuous operation, and efficient methods for quantum error correction. Evered *et al.* showed that an array of Rydberg atoms can now be controlled and manipulated to provide high-entangling gate fidelities that persist for hours. The ability to flexibly create and manipulate entanglement with low gate error rates will be crucial for developing quantum information science and technology. —Ian S. Osborne

Science p. 197, 10.1126/science.aei1312

IN OTHER JOURNALS

Edited by **Corinne Simonti** and **Jesse Smith**

CANCER

Immunotherapy resistance

Checkpoint inhibitor immunotherapies can be highly effective for certain patients with mismatch repair-deficient (MMRd) colorectal cancer. However, mismatch repair-proficient (MMRp) tumors are largely resistant to single-agent therapy. Cattaneo *et al.* investigated how MMRp tumors avoid detection by the immune system. By engineering experimental models in which MMRd and MMRp colorectal cancer cells were subject to similar levels of antigen presentation, the researchers found that MMRp tumor cells actively suppressed recognition and destruction by T lymphocytes. The MMRp tumors displayed glycosylation patterns and produced a secretome-mediated molecular shield that served as an immune evasion mechanism. These new insights may provide a means of improving immunotherapy efficacy for treatment-resistant colorectal cancer. —Priscilla N. Kelly

Cancer Discov. (2026) 10.1158/2159-8290.CD-26-0542

MICROBIOLOGY

A bacterial brake on eukaryotic mRNA

Chemical modifications are a common means of regulating proteins and nucleic acids. Although nucleic acid modifications have been more commonly understood as a process that hosts use to regulate their own gene expression, invading microbes often produce factors that make such posttranslational modifications to proteins in their hosts. McCaffrey *et al.* identified a secreted effector protein from the bacterium *Legionella pneumophila* that binds uracil-rich regions of host mRNA and catalyzes oxo8G modifications, a type of oxidative RNA damage. The modification inhibits ribosomal translation of mRNA and sets up a permissive environment for bacterial replication in the host cell. —Cheri Sirois

Cell Host Microbe (2026) 10.1016/j.chom.2026.07.018

CELL BIOLOGY

A parking place for cell debris

Senescent cells accumulate broken components and danger signals that could lead to their death. Durik *et al.* found that these cells delay their demise by shedding damaged mitochondria, amyloid, and damage-associated molecular patterns into external structures called senescent cell adhesion fragments (SCAFs). The formation of SCAFs depends on adherens junctions, which link cells to each other. When SCAFs eventually burst, their contents promote wound healing and cell proliferation, a hallmark of cancer, whereas amyloid could also promote neurodegeneration. In the future, SCAFs could be used to recognize and eliminate senescent cells, which characterize many aging-related diseases. —Joana Osório

Dev. Cell (2026) 10.1016/j.devcel.2026.08.002

MATERIALS SCIENCE

Dynamic origami

The art of folding paper into complex three-dimensional shapes has been extended to a much broader class of materials, including metals and textiles, and has found value in the design of solar panels and biomedical devices that transform once deployed. However, their shapes are limited by the predetermined folds, or in the case of actuating materials, in the forces embedded

PHOTO: TIM FITZHARRIS VIA MINDEN PICTURES

MUTUALISM

Pollination breakdown

Climate change can disrupt species interactions, such as pollination, that are critical to species' survival. However, few studies have observed directly how interactions change in warming experiments. de Manincor *et al.* warmed plant-pollinator communities of annual wildflowers and a solitary bee species, measuring changes in their interactions, phenology (timing), and traits that influence interactions and resulting fitness. Warming increased seedling mortality and decreased bee cocoon numbers, both directly and by decreasing the number of bee visits to flowers. The decrease in species interactions was due to lowered bee abundance, number of flowers, and nectar volume. These results show that stress-induced trait changes, in addition to phenological mismatch, can weaken mutualisms with climate change. —Bianca Lopez *Glob. Chang. Biol.* (2026) 10.1111/gcb.71074

Hotter temperatures weaken pollinator interactions with plants such as the pictured desert bluebell (*Phacelia campanularia*), with ramifications for fitness under climate change.

in the materials. Huang *et al.* designed a "living" polymer capable of dynamic growth and chain remodeling when supplied with monomers, cross-linkers, and catalysts. This enables control over folding angles and mechanical stiffness, as the authors demonstrated in both rigid and nonrigid patterns that could be repeatedly reconfigured. —Marc S. Lavine

Adv. Mater. (2026) 10.1002/adma.74348

QUANTUM OPTICS

Solar-powered quantum source

The Sun is a blackbody source of largely incoherent photons across the electromagnetic spectrum. Although spatial and wavelength filtering have shown that coherent solar light can be obtained, Li *et al.* now show that sunlight can be used to produce pairs of entangled photons. By spectrally filtering and focusing sunlight into a multimode fiber and then onto a nonlinear optical crystal, pairs of entangled photons can be generated by spontaneous parametric downconversion. Such a solar-generated quantum source could be used to deploy photonic quantum technologies into resource-scarce environments where the traditional vacuum, cooling, and electronic infrastructure would entail expensive overhead. —Ian S. Osborne

Optica (2026) 10.1364/OPTICA.601797

ELECTROLYTES

Stable $LiPF_6$ ethers for lithium metal cells

Lithium-metal batteries could deliver much higher energy densities than at present, but electrolyte instability and poor interphase formation limit their practical deployment. Harks *et al.* revisited the performance of industry-standard $LiPF_6$ salt and showed how tailored cyclic ethers can overcome its long-standing tendency to induce solvent polymerization. The authors identified molecular design rules, including methyl substitution near the ether oxygen and larger ring structures, that suppress ring-opening reactions and allow the formation of stable $LiPF_6$/ether electrolytes. These electrolytes form a distinct inorganic-rich inner and organic-rich outer bilayer solid electrolyte interphase and support highly reversible lithium cycling. —Jack Huang

J. Am. Chem. Soc. (2026) 10.1021/jacs.6c10611

Edited by **Michael Funk**

CELL BIOLOGY

Phosphoinositides in mitochondrial fusion

Mitochondria are organelles within cells that are involved in energy production and cell-fate decisions. They are dynamic structures with shapes and connectivities that are continuously remodeled through fission and fusion events to maintain cellular homeostasis. Although several organelles have been implicated in mitochondrial division, how interorganellar communication regulates mitochondrial fusion remains poorly understood. Aki *et al.* demonstrated a role for Golgi-derived vesicles and phosphoinositide signaling lipids in promoting mitochondrial fusion. Their findings uncover a mechanism linking Golgi-derived membrane trafficking to mitochondrial membrane remodeling and show how distinct vesicle populations coordinate mitochondrial dynamics and maintain cellular homeostasis. —Stella M. Hurtley

Science p. 186, 10.1126/science.adx9628

VOLCANOES

Kīlauea's year of fountains

Lava fountaining is the spectacular and hazardous consequence of magma and gas pushing their way through subvolcanic networks. Over the past year, Hawaii's Kīlauea volcano has experienced more than 50 episodes of lava fountaining, the most ever recorded there. Drawing from extensive summit-monitoring data, Flinders *et al.* documented the start of the eruption and synthesized how the volcano has changed in the lead-up to each fountaining episode. Patterns of summit inflation have allowed quantitative forecasting for the onset of each episode, and early rock and gas analyses suggest that these pressure changes are linked to the behavior of shallow magma and volatiles. —Angela Hessler

Science p. 187, 10.1126/science.aef2931

DOG EPIGENETICS

Epigenetics track life span in dogs

Larger dogs tend to have shorter life spans than small dogs, although the underlying mechanisms remain unclear. Because epigenetic markers track with age in other organisms, Mariner *et al.* generated methylomes for 889 domestic dogs as part of the Dog Aging Project. Using dogs with a known birthdate, they created an age predictor, finding that large, male, and young dogs showed accelerated epigenetic aging patterns, and that divergence from chronological age correlated with mortality risk. Dogs showed increased methylation in promoters of genes down-regulated with age in humans and decreased methylation in transposable elements, with canid-specific genes showing increased changes in large dogs. These results point to potential targets of aging in dogs, with possible translational impacts in humans. —Corinne Simonti

Science p. 188, 10.1126/ science.aeb2986

BIOCATALYSIS

Putting oxygen-based radicals to work

Carbon-phosphorus bonds are rare in nature, but phosphonate-containing small molecules are nonetheless incredibly useful in medicine, agriculture, and other fields. Unfortunately, the few natural enzymes that make molecules with carbon-phosphorus bonds are not amenable for use in bioindustrial processes. Zhou *et al.* thus developed a distinct biocatalytic approach based on radical chemistry in which a flavin enzyme activates molecular oxygen to generate superoxide or hydroxyl radical intermediates that oxidize a phosphite precursor. After addition to an olefin, the enzyme directs hydrogen atom transfer to generate a chiral product. The authors supported their mechanistic proposal with a computational study and optimized enzymes through directed evolution to generate a wide range of potentially useful phosphonate products. —Michael A. Funk

Science p. 189, 10.1126/science.aef3001

PLANT METABOLISM

Pathway to making bitter lupins sweet

Lupins are leguminous crops with high seed protein content, but their adoption as food and feed crops has been limited by their accumulation of bitter-tasting and toxic quinolizidine alkaloids. Using a combination of coexpression analysis and stepwise pathway reconstruction, Mancinotti *et al.* have elucidated the biosynthetic pathway of these alkaloids in narrow-leafed lupins, including the identification and characterization of 13 new biosynthetic genes (see the Perspective by Jiang and Lichman). They then used this information to identify the causal mutation underlying the low-alkaloid content of sweet white lupin and to engineer a low-alkaloid, narrow-leafed lupin. This work offers strategies for the targeted breeding of low-alkaloid, sweet lupins while also advancing our understanding of quinolizidine alkaloid biosynthesis. —Unnati Sonawala

Science p. 190, 10.1126/science.aei5520; see also p. 171, 10.1126/science.ael7880

VACCINATION

Microbiome drives vaccine side effects

Some people develop a fever after receiving a vaccine and others don't. To investigate whether intestinal microbiome differences might explain postvaccination fever responses, Huus *et al.* studied the gut microbial community and inflammation markers in people who received COVID-19 vaccines (see the Perspective by Pulendran). People with measurable fever tended to have higher levels of certain flagellated bacteria in their intestinal samples. Surveys of the participants, combined with mouse and in vitro studies, suggested that diets high in fat or processed food additives may promote the expansion of these motile bacteria and contribute to an overall inflammatory state that primes the body for fever. —Cheri Sirois

Science p. 191, 10.1126/science.aea7733; see also p. 170, 10.1126/science.ael7318

MACHINE LEARNING

Mathematical discoveries by AI

Bridging the generative power of large language models (LLMs) with the rigor of formal verification could fundamentally redefine mathematical research practice. Tsoukalas *et al.* demonstrated this potential through the large-scale deployment of an LLM-powered formal proof search on open research-level problems (see the Perspective by Avigad and Ballard). Their framework, called AlphaProof Nexus, for agents that iterate between LLM-based proof generation and Lean-based proof verification autonomously resolved nine of 353 open Erdős problems, including two that had been open for 56 years, proved 44 of 492 open conjectures from the

Online Encyclopedia of Integer Sequences, resolved an open question in algebraic geometry, and improved a known bound in min-max optimization. —Yury Suleymanov

Science p. 234, 10.1126/science.aej2213; see also p. 166, 10.1126/science.ael6628

OCEAN CARBON

Let it snow

How quickly and at what depth carbon-rich marine particles called "marine snow" are degraded by bacteria as they sink have major implications for global carbon cycling. Inman *et al.* measured the nanoscale viscosity fields of marine snow to show that they are highly structured microhabitats with less-viscous and more-viscous regions, a property that controls the rate of bacterial infiltration. More-viscous aggregates take longer to degrade, sink farther, and contribute more to carbon sequestration, illustrating another way that the physical structure of marine snow can affect the global ocean carbon pump. —Jesse Smith

Science p. 202, 10.1126/science.adx4170

NITROGEN CYCLE

A question of balance

Nitrogen gas (N_2) is produced naturally by either photochemical reactions in the atmosphere or microbial processes in the biosphere, but the magnitudes of those sources are poorly quantified. Liu *et al.* showed that the abundance of the rare $^{15}N^{15}N$ isotopologue can be used to determine the amount of biogenic N_2 production in aquatic environments, thereby allowing physical and biological N_2 sources to be distinguished. This approach, applied to a wide spectrum of surface and ground water bodies, will help to better constrain the global nitrogen cycle. —Jesse Smith

Science p. 207, 10.1126/science.aei0940

IMMUNOLOGY

Fueling T cell inflammation in IBD

Pathogenic T helper 1 (T_H1) cells are T cells that exhibit glycolytic metabolism and produce the cytokine interferon-γ to drive inflammation, for example, in colitis. Mishra *et al.* used methyl pyruvate to force T_H1 cells to instead use oxidative phosphorylation, which caused the cells to become less inflammatory. Glycolytic metabolism in pathogenic T_H1 cells is driven by the kinase glycogen synthase kinase 3β (GSK3β), and treatment of a mouse model of colitis with a GSK3β inhibitor reduced disease severity. These findings highlight a GSK3β-dependent metabolic and gene expression profile that supports pathogenic T_H1 cell function. —John Foley

Sci. Signal. (2026) 10.1126/scisignal.aeb2392

QUANTUM MATERIALS

Surface engineering through premelting

Two-dimensional (2D) materials such as graphene and the 2D transition metal dichalcogenides exhibit a range of quantum transport phenomena that are dependent on the geometric configuration. Tan *et al.* report the fabrication of graphene with reduced symmetry at wafer scale using a Ge(110) surface-premelting approach. Exploiting a giant nonlinear Hall effect at room temperature, the researchers demonstrated that arrays of Hall devices can rectify a 2.4-gigahertz signal sufficiently to trigger a voltage boost converter and power an LED. The approach could be used for energy-harvesting applications to power sensors from a Wi-Fi signal. —Ian S. Osborne

Science p. 222, 10.1126/science.aeh3915

RESEARCH ARTICLE SUMMARY

CELL BIOLOGY

Golgi-derived vesicles containing PI(3,4)P_2 drive mitochondrial fusion

Sho Aki† and Mayuko Segawa† *et al.*

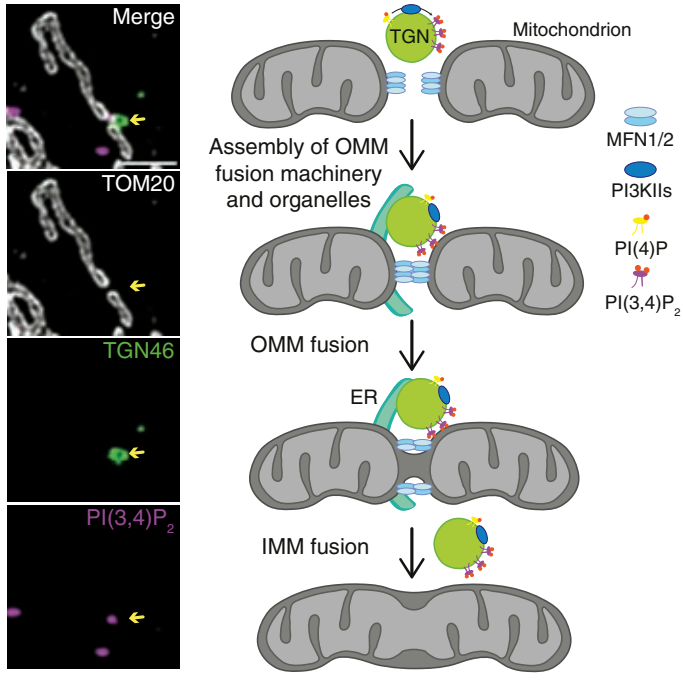
Full article and list of author affiliations: https://doi.org/10.1126/science.adx9628

INTRODUCTION: Mitochondria are double-membrane–bound organelles that generate adenosine triphosphate (ATP) and form a dynamic network remodeled by cycles of fusion and fission. These shape transitions are essential for organelle function, metabolic adaptation, mitochondrial quality control, and cell fate, and defects in these processes are associated with human disease. Although the functions of the core guanosine triphosphatases (GTPases) that drive mitochondrial dynamics are well established, these processes are now recognized to be coordinated by other organelles and regulated by local lipid signaling. However, beyond the endoplasmic reticulum (ER), whether and how other organelles, in particular, Golgi-derived vesicles, regulate mitochondrial fusion remain understudied.

RATIONALE: Phosphoinositides are low-abundance signaling lipids that are compartmentalized and regulate membrane integrity, trafficking, and remodeling. Phosphatidylinositol 4-phosphate [PI(4)P]–containing Golgi-derived vesicles facilitate mitochondrial division, raising the possibility that distinct vesicle populations containing different phosphoinositides may also regulate mitochondrial fusion. Confocal microscopy analysis of the phosphoinositide family revealed that phosphatidylinositol 3,4-bisphosphate [PI(3,4)P_2]–positive structures were closely associated with mitochondria. PI(3,4)P_2 could be generated from PI(4)P by the class II phosphatidylinositol 3-kinases (PI3Ks). We hypothesized that class II PI3Ks and PI(3,4)P_2-containing Golgi-derived vesicles may contribute to mitochondrial dynamics and, in particular, to mitochondrial fusion. To address this, we studied the effects of genetic alterations of PI3K-C2α and/or PI3K-C2β (PI3K-C2α/β) on mitochondrial morphology and examined the organelle dynamics that occur at mitochondrial fusion sites, combining confocal microscopy, live-cell super-resolution imaging, biochemical assays, cell-based and in vitro mitochondrial fusion assays, and in vivo analyses.

RESULTS: Loss of PI3K-C2α/β, or of their kinase activity, caused mitochondrial fragmentation in mammalian cell lines. This phenotype resulted from impaired mitochondrial fusion rather than increased division. Accordingly, PI(3,4)P_2-positive Golgi-derived vesicles were recruited to bona fide mitochondrial fusion events marked by the ER and the outer mitochondrial membrane (OMM) fusion proteins mitofusin 1 and 2 (MFN 1 and 2). Additionally, PI(3,4)P_2 promoted mitochondrial fusion in vitro, supporting a direct role for PI(3,4)P_2 in the fusion process. Investigation of the interplay between PI(3,4)P_2 and the core OMM fusion machinery revealed no direct interaction between mitofusin 2 and PI(3,4)P_2 but showed that loss of PI3K-C2α/β increased MFN1 and 2 oligomer accumulation, suggesting that this regulation occurs during late stages of OMM fusion.

Monitoring mitochondrial fusion and fission events revealed that distinct Golgi-derived vesicle populations were associated with different mitochondrial remodeling events. Whereas PI(4)P-positive vesicles were present at fission sites, PI(4)P and PI(3,4)P_2 double-positive vesicles were associated with fusion events, indicating that PI3K-C2α/β–dependent conversion of PI(4)P to PI(3,4)P_2 generates a subset of vesicles that regulate mitochondrial fusion. Exploring the physiological relevance of this process, we found that PI3K-C2α and PI3K-C2β loss in a cardiomyocyte-specific PI3K-C2α and PI3K-C2β double-deletion mouse model led to mitochondrial fragmentation and caused severe cardiac defects and early lethality, further highlighting the importance of this regulation in vivo.



PI(3,4)P_2-containing Golgi-derived vesicles coordinate mitochondrial fusion. A subset of Golgi-derived vesicles containing PI(3,4)P_2 that was generated from PI(4)P by class II PI3Ks is recruited to ER- and mitofusin-marked mitochondrial fusion sites, facilitating fusion. On the left are super-resolution images of HeLa cells showing mitochondria (visualized with TOM20), Golgi-derived vesicles (TGN46), and PI(3,4)P_2 (PH^{TAPP1}); on the right is a simplified schematic representation of the mitochondrial fusion process. Yellow arrows indicate PI(3,4)P_2-positive Golgi-derived vesicles in close association with mitochondria at potential fusion sites. Scale bar is 2 μm. IMM, inner mitochondrial membrane.

CONCLUSION: Our study identifies PI(3,4)P_2-positive Golgi-derived vesicles as regulators of mitochondrial fusion and class II PI3Ks as important for cardiac function in vivo. These findings support a model in which cells use specific phosphoinositide-containing Golgi-derived vesicles to coordinate either mitochondrial fusion or fission. □

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CELL BIOLOGY

Golgi-derived vesicles containing PI(3,4)P_2 drive mitochondrial fusion

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Mitochondria are dynamic organelles that remodel their shape to regulate cell fate. Mitochondrial division involves interactions with the endoplasmic reticulum (ER), lysosomes, and trans-Golgi network–derived vesicles to facilitate membrane scission. How interorganelle contacts regulate mitochondrial membrane fusion remains largely unknown. Here, we identified a role for Golgi-derived vesicles enriched in phosphatidylinositol 3,4-bisphosphate [PI(3,4)P_2] in regulating mitochondrial fusion. We found that these vesicles were recruited to ER- and mitofusin-marked fusion sites. Accordingly, loss of class II PI3-kinase isoforms α and β (PI3K-C2α and PI3K-C2β), which generate PI(3,4)P_2, led to mitochondrial fragmentation resulting from impaired fusion. Furthermore, cardiomyocyte-specific PI3K-C2α and PI3K-C2β double-deletion mice exhibited mitochondrial fragmentation and heart failure. Thus, subpopulations of Golgi-derived vesicles carrying different phosphoinositides control mitochondrial membrane remodeling and homeostasis.

Mitochondria form a complex network that undergoes continuous cycles of fusion and fission (*1*). These morphological changes are essential for maintaining mitochondrial function and determining cell fate (*2*). Mitochondrial fragmentation, which is often linked to severe cellular stress and dysfunction, plays a role in mitochondrial quality control and programmed cell death (*3*). By contrast, mitochondrial elongation allows the exchange of nucleic acids, proteins, lipids, and metabolites (*4*), and mitochondrial hyperfusion acts as a protective mechanism that enhances metabolic activity and cell survival (*5*). Dysregulation of these processes has been implicated in a range of human diseases, including neurodegeneration and metabolic disorders (*6*).

Mitochondrial morphology is primarily regulated by four large guanosine triphosphatases (GTPases) that control membrane remodeling events. Dynamin-related protein 1 (DRP1) (*7*) governs mitochondrial division, whereas mitofusins 1 and 2 (MFN1/2) and optic atrophy 1 (OPA1) control the fusion of the outer and inner mitochondrial membranes (OMM and IMM), respectively (*8*). Phospholipids, calcium, and other organelles also play a role in controlling mitochondrial shape, highlighting the complexity of these processes beyond the actions of these GTPases alone (*9*).

Mitochondrial division is a multistep process regulated by the recruitment of DRP1 at endoplasmic reticulum (ER) contacts (*10*), mediated by its adaptors, mitochondrial fission factor (MFF) (*11*, *12*), and the mitochondrial dynamics proteins of 49 and 51 kDa (MiD49 and MiD51) (*13*). Actin polymerization between mitochondria and the ER drives preconstriction of mitochondrial membranes, an initial step essential for DRP1 oligomerization (*14*), and subsequent membrane scission (*15*). Lysosomes also establish direct contacts with mitochondria, marking fission sites (*16*). Finally, trans-Golgi network (TGN)–derived vesicles are also recruited to DRP1-induced constriction sites, facilitating efficient mitochondrial division (*17*).

Mitochondrial fusion is less well characterized owing to the inherent challenges of studying membrane fusion events, including optical diffraction limits and the high frequency of intermitochondrial contacts, which can be mistaken for fusion events (*18*). Well-characterized stages of mitochondrial fusion involve OMM tethering and docking, driven by MFN1/2 oligomerization and guanosine triphosphate (GTP) hydrolysis (*8*, *19*). This is followed by IMM tubulation and fusion, mediated by L-OPA1 and S-OPA1 in concert with cardiolipin (*20*–*23*). Although ER tubules accelerate the fusion process (*24*), and ER-associated actin filaments along with mitochondrial actin facilitate fusion (*25*), the role of lysosomes in untethering intermitochondrial contacts (*18*) further illustrates the complex orchestration of cellular organelles in mitochondrial dynamics. Furthermore, the ER serves as a platform for the recruitment and assembly of both fission and fusion machineries (*26*), which suggests a spatiotemporal coupling between these processes. Whether Golgi-derived vesicles contribute directly to mitochondrial fusion remains unclear.

Phosphatidylinositol phosphates (PIPs; also termed phosphoinositides), which are key constituents of cellular membranes, regulate a myriad of functions, including membrane identity, dynamics, trafficking, contact sites, and signal transduction (*27*–*29*). Despite their low abundance, PIPs are highly compartmentalized within specific membrane microdomains, acting as molecular "zip codes" that recruit proteins with lipid-binding domains, thereby facilitating vesicle trafficking and membrane remodeling (*29*–*31*). Generated by phosphatidylinositol (PI) kinases and phosphatases, PIPs are dynamically interconverted in response to cellular signaling cues (*29*). Although PI is present in the OMM (*32*), the roles of PIPs in mitochondrial dynamics have only recently begun to emerge. Golgi-derived vesicles containing phosphatidylinositol 4-phosphate [PI(4)P], produced by phosphatidylinositol 4-kinase IIIβ [PI(4)KIIIβ], are recruited to mitochondria-ER contact sites downstream of DRP1 oligomerization to facilitate the late stages of membrane scission (*17*). Lysosomal PI(4)P may also be transferred to mitochondria (*33*), enhancing mitochondrial fission, which further supports the importance of PIPs and membrane identity in mitochondrial dynamics. However, whether other PIP species control mitochondrial morphology and whether they are involved in mitochondrial fusion remains unknown.

PI3K-C2α/β–generated phosphatidylinositol 3,4-bisphosphate [PI(3,4)P_2] regulates mitochondrial morphology

To investigate the role of PIPs in mitochondrial dynamics, we examined the potential mitochondrial localization of the seven PIPs by

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expressing established green fluorescent protein (GFP)–tagged PIP-specific probes [FYVE^{Hrs}, PI(3)P; PH^{FAPP1}, PI(4)P; PHD^{ING2}, PI(5)P; PH^{TAPP1}, PI(3,4)P_2; MLN1, PI(3,5)P_2; PH$^{PLC\delta}$, PI(4,5)P_2; and PH^{GRP1}, PI(3,4,5)P_3] (*34*–*39*) in HeLa cells (Fig. 1A). As expected, GFP-PH^{FAPP1}, marking PI(4)P, formed foci closely associated with the mitochondrial network, consistent with its established role in mitochondrial division (*17*, *33*) (Fig. 1, A and B). Confocal microscopy further revealed GFP-FYVE^{Hrs}, GFP-PH^{TAPP1} and GFP-MLN1 foci, marking PI(3)P, PI(3,4)P_2, and PI(3,5)P_2, respectively, at or close to mitochondria (Fig. 1, A and B). By contrast, PI(4,5)P_2 predominantly localized to the plasma membrane and PI(3,4,5)P_3 exhibited a nuclear signal, whereas PI(5)P displayed a low-intensity signal and a diffuse distribution, likely reflecting its low abundance (Fig. 1, A and B).

The phosphatidylinositol 3-kinase (PI3K) family comprises eight members divided into three classes, and the phosphatidylinositol 4-kinase (PI4K) family includes four members grouped into two classes, based on their substrate specificity (fig. S1A). Phosphatidylinositol 3-phosphate 5-kinase (PIKfyve) is different from the others because it does not belong to distinct classes or isoforms (*29*). Among the PIPs found associated with mitochondria, PI(3)P, PI(4)P, PI(3,4)P_2, and PI(3,5)P_2 are generated by the coordinated activities of PI3Ks, PI4Ks, and PIKfyve (*29*). PI(4)P is already known to promote mitochondrial fission and so was not reevaluated here (*17*, *33*). To determine whether the remaining PIPs regulate mitochondrial shape, we analyzed the effects of small interfering RNA (siRNA)–mediated knockdown (KD; reduced expression) of various PI3Ks and PIKfyve on mitochondrial morphology in HeLa cells. Silencing of class III PI3K (Vps34) or PIKfyve had no significant impact on mitochondrial morphology (Fig. 1, C and D, and fig. S1, B to I). Confocal and super-resolution (SR) lattice structured-illumination microscopy squared (SIM^2) revealed that single or double silencing of class II PI3Ks [PI3K-C2α and/or PI3K-C2β (PI3K-C2α/β)] led to mitochondrial fragmentation (Fig. 1, C and D, and figs. S1, D and E, and S2, A to C). This fragmented network was characterized by an increased number of mitochondria, reduced mitochondrial length and area, and decreased mitochondrial junctions within regions of interest (ROIs) (Fig. 1, C and D, and figs. S1, D and E, and S2, A to C). Similarly, both single and double knockout (KO and DKO; deletion) of PI3K-C2α and PI3K-C2β in HeLa cells, generated by CRISPR-Cas9 technology, resulted in a fragmented mitochondrial phenotype, further supporting their role in regulating mitochondrial morphology (Fig. 1, E and F, and fig. S3, A and B). Silencing and KO of PI3K-C2α/β reduced cellular levels of PI(3,4)P_2 and increased levels of PI(4)P, confirming their role in generating cellular PI(3,4)P_2 from PI(4)P (fig. S3, C to E). Loss of class I PI3K (p110α) also resulted in mitochondrial fragmentation, although this effect was less pronounced than that observed upon class II PI3K loss and was therefore not pursued further (Fig. 1, C and D, and fig. S1, B to E).

The contribution of PI3K-C2α and PI3K-C2β to mitochondrial dynamics was confirmed across multiple mammalian cell lines, including African green monkey kidney COS-7 cells, lung-cancer A549 cells, normal human dermal fibroblasts (NHDFs), and human umbilical vein endothelial cells (HUVECs). In all these cell lines, single or combined silencing of PI3K-C2α/β induced mitochondrial fragmentation (fig. S4, A to H). In different cell lines analyzed, loss of PI3K-C2α led to a more pronounced phenotype than loss of PI3K-C2β. To explore the role of PI3K-C2α and PI3K-C2β kinase activity, we expressed fluorescently tagged wild-type (WT) PI3K-C2α and PI3K-C2β or kinase-dead (kd) PI3K-C2α-D1268A and PI3K-C2β-D1213A mutants (*40*) in PI3K-C2α and PI3K-C2β DKO HeLa cells. Mitochondrial fragmentation was rescued by WT PI3K-C2α and PI3K-C2β expression but not by their kinase-dead mutants (Fig. 1, G and H, and fig. S5, A to C), demonstrating both the specificity of the phenotype and the nonredundant, cooperative role for PI3K-C2α/β–generated PI(3,4)P_2 in regulating mitochondrial morphology.

PI3K-C2α/β and PI(3,4)P_2 are also involved in clathrin-mediated endocytosis (*41*). Although treatment of HeLa cells with Pitstop 2, a clathrin inhibitor, decreased transferrin internalization, it had no effect on mitochondrial morphology (fig. S6, A to C), which suggests a distinct role for class II PI3Ks in this process. Moreover, although PIPs, including PI(3,4)P_2, can be produced through phosphatase activity (fig. S1A), loss of SHIP1/2, which generates PI(3,4)P_2 from PI(3,4,5)P_3, and silencing of PTEN, a phosphatidylinositol 3′-phosphatase that produces PI(4)P from PI(3,4)P_2, did not alter mitochondrial morphology (fig. S7, A to F), indicating that subcellular microdomains of PI(3,4)P_2 govern distinct cellular functions. Unexpectedly, knockdown of INPP4A, the phosphatase that converts PI(3,4)P_2 to PI(3)P, induced a slight but significant mitochondrial fragmentation of the mitochondrial network, characterized by decreased mitochondrial length and number of junctions in ROIs (fig. S7, D to F). We hypothesized that the accumulated PI(3,4)P_2 owing to INPP4A depletion might be converted to PI(4)P by PTEN, thereby driving fragmentation. Indeed, simultaneous loss of PTEN and INPP4A rescued the mitochondrial fragmentation phenotype observed (fig. S7, D to F). Finally, silencing SHIP1/2 and PTEN in combination also had a minimal effect on mitochondrial morphology compared with the loss of PI3K-C2α/β (fig. S7, G and H). Thus, PI(3,4)P_2 generated by class II PI3Ks is a key regulator of mitochondrial dynamics.

PI3K-C2α/β–generated PI(3,4)P_2 promotes OMM fusion

Mitochondrial morphology is determined by the balance between fusion and fission, with fragmentation resulting from decreased fusion and/or increased fission (*1*). To understand the mechanisms underlying the altered mitochondrial shape upon PI3K-C2α/β depletion, we assessed the expression levels of key effector proteins of mitochondrial fusion and fission. Surprisingly, the levels of these regulators were unchanged in silenced cells (fig. S8A), suggesting a direct role for PI3K-C2α/β in maintaining mitochondrial morphology. Furthermore, the phosphorylation state of DRP1 at Ser^{616} and Ser^{637}, modulating its recruitment to mitochondria (*42*, *43*), remained unaltered (fig. S8A). To confirm this, we analyzed DRP1 recruitment to mitochondria by confocal microscopy and observed no difference between control and PI3K-C2α/β–silenced cells (fig. S8B). Thus, the fragmentation observed did not result from increased DRP1 recruitment or elevated mitochondrial fission.

To assess whether PI3K-C2α/β depletion affects mitochondrial fusion, we performed a polyethylene glycol (PEG)–induced cell fusion assay using two HeLa cell lines expressing different mitochondria-targeted fluorophores (mEmerald-TOM20 and mCherry-TOM20). Confocal live-cell imaging revealed impaired fusion in PI3K-C2α and PI3K-C2β–silenced cells, as shown by a reduced increase of colocalization of mitochondrial probes over time (Fig. 2, A and B). Additionally, decreased diffusion of the mitochondrial matrix–targeted photoactivatable GFP probe [ornithine carbamoyl transferase (OCT)–PA-GFP] (*44*) over time corroborated the impaired fusogenic capacity of mitochondria in cells lacking PI3K-C2α and/or PI3K-C2β (Fig. 2, C and D). To confirm these results, we treated control and class II PI3K–silenced HeLa cells with cycloheximide (CHX), which triggers stress-induced mitochondrial hyperfusion (*5*). Whereas control cells displayed the expected elongation and hyperbranching of mitochondria, mitochondria in PI3K-C2α/β–deficient cells remained fragmented (Fig. 2, E and F, and fig. S8C), further supporting a defect in mitochondrial fusion.

Next, to determine whether PI(3,4)P_2 alone can promote fusion, we developed a NanoLuc Binary Technology (NanoBiT) assay for quantifying mitochondrial fusion. In this system, a split luciferase, comprising a large fragment, LgBiT, and a small high-affinity fragment, HiBiT (*45*), was targeted to the mitochondrial matrix in two separate HeLa cell lines. Isolated mitochondria containing LgBiT or HiBiT were mixed, and luminescence was measured after the addition of various PIPs. The addition of PI(3,4)P_2 enhanced mitochondrial fusion, whereas other PIPs did not enhance fusion (Fig. 2G and fig. S8D). Furthermore, loss of PI3K-C2α and PI3K-C2β decreased mitochondrial fusion in vitro, which was rescued by exogenous PI(3,4)P_2, corroborating its direct role in fusion (Fig. 2H). Finally, PI(3,4)P_2 did not rescue the fusion defects induced by loss of MFN1/2 and OPA1, indicating that they were required for PI(3,4)P_2-dependent fusion (Fig. 2I).

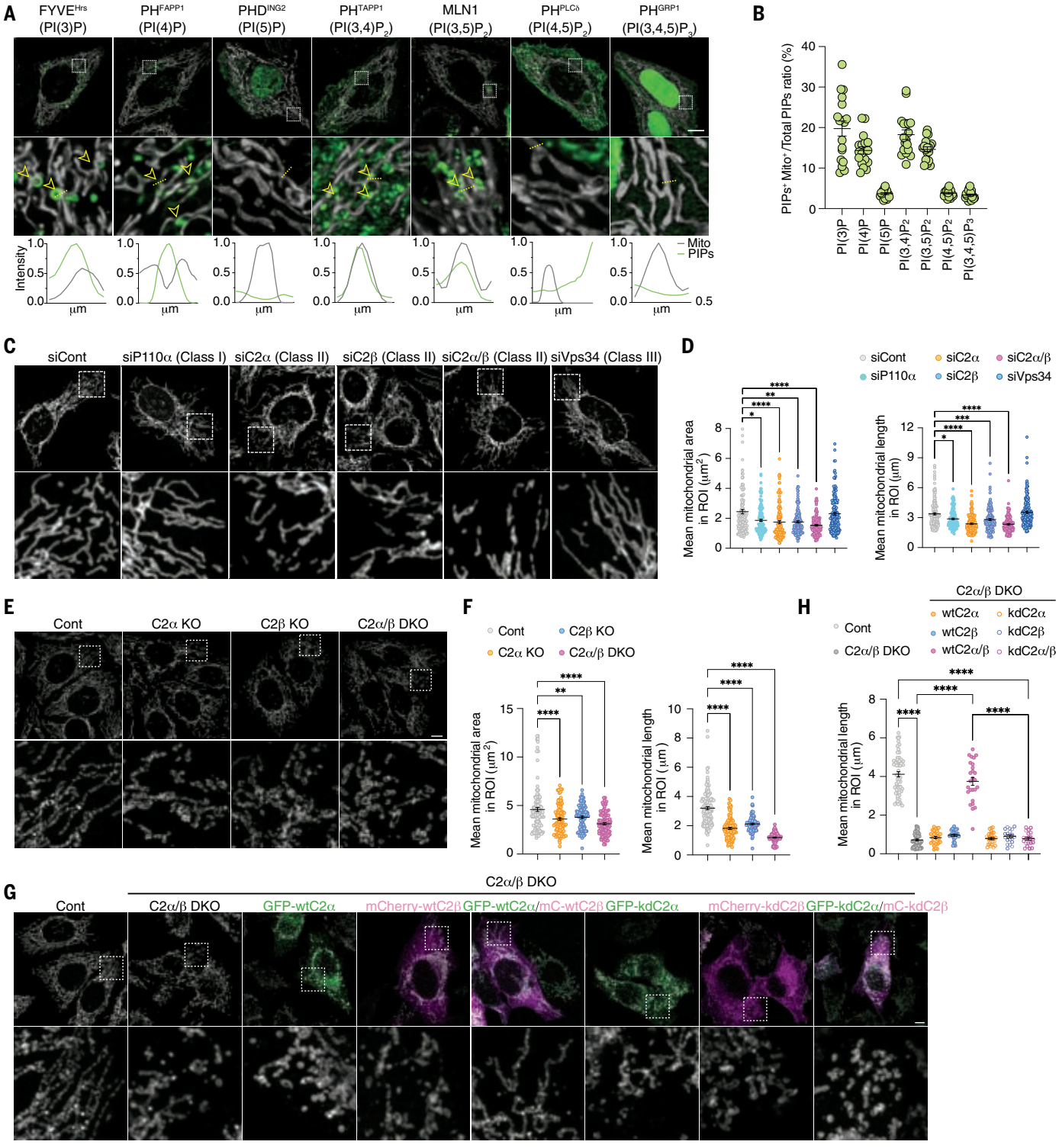

Fig. 1. Loss of class II PI3Ks leads to mitochondrial fragmentation. (**A**) Confocal images of HeLa cells expressing GFP-tagged PIP-specific probes. Mitochondria are visualized with MitoTracker. The line-scan is indicated along dashed yellow lines. Yellow arrowheads indicate GFP-PIPs associated with mitochondria. Scale bar is 10 μm. (**B**) Quantification of mitochondria-associated PIP signal relative to total PIP signal from (A). (**C**) Confocal images of mitochondrial morphology in indicated siRNA-treated HeLa cells (siCont; nontargeting). Mitochondria are visualized with anti-TOM20. Scale bar is 10 μm. (**D**) Quantification of mitochondrial morphology parameters per ROI (225 μm^2) from (C). (**E**) Airyscan 2 images of mitochondrial morphology in WT (Cont) and PI3K-C2α/β KO and DKO HeLa cells. Scale bar is 10 μm. (**F**) Quantification of mitochondrial morphology parameters per ROI from (E). (**G**) Airyscan 2 images of mitochondrial morphology in WT and PI3K-C2α/β DKO HeLa cells expressing GFP and/or mCherry–wild-type (wt) or GFP and/or mCherry-kinase dead (kd)–PI3K-C2α/β. Scale bar is 10 μm. (**H**) Quantification of mitochondrial length per ROI from (G). All values are mean ± SEM from ≥3 independent experiments unless specified otherwise. For (D), (F), and (H), data were analyzed by ordinary one-way ANOVA and Tukey's multiple comparisons test. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; ****$P < 0.0001$; ns is $P > 0.05$.

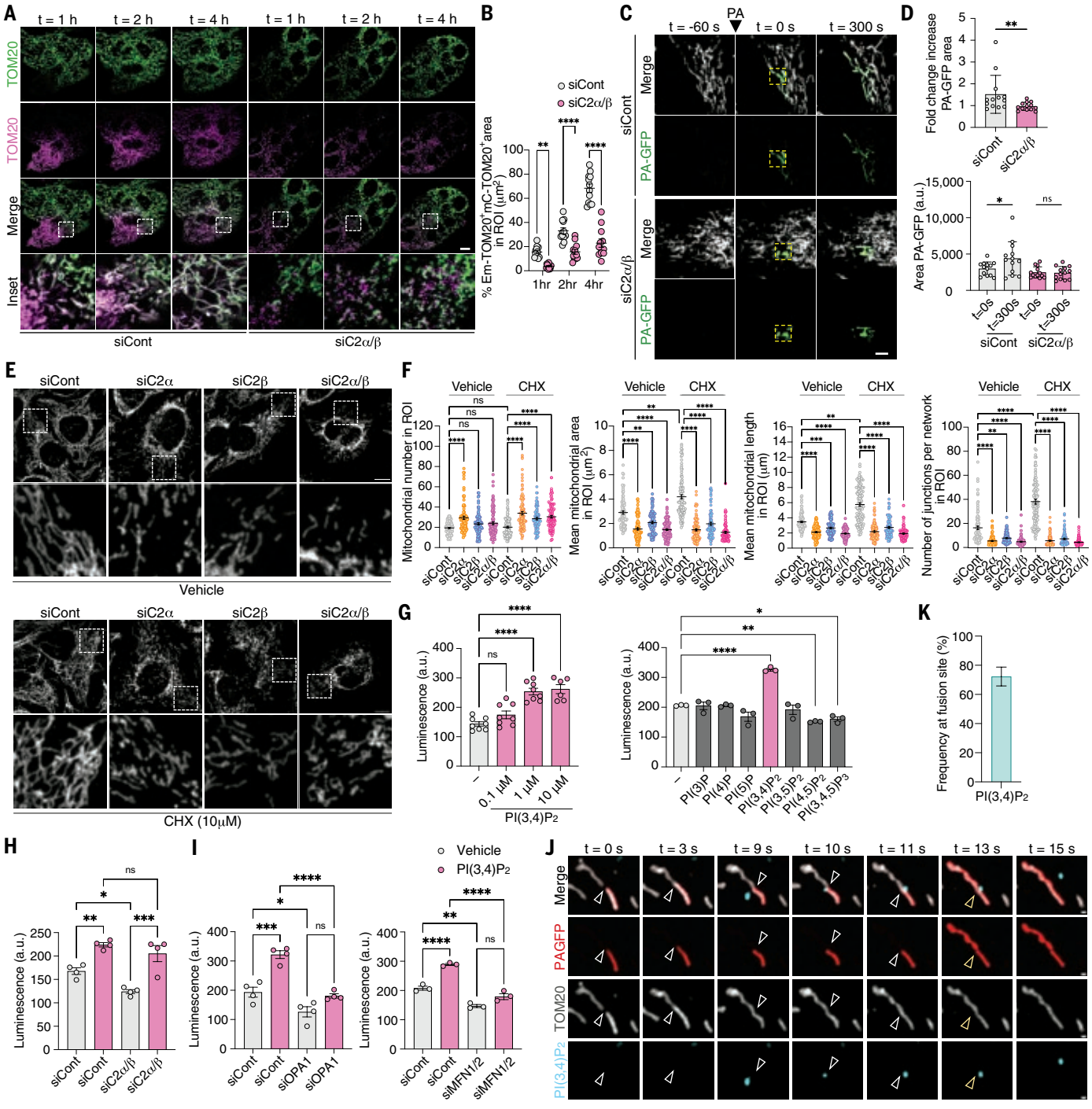

Fig. 2. PI(3,4)P_2 generated by PI3K-C2α/β promotes mitochondrial fusion. (**A**) Time-lapse confocal images of PEG-mediated cell fusion in the indicated siRNA-treated HeLa cells, separately expressing mEmerald-TOM20 or mCherry-TOM20. White indicates fused mitochondria. Scale bar is 10 μm. (**B**) Quantification of mEmerald- and mCherry-TOM20 (Em-TOM20$^+$ mC-TOM20$^+$) colocalization from (A). (**C**) Time-lapse confocal images of indicated siRNA-treated HeLa cells expressing matrix-targeted photoactivatable probe (OCT-PA-GFP) and TOM20-mCherry. Scale bar is 5 μm. PA, photoactivation. (**D**) Quantification of OCT-PA-GFP–positive mitochondrial area 300 s after photoactivation (bottom) and the area ratio at 0 versus 300 s (top) from (C). (**E**) Confocal images of mitochondrial morphology in siRNA-treated HeLa cells incubated with 10 μM CHX or vehicle (DMSO) for 6 hours. Mitochondria are visualized with anti-TOM20. Scale bar is 10 μm. (**F**) Quantification of mitochondrial morphology parameters per ROI from (E). (**G**) NanoBiT assay, measuring mitochondrial fusion, using isolated mitochondria from HeLa cells stably expressing matrix-targeted HiBiT or LgBiT and incubated for 15 min with increasing concentrations of PI(3,4)P_2 (left) or with seven PIPs at 1 μM (right). (**H** and **I**) NanoBiT assay in HeLa cells treated with siCont or siPI3K-C2α/β (H), or siOPA1 or siMFN1/2 (I), with or without PI(3,4)P_2. (**J**) Time-lapse confocal images of HeLa cells expressing TOM20-HaloTag, mCherry-PH^{TAPP1}, and matrix-targeted PA-GFP. White and yellow arrowheads indicate pre- and postfusion, respectively. Scale bars are 500 nm. (**K**) Quantification of mCherry-PH^{TAPP1} marking PI(3,4)P_2 at mitochondrial fusion sites. All values are mean ± SEM except in (D), where values are mean ± SD; data are from ≥3 independent experiments unless specified otherwise. For (B), (D), and (G) to (I), data were analyzed by ordinary one-way ANOVA and Tukey's multiple comparisons test and, for (F), by two-way ANOVA and Tukey's multiple comparisons test. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; ****$P < 0.0001$; ns is $P > 0.05$.

Given that PI(3,4)P_2 is closely associated with mitochondria (Fig. 1, A and B) and directly facilitates fusion in vitro, we hypothesized that it might be actively recruited to fusion sites. To test this, we performed live-cell confocal microscopy on HeLa cells expressing mCherry-PH^{TAPP1} and the matrix-targeted PA-GFP probe, allowing visualization of PA-GFP diffusion from a single mitochondrion to another during a complete fusion event (*46*, *47*). These experiments showed that mCherry-PH^{TAPP1} foci were specifically recruited to mitochondrial fusion sites (Fig. 2, J and K, and movie S1). Thus, class II PI3K–generated PI(3,4)P_2 directly promotes mitochondrial fusion.

PI3K-C2α/β–generated PI(3,4)P_2 drives mitochondrial fusion downstream of MFN1/2 assembly

Mitochondrial fusion is initiated by the tethering and docking of two mitochondria mediated by MFN-cis and MFN-trans oligomerization, before OMM fusion (*8*). We used live-cell SR lattice SIM^2 to examine the temporal recruitment of PI(3,4)P_2 to MFN1-positive fusion sites. Approximately 80% of MFN1-positive mitochondrial fusion events involved recruitment of mCherry-PH^{TAPP1} puncta downstream of MFN1 foci assemblies, which are present at mitochondrial tips (Fig. 3A, fig. S8E, and movie S2). Similar results were obtained using live-cell confocal imaging, where GFP-PH^{TAPP1} foci were present in ~60% of MFN2-positive fusion events (fig. S8F and movie S3). To confirm this, we used confocal microscopy and three-dimensional correlative light and electron microscopy (3D-CLEM) (*48*) to assess MFN1/2 localization in control and PI3K-C2α/β–silenced cells. We observed an increased number of closely apposed, fragmented mitochondria with MFN1/2 foci accumulating at their interfaces (Fig. 3, B and C, and fig. S8G), suggesting that the loss of PI3K-C2α/β–generated PI(3,4)P_2 impaired fusion downstream of the MFN assembly. To further support this, we analyzed MFN complex formation by sucrose gradient fractionation and found that silencing or KO of class II PI3Ks resulted in increased sizes of MFN oligomers (Fig. 3D, and fig. S8, H and I). Notably, the main peaks of both MFN1 and MFN2 were shifted to higher-molecular weight fractions, suggesting increased MFN complex formation or impaired oligomer disassembly in the absence of PI(3,4)P_2. Thus, PI(3,4)P_2 serves as a critical determinant of OMM fusion and potentially acts downstream of MFN assembly.

TGN-derived PI(3,4)P_2 vesicles drive mitochondrial fusion

Although we established that PI3K-C2α/β activity could drive mitochondrial fusion, we also wanted to investigate how PI(3,4)P_2 is recruited to fusion sites to regulate this process. In addition to their localization at the plasma membrane and endolysosomal system, class II PI3Ks can localize to TGN-derived vesicles (*49*, *50*). We hypothesized that, similar to the fission process, TGN-derived vesicles containing PI(3,4)P_2 generated by PI3K-C2α/β could be recruited to fusion sites. Using 3D lattice SIM, we confirmed the presence of PI3K-C2α on TGN vesicles, which often appeared embedded within the vesicle membrane marked by TGN46 (Fig. 3E). Live-cell confocal and SIM imaging further showed that PI3K-C2α/β, and specifically TGN-derived vesicle-associated PI3K-C2α, were recruited to mitochondrial fusion sites (fig. S9, A and B, and movies S4 and S5). Furthermore, 3D SIM and live-cell SR lattice SIM^2 imaging of GFP-PH^{TAPP1} confirmed colocalization of PI(3,4)P_2 with TGN-derived vesicles (fig. S9C) and revealed that PI(3,4)P_2-positive TGN-derived vesicles converge at fusion sites in ~70% of the events analyzed (Fig. 3F and movie S6). Upon CHX-induced mitochondrial hyperfusion, recruitment of both TGN-derived vesicles and PI(3,4)P_2 puncta to mitochondria was significantly enhanced, consistent with their contribution to fusion (Fig. 3G). To complement our in vitro findings (Fig. 2, G to I), we used an FRB-FKBP dimerization system. We used TGN38-FRB to target TGN vesicles and FKBP-tagged INPP4B to locally dephosphorylate PI(3,4)P_2 on these vesicles. Acute depletion of PI(3,4)P_2 from TGN-derived vesicles upon rapamycin treatment induced mitochondrial fragmentation (Fig. 3, H and I, and fig. S9, D to G). Thus, PI(3,4)P_2 generated by PI3K-C2α/β on Golgi-derived vesicles is recruited to mitochondrial fusion sites and allows efficient fusion.

Owing to the challenges associated with expressing PH^{TAPP1} and PI3K-C2α/β for SR and live-cell imaging, we extended our analysis to comprehensively examine the dynamic interplay between mitochondrial fusion and Golgi-derived vesicles. Using the matrix-targeted PA-GFP probe to mark complete fusion events, we corroborated that TGN-derived vesicles are recruited to ~70% of fusion events in both HeLa and COS-7 cells (Fig. 4, A to C; fig. S10, A and B; and movies S7 and S8). This observation was supported by live-cell SR lattice SIM^2 imaging using an additional TGN vesicle marker, confirming the presence of these vesicles at fusion sites (fig. S10, C and D, and movie S9). We also measured the time required for fusion, from initial OMM contact to PA-GFP diffusion, and found that the presence of TGN-derived vesicles accelerated membrane fusion compared with events lacking these vesicles (Fig. 4C). Different types of mitochondrial fusion events have been described, including end-to-end, side-to-side, and end-to-side fusion (*51*). Confocal and PA-GFP live-cell imaging showed that TGN-derived vesicles were observed in all types of fusion events (fig. S10, E to G, and movies S10 and S11). Finally, analysis of the interplay between MFNs and TGN-derived vesicles using live-cell Airyscan imaging showed that TGN-derived vesicles were recruited downstream of MFN1/2 foci accumulation at mitochondrial tips (Fig. 4, D to F, fig. S11A, and movies S12 and S13), consistent with our previous observations (Fig. 3A). To confirm this, we assessed whether blocking OMM fusion by silencing MFN1/2 affects TGN-derived vesicle recruitment to mitochondria, which was enhanced upon CHX-induced fusion. Silencing of MFN1/2 significantly decreased the recruitment and colocalization of TGN vesicles with mitochondria (Fig. 4, G and H). Because MFN2 can bind to phospholipids, including phosphatidylserine and phosphatidic acid (*52*), we tested whether it could bind PI(3,4)P_2, thereby potentially controlling TGN-derived vesicle recruitment to mitochondria. Lipid-binding analysis of immunoprecipitated FLAG-MFN2 did not reveal any interaction between FLAG-MFN2 and PI(3,4)P_2, although binding to other PIP species, including PI(3,5)P_2, PI(4,5)P_2, and PI(3,4,5)P_3, was detected (fig. S11B). Together these results support a critical role of the OMM fusion machinery in the close association between Golgi-derived vesicles and mitochondria and in the sequential events occurring during fusion.

Mitochondrial fusion frequently occurs at sites of ER contact (*24*, *26*), with lysosomes potentially facilitating mitochondrial repositioning before fusion (*24*). However, the temporal coordination of TGN-derived vesicles with these organelles during fusion remains unclear. To address this, we used live-cell lattice SIM^2 imaging, and quantitative analysis showed that ER was present at fusion sites before TGN vesicle recruitment (Fig. 4, I and J, fig. S11C, and movie S14). Specifically, ER was found at fusion sites, marked by a local increase in TOM20-mCherry fluorescence at OMM contact regions, on average 30 s before complete fusion. By contrast, TGN-derived vesicles were recruited later, ~18 s before fusion (Fig. 4J). We further confirmed these sequential observations by confocal imaging, which revealed ER tubules colocalizing with MFN2 foci at interfaces between fragmented mitochondria in PI3K-C2α and PI3K-C2β–silenced cells, without any detectable alteration in overall ER morphology (fig. S12, A and B). Similar live-cell SR lattice SIM^2 analyses showed that lysosomes were rarely present at TGN-derived vesicle-positive fusion sites (fig. S12, C and D, and movie S15).

We next asked whether distinct populations of Golgi-derived vesicles, defined by their PIP composition, may be differentially recruited to mitochondrial fusion or fission sites. Although most TGN-derived vesicles were positive for PI(4)P alone (~60%), ~20% contained both PI(4)P and PI(3,4)P_2, whereas PI(3,4)P_2-only puncta were rarely observed on TGN-derived vesicles (Fig. 4, K and L). Notably, whereas PI(4)P-only puncta were almost exclusively associated with

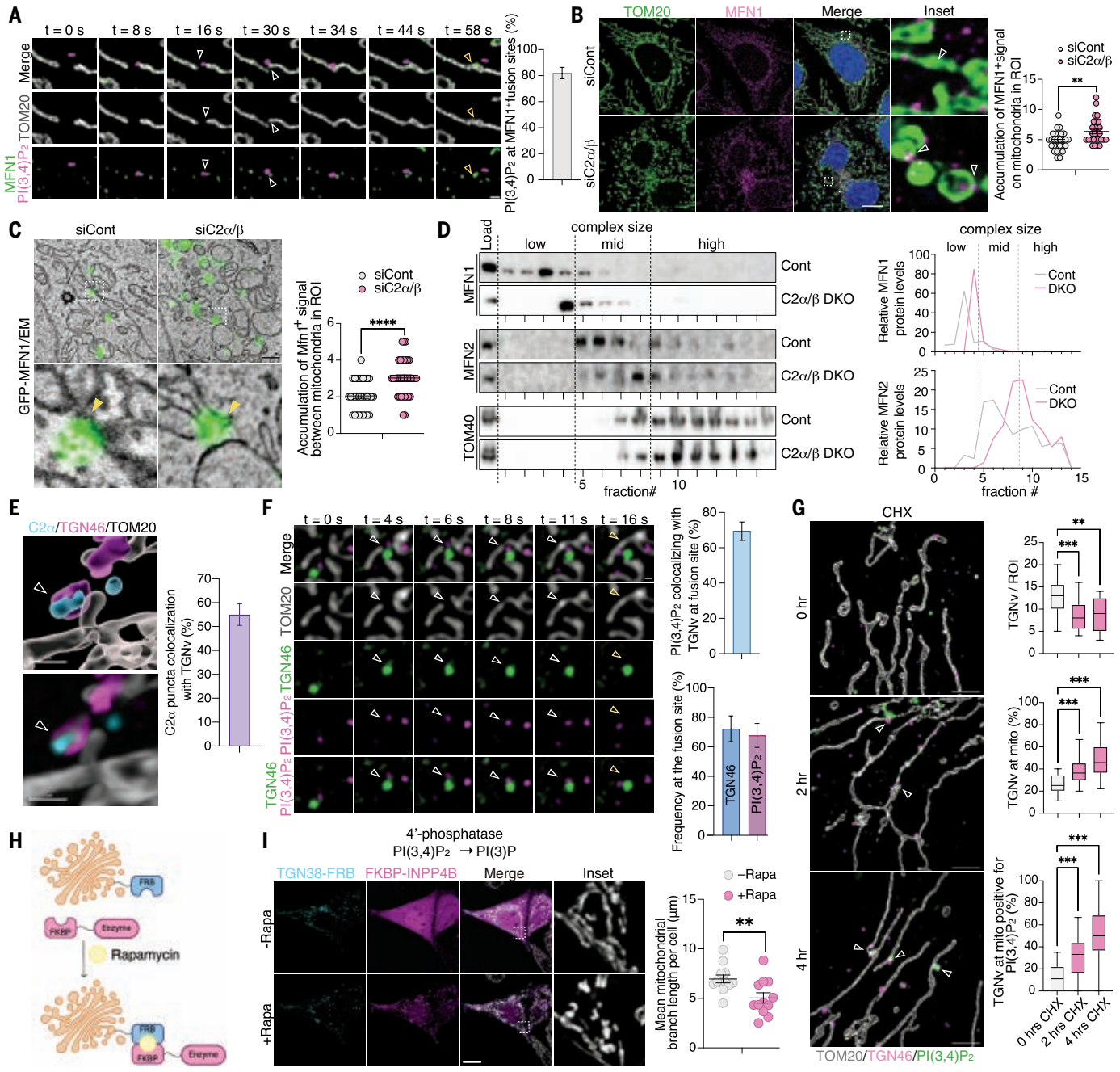

Fig. 3. PI(3,4)P_2-containing TGN vesicles promote mitochondrial fusion at MFN-positive fusion sites. (**A**) SR lattice SIM^2 time-lapse of HeLa cells expressing TOM20-HaloTag, GFP-MFN1, and mCherry-PH^{TAPP1} and quantification of GFP-MFN1–positive fusion events marked by mCherry-PH^{TAPP1}. White and yellow arrowheads indicate pre- and postfusion, respectively. Scale bar is 500 nm. (**B**) Confocal images and quantification of MFN1 localization between fragmented mitochondria in indicated siRNA-treated HeLa cells. Mitochondria and MFN1 are visualized with anti-TOM20 and anti-MFN1, respectively. Nuclei are visualized with DAPI staining. White arrowheads indicate MFN1 foci. Scale bar is 10 μm. (**C**) 3D-CLEM and quantification of indicated siRNA-treated HeLa cells, transiently expressing GFP-MFN1. Yellow arrowheads indicate GFP-MFN1 at the interface between two mitochondria. Scale bar is 1 μm. (**D**) Sucrose-density fractionation and quantification of MFN1/2 oligomeric profiles in control and PI3K-C2α/β DKO HeLa cells. (**E**) SR lattice SIM^2 3D rendering of HeLa cells expressing TOM20-HaloTag, mCherry-TGN46, and GFP-PI3K-C2α and quantification of PI3K-C2α/TGN-derived vesicles (TGNv) colocalization. White arrowheads indicate GFP-PI3K-C2α on mCherry-TGN46. Scale bars are 1 μm. (**F**) Lattice SIM^2 time-lapse of HeLa cells expressing TOM20-HaloTag, TGN46-mEmerald, and mCherry-PH^{TAPP1} and quantification of fusion events marked by TGN46-mEmerald, mCherry-PH^{TAPP1}, or both. White and yellow arrowheads indicate pre- and postfusion, respectively. Scale bar is 500 nm. (**G**) SR lattice SIM^2 images of HeLa cells stably expressing TOM20-StayGold, transfected with PI(3,4)P_2 probe mNES-mCherry-cPHx3, and treated with CHX. TGNv is visualized with anti-TGN46. Also shown is a quantification of TGNv number per ROI (top), the percentage of mitochondrial-associated TGNv (middle), and the percentage of mitochondrial-associated PI(3,4)P_2-positive TGNv (bottom). White arrowheads indicate PI(3,4)P_2 on TGN46 associated with mitochondria. Scale bars are 5 μm. (**H**) Rapamycin-inducible TGNv PI(3,4)P_2 depletion model. (**I**) Confocal images and quantification of mitochondrial branch length in HeLa cells expressing mCherry-FKBP-INPP4B and TGN38-FRB-CFP, with or without rapamycin for 10 min. Mitochondria are visualized with MitoTracker. Scale bar is 10 μm. All values are mean ± SD, except in (B), (C), and (I), where values are mean ± SEM; data are from ≥3 independent experiments unless specified otherwise. For (G), data are presented as box-and-whisker plots; the center line represents the median, the box indicates the interquartile range (25th to 75th percentiles), and the whiskers extend from the minimum to the maximum value. For (B), (C), and (I), data were analyzed by unpaired two-tailed *t* test and, for (G), by ordinary one-way ANOVA and Tukey's multiple comparisons test. $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$; ns is $P > 0.05$.

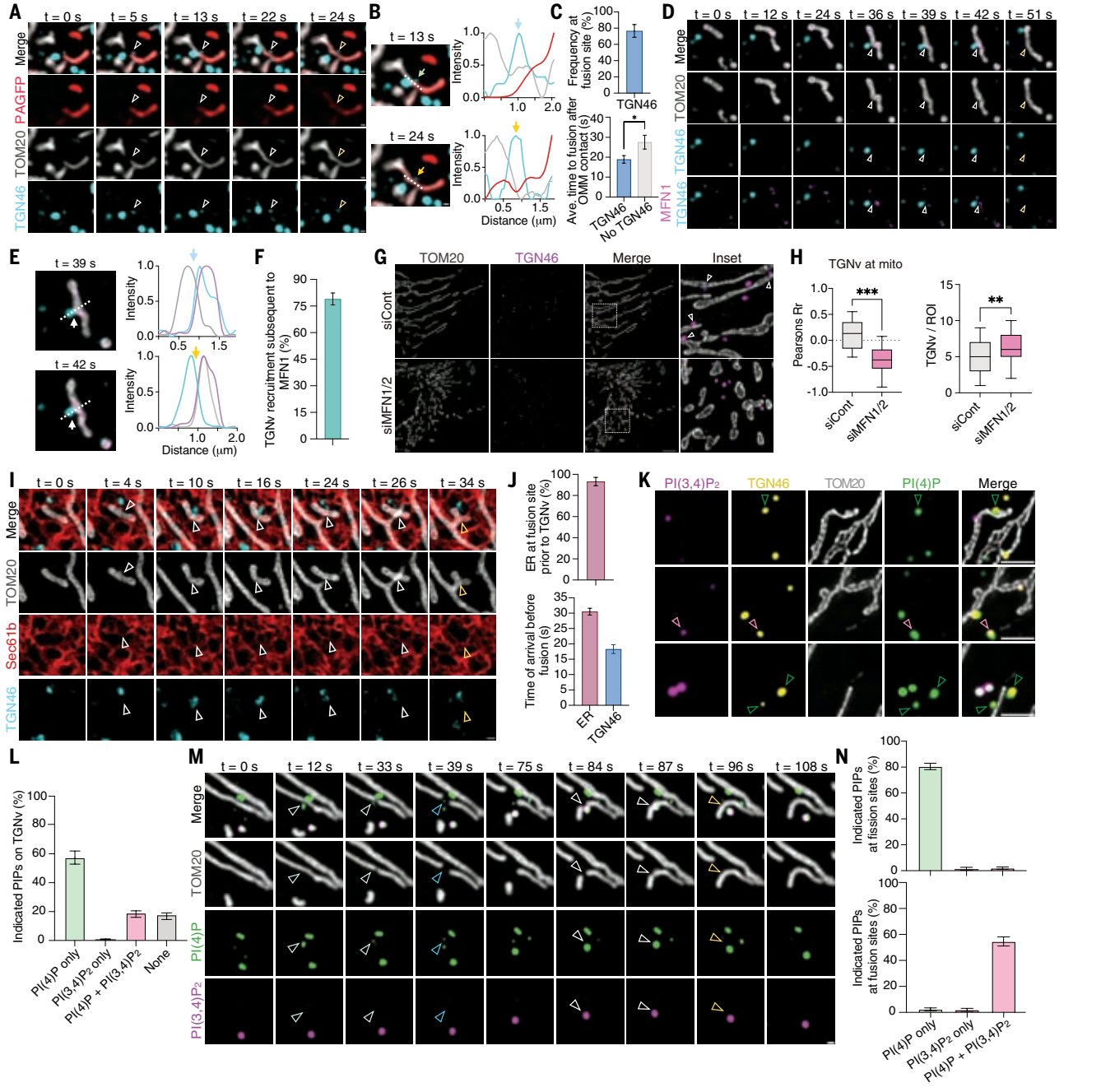

Fig. 4. Characterization of Golgi-derived vesicles and mitochondrial fusion. (**A**) Time-lapse confocal images of HeLa cells expressing TOM20-mCherry, TGN46-iRFP, and matrix-targeted PA-GFP. White and yellow arrowheads indicate pre- and postfusion, respectively. Scale bars are 500 nm. (**B**) Line-scan along dashed lines before and after mitochondrial fusion. White and yellow arrows indicate before and after fusion, respectively. Scale bar is 500 nm. (**C**) Quantification of PA-GFP–positive mitochondrial fusion events marked by TGN46-iRFP (top) and the average time from OMM contact to PA-GFP diffusion with or without TGNv (bottom). (**D**) Airyscan 2 time-lapse of HeLa cells expressing TOM20-mCherry, GFP-MFN1, and TGN46-iRFP. White and yellow arrowheads indicate pre- and postfusion, respectively. Scale bars are 500 nm. (**E**) Line-scan along dashed lines during fusion from (D). (**F**) Quantification of fusion events marked by TGN46-iRFP downstream of GFP-MFN1 foci at mitochondrial tips. (**G**) Lattice SIM^2 images of HeLa cells expressing TOM20-StayGold, mCherry-PH^{TAPP1}, and TGN46-iRFP treated with siMFN1/2 or siCont. White arrowheads indicate TGN46 associated with mitochondria. Scale bar is 5 μm. (**H**) Quantification of mitochondria-TGNv colocalization (left) and TGNv number per ROI (right) from (G). Pearsons Rr stands for Pearson's correlation coefficient. (**I**) Lattice SIM^2 time-lapse of HeLa cells expressing TOM20-mCherry, TGN46-iRFP, and Sec61b-GFP. White and yellow arrowheads indicate pre- and postfusion, respectively. Scale bar is 500 nm. (**J**) Quantification of fusion events marked by TGN46-iRFP downstream of Sec61b-GFP (top) and arrival times of Sec61b-GFP and TGN46-iRFP before fusion (bottom). (**K**) 3D-SIM images of HeLa cells stably expressing TOM20-StayGold, transfected with PI(3,4)P_2 probe mNES-mCherry-cPHx3 and PI(4)P probe iRFP-P4M-SidM. TGNv are visualized with anti-TGN46 antibody. Arrowheads show vesicle populations. Scale bars are 2 μm. (**L**) Quantification of PI(4)P- and PI(3,4)P_2-associated TGNv populations. (**M**) SR lattice SIM^2 time-lapse of HeLa cell stably expressing TOM20-StayGold, transfected with mNES-mCherry-cPHx3 and iRFP-P4M-SidM. Light blue and blue arrowheads indicate pre- and postfission, respectively; white and yellow arrowheads indicate pre- and postfusion, respectively. Scale bar is 500 nm. (**N**) Quantification of PIP recruitment to fission (top) or fusion (bottom) sites from (M). All values are mean ± SEM except in (H), (L), and (N), where values are mean ± SD; data are from ≥3 independent experiments unless specified otherwise. For (H), data are presented as box-and-whisker plots; the center line represents the median, the box indicates the interquartile range (25th to 75th percentiles), and the whiskers extend from the minimum to the maximum value. For (C) and (H), data were analyzed by unpaired two-tailed *t* test. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; ns is $P > 0.05$.

mitochondrial fission events, as expected (*17*, *33*), vesicles containing both PI(4)P and PI(3,4)P_2 were predominantly recruited to fusion sites (Fig. 4, M and N, fig. S13A, and movies S16 and S17), indicating that functionally distinct TGN-derived vesicle subpopulations selectively associate with mitochondria depending on the type of membrane remodeling event. Indeed, loss of PI3K-C2α and PI3K-C2β reduced PI(3,4)P_2 levels and decreased the number of TGN-derived vesicles in contact with mitochondria (fig. S13, B and C).

Thus, distinct phosphoinositide-defined TGN-derived vesicle populations differentially regulate fission and fusion, and class II PI3K–dependent conversion of PI(4)P to PI(3,4)P_2 predetermines which vesicles are to be recruited to MFN-dependent fusion sites, thereby promoting efficient mitochondrial fusion.

Loss of PI3K-C2α/β impairs mitochondrial function and leads to cardiac dysfunction

Impairment of mitochondrial fusion leads to fragmentation and cellular dysfunction in diseases affecting the heart, brain, muscle, and liver (*53*). We analyzed mitochondrial function in cells by silencing individual or combined PI3K-C2α and PI3K-C2β isoforms. Loss of PI3K-C2α/β reduced mitochondrial membrane potential, increased reactive oxygen species, and affected oxidative phosphorylation (fig. S14, A to E), indicating a role for PI3K-C2α and PI3K-C2β in mitochondrial homeostasis, consistent with their role in facilitating mitochondrial fusion.

To investigate the physiological relevance of PI3K-C2α/β–generated PI(3,4)P_2 in vivo, we generated a cardiomyocyte-specific PI3K-C2α and PI3K-C2β DKO mouse model (Fig. 5, A and B). Unlike PI3K-C2α and PI3K-C2β, which are ubiquitously expressed, PI3K-C2γ is primarily expressed in liver, breast, testes, and prostate (*50*, *54*). Global KO of PI3K-C2α is embryonically lethal owing to angiogenesis defects (*55*), whereas PI3K-C2β KO mice exhibit no overt abnormalities (*56*, *57*). Despite normal delivery, more than 60% of cardiomyocyte-specific DKO pups died within the first 1 to 2 days after birth (Fig. 5, C and D). Cardiac function was evaluated immediately after birth using small animal echocardiography, which revealed severe cardiac abnormalities in DKO pups compared with controls (Fig. 5E). Motion-mode (M-mode) echocardiography, which assesses cardiac contractility, showed a significant reduction in ejection fraction and fractional shortening in DKO pups, confirming severe cardiac dysfunction (Fig. 5F). Moreover, immunofluorescence staining for sarcomere markers, such as α-actinin, showed disorganized sarcomeres in cardiomyocytes isolated from DKO pups (Fig. 5G). Additionally, mRNA levels of heart failure markers, *Nppa* and *Nppb*, were significantly up-regulated in heart tissues from DKO pups compared with control mice (Fig. 5H). Consistent with our cellular data, mitochondrial morphology in DKO mouse hearts, as evaluated by transmission electron microscopy and confocal microscopy, exhibited a reduction in size and aspect ratio, indicative of network fragmentation (Fig. 5, I to L). Thus, PI3K-C2α/β activity is essential for maintaining mitochondrial activity and cardiac function in vivo.

Discussion

Here, we found that loss of PI3K-C2α/β causes mitochondrial fragmentation and that this phenotype depends on a loss of kinase activity and PI(3,4)P_2 production. Complementary fusion assays and imaging approaches revealed that this phenotype reflects impaired mitochondrial fusion rather than enhanced fission. Mechanistically, SR lattice SIM^2 imaging revealed that TGN-derived vesicles enriched in both PI3K-C2α and PI(3,4)P_2 associate with MFN-marked mitochondrial fusion sites. Consistent with this, loss of PI3K-C2α and PI3K-C2β led to the accumulation of MFN oligomers, supporting the idea that PI(3,4)P_2 may promote progression from MFN assembly to productive OMM fusion. Our in vitro fusion assay demonstrated that the addition of PI(3,4)P_2, but not other PIPs, promoted mitochondrial fusion, indicating a specific role for this lipid in the fusion process. However, although live-cell SR imaging indicated that PI(3,4)P_2-containing TGN-derived vesicles associate with MFN-marked fusion sites, we could not define precisely at which stage of MFN oligomerization these vesicles act.

Recent studies proposed that during mitochondrial division, lysosomal PI(4)P may be transferred from lysosomes to mitochondria via lysosome-localized oxysterol-binding protein-related protein 1L (*33*, *58*), thereby facilitating division. Perhaps PI(3,4)P_2 is similarly transferred locally from TGN-derived vesicles to mitochondria at fusion sites. At these sites, PI(3,4)P_2 may act as a local signaling platform, similarly to its role in endocytosis (*41*), to facilitate mitochondrial fusion. MFN2 did not detectably bind PI(3,4)P_2 under the conditions tested here. This suggests that PI(3,4)P_2 may instead control fusion through additional effector proteins, the local lipid environment, or membrane remodeling mechanisms.

We also assessed the spatiotemporal organization of organelles at fusion sites. The ER was recruited first and then TGN-derived vesicles. This supports a model in which the ER marks future fusion sites, whereas TGN vesicles promote later stages of the process. Furthermore, TGN-derived vesicles carrying different phosphoinositide compositions are differentially associated with mitochondrial fission or fusion: Vesicles positive for PI(4)P only are linked to fission, whereas vesicles containing both PI(4)P and PI(3,4)P_2 are preferentially recruited to fusion sites. These findings suggest that cells use a coordinated phosphoinositide-based code to direct distinct membrane remodeling events and maintain mitochondrial homeostasis.

Induction of fusion through CHX treatment promoted PI(3,4)P_2-positive TGN-derived vesicle recruitment to mitochondria, indicating that their association during mitochondrial fusion is also relevant during adaptations to environmental stresses. Decreased PI(3,4)P_2 levels in TGN-derived vesicles upon silencing of PI3K-C2α and PI3K-C2β, as well as loss of mitofusins, reduced the close association between TGN-derived vesicles and mitochondria. This indicates that the OMM fusion machinery is required to signal or stabilize vesicle recruitment at fusion sites. However, whether MFNs, which are involved in other mitochondria-organelle membrane contact sites (*59*–*61*), or other components directly contribute to vesicle tethering remains to be elucidated.

Impaired fusion results in fragmented mitochondrial networks and correlates with a broad spectrum of diseases, including neurodegenerative disorders, cardiomyopathy, and cancer (*53*). Consistent with this, KO of key fusion regulators such as MFNs and OPA1 is embryonically lethal in mice (*62*, *63*), and mutations in these genes have been linked to conditions such as Charcot-Marie-Tooth disease and optic atrophy (*64*–*66*), respectively. Loss of PI3K-C2α and PI3K-C2β caused mitochondrial dysfunction in cultured cells, and cardiomyocyte-specific DKO mice exhibited severe cardiac dysfunction accompanied by mitochondrial fragmentation, indicating that class II PI3Ks are essential for cardiac function in vivo. However, whether mitochondrial fragmentation is a cause or consequence of cardiac dysfunction remains to be determined.

Our study expands our understanding of how interorganelle communication and phosphoinositide signaling coordinate mitochondrial membrane remodeling and homeostasis. It also identifies Golgi-derived PI(3,4)P_2-positive vesicles as key regulators of mitochondrial fusion.

Materials and methods

Cell culture

All cell lines were obtained from American Type Culture Collection (ATCC) and maintained at 37°C, 5% CO_2 and 95% relative humidity. HeLa, HeLa cells stably expressing TOM20-StayGold, COS-7, and A549 were cultured in Dulbecco's Modified Eagle's Medium (DMEM) (Gibco, cat. no. 31966047), and normal human dermal fibroblasts in Roswell Park Memorial Institute medium (Nacalai Tesque, cat. no. 30264-85), with both media containing high glucose and supplemented with 10%

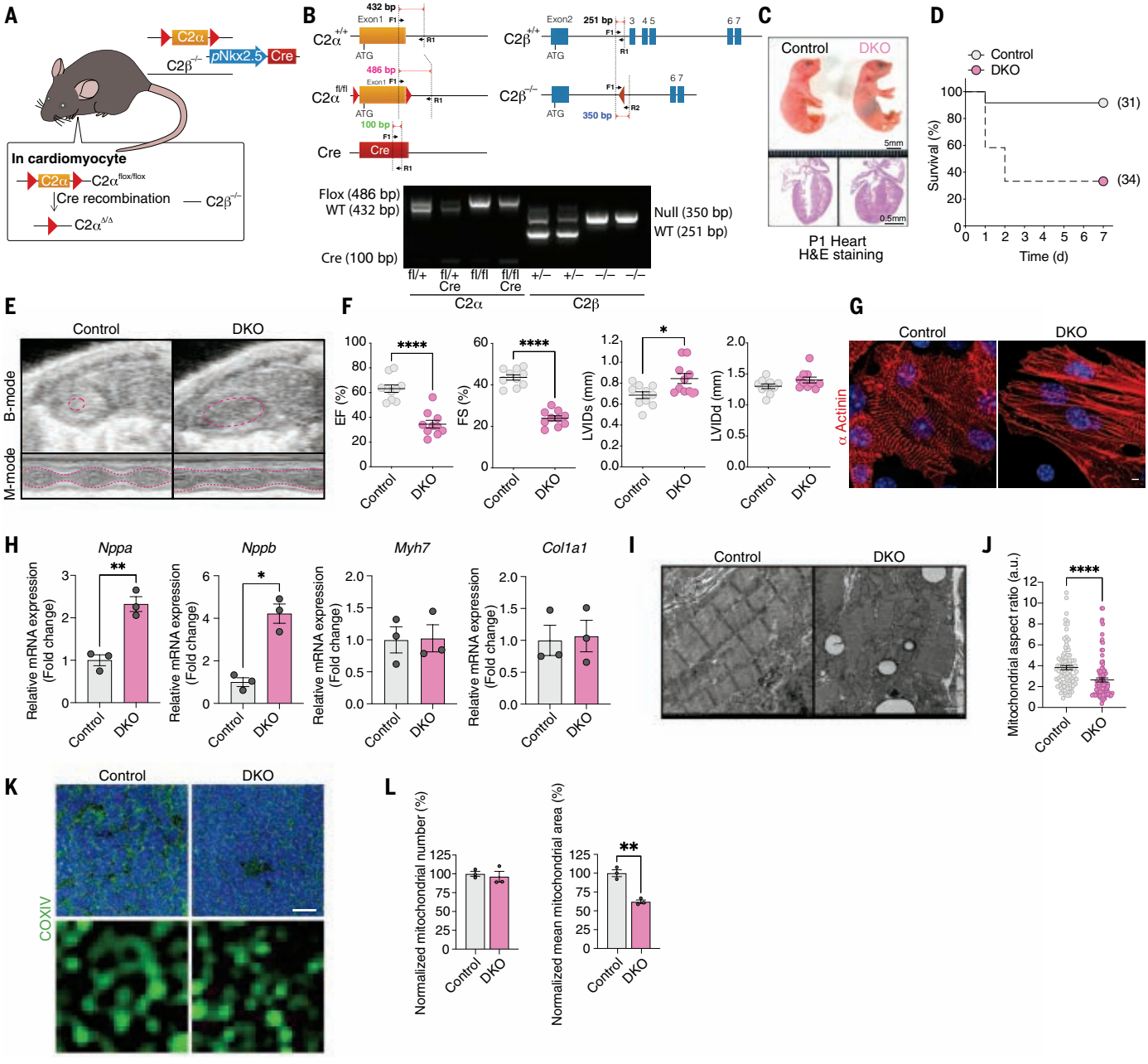

Fig. 5. Cardiomyocyte-specific PI3K-C2α/β DKO induces mitochondrial fragmentation and heart failure. (**A**) Schematic of PI3K-C2α/β cardiomyocyte-specific, Nkx2.5-Cre, DKO mice. (**B**) PCR genotyping of mouse tails confirming PI3K-C2α/β DKO: PI3K-C2α flox and WT alleles, 486 and 432 bp, respectively; PI3K-C2β WT and null alleles, 251 and 350 bp, respectively; and Nkx2.5-Cre, 100 bp. (**C**) Stereomicroscopy of postnatal day 1 (P1) (top) and hematoxylin and eosin (H&E) staining of P1 heart (bottom) from PI3K-C2α/β DKO and control mice. Scale bars are 5 mm (top) and 0.5 mm (bottom). (**D**) Cumulative survival curve of DKO and control mice up to day 7. (**E**) B- and M-mode echocardiography of P1 PI3K-C2α/β DKO and control mice. (**F**) Cardiac function indexes in P1 PI3K-C2α/β DKO and control mice. EF, ejection fraction; FS, fractional shortening; LVIDs and LVIDd, left ventricular dimension at systole and diastole. (**G**) Confocal images of α-actinin–labeled (red) isolated cardiomyocytes from PI3K-C2α/β DKO and control mice. α-Actinin is visualized with anti–α-actinin and nuclei with DAPI. Scale bar is 10 μm. (**H**) Relative mRNA expression of heart-failure markers (*Nppa*, *Nppb*, *Myh7*, and *Col1a1*) in heart tissues from PI3K-C2α and PI3K-C2β DKO and control mice. *Nppa* and *Nppb*, natriuretic peptide types A and B; *Myh7*, myosin heavy polypeptide 7; *Col1a1*, collagen type I alpha 1 chain. (**I**) Transmission electron microscopy of P1 PI3K-C2α/β DKO and control hearts. Scale bar is 1 μm. (**J**) Quantification of mitochondrial morphology from (I). (**K**) Confocal images of mitochondrial morphology from PI3K-C2α/β DKO and control cardiac sections. Mitochondria are visualized with anti-COXIV and nuclei with DAPI. Scale bar is 10 μm. (**L**) Quantification of mitochondrial number (left) and area (right) from (K). All values are mean ± SEM from ≥3 independent experiments unless specified otherwise. For (F), (H), (J), and (L) data were analyzed by unpaired two-tailed *t* test. $*P < 0.05$; $**P < 0.01$; $****P < 0.0001$; ns is $P > 0.05$.

fetal bovine serum (FBS). Human umbilical vein endothelial cells were cultured in complete endothelial growth medium containing 2% FBS and growth factor supplements (EGM-2; Lonza, cat. no. CC3156). Human umbilical vein endothelial cells between passages four and six were used for experiments.

siRNA and plasmid transfection

Plasmid transfections were performed for 24 hours using Lipofectamine 2000 (Invitrogen) or FuGene HD (Promega, cat. no. E2311) reagents, according to the manufacturers' instructions. For siRNA transfection, cells were reverse transfected using Lipofectamine RNAiMAX

(Invitrogen, cat. no. 13778150) with 20 nM siRNA for 3 or 6 days, following the manufacturer's instructions.

List of siRNAs

Individual siRNAs targeting *PIK3C2A* (PI3K-C2α), *PIK3C2B* (PI3K-C2β), *PIK3CA* (P110α), *PIK3C3* (Vps34), and *INPP4A* were synthesized using a Silencer siRNA construction kit (Ambion, cat. no. AM1620) according to the manufacturer's instructions. Target sequences were as follows: *INPP4A* #1: 5′-GGCTTTATCATCAAGCTGA-3′, #2 5′-CTCCATCGCTAGATCGAAA-3′; *PI3K-C2*α #1: 5′-AAGGTTGGCACTTACAAGAAT-3′, #2: 5′-AAGTAAGCCTAAGGTGGATAA-3′; *PI3K-C2*β #1: 5′-AAGCCGGAAGCTTCTGGGTTT-3′, #2: 5′-TCAGGGCAATGGGGAACAC-3′; *PI10*α: 5′-GGACAACTGTTTCATATAG-3′; *Vps34*: 5′- AAACTCAACACTGGCTAATTA-3′. siRNAs for *INPP5D* (SHIP1) (#1 s7459, #2 s7460), *INPPL1* (SHIP2) (#1 s7463, #2 s7464), *OPA1* (#1 s534147, #2 s534148), and *PTEN* (#1 s325, #2 s326) were purchased from Thermo Fisher Scientific (Silencer Select). *INPPL1* (SHIP2) (ON-TARGETplus SMARTpool L-004152-00-0005), *INPP5D* (SHIP1) (ON-TARGETplus SMARTpool L-003013-00-0005), *MFN1* (ON-TARGETplus SMARTpool L-010670-01-0005), *MFN2* (ON-TARGETplus SMARTpool L-012961-00-0005), *OPA1* (ON-TARGETplus SMARTpool L-005273-00-0020), *PIK3CA* (P110α) (ON-TARGETplus SMARTpool L-003018-00-0005), *PIK3C2A* (PI3K-C2α) (ONTARGETplus SMARTpool L-006771-00-0005), *PIK3C2B* (PI3K-C2β) (ON-TARGETplus SMARTpool L-006772-00-0005), *PIK3C3* (Vps34) (ON-TARGETplus SMARTpool L-005250-00-0005), *PIKFYVE* (ON-TARGETplus SMARTpool L-005058-00-0005), and *PTEN* (ON-TARGETplus SMARTpool L-003023-00-0005) siRNAs were purchased from Dharmacon.

List of plasmids

The GFP-PI3K-C2α expression vector (provided by H. Naito) was described previously (*49*). mCherry-PI3K-C2β (WT) was a gift from H. Naito. The kinase-deficient GFP-PI3K-C2α mutant (D1268A) and mCherry-PI3K-C2β mutant (D1213A) were generated using standard polymerase chain reaction (PCR)–based methods. For siRNA-resistant *PI3K-C2*α^r and *PI3K-C2*β^r constructs, specific codons were mutated (*PI3K-C2*α^r 5′-AAG-GTT-GGC-ACT-TAC-3′ for 5′-AAA-GTC-GGT-ACC-TAT-3′, and *PI3K-C2*β^r 5′-GGC-CGG-AAG-CTT-CTG-GGT-3′ for 5′-GGG-CGC-AAA-TTG-TTA-GGT-3′) while encoding identical amino acids. pEGFP-2xFYVE was a gift from H. Stenmark (Addgene no. 140047), GFP-PH^{FAPP1} was a gift from T. Balla and D. Bryant (Addgene no. 161986), GFP-ING2(pHD) was a gift from J. Yuan (Addgene no. 21589), mEGFP-PI35P2 was a gift from G. van den Bogaart (Addgene no. 92419), GFP-C1-PLCdelta-PH was a gift from T. Meyer (Addgene no. 21179), Grp1-PH pEGFP-C1 was a gift from M. Lemmon (Addgene no. 71378), ER-mRFP was a gift from E. Snapp (Addgene no. 62236), CLC-pmCherryC1 was a gift from C. Merrifield (Addgene no. 27680), pLAMP1-mCherry was a gift from A. Palmer (Addgene no. 45147), mEmerald-TGNP-N-10, mCherry-TOMM20-N-10, mEmerald-TOMM20-N-10, and mCherry-TGNP-N-10 were gifts from M. Davidson (Addgene nos. 54279, 55146, 54282, and 55145), BFP-KDEL, Mito-BFP, GFP-Mfn1, and mCherry-Mfn2 were gifts from G. Voeltz (Addgene nos. 49150, 49151, 141154, and 141156), pLAMP1-emiRFP670 was a gift from V. Verkhusha (Addgene no. 136570), pAc-GFPC1-Sec61beta was a gift from T. Rapoport (Addgene no. 15108), and pOCT-PA-GFP was described previously (*44*). Matrix-HiBiT/LgBiT and FLAG-MFN2 were gifts from S. Yanagi and FLAG-MFN2 was generated by cloning the PCR-amplified MFN2 coding sequence from MFN2-FLAG (*67*) into pCMV2-FLAG. NES-EGFP-cPHx3, NES-mCherry-cPHx3 and mCherry-FKBP-INPP4B were gifts from G. Hammond (Addgene nos. 116855, 116862, and 116864), and GFP-PH^{TAPP1} was a gift from D. Alessi and D. Bryant (Addgene no. 161985). PM-FRB-CFP, GFP-, mCherry-, and iRFP-P4M-SidM were gifts from T. Balla (Addgene nos. 7517, 51469, 51471, and 51470), and TGN38-FRB-CFP was obtained from T. Balla. Halo-TOMM20-N-10 was a gift from K. McGowan (Addgene no. 123284). GRIP-mStayGold was generated by subcloning the GRIP domain from mCherry-C1-GOLGA4-GRIP (Addgene no. 220068) into pcDNA3-mStayGold (Addgene no. 211363) by the Infusion Cloning method via AgeI and KpnI restriction sites. TGN46-iRFP was generated by first creating TGN46-FRB-iRFP by linearizing the Akap1(31)-FRB-iRFP vector, a gift from G. Hammond (Addgene no. 139315), using forward primer 5′-ATCCTCTGGCATGAGATGTGG-3′ and reverse primer 5′-GCTAGCGGATCTGACGGT-3′, and amplifying the TGN46 sequence using forward primer 5′-GTCAGATCCGCTAGCACCATGCGGTTCG-3′ and reverse primer 5′-CTCATGCCAGAGGATGGACTTCTGGTCCAAACGT-3′ from the mCherry-TGNP-N-10 vector as a template by PCR. TGN46-specific primers were designed with 15–base pair (bp) extensions homologous to linearized vector ends. TGN46 sequence was inserted in a linearized vector backbone using In-fusion cloning kit (Takara).

Generation of PI3K-C2α/β KO cells

To generate single *PI3K-C2*α and *PI3K-C2*β, and double *PI3K-C2*α and *PI3K-C2*β CRISPR-Cas9–mediated KO cells, HeLa cells were seeded at a density of 2×10^4 cells per well in 24-well plates and transfected with the pSpCas9(BB)-2A-Puro (PX459) vector (Addgene no. 62988) encoding SpCas9 and gene-specific guide RNAs (gRNAs) using FuGENE 4K transfection reagent (Promega). gRNA sequences were designed using the online CRISPR software (https://chopchop.cbu.uib.no) and selected based on predicted efficiency and minimal off-target effects. For *PI3K-C2*α, the following gRNA sequence targeting exon two was used: 5′-CGGAGAAAATGCTAGTGTGAAGG-3′. For *PI3K-C2*β, the following gRNA sequence targeting exon three was used: 5′-GTATCCCAGATAGAAGCTCGGGG-3′. Twenty-four hours after transfection, cells were subjected to puromycin selection (0.5 μg/ml) to eliminate nontransfected cells. After 48 hours, cells were washed with phosphate buffer saline (PBS), trypsinized, and centrifuged. Cell pellets were resuspended in culture medium and filtered through a 70-μm cell strainer. Surviving cells were clonally isolated by limiting dilution in 96-well plates at approximately one cell per well and cultured in complete medium supplemented with 20% FBS. After 2 to 3 weeks, single-cell–derived colonies were expanded and transferred to 24-well plates. Gene KO was confirmed by immunoblotting.

Generation of WT-HeLa cells stably expressing TOM20-StayGold

TOM20 and StayGold were amplified from previously generated plasmids (*68*) and subcloned into the pWPXLd-IRES-Puro vector (obtained from M. Zeviani, Mitochondrial Biology Unit, University of Cambridge, UK) using the In-Fusion cloning method (Takara Clontech, cat. no. 638943), to generate the pWPXLd-TOM20-StayGold-puromycin lentiviral vector. Lentiviral particles were produced in HEK293T cells by cotransfection of the transfer vector together with the packaging plasmid psPAX2 (Addgene no. 12260) and the envelope plasmid pMD2.G (Addgene no. 12259). Viral supernatants were collected 24 hours post-transfection and used for transduction in the presence of 8 μg/ml polybrene (Sigma-Aldrich, cat. no. TR-1003). WT HeLa cells transduced with pWPXLd-TOM20-StayGold-puromycin lentivirus were selected with puromycin 24 hours after transduction. Localization of the StayGold signal to the outer mitochondrial membrane was validated by confocal and SR microscopy.

Chemical treatment and reagents

Pitstop 2 (Abcam, cat. no. ab1206871) was used at a final concentration of 30 μM for 30 min to inhibit clathrin-dependent endocytosis. Cells were stained with the mitochondria-specific dye MitoTracker Deep Red FM (5 nM; Invitrogen, cat. no. M22426) for 30 min at 37°C. MitoSOX (5 μM; Invitrogen, cat. no. M36008) was used for 10 min at 37°C to detect mitochondrial superoxide. For mitochondrial membrane potential detection, MT-1 (MT13, Dojindo laboratories, cat. no. 343-09781) was used according to the manufacturer's instructions and incubated 30 min at 37°C. CHX (Sigma-Aldrich, cat. no. C4859) was used at a final concentration of

10 μM for indicated times to promote stress-induced mitochondrial hyperfusion. Dimethyl sulfoxide (DMSO) was used as a control vehicle.

Antibodies

Immunoblotting: Mouse monoclonal anti–β-actin (cat. no. A5316) antibody was purchased from Sigma-Aldrich. Mouse monoclonal anti-TOM20 (cat. no. sc-17764), anti-PTEN (cat. no. sc-7974), and anti-FIS1 (cat. no. sc-376469) antibodies were obtained from Santa Cruz Biotechnology. Mouse monoclonal anti–PI3K-C2β (cat. no. 611342), and anti-DRP1 (cat. no. 611113) antibodies were purchased from BD BioSciences. Rabbit polyclonal anti–PI3K-C2α (cat. no. 12402), anti-p110α (cat. no. 4902), anti-Vps34 (cat. no. 4263), anti-COXIV (cat. no. 4850), anti-MFN1 (cat. no. 14739), anti-MFN2 (cat. no. 11925), anti-OPA1 (cat. no. 80471), anti-pDRP1 (Ser^{616}) (cat. no. 3455), and anti-pDRP1 (Ser^{637}) (cat. no. 4867) antibodies were obtained from Cell Signaling Technology. Rabbit polyclonal anti-MFF (cat. no. 17090-1-AP), anti-TOM40 (cat. no. 18409-1-AP), anti-INPP4A antibody (cat. no. 12665-1-AP), and mouse monoclonal anti-HSP60 (cat. no. 66041-1-Ig) antibodies were purchased from Proteintech. Mouse monoclonal anti-FLAG M2 was obtained from Sigma-Aldrich (cat. no. F1804).

Immunofluorescence: Mouse monoclonal anti-TOM20 (cat. no. sc-17764) antibody was purchased from Santa Cruz Biotechnology. Rabbit polyclonal anti-calnexin (cat. no. 10427-2-AP) was purchased from Proteintech. Mouse monoclonal anti-Sarcomeric α-actinin (cat. no. ab9465), anti-TOM20 (cat. no. ab56783), and rabbit polyclonal anti-KDEL (cat. no. ab176333) antibodies were obtained from Abcam. Mouse monoclonal anti-TGN46 (cat. no. SAB4200355) antibody was purchased from Sigma-Aldrich. Rabbit polyclonal anti-COXIV (cat. no. 4850), anti-MFN1 (cat. no. 14739), and anti-MFN2 (cat. no. 11925) antibodies were obtained from Cell Signaling Technology. Goat anti-mouse, and goat anti-rabbit Alexa Fluor 488, 568, or 647 were used as secondary antibodies (Thermo Fisher Scientific).

Gene expression analysis using real-time PCR

Total RNA was extracted from cells using Isogen reagent (Nippon Gene), converted to cDNA using Prime Script reverse transcriptase (Takara), according to the manufacturer's instructions. Quantitative real-time PCR was performed using SYBR Green (Toyobo) and the indicated primers. All mRNA expression level results were normalized to the expression level of *β-actin* (*ACTB*) or *glyceraldehyde-3-phosphate dehydrogenase* (*GAPDH*). The following pairs of primers were used: *ACTB*: forward (Fwd): 5′-AGAAGGAGATCACTGCCCTGGCACC-3′, reverse (Rev): 5′-CCTGCTTGCTGATCCACATCTGCTG-3′; *GAPDH*: Fwd: 5′-GGTGAAGGTCGGAGTCAACG-3′, Rev: 5′-GAGGGATCTCGCTCCTGGAAG-3′; *PIK3CA*: Fwd: 5′-CCACGACCATCATCAGGTGAA-3′, Rev: 5′-CCTCACGGAGGCATTCTAAAGT-3′; *PIK3C2A*: Fwd: 5′-TGCTGGGTACATGATGACTTGA-3′, Rev: 5′- TGTCCCATTTTCGACAGTTTTGA-3′; *PIK3C2B*: Fwd: 5′-CGACCTCAAGCTGTTACGC-3′, Rev: 5′-GAGCCCTCTATGTCCCCTGG-3′; *PIK3C3*: Fwd: 5′-GTCTGGCCTAATGTAGAAGCAG-3′, Rev: 5′-GGCAAGACGGCTCATCTGAT-3′; *PIKFYVE:* Fwd 5′-CTGAGTGATGCTGTGTGGTCAAC-3′, Rev: 5′-CAAGGACTGACACAGGCACTAG-3′; *INPP5D*: Fwd: 5′-GCGTGCTGTATCGGAATTGC-3′, Rev: 5′-TGGTGAAGAACCTCATGGAGAC-3′; *INPPL1*: Fwd: 5′-GCACACGTATCGCATTCTGC-3′, Rev: 5′- CTCGCTCACCCTCTACAGGAA-3′; *Nppa*: Fwd: 5′-GCTTCCAGGCCATATTGGAG-3′, Rev: 5′-GGGGGCATGACCTCATCTT-3′; *Nppb*: Fwd: 5′-GAGGTCACTCCTATCCTCTGG-3′, Rev: 5′-GCCATTTCCTCCGACTTTTCTC-3′; *Myh7*: Fwd: 5′-ACTGTCAACACTAAGAGGGTCA-3′, Rev: 5′-TTGGATGATTTGATCTTCCAGGG-3′; *Col1a1*: Fwd: 5′-GCTCCTCTTAGGGGCCACT-3′, Rev: 5′-CCACGTCTCACCATTGGGG-3′.

Immunoblotting

Cells were lysed in RIPA buffer [20 mM Tris pH 8.0, 150 mM NaCl, 0.1% sodium dodecyl sulphate (SDS), 1% deoxycholic acid, 1% NP-40] or cell lysis buffer (Cell Signaling Technology, cat. no. 7018S), complete with protease inhibitor cocktail or phosphatase inhibitor (Sigma-Aldrich, cat. no. 4906845001) when phosphorylated proteins were studied. Samples were normalized for protein concentration using a Bradford assay (Bio-Rad, cat. no. 500006). Proteins were resolved by SDS–polyacrylamide gel electrophoresis (SDS PAGE) using NuPAGE Novex 4–12% Bis-Tris Protein Gels (Invitrogen) and transferred to nitrocellulose membranes (0.2 μm) (Amersham Protran Western blotting membranes) using a wet tank transfer system (Bio-Rad). For immunoblot analysis, membranes were blocked with 5% nonfat milk in PBS for 60 min at room temperature (RT). Membranes were incubated with primary antibody at the appropriate dilution [in 2% milk or 2% bovine serum albumin (BSA) in 0.01% Tween-20 in PBS] at 4°C overnight. Membranes were then washed three times in PBS containing 0.05% Tween-20 for 15 min and incubated with appropriate horseradish peroxidase (HRP)–conjugated secondary antibodies diluted in 2% milk or 2% BSA in PBS containing 0.01% Tween-20. The secondary antibodies used were peroxidase-conjugated affinity-purified donkey anti-mouse (Sigma-Aldrich, cat. no. AP192P, 1:1000), anti-rabbit (Cell Signaling Technology, cat. no. 7074, 1:1000; or Amersham ECL Rabbit immunoglobulin G (IgG), HRP-linked whole antibody, cat. no. NA934, 1:3000), and sheep anti-mouse (Amersham ECL Mouse IgG, HRP-linked whole antibody, cat. no. NA931, 1:3000). Membranes were treated with ECL Western Blotting Detection Reagent Kit (Merck-Cytiva, cat. no. GERPN2106), SuperSignal West Dura Extended Duration Substrate (Thermo Fisher Scientific, cat. no. 34075) or ECL Prime Western Blotting Detection Reagent Kit (Thermo Fisher Scientific, cat. no. RPN2232) and exposed to chemiluminescence either in films (PROTEC) or on digital ECL machines (Fusion FX, Vilber; or Amersham Imager 680). Uncropped immunoblots are shown in figs. S15 to S17.

Subcellular and sucrose gradient fractionation

Cell pellets were resuspended in PBS and partially solubilized by treatment in 4 mg/ml digitonin (Sigma-Aldrich, cat. no. D141) for 10 min on ice. The reaction was stopped by adding five volumes of PBS, and heavy membrane fractions were collected by centrifugation at 10,000*g* for 5 min at 4°C. Heavy membrane fractions were solubilized with 1% (w/v) n-dodecyl-β-D-maltoside (DDM; Sigma-Aldrich, cat. no. D4641) on ice for 5 min and centrifuged at 20,000*g* for 30 min at 4°C. Five percent of the supernatant volume was saved as an input control. The remaining supernatant was layered on top of 12 ml of a linear 5 to 25% sucrose gradient in solubilization buffer [50 mM Tris HCl pH 7.4, 150 mM NaCl, 1x cOmplete EDTA-free Protease Inhibitor Cocktail (Roche, cat. no. 11873580001)] containing 0.1% (w/v) DDM in open-top thin wall ultraclear centrifuge tubes (Beckman Coulter, cat. no. 344060). Samples were centrifuged at 174,000*g* for 14 hours at 4°C using a swinging bucket (Beckman Coulter, SW 40 Ti). The centrifuged solution was manually separated into 500-μl fractions from top to bottom. Proteins from each fraction were precipitated with 10% (v/v) trichloroacetic acid (TCA) on ice for 10 min, followed by centrifugation at 20,000*g* for 10 min at 4°C. Protein pellets were washed once with ice cold acetone, resuspended in Laemmli buffer and incubated at 45°C for 20 min. Protein fractions were analyzed using SDS-PAGE as described above.

Immunoprecipitation of FLAG-MFN2

HeLa Cells were transiently transfected with Flag-MFN2 for 24 hours. Cells were trypsinized, pelleted by centrifugation, and resuspended in PBS. Cell pellets were partially solubilized with 4 mg/ml digitonin (Sigma-Aldrich no. D141) for 10 min on ice. The reaction was stopped by dilution with five volumes of PBS, and heavy membrane fractions were collected by centrifugation at 10,000*g* for 5 min at 4°C. Heavy membrane pellets were washed once with PBS and resuspended in solubilization buffer [50 mM Tris HCl pH7.4, 150 mM NaCl, 1x cOmplete EDTA-free Protease Inhibitor Cocktail (Roche, cat. no. 11873580001)]. Anti-FLAG M2 Magnetic Beads (Sigma-Aldrich, M8823) were equilibrated and washed with solubilization buffer using a magnetic stand. Heavy membrane fractions from

HeLa cells expressing Flag-MFN2 were solubilized with 1% (w/v) DDM (Sigma-Aldrich, cat. no. D4641) for 5 min on ice and centrifuged at 20,000*g* for 30 min at 4°C. A 10% volume of the supernatant was saved as input, and the remaining supernatant was incubated with equilibrated anti-FLAG M2 beads overnight at 4°C on a rotating wheel. Beads were then collected using a magnetic stand, the supernatant was discarded, and beads were washed three times with solubilization buffer. FLAG-MFN2 was eluted from the beads using 100 μg/ml 3xFLAG peptide (Sigma-Aldrich, F4799) in solubilization buffer for 2 hours at RT. Protein concentration in the eluate was quantified using a Bradford assay (Bio-Rad, cat. no. 500006). For immunoblot validation, 5% of the eluate was mixed with Laemmli sample buffer and incubated for 20 min at 45°C before separation by SDS-PAGE.

Lipid-strip binding

Immunoprecipitated FLAG-MFN2 was incubated with PIP Strips membranes (Echelon Biosciences, cat. no. P-6001,) according to the manufacturer's instructions and adapted from previous work (*52*). Briefly, PIP Strips were preblocked in 3% BSA in PBS containing 0.1% Tween-20 (PBS-T) for 1 hour at RT with gentle agitation. Approximately 500 ng/ml of immunoprecipitated FLAG-MFN2 was diluted in 10 ml of 3% BSA PBS-T and incubated with the PIP Strips for 1 hour at RT. For detection of protein binding, PIP Strips were washed six times in PBS-T and incubated with an anti-MFN2 antibody in 3% BSA PBS-T. Membranes were then washed three times in PBS-T and incubated with the appropriate HRP secondary antibody, and analyzed by chemiluminescence.

Immunofluorescence

Cells were fixed in prewarmed 4% or 5% paraformaldehyde (PFA) in PBS for 10 or 15 min at 25° or 37°C and then washed three times with PBS. Cells were subsequently treated with 50 mM ammonium chloride in PBS to quench PFA-induced background fluorescence, washed three times with PBS, and permeabilized using 0.1% Triton X-100 (Sigma-Aldrich, cat. no. T8787) in PBS for 15 min, followed by an additional three PBS washes. Cells were blocked with Protein Block Serum-Free Ready-To-Use solution (Dako, cat. no. X0909) or 10% FBS in PBS and subsequently incubated overnight at 4°C with the indicated primary antibodies. Cells were washed three times in PBS containing 5% FBS and incubated for 1 hour at RT with Alexa Fluor–conjugated secondary antibodies diluted to 1:1000 in PBS. Cells were mounted using Fluoromount-G Mounting Medium with or without 4′,6-diamidino-2-phenylindole (DAPI; Invitrogen, cat. no. 0100-01 or 0100-20) or ProLong Gold Antifade reagent (Invitrogen, cat. no. P36965).

Live-cell imaging

Cells were imaged in culture media at 37°C. Experiments were conducted on a confocal microscope (STELLARIS 5, Leica), a Dragonfly 500 confocal spinning disk microscope (Andor Technology Inc.), a Zeiss Elyra7 Lattice structured illumination microscopy (SIM), or a Zeiss LSM 980 with Airyscan 2.

Light microscopy

Spinning disk confocal microscopy: Spinning disk confocal imaging was performed on a Nikon Eclipse Ti-E microscope with a Dragonfly 500 confocal spinning disk system (Andor Technology Inc.) equipped with a Zyla 4.2 PLUS sCMOS camera. Cells were excited with 488-, 561-, and 640-nm lasers, and fluorescence emissions were collected with a 100X/1.45 NA Plan-apochromat oil-immersion objective using standard emission filters. Images were acquired with Fusion software. For fixed-cell imaging (e.g. mitochondria and ER morphology, DRP1 colocalization with mitochondria, MFN2, ER, and mitochondria), each dataset consisted of seven z-stacks acquired at 0.2-μm intervals. All images from the same experiment were acquired with the same parameters, including exposure time and laser intensities.

Photoactivation experiments were performed using the Andor iQ software FRAPPA module integrated with a Mosaic digital mirror device within the Andor Dragonfly system. Photoactivation of PA-GFP in individual mitochondria was slightly optimized for each cell: One baseline scan was taken before photoactivation, using a 405-nm laser at 40% power, followed by acquisition at 2-s intervals for 300 frames with a Zyla 4.2 PLUS sCMOS camera. Acquisition was controlled using the Andor IQ3 multidimensional image acquisition software. All live-cell confocal imaging was performed at 37°C, 5% CO_2, and 95% relative humidity within an H301-KFRAME stage-top incubator (OKOLAB, Italy).

The mitochondrial fusion rates in HeLa cells treated with nontargeting or PI3K-C2α and PI3K-C2β–targeting siRNAs were quantified by analysis of the PA-GFP probe diffusion. ROIs were defined within Fusion software and activated using the 405-nm laser line (power optimized per experiment). Five baseline images were acquired before activation, followed by time-lapse acquisition at 10-s intervals for 31 frames using a Zyla 4.2 PLUS sCMOS camera. Imaging settings remained constant within each experiment. Time-lapse photoactivation experiments were analyzed using a custom macro written in Fiji (ImageJ). For each dataset, a single-pass smoothing filter ("Smooth", 3×3 mean filter) was applied across all channels, slices, and time points. Quantitative measurements were extracted at defined time points (frames 1, 6, and 36, equivalent to $t = -60$ s, $t = 0$ s, and $t = 300$ s, respectively). Whole-frame intensity measurements were obtained using integrated density (IntDen) without spatial masking. Binary masks were analyzed to determine total foreground area (in pixels) and the percentage area relative to the full image field. Processing was performed in batch mode to ensure consistency across datasets.

SIM: Lattice SIM^2 images were acquired using a Zeiss Elyra7 system (Carl Zeiss Microscopy) equipped with dual sCMOS cameras. Imaging was performed using a 13-phase acquisition scheme for live-cell and a 15-phase scheme for fixed cells across all channels. Excitation was provided by 488-, 561-, and 642-nm lasers, and emission was collected using an α Plan-Apochromat 63×/1.4 NA oil immersion objective. For each fixed cell acquisition, 15 z-plane images were acquired at 0.1-μm intervals. Live-cell imaging was performed at an acquisition rate of one frame every 1 or 2 s (one frame every 3 s for 20 to 40 cycles for Fig. 4M and figs. S10C and S13A). All live-cell SR lattice SIM^2 imaging was performed at 37°C, 5% CO_2, and 95% relative humidity using an incubation system S (Carl Zeiss Microscopy). Raw data were reconstructed using the two-step SIM^2 algorithm in ZEN Black (Carl Zeiss Microscopy) with experimentally optimized parameters validated using 100-nm TetraSpeck fluorescent microspheres (Thermo Fisher Scientific, cat. no. T7279). Reconstruction parameters were kept constant within each experiment.

Additional fixed-cell 3D-SIM imaging was performed using a Nikon N-SIM microscope equipped with an SR Apo TIRF 100×/1.49 NA oil immersion objective and a DU897 iXon EMCCD camera (Andor). For each acquisition, nine z-planes were collected at 0.2-μm intervals. 3D-SIM reconstruction was performed in NIS-Elements (Nikon) using the slice reconstruction module with consistent parameters applied within each experiment. Where applicable, fixed-cell staining was performed before SIM imaging.

Laser-scanning confocal microscopy: Images were acquired with a STELLARIS 5 confocal microscope (Leica) equipped with HCX PL APO CS 63X/NA1.4 oil or HCX PL APO CS 100X/NA1.4 oil objectives. Image acquisition and processing were controlled by LASX software (Leica). For fixed-cell imaging, datasets consisted of seven z-stacks acquired at 0.2-μm intervals. All images shown have been subjected to background subtraction (rolling ball algorithm, radius = 50 pixels) and processed with the "smooth" function once in FIJI (ImageJ) software.

Airyscan 2 microscopy: Airyscan 2 microscopy was performed using a Zeiss LSM980 microscope equipped with an Airyscan 2 detector. Cells were excited using 405-, 488-, 561-, and 633-nm lasers, and emission was collected using a Plan-Apochromat 63X/1.40 Oil (DIC M27) objective. Images were acquired at an acquisition interval of ~2 s using the multiplex (2Y) mode. Raw images were processed using Airyscan SR deconvolution software in Zen Blue (Carl Zeiss Microscopy).

Image analysis

Images were converted to maximum-intensity projections and mitochondrial morphology was manually analyzed and presented as intermediate, elongated, or fragmented, as previously described (*68*). We classified mitochondria as "fragmented" when most of the mitochondria of the cell were short and spherical, "elongated" when most of the mitochondria of the cell presented highly elongated mitochondria with fewer than 10 free ends, and "intermediate" when most of the mitochondria of the cell were tubular, neither connected nor spherical. At least 30 cells per condition were analyzed from at least three independent experiments. All images shown have been subjected to background subtraction (rolling ball algorithm, radius = 50 pixels) and processed with the "smooth" function once in Fiji software.

For mitochondrial morphology parameters analysis, the Fiji macro MitoMAPR (*69*) was used to measure specific morphological parameters of mitochondrial architecture. ROIs (225 μm^2) were selected from the cell periphery and converted to maximum-intensity projections, where mitochondrial networks were better defined. At least 30 cells per condition were analyzed from at least three independent experiments.

In Fiji, live-cell images were corrected for photobleaching, background was subtracted using a rolling ball filter equal to 50, and a median filter was applied before quantification. To identify mitochondrial fusion events occurring at TGN-derived vesicles, PI(3,4)P_2, lysosomes, ER, and MFN1/2 foci, cells were imaged using the corresponding organelle, protein, or lipid markers indicated in the figure legends. For live-cell confocal microscopy using the matrix-targeted PA-GFP probe, complete mitochondrial fusion events were defined as the diffusion of photoactivated PA-GFP from one mitochondrion into an adjacent mitochondrion after OMM contact and increased fluorescence intensity (FI) at these sites. For live-cell SR imaging, mitochondrial fusion events were defined as events in which two mitochondria made contact with each other, showed a transient increase in local FI of the OMM marker at the contact site and followed by decreased local FI at the point of contact associated with relaxation of the constricted region and coalescence of the membranes (*24*), with no division event observed for an additional 1 min. Recruitment of TGN-derived vesicles, PI(3,4)P_2, ER, lysosomes, or specific proteins to mitochondrial fusion site was defined as the presence of the corresponding fluorescence signal within a 500-nm diameter ROI centered on the fusion sites and within 45 s before PA-GFP diffusion, or membrane relaxation and coalescence.

Colocalization and proximity of fluorescent markers to mitochondrial fusion sites were determined from thresholded images. Overlap was defined as direct pixel-wise intersection between the two signals, whereas proximity was defined as the presence of the nonmitochondrial signal immediately adjacent (within one pixel) to mitochondrial pixels at the fusion site. Line-profile fluorescence analysis was calculated using the formula: $zi = [xi - \min(x)]/[\max(x) - \min(x)]$. Raw FI was determined in Fiji by drawing a line over the relevant area and measuring the FI.

For 3D rendering and quantification, z-stacks of individual cells were cropped in Fiji and converted into .ims files using the Imaris File Converter. Using Imaris v10.1.0 (Oxford Instruments), objects from separate channels were segmented and rendered in 3D using the "surfaces function." Segmentation parameters included smoothing (surface detail: 0.0986 μm) and background subtraction (local contrast: 0.370). Manual thresholding was applied for TOM20, TGN-derived vesicles, and PI3K-C2α or PI(3,4)P_2 channels. The same segmentation criteria were applied consistently across comparable datasets. Object statistics generated from Imaris surfaces were exported to Microsoft Excel for analysis.

SIM colocalization analysis

Pearson's correlation coefficient (PCC) for TGN-derived vesicles and mitochondria or PI(3,4)P_2 and mitochondria was calculated on background-subtracted images using Fiji "colocalization finder" function to assess signal correlation where indicated. Fifty ROIs [20 ROIs for PI(3,4)P_2 versus Mito PCC] per condition were analyzed across a minimum of three independent biological replicates. To quantify spatial overlap between TGN-derived vesicles, mitochondria, and PI(3,4)P_2 reporter signals, channels were split and converted to 8-bit after background subtraction (rolling ball radius = 50 pixels). Otsu thresholding was applied to generate binary masks of TGN-derived vesicles (T), mitochondrial signal (M), and, where applicable, phosphoinositide reporter signals (P). The total number of TGN-derived vesicles within each ROI (T) was quantified using the "analyze particles" function in ImageJ, with area (0.01 to 0.12 μm^2) and circularity (0.40 to 1.00) thresholds held constant across all conditions. Mask overlays were generated using logical AND operations to define spatial subsets: $T \cap M$, mitochondria-associated TGN vesicles; and $(T \cap M) \cap P$, mitochondrial-associated TGN vesicles positive for PI(3,4)P_2 reporter. Percentages were calculated as mitochondria-associated TGN vesicles, $[(T \cap M)/T] \times 100$; and PI(3,4)$P_2$-positive mitochondrial TGN vesicles, $\{[(T \cap M) \cap P]/(T \cap M)\} \times 100$.

For phosphoinositide association analyses independent of mitochondria, TGN vesicles were classified as positive for PI(3,4)P_2 only, PI(4)P only, both, or neither based on binary mask intersection with respective reporter channels. Data were reported as percentage of total TGN vesicles per ROI.

Analysis of DRP1 recruitment to mitochondria

Colocalization analysis between DRP1 and mitochondrial signals from images obtained with the ANDOR Dragonfly spinning disk was performed in an automated manner using a macro written for FIJI. Background subtraction (rolling ball = 50) was performed, and images were converted to 8-bit. The DRP1 signal was subjected to thresholding in all images using Otsu's threshold technique. The degree of colocalization between DRP1 and mitochondrial signals was determined via PCC and Overlap, generated by calling the "colocalization finder" function in Fiji.

Imaging and quantitative analysis of PIP and mitochondrial colocalization

z-Stack images were acquired using a laser-scanning confocal microscope (Leica, STELLARIS5) equipped with a ×63/1.4 NA oil-immersion objective, with a z-step size of 0.2 μm. Imaging settings (laser power, gain, and pinhole) were kept constant across conditions within each experiment. Maximum-intensity projections were generated for visualization purposes only. All quantitative analyses were performed using Fiji on full z-stack data. For each cell, ROIs encompassing the entire mitochondrial network or defined subcellular areas were manually selected. Background subtraction was applied before analysis. PIP signals were quantified by detecting discrete puncta using a fixed threshold applied identically across all conditions. Particle detection was performed using the "analyze particles" function in Fiji with predefined size and circularity thresholds (size: 0.1 to 5.0 μm^2; circularity: 0.00 to 1.00) to exclude background noise. Line-scan analyses were performed on individual optical sections where mitochondria and reporter signals were present within the same focal plane. FI profiles were extracted along manually drawn lines using Fiji and normalized to the maximum intensity within each profile. Colocalization and spatial distribution analyses were restricted to z-planes or z-subvolumes where both signals

overlapped in the same optical sections, thereby avoiding artifacts arising from projection of signals from different axial planes.

Quantification of total cellular levels of PIPs

Cells expressing the indicated PIP probes were imaged using a Zeiss LSM980 confocal microscope, equipped with an Airyscan 2 detector, with a ×63/1.4 NA oil-immersion objective and a z-step size of 0.2 μm. Total cellular levels of PIPs were quantified based on the number of fluorescent puncta using Fiji. For each cell, ROIs encompassing the entire cell area were manually defined. Images were subjected to background subtraction followed by thresholding using identical parameters across all conditions. Puncta were identified using the "analyze particles" function with predefined size and circularity thresholds (size: 0.1 to 5.0 μm^2; circularity: 0.00 to 1.00) to exclude background and noise.

PEG fusion assay

Cells were separately transfected with mEmerald-TOM20 and mCherry-TOM20. After 16 hours, cells were trypsinized and co-plated into 35-mm glass-bottom dishes. After an additional 24 hours, cells were fused with 50% (w/v) PEG (Roche; molecular weight: 1500 Da) for 1 min. After three washes with complete DMEM, cells were monitored by spinning disk confocal microscopy, starting 1 hour after whole-cell fusion procedure (*70*).

3D- CLEM

A glass-based dish with 150-μm grids (Iwaki, cat. no. TCI-3922-035R-1CS,) underwent the following pretreatment: A glass coverslip was coated with carbon using a vacuum evaporator (JEOL, cat. no. IB-29510VET), exposed to ultraviolet irradiation for 15 min, and then coated with 0.1% gelatine. HeLa cells transfected with the indicated siRNAs and expression vectors were cultured in DMEM with high glucose supplemented with 10% FBS in the pretreated glass-based dish for 2 days. Cells were washed twice with 0.1 M phosphate buffer at pH 7.4 (LSI Medience, cat. no. RM102-5L) and fixed with 2% PFA (Nacalai Tesque, cat. no. 26126-54) and 0.5% glutaraldehyde (TAAB, cat. no. G018/1) in 0.09 M phosphate buffer (pH 7.4) for 1 hour at 4°C. Subsequently, cells were washed three times with 0.1 M phosphate buffer (pH 7.4), and z-section images were acquired using a confocal microscope system (STELLARIS 5, Leica) equipped with a 63× oil-immersion objective lens (HCX PL APO CS ×63/NA1.4-oil, Leica).

After fluorescence microscopy, cells were fixed overnight at 4°C with 2.5% glutaraldehyde in 0.1 M cacodylate buffer pH 7.4 (Nacalai Tesque, cat. no. 37237-35). Cells were then fixed for 2 hours at 4°C with 1% osmium tetroxide (Nisshin EM, cat. no. 3020-4) and 1.5% potassium ferrocyanide (Wako, cat. no. 161-03742) in 0.065 M cacodylate buffer. Samples were washed five times with Milli-Q water, treated with 3% uranium acetate for 1 hour, and dehydrated in an ascending series of ethanol concentrations. After dehydration in 100% ethanol for 1 hour, cells were embedded in Epon resin (TAAB, cat. no. EPON 812) and polymerized first at 40°C for 12 hours and then at 60°C for 48 hours. The resulting resin blocks were trimmed into ~100-μm–by–100-μm cuboids using razor blades. Ultra-thin serial sections (25-nm thickness) were cut with a diamond knife equipped with an ultrajumbo boat (Ultrajumbo 35°, Diatome), mounted on an ultramicrotome (Leica, UC7). The serial sections formed ribbons of 100 sections, which were transferred onto silicon wafers (Mitsubishi Materials Trading Corporation) using a micromanipulator (Märzhäuser Wetzlar). Samples were stained with 3% uranium acetate and Reynolds' lead citrate.

Scanning electron microscopy images were acquired using an electron microscope (JEOL, cat. no. JSM7900F) at a 5000× magnification (each image covered an area of 18 μm by 24 μm). Image acquisition was performed semiautomatically with Array Tomography Supporter 1.0.0.0 software (System In Frontier). The images were automatically tiled using Measurement Advisor 4.1.14.0 software (System In Frontier) and stacked using Stacker NEO TEMography.com 3.3.4.0 software (System In Frontier). Correlation analysis of fluorescence and electron microscopy images was performed with Fiji. Mitochondria were identified based on the presence of a double membrane and cristae, and their area and length were measured from the electron micrographs using Fiji.

HiBiT-LgBiT mitochondrial fusion assay

Crude mitochondria were isolated from HeLa cells stably expressing mitochondrial matrix-targeted HiBiT or LgBiT, treated or not with indicated siRNAs. Fifty micrograms of isolated mitochondria were solubilized in 50 μl of homogenization buffer (10 mM HEPES-KOH pH 7.4, 1 mM EDTA pH 8.0 and 250 mM sucrose). These mitochondria were mixed and incubated for 15 min at 37°C, with or without additional reagents. Short-chain phosphoinositides including diC8-PI(3)P (cat. no. P-3008), diC8-PI(4)P (cat. no. P-4008), diC8-PI(5)P (cat. no. P-5008), diC8-PI(3,4)P_2 (cat. no. P-3408), diC8-PI(3,5)P_2 (cat. no. P-3508), diC8-PI(4,5)P_2 (cat. no. P-4508), and diC8-PI(3,4,5)P_3 (cat. no. P-3908) were obtained from Echelon Biosciences, Inc. Valinomycin (cat. no. 10009152) was obtained from Cayman, and Wortmannin (cat. no. 681675) was from Calbiochem. Subsequently, 25 μl of Nano-Glo Live Cell Reagent (Promega, cat. no. N2011) were added to each well containing the mitochondrial mixture, followed by incubation for 15 min at 37°C. Luminescence was then measured to assess mitochondrial fusion.

Transferrin uptake assay

HeLa cells were seeded on glass-bottom dishes or coverslips and serum-starved in Leibovitz's medium for 1 hour at 37°C. Cells were then placed on ice for 15 min to inhibit endocytosis and incubated with Alexa Fluor 488–labeled transferrin (25 μg/ml; Thermo Fisher Scientific, cat. no. T-13342) in ice-cold starvation medium for 30 min at 4°C to allow surface binding. After washing with ice-cold medium, cells were incubated at 37°C for the indicated times (0 to 30 min) to allow internalization. Surface-bound transferrin was removed by acid stripping using citrate buffer (pH 4.6; 25.5 mM citric acid, 24.5 mM sodium citrate, 280 mM sucrose, 0.01 mM deferoxamine mesylate) for 2 min on ice. Cells were washed with PBS and fixed in 4% paraformaldehyde for 15 min on ice, followed by DAPI staining and mounting.

Phosphoinositide phosphatase translocation assay

HeLa cells were cotransfected with mCherry-FKBP-INPP4B together with TGN38-FRB-CFP. After 24 hours, cells were stained with MitoTracker, and confocal microscopy was used to monitor cells before and 10 min after the addition of rapamycin (100 nM).

Oxygen consumption experiments (Oroboros)

Cells were transfected with the indicated siRNAs and seeded onto 10-cm dishes. Cells were rinsed with PBS, trypsinized, and collected by centrifugation at 300*g* for 5 min at 25° ± 2°C. Cell pellets were resuspended in mitochondrial respiration medium (MiR05), and cell density was determined using a Countess 3 automated cell counter (Thermo Fisher Scientific). Oxygen consumption in intact cells was monitored polarographically with an Oxygraph-2k (Oroboros Instruments) at 37°C in a 2-ml chamber containing MiR05. The instrument was calibrated daily with air-saturated reaction medium using the SUIT-008 O_2 ce-pce D025 protocol. Cell suspensions were added to the chamber and allowed to equilibrate for 2 to 3 min before sealing the chamber and starting the record. After stable basal respiration was achieved (5 min), compounds were added sequentially. Respiration rates were calculated as the average value over a 30-s window using DatLab 7.4 software (Oroboros Instruments) and expressed as pmol O_2 per second per 2.5×10^6 cells.

Mice

Mice were housed in a temperature-controlled (24°C) room under a 12-hour light-dark cycle with free access to regular chow and water

under specific pathogen-free conditions in the animal facility of RCAST, University of Tokyo. All mouse experiments were performed according to the "Fundamental guidelines for the proper conduct of animal experiment and related activities in academic research institutions" under the jurisdiction of the Ministry of Education, Culture, Sports, Science, and Technology of Japan and were approved by the Committee on Animal Experimentation at the University of Tokyo. PI3K-C2α-floxed ($C2\alpha^{flox/flox}$) mice and PI3K-C2β-null ($C2\beta^{-/-}$) mice were described previously (*55*, *56*). To generate cardiomyocyte-specific conditional *PI3K-C2*α and *PI3K-C2*β DKO mice, $C2\alpha^{flox/flox}$; $C2\beta^{-/-}$ mice were crossed with Nkx2.5-Cre transgenic mice (Jackson Lab., cat. no. 030047). $C2\alpha^{flox/flox}$; $C2\beta^{+/+}$; Cre^{-} littermates were used as controls. Mice were euthanized using intraperitoneal injections of an overdose of avertin, in accordance with acceptable euthanasia guidelines. Mice were genotyped by PCR analysis of genomic DNA extracted from tail biopsies. Cardiac function was assessed in postnatal day 1 (P1) mice using a Vevo 2100 high-resolution ultrasound imaging system equipped with a 40-MHz probe. Echocardiography was performed without anesthesia. B-mode and M-mode images were acquired, and cardiac parameters, including ejection fraction (EF), fractional shortening (FS), and left ventricular internal dimensions at systole and diastole (LVIDs and LVIDd), were analyzed using Vevo LAB software. For histological analysis, neonatal mouse hearts were collected and fixed in 4% PFA in PBS at 4°C. After fixation, hearts were dehydrated through a graded ethanol series, cleared, and embedded in paraffin. Hearts were sectioned at 3 to 5 μm and stained with hematoxylin and eosin (H&E) according to standard procedures. For transmission electron microscopy, neonatal mouse hearts were fixed in 2.5% glutaraldehyde in 0.1 M phosphate buffer (pH 7.4) at 4°C and postfixed with 1% osmium tetroxide. Samples were dehydrated through a graded ethanol series and embedded in Epon 812 resin. Ultrathin sections were prepared using an ultramicrotome, stained with uranyl acetate and lead citrate, and examined using an H-7650 transmission electron microscope (Hitachi High-Technologies, Tokyo, Japan) operated at 80 kV. Myocardial cells were isolated and cultured from excised heart of neonatal mice. Heart tissues were incubated in a 1.5 mg/ml collagenase/dispase mixture and 20 mM 2,3-butanedione monoxime (BDM) in L-15 solution for 30 min at 37°C. Cell suspensions were centrifuged for 5 min at 100*g* at 20°C and plated onto type I collagen (Nitta Gelatin)–coated dishes in 10% FBS-containing DMEM.

Number of cells and events analyzed by microscopy

For quantification, cell numbers and fusion events are reported from three independent experiments unless stated otherwise. For mitochondrial morphology and colocalization analyses, at least one ROI per cell was used for quantification. If not specified, a minimum of 10 cells were counted per experiment.

Figure 1: Panel B: $n = 18$ cells for PI(3)P, PI(4)P, PI(5)P, PI(3,4)P_2, PI(3,5)P_2, PI(4,5)P_2 and PI(3,4,5)P_3 (three independent experiments); panel D: $n = 168, 155, 136, 145, 143, 159$ cells for siCont, si*PI10*α, si*PI3K-C2*α, si*PI3K-C2*β, si*PI3K-C2*α/β, si*Vps34*, respectively (three independent experiments); panel F: $n = 138, 100, 97, 94$ cells for Control, *PI3K-C2*α KO, *PI3K-C2*β KO, and *PI3K-C2*α/β DKO, respectively (three independent experiments); panel H (and panels B and C from fig. S5): $n = 60, 60, 29, 26, 26, 30, 20, 20$ cells for Control, *PI3K-C2*α/β DKO, *PI3K-C2*α/β DKO + WT-PI3K-C2α, *PI3K-C2*α/β DKO + WT-PI3K-C2β, *PI3K-C2*α/β DKO + WT-PI3K-C2α/β, *PI3K-C2*α/β DKO + KD-PI3K-C2α, *PI3K-C2*α/β DKO + KD-PI3K-C2β, *PI3K-C2*α/β DKO + KD-PI3K-C2α/β, respectively (three independent experiments).

Figure 2: Panel B: $n = 12$ cells for siCont and si*C2*α/β (three independent experiments); panel D: $n = 13, 13$ cells for siCont and siPI3K-*C2*α/β, respectively (three independent experiments); panel F: $n = 90, 90, 90, 98, 113, 104, 82, 90$ cells for siCont, si *PI3K*-C2α, si *PI3K*-C2β, si *PI3K*-C2α/β, treated or not with CHX respectively (three independent experiments); panel K: $n = 21$ events from 15 cells (four independent experiments).

Figure 3: Panel A: $n = 45$ events from 20 cells (three independent experiments); panel B: $n = 28, 28$ cells for siCont and si*PI3K-C2*α/β (three independent experiments); panel C: $n = 32, 35$ events for siCont and si*PI3K-C2*α/β, respectively (three independent experiments); panel E: $n = 18$ cells (four independent experiments); panel F: $n = 65$ events from 42 cells (four independent experiments); panel G: $n = 50, 50, 50$ cells for CHX treatment at 0, 2, and 4 hours, respectively (three independent experiments); panel I: $n = 12, 12$ cells for DMSO and rapamycin treatment, respectively (three independent experiments).

Figure 4: Panel C: $n = 101$ events from 47 cells (four independent experiments); panel F: $n = 34$ events from 17 cells (four independent experiments); panel H: $n = 52, 44$ cells for siCont and si*MFN1/2*, respectively (three independent experiments); panel J (top): $n = 86$ events from 38 cells (four independent experiments); panel J (bottom): $n = 34$ events from 38 cells (four independent experiments); panel L: 18 cells (three independent experiments); panel N: $n = 89$ fission and 63 fusion events from 40 cells (eight independent experiments).

Figure 5: Panel D: $n = 31, 34$ mice for WT and *PI3K-C2*α/β-DKO; panel F: $n = 10$ mice for WT and *PI3K-C2*α/β-DKO; panel H: $n = 3$ mice for WT and *PI3K-C2*α/β-DKO (three independent experiments); panel J: $n = 99, 99$ mitochondria for Control and *PI3K-C2*α/β-DKO, respectively (six independent experiments); panel L: $n = 3$ mice for Control and *PI3K-C2*α/β-DKO (three independent experiments).

Figure S1: Panels D and E: same number of cells from Fig. 1D; panels H and I: $n = 116, 207$ cells for siCont and si*PIKfyve*, respectively (three independent experiments).

Figure S2: Panel C: $n = 92, 93, 90, 93$ cells for siCont, si*PI3K-C2*α, si*PI3K-C2*β, si*PI3K-C2*α/β, respectively (three independent experiments).

Figure S3: Panels A and B: same number of cells from Fig. 1, F and H, respectively; panel C: $n = 24, 24$ cells for GFP-PH^{FAPP1} and GFP-PH^{TAPP1} in siCont and si*PI3K-C2*α/β (three independent experiments); panel D: $n = 24, 24, 24, 24$ cells for GFP-PH^{FAPP1} and GFP-PH^{TAPP1} in Control, *PI3K-C2*α KO, *PI3K-C2*β KO, and *PI3K-C2*α/β DKO, respectively (three independent experiments).

Figure S4: Panel B: $n = 100, 111, 119, 107$ cells for siCont, si*PI3K-C2*α, si*PI3K-C2*β, si*PI3K-C2*α/β, respectively (three independent experiments); panel D: $n = 73, 68, 59, 72$ cells for siCont, si*PI3K-C2*α, si*PI3K-C2*β, si*PI3K-C2*α/β, respectively (three independent experiments); panel F: $n = 80, 80, 80, 80$ cells for siCont, si*PI3K*-C2α, si*PI3K*-C2β and si*PI3K*-C2α/β (four independent experiments); panel H: $n = 80, 80, 80, 80$ cells for siCont, si*PI3K-C2*α, si*PI3K-C2*β, and si*PI3K-C2*α/β, respectively (four independent experiments).

Figure S6: Panel A: $n = 36, 36$ cells for vehicle and Pitstop, respectively (three independent experiments); panel C: $n = 90, 90$ cells for Vehicle and Pitstop 2 (three independent experiments).

Figure S7: Panel C: $n = 73, 79, 68, 81, 57$ cells for siCont, si*SHIP1#1*, si*SHIP1 #2*, si*SHIP2 #1* and siSHIP2 #2, respectively (three independent experiments); panel F: $n = 71, 50, 71, 67, 61, 75, 78$ cells for siCont, si*INPP4A#1*, si*INPP4A#2,* si*PTEN#1*, si*PTEN#2,* si*INPP4A*/*PTEN#1,* and si*INPP4A*/*PTEN#2,* respectively (three independent experiments); panel H: $n = 268, 298$ cells for siCont and si*PTEN*/*SHIP1/2*, respectively (three independent experiments).

Figure S8: Panel B: $n = 104, 88$ cells for siCont and si*PI3K-C2*α/β, respectively (three independent experiments); panel C: same number of cells from Fig. 2F; panel E: same number of events and cells from Fig. 3A; panel F: $n = 18, 18$ events for mCherry-PH^{TAPP1-} and mCherry-PH^{TAPP1+} (four independent experiments); panel G: $n = 28, 28$ cells for siCont and si*PI3K-C2*α/β (three independent experiments).

Figure S9: Panel A: $n = 18, 18$ events for PI3K-C2α and PI3K-C2β, respectively (three independent experiments); panel C: 19 cells (three independent experiments); panel D: $n = 24, 24$ (left) and 12, 12 (right) cells for −rapa and + rapa, respectively (three independent experiments); panel E: same number of cells from Fig. 3I; panel G: $n = 12, 12$ cells for DMSO and rapamycin treatment, respectively (three independent experiments).

Figure S10: Panel B: $n = 31$ events from 15 cells (three independent experiments); panel D: $n = 81$ events from 50 cells (three independent experiments); panel G: $n = 101$ events from 47 cells (four independent experiments).

Figure S12: Panel B: $n = 262$, 255 cells for siCont and si*PI3K-C2*α/β, respectively (three independent experiments); panel D: $n = 32$ events from 16 cells (four independent experiments).

Figure S13: Panel C: $n = 50$, 50 cells for siCont and si*PI3K-C2*α/β, respectively (three independent experiments).

Figure S14: Panel B: 48, 48, 48, 48 cells for siCont, si*PI3K-C2*α, si*PI3K-C2*β and si*PI3K-C2*α/β, respectively (four independent experiments); panel D: 48, 48, 48,48 cells for siCont, si*PI3K-C2*α, si*PI3K-C2*β and si*PI3K-C2*α/β, respectively (four independent experiments).

Statistical analysis

Data are presented as the mean ± SD or SEM and expressed as the percentages or fold changes relative to control values. Statistical significance was analyzed using Prism 10 software (GraphPad Software). Student's *t* tests were used for pairwise comparisons, whereas one-way analysis of variance (ANOVA) was performed with Dunnett's test for comparisons to the control group or Tukey's test for comparisons between multiple groups other than the control. For two-way ANOVA, Bonferroni's test was used for multiple comparisons or Tukey's test was applied for post hoc pairwise comparisons, as appropriate. *P* values are indicated as $*P < 0.05$; $**P < 0.01$; $***P < 0.001$; $****P < 0.0001$; and n.s. is not significant.

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SUPPLEMENTARY MATERIALS

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RESEARCH ARTICLE SUMMARY

VOLCANOES

Insights from more than a year of ongoing episodic lava fountaining at Kīlauea

A. F. Flinders* *et al.*

Full article and list of author affiliations: https://doi.org/10.1126/science.aef2931

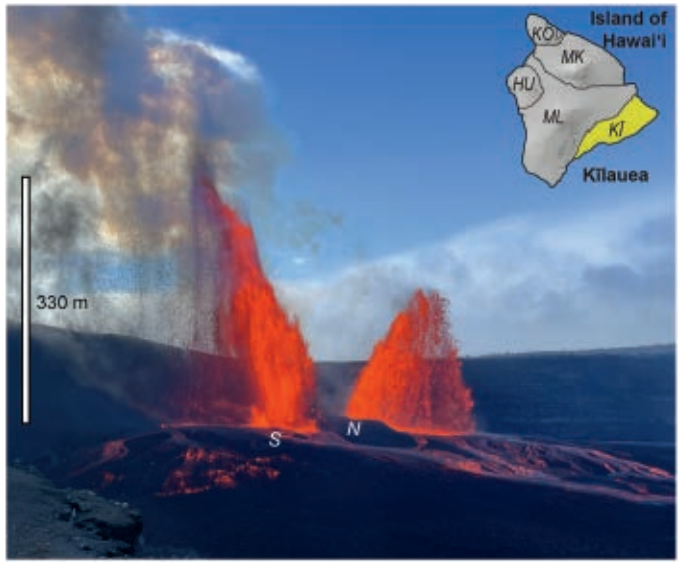
INTRODUCTION: Lava fountains are among the most striking expressions of basaltic volcanism, ejecting gas and molten magma hundreds of meters above a vent and often feeding lava flows. Lava fountaining is best known from Hawaii, Italy, and Iceland, although analogous eruptions have been inferred on the Moon and Jupiter's moon Io. Fountaining occurs at basaltic volcanoes worldwide, but protracted episodic fountaining, in which hours-long episodes alternate with pauses lasting days to weeks, is exceptionally rare. Before 2024, only three episodic fountaining eruptions had been documented at Kīlauea, Hawaii, since 1823. On 23 December 2024, a new sequence began within Halema'uma'u at Kīlauea's summit. By September 2026, it had produced 54 episodes, the most recorded during any Kīlauea eruption.

RATIONALE: The processes that initiate, sustain, and terminate individual episodes remain uncertain. Two leading mechanisms have been proposed: pressure-driven magma ascent followed by water-rich (H_2O) volatile exsolution in a shallow conduit, and accumulation and collapse of carbon dioxide–rich (CO_2) foam near a magma-reservoir roof. This is the first Kīlauea episodic fountaining sequence observed with a dense, modern monitoring network. We integrated seismic, infrasound, deformation, gas, satellite, visual, thermal, and geochemical observations from the first 39 episodes to document the eruption's evolution in the context of these proposed mechanisms.

RESULTS: Each episode produced rapid summit deflation, whereas pauses were characterized by slower reinflation as pressure rebuilt within the shallow Halema'uma'u magma reservoir. Fountaining repeatedly began as summit tilt approached similar but gradually evolving thresholds. Eruption forecasts based on the intersection of the long-term threshold trend with ongoing reinflation provided quantitative estimates of episode onset and supported public hazard communication.

Beginning with episode 14, sustained fountaining was commonly preceded by gas-piston activity, marked by repetitive seismic and infrasonic bursts as magma cyclically rose and fell within the vents. Sulfur dioxide (SO_2) emissions decreased as magma rose and increased during drainback, which is consistent with shallow gas accumulation and release. Fountain heights varied over months and reached ~480 m. Tephra affected National Park Service infrastructure, roads, and communities as far as ~70 km downwind.

The erupted magma was overall more mafic than summit magma erupted during 2020–2023. Cyclic changes in glass magnesium content indicate periodic flushing of the Halema'uma'u reservoir by hotter, more mafic magma, whereas no discernible compositional changes occurred during individual fountains. Persistent SO_2 emission during pauses suggests open-system degassing as reservoir pressure rebuilt. Preliminary measurements yielded low interepisode CO_2/SO_2 ratios, which is consistent with magma equilibration within the reservoir without formation and collapse of CO_2-rich foam.



Map of the Island of Hawai'i and representative photograph of summit fountaining episodes during the 2024–2025 Kīlauea eruption. (Upper right) The locations of the five volcanoes that form the Island of Hawai'i: Kohala (KO), Mauna Kea (MK), Hualālai (HU), Mauna Loa (ML), and Kīlauea (Kī). (Main) Dual fountaining from the north (N) and south (S) vents within Halema'uma'u, at Kaluapele (Kīlauea caldera), during episode 36 on 9 November 2025, with peak heights of 385 m. View is from the south rim of Halema'uma'u. Scale bar shows the fountain height at the time the photograph was taken. [Photograph by USGS]

CONCLUSION: Fountaining began as summit reinflation approached evolving tilt thresholds, enabling useful forecasts despite no appreciable increases in summit seismicity before most episodes. These thresholds imply repeatable conditions for initiating fountaining, although whether they depend on reservoir pressure alone remains unclear. Gas observations may favor a major role for shallow H_2O-rich bubble clusters over deep CO_2-rich foam collapse, but do not distinctively identify the processes that initiate or terminate individual episodes. The eruption's outcome remains unknown. Continued monitoring will help constrain how magma and gas recharge, reservoir pressure, conduit connectivity and permeability, and volatile segregation and outgassing produce episodic fountaining, while underscoring Kīlauea's fundamental role as a globally accessible natural laboratory for developing new techniques in volcano monitoring and eruption forecasting. □

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VOLCANOES

Insights from more than a year of ongoing episodic lava fountaining at Kīlauea

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Episodic lava fountaining is rare globally, with only three occurrences from Kīlauea (Hawaii) since 1823. Beginning in December 2024, an eruption within the summit crater, Halemaʻumaʻu, has produced 54 fountaining episodes to date, each lasting typically for hours and separated by days to weeks. Although multiple mechanisms have been proposed, fundamental questions remain about how magma reservoir pressure, conduit permeability, and volatile exsolution control fountaining onset. We present a multiparameter record of episodic fountaining at Kīlauea using modern geochemical, geophysical, and geologic monitoring. These observations revealed systematic correlations between fountaining and summit inflation, enabling unprecedented quantitative forecasts. This eruption will provide new constraints on the timing, evolution, and processes driving episodic fountaining globally, and highlights Kīlauea's role as a natural laboratory for studying volcanic processes.

Lava fountaining is a distinctive form of volcanism that can eject lava hundreds of meters into the air, and when molten fallout accumulates outside of eruptive vents, can initiate spatter-fed lava flows (*1*). Fountains can also disperse ejecta tens of kilometers downwind and release large volumes of gas (e.g., H_2O, SO_2, and CO_2), posing hazards to nearby populations [e.g., (*2*, *3*)]. Although lava fountaining is most associated with volcanoes in Hawaii (*1*, *4*–*6*), Italy (*2*, *7*), and Iceland (*8*), its occurrence has been inferred elsewhere in the solar system, including the Moon (*9*) and Jupiter's moon Io (*10*).

Episodic lava fountaining—in which individual fountaining episodes are separated by pauses lasting minutes to weeks—is far rarer than sustained fountaining. At Kīlauea, it has only been documented three times: during the 1959 Kīlauea Iki Crater eruption (*4*) and in the early phases of the Maunaulu [1969–1974; (*5*)] and Puʻuʻōʻō eruptions [1983–2018; (*6*)]. Despite the spectacular nature and hazard potential, the conditions governing fountaining onset, dynamics, and cessation are debated and highlight uncertainties in how magma recharge, volatile behavior, and shallow conduit dynamics interact.

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On 23 December 2024, Kīlauea began its first episodic fountaining eruption in 38 years and the first observed within its summit crater, Halemaʻumaʻu (Fig. 1). To date (September 2026), 54 fountaining episodes have occurred, the most recorded for any Kīlauea eruption, with a median duration of 9 hours (interquartile range: 5 hours), separated by pauses of days to weeks. Compared with previous eruptions, this eruption has been documented by a comprehensive summit monitoring network of more than 35 stations, including seismic, infrasound, geodetic, gas, and visual and thermal cameras (fig. S1). Additionally, repeated sampling of tephra from all episodes, and infrequent lava samples collected on the caldera floor, provide a chronological record of magma composition and its evolution.

These integrated, multiparameter observations will offer an unprecedented opportunity to observe and constrain the processes governing this unusual eruptive style. In this study, we present a synthesis of observations from the first year of the eruption (episodes 1 to 39), including early eruption forecasting efforts, previously proposed mechanisms for episodicity, and open questions about the controls of fountaining episodes.

Buildup to the December 2024 eruption

Kīlauea is the youngest and most active of the five volcanoes that form the Island of Hawaiʻi (Fig. 1). The volcano is composed of a summit caldera (Kaluapele) overlying two magma storage zones: the Halemaʻumaʻu magma reservoir (HMR) at 1- to 2-km depth and the deeper south caldera reservoir (SCR) at 3- to 5-km depth (*11*). Extending from the summit are two rift zones–the East Rift Zone and the Southwest Rift Zone (Fig. 1).

The 2018 eruption of Kīlauea partially drained the HMR, leading to one of the largest basaltic caldera collapses in recorded history, expanding and deepening Halemaʻumaʻu in the south end of Kaluapele (Fig. 1) by ~0.8 km^3 (*12*). Between December 2020 and September 2023, Kīlauea entered a period of intermittent activity marked by five small- to moderate-volume eruptions (~0.01 to 0.1 km^3), all confined to within the collapse area [Fig. 1 and table S1; (*13*)]. These eruptions were characterized by low sustained fountains (10 to 100 m high), the formation of lava lakes (*14*), and the collective production of ~0.2 km^3 of lava. None produced episodic fountaining.

Between January 2019 and March 2023, increasing distance between the Global Navigation Satellite System (GNSS) station pair UWEV-CRIM [Fig. 1; (*15*, *16*)] indicated that the summit was inflating at more than twice its pre-2018 rate (22 cm/year versus 9 cm/year). After March 2023, this elevated rate doubled (57 cm/year). Over the same period, uplift at GNSS station OUTL (*17*), a proxy for south caldera inflation, increased sevenfold (8 cm/year to 56 cm/year). These increases, similar to those historically interpreted as pressurization of Kīlauea's magma reservoirs [e.g., (*18*)], coincided with intrusions into the Southwest and upper East Rift Zones, culminating in short-lived eruptions on 3 June 2024, and 15 to 20 September 2024, respectively [Fig. 1; (*13*)].

Major element geochemical data (glass MgO, olivine Mg content) from December 2020 lavas show that the 2018 collapse disrupted shallow magma pathways but did not strongly affect the stability of the HMR or SCR (*19*, *20*). Minor elements in glasses and major elements in whole-rock compositions from 2017 to 2021 indicate that magma from pre- and postcollapse summit eruptions have distinct compositions (*19*, *21*), but geobarometry on September 2023 olivine phenocrysts sill places crystallization within the HMR (*22*). The 2020–2024 period may represent a transitional phase of magmatic recharge, during magma pathway reorganization.

A new eruption begins

In early December 2024, seismicity was concentrated between Kaluapele and the upper East Rift Zone (Figs. 1 and 2), before shifting to scattered

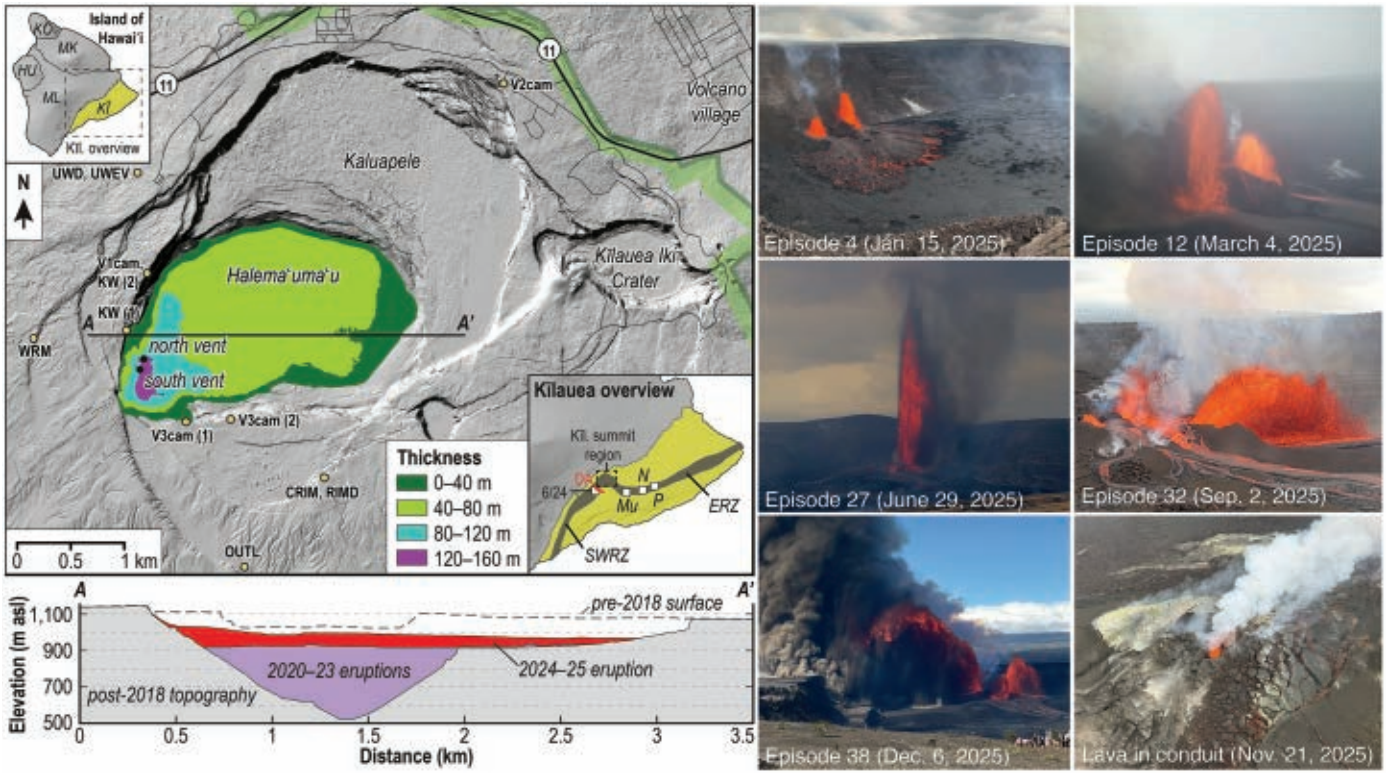

Fig. 1. Map of Kīlauea volcano on the Island of Hawai'i and representative photographs of summit fountaining episodes during the 2024–2025 eruption. (**Left**) On the map, yellow circles indicate the locations of selected Hawaiian Volcano Observatory (HVO) monitoring stations (fig. S1), and colored areas show the accumulated thickness of tephra and lava flows within Halema'uma'u from fountaining episodes 1 to 39; accumulated tephra outside of Halema'uma'u is shown in Fig. 3. Upper-left map inset shows the locations of the five volcanoes that form the Island of Hawai'i: Kohala (KO), Mauna Kea (MK), Hualālai (HU), Mauna Loa (ML), and Kīlauea (KI). Lower-right map inset shows the locations of Kīlauea's previous episodic fountaining eruptions at Maunaulu (Mu) and Pu'u'ō'ō (P), as well as the 2024 Southwest Rift Zone (SWRZ) and East Rift Zone (ERZ) [Nāpau Crater (N)] eruptions (*57*), and the location of the summit UV differential optical absorption spectrometer (DOAS) array [red line (DA)]. Lower cross-section (A to A') depicts the infilling of the 2018 collapse area from the previous five summit eruptions in 2020–2023 (purple) and the 2024–2025 eruption discussed here [red; (*58*)]. (**Right**) Representative photographs of selected fountaining episodes and lava ponded within the north vent conduit. Episodes 4 (<100 m), 12 (100 to 200 m), and 27 (>300 m) represent a range of increased maximum sustained fountaining heights throughout the eruption. Episode 32 shows an inclined fountaining (~100 m high) from the north vent to the northeast. Episode 38 shows a highly inclined fountain >380 m tall that destroyed the HVO V3cam webcam. Lava ponded in the north vent conduit, ~10 m below the vent rim, as observed during a helicopter overflight 4 days before episode 37. USGS photographs are taken from several locations around Kaluapele.

events beneath the caldera on 13 December. Seismicity persisted for several days, with the number of volcano-tectonic (VT) earthquakes ranging from 80 to 100/day, before declining to fewer than 40/day by 18 December. GNSS station-pair distances showed increasing summit inflation beginning on 22 December (Fig. 2). Seismicity remained subdued until 23 December at 1:42 a.m. (Hawaii Standard Time), when an intense swarm of VT earthquakes (>10/hour) began under the north rim of the caldera, then migrated south along its western edge, while shallowing from 2 to 1 km depth. Rates increased more than fivefold in the span of ~45 min, signaling dike ascent, before giving way to rapid tilt changes, continuous eruptive tremor, a steep decline in VT earthquakes, and the onset of lava fountaining at 2:20 a.m.

The eruption began with the opening of an ~900-m-long fissure system comprising a primary vent (later referred to as the "south" vent) and three northeast to southwest (NE to SW)–oriented fissure segments along the southwest corner of Halema'uma'u (movies S1 and S2). Satellite interferometric synthetic aperture radar (InSAR) captured the coeruptive opening of an inferred subvertical shallow dike, trending southwest (Fig. 3A). This first episode lasted for ~13 hours (table S2) and, like later episodes, produced intense high-amplitude seismic tremor (<15 Hz) and high real-time seismic amplitudes (RSAM). Fountaining reached heights of up to ~160 m [fig. S2; (*23*)].

The second fountaining episode began just 22 hours later, and by its end, activity had largely concentrated to two vents ("south" and "north"; Fig. 1) ~150 m apart. Fountain heights remained low (<~100 m) over the ensuing 6 weeks (episodes 2 to 7). Erupted material accumulated predominantly on the caldera floor, producing spatter-fed lava flows, and minor tephra was ejected over the caldera rim (movie S3 and fig. S3). From 3 February 2025 (episode 8) onward, fountain heights began to exceed 100 m [figs. S4 and S5 and table S2; (*23*)].

Virtual engagement

As the first episodic fountaining eruption in 38 years, and one in an easily accessible area of Hawai'i Volcanoes National Park (HAVO), the eruption drew widespread in-person and virtual attention. Before the eruption, the Hawaiian Volcano Observatory (HVO) operated one live stream webcam on the caldera rim; two more were added mid-eruption [Fig. 1; (*24*)]. Over the first year of the eruption, virtual engagement exceeded millions of watch hours and reached more than ~9000 concurrent live viewers across the United States and internationally. These feeds were rebroadcast on multiple non-HVO live streams; therefore, viewership was likely substantially higher. Although the live streams reached a global audience, they were used most heavily by Hawaii residents, particularly those on the Island of Hawai'i, where per-capita

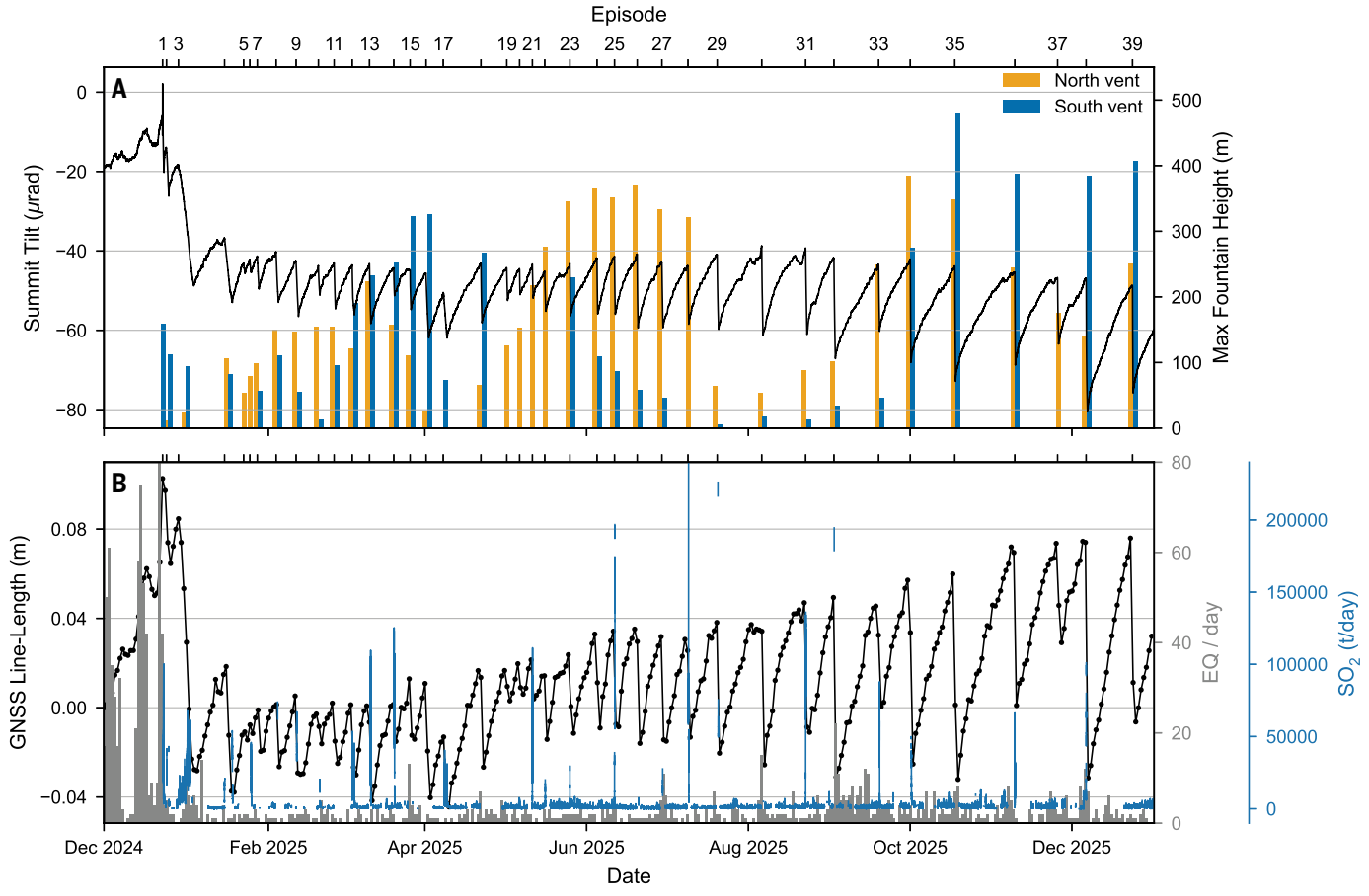

Fig. 2. Representative geologic and geophysical Kīlauea summit time-series data during the 2024–2025 episodic fountaining eruption (episodes 1 to 39). (**A**) Maximum lava fountain height for each episode (*23*), colored by vent, plotted alongside caldera-radial summit ground tilt from station UWD [azimuth, 300°; (*25*)], showing fountaining onsets and interepisode reinflation. Bars indicate episode-maximum fountain heights; horizontal positions are not proportional to episode duration. (**B**) GNSS line-length between stations UWEV and OUTL (*15*,*17*) across the caldera (black), demonstrating that summit deformation, which is associated with HMR pressure, largely returns to a consistent threshold between episodes. Also shown are daily summit earthquake counts (gray) from the HVO's island-wide seismic network (fig. S1), showing no appreciable changes in summit seismicity leading to fountaining onset. Also shown are 1-min averaged SO_2 emission rates (blue) from an array of summit UV differential optical absorption spectrometers [DOAS; (*30*)]. Interepisode emission rates are low and peak during fountaining; however, accurate data depended on wind and plume direction.

viewership far exceeded that of any other region, highlighting the webcams' value as a near-real-time communication tool for local communities (*24*). Notably, on 6 December 2025 (episode 38), shallow collapse of the south vent produced a south-inclined fountain (fig. S6), burying one of HVO's live stream webcams under 10 m of agglutinated spatter, ~630 m away (Fig. 1). A video clip of the webcam being destroyed was viewed more than 500,000 times across social media platforms (*24*).

Initial operational forecasting

Early in the eruption (episode 3), it was recognized that at fountaining onset, caldera-radial summit ground tilt (*25*, *26*)—a proxy for HMR pressure (*18*)—transitioned from inflation to deflation and reversed when fountaining ended (Fig. 2). Additionally, the pattern of surface deformation observed in InSAR spanning individual episodes was similar to previous interpretations of pressure loss from the HMR [Fig. 3B; (*11*)]. Because summit inflation, and likely HMR pressure, largely returned to consistent thresholds between episodes, cumulative deformation was limited (Figs. 2 and 3C).

Eruption forecasting is inherently challenging, with prior quantitative approaches largely focused on identifying accelerating changes in monitoring signals (*27*). However, consistent correlation between summit tilt changes and fountaining onsets enabled the implementation of operational eruption forecasts. Forecast windows were used by HAVO for staffing, traffic control, and general planning and by the Hawai'i County Civil Defense Agency to inform communities of eruption-related hazards.

Summit tilt maxima at station UWD (azimuth, 300°; Fig. 1) were modeled using exponential decay functions anchored to the eruption onset in December 2024, whereas interepisode reinflation followed power-law–like recoveries (Fig. 4). Initial quantitative forecasts after each episode (episode 12 onward) were generated from the intersection of these decay and recovery functions (Fig. 4). By modeling the mean behavior of each previous interepisode reinflation, an early forecast could be provided immediately after fountaining ended (tables S3 and S4). As interepisode reinflation continued, episode-specific recovery functions were implemented, with forecasts becoming increasingly precise in time (2 to 5 days; Fig. 4). This initial approach was used through episode 24, since which a more integrated methodology has provided forecasts in the form of interpreted suites.

Ongoing observations

Whereas episodes 2 to 13 were preceded by lava flows and or intermittent spatter, a change in precursory behavior occurred around

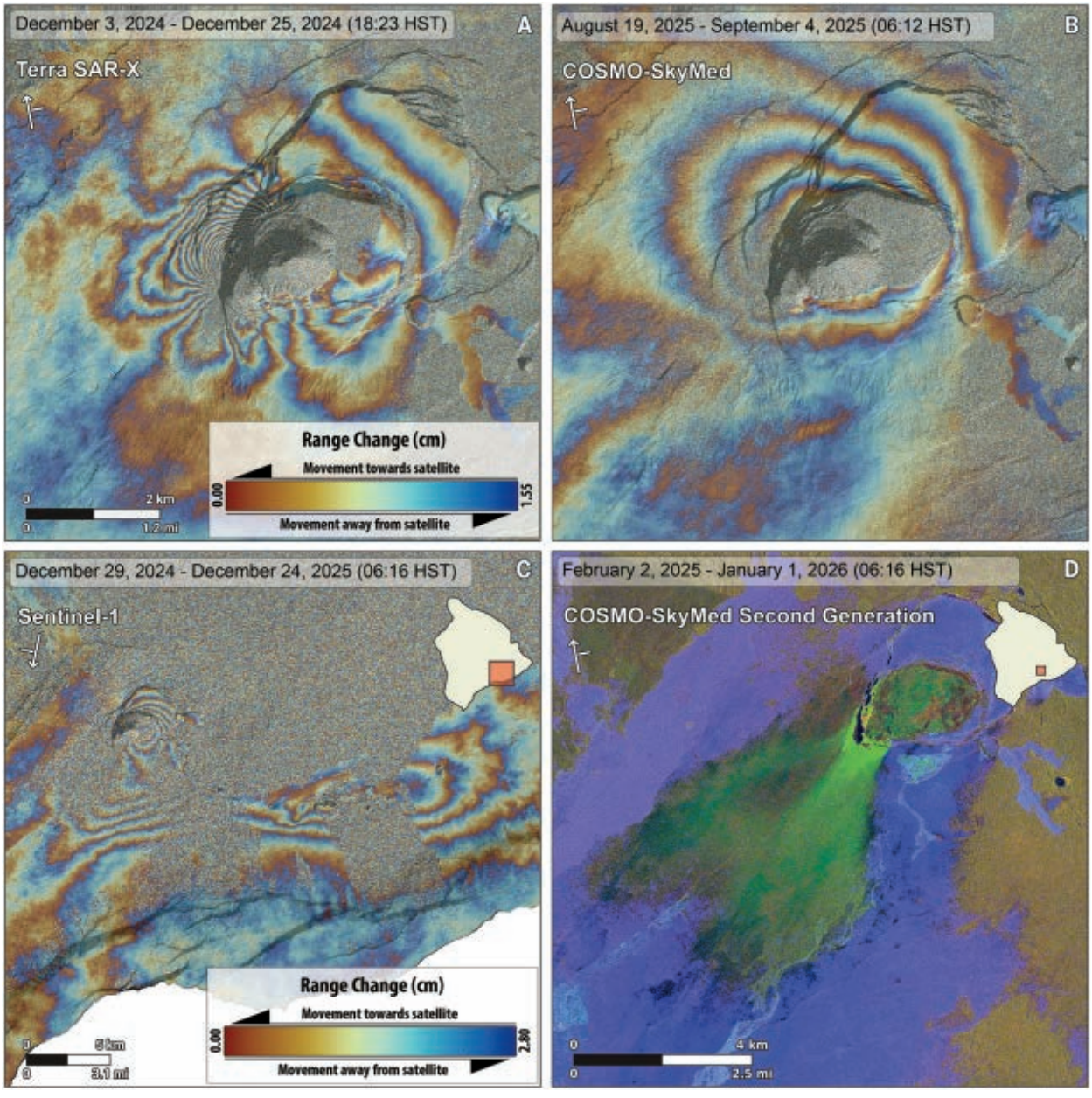

Fig. 3. InSAR interferograms of the summit of Kīlauea from 4 December 2024 to 1 January 2026. (**A**) Interferogram from the German Aerospace Center's (DLR) TerraSAR-X, spanning the eruption onset and episodes 1 to 3. Although the image span episodes 1 to 3, the observed deformation is predominantly associated with a syn-eruptive dike opening along the southwest margin of Halema'uma'u during episode 1. (**B**) Interferogram from the Italian Space Agency's (ASI) COSMO-SkyMed, spanning episodes 31 and 32, showing summit deflation associated with magma withdrawal during fountaining. (**C**) Interferogram from the European Space Agency's (ESA) Sentinel-1, spanning 1 year and episodes 3 to 39, showing minor subsidence across Kīlauea summit, the upper Southwest Rift Zone, and the upper-to-middle East Rift Zone. The magnitude of range change per color cycle differs between the TerraSAR-X/COSMO-SkyMed and the Sentinel-1 datasets. (**D**) Lava flow and tephra accumulation at the summit from episodes 4 to 39, using data from COSMO-SkyMed. Unchanged and barren areas are represented in blue-purple, vegetated areas appear in yellow-brown, and accumulated tephra deposits over barren land in green. Arrows indicate satellite flight direction, and perpendicular bars are look directions. The red square over map of the Island of Hawai'i shows the extent of image [same extent for (A) and (B)].

19 March 2025 (episode 14). Minutes-long oscillatory bursts of sub–6 Hz seismic energy, superimposed on lower-amplitude interepisode seismic tremor and infrasound, correlated with visual observations of rising magma within the vent conduits. This repetitive degassing was likely driven by buoyant gas slugs or high concentrations of bubbles [i.e., "gas pistoning"; (*5*, *28*, *29*)] and became consistent precursors to sustained fountaining (movie S4). During gas pistoning, SO_2 emissions fell below average interepisode levels (~1500 tons/day) as magma in the conduits rose, then peaked at ~4000 tons/day during magma drainback and degassing. Episodes 14 and 15 marked the first occurrence of this pattern, with days of gas pistoning preceding fountaining (movie S5).

Subsequent episodes followed a similar sequence as magma refilled the conduits. Days- to week-long pauses were characterized by

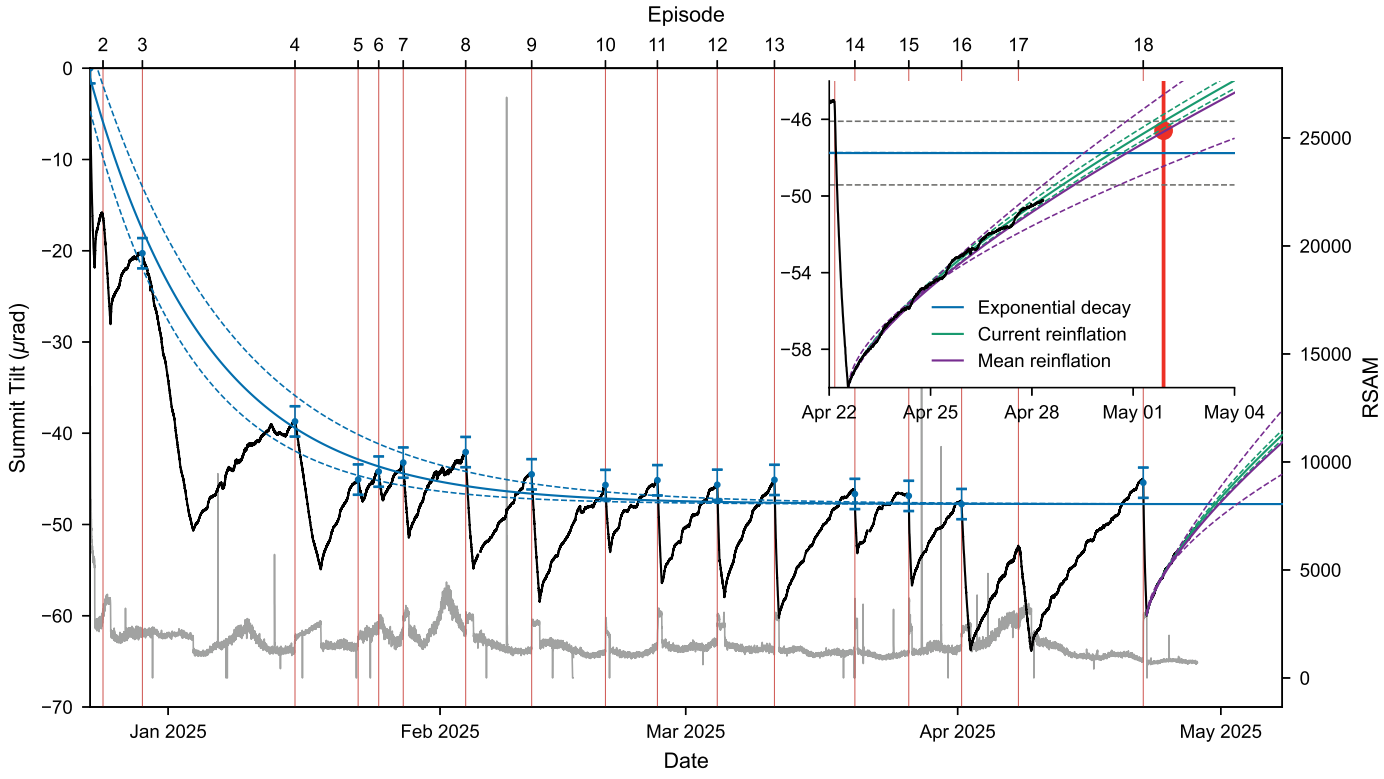

Fig. 4. Real-time eruption onset forecasting before episode 19. Daily forecast windows for lava fountaining onset were provided to Hawai'i Volcanoes National Park and Hawai'i County Civil Defense Agency partners and the public through HVO updates. (**Main**) The figure illustrates one example of the forecasting framework used internally to generate these onset estimates. Forecasts were derived from intersections between a mean exponential decay fit (blue lines) to summit radial tilt maxima at station UWD (blue circles; bars are 1 SD) and both the previous mean power-law reinflation model (purple) and the evolving interepisode. The dashed lines represent 95% confidence intervals for the exponential decay (blue) and current reinflation model (green), and 1 SD for the mean of previous reinflation (purple; table S3). In this example, the tilt maximum for episode 17 was considered anomalous and excluded in real time from the exponential decay fit. This forecast was generated on 28 April 2025, at 08:00 (Hawai'i Standard Time), ~85 hours before the onset of episode 19 (table S4). For comparison, also shown are real-time seismic amplitudes (RSAM) from the summit seismic station RIMD (gray curve). No clear precursory seismic behavior was observed. (**Inset**) At the time the forecast was made, the model predicted the most likely eruption onset window between 28 April at 16:00 and 2 May at 06:30. Episode 19 fountaining began on 1 May 2025, at 21:35 (vertical red line).

low-amplitude seismic tremor (<12 Hz) and vents that ranged from appearing dark at night (e.g., no near-surface magma) to intermittently incandescent (fig. S7), with occasional overflight observations revealing lava ponded within the vent conduits (Fig. 1 and fig. S8). As magma reached the surface, early activity included low dome fountaining, weak strombolian-like bursts, and occasional drainback as trapped gas was released, followed by vent overflows lasting hours to days (movie S6). This activity typically began at the north vent as gas-dominated bursts and was followed by more effusive activity from the south vent. Fountaining then initiated rapidly over minutes to tens of minutes, although both vents were not always active during a single episode (table S2).

Between late April and early June 2025 (episodes 19 to 24), fountain heights generally increased, after which fountains remained high (>300 m) but variable through early July (episode 28; Fig. 4 and table S2). Throughout this period (episodes19 to 28), extensive spatter-fed flows resurfaced the floor of Halema'uma'u (fig. S9). By late July through early September (episodes 29 to 32) fountain heights were again low (<~100 m, although episodes 31 and 32 were inclined) but then progressively increased. Several episodes in October through December exceeded 350 m, with the maximum fountain height during the first year of the eruption reaching 479 m [episode 35; Figs. 2 and 5 and movie S7; (*23*)]. Fountain heights and syn-fountaining summit deflation exhibited months-long variability throughout the eruption (Fig. 2 and fig. S10), although whether this reflected changes in magma supply, volatile abundance, conduit or vent geometries, or some combination thereof remains unclear. A moderate correlation was observed between syn-fountaining summit deflation and subsequent interepisode pause duration (fig. S10).

Peak SO_2 emission rates during fountaining [~100,000 tons/day; (*30*, *31*)] were similar to recent Kīlauea eruptions (*32*). However, comprehensive syn-eruptive characterization was limited by hazardous conditions restricting campaign measurements, burial of monitoring stations by tephra, plume rise and dispersion above the planetary boundary layer, optical interference from fine ash, inability of ultraviolet (UV) techniques to measure nocturnal fountaining, and weak winds that inhibited plume advection. Interepisode emission rates were consistent near 1500 tons/day, compared with <100 tons/day during previous nonerupting times (*31*).

As early as January 2025 (episode 4), tephra fall affected public areas of HAVO, Hawaii State Route 11, and nearby communities (Fig. 1 and fig. S11). Deposition increasingly affected monitoring stations within ~6 km west and southwest of the vents, requiring repeated excavation, maintenance, and occasional relocation (Fig. 3D and fig. S12). Beginning in late May (episodes 21 and 22), repeated deposition southwest of Halema'uma'u built a growing hill on the caldera rim (>40 m high), with individual episodes depositing up to ~8 m of spatter and tephra (episode 24; fig. S13). From May 25 (episode 23) onward (fig. S14), windborne tephra was intermittently reported in communities as far as ~70 km to the southwest.

Fig. 5. Representative photographs of summit fountaining episodes during the 2024–2025 Kīlauea eruption. (**A**) Fountaining from the north (*N*) and south (*S*) vents during episode 25 on 11 June 2025. Fountaining from the north vent reached 350 m, whereas the south vent remained less than 100 m. View is from the western rim of Halema'uma'u. (**B**) Fountaining from the north vent during episode 28 on 9 July 2025, which reached 320 m, whereas the south vent was inactive. View is from the publicly accessible trail on the north rim of Kaluapele. (**C**) Dual fountaining (*N*, *S*) during episode 36 on 9 November 2025, with peak heights of 385 m. View is from the south rim of Halema'uma'u. Scale bars show the fountain height at the time the photographs were taken. Photographs by USGS.

Preliminary geochemical analyses

Tephra and fountain-fed lava flows sampled throughout the first 39 episodes were analyzed for whole-rock chemistry in near real time by energy dispersive x-ray fluorescence (XRF) and later for whole-rock chemistry by wavelength-dispersive XRF and for glass chemistry by electron probe microanalysis (*33*). These episodes were overall more mafic (e.g., Mg-rich) than 2020–2023 summit eruptions [Fig. 6; (*21*)]. Tephra glass MgO (a proxy for magma temperature) increased from December 2024 through February 2025 (episodes 1 to 11; 7.0 to 7.7 wt %; ~1165° to 1180°C), decreased through June (episode 25; to 7.2 wt %; ~1170°C) and increased again through October (episode 35; to 7.6 wt %; ~1178°C) before starting to decline again, indicating periodic flushing of the HMR with inputs of more mafic magma. However, there were no discernable compositional changes during individual fountains (i.e., episode 14; Fig. 6), indicating that syn-eruptive

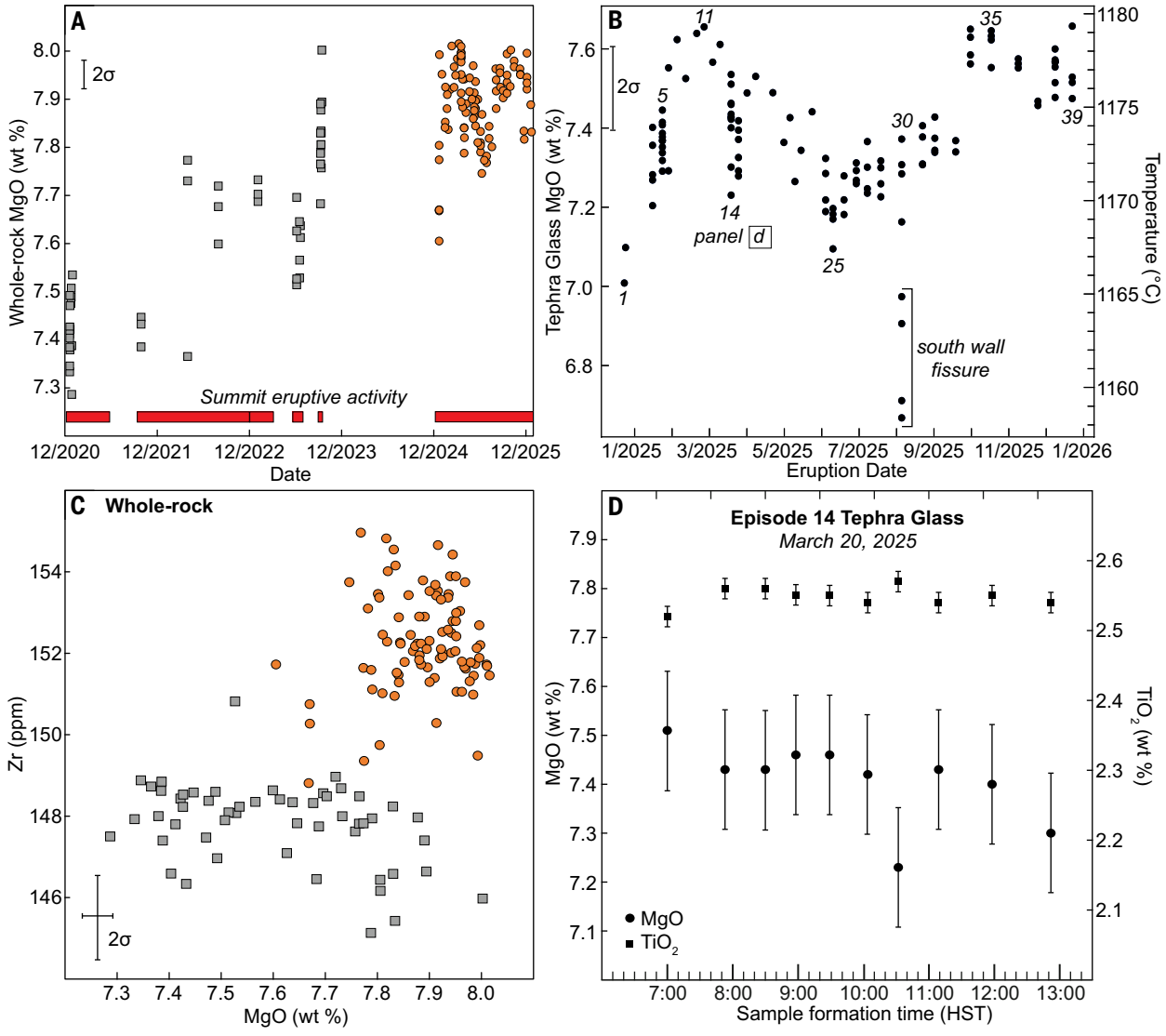

Fig. 6. Geochemical variations of 2020–2025 Kīlauea eruption samples. (**A**) Whole-rock MgO (wt %) through time (*21*,*33*), including the five summit eruptions (gray squares) before the ongoing episodic activity (orange circles). Red bars indicate the durations of summit eruptive activity since December 2020. (**B**) Glass MgO (wt %) of tephra samples from episodes 1 to 39 (*33*), showing cyclical behavior with max MgO at episodes 11 and 35 (February and October, respectively) a trough at episode 25 (June) and decreasing MgO at the end of 2025. Equivalent eruptive temperatures were calculated using Shea *et al*. [2022; (*59*)]. Also shown is glass MgO from a fissure along the south caldera wall that was active during the first few hours of episode 30. (**C**) Whole-rock MgO (wt %) versus Zr (ppm), showing elevated incompatible-element concentration for the ongoing episodic eruption (orange circles) compared with the 2020–2023 summit eruptions [gray squares; (*21*,*33*)]. (**D**) Time-series glass MgO (wt %; circles) and TiO_2 (wt %; squares) from sequential sampling of tephra during episode 14 [20 March 2025; (*33*)]. Error bars for (A) to (D) (2σ) are precision on individual analyses based on replicates of a known reference material (see Materials and methods).

dynamics were not influenced by intraepisode fluctuations in melt chemistry.

Whole-rock chemical trends provided additional constraints on fluctuations in MgO. Fountains recorded higher incompatible elements (e.g., K_2O, TiO_2, and Zr) at equivalent or higher MgO contents than HMR magmas erupted from 2020 to 2023 (Fig. 6). Although MgO fluctuated throughout the eruption, K_2O, TiO_2, and Zr continued to increase over time, suggesting that the end-member composition of this new magma is still in the process of infiltrating the HMR.

Previous models for episodic fountaining

Fountaining has been explained by two end-member models that differ in where and what gas species exsolves to drive the eruption [i.e., H_2O-rich conduit and CO_2-rich reservoir; (*34*)]. In the first model, reservoir overpressure initially drives ascent of bubble-free magma in the conduit until the pressure-depth of H_2O saturation is reached, after which ascent is controlled by vigorous volatile exsolution. Once the gas-charged magma begins to fragment, the resulting pressure decrease and reduced friction produce jet-like acceleration out of the vent and fountaining (*9*, *34*). In the second model, deeply exsolved CO_2 accumulates as a foam layer or pockets at the roof of a magma reservoir; once a critical thickness is reached, collapse of the foam drives fountaining until the trapped gas is exhausted (*35*, *36*). Intermediate and mixed-gas models can produce transitions between effusive activity, strombolian activity, and fountaining, depending on magma rise rate, gas flux, and shallow system geometry (*34*, *37*–*40*).

Episodicity at both Kīlauea Iki Crater [1959; 17 episodes over 36 days (*4*, *41*)] and Geldingadalir, Iceland [2021; 7058 episodes with ~3- to 17-min pauses (*42*)] has been attributed to cyclic formation of a viscous, low-permeability plug at the top of a shallow conduit, produced by drainback of degassed lava that overlies continuously replenished, gas-rich magma. For both Maunaulu [1969; 12 episodes over 7 months (*5*)] and Pu‘u‘ō‘ō [1983–1986; 47 episodes (*6*, *43*)], proposed models invoke CO_2-rich episodicity when gas flux reached a viscosity-dependent critical threshold for foam collapse (*36*). Similar models have been proposed for fountaining at Mount Etna (Italy), which has produced more than 220 episodes over the past three decades (*2*, *35*, *44*–*46*). At Kīlauea, summit gas-emission data (*47*) and reservoir-scale constraints (*34*, *47*–*50*) are difficult to reconcile with deep CO_2 accumulation and instead may favor the formation of H_2O-rich bubble clusters at shallower depth (*34*, *39*, *40*, *51*). Mixed H_2O–CO_2 or CO_2-rich degassing could be important if parental magma bypasses summit reservoirs or transits them rapidly, as proposed for the 1959 Kīlauea Iki Crater eruption (*34*).

Open questions

Throughout the eruption, fountaining began approximately when interepisode summit reinflation recovered to tilt thresholds that proved useful for forecasting (Fig. 4). Although this implied repeatable conditions necessary to initiate fountaining (*52*), whether these conditions depend on HMR pressure alone is unclear. What were the physical processes that control these conditions? Did they reflect the decoupling of accumulated gas, evolving connectivity within the shallow reservoir-conduit systems, or the opening of a mechanically resistive segment of the conduits (*37*)? Did episodicity reflect transient reductions in conduit permeabilities, potentially from the temporary accumulation of degassed, more viscous magma that suppressed fountaining without fully sealing the conduits (*41*, *42*)?

Emission rates of SO_2 during interepisode pauses (1500 to 4000 tons/day) were comparable to those during Kīlauea's 2008 to 2018 Halema‘uma‘u lava lake period (*53*), suggesting open-system degassing even as HMR pressure rebuilt toward the next episode. Measurements of interepisode CO_2/SO_2 ratios were low (~0.2 to 0.3), which is consistent with magma equilibration in the HMR (*54*), without formation of a CO_2-rich foam. However, key uncertainties remain, including when and where exsolved volatiles segregated before eruption, their relative contributions to fountaining, and whether the gas budget to sustain fountaining could be supplied by conduit-level exsolution and growth of bubbles. Constraining gas compositions remains challenging because Fourier transform infrared spectroscopy (FTIR) measurements were hindered by complex acquisition geometries and interference from persistent summit CO_2 sources (*55*). Nevertheless, preliminary FTIR observations suggest that, at least during the later stages of each episode, the gas phase was H_2O-dominated.

Each of Kīlauea's previous episodic fountaining eruptions ended differently. At Kīlauea Iki Crater, fountaining ended abruptly after 36 days and was followed less than a month later by an intrusion 40 km downrift into the lower East Rift Zone and an eruption that destroyed the village of Kapoho (*4*). For both Maunaulu and Pu‘u‘ō‘ō, fountaining transitioned into years to decades, respectively, of nearly continuous lava effusion (*5*, *6*, *43*). Whereas past eruptions provide useful points of comparison, the outcome of the current eruption remains unknown. The ongoing eruption underscores Kīlauea's fundamental role as a globally accessible natural laboratory for developing new techniques in volcano monitoring and eruption forecasting.

Materials and methods

InSAR processing

InSAR observations were used from three satellite missions: the TerraSAR-X and COSMO-SkyMed satellites all operating in the X-band range (3.1-cm wavelength) and the Sentinel-1 satellites operating in the C-band range (5.6-cm wavelength). The InSAR processing was performed using the USGS Volcano Science Center processing system and based on GAMMA Software (*56*). Interferograms are geocoded to the WGS84 UTM (Zone 5N) coordinate system with a pixel spacing of 4.55 m for CSK and TSX and of 30 m for S1. Surface displacements are measured in the line-of-sight (LOS) direction between the satellite and the ground.

Physical sampling and geochemical analysis

The geochemical dataset used in this study comprises tephra and spatter samples collected from eruptive episodes 1 to 39 by HVO field teams at locations in and around Halema‘uma‘u. Whole-rock major- and trace-element compositions were measured by energy-dispersive and wavelength-dispersive XRF, with a subset analyzed by quadrupole inductively coupled plasma mass spectrometry; samples were cleaned as appropriate, powdered, and prepared as pressed pellets, fused beads, or acid digestions. Glass compositions were measured on polished grain mounts by electron probe microanalysis. Analyses used BHVO-2, AGV-2, BCR-2, GSP-2, and appropriate mineral and glass reference materials, and included replicate analyses and data-quality screening. Full sample metadata and detailed descriptions of sample preparation, instrumentation, analytical conditions, reference materials, and analytical precision and accuracy are provided in the associated USGS data release (*32*).

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SUPPLEMENTARY MATERIALS

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Supplementary Text; Figs. S1 to S14; Tables S1 to S4; Movies S1 to S7

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RESEARCH ARTICLE SUMMARY

DOG EPIGENETICS

Epigenetic aging and transposon dysregulation reflect size-related lifespan compression in dogs

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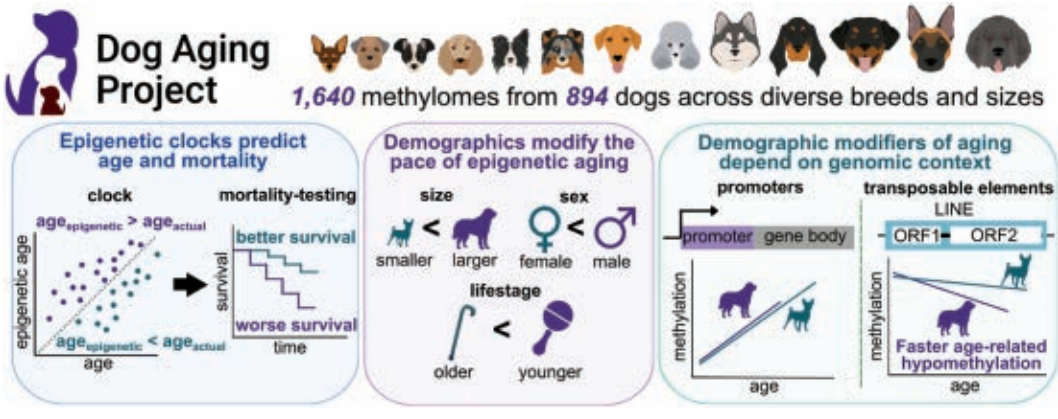
INTRODUCTION: Aging is a universal biological process, yet individuals within a species can differ dramatically in lifespan and disease risk. Whether shorter-lived individuals simply die earlier or instead experience accelerated biological aging throughout life remains unclear. Resolving this question is central to understanding the mechanisms that drive variation in healthspan and lifespan. Domestic dogs offer a particularly advantageous model to ask and answer these fundamental questions, as dogs exhibit extraordinary phenotypic variation, not only in size, shape, and behavior but also with respect to lifespan and risk of specific diseases. In this study, we used dogs as a model to test the association between life expectancy and biological aging. We found that size and sex influence the aging methylome and identified the epigenetic architecture that mediates these demographic aging differences.

RATIONALE: Comparative studies across species have identified mechanisms associated with exceptional longevity, but they are less well suited to identifying the molecular drivers of aging variation among individuals within a species. Domestic dogs provide a model for addressing this challenge because body size strongly predicts lifespan, with larger breeds living substantially shorter lives than smaller breeds despite sharing many physiological and environmental conditions. To directly estimate biological aging rather than relying on lifespan end points, we used DNA methylation (DNAm), an established molecular biomarker of age, to test how demographic factors associated with lifespan shape molecular aging within a species.

RESULTS: We generated 1640 methylomes from 894 companion dogs enrolled in the Dog Aging Project and paired these data with detailed demographic and genetic information. Using these data, we developed highly accurate DNAm age predictors [coefficient of determination (R^2) = 0.86, median average error = 0.82 years]. We found that dogs with epigenetic ages older than expected exhibited elevated mortality risk, demonstrating that DNAm captures biologically meaningful variation in aging. Larger dogs, males, and younger individuals showed molecular profiles consistent with accelerated epigenetic aging. Age-associated differential methylation was highly structured by genomic context. Promoters generally gained methylation with age, whereas introns and transposable elements (TEs) predominantly lost methylation with age. Sex differences in age effects were most pronounced on the X chromosome, whereas size-associated differences were greatest in TEs: Larger dogs had 31% stronger age-associated TE demethylation than smaller dogs. This amplification was widespread across TE classes and evolutionary lineages, including long interspersed nuclear elements (LINEs), short interspersed nuclear elements, LINE2s, and DNA transposons, but was most pronounced in evolutionarily young LINE1s.

CONCLUSION: Our findings broadly establish companion dogs as a translational model for uncovering molecular mechanisms of lifespan variation, epigenomic instability, and age-related disease. We demonstrate that shorter-lived dogs do not merely die earlier; they undergo accelerated molecular aging specific to genomic locations and structures. Thus, lifespan variation within a species is reflected not only in the pace of epigenetic aging but also in the genomic architecture through which aging unfolds. □

Corresponding author: Noah Snyder-Mackler (nsnyderm@asu.edu) †These authors contributed equally to this work. **Cite this article as B. L. Mariner *et al.*, *Science* 394, eaeb2986 (2026). DOI: 10.1126/science.aeb2986**



Domestic dogs provide a powerful natural experiment for understanding how intrinsic rates of biological aging shape lifespan variation within a species. Analysis of 1640 DNA methylomes from 894 companion dogs in the Dog Aging Project, spanning diverse breeds and body sizes, reveals accelerated epigenetic aging in larger, shorter-lived dogs and distinct genomic aging signatures, including greater promoter hypermethylation and TE demethylation. These findings demonstrate that lifespan variation is reflected in both the pace and genomic architecture of biological aging. ORF, open reading frame. [Dog illustrations: ©dmytrobosnak, https://www.canva.com]

DOG EPIGENETICS

Epigenetic aging and transposon dysregulation reflect size-related lifespan compression in dogs

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The extraordinary lifespan variation in domestic dogs provides a natural experiment for testing how intrinsic rates of biological aging shape lifespan. Using 1640 methylomes from 894 dogs, we developed an epigenetic clock that predicted mortality and demonstrated that epigenetic aging is fastest early in life. At orthologs of human age-associated genes, dogs exhibited concordant age effects on promoter methylation, highlighting conserved remodeling of immune pathways. We found that larger and male dogs, which are shorter lived, exhibit accelerated molecular aging. Distinct epigenetic architectures mediated these effects: Sex-dependent methylation changes were concentrated on the X chromosome, whereas size-associated methylation was especially pronounced at transposable elements (TEs). These findings show that epigenetics reflects lifespan differences in dogs and identifies TEs as potential mediators of size-associated lifespan.

Aging is a universal biological process, yet its timing, progression, and underlying drivers vary widely both within and across species (*1*, *2*). Although it is well established that aging is associated with physiological decline and increased disease risk, the molecular mechanisms underlying these changes remain poorly understood. Comparative studies across species have begun to fill this gap, revealing mechanisms of exceptional longevity in species such as naked mole rats and bowhead whales (*3*, *4*). These discoveries have not only advanced our understanding of the evolutionary biology of aging but also inspired translational strategies, such as senolytics and caloric restriction mimetics (*5*, *6*).

However, cross-species comparisons obscure the mechanisms that drive aging variation among individuals within a species. The domestic dog (*Canis familiaris*) provides a powerful system in which to address this gap (*7*, *8*). Dogs provide a within-species model of lifespan scaling that isolates systemic drivers of aging while holding evolutionary background largely constant. For example, breed size is a strong and continuous predictor of lifespan, with smaller breeds living up to twice as long as larger ones (*2*, *9*, *10*). This size-longevity tradeoff is associated with differential disease risk, including increased incidence of cancer and musculoskeletal disease in large breeds and increased incidence of cardiac disease in small breeds (*10*). However, the molecular basis of these patterns remains unclear (*11*). Unlike laboratory models, where lifespan variation is often induced through genetic or environmental perturbations, dogs exhibit differences in lifespan under shared physiological constraints and much shorter timescales than in humans. Dogs also live in human environments, consume commercial diets, and receive routine medical care, making them a relevant translational model for studying how genetic background and environmental exposures interact to shape aging (*8*).

To move beyond lifespan comparisons and resolve how demographic and environmental factors shape aging trajectories, we need sensitive molecular measures of biological age, the intrinsic physiological state of an organism that is a better predictor than chronological age of age-associated morbidity and mortality (*12*). DNA methylation (DNAm) has emerged as a robust molecular biomarker of age, with "epigenetic clocks" offering a quantitative framework for identifying factors that accelerate or decelerate biological age (*13*–*16*). It is well established that size and sex influence lifespan in dogs, but it remains unclear whether shorter-lived dogs experience accelerated epigenetic aging (*17*).

In this work, we generated 1640 methylomes from 894 dogs paired with detailed genetic and demographic data to dissect variation in biological age within a species (Fig. 1A). We tested how demographic factors that strongly predict lifespan shape molecular age trajectories and identified shared and distinct epigenetic signatures underlying these differences. We show that larger dogs exhibit more pronounced age-related epigenetic changes than smaller dogs, reflected not only in earlier epigenetically predicted mortality risk, but also in pronounced age-related remodeling of immune gene promoters and transposable elements (TEs). Larger dogs show greater methylation loss at TEs with age, consistent with increased immune remodeling and genomic instability. Collectively, our results reveal that demographic variation in lifespan corresponds to systematic differences in the rate and molecular features of age, linking size- and sex-associated mortality risk to distinct epigenetic age trajectories within a single species.

Results

Canine epigenetic clock predicts age and mortality.

We generated DNAm profiles from 1640 peripheral blood mononuclear cell (PBMC)–enriched blood samples collected from 894 dogs participating in the Precision Cohort of the Dog Aging Project (fig. S1) (*7*, *8*). This cohort is demographically, genetically (53% single breed, 47% mixed breed), and geographically diverse (Fig. 1, B and C), and we sampled annually for up to 4 years per dog (Fig. 1D) (*7*). Given published estimates of average canine lifespan of 12.7 years, which vary by size, sex, and reproductive status, including spay or neuter status (*7*, *18*), our dataset captured up to ~25% of the expected average adult lifespan.

We first used these DNAm profiles (fig. S2, A to C) to train an epigenetic clock to predict the chronological age of dogs with known birthdays using only the first sample collected ($n = 809$ dogs; see Materials and methods) (Fig. 2A). Our leave-one-out cross-validation (LOOCV) model predicted chronological age with exceptional accuracy (coefficient of determination, $R^2 = 0.86$; median average error, MAE = 0.82 years), demonstrating that the clock performs well across dogs. Details on model optimization, including parameter selection, feature stability, the effects of age transformations, and performance comparisons, are

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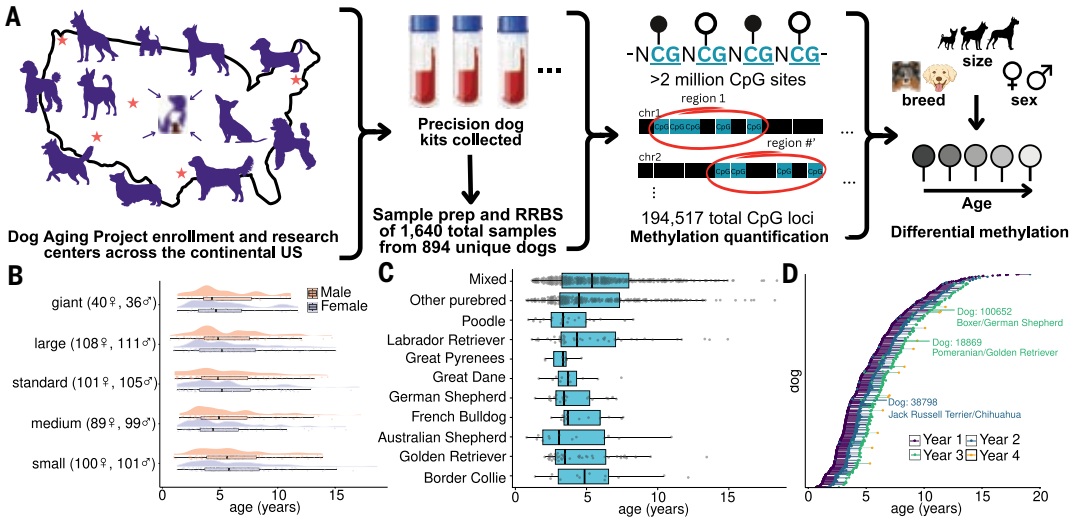

Fig. 1. Sample collection and cohort characteristics of the Dog Aging Project Precision Cohort. (**A**) Dogs enrolled in the Precision Cohort are geographically distributed across the United States. Sample collection kits are sent to dog owners and their veterinarians, who collect whole blood for processing. Blood samples are then enriched for PBMCs and subjected to RRBS to quantify DNAm at the regional level (Materials and methods). Stars indicate many of the Dog Aging Project institutions vital for the data collection, processing, and analysis in the present study. (**B** to **D**) The cohort includes dogs with a wide range of body sizes (categorically classified here in accordance with the American Kennel Club), sexes, and breeds, enabling investigation across diverse demographic strata. Longitudinal sampling is ongoing, with multiple timepoints collected for many individuals to capture molecular changes associated with aging.

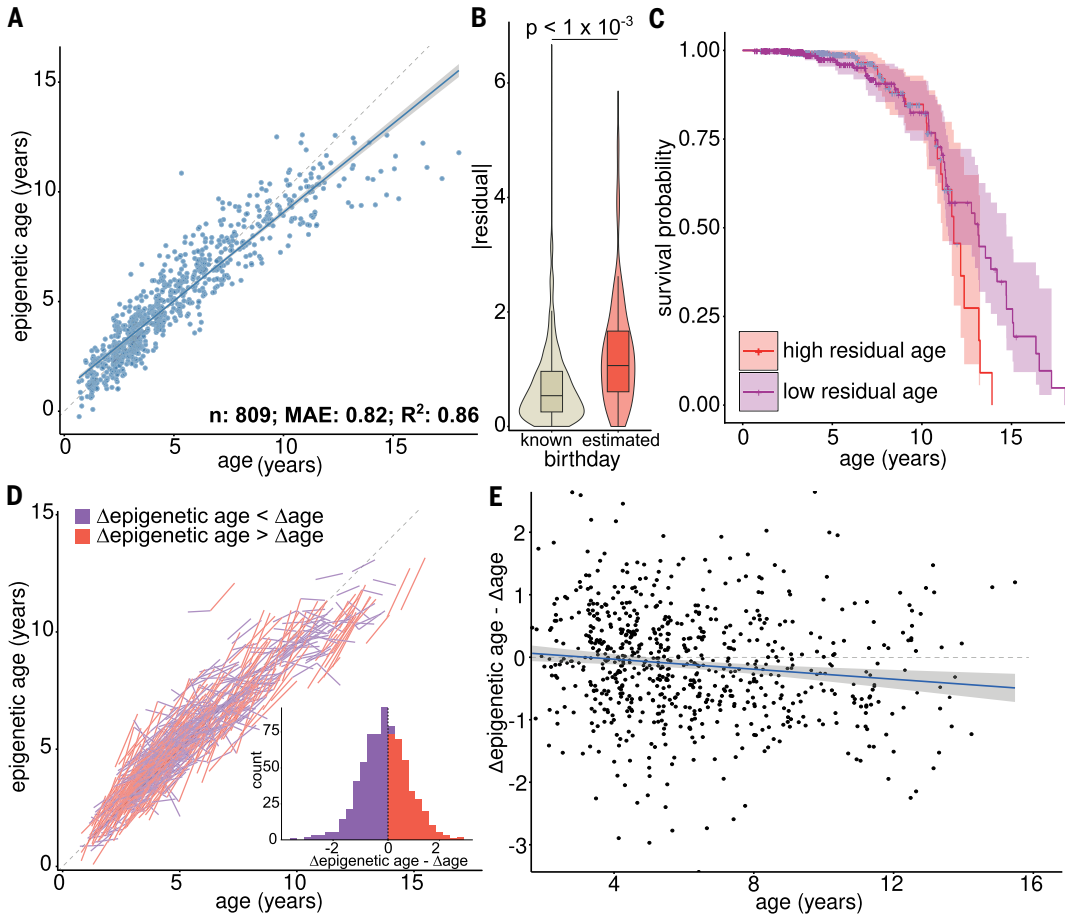

Fig. 2. Epigenetic clock predicts age and mortality in companion dogs. (**A**) Epigenetic clock predicts dog age to within a year [median average error (MAE) = 0.78]. (**B**) Age prediction error, as measured as the absolute value of the difference between predicted epigenetic age and reported chronological age, |residual|, is lowest for dogs with known birth dates, which likely reflects imprecise owner-reported age estimates. (**C** and **D**) Longitudinal samples reveal that epigenetic aging is faster earlier in life.

provided in supplementary note 1 (fig. S3). The model also produced accurate age estimates for dogs with owner-reported birth dates. However, the prediction error for these dogs with owner-reported birth dates was higher than for dogs with verified birth dates (Fig. 2B), likely owing to inaccuracies in the estimated ages. This not only highlights the accuracy of our age predictor but also demonstrates a potential veterinary application of our clock, enabling more accurate estimation of dog age when the birth date is unknown.

To investigate whether demographic factors modify epigenetic age, we modeled predicted epigenetic age as a function of sex, heterozygosity, and a genetically predicted continuous measure of adult size (see Materials and methods) while controlling for chronological age as a covariate [as recommended by (*19*)]. We identified significant age × adult size and age × sex interactions in epigenetic age ($\beta_{age:adult\ size} = 0.080$, $P = 1.2 \times 10^{-4}$; $\beta_{age:sex\ Male} = 0.050$, $P = 2.8 \times 10^{-2}$; fig. S4, A to C). These results support a model in which lifespan differences among dogs reflect compression of the epigenetic age trajectory in larger and male dogs.

Epigenetic age is thought to be a measure of an individual's biological age, which can differ from their chronological age (*15*). This means that the difference between epigenetic age and chronological age should predict health and survival [as has been shown in humans and other animals (*20*–*24*)]. Thus, we tested whether residual epigenetic age predicted survival in a Cox proportional hazards model. Because epigenetic age, and biological age, more generally, is not a fixed trait and can change across the lifespan, we modeled mortality for up to 2 years after each epigenetic age measurement, which included 121 deaths across the 894 dogs (alternative follow-up windows, evaluated at 1-month intervals through 3 years after sampling, yielded consistent results; supplementary note 2 and fig. S5). We included adult size and sex as covariates, both of which significantly predicted mortality hazard [$HR_{adult\ size} = 2.40$, 95% confidence interval (CI) = 1.67 to 3.44, $P = 2.0 \times 10^{-6}$; $HR_{male} = 1.95$, 95% CI = 1.32 to 2.89, $P = 9.8 \times 10^{-4}$; HR, hazard ratio]. Controlling for size and sex, residual epigenetic age significantly predicted mortality (HR = 1.15, 95% CI = 1.01 to 1.31, $P = 0.037$), such that each 1-year increase in residual epigenetic age was associated with a 15% increase in mortality hazard.

Cross-sectional aging analyses are susceptible to survival bias (*25*). In this study, older individuals disproportionately represent dogs healthy enough to survive and be sampled. To correct for this bias, we leveraged our longitudinal dataset to estimate within-individual epigenetic aging trajectories. We analyzed 764 follow-up samples from 547 dogs (353 dogs with one sample, 339 with two, 199 with three, and 9 with four samples; Fig. 2C). On average, dogs showed a 0.88-year increase in epigenetic age for every chronological year of follow-up. Using these longitudinal data, we next tested whether variation in within-individual aging trajectories could be explained by sex, heterozygosity, adult size, and age (defined as the midpoint of the sampling interval). Age was significantly associated with the rate of epigenetic aging ($\beta = -0.038$, $P = 1.5 \times 10^{-3}$) such that younger dogs had a faster rate of epigenetic aging compared with that of older dogs (Δepigenetic age − Δage ~ age + sex + adult size) (Fig. 2D) (*26*). We observed the same significant and negative association between chronological age and the relative change in epigenetic aging in age-transformed clocks ($P_{\sqrt{(age)}} = 4.7 \times 10^{-3}$; $P_{log(age)} = 1.5 \times 10^{-4}$), demonstrating the robustness of our finding that older dogs show less year-over-year change in epigenetic age.

Epigenetic age in dogs involves genome-wide shifts in methylation at TEs and promoters

Although our epigenetic clock serves as a reliable biomarker for chronological age and predicts mortality, it does not provide insight into the molecular mechanisms underlying variable aging in dogs. To better understand how methylation patterns accumulate over time, we quantified age effects on DNAm across the genome (194,517 loci, spanning 2.3 million CpG sites).

As expected, age was the primary driver of DNAm variation across all loci tested. On average, age explained 16.0% of the total variance in DNAm, substantially more than other biological factors, including sex (5.2%), adult size (4.9%), and heterozygosity (0.003%) (fig. S6A). Overall, 42% (82,447 loci covering 921,156 CpG sites) of loci were significantly differentially methylated with age [false discovery rate (FDR) < 0.05]. Consistent with blood-based DNAm studies in other species, the majority of these age-associated loci (64,101/82,447, 78%) were hypomethylated in older dogs, reflecting a global loss of methylation with age (*27*–*29*). Nonlinear transformations did not substantially differ from the linear age effects (fig. S6B). This suggests that age-related methylation patterns are, on the whole, robust and reflected accurately in linear models of age, which we focus on for all subsequent analyses.

Age-related changes were not randomly distributed throughout the genome and instead showed strong context dependence (Fig. 3A). We found greater hypermethylation with age in promoters [odds ratio (OR) = 2.30, FDR < 2.2×10^{-16}], consistent with studies in humans (*30*), exons (OR = 2.14, FDR < 2.2×10^{-16}), CpG islands (which are CpG-rich loci in the genome; OR = 3.55, FDR < 2.2×10^{-16}), and CpG shores (which flank CpG islands; OR = 1.28, FDR < 2.2×10^{-16}) as well as greater hypomethylation with age in CpG shelves (which reside next to CpG shores; OR = 1.19, FDR < 2.2×10^{-16}), introns (OR = 1.13, FDR < 2.2×10^{-16}), and TEs (OR = 1.40, FDR < 2.2×10^{-16}). These directional age effects are constrained by baseline methylation states, which differ across genomic contexts (fig. S7A), such that loci that begin hypomethylated predominantly gain methylation with age, whereas highly methylated loci tend to lose methylation with age (fig. S7B and supplementary note 3).

To quantify the translational relevance of these patterns, we compared our results to genes differentially expressed with age in humans [GenAge Human Genes; (*31*)]. We found that promoters hypermethylated with age in dogs were significantly enriched for genes that are down-regulated with age in humans (OR = 1.68, 95% CI: 2.63 to 1.08). Conversely, hypomethylated promoters in dogs were enriched for age–up-regulated genes in humans (OR = 1.79, 95% CI: 2.73 to 1.18; Fig. 3B), consistent with the regulatory link between promoter DNAm and gene expression. To assess the molecular processes linked to age effects on promoter methylation, we tested for enrichment of gene sets in age-associated promoters (*32*), focusing on immune-related gene sets given our tissue (while blood cells) and the central role of the immune system in aging (*33*). We focused on biological process terms under immune response [Gene Ontology ID (GO):0006955] and immune system

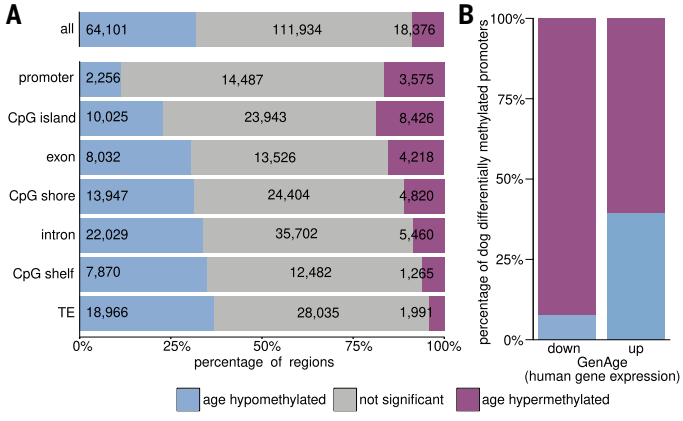

Fig. 3. Age effects on DNAm vary across genomic context and shape immune regulatory pathways in dogs. (**A**) Loci differentially methylated with age are lost with age more rapidly in larger dogs. Blue, hypomethylated with age (FDR < 0.05); gray, no significant association with age (FDR > 0.05); purple, hypermethylated with age (FDR < 0.05). (**B**) Promoters hypermethylated with age in dogs were significantly enriched for genes that are down-regulated with age in humans, and conversely, hypomethylated promoters were enriched for human age–up-regulated genes (*31*).

process (GO:0002376). Our enrichment analysis revealed changes consistent with the remodeling of circulating immune function in aging dogs (*34*, *35*). Age-hypermethylated promoters were enriched for pathways regulating adaptive immunity, including B cell differentiation (GO:0045578), T cell activation and effector function (GO:0046639, GO:0002841), and immune defense processes (GO:0043311), consistent with age-related suppression of adaptive immune function observed in humans (table S1) (*36*–*38*). By contrast, age-hypomethylated promoters were enriched for pathways associated with inflammation, innate immune activation (GO:0002415, GO:0002523), and features of dysregulated adaptive immunity (GO:2000319, GO:1903238, GO:0006957, GO:0043379, GO:0001770). These signatures mirror the heightened innate activation and inflammatory skewing ["inflammaging" (*39*)] characteristic of aging (table S1) (*40*, *41*). These results show conserved epigenetic signatures of age-related immune remodeling in dogs and humans, highlighting the translational relevance of companion dogs for understanding human aging.

Adult size but not sex mimics age in the dog methylome

Dog demographics are closely tied to lifespan: Single breeds, males, and larger dogs generally have shorter lives than mixed breeds, females, and smaller dogs, respectively (*1*, *2*, *9*, *11*). These consistent lifespan differences suggest that certain demographic groups may also differ in their molecular aging processes.

To investigate how these demographic factors shape epigenetic age, we first needed to examine their effects on locus-specific DNAm independent of age. We found that adult size and sex, but not heterozygosity, significantly predicted DNAm across the genome when controlling for age. Although adult size was associated with DNAm at 447 loci (0.23% of tested), sex had a far larger impact, with 8476 (4.4% of tested) sex-associated loci (supplementary note 4). The majority of these sex-associated loci (53%) were located on the X chromosome (OR = 86.3; 95% CI: 81.4 to 91.3), likely reflecting X-inactivation dynamics (*42*).

Having identified sex- and adult size–associated loci, we next asked whether these effects overlap with age effects. To do so, we tested whether loci associated with sex or adult size in the additive models were also more likely to be age associated and whether their direction of effect aligned with known lifespan differences in sex and/or size. This framework allowed us to evaluate whether demographic factors independently influence the aging methylome.

We found that loci associated with both age and sex were overrepresented on the X chromosome compared with on the autosomes (Fisher's exact test, OR = 40.8, $P < 2.2 \times 10^{-16}$). On autosomes, sex-associated sites were significantly depleted for age-associated differential methylation (OR = 0.024, $P < 2.2 \times 10^{-16}$). Within the X chromosome, age-associated differential methylation was specifically enriched at TEs, where 53.0% of TE loci were age associated (1628/3071; OR = 1.26, 95% CI = 1.14 to 1.38, $P = 1.9 \times 10^{-6}$) and 26.4% were depleted at promoters (152/575; OR = 0.33, 95% CI = 0.27 to 0.41, $P < 2.2 \times 10^{-16}$). Other tested annotations on the X chromosome, including CpG islands, CpG shores, exons, and introns, did not have a significant age association. These results indicate that X-linked age-associated methylation changes are preferentially concentrated at TEs rather than broadly distributed across regulatory or genic annotations.

In contrast with the modest overlap of age and sex effects, adult size–associated loci showed 2.9-fold enrichment for age associations (OR = 2.9, $P < 2.2 \times 10^{-16}$). Among these loci associated with both adult size and age, 81% (247/304) showed concordant directionality (increasing size affects loci in the same direction as increasing age) (OR = 4.0, $P < 2.2 \times 10^{-16}$). This pattern was even more pronounced in promoters: Of the 38 promoters associated with both adult size and age, 97% (37/38) showed increased methylation in larger and older dogs, consistent with transcriptional silencing and with larger dogs exhibiting DNAm patterns typically seen at older ages (table S2). Several of these promoters are linked to genes associated with Insulin-like Growth Factor 1 (*Igf1*), which has been shown to largely stratify with adult size in dogs (*43*) and which impacts longevity across many species (*44*–*48*). For example, among these loci is the promoter for Forkhead box protein E1 (*Foxe1*), which enhances *Igf1*'s DNA-binding and transcriptional activity (*49*), and the promoter for the transcription factor *Gata4*, which has expression associated with *Igf1* and activates in response to IGF1 (*50*, *51*). Even though many of the promoters differentially methylated with age, many are promoters for genes that are not necessarily expressed in blood; this patterning could reflect systemic changes in DNAm across multiple cell types in the body, including those where these genes are expressed (*52*).

Distinct size- and sex-dependent age effects in the dog methylome

Adult size and sex were associated with DNA methylation and, in the case of adult size, overlapped substantially with age-associated loci. We next tested whether sex or adult size modifies the trajectory of epigenetic age. To do so, we focused on loci that were significantly associated with age (FDR < 0.05) and then compared the magnitude of age effects between demographic subgroups by estimating separate age coefficients for males and females and for larger versus smaller dogs (Materials and methods). This approach allowed us to ask whether certain groups with shorter expected lifespans exhibited steeper age-associated molecular trajectories.

We found that sex differences in age effects on DNAm were most pronounced on the X chromosome. Age-associated X-linked loci showed 28.1% stronger age effects in males than in females (Wilcoxon $p < 2.2 \times 10^{-16}$). In contrast, age-associated loci on the autosomes exhibited 5.7% stronger effects in females (Wilcoxon $p < 2.2 \times 10^{-16}$). When we compared the age effects by size, we found pronounced differences between larger versus smaller dogs (defined by a median split on predicted adult height; Methods Fig. 4A). On average, age effects were 27% stronger in larger dogs compared to smaller dogs (Wilcoxon $p < 2.2 \times 10^{-16}$). These differences were most pronounced in TEs, where age effects were 31.3% stronger in larger dogs (FDR $< 2.2 \times 10^{-16}$; Fig. 4B and fig. S8). Promoters (12.4%), exons (21.9%), introns (27.2%), CpG islands (16.1%), CpG shelves (30.8%), and CpG shores (24.6%) all showed stronger age effects in larger dogs compared to smaller dogs. This difference was also evident when we modeled the effect of adult size on the average

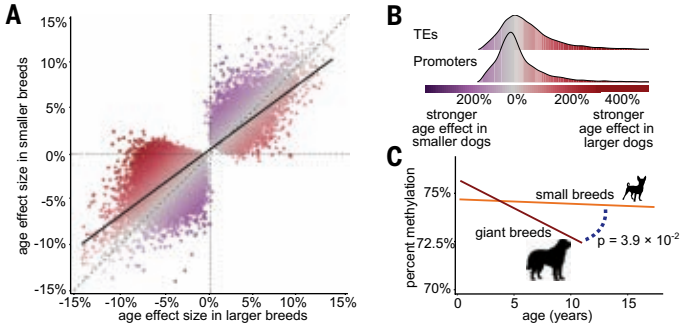

Fig. 4. Larger dogs show more age-related hypomethylation at TEs than smaller dogs. (**A**) Nested effect sizes of age-associated loci in larger versus smaller dogs and the difference between them (materials and methods). The black line is the regression line estimate ($\beta_{age:smaller} \sim \beta_{age:larger}$), and the gray dotted line represents x = y. (**B**) The distribution of the difference (percentage) in age effect sizes between larger and smaller dogs ($|\beta_{age:larger}|/|\beta_{age:smaller}| - 1$) of promoters and TEs (also see fig. S8). (**C**) Mean age-associated TE methylation varies with adult size. Regression lines for visualizing the methylation of age-associated TEs in small and giant dogs [American Kennel Club classifications (*105*)] are in orange and red, respectively. TEs with FDR < 0.05 from PQLSeq age modeling were used for this analysis, and the *P* value is from a linear model: age-associated TE mean methylation ~ age × adult size + sex among all dogs, where adult size is the continuous value as described in the Materials and methods.

age-associated TE methylation across all dogs ($\beta_{age:adult\ size}$= -1.4 × 10^{-3}, p = 3.9 × 10^{-2}, R^2 = 0.067) (Fig. 4C).

We next asked whether specific TE classes and families showed stronger age-associated methylation changes in larger dogs. Across the major TE classes we tested, age effects were consistently steeper in larger dogs, including long interspersed nuclear elements-1 (LINE1s) (35.0%; Wilcoxon FDR < 2.2 × 10^{-16}), LINE2s (32.2%; FDR = 4.2 × 10^{-6}), short interspersed nuclear elements (30.7% stronger; FDR < 2.2 × 10^{-16}), and DNA transposons (29.1%; FDR = 7.1 ×10^{-5}). TE families reflect distinct evolutionary histories, and evidence suggests that the activity of more recent insertions more strongly impacts aging (*20*). We therefore examined whether particular LINE1 families were especially enriched for these effects (*53*, *54*). Within LINE1s, the strongest difference between large and small dog age effects was observed in the Canid LINE1 family (43.5%; *n* = 146 Canid family LINE1-associated regions) (*55*–*58*). The Canid LINE1 family is composed of elements that expanded across the Canidae lineage. The next-strongest difference was observed in the *Canis* LINE1 family (37.4%; *n* = 1636 *Canis* family LINE1-associated regions), which expanded across the genus *Canis*. By contrast, older LINE1 families showed effect sizes closer to the overall LINE1 average, including L1 Carn (36.2%; *n* = 401 L1 Carn LINE1-associated regions), which expanded across a broader carnivore clade, and L1M (33.3%; *n* = 2540 L1M LINE1-associated regions), the oldest mammalian LINE1 family in the dataset. Together, these results suggest that the strongest age-associated demethylation among LINE1s occurs in evolutionarily younger LINE1s in larger dogs.

Discussion

We generated the largest genome-wide DNA methylation dataset in a companion animal species, encompassing a diverse, well-characterized cohort of dogs enrolled in the Dog Aging Project. By integrating demographic, genetic, and longitudinal data, we identified how age and intrinsic factors, such as adult size and sex, intersect to shape the aging canine methylome. These data address a major gap in comparative epigenomics by providing evidence that, within a species, demographic groups with shorter expected lifespans show faster epigenetic aging. We found that TEs have widespread age-associated demethylation, with the strongest size-nested age effects in evolutionarily young TE families, implicating TE dysregulation as a potential contributor to size-linked variation in dog aging. (*20*, *59*).

One implication of our findings is that lifespan compression in dogs does not appear to reflect a single, uniform acceleration of the aging methylome. Adult size and sex both track shorter expected lifespan, but through different molecular routes. Dog size generally mimicked older methylation patterns, whereas sex-dependent age effects were concentrated on the X chromosome, suggesting that size and sex compress canine aging through partially distinct epigenetic architectures. This distinction is only visible at the scale of this cohort, where size, sex, genotype, longitudinal sampling, and mortality can be modeled in the same individuals rather than inferred from breed averages or cross-species comparisons.

Our DNAm-based epigenetic clock predicted chronological age to within a year and captured individual differences in the pace of aging. Residuals from the clock predict mortality risk, consistent with the hypothesis that these residuals reflect accelerated or decelerated biological age. Although this mirrors similar observations in humans and other mammals (*20*–*24*), the magnitude of this association in dogs is larger than that reported in humans. First-generation human epigenetic clocks typically show a ~1.5 to 3% increase in mortality risk per year increase in residual epigenetic age (*60*), whereas we found a 15% increase per year in dogs. This 5- to 10-fold difference in effect sizes is likely due to the much shorter canine lifespan. We further found that older dogs have less year-over-year change in epigenetic age than younger dogs, suggesting that the pace of epigenetic aging may diverge across life course. These results align with observations across mammals while underscoring the distinctive value of dogs as a translational model for linking demographic variation, biological aging, and survival on an accelerated time scale (*61*, *62*).

We found strong genome-wide effects of age on DNAm, with three patterns in particular that highlight the evolutionary and translational relevance of the companion dog as a model of aging. First, epigenetic drift is context dependent: Loci that are typically unmethylated become methylated with age and vice versa, revealing a global trend toward intermediate methylation (fig. S7B), consistent with findings in mice and humans (*20*, *63*). Second, age-associated promoters are enriched for genes linked to immune function and genome stability, consistent with their known roles in aging and disease vulnerability (*29*, *64*–*66*). These age–differentially methylated promoters were also significantly associated with genes that are differentially expressed with age in humans (Fig. 3B) (*31*). Third, age-associated loci are strongly structured by genomic context: Regulatory regions, such as promoters and CpG islands, tend to gain methylation with age (*30*), whereas TEs tend to lose methylation (*20*). Within this broader age patterning, we found that the demethylation of TEs was especially pronounced in the evolutionarily younger LINE1s in larger, shorter-lived dogs, indicating that more recently expanded TE lineages are particularly sensitive to age-related epigenetic erosion.

TE demethylation is associated with increased TE transcription, which has been linked to cellular senescence, inflammation, genomic instability and transcriptional dysregulation in many species (*20*, *59*, *67*–*69*). TEs also have various roles across an individual's lifespan, contributing to development (*70*–*72*), immune regulation (*73*), and brain function (*74*) while also being linked to cancer (*75*–*83*) and aging (*20*, *67*, *68*, *84*). Although age-associated TE hypomethylation has been described in other systems (*20*, *29*), we show that this feature of age is not uniform but can scale with size. Size is highly predictive of dog health and lifespan (*43*, *85*), and the discovery that age-associated TE demethylation is particularly strong in large dogs points toward a TE mechanism of aging in dogs. Because we measure methylation rather than TE transcription or insertions, this model remains speculative. However, it makes a clear prediction: Methylation loss at evolutionarily young LINE1s should covary with or precede inflammation, cancer risk, morbidity, or mortality, and this would be most pronounced in larger dogs. Future longitudinal multiomic analyses can directly test whether the demethylation of TEs is merely a biomarker of size-compressed aging or a contributor to the disease trajectories that shorten lifespan in large breeds.

Prior work on the canine epigenome focused primarily on clock construction. Earlier work provided important age-associated insights using smaller cohorts (*86*), specific breeds (*18*), or breed-average lifespan (*23*). More recent comparative and wildlife studies have extended epigenetic clocks across mammals and into nonlaboratory species (*61*, *87*–*89*), demonstrating the broad utility of methylation-based age estimation. However, most clocks remain optimized for age prediction rather than for explaining why individuals within a species differ in lifespan. In this study, we leveraged the Dog Aging Project to move from clock construction to mechanistic inference. The model that emerged is more nuanced than a single "fast-aging" signature: Larger dogs show broadly age-concordant methylation changes and amplified TE demethylation, whereas sex-dependent aging effects are concentrated on the X chromosome. Thus, size and sex, two axes associated with shorter expected lifespan in dogs, appear to shape aging through partially distinct epigenetic architectures. This within-species design is what makes dogs especially powerful: Extreme variation in adult size can be modeled across a shared evolutionary background, allowing growth-associated genetics to be linked to age-related erosion of genome regulation. Together, these findings support a size-truncated aging model (*17*) in which selection for large adult body size may compress lifespan through molecular pathways that include but are not limited to TE dysregulation.

Materials and methods

Blood sample collection

We focused on whole blood samples collected annually from the Dog Aging Project's Precision cohort, starting in 2021 up until February 2024. Dogs were enrolled in the Precision cohort on a rolling basis, with veterinarians collecting samples annually based on each dog's enrollment date (fig. S1) (*7*). These blood samples were refrigerated and shipped overnight to the DAP team members at Texas A&M University, where they were enriched for PBMCs, and stored in Zymo DNA/RNA Shield (Zymo Research, Irvine, California, USA) at −80°C until processing. Details of the Precision cohort, collection, and processing can be found in (*7*).

RRBS preparation, sequencing, and data preprocessing

DNA was then extracted from PBMCs for reduced-representation bisulfite sequencing (RRBS) using the Quick-DNA/RNA Magbead kit (*90*). The Translational Genomics Research Institute in Phoenix, Arizona, sequenced the samples on the Illumina NovaSeq S4 platform in collaboration with members of the ASU DAP team.

We trimmed reads with Trim Galore! (v0.6) and aligned the reads to the NCBI UU_Cfam_GSD_1.0 (CanFam4) reference genome (*91*). To extract methylation and read count data, we used Bismark (v0.24) (*92*). We excluded samples with fewer than one million mapped reads and re-sequenced them when sufficient material was available. We excluded RRBS samples with low genotype concordance to the expected dog, consistent with potential sample swaps, leaving a final dataset containing 894 dogs.

All CpG sites (26,629,073 total) were first filtered to require 5x coverage in ≥33% of samples, while also excluding those that were either constitutively methylated or unmethylated (sites required 95% ≥ mean percent methylation ≥ 5%), resulting in 1,379,948 CpG sites. We expect neighboring CpG sites to have similar DNAm patterns, given their relative proximity in the genome (as shown by the literature), so we grouped these CpG sites into genomic loci using BSseq (*76, 77*) (fig. S2A). In our framework, we required CpG sites to be within 250 base pairs of each other to be grouped into the same region (fig. S2B). After grouping CpG sites into loci, we ensured they were not constitutively methylated or unmethylated by requiring mean locus methylation to be greater than 5% and less than 95% (fig. S2C). After both layers of this filtering, the mean per-locus methylation SD was 0.147 (median = 0.135), indicating moderate inter-individual variability at most sites. Samples had a mean methylation level of ~0.72 and an average per-sample SD of 0.28, indicating that each profile captures a wide dynamic range of methylation states. Along with these genome-wide loci, we also calculated methylation at a priori-defined promoters, defined as 2 kb upstream and 200 bp downstream of transcription start sites of annotated genes in the CanFam4 reference genome. In total, we included 194,517 loci for downstream analyses. Approximately eighty-six percent of all loci had ≥5x coverage across ≥75% of all samples. Moreover, ~94% of all samples had ≥5x coverage across ≥75% of all loci.

DNAm clocks

To build our epigenetic clock, we first calculated percent methylation for each sample at each locus and imputed missing values using the methyLImp2 R package (*93*). We used an elastic net regression to construct all clocks [glmnet package in R, (*94*)]. The LOOCV clock was built by using all but one of the samples to predict the age of the left-out sample. The model has two parameters, a mixing parameter alpha (α), and a regularization parameter lambda (λ). The LOOCV and longitudinal clocks presented in the main text were trained on standard chronological age with an αof 0.5. This α value, the midpoint between ridge (α = 0) and lasso (α = 1) regression types, was selected based on both empirical model performance and prior literature (*15*, *16*, *95*), although many α values were tested (fig. S3 and supplementary note 1).

Dog adult size

Dog adult size was predicted from genotypes using a random forest regression model that we have previously published (*96*). Using adult size as a proxy for dog size offers several advantages: (i) It reduces subjectivity in breed size classification by positioning dog size on a continuous scale, (ii) it accounts for pre-adult dogs by consistently reflecting predicted adult size throughout the dog's lifespan, allowing for the inclusion of dogs that are not fully grown, and (iii) it captures the inherent nonlinearity of breed size categories. For locus-specific age-size nested analyses, categorized dogs as "smaller" or "larger" based on whether they were smaller or larger than the median adult size of all 894 dogs in the dataset. Effectively, this gives us the ability to compare the differences and similarities of age effects at the locus-specific level.

Genome annotation

We obtained gene information from the CanFam4 gene annotation file (GCF_011100685.1). We defined promoters as regions 2kb upstream of transcriptional start sites, and when annotation was available, we identified introns. We downloaded the CpG islands and TE information (coordinates, classes, and family annotations) from the CanFam4 UCSC genome browser (*97*). We generated annotations for CpG shores and CpG shelves in R, where CpG shores are the 2kb regions flanking CpG Islands, and CpG shelves are the 2 kb regions flanking CpG shores (*98*). X chromosome gene status retrieved from (*99*).

Differential methylation modeling

PQLSeq, a binomial mixed modeling framework, was used to model differential methylation as a function of age, while controlling for sex, predicted adult height, genetic heterozygosity, and sequencing batch (additional models were fit using log or square root-transformed age, fig. S6B), and genetic relatedness across 894 dogs (*100*, *101*). Genetic relatedness matrices were generated from genome-wide single-nucleotide polymorphism genotypes using GCTA, which allowed us to account for both population structure and genetic relatedness. These matrices were incorporated as random effects in our binomial mixed-effect models (*102*, *103*). Nested models were fit using the same functional framework, where estimates and p-values were obtained for both demographic categories, respectively (e.g., males versus females, larger versus smaller dogs). For longitudinal analyses, to maximize statistical power, we leveraged the longitudinal design by comparing the oldest and youngest available samples for each dog, calculating the rate of methylation change at each locus as the difference in percent methylation divided by the difference in age (Δpercent methylation/Δage). We modeled locus-specific year-to-year change in methylation by age, sex, and adult size to assess if there were loci with detectable differences in year-to-year changes in dogs of different demographics (Δ percent methylation / Δ age ~ 1 + sex + adult size). Percent differences in effect size in larger versus smaller dogs was calculated by using the median age effect in larger dogs/(median age effect in smaller dogs + median age effect in larger dogs).

Functional enrichment

We performed GO enrichment using topGO (*32*) in R with age-associated promoters from the differential methylation modeling. To focus on immune-relevant biology, since this study used PBMCs, analyses were restricted to GO terms descending from immune response (GO:0006955) and immune system process (GO:0002376). Statistical significance was assessed using Fisher's exact test with topGO's weight algorithm to account for GO topology; the results are provided in Table S1.

Statistical programs and code

All bioinformatics analyses were conducted on the Sol Supercomputer at ASU (*104*) and with R (v4.2.2). We corrected for multiple hypothesis tests using the Benjamini-Hochberg FDR method. Code to construct

the clock and run PQLseq (*101*) is available on GitHub at https://github.com/smacklab/DAP_DNAm.

Human and animal subjects research approval

The University of Washington IRB deemed that recruitment of dog owners for the Dog Aging Project, and the administration and content of the DAP Health and Life Experience Survey, are human subjects research that qualify for Category 2 exempt status (IRB ID no. 5988, effective 10/30/2018). All study-related procedures involving privately owned dogs were approved by the Texas A&M University IACUC, under AUP 2021-0316 CAM (effective 12/14/2021).

REFERENCES AND. NOTES

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aeb2986
Figs. S1 to S8; Tables S1 to S2; Supplementary Notes 1 to 4; Dog Aging Project Consortium Authors; References (*107*,*108*); MDAR Reproducibility Checklist

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RESEARCH ARTICLE SUMMARY

BIOCATALYSIS

Asymmetric enzymatic hydrophosphorylation through O_2 activation

Yi Zhou, Yifei Ge, Wesley Harrison, Huimin Zhao*

Full article and list of author affiliations: https://doi.org/10.1126/science.aef3001

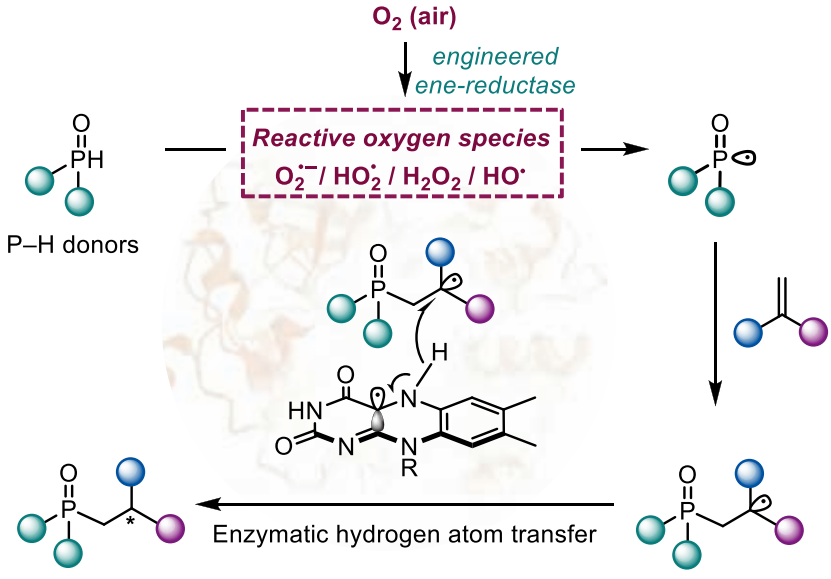
INTRODUCTION: Carbon–phosphorus (C–P) compounds are widely used in medicines, agriculture, and advanced materials. However, enzymatic C–P bond formation in nature is extremely rare. Chemical methods can construct these bonds through P-centered radicals but often require harsh oxidants or specially activated starting materials and provide limited selectivity. Enzymes could overcome these limitations by guiding highly reactive radical intermediates with precision. A biocatalytic method that directly incorporates common P compounds to alkenes would therefore provide a mild and selective route to value-added, structurally diverse products.

RATIONALE: Molecular oxygen (O_2) is an attractive, abundant oxidant, but its partial reduction generates reactive oxygen species (ROS) that usually damage enzymes and disrupt radical reactions. Flavin-dependent enzymes naturally manage O_2 activation within confined active sites, suggesting that these reactive species might be redirected toward new reactivity. We reasoned that an engineered flavin enzyme could harness ROS to generate P-centered radicals from phosphorus–hydrogen (P–H) donors and control product enantioselectivity through enzymatic H atom transfer. By combining condition optimization, directed evolution, and mechanistic studies, we sought to establish a mild biocatalytic route to diverse chiral organophosphorus compounds, unlocking new reactivity of enzymatic O_2 activation.

RESULTS: Through semirational and machine-learning–guided directed evolution, we developed variants of the flavin-dependent enzyme OYE1 to catalyze asymmetric enzymatic hydrophosphorylation through O_2 activation under both light and dark conditions. Experimental and computational studies revealed the role of the ROS generated in situ. Three pathways are proposed to provide P-centered radicals from P–H donors in the active site or in bulk solution. Computational results further revealed how enzymatic H atom transfer controls enantioselectivity and how the mutations affect the process. The evolved top three enzymes showed excellent activity and stereoselectivity across a broad range of alkenes and P–H substrates, including phosphites, phosphorous acid, and H-phosphinates Value-added organophosphorus compounds, such as the broad-spectrum herbicide glufosinate-ammonium, could be synthesized through this one-step biocatalytic transformation.

CONCLUSION: This project establishes a biocatalytic strategy for synthesizing C–P compounds from readily available alkenes and P–H compounds under mild conditions. By engineering a flavin-dependent enzyme, we redirected ROS from damaging side reactions toward productive P-centered radical formation and used enzymatic H atom transfer to control stereochemistry. The resulting system provides access to structurally diverse chiral organophosphorus compounds with high enantioselectivity. More broadly, this work shows that ROS can be used, rather than merely suppressed, to initiate enzymatic radical reactions and expand the synthetic capabilities of biocatalysis. □

*Corresponding author. Email: zhao5@illinois.edu Cite this article as Y. Zhou *et al.*, *Science* **394**, eaef3001 (2026). DOI: 10.1126/science.aef3001



Asymmetric enzymatic hydrophosphorylation enabled by ROS. Engineered flavin-dependent enzymes use molecular O_2 to generate reactive radical intermediates that form C–P bonds with high enantioselectivity.

BIOCATALYSIS

Asymmetric enzymatic hydrophosphorylation through O_2 activation

Yi Zhou[1,2,3], Yifei Ge[1,2,3,4], Wesley Harrison[1,2,3], Huimin Zhao[1,2,3,5,6]*

Enzymatic carbon-phosphorus bond formation is extremely rare in nature, limiting biocatalytic access to phosphorus-containing compounds that are widely used in pharmaceuticals and agrochemicals. Here, we report an asymmetric enzymatic hydrophosphorylation through oxygen activation using a repurposed flavin-dependent enzyme. Mechanistic studies revealed that reactive oxygen species are converted into productive phosphorus-centered radicals, followed by radical addition and enzymatic hydrogen atom transfer, achieving high enantioselectivity. The enzyme accommodates diverse phosphorus-hydrogen donors that pose challenges to chemical catalysis, enabling the biosynthesis of valuable phosphorus-containing scaffolds. This work expands the scope of biocatalysis to programmable carbon-phosphorus bond formation and establishes a paradigm for channeling oxygen reactivity in enzymes.

Phosphorus plays essential roles in biology, mediating energy transduction, signal regulation, and metabolic transformations. Among these, phosphonates containing C–P bonds have been widely applied in pharmaceuticals, agrochemicals, and materials science due to their unique physicochemical properties (*1*–*4*). However, naturally occurring enzymatic C–P bond formation is confined to highly specialized transformations, including the phosphoenolpyruvate mutase–catalyzed conversion of phosphoenolpyruvate to phosphonopyruvate (Fig. 1A), as well as the methylation reactions catalyzed by P-methyltransferases (*5*). As a result, no enzymatic strategy currently available enables direct incorporation of diverse P motifs into complex molecular scaffolds through C–P bond formation, thereby constraining structural diversity and hindering the biosynthesis of value-added phosphonates.

Chemical synthesis offers complementary solutions for broad C–P bond formation through P-centered radical transformations (Fig. 1B) (*6*–*9*). However, these methods typically require strong oxidants for common phosphites or rely on preactivated P precursors, thus limiting chemoselectivity and rarely achieving high enantioselectivity. These challenges stem from the narrow activation mode of common P precursors and the difficulty of controlling reactive radical intermediates.

Photoenzymatic catalysis has emerged as a powerful platform for enantioselective radical reactions (*10*–*12*). Among these photoenzymes, flavin-dependent enzyme catalysis has been used to achieve C–X bond formation through single-electron transfer between flavin and preactivated radical precursors (Fig. 1B) (*13*–*16*). However, extending photoenzymatic radical chemistry to C–P bond formation remains unexplored. Stable P-centered radical precursors are scarce, and existing photoenzymatic systems show poor tolerance to oxidative conditions, highlighting the need for a different biocatalytic strategy.

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Molecular oxygen (O_2) could be an attractive oxidant for enzymatic reactions because of its abundance and high reduction potential, although uncontrolled reduction of O_2 often generates toxic reactive oxygen species (ROS) that damage enzymes or quench radical reactions (*17*). Naturally occurring flavin-dependent monooxygenases and oxidases often activate O_2 within organized active sites through stepwise two-electron transfer from reduced flavin, enabling monooxygenation, hydroxylation, and halogenation, but have not been reported to generate non–O-centered radicals (*18*–*20*).

Inspired by C–P bond formation through O-centered radical chemistry (*6*–*9*), we envisioned that an engineered enzyme capable of regulating O_2 activation could enable C–P bond formation under biocompatible conditions. Here, we report the discovery and directed evolution (*21*, *22*) of a flavin-dependent ene-reductase capable of catalyzing C–P bond formation for a highly enantioselective hydrophosphorylation of alkenes through controlled reactive oxygen species (ROS) generation (Fig. 1C). This radical relay strategy (*23*) redirects ROS into productive P-centered radicals through H atom transfer (HAT), overcoming the inherent O_2 sensitivity of enzymatic radical reactions. The resulting system enables efficient activation of diverse P–H compounds with high enantioselectivity, providing access to valuable C–P compounds relevant to medicinal and agrochemical applications.

Discovery and directed evolution of O_2-activating enzymes

Based on our initial design shown in Fig. 1C, we first screened a number of wild-type, flavin-dependent ene-reductases using diethyl phosphite **1a** and α-methyl styrene **2a** as model substrates (table S1) under aerobic conditions. Fortunately, using an anaerobic chamber with a limited O_2 level and blue light, these enzymes produced detectable amounts of product ***3*** (1 to 2% yield), and old yellow enzyme 1 (OYE1) from *Saccharomyces pastorianus* exhibited the highest enantioselectivity [75% enantiomeric excess (ee)] (Fig. 2A). Optimization of various reaction parameters, including buffer, pH, cosolvent, and substrate equivalents (tables S2 to S4), increased the yield to 9% under a controlled O_2 level [15,000 parts per million (ppm) or by adding 200 μl of air], whereas only 1% yield was obtained under ambient air. In the absence of enzyme, light, regeneration system, or O_2, the desired product was not formed (entries 2 to 7; Fig. 2A). Other oxidants were also tested in the absence of air, including various peroxides and photocatalysts; H_2O_2 and *tert*-butyl hydroperoxide gave lower yield while maintaining selectivity, whereas other oxidants gave no product or reduced selectivity (tables S5 and S6). $Ru(bpy)_3Cl_2$ did not produce the product without O_2 but gave a slightly improved yield under aerobic conditions, probably due to its ability to activate O_2 (*24*) and its affinity with the enzyme (*25*).

Given that precise control of the O_2 level may be impractical and compromise reproducibility, we performed directed evolution (*21*–*23*) to improve the yield and enantioselectivity under air. Using a soft docking study (*26*) with the key radical intermediate **R1** in the active site of OYE1, 12–amino acid residues in the first sphere [within 5 Å around flavin mononucleotide (FMN)] were selected for site-directed mutagenesis (Fig. 2B). A library of 137 single mutants was constructed and purified for characterization, and 21 of them showed improved enantioselectivity (>85% ee). Iterative recombination of the most promising mutations established F250K/H191Q as a beneficial core (97% ee), with the subsequent incorporation of F296A and Y82F further improving reactivity. This directed evolution strategy ultimately yielded OYE1-F250K/H191Q/F296A/Y82F, which afforded 46% yield and 97% ee under air. However, additional fifth-site combinations provided no further improvement, indicating diminishing returns from continued recombination of the initially identified first-sphere mutations.

To overcome this limitation, we incorporated machine learning to guide the identification of mutational hotspots beyond the first coordination sphere. Three state-of-the-art zero-shot models (*27*–*30*) were benchmarked against our experimental dataset (137 single mutants),

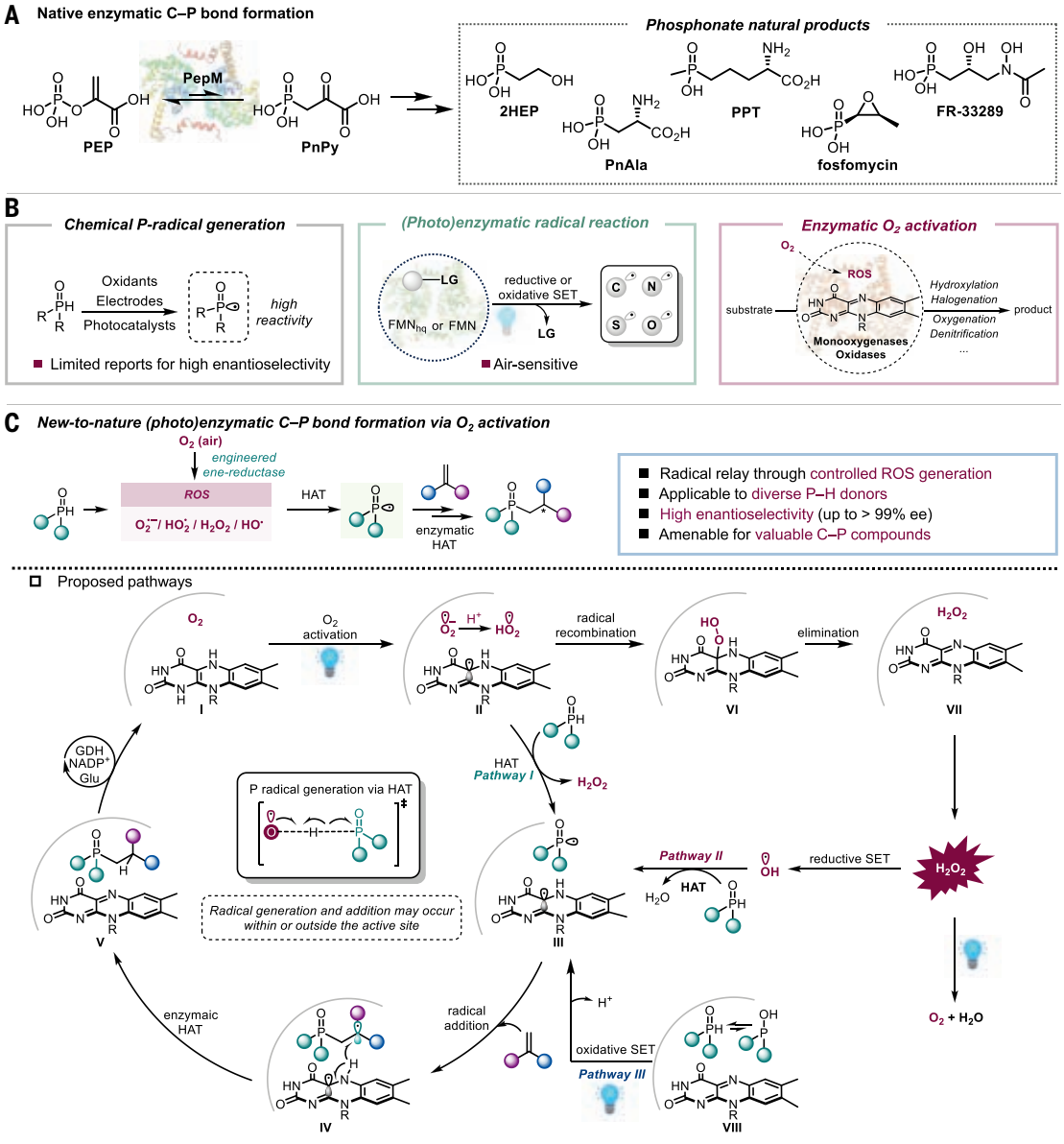
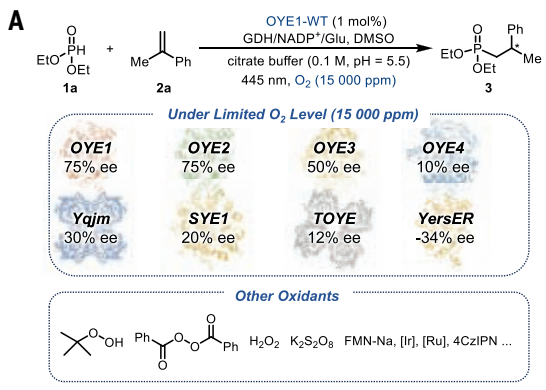


Fig. 1. Asymmetric enzymatic hydrophosphorylation through O_2 activation. (**A**) Native enzymatic C–P bond formation found in phosphonate biosynthesis. PEP, phosphoenolpyruvate; PepM, phosphoenolpyruvate mutase; PnPy, phosphonopyruvate; HEP, 2-hydroxyethyl phosphate; PPT, phosphinothricin. (**B**) Challenges in chemical and (photo) enzymatic C–P bond formation and the present application of enzymatic O_2 activation. LG, leaving group; SET, single-electron transfer. (**C**) Proposed strategy for the asymmetric enzymatic hydrophosphorylation enabled by ROS generation.

with VespaG showing the highest correlation between predicted and measured mutational effects (fig. S1). This model was then applied to predict beneficial mutations within the second-sphere active site (within 10 Å of FMN) using OYE1-F250K/H191Q/F296A as the parent mutant (fig. S2). Experimental evaluation of 23 predicted variants revealed S192A as the most effective mutation, increasing the yield to 58% (table S7). Finally, incorporation of S192A together with additional beneficial mutations P295S and Y82F yielded three top-performing variants, OYE1-M5a, OYE1-M5b, and OYE1-M6. After reoptimizing the conditions (tables S8 and S9), OYE1-M5a catalyzed the photoenzymatic C–P bond formation to give ***3*** in 76% yield and 98% ee under air and blue light. By contrast, in the absence of light, OYE1-M5a afforded only 22% yield after 24 hours (31% yield after 48 hours), whereas OYE1-M6 delivered ***3*** in 75% yield and 97% ee after 24 hours without light.

Mechanism of the asymmetric enzymatic hydrophosphorylation through O_2 activation

We considered several potential pathways for O_2-mediated enzymatic C–P bond formation (Fig. 1C). On the basis of a previous study on the O_2 activation of flavin-dependent enzyme (*18*), we hypothesized that the reaction is initiated by single-electron transfer from flavin hydroquinone (FMN_{hq}) to O_2 (**I**) to generate the flavin semiquinone (FMN_{sq}) and a diffusible superoxide radical anion (**II**) (*31*). Subsequently, protonation of superoxide occurs to give a more reactive hydroperoxyl radical (HO_2·), which abstracts an H atom from the P–H bond (*32*, *33*) to form a P-centered radical (**III**) and release H_2O_2 either in solution or within the enzyme. The P-centered radical then attacks the olefin to give a C-centered radical **IV**, which receives an H atom from the FMN_{sq} to afford the product **V** with high enantioselectivity (*25*, *34*). Conversely, the superoxide can also combine with FMN_{sq} to form an



entry	conditions	Yield (%)	ee (%)
1	**Standard**	**9**	**75**
2	No OYE1	0	-
3	FMN-Na instead of enzyme	1	0
4	Air atmosphere	1	-
5	No GDH, $NADP^+$ or Glu	1	-
6	No light	1	-
7	N_2 (O_2 = 10 ppm)	2	75
8	H_2O_2 (5.0 eq), N_2 (O_2 = 10 ppm)	3	75
9	TBHP (5.0 eq), N_2 (O_2 = 10 ppm)	4	75
10	$Ru(bpy)_3Cl_2$ (1 mol%), N_2 (O_2 = 10 ppm)	2	75
11	$Ru(bpy)_3Cl_2$ (1 mol%), O_2 = 15000 ppm	15	75
12	$Ru(bpy)_3Cl_2$ instead of OYE1-WT	0	-

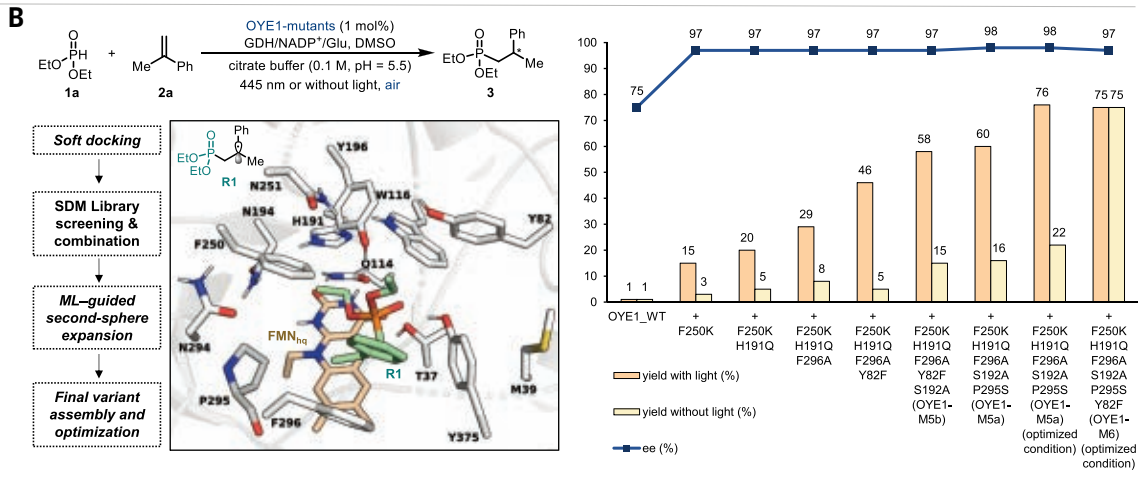


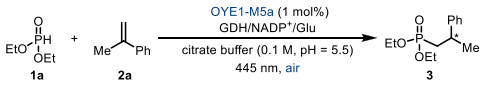
Fig. 2. Asymmetric enzymatic hydrophosphorylation of α-methyl styrene with diethyl phosphite. (**A**) Discovery of enzymatic hydrophosphorylation through O_2 activation. Conditions: **1a** (0.020 mmol), **2a** (0.002 mmol), ene-reductase (1 mol %), GDH-105 (5 U/ml), $NADP^+$ (2.5 mol %), glucose (0.04 mmol), and 80 μl of DMSO as additives (16% v/v) in 0.5 ml (total) of sodium citrate buffer (100 mM, pH 5.5), 445 nm light-emidding diode (LED), O_2 (15,000 ppm), 25°C, and 16 hours. (**B**) Directed evolution of OYE1 for enzymatic hydrophosphorylation and soft docking of wild-type OYE1 [Protein Data Bank (PDB) ID: 3TN9] with the carbon radical **R1**. SDM, site-directed mutagenesis; ML, machine learning. Yields of the directed evolution are reported under aerobic conditions. Yields without light were determined under identical conditions for 24 hours. Optimal reaction conditions were **1a** (0.160 mmol, 20.0 equivalents), **2a** (0.008 mmol, 1.0 equivalents), OYE1-M5a (1 mol %), GDH-105 (5 U/ml), $NADP^+$ (5 mol %), glucose (0.04 mmol, 5.0 equivalents), and 320 μl of DMSO as additives (16% v/v) in 2 ml (total) of sodium citrate buffer (100 mM, pH = 5.5), 445 nm LED, air, 25°C, and 16 hours. The yields were determined by GC-MS. The ee values were determined by chiral high-performance liquid chromatography (HPLC).

FMN_{OOH} intermediate (**VI**) and then release H_2O_2 (*18*). Two possible pathways may account for the consumption of H_2O_2. First, H_2O_2 may undergo a reductive single-electron transfer mediated by flavin (*35*, *36*) or trace metal ions (*37*), generating hydroxyl radicals (·OH). These species can then promote HAT to form the P-centered radical (pathway II) (*38*, *39*). Second, H_2O_2 may decompose into molecular O_2 and H_2O, a process that can be accelerated by light (*37*). Blue light may also increase ROS accumulation (*40*), potentially through excitation of the FMN_{hq}, to generate a highly reducing excited state [$E_{1/2}^* = -2.26$ V versus saturated calomel electrode (SCE)] (*10*), leading to faster O_2 activation. In addition, photoexcited oxidized flavin mononucleotide (FMN_{ox}), which is a strong oxidant ($E_{1/2}^* = +1.76$ V versus SCE), could provide an alternative pathway for P-centered radical generation by direct oxidation (*12*).

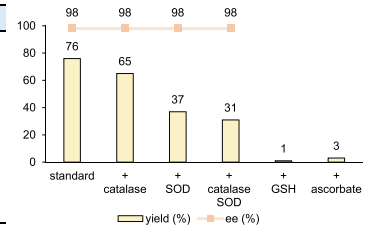
To validate the proposed pathways, we performed detailed mechanistic studies using both experiments and computational tools. Control experiments (Fig. 3A and table S10) showed that the OYE1-M5a enzyme, cofactor regeneration, and air are essential (entries 2 to 5; Fig. 3A), excluding the possible catalytic activity of nicotinamide adenine dinucleotide phosphate ($NADP^+$), whereas pathway III represents a minor route involving FMN_{ox}. Using free FMN-Na instead of enzyme gave a trace amount of product (entry 6; Fig. 2A), indicating the necessity for an enzymatic environment. $Ru(bpy)_3Cl_2$ was also tested but gave no improvement (entry 7; Fig. 2A). To investigate the role of in situ–formed H_2O_2 and superoxide, catalase (1000 U/ml) and superoxide dismutase (SOD, 1000 U/ml) were added to continuously remove H_2O_2 and superoxide in solution (*41*). Removing H_2O_2 only caused a small drop in yield (Fig. 3A), and removal of superoxide afforded 37% yield. The combined addition of catalase and SOD led to a lower yield of 31%. Superoxide can thus diffuse into solution and plays a critical role in the reaction (pathway I). The incomplete suppression upon the addition of SOD and catalase may reflect minor contributions from pathway III, as well as the limited accessibility of superoxide and H_2O_2 present within the enzyme active site, where they contribute to the formation of P-centered radicals. Furthermore, the radical scavengers glutathione (5.0 equivalents) and ascorbate (20.0 equivalents) strongly inhibited the reaction, supporting the involvement of radical intermediates.

The generation of ROS was then evaluated under different conditions (Fig. 3B and fig. S3) (*41*). Upon irradiation of 1 hour, 176 μM H_2O_2 was detected with OYE1-M5a, which increased to 219 μM at pH 9.0. However, after 16 hours of irradiation, the H_2O_2 concentration decreased to 61 μM, indicating its continuous consumption in the system. By contrast, under dark conditions, H_2O_2 production was slower (fig. S3), with only 28 μM H_2O_2 detected for OYE1-M5a at pH 5.5, and OYE1-M6 produced 92 μM H_2O_2. Superoxide formation was further evaluated by SOD inhibition experiments under various conditions. The addition of SOD altered the distribution of ROS

A



entry	conditions	Yield (%)	ee (%)
1	**Standard**	**76**	**98**
2	No OYE1-M5a	0	-
3	N_2 instead of air (O_2 = 10 ppm)	2	-
4	No GDH, $NADP^+$ or Glu	3	-
5	N_2, No GDH/$NADP^+$/Glu	7	98
6	FMN-Na instead of enzyme	1	0
7	With $Ru(bpy)_3Cl_2$ (1 mol%)	75	98



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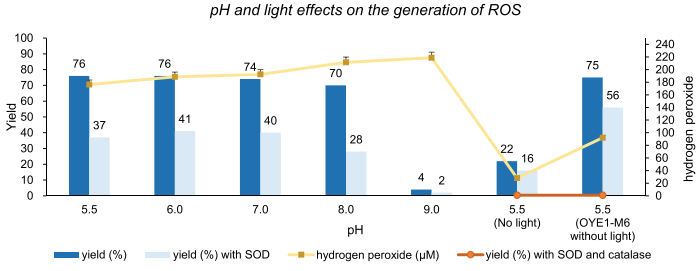

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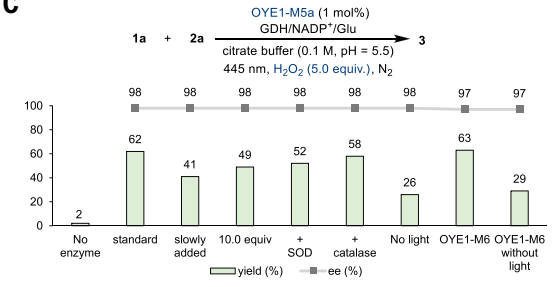

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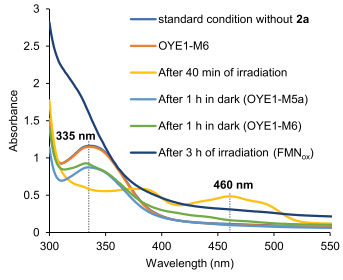

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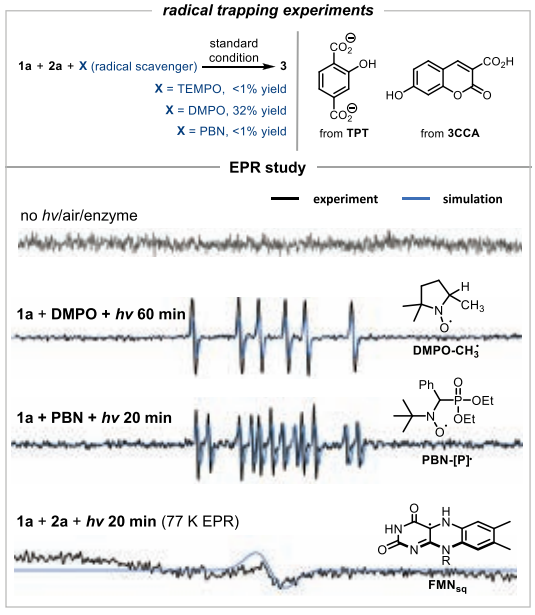

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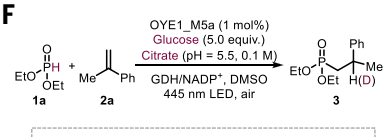


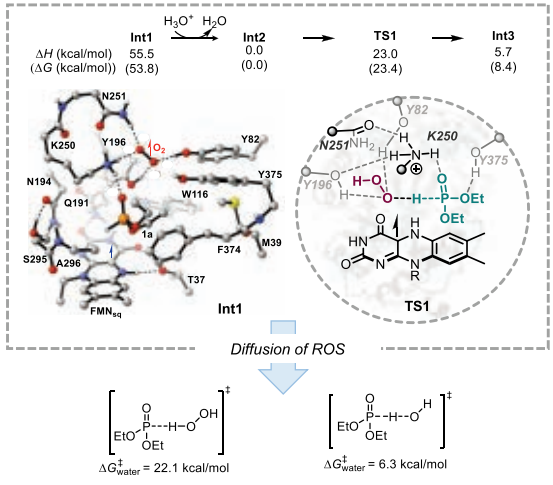
Citrate (D_2O) + Glucose-d_7 + **1a**-d_1 : 16% yield, 93% D

Citrate (H_2O) + Glucose-d_7 + **1a**: 32% yield, 16% D

Citrate (D_2O) + Glucose + **1a**-d_1: 37% yield, 25% D

Citrate (H_2O) + Glucose + **1a**-d_1: 67% yield, 5% D

G



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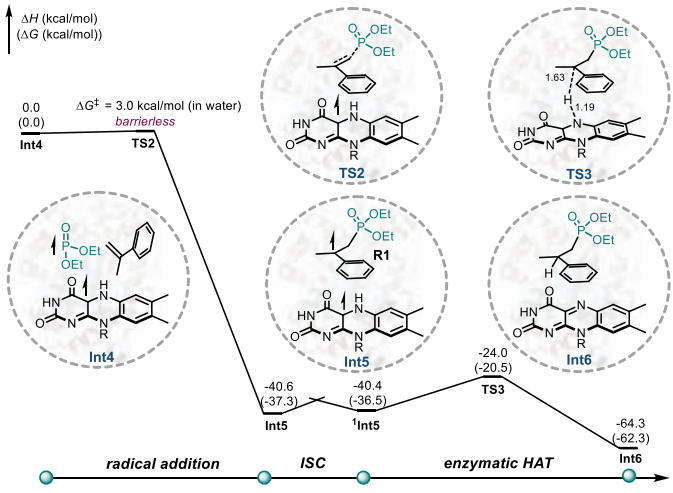

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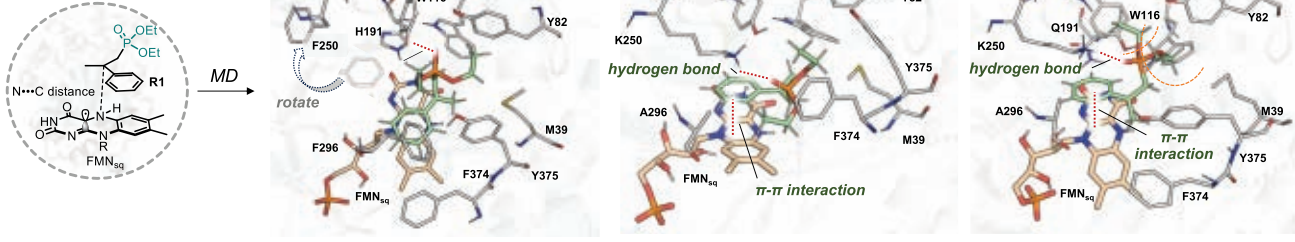

Fig. 3. Experimental and computational studies of the reaction mechanism and enantioselectivity. (**A**) Control experiments of enzymatic hydrophosphorylation through O_2 activation. Conditions: standard conditions with catalase (1000 U/ml), SOD (1000 U/ml), glutathione (GSH) (5.0 equivalents), and ascorbate (20.0 equiv). (**B**) pH and light effects on the generation of ROS. Conditions: standard conditions using citrate buffer (pH 5.5 to 7.0), citrate-Tris buffer (pH 8.0), Tris buffer (pH 9.0), SOD (1000 U/ml), and catalase (1000 U/ml). The H_2O_2 concentration was measured in three independent experiments under standard conditions without substrates after stirring for 1 hour. (**C**) Control experiments of enzymatic hydrophosphorylation through H_2O_2 activation. Conditions: standard conditions using 5.0 equivalents of H_2O_2 under N_2 atmosphere, SOD (1000 U/ml), and catalase (1000 U/ml). (**D**) Ultraviolet-visible spectra of OYE1-M5a and OYE1-M6 in the presence of substrate **1a** and GDH/$NADP^+$/glucose under aerobic conditions under light or dark conditions. (**E**) Radical-trapping experiments and EPR study of enzymatic hydrophosphorylation with different trapping reagents. TPT, disodium terephthalate; 3CCA, coumarin-3-carboxylic acid. (**F**) Isotopic labeling experiments for enzymatic hydrophosphorylation. (**G**) DFT calculation for P-centered radical generation through O_2 activation in enzyme and solution. (**H**) DFT calculation for the radical addition and enzymatic HAT. (**I**) MD simulation of wild-type OYE1 and OYE1-M5a with the radical **R1**.

in solution, decreasing superoxide while increasing H_2O_2. The inhibitory effects of SOD were similar at pH 5.5 to 7 but became more pronounced at pH 8, indicating an increased contribution from superoxide at higher pH. Under dark conditions, the suppression by SOD was weaker, but the combined addition of SOD and catalase led to strong inhibition of the reaction (1% yield), accompanied by recovery of the starting alkene. These results suggest that under dark conditions, the reaction relies predominantly on the coupled superoxide/H_2O_2 network for P-centered radical generation (pathways I and II). Thus, the lower steady-state concentration of ROS under dark conditions allows efficient scavenging by SOD and catalase before productive interception by **1a**.

In the absence of enzyme, H_2O_2 alone led to only trace product formation, whereas adding H_2O_2 (5.0 equivalents) to an anaerobically prepared reaction afforded the product in 62% yield (Fig. 3C). Slowly added H_2O_2 and higher loadings led to decreased yields. Although SOD activity may decrease under 20 mM H_2O_2 (*42*), the addition of SOD (1000 U/ml) suppressed the reaction, indicating the involvement of superoxide derived from H_2O_2. Under dark conditions, OYE1-M6 only afforded 35% yield after 24 hours, whereas irradiation increased the yield to 63%. Control experiments showed that in the absence of enzyme, H_2O_2 was completely consumed after 16 hours of irradiation, but only partially depleted (to 7.8 mM) in the dark. Consistently, in situ generation of O_2 from H_2O_2 using catalase provided similar reactivity. These results support a pathway in which H_2O_2 generates O_2 in the presence of light, and the generated O_2 is reduced by FMN_{hq} to produce superoxide. Alternatively, superoxide may also arise from reactions involving putative ·OH and H_2O_2 (*37*). Kinetic studies (fig. S4) showed that the reaction using OYE1-M5a with H_2O_2 proceeds faster than under air, consistent with the multiple pathways of H_2O_2.

To further study the effect of light, we monitored the ultraviolet-visible spectroscopy of OYE1-M5a in the presence of **1a** and a regeneration system [glutamate dehydrogenase (GDH)/$NADP^+$/glucose] (Fig. 3D). The spectroscopic data indicated that no electron donor-acceptor (EDA) complex formed (fig. S5) with FMN_{ox} or FMN_{hq}. Irradiation under air produced a clear increase at 460 nm after 40 min, indicating oxidation of FMN_{hq} to FMN_{ox}. However, direct irradiation of FMN_{ox} in air for 3 hours lead to deactivation of the flavin cofactor, suggesting that FMN_{ox} is not sufficiently stable under light and aerobic conditions, consistent with pathway III being a minor pathway (entries 4 and 5; Fig. 3A). In the dark, only small peaks at 460 nm were observed after 1 hour. The above results indicate that illumination accelerates the O_2-dependent oxidation of FMN_{hq} and increases ROS accumulation. These effects may arise from the increased thermodynamic driving force for electron transfer after photoexcitation of FMN_{hq} ($E_{1/2}^* = -2.26$ V versus SCE) (*43*), together with the light-promoted conversion of H_2O_2 into other ROS. Alternativity, photoexcitation of FMN_{ox} can also achieve the direct oxidation of phosphite through pathway III as a minor route. Conversely, the higher reactivity of OYE1-M6 compared with OYE1-M5a under dark conditions can be attributed to the 30 mV lower redox potential of FMN_{hq} in OYE1-M6 (−510 mV versus SCE, compared with −480 mV for OYE1-M5a; fig. S6), as well as a more accessible active site that may be favorable for O_2 access, as indicated by cavity analysis (354.9 versus 125.1 $Å^3$; fig. S21), although OYE1-M6 under dark conditions still has lower efficiency than OYE1-M5a under light (fig. S4).

Radical-trapping experiments provided direct evidence for the proposed radical species (Fig. 3E). 2,2,6,6-Tetramethylpiperidine 1-oxyl (TEMPO) and *N*-tert-butyl-α-phenylnitrone (PBN) completely inhibited the reaction, whereas 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) lowered the yield to 32%. Room temperature electron paramagnetic resonance (EPR) was used to investigate the radical species (Fig. 3E and figs. S7 and S8). No signal was observed without enzyme, air, or light with OYE1-M5a after 20 min. Under standard conditions, a methyl radical was detected by DMPO after 3 min of irradiation, whereas no P-centered radical signal was observed even after 60 min. This was likely due to rapid quenching of the P-centered radical by O_2 to form diethyl phosphate, as confirmed by gas chromatography–mass spectrometry (GC-MS). With the more reactive trapping reagent PBN, the P-centered radical appeared after 20 min of irradiation (*44*), whereas the methyl radical emerged after 40 min. The observation of methyl radicals is attributed to the HO· radical being scavenged by the cosolvent dimethylsulfoxide (DMSO) (*45*), whereas the HO_2· radical reacts much more slowly with DMSO (calculated $\Delta G^{\ddagger} = 10.8$ and 36.5 kcal/mol for HO· and HO_2· in water, respectively). Consistently, the side product ***3′***, generated by methyl radical addition to alkene **2a**, was obtained in 8% GC yield under both light (OYE1-M5a) and dark (OYE1-M6) conditions, although fully excluding DMSO led to no product due to poor substrate solubility. We also tried to directly detect the HO· radical and the HO_2· radical in the absence of DMSO, but the DMPO–OOH/OH radical adduct was not stable enough under light; only complex signals containing DMPO–OH adducts were observed under dark conditions with OYE1-M6 (fig. S7). The formation of HO· was further supported by trapping experiments using disodium terephthalate and coumarin-3-carboxylic acid, which produced characteristic HO· adducts detected by liquid chromatography–MS and fluorescence (*46*). To evaluate the potential contribution of trace metal ions, metal chelators ethylenediaminetetraacetic acid (2 mM) and pentetic acid (2 mM) were added (*47*). No change in HO· formation was observed, as evaluated by the yield of the methylation product ***3′*** (table S10), suggesting that the HO· may originate from flavin rather than trace metal–mediated pathways. A low-temperature EPR experiment revealed a signal consistent with FMN_{sq} (*48*), providing additional support for the proposed radical mechanism.

Isotopic labeling was then used to identify the H donor in the product (Fig. 3F and fig. S6). The use of deuterated glucose, deuterated buffer, or **1a**-d_1 resulted in low levels of D incorporation and reduced yields. This could be attributed to the rapid H–D exchange of FMN_{sq} at pH 5.5 (*49*), which competes with the final HAT step (figs. S9 and S10). The introduction of D also increased the activation energy of the relevant HAT steps, leading to side reactions. Despite the low D incorporation, these results remain consistent with FMN_{sq} serving as the H-atom donor in the final HAT step.

Computational study

To gain insights into the key steps of the reaction pathway, we performed a series of computational studies. A theozyme model (*50*, *51*) containing 12 residues was built from a selected trajectory of molecular dynamics (MD) simulation of OYE1-M5 with the putative catalytically

relevant intermediates (figs. S11 and S12). The active site in this model is highly organized and enriched with H bonds through K250, N251, Y196, and Y82 residues. In particular, the F250K mutation plays an essential role by forming strong H bonds with the O atom of the P–H substrate and with ROS.

Density functional theory (DFT) calculations showed that electron transfer from the anionic FMN_{hq} to O_2 is thermodynamically favorable, as supported by the spin density analysis of **Int1** (fig. S14). A possible proton-coupled electron transfer process (*52*) from K250 to O_2 was explored, but no viable pathway was located (fig. S15) because the lysine side chain is more basic than the superoxide anion. Protonation of the superoxide radical anion ($O_2^{\cdot-}$) mediated by the solvent then generates the $HO_2\cdot$ radical (Fig. 3G), which may occur within the enzyme, in solution, or in both environments. This is reasonable because the active site of OYE1 is solvent exposed and the pK_a of $HO_2\cdot$ (4.8) is close to the pH of the citrate buffer (5.5) (*33*). We also propose that protonation of superoxide may be facilitated within the enzyme through the stabilization of $HO_2\cdot$ by various H-bonding interactions with enzyme residues in **Int2** (*18*, *52*). This is supported by the comparable yields and inhibitory effects of SOD observed at pH 5.5 and pH 8.0 despite the negligible fraction of $HO_2\cdot$ in bulk solution at pH 8.0. Subsequently, $HO_2\cdot$ abstracts the H atom from the P–H compound through **TS1** in the enzyme active site, requiring an activation enthalpy of 23.0 kcal/mol. The radical recombination of $HO_2\cdot$ and FMN_{sq} involves intersystem crossing and radical recombination, requiring a higher overall activation enthalpy than **TS1** by 7.0 kcal/mol (fig. S14). This could be attributed to the rich H-bonding network in the active site, preventing the approach of $HO_2\cdot$ to the plane of FMN_{sq}. The alternative pathway for P-centered radical generation from hydroxyl radical (pathway II) in the enzyme was studied under the free model without considering surrounding residues (fig. S16). The HO· generated from H_2O_2 abstracts the H of **1a** through the HAT transition state **TS1′** with an activation free energy of 5.4 kcal/mol. This pathway is 1.6 kcal/mol more favorable than the reaction with DMSO, which accounts for the minor formation of the methyl radical and agrees well with the experimental observations. Conversely, ROS generated within the enzyme can diffuse into solution (*31*), where they could be captured by SOD and catalase, so P-centered radical generation is not confined to the enzyme active site. DFT calculations found that the activation free energies of P-centered radical generation from $HO_2\cdot$ and HO· radicals in water were 22.1 and 6.3 kcal/mol, respectively, which are comparable to those within the active site. Given the high reactivity and short lifetime of HO·, these radicals may also be quenched by nearby amino acid residues such as tyrosine, potentially leading to enzyme deactivation. However, the high concentration (80 mM) of **1a** and the relatively low bond dissociation energy of the P–H bond (87 kcal/mol) favor productive radical formation rather than an enzyme deactivation pathway. DFT calculations showed that the P-centered radical could also be generated from tyrosine (the phenol radical), requiring an activation free energy of 20.5 kcal/mol.

After the generation of the P-centered radical, C–P bond formation proceeded through addition to the alkene (Fig. 3H). This step is barrierless and irreversible within the enzyme active site (fig. S17), whereas calculations in aqueous solution gave a low activation free energy of 3.0 kcal mol^{-1}, consistent with a diffusion-controlled process. These results reflect the high reactivity of P-centered radicals toward alkenes (*53*), which minimizes competing quenching by ROS. Finally, a singlet ^{1}**Int5** was obtained through intersystem crossing, and the enzymatic HAT through **TS3** afforded the desired product (*R*)-***3***, requiring an activation enthalpy of 16.6 kcal/mol. Although the final radical termination must occur within the enzyme to achieve high enantioselectivity, it remains unclear whether the radical addition occurs within the enzyme. Given that P-centered radical generation can occur outside of the enzyme and the radical addition to the alkene is diffusion controlled, P-centered radicals formed outside of the enzyme may also add to the alkenes in solution. The lifetime of the resulting C radical **R1** was assessed by evaluating its tendency to form dimers (*54*). DFT calculation revealed that the dimer of **R1** has a relatively weak C–C bond (1.64 Å, 37.6 kcal/mol; fig. S19), indicating modest reactivity of the tertiary benzyl radical (*25*). However, the transient concentration of **R1** is expected to be very low under catalytic conditions, thereby disfavoring bimolecular coupling. Considering the relatively high enzyme concentration (1 mol %, 40 μM), we propose that if the radical addition occurs in solution, then **R1** could diffuse into enzyme and undergo the enantioselective HAT step.

We also used classical MD simulations to examine the effects of the mutations and to provide insights into the origin of the enantioselectivity (Fig. 3I). In wild-type OYE1, the large C radical intermediate **R1** rarely entered the active site because F250 blocked access (fig. S20), and restrained simulations showed that only rotation of F250 allowed **R1** to approach the FMN_{sq}, leading to higher activation free energy of HAT process. Therefore, under air conditions with a higher ROS concentration, OYE1-WT only gave trace amount of product due to the faster quenching of productive radicals. By contrast, in OYE1-M5a and OYE1-M6, the F250K mutation created a stable binding mode in which **R1** is fixed by an H bond with K250 and a π–π interaction with FMN_{sq}, closely matching the geometry of the H transfer transition state. Cavity analysis (*55*) also showed that the active site of OYE1-M5a and OYE1-M6 is larger and more flexible than the OYE1-WT (125.11 versus 354.9 versus 102.11 Å; fig. S21), which may be more beneficial for the access of both O_2 and C-centered radicals to the cofactor. The nonrestrained MD simulations of OYE1-M5a with **R1** provided near-attack conformations (NACs) (*56*) for both enantiomers and predicted a strong preference for the trajectory leading to the (*R*)-***3*** consistent with the observed 98% ee (fig. S22). Restrained MD simulations generated more strict NACs with N–C distances of 2.8 to 3.5 Å, leading to (*R*)- and (*S*)-product. The NAC analysis (fig. S23) indicated that two factors contribute to the high enantioselectivity. The first factor is the highly organized active site that positions **R1** and FMN_{sq} in a reactive orientation and is mainly shaped by residues K250, Q191, Y196, N251, and T37. By contrast, a hydrophobic region (fig. S21) containing A296, F374, M39, and Y375 is sufficiently flexible to accommodate the bulky phosphite group. The second factor is steric control from residues W116 and Y196, weakening the H bonding and π–π interactions in the NACs of the (*Si*)-face HAT step. Consistent with this interpretation, the single mutants W116A and Y196A derived from wild-type OYE1 exhibited reduced enantioselectivities (44 and 60% ee, respectively).

Substrate scope and practical application

Having established the mechanistic basis for O_2-driven P-centered radical formation, we next evaluated the generality of this strategy across a broad range of olefins and P–H precursors using the top three mutants (Fig. 4). We first tested various α-methyl vinyl arenes for the reaction with diethyl phosphite **1a**. Moderate to high yields (24 to 90%) with excellent enantioselectivities (86 to 98% ee) (compounds ***3*** to **15**) were achieved for substrates with diverse substituents. The heterocyclic alkenes were also compatible in this reaction, giving high enantioselectivities (95 to 97% ee) with reduced yields (25 to 37%) (compounds **16** to **17**). α-Ethyl styrene (**18**) gave 34% yield and 81% ee due to the enlarged steric hinderance. A smaller α-trifluormethyl styrene (**19**) afforded a 71% yield and >99% ee. The defluorinated product **19′** was obtained in 1% yield, further supporting the HAT mechanism rather than a stepwise electron transfer/proton transfer mechanism from intermediate **IV**. Unactivated alkyl-substituted alkenes were also examined (**20** and **21**), affording yields of 65 and 83%, respectively, although enantioselectivity could not be determined because of the lack of a suitable separation method.

We then studied the scope of the P-centered radical precursors. Inorganic phosphite is used by phosphite dehydrogenases for cofactor regeneration (*57*) but is rarely used as a radical precursor in chemistry due to its low solubility in organic solvent. We found that phosphorous

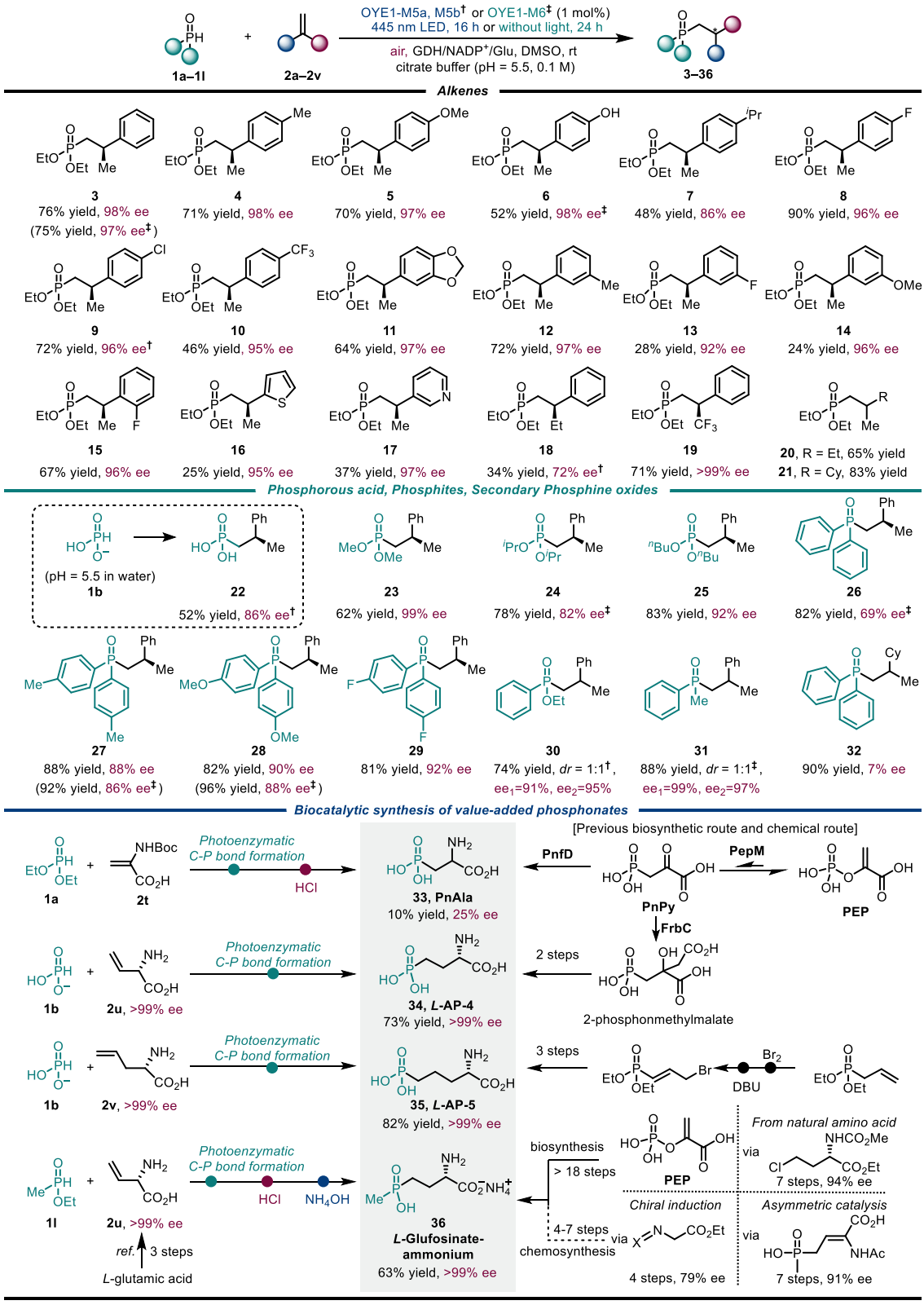

Fig. 4. Substrate scope and application of enzymatic hydrophosphorylation. Reaction conditions were **1** (0.160 mmol for **1a** to **1e** and **1l** or 0.04 mmol for **1f** to **1k**), **2** (0.008 mmol), OYE1-M5a (1 mol %), GDH-105 (5 U/ml), $NADP^+$ (5 mol %), glucose (0.04 mmol), and 320 μl of DMSO as additives (16% v/v) in 2 ml (total) of sodium citrate buffer (100 mM, pH 5.5), 445 nm LED, air, 25°C, and 16 hours. Each substrate was screened against the three top mutants, and the result with the highest enantioselectivity is reported. †OYE1-M5b (1 mol %) instead of OYE1-M5a (1 mol %). ‡OYE1-M6 (1 mol %) instead of OYE1-M5a (1 mol %), 24 hours without light. Yields were determined by ^{31}P–nuclear magnetic resonance (^{31}P-NMR) or ^{19}F-NMR spectroscopy using triphenyl phosphate or trifluorotoluene, respectively, as internal standards. The ee values were determined by HPLC. Me, methyl; Et, ethyl; Cy, cyclohexyl; ^{i}Pr, isopropyl; ^{n}Bu, n-butyl.

acid (**1b**) solubilized in aqueous buffer (pH 5.5) was compatible with this reaction, delivering 52% yield and 86% ee (**22**). Various phosphite esters also furnished moderate to good yields (62 to 83%) with commendable enantioselectivities (82 to 99% ee) (**23** to **25**), although OYE1-M6 with a larger active site was used to accommodate a larger phosphite (**25**). The more reactive secondary phosphine oxides bearing different aryl substituents (**26** to **29**) were converted efficiently, providing the corresponding products in yields of up to 96% and an ee value of up to 92%, with only 5.0 equivalents of **1**. Ethyl phenylphosphinate provided **30** in 74% yield with high enantioselectivity (91 and 95% ee) and low diastereoselectivity (1:1). An alkyl-substituted P–H compound was also compatible, giving 88% yield (**31**) without erosion of enantioselectivity (99 and 97% ee). The different P–H reagents could also be paired with various alkenes. As an example, **32** was obtained in 90% yield from **1f** and **2s** with low enantioselectivity (7% ee) but highlighted the board reactivity of this strategy.

To demonstrate the synthetic potential of this enzymatic C–P bond formation, we prepared several bioactive P-containing molecules. Phosphonoalanine (PnAla) is a key intermediate in the biosynthesis of natural phosphonates, including phosphonoglycans and phosphonalamides (*58*). Here, we showed that PnAla (**33**) could be obtained through the enzymatic C–P bond formation, followed by hydrolysis and deprotection in a single reaction vessel, albeit the yield (10%) and enantioselectivity (25% ee) were low. We next targeted *L*-AP-4 **34** and *L*-AP-5 **35**, glutamate antagonists with previous biosynthetic or chemical routes that required four to five steps (*5*, *59*–*61*). Using unprotected chiral amino acids (**2u** and **2v**), the corresponding phosphonates were obtained in high yields with full retention of enantioselectivity using OYE1-M5a. Finally, we evaluated the synthesis of *L*-glufosinate (**36**), a nonselective, broad-spectrum herbicide and one of the most important P-containing agrochemicals (*62*). Three chemical strategies have been developed for its enantioselective preparation, including chiral induction, chiral substrates, and asymmetric catalysis, but these routes typically require multiple steps and strict conditions for C–P bond formation through the Arbuzov reaction and Michael addition (*62*). A fully biosynthetic route to *L*-glufosinate has also been proposed, but it requires >8 steps from phosphoenolpyruvate (*5*, *62*). Using *L*-vinylglycine (**2u**), which could be synthesized from the natural amino acid *L*-glutamate in three steps (*63*), the present enzymatic reaction furnished *L*-glufosinate ammonium in 63% yield with >99% ee using OYE1-M5a under light. This reaction represents an important step toward a fully green, efficient biosynthesis of *L*-glufosinate.

Conclusions

We have discovered and engineered a flavin-dependent ene-reductase to enable unprecedented C–P bond formation under both light and dark conditions, providing access to organophosphorus compounds with exquisite stereoselectivity. The repurposed enzyme activates molecular O_2 to generate various ROS, which are funneled into productive P-centered radical formation. Computational studies revealed how the protein environment directs O_2 activation and the origin of enantioselectivity from the enzymatic HAT step. This work provides a simple synthesis route to important phosphonate products in medicine and agriculture, and also shows how enzymes can channel O_2 reactivity toward selective radical steps, opening new directions for sustainable biocatalysis.

Materials and methods are available as supplementary materials.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S23; Tables S1 to S12; References (*65*–*96*); MDAR Reproducibility Checklist

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RESEARCH ARTICLE SUMMARY

PLANT METABOLISM

The genetic basis for the production of toxic quinolizidine alkaloids in lupins

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INTRODUCTION: Quinolizidine alkaloids (QAs) are specialized metabolites produced by several species of legumes, including the protein-rich pulses called lupins (*Lupinus* spp.). Given their bitter taste and anticholinergic toxicity, reducing alkaloid levels by breeding is crucial to meet food safety standards. However, the use of new breeding technologies is hampered by a limited understanding of the underlying biosynthetic pathway. The pathway starts with the amino acid lysine, and a few enzymes in the early and late stages have been discovered. However, the core of the pathway remains unknown, including the formation of the quinolizidine scaffold that structurally defines this class of alkaloids.

RATIONALE: Genes coding for enzymes in QA biosynthesis are potential targets for improvement of lupin crops and domestication of related legume species. Moreover, key chemical transformations predicted to occur in the pathway lack clear enzymatic precedence. Thus, elucidating the pathway offers the opportunity to discover novel enzyme capabilities. Lastly, QAs have complex polycyclic structures, some of which have useful bioactivities. Expression of the pathway in heterologous hosts could make these chemical scaffolds more easily accessible.

RESULTS: We leveraged available gene expression data from narrow-leafed lupin (*L. angustifolius*) to identify nine enzymes that, in addition to six known ones, enable the production of a variety of QAs in the model plant *Nicotiana benthamiana*. The quinolizidine scaffold is assembled in acetylated form, and the acetyl group is later removed. Several major latex protein–like proteins are involved in at least four key steps including C–C bond formation and intramolecular rearrangement. We also identified an oxidoreductase-like protein that determines the stereoselectivity of the pathway and a cytochrome P450 that catalyzes ring cleavage late in the pathway. We complemented the work in *N. benthamiana* with in vitro assays of a key step previously proposed to be nonenzymatic. Finally, we validated our biochemical hypothesis through mutagenesis in narrow-leafed lupin.

We used the new understanding of QA biosynthesis to engineer a novel low-alkaloid line of narrow-leafed lupin by inactivating the hydrolase that removes the cryptic acetyl group. We also reviewed the pangenome of white lupin (*L. albus*) and observed that a low-alkaloid variety of the *exiguus* type (mutant gene unknown) carries a splice-site mutation in a gene coding for one of the major latex protein–like proteins. When expressed in *N. benthamiana*, the mutant allele was nonfunctional, explaining the low-alkaloid phenotype.

CONCLUSION: We have uncovered the pathway toward most of the QAs found in narrow-leafed lupin. This knowledge can be used to develop lupin varieties with reliably low alkaloid levels and may serve as starting point in the domestication of other QA-containing legumes. The pathway involves numerous proteins whose exact function is unclear. Elucidation of the precise catalytic role of these enzymes will deepen our insight into the formation of complex chemical scaffolds and into the evolution of stereoselectivity in specialized metabolite pathways. □

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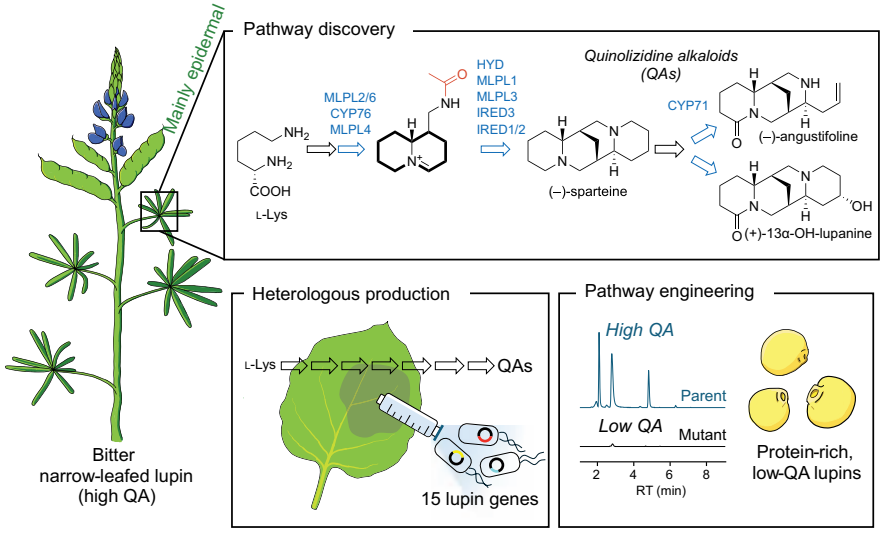
Discovery and applications of QA biosynthesis in lupins. Lupins are protein-rich pulses that accumulate toxic QAs. We have completed the elucidation of the QA biosynthetic pathway in narrow-leafed lupin comprising 15 enzymes predominantly expressed in epidermal cells. We engineered mutants with low QA levels that may be used in breeding. We also reconstructed the pathway in the model plant *N. benthamiana*, paving the way for heterologous production of QAs with useful bioactivities.



PLANT METABOLISM

The genetic basis for the production of toxic quinolizidine alkaloids in lupins

Davide Mancinotti[1], Hajar Golshadi Ghalehshahi[1], Isabella Kruse-Andersen[1]†, Louise Kjaerulff[2], Ting Yang[1], Dan Stærk[2], Fernando Geu-Flores[1]*

Lupins (*Lupinus* spp.) are promising protein crops that accumulate bitter and toxic quinolizidine alkaloids (QAs). The effective removal of QAs through new breeding technologies is hampered by a poor understanding of their biosynthesis. In this study, we used pathway reconstruction and mutant analysis to elucidate the full QA pathway in *L. angustifolius* [narrow-leafed lupin (NLL)] comprising 15 enzymatic steps. Major latex protein–like proteins play a prominent role in the pathway by accelerating spontaneous chemical equilibria, and an oxidoreductase-like protein ensures stereoselectivity. We also engineered a new low-alkaloid NLL line and identified the causal mutation in an existing low-alkaloid line of *L. albus*. Our work reveals the genetic and biochemical basis of toxic QA production and streamlines the de novo domestication of wild lupins and other QA-containing legumes.

Alkaloids are a large family of bioactive, nitrogen-containing compounds commonly used as medical drugs or stimulants. In food and feed crops, however, the accumulation of alkaloids is often undesirable, as they typically confer toxicity or act as antinutritional compounds. This is true of the quinolizidine alkaloids (QAs), which are anticholinergic compounds produced by the protein crops collectively called lupins (*Lupinus* spp.). Lupins are the legume crops with the highest seed protein content (up to 44% protein for *L. mutabilis*); however, their accumulation of QAs limits their use as vegetable protein sources. Traditional breeding has produced low-alkaloid varieties (sweet varieties), but the remaining QA levels are seasonally variable and often exceed the safety limits established by the food and feed industries (*1*–*5*).

The QAs that accumulate in lupins are mostly tetracyclic, the simplest of which is sparteine (**6**) (Fig. 1). The sole precursor for the biosynthesis of QAs is the amino acid L-lysine (**1**), which is decarboxylated by a lysine decarboxylase (LDC) (*6*) and oxidized by a copper amine oxidase (CAO) (*7*) to give Δ^1-piperideine (**3**) and its spontaneously formed dimer, tetrahydroanabasine (**4**) (Fig. 1 and fig. S1). Recently, we showed that the *N*-acetylation of **4** by a BAHD acetyltransferase (AT) is an integral part of early QA biosynthesis (Fig. 1) (*8*). Further conversion to (–)-**6** remains obscure but must include loss of the acetyl group, rearrangement to form a quinolizidine ring, addition of a Δ^1-piperideine (**3**) unit, and reduction (Fig. 1). (–)-**6** is then oxidized to (+)-lupanine [(+)-**8**] [the most common QA in lupins (*9*)] by the concerted action of a cytochrome P450 (CYP71D189) and a short-chain dehydrogenase/reductase (SDR1) (*10*). In the commonly cultivated lupin species, narrow-leafed lupin (NLL; *L. angustifolius*), a portion of (+)-**8** is converted to (+)-13-hydroxylupanine [(+)-**9**] and (–)-angustifoline [(–)-**10**] by unknown enzymes. Furthermore, a portion of (+)-**9** is esterified with a variety of CoA-activated acids, such as tiglic acid, by at least one known acyltransferase [13-hydroxylupanine *O*-tigloyltransferase (HMT/HLT)] (*11*–*14*). At least in NLL, QAs are produced in green organs (leaves, stems, and pods), with the epidermis being a major site of biosynthesis (*15*). Esterified QAs accumulate predominantly in the epidermis of biosynthetic organs, whereas core QAs accumulate in most tissues (*15*) and are transported via the phloem to the developing seeds (*16*). The seeds themselves produce very little if any QAs (*16*).

Elucidation of the QA pathway up to (–)-sparteine

We first set out to identify the missing enzymes in the pathway toward (–)-sparteine [(–)-**6**] in NLL. (–)-**6** is commercially valuable as an auxiliary molecule in chiral synthesis (*17*); however, it does not normally accumulate in NLL owing to oxidative conversion to (+)-lupanine [(+)-**8**] (*9*, *18*). Since biosynthetic genes in plant specialized metabolism are typically coexpressed (*19*), we used available transcriptomic datasets (*7*, *20*) to carry out coexpression analysis with *LDC* as a bait. The analysis included both biosynthetically active and inactive organs and cultivars to ensure a wide range of expression levels of the relevant genes (*21*). We found that *AT*, *CAO*, and *CYP71D189* were among the 56 most tightly coexpressed transcripts (Pearson correlation coefficient > 0.88), suggesting co-regulation of the core biosynthetic genes at least up to the oxidation of (–)-**6** (table S1). From this set of transcripts, we further selected 31 with a high degree of phylogenetic specificity (low identity to their respective top BLASTX hits) (table S1). Among these were transcripts encoding putative oxidases, reductases, and hydrolases as well as a number of major latex protein–like proteins (MLPLs; table S1).

MLPLs typically catalyze energetically undemanding reactions with appreciable spontaneous rates (*22*, *23*). One such reaction occurs early in the QA pathway, namely, the dimerization of Δ^1-piperideine (**3**) to give tetrahydroanabasine (**4**) (Fig. 1). The reaction has been observed in vitro over a range of physiologically relevant pHs, where **3** exists in equilibrium with **4** (*24*). In the model plant *Nicotiana benthamiana*, coexpression of LDC, CAO, and AT yields ammodendrine (**5**), which is an *N*-acetylated form of **4** (*8*). We tested the involvement of MLPLs in the same system and observed an ~3-fold increase in **5** upon coexpression of either MLPL2 or MLPL6 [93% identity at the protein level (fig. S2A)], suggesting a role in the dimerization step (Fig. 2, A and C). Chiral gas chromatography–mass spectrometry (GC-MS) analysis revealed that the produced **5** was close to enantiomerically pure [~96% enantiomeric excess (ee) for (+)-**5**] (figs. S3 and S4), confirming our previous suggestion that this chiral center is established early in the QA pathway (*25*) (fig. S3B). Notably, the ee did not change in the absence of MLPL2 or MLPL6 (fig. S3A), suggesting that AT is the enzyme conferring the stereoselectivity (and not MLPL2 or MLPL6). To observe MLPL function directly, we produced MLPL2 in *Escherichia coli* (fig. S5) and conducted in vitro time course experiments with either **3** or **4** as starting points, including reduction and derivatization of both **3** and **4** to give stable analytes (figs. S6 and S7). The experiments revealed that MLPL2 could accelerate the establishment of the equilibrium from either direction (fig. S7). Together with the observation that AT determines the stereoselectivity, this result suggests that MLPL2 enables a dynamic kinetic resolution of **4** to give (+)-**5** specifically.

Prior to uncovering the role of AT, tetrahydroanabasine (**4**) was thought to undergo rearrangement to a bicyclic QA intermediate via a series of steps in which oxidation next to the imine nitrogen was key (*25*, *26*). We speculated that an oxidase could act directly on the product of AT to enable an analogous rearrangement yielding an *N*-acetylated bicyclic QA intermediate. To test this hypothesis, we coexpressed each of three oxidase candidates together with LDC, CAO, MLPL2, and AT in *N. benthamiana*. The coexpression of cytochrome P450 CYP76E36 led to a decrease in ammodendrine (**5**) accumulation as well as the appearance of three previously unknown compounds, two of which eluted very closely in our chromatographic separation system (Fig. 2B). We purified these two compounds and determined their structures by mass spectrometry and two-dimensional nuclear magnetic resonance

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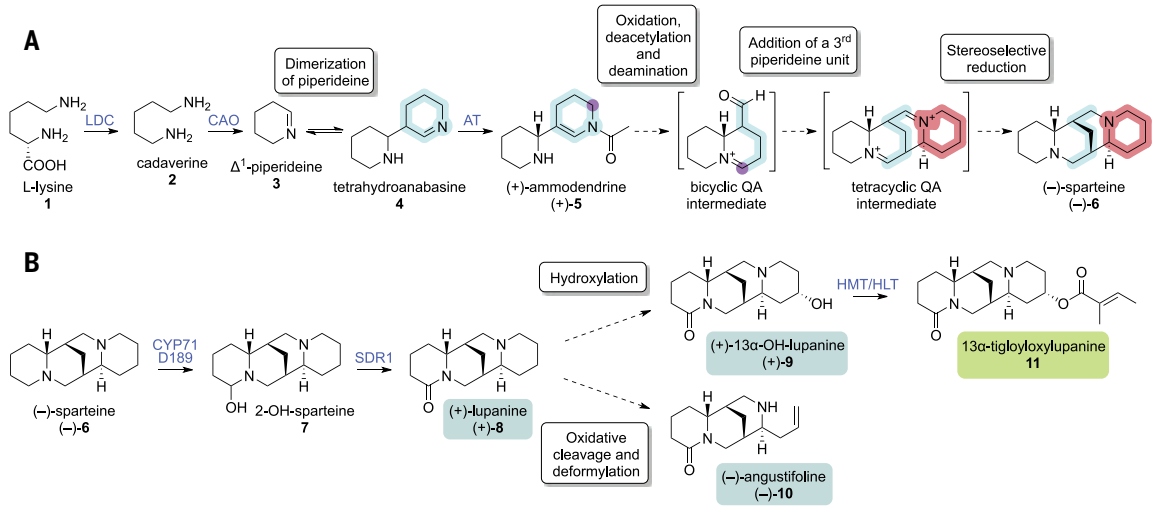

Fig. 1. Known and proposed reactions in the biosynthesis of QAs in NLL. Plain arrows indicate known steps, with the responsible enzymes shown in violet. Dashed arrows indicate unknown, possibly multiple steps, with the predicted chemical transformations described in boxes. Putative intermediates are shown in brackets. Intermediates containing imine, enamine, and/or aldehyde functionalities such as Δ^1-piperideine (**3**) or tetrahydroanabasine (**4**) are depicted only in one form although they occur in a complex equilibrium of hydrated and tautomeric forms (fig. S1). (**A**) Proposed pathway up to (−)-sparteine [(−)-**6**]. The backbone of (−)-**6** is assembled from three piperideine (**3**) units derived from L-lysine (**1**). Fragments originating from the second and third piperideine units are highlighted in cyan and red, respectively. Purple circles indicate the position of a key carbon atom during the formation of the quinolizidine core. (**B**) Proposed steps leading from (−)-**6** to the major QAs that accumulate in NLL. Gray-shaded name labels mark core QAs (**8** to **10**), and the green label indicates one of several possible esterified QAs (**11**).

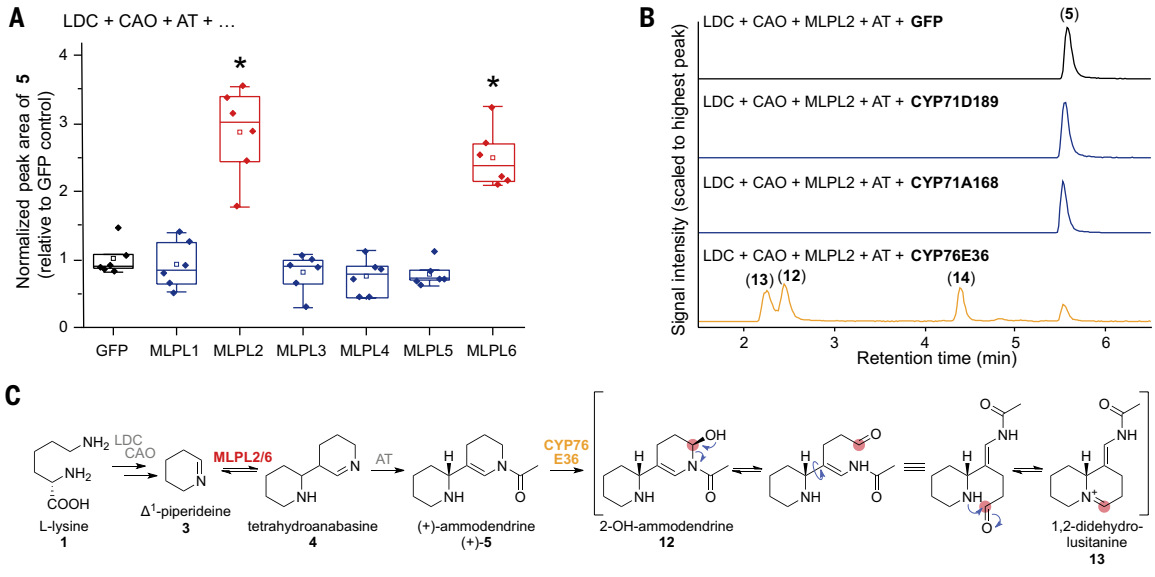

Fig. 2. Formation of the quinolizidine core by coexpression of five NLL enzymes in *N. benthamiana*. All *m/z* values are provided in the Materials and methods. (**A**) Abundance of ammodendrine (**5**) in leaves of *N. benthamiana* transiently coexpressing LDC, CAO, AT, and either GFP or one of six candidate MLPLs. Agroinfiltrated leaves were harvested at 7 days post-infiltration (dpi). Boxes in the plot span the interquartile range; whiskers extend to all data points that fall within the inner fences; lines within the boxes indicate median values; and hollow squares indicate mean values. Asterisks indicate significantly different mean peak areas compared with the GFP control (two-tailed Welch's *t* test with Bonferroni correction for $m = 6$ multiple comparisons; $n = 6$ or 7; $P < 0.0084$). See fig. S3 for the chiral analysis of **5** in the presence or absence of MLPL2 or MLPL6. (**B**) Representative extracted ion chromatograms from the LC-MS analysis of leaf extracts of *N. benthamiana* transiently coexpressing LDC, CAO, MLPL2, and AT as well as either GFP or one of three candidate oxidases ($n = 3$; harvested at 5 dpi). Of the three compounds that appeared upon coexpression of CYP76E36, the structures of 2-OH-ammodendrine (**12**) and of 1,2-didehydrolusitanine (**13**) were determined by mass spectrometry and NMR (figs. S8 to S20). (**C**) Proposed mechanism for the formation of **12** and **13** upon oxidation of **5** by CYP76E36.

(2D NMR) spectroscopy (figs. S8 to S20). One of them corresponded to a hydroxylated ammodendrine [2-hydroxyammodendrine (**12**)], and the other corresponded to the expected *N*-acetylated bicyclic QA intermediate (Fig. 2C). We named this compound 1,2-didehydrolusitanine (**13**) in reference to the known QA lusitanine (*27*). We propose that CYP76E36 hydroxylates ammodendrine (**5**), thus triggering a spontaneous rearrangement leading to the quinolizidine core scaffold in *N*-acetylated form (Fig. 2C).

As the precise order of the remaining biochemical steps was uncertain, we decided to attempt full pathway reconstruction up to (–)-sparteine

[(–)-**6**] by coexpressing all of the most promising candidate enzymes according to our pathway conjecture (Fig. 1). Among these were the four remaining MLPLs, as we suspected them of aiding several downstream reactions. These included the conversion of 2-hydroxyammodendrine (**12**) to 1,2-didehydrolusitanine (**13**), the addition of the third Δ^1-piperideine unit, and the isomerization of the double bonds of the tetracyclic QA intermediate before reduction (*23*, *25*). We also chose one candidate for the deacetylation reaction (the carboxylesterase HYD) and two candidates for the reduction steps (the iminium reductases IRED1 and IRED2). Coexpression of these seven candidates with LDC, CAO, MLPL2, AT, and CYP76E36 in *N. benthamiana* led to the production of **6** (Fig. 3, A and C).

To determine whether each of the newly selected enzymes was necessary for the efficient production of **6**, we carried out a similar *N. benthamiana* experiment wherein we replaced each of these enzymes with a noncatalytic protein, green fluorescent protein (GFP; dropout experiment). We also tested the already implicated MLPL2 and CYP76E36 in a similar way. The individual absence of MLPL2, CYP76E36, HYD, and MLPL1 led to trace or undetectable levels of **6** (Fig. 3B), suggesting distinctive involvement in the reconstructed pathway. Furthermore, the individual absence of MLPL3 and MLPL4 led to reduced amounts of **6** compared with the full enzyme set (Fig. 3B). However, the individual absence of IRED1 and IRED2 did not lead to any changes in

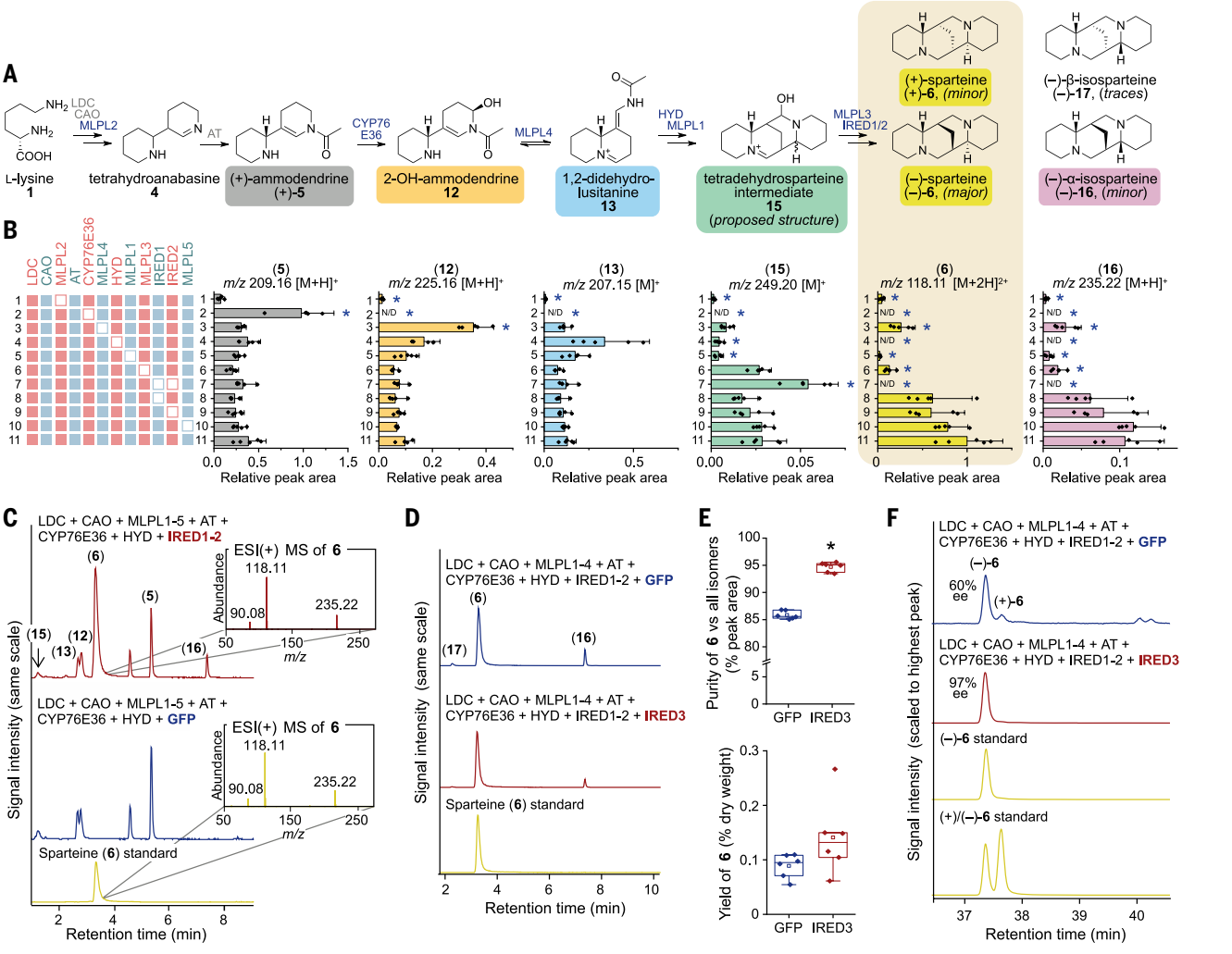

Fig. 3. Production of (–)-sparteine [(–)-6] by coexpression of at least 10 NLL enzymes in *N. benthamiana*. Additional *m/z* values are provided in the Materials and methods. (**A**) Proposed biosynthetic pathway leading to sparteine (**6**) and its isomers α-isosparteine (**16**) and β-isosparteine (**17**) [drawn as the (–) enantiomers for simplicity]. (**B**) Gene dropout experiment to assess the effect of each candidate gene on the production of **6** as well as on the accumulation of intermediates (+)-**5**, **12**, **13**, **15**, and side product **16** at 6 dpi. Hollow boxes on the left indicate genes that were dropped out (replaced by *GFP*) in each infiltrated combination (row). Bar charts on the right show the abundance of metabolites relative to the mean peak area of **6** in the "no dropout" combination (bottom row). Bars indicate mean values; whiskers extend 1.5 standard deviations beyond the mean values; asterisks indicate significantly different mean peak areas compared to the "no dropout" combination (two-tailed Welch's *t* test with Bonferroni correction for $m = 10$ multiple comparisons; $n = 4$ or 5; $P < 0.005$). (**C**) Representative LC-MS extracted ion chromatograms (EICs) showing the accumulation of **6**, **16**, and putative QA pathway intermediates upon transient expression of candidate biosynthetic genes in the leaves of *N. benthamiana* ($n = 5$; harvested at 6 dpi). (**D**) Representative LC-MS EICs showing the accumulation of **6** and its isomers **16** and **17** upon reconstruction of the pathway in the absence (GFP) and in the presence of IRED3 ($n = 6$; harvested at 7 dpi). (**E**) Purity of **6** relative to all three diastereomers of sparteine (upper graph) and yield of **6** (lower graph) in leaves from (D). Boxes in the plots span the interquartile range; whiskers extend to all data points that fall within the inner fences; lines within the boxes indicate median values; and hollow squares indicate mean values. The asterisk in the upper graph indicates a significant difference (two-tailed Welch's *t* test; $n = 6$; $P < 10^{-8}$). Levels in the lower graph were not significantly different (two-tailed Welch's *t* test; $n = 6$; $P > 0.05$). (**F**) EICs from the chiral GC-MS analysis of leaf extracts of *N. benthamiana* showing the percentage enantiomeric excess (%ee) of (–)-**6** upon reconstruction of the pathway in the absence (GFP) and in the presence of IRED3 at 7 dpi.

the levels of **6**. As we suspected functional redundancy in this case, owing to the similarity between the two proteins [91% identity at the protein level (fig. S2B)], we also included a double absence test. The combined absence of IRED1 and IRED2 led to a complete loss of **6** (Fig. 3B), thus demonstrating necessary and redundant functions. Finally, the individual absence of MLPL5 did not lead to any change in the levels of **6** (Fig. 3B), suggesting a lack of involvement or a negligible role in its biosynthesis.

To gain insights into the activity of the implicated enzymes, we analyzed the relative levels of the putative pathway intermediates ammodendrine (**5**), 2-hydroxyammodendrine (**12**), and 1,2-didehydrolusitanine (**13**) in the same *N. benthamiana* leaves expressing the different gene combinations. In addition to these, we also quantified a tetradehydrosparteine species (**15**) (likely a version of the tetracyclic QA intermediate depicted in Fig. 1) that accumulated with particular gene combinations. The absence of MLPL2 virtually eliminated the production of any of these intermediates (as well as the end product) (Fig. 3B), confirming its proposed role in the early QA pathway. The absence of MLPL4 led to the highest amounts of **12** (Fig. 3B) among all enzyme combinations. This chemotype suggests a role for MLPL4 in aiding the rearrangement of **12** to **13**. Omitting HYD led to the virtual absence of both **15** and **6**, a profile that closely resembled that of omitting MLPL1 (Fig. 3B). This result indicates that HYD and MLPL1 are responsible for deacetylation coupled to the addition of the third Δ^1-piperideine unit. Omitting both IRED1 and IRED2 led to accumulation of **15** in the absence of **6**, suggesting a direct role in the reduction of the former to the latter. In turn, the absence of MLPL3 gave a profile resembling that of the double IRED dropout; however, trace amounts of **6** were detected (Fig. 3B). This result suggests an auxiliary role for MLPL3 in said reduction, possibly aiding the dehydration of **15** to a form that is directly reducible by IRED1 or IRED2. Apart from the gene dropout experiment, we also conducted two partial pathway reconstruction experiments in *N. benthamiana*, which supported the proposed roles of MLPL4 and MLPL1 as shown in figs. S21 and S22. Phylogenetic analysis showed that the lupin MLPLs involved in QA biosynthesis belong to a clade of the Bet v 1 protein family (PF00407) that includes other plant enzymes and is separate from the known pollen and foodborne allergens, further supporting their biosynthetic roles (fig. S23).

When producing sparteine (**6**) in *N. benthamiana* as described above, we noticed smaller amounts of two sparteine isomers accumulating in the agroinfiltrated *N. benthamiana* leaves, possibly α- and β-isosparteine (Fig. 3D). Inspired by the discovery that oxidoreductase-like proteins are able to improve the stereoselectivity of the iridoid pathway in *Nepeta* species (*22*), we decided to test our last oxidoreductase candidate enzyme—the putative reductase IRED3—for a similar function in the QA pathway. Coexpression of QA pathway enzymes with IRED3 in *N. benthamiana* leaves (Fig. 3D) significantly improved the relative amount of sparteine (**6**) compared with both α-isosparteine (**16**) and β-isosparteine (**17**) from ~86% to ~95% (Fig. 3E, top graph). Finally, we analyzed the enantiomeric purity of **6** in the presence or absence of IRED3 using chiral GC-MS. We found that IRED3 was able to improve the ee of (−)-**6** from ~60% to ~97% (Fig. 3F). The levels of (−)-**6** produced in *N. benthamiana* in the presence of IRED3 averaged 0.14% on a dry weight basis (Fig. 3E, bottom graph). Coexpression of IRED3 (and MLPL3) in *N. benthamiana* leaves producing **15** did not lead to production of **6**, indicating a distinct function from that of IRED1 and IRED2 (fig. S24; compare gene combinations 11 and 12). Phylogenetic analysis supported the distinct function of IRED3 while showing that the three IREDs belong to the epimerase protein family (PF01370) (fig. S25).

Completion of the pathway toward NLL QAs

We continued our pathway elucidation efforts by feeding (−)-sparteine [(−)-**6**] to *N. benthamiana* and coexpressing the known enzymes CYP71D189 and SDR1 (*10*), upon which we observed the production of (+)-lupanine [(+)-**8**] (Fig. 4A). In the same setting, we then added the only oxidase left in our list of co-regulated enzymes, CYP71A168, and we observed the production of both (+)-13α-hydroxylupanine [(+)-**9**] and (−)-angustifoline [(−)-**10**] (Fig. 4). The relative intensities of (+)-**9** and (−)-**10** between both QAs in our liquid chromatography–mass spectrometry (LC-MS) analysis matched those observed in NLL leaf extracts (Fig. 4A), suggesting that only one enzyme, CYP71A168, is responsible for the production of both QAs in NLL. A suggested mechanism for the reactions catalyzed by CYP71A168 is shown in fig. S26. Finally, addition of the NLL homologs of the known acyltransferase HMT/HLT (*13*) (HLT1 and HLT2) resulted in the production of both tigloyl and benzoyl esters of (+)-**9**, giving a profile that resembled that of NLL leaves (Fig. 4 and fig. S27). We complemented these results by reconstructing the entire pathway stepwise from LDC to HMT/HLT (fig. S24).

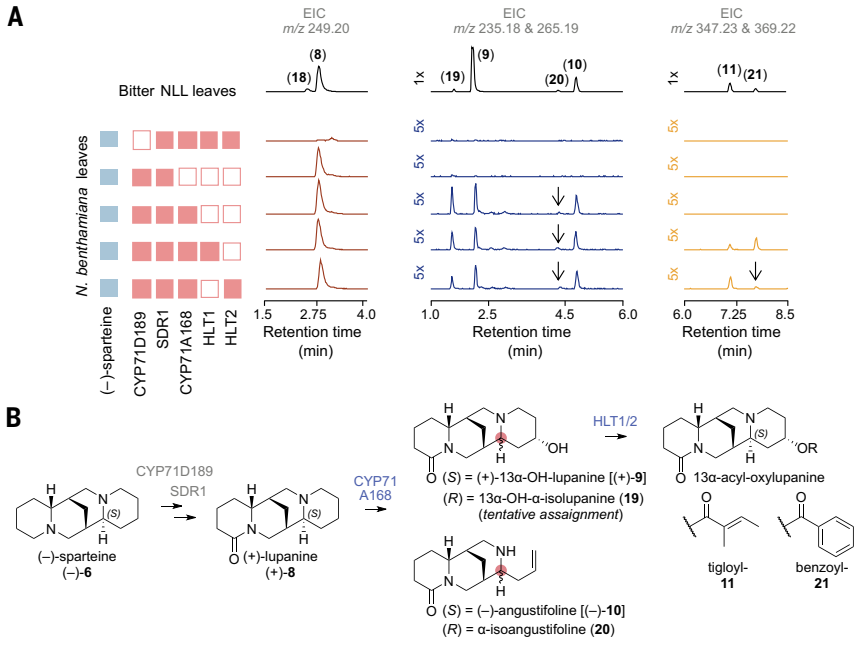

Fig. 4. Reconstruction of the late QA pathway from NLL in *N. benthamiana*. (**A**) Representative LC-MS EICs ($m/z \pm 0.01$) of extracts of *N. benthamiana* leaves coexpressing late pathway candidate enzymes and fed with 50 μg/ml (−)-sparteine [(−)-**6**] ($n = 5$). Leaves were fed with (−)-**6** at 4 dpi and collected 4 days later. EICs from leaves of the bitter NLL cultivar Oskar are included as reference. The oxidation of (+)-lupanine [(+)-**8**] by CYP71A168 leads to the accumulation of (+)-13α-hydroxylupanine [(+)-**9**], (−)-angustifoline [(−)-**10**], and two other compounds, **19** and **20**, which may be their respective α-iso isomers (see fig. S26 for a mechanistic proposal). The acyltransferases HLT1 and HLT2 produce multiple esterified QAs when expressed in *N. benthamiana*, of which 13α-tigloyloxylupanine (**11**) and 13α-benzoxyloxylupanine (**21**) are also abundant in the leaves of NLL (see also fig. S27). The full pathway was also reconstructed separately in *N. benthamiana* leaves demonstrating de novo production of all the QAs shown here from endogenous L-lysine (**1**) (fig. S24). Hollow boxes on the left signal that the corresponding gene was replaced by *GFP* in that combination (row). The indicated chromatogram intensity scale factors are relative to the leftmost chromatograms. (**B**) Proposed scheme for the conversion of (−)-**6** to the three major lupin QAs (+)-**8**, (+)-**9**, and (−)-**10** as well as two esterified QAs, **11** and **21**.

Identification of the sweet gene *exiguus* and its causal mutation in white lupin

We recently reported that an inactivating mutation in the gene encoding AT is responsible for the low-alkaloid (sweet) phenotype of white lupin (*L. albus*) cultivars carrying the sweet gene commonly known as *pauper* (*8*). Other sweet lupin genes exist, but their identity remains unknown. We mined the recently published pangenome of white lupin (*28*, *29*), where we looked for inactivating mutations in the newly discovered biosynthetic genes among sweet cultivars. Cultivars Neuland, Gyulatanya, and Neutra stood out for having such mutations in *MLPL1*. Neuland appeared to have a biallelic mutation, with one allele featuring an 11–base pair deletion in the coding region and the other one featuring a mutation at a splice site (G/GT to C/GT) (fig. S28A). By contrast, Neutra was homozygous for the deletion, and Gyulatanya was homozygous for the splice site mutation (fig. S28A). The unknown sweet gene in cultivar Neuland is referred to as *exiguus* (*30*). Upon sequencing MLPL1 at the genomic DNA level in Neuland, which should have carried both mutations, we found that it was homozygous for the splice site mutation (fig. S28B). cDNA amplification revealed that the main transcript observed in high-QA ("bitter") lines corresponded to two transcripts in Neuland, one shorter and one longer (fig. S28C). Examination of the genomic sequences revealed two possible splice sites (G/GT) upstream and downstream of the mutation that could explain the two transcripts produced instead of the wild-type one (Fig. 5A). To confirm that the mutation led to a nonfunctional gene, we cloned the genomic sequence from Neuland ($MLPL1^{exiguus}$) and from a bitter line ($MLPL1^{WT}$) and expressed them in *N. benthamiana* together with the rest of the QA pathway genes up to (−)-sparteine [(−)-**6**]. Whereas $MLPL1^{WT}$ led to full reconstitution of (−)-**6** biosynthesis, $MLPL1^{exiguus}$ led to results resembling the replacement of the gene by *GFP*, namely small amounts of (−)-**6** concomitant with a larger proportion of α-isosparteine (**16**) (Fig. 5B). Furthermore, the accumulation of pathway intermediates observed for $MLPL1^{exiguus}$ and *GFP* resembled the one observed in leaves of Neuland (Fig. 5C). Bitter white lupin accumulates mostly lupanine (**8**) and only trace amounts of α-isolupanine (**18**); by contrast, **8** in Neuland is reduced to levels similar to **18** (Fig. 5D), mirroring the results seen in *N. benthamiana* for sparteine (**6**) and α-isosparteine (**16**) (Fig. 5B). This chemotype suggests that MLPL1 adds the third Δ^1-piperideine (**3**) unit in a stereoselective way and that the addition occurs slowly and nonstereoselectively in the absence of MLPL1. Alternatively, MLPL1 may enable a dynamic kinetic resolution of **15** to give predominantly **6** (relative to **16**) as result of the specificity of downstream enzymes, as proposed in the case of MLPL2 and MLPL6.

Engineering a sweet gene in NLL

We have previously used mild chemical mutagenesis to generate a very large mutant library of the bitter NLL cultivar Oskar (*8*), from which individual mutants can be isolated by high-throughput genetic screening (FIND-IT technology) (*8*, *31*). To further exemplify the utility of elucidating the QA pathway, we isolated an *HYD* knockout mutant (HYD^{KO}) (fig. S29). HYD is the hydrolase likely responsible for removing the acetyl group from the rearranged dipiperidine intermediate now containing the quinolizidine core scaffold (Fig. 3A). The homozygous HYD^{KO} knockout plant (HYD^{KO}) did not accumulate virtually any of the QAs present in the bitter parent cultivar, with only (+)-13α-hydroxylupanine [(+)-**9**] being just above detection levels in seeds (Fig. 5, E and F). Instead, HYD^{KO} accumulated small amounts of early pathway intermediates and shunt products (Fig. 5, E and F) with predominance of a compound with the spectral characteristics of lusitanine (**25**), a bicyclic QA reported to accumulate in *Maackia* species (*32*). We had detected this compound previously in the gene dropout experiment described above when exchanging *HYD* for *GFP* (fig. S30, A and B). LC-MS analysis of *M. fauriei* leaves confirmed that it contained the same compound (fig. S30, A and B). We speculate that, in the absence of HYD, its substrate undergoes imine reduction to give small amounts of **25** (fig. S31B). Unfortunately, quantification of **25** was not possible owing to the absence of a standard. Given the accumulation of epilupinine (**23**) and lupinine (**24**) (fig. S30, C and D) observed in $MLPL1^{exiguus}$ and not in HYD^{KO} plants (Fig. 5, C, E, and F), we suggest that the catalytic role of HYD precedes that of MLPL1 in the QA pathway (fig. S31B).

Isolation and analysis of additional QA biosynthetic mutants in NLL

To test our biosynthetic hypothesis, we isolated seven additional knockout (KO) lines from our NLL mutant library. The homozygous mutants showed chemotypes that were consistent with the proposed QA pathway (fig. S31). For example, individual ablation of CYP76E36 and CYP71A168 led to accumulation of their respective substrates, ammodendrine (**5**) and lupanine (**8**), as well as drastic decreases in downstream metabolites. Knocking out MLPL4—for which we propose a role in the rearrangement of 2-OH-ammodendrine (**12**) into 1,2-didehydrolusitanine (**13**)—resulted in a mild reduction in QA levels consistent with the catalyzed equilibrium also being spontaneous (fig. S31B). Furthermore, $MLPL4^{KO}$ plants accumulated both **12** and **13**, with a higher proportion of **12** compared with HYD^{KO} plants (HYD is the enzyme immediately downstream of MLPL4). The chemotype of $MLPL4^{KO}$ also resembles the MLPL4 dropout in the pathway reconstruction experiment in *N. benthamiana* (Fig. 3B). As a final example, knocking out IRED3 led to a mild reduction in QA levels and, in this case, a marked accumulation of (+)-sparteine [(+)-**6**] and (−)-β-isosparteine [(−)-**17**] (fig. S32). For a full discussion on the chemotypes of all mutants, see the supplementary text in the supplementary materials.

Genomic location of biosynthetic genes and spatial localization of the pathway

Genes involved in the biosynthesis of plant specialized metabolites sometimes occur in physical proximity on genomes, forming gene clusters that streamline genetic inheritance and aid coexpression (*33*). We located the QA biosynthetic genes in the reference genome of NLL. Apart from a few local duplications leading to redundancy, the genes are scattered throughout the genome (fig. S33). Nevertheless, tight coexpression can be observed up to the oxidation of (−)-sparteine [(−)-**6**] by CYP71D189, with predominant expression in the epidermis of green organs (as noted previously for *LDC*, *CAO*, and *AT*) (fig. S34) (*15*). The downstream biosynthetic genes—*SDR1*, *CYP71A168*, and *HLT1/2*—are also expressed in the epidermis; however, they are additionally expressed in the neighboring mesophyll as well as the vasculature (fig. S34). With respect to predictions of subcellular localization, the pathway appears to be mostly cytosolic, with the exception of LDC being chloroplastic, CAO being peroxisomal, and HLT1/2 also being peroxisomal (table S3). A confident establishment of cellular and subcellular localization will require dedicated work (e.g., single-cell transcriptomics and localization of fluorescent protein fusions). We present a preliminary model of the cellular and subcellular localization of QA biosynthesis in fig. S35.

Discussion

In this study, we have uncovered the pathway toward most of the QAs that accumulate in NLL, including nine newly identified biosynthetic enzymes (Fig. 6). Among the newly described pathway steps, four of them are catalyzed by MLPLs. Although the results presented here do not allow us to propose exact mechanisms, it is notable that all MLPL-catalyzed steps are spontaneous to some degree and involve reactivity at imine centers. In aqueous solution, imines are likely in equilibrium with several other forms, including open aminoaldehyde forms and hemiaminal forms (fig. S1). Thus, it is likely that MLPLs aid the interconversion between forms, effectively accelerating the establishment of chemical equilibria as demonstrated in vitro for MLPL2. We have also shown that IRED3 improves the overall selectivity toward

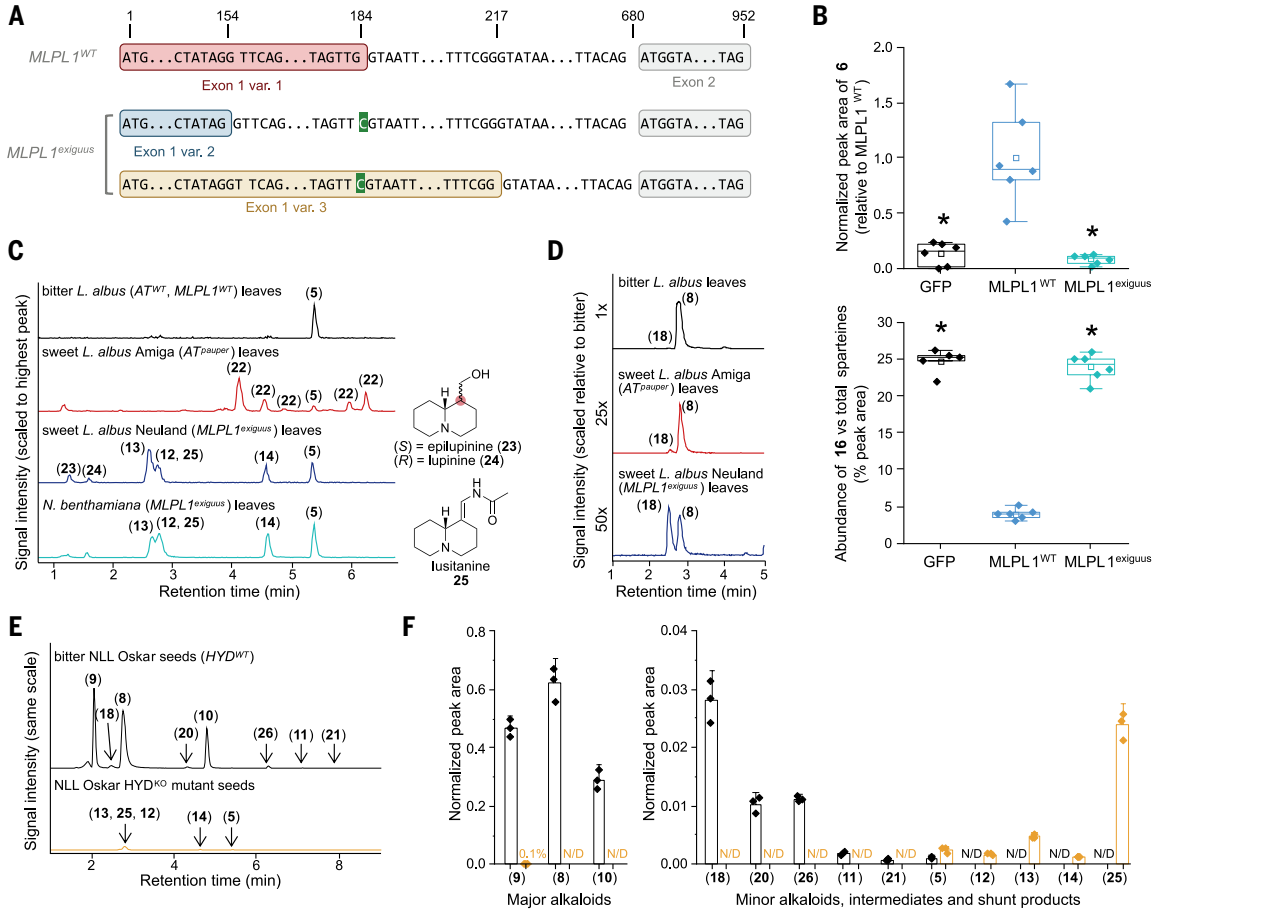

Fig. 5. Characterization of two lupin QA biosynthetic mutants. All *m/z* values are provided in the Materials and methods. (**A**) Gene models of *MLPL1* from bitter white lupin (*MLPL1*^{WT}) and from the sweet white lupin cultivar Neuland (*MLPL1*^{exiguus}) as observed by sequencing of genomic DNA and cDNA (fig. S28, B to D). The *exiguus* splice site mutation (G184C) is highlighted in green. (**B**) Abundance of sparteine (**6**) and α-isosparteine (**16**) in the leaves of *N. benthamiana* transiently coexpressing LDC, CAO, AT, CYP76E36, HYD, MLPL2-4, and IRED1-3 in combination with either GFP, MLPL1^{WT}, or MLPL1^{exiguus}. Leaves were collected 7 dpi. Boxes in the plots span the interquartile range; whiskers extend to all data points that fall within the inner fences; lines within the boxes indicate median values; and hollow squares indicate mean values. Asterisks indicate a significant difference compared with the MLPL1^{WT} control (two-tailed Welch's *t* test with Bonferroni correction for $m = 2$ comparisons; $n = 6$; $P < 0.025$). (**C**) Representative LC-MS EICs of bitter white lupin leaves compared with the sweet cultivar Neuland (*MLPL1*^{exiguus}) and the sweet cultivar Amiga, which carries a different sweet gene called *pauper* (*AT*^{pauper}) (*8*) ($n = 5$). *N. benthamiana* leaves expressing MLPL1^{exiguus} from (B) (*exiguus* chemotype mimic) are included for comparison. The structure of *pauper* metabolites (**22**) (*8*) and ammodendrine metabolite (**14**) (Fig. 2B and fig. S21) is unknown. (**D**) Representative LC-MS EICs showing the accumulation of lupanine (**8**) in comparison to its isomer α-isolupanine (**18**) in the leaves of bitter white lupin, sweet cultivar Amiga (*AT*^{pauper}), and sweet cultivar Neuland (*MLPL1*^{exiguus}) ($n = 5$). Chromatogram intensity scale factors are indicated next to the *y* axis. (**E**) Representative LC-MS EICs showing the accumulation of QAs and QA pathway metabolites in the seeds of NLL cultivar Oskar carrying the *HYD*^{WT} and *HYD*^{KO} alleles ($n = 3$). Peak (**26**) corresponds to the esterified QA 13-[3-(4-OH-phenyl)lactoyl]oxylupanine (structure not shown). The sum of peak areas of QA pathway metabolites in the mutant was ~2.3% of the sum of peak areas of QAs in the parent line. (**F**) Distribution and abundance of individual QAs and of QA pathway metabolites in seed extracts of NLL cultivar Oskar carrying the *HYD*^{WT} and *HYD*^{KO} alleles. Peak areas are normalized to the area of the internal standard (average intensity ~1.1×10^6). Bars indicate mean values; whiskers extend 1.5 standard deviations beyond the mean values ($n = 3$). The percentage indicates residual levels of 13α-hydroxylupanine (**9**) in HYD^{KO} relative to HYD^{WT} seeds. "N/D" denotes metabolite levels below detection limit.

(−)-sparteine [(−)-**6**], both with respect to its diastereomers α-isosparteine (**16**) and β-isosparteine (**17**) as well as its enantiomer (+)-**6**. We speculate that IRED3 ensures preponderance of the stereochemically correct tetradehydrosparteinium intermediate (**28**), either by aiding MLPL3 in its selective production or by bringing spontaneously produced alternative forms back into the correct form, possibly in a redox-neutral way (fig. S31C). Elucidation of the precise catalytic roles of IRED3 and the many MLPLs involved in the QA pathway will require dedicated experiments. Finally, the quinolizidine core common to all QAs arises in an acetylated state, and the acetyl group is removed in the next step. It remains unknown whether there is a mechanistic reason behind this cryptic acetylation, or whether it reflects the stepwise evolution of a long pathway in which acetylated precursors may have served an ancestral biological function.

The elucidation of the QA pathway in NLL has several practical applications. First, it will enable the heterologous production of (−)-**6**, a valuable auxiliary molecule in chiral chemical synthesis for which there is an unmet demand (*34*). Production is envisioned either as shown here in *N. benthamiana* or alternatively in microbes such as yeast (*Saccharomyces cerevisiae*), where it may be more readily scalable. Whether heterologous production can compete with production in the recently created *CYP71D189* knockout NLL plants [which accumulate (−)-**6** themselves] (*10*) remains to be determined. Second, knowledge of the QA pathway in lupins can facilitate the identification

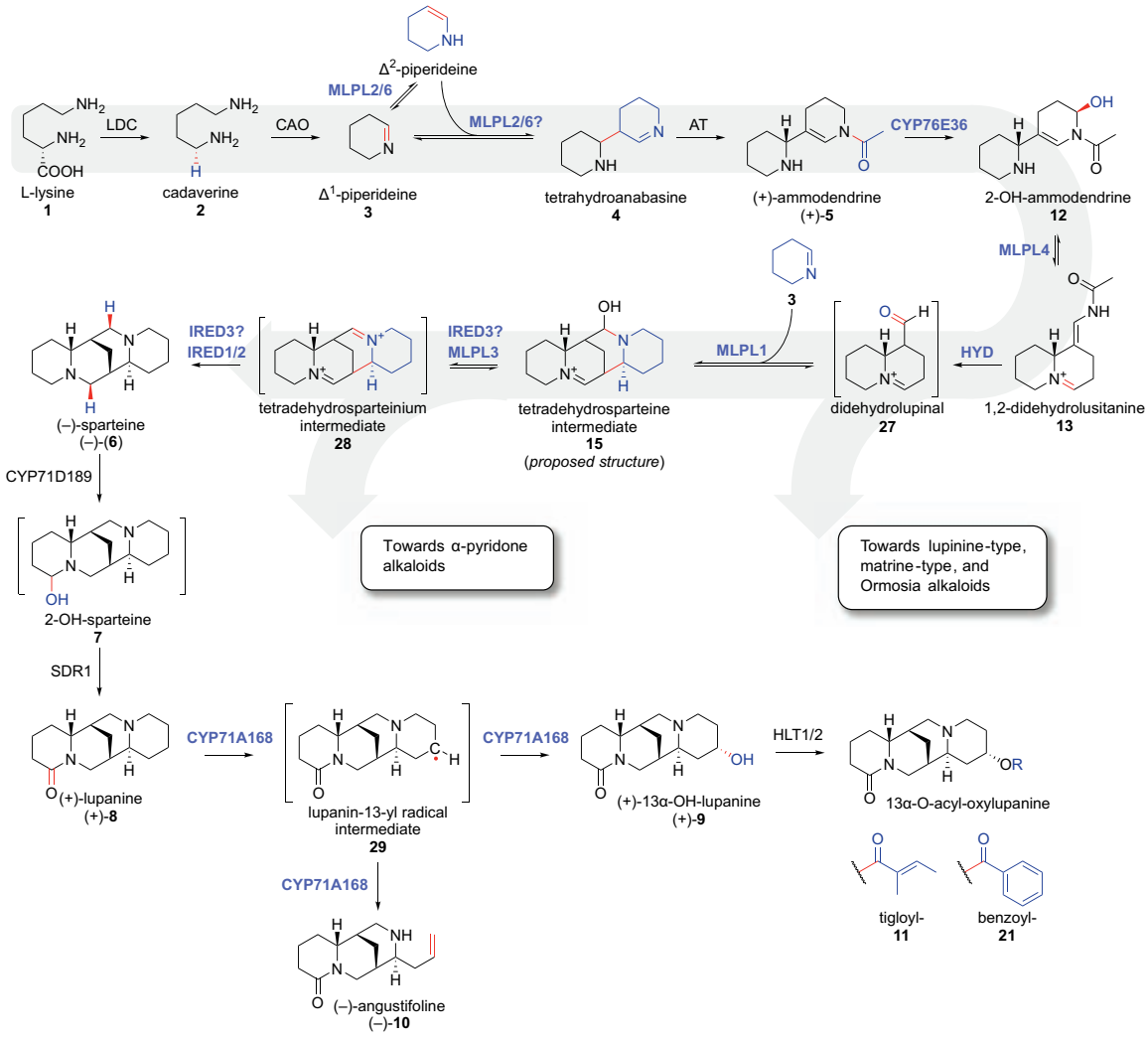

Fig. 6. Proposed biosynthetic pathway of QAs. We have identified key enzymes in the biosynthesis of QAs, which are not only produced by lupins (*Lupinus* spp.) but also by most other members of the genistoid *sensu lato* clade of legumes (*61*) [~10% of all known legume species; number of species retrieved from Plants of the World Online (*62*)]. The gray arrows in the background highlight the predicted main branching points in QA biosynthesis within genistoid legumes. Hypothetical intermediates are surrounded by square brackets. Enzymes newly identified in this study are shown in violet. Question marks indicate tentative functional assignments. New bonds formed after each enzymatic step are drawn in red. New atoms introduced after each enzymatic step are drawn in blue.

of existing sweet lupin genes, of which only one has been unambiguously identified (mutant allele AT^{pauper}) (*8*). As an example, we here unveiled the *exiguus* sweet gene as a mutant allele of *MLPL1*. Notably, the causal mutation (splice site polymorphism) represents a so-called perfect marker that white lupin breeders can use for efficient introgression into bitter germplasm or for stacking on top of the *pauper* gene (*35*). Third, knowledge of several newly identified QA biosynthetic genes provides opportunities for the generation of novel sweet lupins as exemplified by the HYD^{KO} NLL mutant. This line did not accumulate any of the QAs normally present in NLL seeds (below detection limits) but did accumulate low levels of a few pathway intermediates and shunt products, notably lusitanine (**25**). Pharmacological tests in mice (*36*) and brine shrimp (*37*) suggest that the toxicity of **25** lies in between that of (+)-lupanine [(+)-**8**] and (+)-13α-hydroxylupanine [(+)-**9**] (*38*). Thus, quantification of lusitanine (**25**) in this mutant will be crucial to determine its safety compared with already existing sweet NLL varieties (which do accumulate low levels of **8**, **9**, and other intermediates and shunt products). In any case, stacking of the sweet HYD^{KO} allele on top of the existing sweet *iucundus* allele (*39*, *40*) has the potential to produce NLL lines with ultra-low QA levels that fall consistently below food safety thresholds (*35*). We envision that the generation of such sweet lines by knocking out selected QA biosynthetic genes will be possible not only in various lupin species but also in other QA-containing legumes, some of which have agricultural potential.

Materials and methods

Analytical and preparative methods

Metabolite analysis by LC-MS: Analytical LC-MS was carried out on a Thermo Fisher Dionex 3000 RS HPLC/UPLC (high-performance liquid chromatography/ultrahigh-performance liquid chromatography) system [Thermo Fisher Scientific (Waltham, MA, USA)] interfaced to a Bruker compact QqTOF mass spectrometer [Bruker (Billerica, MS, USA)] through an electrospray ionization (ESI) source.

LC method 1 [used only in fig. S30, A and B, and to test the purity of synthesized ammodendrine (**5**) (fig. S4A)]: Analytes were separated on a Kinetex XB-C18 column (100 mm by 2.1 mm, 1.7 μm, 100 Å) [Phenomenex (Torrance, CA, USA)] kept at 40°C. Mobile phases A and B consisted of, respectively, 0.05% formic acid in water and 0.05% formic

acid in acetonitrile. Analytes were eluted using the following gradient at a constant flow rate of 0.3 ml/min: 0 to 1 min, 2% B (constant); 1 to 16 min, 2 to 25% B (linear); 16 to 24 min, 25 to 65% B (linear); 24 to 26 min, 65 to 100% B (linear); 26 to 27 min, 100% B (constant); 27 to 27.5 min, 100 to 2% B (linear); and 27.5 to 33 min, 2% B (constant).

LC method 2 (used everywhere else): Analytes were separated on a Luna C18(2) column (150 mm by 2 mm, 3 μm, 100 Å) [Phenomenex (Torrance, CA, USA)] kept at 40°C. Mobile phases A and B consisted of, respectively, 0.05% formic acid in water and 0.05% formic acid in acetonitrile. Analytes were eluted using the following gradient at a constant flow rate of 0.3 ml/min: 0 to 0.5 min, 2% B (constant); 0.5 to 2.375 min, 2 to 6% B (linear); 2.375 to 7 min, 6 to 25% B (linear); 7 to 13 min, 25 to 100% B (linear); 13 to 14 min, 100% B (constant); 14 to 14.5 min, 100 to 2% B (linear); and 14.5 to 20 min, 2% B (constant).

ESI mass spectra [mass/charge ratio (*m/z*) 50 to 1000] were acquired in positive ionization mode with automatic MS^2 acquisition using the following parameters: capillary voltage, 4500 V; end plate offset, −500 V; source temperature, 250°C; desolvation gas flow, 8.0 liters/min; and nebulizer pressure, 2.5 bar. N_2 was used as desolvation, nebulizer, and collision cell gas. Caffeine was used as internal standard for quantification and peak area normalization. The identity of sparteine (**6**) and lupinine (**24**) was confirmed by comparison with commercial standards from Sigma-Aldrich (St. Louis, MO, USA) and that of lupanine (**8**) with commercial standards from Innosil Sp. z o.o. (Poznan, Poland). See the section "Identity and structure of the numbered compounds detected in plant extracts" in the supplementary text for details on the identification of other metabolites.

QA analysis and analysis of the MLPL2 in vitro assays by GC-MS: GC-MS was carried out on a Shimadzu GC-2010 gas chromatograph equipped with a Shimadzu AOC-5000 autosampler and coupled to a Shimadzu GCMS-QP2010 Plus single quadrupole mass spectrometer [Shimadzu (Kyoto, Japan)]. Analytes were separated on an Agilent J&W HP-5MS UI capillary column (30 m by 0.25 mm by 0.25 μm) [Agilent Technologies (Santa Clara, CA, USA)] using He as carrier gas.

For the determination of QAs in *N. benthamiana* leaf samples, ethyl acetate extracts were injected in splitless mode (1 μl) at an inlet temperature of 250°C. QAs were separated using the following column temperature program at a constant gas linear velocity of 33.7 cm/s: initial 80°C, hold for 3 min; ramp to 150°C at 30°C/min; ramp to 300°C at 6°C/min; hold for 10 min. The GC-to-MS interface temperature was set at 250°C. The separated analytes were ionized using an electron impact ion source at 250°C. MS spectra were acquired in scan mode (*m/z* 30 to 600) at an energy of 70 eV. QAs were identified by comparison to spectral library data and to previously reported fragmentations (*9*).

For the analysis of the MLPL2 in vitro assays, ethyl acetate extracts were injected in splitless mode (1 μl) at an inlet temperature of 290°C. The reduced and benzoylated products were separated using the following column temperature program at a constant gas linear velocity of 33.7 cm/s: initial 80°C, hold for 3 min; ramp to 150°C at 30°C/min; ramp to 300°C at 20°C/min; hold for 15 min. To reduce condensation, the GC-to-MS interface temperature was set at 325°C. The separated analytes were ionized using an electron impact ion source at 250°C. MS spectra were acquired in scan mode (*m/z* 10 to 500) or in SIM mode (*m/z* 175, 174, and 146 for *N*-benzoylpyrrolidine, RT 9.27 min; *m/z* 189 and 188 for *N*-benzoylpiperidine, RT 9.44 min; and *m/z* 271, 189, and 188 for *N,N'*-dibenzoyl-2,3′-bipiperidines, RT 20.02 min and 20.79 min) at an energy of 70 eV. Quantitative analysis was done in SIM mode using *m/z* 174 for *N*-benzoylpyrrolidine (internal standard) and *m/z* 188 for *N*-benzoylpiperidine and *N,N'*-dibenzoyl-2,3′-bipiperidines. *N*-benzoylpiperidine was identified by comparison with a commercial standard [Tokyo Chemical Industry (Tokyo, Japan)]. *N*-benzoylpyrrolidine was identified by comparison with spectral library data. *N,N'*-dibenzoyl-2,3′-bipiperidines were identified on the basis of their MS spectrum (fig. S6, C and F).

Chiral analysis by GC-MS: Chiral GC-MS was carried out on a Shimadzu Nexis GC-2030 gas chromatograph equipped with a Shimadzu AOC-6000 autosampler and coupled to a Shimadzu GCMS-QP2020 NX single quadrupole mass spectrometer [Shimadzu (Kyoto, Japan)]. Analytes were separated on an Agilent J&W CycloSil-B capillary column (30 m by 0.25 mm by 0.25 μm) [Agilent Technologies (Santa Clara, CA, USA)] using He as carrier gas.

Ethyl acetate extracts were injected in splitless mode (1 μl) at an inlet temperature of 250°C. (+)- and (−)-sparteine (**6**) were separated using the following column temperature program at a constant carrier gas pressure of 51.0 kPa: initial 80°C, hold for 3 min; ramp to 125°C at 30°C/min; ramp to 240°C at 2°C/min. (+)- and (−)-ammodendrine (**5**) were separated using the following column temperature program at a constant carrier gas pressure of 51.0 kPa (fig. S4B) or at a constant carrier gas linear velocity of 33.7 cm/s (fig. S3A): initial 80°C, hold for 3 min; ramp to 125°C at 30°C/min; hold for 540 min (fig. S3A) or 120 to 540 min (fig. S4B); ramp to 210°C at 1°C/min. The separated analytes were ionized using an electron impact ion source at 250°C. MS spectra were acquired in scan mode (*m/z* 10 to 300) at an energy of 70 eV. (+)- and (−)-sparteine (**6**) were identified by comparison with commercial standards [Sigma-Aldrich (St. Louis, MO, USA)]. Racemic ammodendrine [(+)/(−)-**5**] was synthesized from isotripiperideine as described below in the section_"Synthesis and resolution of (+)/(−)-ammodendrine [(+)/(−)-**5**]."

Preparative HPLC: Purification of selected metabolites was carried out on a Shimadzu preparative HPLC system with ultraviolet detection (190 to 350 nm) and automatic fraction collection [Shimadzu (Kyoto, Japan)]. The system entailed a DGU-20A_5 degasser, a SIL-10AP autosampler, an LC-20AT pump, a CTO-10ASvp column oven, an SPD-M20A photodiode array detector, and an FRC-10A fraction collector. Metabolites were separated on a Luna C18(2) semi-preparative column (250 mm by 10 mm, 5 μm, 100 Å) [Phenomenex (Torrance, CA, USA)] kept at 30°C. Mobile phases A and B consisted of, respectively, 0.05% formic acid in water and 0.05% formic acid in acetonitrile.

Metabolites were eluted using the following gradient at a constant flow rate of 2 ml/min: 0 to 2 min, 2% B (constant); 2to 37 min, 2to 20% B (linear); 37 to 42 min, 20 to 95% B (linear); 42 to 45 min, 95% B (constant); 45 to 50 min, 95 to 5% B (linear); and 50–62 min, 2% B (linear). One chromatographic peak (λ_{max} = 235 nm) was collected in the fraction eluting between 18.80 and 19.13 min. The peak could be resolved on analytical LC-MS, confirming that it consisted of the two closely eluting intermediates 2-hydroxyammodendrine (**12**) and 1,2-didehydrolusitanine (**13**), with trace amounts of the third metabolite of ammodendrine (**14**) possibly arising from equilibration with **12** and/or **13** (fig. S8).

m/z* values and scaling of chromatograms in main figures:** In Fig. 2A, ***5 was quantified from extracted ion chromatograms (EICs) of *m/z* 209.16 ± 0.01 ($[M+H]^+$ ion of **5**). In Fig. 2B, traces are combined EICs of *m/z* 207.15 ± 0.01 ($[M]^+$ of **13**, $[M\text{-}H_2O+H]^+$ of **12**, and prominent ion of **14**), 209.16 ± 0.01 ($[M+H]^+$ of **5**), and *m/z* 225.16 ± 0.01 ($[M+H]^+$ of **12** and prominent ion of **14**). Signal intensities were scaled to the highest peak.

In Fig. 3C, traces are combined EICs of *m/z* 207.15 ± 0.01 ($[M]^+$ of **13** and $[M\text{-}H_2O+H]^+$ of **12**), *m/z* 209.16 ± 0.01 ($[M+H]^+$ of **5**), *m/z* 235.22 ± 0.01 ($[M+H]^+$ of **6** and **16**), and *m/z* 249.20 ± 0.01 ($[M]^+$ of **15**). The three chromatograms have the same intensity scale. In Fig. 3D, traces are combined EICs of *m/z* 118.11 ± 0.01 ($[M+2H]^{2+}$ of **6**, **16**, and **17**) and *m/z* 235.22 ± 0.01 ($[M+H]^+$ of **6**, **16**, and **17**). The three chromatograms have the same intensity scale. In Fig. 3F, traces are EICs of *m/z* 98 ± 0.5 (prominent EI peak of **6**). The chromatograms are on

different intensity scales, and each chromatogram is scaled to the height of the highest peak.

In Fig. 5B, **6**, **16**, and **17** were quantified from combined EICs of *m/z* 118.11 ± 0.01 ($[M+2H]^{2+}$ of **6**, **16**, and **17**), and *m/z* 235.22 ± 0.01 ($[M+H]^+$ of **6**, **16**, and **17**). In Fig. 5C, traces are combined EICs of *m/z* 170.15 ± 0.01 ($[M+H]^+$ of **23** and **24**), 207.15 ± 0.01 ($[M]^+$ of **13**, $[M-H_2O+H]^+$ of **12**, and prominent ion of **14**), 209.16 ± 0.01 ($[M+H]^+$ of **5** and **25**, and prominent ion of **22**), and 225.16 ± 0.01 ($[M+H]^+$ of **12** and prominent ion of **14**). The four chromatograms are on different intensity scales, and each chromatogram is scaled to the height of the highest peak. In Fig. 5D, traces are EICs of *m/z* 249.20 ± 0.01 ($[M+H]^+$ of **8** and **18**). In Fig. 5E, traces are combined EICs of *m/z* 207.15 ± 0.01 ($[M]^+$ of **13**, $[M-H_2O+H]+$ of **12**, and prominent ion of **14**), 209.16 ± 0.01 ($[M+H]^+$ of **5** and **25**), 225.16 ± 0.01 ($[M+H]^+$ of **12** and prominent ion of **14**), 235.18 ± 0.01 ($[M+H]^+$ of **10** and **20**), 249.20 ± 0.01 ($[M+H]^+$ of **8** and **18**), 265.19 ± 0.01 ($[M+H]^+$ of **9**), 347.23 ± 0.01 ($[M+H]^+$ of **11**), 369.22 ± 0.01 ($[M+H]^+$ of **21**), and 429.24 ± 0.01 ($[M+H]^+$ of **26**). The two chromatograms have the same intensity scale. In Fig. 5F, compounds were quantified from the EICs of their respective $[M]^+$ or $[M+H]^+$ ions (see previous sentence for *m/z* values).

Selection of candidate biosynthetic genes

We selected candidate QA biosynthetic genes by coexpression analysis with *LDC* in NLL. We used a previously published NLL transcriptome as reference (*7*) and obtained organ-specific transcript abundances [in transcripts per million (TPM)] from two publicly available RNA sequencing (RNA-seq) datasets (*7*, *20*) using Kallisto (*41*) (v0.43.0; kallisto quant was run with default parameters). Specifically, transcript abundances were quantified in eight different organs of the bitter cultivar Oskar (NCBI BioProject PRJNA386115) and five different organs of the sweet cultivar Tanjil (NCBI BioProject PRJNA248164). Tanjil is thought to carry a deleterious mutation in a transcriptional activator of QA biosynthetic genes (*39*). Organs that are normally biosynthetically active (leaves, stems, pedicels, and pods) as well as inactive (flowers, roots, and seeds) were included in the analysis (*15*, *16*, *42*). The transcript encoding *LDC* was identified by sequence homology, and Pearson correlation coefficients (*43*, *44*) were calculated between the expression vector of the *LDC* transcript and that of all the other transcripts (Pearson correlation coefficient is a general measure of correlation and is often used to assess coexpression between genes involved in specialized metabolism in plants) (*45*–*47*).

Transcripts with low expression levels (<10 TPM) in all organs were not included in the analysis. The 56 transcripts with Pearson correlation coefficient to *LDC* > 0.88 were subjected to BLASTX against green plant protein sequences from NCBI (taxid: 33090), and the annotation and percentage identity (%ID) of the first 40 results were recorded for each transcript. Transcripts were separated into specific and general transcripts on the basis of the %ID of the first 40 BLASTX results, with 31 transcripts encoding specific proteins with six or more BLASTX results with %ID < 70%. Of these, 10 were putative enzymes, 8 were uncharacterized proteins, 1 was a putative porphyrin-sequestering protein, 1 was a putative phospholipid transfer protein, 1 was a putative chloroplast import protein, 1 was a putative sugar phosphate translocator, 1 was a putative RNA-binding protein, and the remaining 8 encoded proteins of the major latex protein family. After removal of redundant transcripts, eight unique enzymes (three CYP450s, three reductases, one hydrolase, and one acyltransferase) and six MLPLs were left (table S1). Notably, one of the three CYP450s was CYP71D189, which catalyzes the first step in the oxidation of (−)-sparteine [(−)-**6**] to (+)-lupanine [(+)-**8**] (*10*). In turn, the acyltransferase was AT, which catalyzes the acetylation of tetrahydroanabasine (**4**) to ammodendrine (**5**) (*8*).

The coding sequences of all 14 genes were cloned and tested for activity in the QA biosynthesis pathway in addition to the remaining four known biosynthetic genes *LDC* (*6*), *CAO* (*7*), *SDR1* (*10*), and *HMT/HLT* (*13*), as described below in the section "Transient expression of biosynthetic gene candidates in *N. benthamiana*." Note that in our distinction between specific and general transcripts, the two first pathway genes, *LDC* and *CAO*, were classified as general. Both genes are derived from biosynthetic genes in polyamine biosynthesis, and the respective enzymes still metabolize substrates that are highly similar to their polyamine pathway cognates [lysine (**1**) and cadaverine (**2**), instead of ornithine and putrescine]. Accordingly, we expected the close homologs of *LDC* and *CAO* from non–QA-producing legumes to encode enzymes in polyamine biosynthesis. The next known gene in the pathway, *AT*, was classified as specific, and so was *CYP71D189*; thus, we were confident in applying the criterium of phylogenetic specificity in our search for pathway genes downstream of *CAO*.

HMT/HLT was originally described in white lupin (*13*) (gene ID Chr15 g0077591). We used BLASTP to find the two closest homologs in NLL cultivar Oskar, which we called HLT1 (75%ID to Chr15 g0077591) and HLT2 (86%ID to Chr15 g0077591) (table S1).

Transient expression of biosynthetic gene candidates in *N. benthamiana*

Total RNA was extracted from young leaves of the bitter NLL cultivar Oskar using the Spectrum Plant Total RNA Kit [Sigma-Aldrich (St. Louis, MO, USA)] and was converted to cDNA using the iScript cDNA Synthesis Kit [Bio-Rad (Hercules, CA, USA)]. The full-length coding regions of *LDC*, *CAO*, *AT*, *CYP71D189*, *SDR1*, and *HLT1-2* and of the gene candidates *MLPL1-6*, *CYP76E36*, *HYD*, *IRED1-3*, and *CYP71A168* were amplified from young leaf cDNA of cultivar Oskar using the primers specified in table S2. The genes were cloned into the plant expression vector pEAQ-USER (*48*) by USER cloning and transformed into *E. coli* strain Top10. Clones were sequenced by Sanger sequencing, and correct clones were transformed into electrocompetent *Agrobacterium tumefaciens* strain AGL-1. For agroinfiltration, AGL-1 strains were grown on YEP liquid medium supplemented with 50 μg/ml kanamycin, 25 μg/ml rifampicin, and 50 μg/ml carbenicillin at 28°C and 220 rpm until $OD_{600} \approx 3$ to 4. The bacteria were collected by centrifugation and resuspended in ultrapure H_2O to $OD_{600} = 1$. The diluted AGL-1 strains were mixed in equal portions to obtain the desired gene combination for coexpression in leaves of *N. benthamiana*. After a 1-to-3-hour incubation at room temperature, the mixtures were infiltrated into the abaxial side of young leaves of 4-week-old *N. benthamiana* plants. The agroinfiltrated plants were allowed to recover overnight in the dark before being brought back to the greenhouse. In selected experiments, exogenous (−)-sparteine [(−)-**6**] was supplied as precursor by re-infiltrating previously agroinfiltrated leaves 3 to 4 days after agroinfiltration with an aqueous solution containing 50 μg/ml of the alkaloid, 10 mM $MgCl_2$, and 10 mM 2-(*N*-morpholino)ethanesulfonic acid (MES) [adjusted to pH 5.6 with $NaOH_{(aq)}$]. In all cases, agroinfiltrated leaves were harvested 5 to 12 days after agroinfiltration. Two 1-cm leaf discs were punched out of the agroinfiltrated (discolored) portion of the leaves, collected in 1.5 ml screw-cap microcentrifuge tubes, and either flash-frozen in liquid nitrogen (Fig. 2B) or dried for 1 to 4 days at 40°C (all other figures). In both cases, the leaf discs were then pulverized using a steel bead and a TissueLyzerII bead beater [Qiagen (Venlo, The Netherlands)] and extracted with 250 μl of extractant [60% methanol, 0.06% formic acid, and 5 parts per million (ppm) caffeine in water] for 3 hours by shaking on an orbital shaker at 400 rpm. The extracts were briefly centrifuged to remove large particulates, diluted five times with ultrapure water, and filtered through a 0.22 μm polyvinylidene fluoride (PVDF) filter. The extracts were then analyzed by analytical LC-MS as described above in the "Analytical and preparative methods" section. Qualitative analysis of QAs produced in *N. benthamiana* was also done by GC-MS, as described below in the paragraph regarding the full reconstruction of the pathway.

In the gene dropout experiment for the biosynthesis of sparteine (**6**) (Fig. 3B), we compared leaves of *N. benthamiana* that express 12

QA biosynthetic enzyme candidates (LDC, CAO, AT, CYP76E36, HYD, IRED1-2, and MLPL1-5) to leaves where selected candidates are replaced by biosynthetically inactive GFP (to maintain the relative proportions of each gene in all combinations). Replacing a biosynthetic enzyme with GFP is expected to create a bottleneck in the pathway, creating a metabolic effect that helps infer the role of the dropped-out enzyme in the pathway.

In the stepwise reconstruction of the entire pathway (fig. S24), we transiently expressed different combinations of 17 QA biosynthetic genes in leaves of *N. benthamiana*. The combinations were selected in such a way as to extend the pathway by one step (from LDC to HLT1/2) with each subsequent combination according to the proposal on Fig. 6. To prepare each combination, *Agrobacterium* strains for expression of biosynthetic genes and of GFP were cultured and diluted to $OD_{600} = 1$ as described above. To prepare the different gene combinations, equal amounts of diluted cultures for expression of the required biosynthetic genes were first mixed. As the total number of biosynthetic genes in each combination varied, the mixed cultures were diluted to the same final volume using an equally diluted culture for expression of biosynthetically inactive GFP. This ensured that the dosage of every biosynthetic gene was the same in all combinations. The experiment also includes a dropout of IRED3 (combination 17 in fig. S24). For qualitative analysis of the QAs produced upon full reconstruction of the pathway, *N. benthamiana* leaves expressing all proposed biosynthetic genes (combination 16 in fig. S24) were dried and pulverized as described above. Approximately one hundred milligrams of pulverized leaf was extracted in 2 ml 1 M $HCl_{(aq)}$ by shaking for 1 hour on an orbital shaker at 1200 rpm. The extracts were clarified by centrifugation. The clarified extracts were basified with 1 ml 2 M $NaOH_{(aq)}$ and extracted with 3 × 2 ml portions of ethyl acetate. The extracts were dried with anhydrous Na_2SO_4, and the solvent was evaporated under reduced pressure. The residue was reconstituted in 100 µl of ethyl acetate and analyzed GC-MS as described under "Analytical and preparative methods."

Agrobacterium-mediated transient expression of genes in *N. benthamiana* leaves is the most efficient and versatile method to test the function of plant genes and reconstruct a plant biochemical pathway in vivo. However, transient gene expression levels in *N. benthamiana* leaves are difficult to control, and thus, a large variation is often observed between individual samples. To mitigate this, we typically used five biological replicates in our experiments. We indicate the actual number of replicates in the relative figure legends.

Heterologous expression and purification of MLPL2

The full-length coding sequence of *MLPL2* from NLL cultivar Oskar was codon-optimized for expression in *E. coli* (see data S1) and cloned between the *Bam*H I and *Not* I restriction sites of pET29b(+) by GenScript (Nanjing, PRC). The resulting construct, pET29b-MLPL2, encodes an N-terminally S-tagged and C-terminally His-tagged version of MLPL2 under regulation of an IPTG-inducible promoter. Following validation by Sanger sequencing, pET29b-MLPL2 was transformed into *E. coli* strain ArcticExpress (DE3) RIL. For expression, single colonies were inoculated into 5 ml Luria broth supplemented with 20 µg/ml gentamicin, 75 µg/ml streptomycin, 10 µg/ml tetracycline, and 50 µg/ml kanamycin, and grown to saturation at 37°C and 220 rpm in 50-ml centrifuge tubes. Five milliliters of saturated culture was added to 110 ml of terrific broth without antibiotics in a 1-liter conical flask. The culture was then incubated at 32°C and 200 rpm to $OD_{600} = 0.4$ to 0.5 (≈ 1 hour) and then chilled on ice. The expression of MLPL2 was induced with 0.25 mM of isopropyl-β-D-thiogalactopyranoside (IPTG) and the culture was shaken at 16°C and 200 rpm for ≈ 70 hours. The bacterial pellet was harvested by centrifugation at 3900 relative centrifugal force (rcf) for 30 min at 4°C and resuspended into 5 ml of ice-cold lysis buffer (0.3 M NaCl, 10% glycerol, 0.1 mM Na_2HPO_4, pH 8.0) by vortexing. The bacteria were lysed with a Branson 250 Sonifier [Branson Ultrasonics (Brookfield, CT, USA)] using a 3-mm microtip (six 10-s pulses at a constant 40% amplitude were applied, and 60-s pauses were held between pulses; the tube was kept in an ice bath at all times). The lysate was transferred to 2-ml microcentrifuge tubes and centrifuged at 20,000 rcf for 30 min at 4°C. The supernatant was added to 1.5 ml 50% Ni-NTA agarose slurry [Qiagen (Venlo, The Netherlands)], which had been equilibrated in ice-cold lysis buffer in a 5-ml centrifuge tube. The slurry was incubated overnight at 6°C with gentle mixing on a rotary mixer. After binding, the mixture was transferred to a fritted column and allowed to drain. The resin was washed with 14 ml of ice-cold lysis buffer supplemented with 25 mM imidazole. Bound protein was eluted with three 1.5 ml portions of lysis buffer supplemented with 0.5 M imidazole. The eluates were pooled, gently mixed, and stored at 4°C until use. Enzymatic activity was retained for at least 3 weeks under these conditions. We consistently obtained yields of purified, tagged MLPL2 in excess of 1 g per liter of culture.

Time-course equilibration of piperideine (3) and tetrahydroanabasine (4) by MLPL2 in vitro

We previously confirmed (*8*) that piperideine (***3***) and tetrahydroanabasine (**4**) exist in equilibrium with one another and that the pure compounds equilibrate spontaneously at a non-negligible rate in phosphate buffer at nearly neutral pH (*24*). To probe the possible involvement of MLPL2 in facilitating this reaction, we tested whether equilibration occurred more quickly in the presence of the enzyme compared with the spontaneous reaction. Briefly, we first prepared a mixture containing either ***3*** (≈ 0.5 mM; forward reaction) or **4** (≈ 0.25 mM; reverse reaction) in phosphate buffer (final pH ≈ 7.2). We then rapidly split the mixture into two portions and added purified MLPL2 (final concentration 1.25 µg/µl) to one portion and the same amount of heat-inactivated, purified MLPL2 to the other portion. We incubated the mixtures at 25°C with shaking, and we monitored the progress of the reactions by taking aliquots at 15 s and up to 24 hours of incubation. The aliquots were immediately reduced with $NaBH_4$ to halt the reactions and stabilize the compounds. Finally, the samples were treated with benzoyl chloride as derivatizing agent, extracted with ethyl acetate, and analyzed by GC-MS as described under "Analytical and preparative method."

Specifically, for the experiment shown in fig. S7, A and B (forward reaction), we first prepared a 0.5 M solution of ***3*** by dissolving 1.9 mg of its acid-labile trimer, α-tripiperideine [synthesized following a previously described procedure (*49*)] in 45.7 µl of 2 M $HCl_{(aq)}$. We then added 2.95 µl of this solution to a mixture composed of 3 ml of a 0.1 M sodium phosphate buffer (pH 8), 28 µl of 2 M $HCl_{(aq)}$, 2.95 µl of dimethyl sulfoxide, and 3.16 µl of aqueous 0.2 M pyrrolidine HCl (Tokyo Chemical Industry, Tokyo, Japan) as internal standard. We quickly drew two 1330 µl aliquots of this substrate solution into two 1.5-ml microcentrifuge tubes. We then started the spontaneous reaction in one of the 1.5-ml tubes by adding 70 µl of heat-inactivated MLPL2 (25 µg/µl purified MLPL2, 0.5 M imidazole, 0.3 M NaCl, 10% glycerol, 0.1 mM Na_2HPO_4, pH 8.0; heat inactivation was carried out previously at 95°C for 5 min) and placing the tube in a heat block at 25°C with shaking (1000 rpm). After 15 s, we started the enzymatic reaction in the other 1.5-ml tube by adding 70 µl of active MLPL2 (same MLPL2 preparation but without heat inactivation) and placing it in the same heat block. After 15 s, we collected the first sample of the enzymatic reaction in triplicate by using a multistep electronic pipette to transfer 50-µl aliquots into corresponding glass vials, each containing ≈ 4 mg of solid $NaBH_4$. The vials were immediately vortexed to ensure complete dissolution of the hydride. Immediately afterwards, we collected the first sample of the spontaneous reaction in a similar way (also in triplicates). In total, samples were collected for the enzymatic reaction at the following time points: 0.25 min, 3.25 min, 7.5 min, 15 min, 30 min, 1 hour, 2 hours, 6 hours, and 24 hours. Samples of the spontaneous reaction followed the same time plan, resulting in time points that

were 60 s behind owing to the time needed to process the enzymatic reaction samples.

To test the reverse reaction (fig. S7, C and D), we followed the same procedure described above but replaced the initial piperideine (**3**) substrate solution with a 0.25 M solution of tetrahydroanabasine (**4**) in dimethyl sulfoxide, which we prepared by dissolving 0.8 mg of the hydrated dihydrobromide salt of **4** [synthesized previously (*50*)] into 9.25 µl of dimethyl sulfoxide. Furthermore, during the next step, we replaced the 2.95 µl of dimethyl sulfoxide with 2.95 µl of 2 M $HCl_{(aq)}$. Processing and time-course sampling were done as described above for the forward reaction.

To prepare the time-course samples for GC-MS analysis, we first capped the $NaBH_4$-treated samples and allowed them to react for at least 1 hour at room temperature with occasional vortexing. We then carefully added 100 µl of 2 M $HCl_{(aq)}$ to destroy the unreacted hydride. After basifying with 200 µl of 2 M $NaOH_{(aq)}$, we added 4 µl benzoyl chloride and vortexed vigorously for at least 30 s. We then let the samples react at room temperature for at least 1 hour with occasional vortexing. When benzoyl chloride droplets were no longer visible, we added 300 µl of a 1% (vol) solution of triethylamine in ethyl acetate and 200 µl of 2 M $Na_2CO_{3(aq)}$ and vortexed vigorously for at least 30 s. We separated the ethyl acetate layer by centrifugation and transferred 200 µl of it into clean glass vials for GC-MS analysis (described above under "Analytical and preparative methods"). By using large amounts of aqueous base, this procedure appears to remove most benzoic acid and unwanted reaction products from the final extracts while ensuring benzoylation of the amines.

Synthesis and resolution of (+)/(−)-ammodendrine [(+)/(−)-5]

(+)/(−)-ammodendrine [(+)/(−)-**5**] was prepared by acetylation of racemic isotripiperideine with acetic anhydride followed by acid hydrolysis of the product. Sixty-six milligrams of racemic isotripiperideine, prepared from piperidine as previously described (*49*), was dissolved in 0.5 ml dichloromethane in a 10-ml glass vial supplied with a small stir bar. Triethylamine (90.6 mg) and 4-dimethylaminopyridine (3.2 mg) were added, and the vial was cooled to 0°C in an ice-water bath. One hundred forty milligrams of acetic anhydride was added slowly, and the mixture was stirred at room temperature overnight.

The next day, the stir bar was removed from the dark brown reaction mixture and washed with 0.5 ml of dichloromethane. After adding the wash to the reaction mixture, the mixture was quenched with 1 ml of 2 M $NaOH_{(aq)}$ and washed with three more 1-ml portions of 2 M $NaOH_{(aq)}$. The dichloromethane fraction was then transferred to a clean 10-ml vial and dried over anhydrous Na_2SO_4. The dried dichloromethane fraction was transferred to a clean 10-ml vial, and the wet Na_2SO_4 was washed with two 0.5-ml portions of dichloromethane. The washes were combined with the dried dichloromethane fraction, and the solvent was stripped under reduced pressure using a rotary evaporator. The dark, viscous residue was redissolved in 1 ml of 2 M $HCl_{(aq)}$ and allowed to sit overnight at room temperature.

The next day, the mixture was cooled to 0°C in an ice-water bath and 1.5 ml of 2 M $NaOH_{(aq)}$ were slowly added while stirring. The mixture was then extracted with five 1-ml portions of ethyl acetate. The extracts were combined and dried over anhydrous Na_2SO_4. The dried extract was transferred to a clean 10-ml glass vial. The wet Na_2SO_4 was washed with two 0.5-ml portions of ethyl acetate, which were combined with the extract and stored at −20°C under nitrogen. To test for purity, 2 µl of the crude product was reconstituted in 200 µl of an aqueous solution containing 12% methanol and 0.012% formic acid, filtered through a 0.22 µm PVDF filter, and analyzed by LC-MS-ESI(+) using LC method 1. The crude product contained mostly ammodendrine (**5**) in 82% purity (% peak area of **5** in the total ion chromatogram). The main contaminants (14% of the total) were piperideine (*m/z* 84.0808), hydroxypiperidine (*m/z* 102.0913), and 4-dimethylaminopyridine (*m/z* 123.0915), as inferred from their *m/z* values. Acetylated isotripiperideines (*m/z* 292.2383) were also present in very small amounts (<1%). Ammodendrine (**5**) was the only mono- or bis-acetylated derivative of tetrahydroanabasine (**4**) that was detected in the crude product.

The two enantiomers (+)- and (−)-**5** were resolved by chiral GC-MS on a CycloSil-B capillary column [Agilent Technologies (Santa Clara, CA, USA)] as described above in "Analytical and preparative methods." Derivatization with the pentafluorophenyl ester of Fmoc-conjugated D-alanine (Fmoc-D-Ala-OPfp) was also successful, and the resulting diastereomers were separated by reversed-phase LC-MS on a Kinetex XB-C18 column [Phenomenex (Torrance, CA, USA)], confirming a previous result (*51*). See fig. S4 for more details on the derivatization and separation methods.

Purification and structure elucidation of 2-hydroxyammodendrine (12) and 1,2-didehydro-lusitanine (13) from *N. benthamiana*

LDC, *CAO*, *AT*, *MLPL2*, and *CYP76E36* were coexpressed in young leaves from 4-week-old *N. benthamiana* plants by agroinfiltration as described above in the section "Transient expression of biosynthetic gene candidates in *N. benthamiana*." Whole infiltrated leaves were harvested 7 days after the agroinfiltration, quickly frozen in liquid nitrogen, and pulverized. Five milliliters of extractant (60% methanol and 0.06% formic acid in water) was added to 800 mg of frozen leaf powder. The mixture was shaken for 2 hours at room temperature. The light green mixture was centrifuged, and the supernatant was concentrated under vacuum at room temperature to a viscous yellow oil. The oily residue was diluted with 2 ml of 20% methanol in water and passed through a 0.22 µm PVDF filter.

We used preparative HPLC (see "Analytical and preparative methods") to co-isolate the formate salts of 2-hydroxyammodendrine (**12**) and 1,2-didehydrolusitanine (**13**). The fractions from up to 40 consecutive 200-µl injections were pooled and dried at 35°C under vacuum to a white powdery residue. The residue was dissolved in D_2O with 0.05% sodium trimethylsilylpropanesulfonate (DSS) as internal reference (calibrated to δ_H 0.0 ppm and δ_C 0.0 ppm) and analyzed by NMR spectroscopy and LC-MS. NMR spectra were acquired at 300 K on a Bruker Avance III spectrometer (^{1}H operating frequency of 600.13 MHz) equipped with a Bruker SampleJet sample changer and a cryogenically cooled gradient inverse triple-resonance 1.7 mm TCI probe-head [Bruker (Billerica, MA, USA)] using standard 1D and 2D experiments (^{13}C NMR data was assigned using HSQC and HMBC experiments). Data were processed using Topspin ver. 4.1.4 [Bruker (Billerica, MS, USA)].

LC-MS analysis showed that the sample contained mostly **13** and **12** in a ~1:4 ratio together with smaller amounts of **14** and minor unknown impurities (fig. S8A). Mass spectral analysis of **12** indicated addition of an oxygen atom compared with ammodendrine (**5**) and presence of a secondary *N*-acetyl group, suggesting that **12** be a hydroxylated form of **5** (fig. S8B). In contrast, analysis of **13** indicated abstraction of dihydrogen compared with **5** and presence of a primary *N*-acetyl group, suggesting that **13** be an oxidized quinolizidine (fig. S8B). Further analysis of the mixture by NMR confirmed and helped refine this structural assignment (figs. S9 to S20). **12** existed as an equimolar mixture of cis:trans rotamers about the amide bond in D_2O. In addition, each rotamer appeared as a mixture of a major and a minor conformer (~7:3 ratio) identified by their different ^{1}H and ^{13}C chemical shifts at position 6 (figs. S9 and S10). **13** was present in a ~1:4 ratio to **12** (fig. S9), comparable to the LC-MS–based estimate (fig. S8A). Highly overlapping signals in the aliphatic range made it impossible to identify all the resonances of **13**. Nevertheless, 2D experiments (figs. S10 to S20), particularly ^{1}H-^{13}C-HMBC correlations (figs. S17 to S19), combined with mass spectral data (fig. S8B) and mechanistic logic (considering the reactivity of **12**; see also fig. S21) strongly suggest a quinolizidine iminium core scaffold.

trans-*2-OH-ammodendrine (*trans-***12***).* ^{1}H NMR (600 MHz, D_2O, 300 K, DSS): δ_H 6.87 (0.7H, s, H6 – major conformer), 6.83 (0.3H, s, H6 – minor conformer), 5.98 (1H, d, J = 4 Hz, H2), 3.73 (1H, t, J = 11 Hz, H2'), 3.44

(1H, br, H6'B), 3.08 [1H, t(dd), J = 13, H6'A], 2.29 (1H, H4B), 2.24 (3H, s, H8), 2.12 (H3B), 2.12 (1H, H4A), 1.96 (1H, H4'B), 1.95 (1H, H3′B), 1.90 (1H, H5′B), 1.81 (1H, H3′A), 1.68 (1H, H3A), 1.67 (1H, H5′A), 1.60 (1H, H4'A). ^{13}C NMR (150 MHz, D_2O, 300 K, DSS): δ_C 175.3 (C7), 127.4 (C6 – major conformer), 126.8 (C6 – minor conformer), 118.8 (C5), 73.2 (C2), 63.8 (C2'), 48.1 (C6'), 29.9 (C3′), 29.0 (C3), 24.8 (C4'), 24.5 (C5′), 23.8 (C8), 18.8 (C4).

cis-*2-OH-ammodendrine (*cis-***12***)*. ^{1}H NMR (600 MHz, D_2O, 300 K, DSS): δ_H 7.22 (0.7H, s, H6 – major conformer), 7.19 (0.3H, s, H6 – minor conformer), 5.73 (1H, d, J = 3, H2), 3.75 (1H, d, J = 9, H2'), 3.44 (1H, br, H6'B), 3.08 [1H, t(dd), J = 13 Hz, H6'A], 2.34 (3H, s, H8), 2.31 (1H, H4B), 2.15 (1H, H3B), 2.12 (1H, H4A), 1.96 (1H, H4'B), 1.95 (1H, H3′B), 1.90 (1H, H5′B), 1.81 (1H, H3′A), 1.77 (1H, H3A), 1.67 (1H, 5′A), 1.60 (1H, H4'A). ^{13}C NMR (150 MHz, D_2O, 300 K, DSS): δ_C 175.3 (C7), 124.3 (C6 – major conformer), 123.6 (C6 – minor conformer), 120.5 (C5), 77.4 (C2), 63.8 (C2'), 48.1 (C6'), 29.9 (C3′), 29.8 (C3), 24.8 (C4'), 24.5 (C5′), 23.3 (C8), 18.8 (C4).

High-resolution mass calculated for $C_{12}H_{21}N_2O_2$ ($[M+H]^+$ of **12**): 225.159754; found 225.1598 (err. −0.2 ppm).

*1,2-didehydrolusitanine (***13***)*. ^{1}H NMR (600 MHz, D_2O, 300 K, DSS): δ_H 8.63 (1H, s, H2), 6.80 (1H, s, H11), 4.45 (1H, d, J = 11, H6), 4.17 (1H, d, J = 12, H10B), 3.90 [1H, t(dd), J = 12, H10A], 2.73 (1H, d, J = 14 Hz, H4B), 2.33 (1H, H4A), 2.10 (3H, s, H14), 2.06 (1H, H9B), 2.02 (1H, H7B), 1.98 (1H, H7A), 1.81 (1H, H9A). Signals for H3 and H8 could not be unambiguously located. ^{13}C NMR (150 MHz, D_2O, 300 K, DSS): δ_C 178.8 (C2), 175.5 (C13), 122.2 (C11), 118.5 (C5), 68.5 (C6), 64.4 (C10), 37.6 (C7), 30.9 (C3), 29.1 (C9), 24.7 (C14), 17.6 (C4). The signal for C8 could not be unambiguously located.

High-resolution mass calculated for $C_{12}H_{19}N_2O$ ($[M]^+$ of **13**): 207.149190; found 207.1489 (err. +1.4 ppm).

Determination of the isomeric purity and yield of sparteine (6) produced in *N. benthamiana*

Agroinfiltrated leaves of *N. benthamiana* transiently coexpressing LDC, CAO, AT, MLPL1-4, HYD, and IRED1-2 together with either GFP or IRED3 were harvested 7 days after agroinfiltration. Two leaf discs of 1-cm diameter were punched out of each infiltrated leaf ($n = 6$) and collected in 1.5-ml screw-cap microcentrifuge tubes. The remaining agroinfiltrated (discolored) portions of the leaves were clipped out and pooled in 50-ml centrifuge tubes. The leaf discs and the leaf clippings were dried for 4 days at 40°C after which they were pulverized using steel beads and a TissueLyzerII [Qiagen (Venlo, The Netherlands)] bead beater (in the case of the leaf discs) or a benchtop vortex (in the case of the leaf clippings).

The pulverized leaf discs were extracted with 250 µl extractant (60% methanol, 0.06% formic acid, and 5 ppm caffeine in water) for 3 hours at 400 rpm. Leaf debris was separated by centrifugation (1 min at 21,000 rcf) and the extracts were diluted five times in water, filtered through a 0.22-µm PVDF filter, and analyzed by LC-MS (see "Analytical and preparative methods"). We searched for sparteine (**6**) and its isomers by producing extracted ion chromatograms of combined *m/z* ratios of 235.22 ± 0.01 and 118.11 ± 0.01, which corresponded to the $[M+H]^+$ and $[M+2H]^{2+}$ ions, respectively. Three peaks were detected, of which the largest one corresponded to sparteine (**6**) by comparison to a known standard. The two smaller peaks were tentatively identified as the other two possible diastereomers of sparteine: α-isosparteine (**16**) and β-isosparteine (**17**). To determine the purity of sparteine compared with the isosparteines, the three peaks were integrated from the extracted ion chromatograms and added together to give a total peak area. The % purity of sparteine was evaluated by comparing the area of the sparteine peak to the total peak area. Quantification of sparteine in the extracts was done using the internal standard method, with caffeine as internal standard.

To determine the enantiomeric purity of sparteine, 50 mg of the pulverized leaf clippings was transferred to 12-ml glass vials and extracted with 3 ml of 1 M $HCl_{(aq)}$ by incubating for 1 hour at 400 rpm. Leaf debris was separated by centrifugation, and the cleared extracts were transferred to clean 12-ml glass vials. The extracts were defatted with *n*-hexane (3x 1.5-ml portions), and the aqueous extracts were transferred to clean 12-ml glass vials. The extracts were basified with 2 ml of 2 M $NaOH_{(aq)}$ and the alkaloids were extracted with dichloromethane (3x 1.5-ml portions). The dichloromethane extracts were transferred to clean 12-ml glass vials and dried with anhydrous $MgSO_4$. The dried extracts were transferred to clean vials and evaporated under reduced pressure. The residues were reconstituted in 100 µl ethyl acetate and analyzed by chiral GC-MS as described above under "Analytical and preparative methods." (−)-sparteine and (+)-sparteine were identified on the basis of comparison with commercial standards [Sigma-Aldrich (St. Louis, MO, USA)]. The ee was calculated from extracted ion chromatograms of *m/z* 98 ± 0.5 (prominent EI peak of **6**).

Determination of the isomeric purity of ammodendrine (5) produced in *N. benthamiana*

The procedure is analogous to the one used for sparteine (**6**) (see previous paragraph). Briefly, *N. benthamiana* plants transiently expressing LDC, CAO, AT, and either GFP, MLPL2, or MLPL6 were harvested 8 days after agroinfiltration ($n = 6$). The agroinfiltrated (discolored) portions of the leaves were clipped out, pooled into 50-ml centrifuge tubes, and dried at 40°C for 4 days. The dried leaves were crushed, and ~25 mg of powder was transferred to 12-ml glass vials. Two milliliters 1 M $HCl_{(aq)}$ was added to the leaves, and the tubes were shaken at 400 rpm for 1 hour. Leaf debris was removed by centrifugation, and the extracts were basified with 300 µl 10 M $NaOH_{(aq)}$ and extracted twice with 2 ml ethyl acetate. The extracts were combined and dried over anhydrous Na_2SO_4. One milliliter of extract was evaporated under a stream of N_2, reconstituted in 100 µl ethyl acetate, and analyzed by chiral GC-MS as described above under "Analytical and preparative methods." Ammodendrine extracted from *N. benthamiana* was compared with a sample of racemic ammodendrine synthesized in house {see under "Synthesis and resolution of (+)/(−)-ammodendrine [(+)/(−)-**5**]"}. The ee was calculated from extracted ion chromatograms of *m/z* 165 ± 0.5 (EI base peak of **5**).

Discovery of the *exiguus* low-alkaloid mutation in white lupin

White lupin homologs of all QA biosynthetic genes up to (−)-sparteine [(−)-**6**] were identified by BLASTing the NLL coding sequences against the white lupin reference genome (*28*). *LDC*, *CAO*, *AT*, *CYP76E36*, *HYD*, *MLPL1*, and *MLPL4* only had one homolog in white lupin. Using the white lupin genome browser (*28*), the sequence of the seven unique genes was inspected for the presence of nonsense mutations or splice site mutations in any of the 39 white lupin accessions included in the recent pangenome project (*28*, *29*). The accessions Neuland, Neutra, and Gyulatanya appeared to carry mutations in the *MLPL1* gene (Lalb_Chr14g0362281; GenBank accession number of corresponding protein: KAE9599443.1). Neutra carried a frameshift mutation, Gyulatanya carried a splice site mutation, and Neuland carried both mutations (fig. S28A). Neuland was expected to carry the low-alkaloid mutation named *exiguus* (*30*).

Seeds of white lupin cultivar Neuland were obtained from the Genebank of the Leibniz Institute of Plant Genetics and Crop Research (Gatersleben, Germany) (accession no. LUP 6446) and from the seed collection of the University of Reading (Reading, Great Britain) (accession no. 22). Ten seeds from each source were sown individually in 16-cm-wide, 20-cm-deep pots containing a light peat-based potting soil mix and grown in a growth cabinet with day/night temperature of 21°/18°C, day/night length of 16 hours/8 hours, relative humidity of 60 to 65%, and light intensity of 220 µmol m^{-2} s^{-1}. Plants of the commercial cultivar Amiga [Florimond Desprez (Chappelle-en-Pévèle, France)], that carry the low-alkaloid mutation *pauper* (*8*), were grown alongside Neuland for comparison. When the plants were 3 weeks old, the central leaflets of two young leaves were harvested, flash frozen

in liquid nitrogen, and ground to a fine powder. Metabolites were extracted from ~30 mg of frozen powder in 1 ml of extractant (60% methanol, 0.06% formic acid, and 5 ppm caffeine in water) by shaking in a 1.5-ml microcentrifuge tube for 2 hours at 850 rpm. The extracts were briefly centrifuged to separate solid debris, and the supernatants were diluted five times with water. The diluted extracts were filtered through 0.22-µm PVDF filters and analyzed by LC-MS. The plants were classified on the basis of chemotype into bitter (high in QAs and without detectable intermediates), *pauper* [low in QAs and accumulating *pauper* metabolites (**22**)], *exiguus* [low in QAs and accumulating 2-OH-ammodendrine (**12**) and 1,2-didehydrolusitanine (**13**)], and uncertain [low in QAs and accumulating low but detectable amounts of **12**, **13**, and **22**). Of 10 tested plants from the Leibniz Institute, eight were found to be bitter, one was *pauper*, and one was *exiguus*. Of nine tested plants from the University of Reading, six were found to be *exiguus* and three were uncertain. Two bitter plants from the Leibniz institute and the six *exiguus* plants from the University of Reading were selected for further analysis.

Total RNA was extracted from young leaves of the white lupin plants using the Spectrum Plant Total RNA Kit [Sigma-Aldrich (St. Louis, MO, USA)] and was converted to cDNA using the iScript cDNA Synthesis Kit [Bio-Rad (Hercules, CA, USA)]. Genomic DNA (gDNA) was extracted from the leaves using the E.Z.N.A. Plant DNA DS Kit [Omega Bio-tek (Norcross, GA, USA)]. To clarify the genotype underlying the different white lupin chemotypes, the open reading frame of MLPL1 was amplified from gDNA of two bitter types, two *exiguus* types, and of cultivar Amiga (*pauper* type, which served as reference) using primers LalbMLPL1_pEAQ_FW and LalbMLPL1_pEAQ_RV, and the resulting polymerase chain reaction (PCR) band was sequenced by Sanger sequencing using the same two primers (see table S2). Both *exiguus* plants were homozygous for the splice site mutation identified from sequencing reads, which was named *MLPL1*^{exiguus}. To examine the effect of the splice site mutation on the splicing of MLPL1 mRNA, the coding sequence of MLPL1 was amplified from cDNA of a bitter type and an *exiguus* type plant using primers LalbMLPL1_pEAQ_FW and LalbMLPL1_pEAQ_RV. The resulting PCR products were separated and purified from agarose gels and sequenced by Sanger sequencing. To validate the function of different MLPL1 variants in vivo, the genomic open reading frames (i.e., with intron) of MLPL1 were cloned into pEAQ-USER and transiently expressed in *N. benthamiana* in combination with biosynthetic genes from NLL as described above in the section "Transient expression of biosynthetic gene candidates in *N. benthamiana*."

Seeds from a single genotyped bitter plant from the Leibniz Institute and a single genotyped *exiguus* plant from the University of Reading were propagated and used in later experiments. For metabolite characterization in leaves, five MLPL1^{exiguus} white lupin plants were grown alongside five bitter and five sweet Amiga (AT^{pauper}) plants in a growth cabinet, as described above. Five weeks after sowing, a young leaf was harvested from each of these plants and dried for 4 days at 40°C. The dried leaves were pulverized, and ~12 mg of each leaf were extracted with 1 ml of extractant (60% methanol, 0.06% formic acid, and 15 ppm caffeine in water) by shaking vigorously for 2 hours in 2-ml screw-cap microcentrifuge tubes. The extracts were clarified by centrifugation and were then diluted 15 times with water. After filtration through a 0.22-µm PVDF filter, the extracts were analyzed by LC-MS as described under "Analytical and preparative methods."

Isolation and characterization of HYD^{KO} low-alkaloid NLL plants

The previously constructed mutant library of bitter NLL cultivar Oskar (*8*) was screened essentially as described in Knudsen *et al.* (2022) (*31*) to identify a mutant seed harboring the specific G-to-A nucleotide change at position 188 in the coding region of *HYD* (corresponding to the W63Stop amber mutation; see also gene sequences in data S1). For TaqMan assays, we used the following oligonucleotides: a target-specific forward primer (HYD_TaqMan_FW), a target-specific reverse primer (HYD_TaqMan_RV), a wild-type–specific probe containing a HEX fluorophore and a BHQ1 quencher (HYDWT_TaqMan_HEX), and a mutant-specific probe containing a FAM fluorophore and a BHQ1 quencher (HYDKO_TaqMan_FAM) (table S2). One homozygous mutant seed was retrieved. The seed was sown in a 16-cm-wide, 20-cm-deep pot containing a light, peat-based potting soil mix and grown in a growth cabinet with day/night temperature of 21°/18°C, day/night length of 16 hours/8 hours, relative humidity of 60 to 65%, and light intensity of 220 µmol m^{-2} s^{-1}. Genomic DNA was extracted from the leaves of this plant using the E.Z.N.A. Plant DNA DS Kit [Omega Bio-tek (Norcross, GA, USA)], and the W165Stop mutation (*HYD*^{KO}) was confirmed by sequencing a fragment of *HYD*, which was amplified by PCR using primers HYDKO_GnTp_FW and HYDKO_GnTp_RV (table S2). Seeds of this homozygous mutant were collected and used for further propagation and characterization.

For metabolite characterization in seeds, six NLL HYD^{KO} mutant plants were grown in in the field at Nørre Aaby, Fyn, Denmark, between April and late August of 2022. Seeds from all six plants were pooled, and 3x 12 seeds were pulverized using a steel ball and a TissueLyzer bead beater [Qiagen (Venlo, The Netherlands)]. Approximately 15 mg of seed flour was extracted in 1 ml of extractant (60% methanol, 0.06% formic acid, and 15 ppm caffeine in water) by shaking for 3 hours on a small tabletop orbital shaker at 1200 rpm. The extracts were clarified by centrifugation, diluted 15 times with ultrapure water, filtered through a 0.22-µm PVDF filter, and analyzed by LC-MS as described under "Analytical and preparative methods." Seeds of wild-type Oskar (3x 12) from a supplier [Hodowla Roślin Smolice Sp. z o. o. Grupa IHAR (Smolice, Poland)] (labeled HYD^{WT} in Fig. 5) were extracted in the same way as the HYD^{KO} seeds and used as controls to assess the effects of the mutation. For metabolite characterization in leaves, see the next section, "Isolation and characterization of other biosynthetic mutant NLL plants."

Isolation and characterization of other biosynthetic mutant NLL plants

To obtain genetic evidence of the unique involvement of the newly identified biosynthetic genes in the QA pathway, we identified NLL early-stop codon mutants of CYP76E36, MLPL4, MLPL1, and CYP71A168 and a splice site mutant of IRED3. We also identified early-stop codon mutants of functionally redundant MLPL2 and MLPL6, which we intended to cross in order to obtain a functional knockout phenotype. Unfortunately, genetic analysis of the MLPL6 mutant revealed the presence of an additional, identical copy of the gene (fig. S29), which nullified our efforts. We did not attempt to isolate mutants of MLPL3/7 and IRED1/2 owing to redundancy and, at least in the case of IRED1/2, genetic linkage.

Mutant seeds were extracted from our previously constructed bitter NLL mutant library (*8*) essentially as described for HYD^{KO} in the previous section. Oligonucleotide sequences for the TaqMan assays are included in table S2. We sowed the mutant seeds and grew them to mature plants in a growth cabinet, as described above for HYD^{KO}. We extracted genomic DNA from these plants and confirmed their genotype by Sanger sequencing (see primers in table S2). We allowed heterozygous mutant plants to self-pollinate and then grew 10 to 30 of their seeds into mature plants in a growth cabinet. We used Sanger sequencing to detect homozygous mutant plants among the offspring and pooled the seeds from those plants. Subsequent generations of homozygous mutant plants were generated by self-pollination.

To compare the different genotypes, we grew five or six homozygous mutant plants of each genotype and of wild-type NLL Oskar alongside each other under greenhouse conditions. We sowed the seeds individually in the center of 16-cm-wide, 20-cm-deep pots. We used a potting mix containing equal parts of sterilized, peat-based potting soil, sand, and 0-to-4-mm expanded clay pebbles. Growing conditions were: day/night temperatures of 21°/18°C, light/dark photoperiod of 16 hours/8 hours, and relative humidity of 60%. Artificial light was supplied during the day to keep the photosynthetic photon flux density above 150 µmol m^{-2} s^{-1}. For a complete comparison, we included HYD^{KO} (described

in the previous section) as well as AT^{KO} (*8*) and $CYP71D189^{KO}$ (*10*) isolated previously. To profile the metabolic effect of the $CYP71A168^{KO}$ mutation, we grew three homozygous mutant and three wild-type plants under growth cabinet conditions rather than greenhouse conditions. Controllable conditions (photoperiod, relative humidity, light intensity, and growth substrate) used in the cabinet were the same as in the greenhouse except that light was supplied with fluorescent tubes instead of LED arrays. Cabinet-grown plants accumulated tetrahydrorhombifoline (**31**), which was not detected in greenhouse-grown plants. Thus, we were able to test whether **31** is a downstream product of CYP71A168.

For metabolite characterization, young leaves of all mutant and wild-type control plants were harvested 4 weeks after sowing and dried for 3 days at 35°C. The dried leaves were pulverized, and ~6 mg of each leaf was extracted with 0.5 ml of extractant (60% methanol, 0.06% formic acid, and 15 ppm caffeine in water) by shaking vigorously for 3 hours in 2-ml screw-cap microcentrifuge tubes. The extracts were clarified by centrifugation and were then diluted 15 times with water. After filtration through a 0.22-µm PVDF filter, the extracts were analyzed by LC-MS as described under "Analytical and preparative methods."

For metabolite characterization in seeds of $IRED3^{KO}$ and $CYP71D189^{KO}$, 12 seeds from a pool of homozygous plants were pulverized using a steel ball and a TissueLyzer bead beater [Qiagen (Venlo, The Netherlands)]. Approximately 150 mg of seed flour was extracted in 2 ml of 1 M $HCl_{(aq)}$ by shaking for 1 hour on a tabletop orbital shaker at 1200 rpm. The extracts were clarified by centrifugation. The clarified extracts were basified with 1 ml of 2 M $NaOH_{(aq)}$ and extracted with three 2-ml portions of ethyl acetate. The extracts were combined and dried with anhydrous Na_2SO_4, and the solvent was evaporated under reduced pressure. The residue was reconstituted in 100 µl of ethyl acetate, resulting in a concentrated extract that was diluted as needed with ethyl acetate for analysis by chiral GC-MS as described under "Analytical and preparative methods."

Tissue-level expression and predicted subcellular localization of QA biosynthetic enzymes

We recently developed a tissue-level transcriptomic dataset of leaves, stems, and pods of bitter NLL using laser-capture microdissection followed by RNA-seq (*15*). We accessed this dataset from the publicly available lupin eFP browser (*52*), searched for the transcripts encoding the proposed QA biosynthetic genes (see table S1 for transcript IDs), and collected their expression levels across tissues (fig. S34).

The subcellular localization of the proposed QA biosynthetic enzymes was predicted using the DeepLoc - 2.1 (*53*) and TargetP - 2.0 (*54*) servers. Secondary structures were predicted using the NetSurfP - 3.0 server (*55*). The N- and C-terminal amino acid sequence and predicted secondary structure of each protein were also inspected manually for notable structural features that may support or oppose a predicted localization. The subcellular localization of LDC (*6*) and CAO (*7*) has been determined experimentally before.

Phylogenetic analysis of MLPLs and IREDs

We searched the UniProt database (*56*) to determine which protein family MLPL1-7 and IRED1-3 belong to. MLPL1-7 were found to belong to the Bet_v_1 (PF00407) family and IRED1-3 to the epimerase (PF01370) family of the Pfam database (*57*). We then used the InterPro database (*57*) to retrieve the sequences of all NLL reference proteins (taxon ID 3871, cultivar Tanjil) predicted to belong to the two protein families. We retrieved 44 sequences for the Bet_v_1 family and 56 sequences for the epimerase family. We then subjected the sequences to BLASTP against the database of NLL reference proteins from NCBI (refseq_protein database, limited to taxid:3871). We then checked each result manually for matches. When we found inconsistencies between the sequences from InterPro and those from NCBI, such as deletions or insertions of multiple amino acids, we amended the sequence from InterPro by substituting it with the sequence from NCBI, but we maintained the identifier from InterPro for easier access. When we found major mismatches (such as %ID < 95% over large stretches of protein), we deleted the sequence from our database altogether. We also deleted redundant sequences. Among the Bet_v_1 family, we amended three sequences and deleted two redundant sequences and two mismatch sequences. Among the epimerase family, we amended 17 sequences and deleted 3 mismatch sequences.

We expanded our database of Bet_v_1 and epimerase family proteins by including other plant proteins with proposed functions belonging to these two families. To find the proteins, we searched the UniProt database using PF00407 and PF01370 as queries and restricted the results to plants (taxonomy_id:33090) and to well-annotated proteins (i.e., to proteins whose existence was supported by experimental evidence at the protein level and whose entry had an annotation score ≥ 3, and to proteins whose existence was supported by experimental evidence at the transcript level and whose entry had an annotation score of 5). This search yielded 47 Bet_v_1 and 35 epimerase family proteins.

We further expanded our list of Bet_v_1 family proteins, by including MLPL7 from NLL (which did not appear in databases of NLL proteins) and two MLPLs from catnip, which have been proposed to act as scaffolding enzymes in the biosynthesis of iridoids in conjunction with dehydrogenases (*22*). The final list of Bet_v_1 proteins for alignment consisted of 91 sequences (41 from NLL and 50 from other plants). We also added one more proposed enzyme to our list of epimerase family proteins: the dehydrogenase noscapine synthase from opium poppy (*58*). The final list of epimerase proteins for alignment consisted of 89 sequences (53 from NLL and 36 from other plants).

We used MEGA (version 12.1.2) (*59*) to perform multiple sequence alignment of the protein sequences by Clustal Omega and to generate a maximum likelihood phylogeny tree on the basis of those alignments. We also calculated the pairwise percentage identity [(1 – *p*-distance) × 100] between MLPL1-7 and IRED1-3 on the basis of the Clustal Omega alignment.

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availability: Coexpression analysis was conducted using publicly available RNA-seq datasets from the National Center for Biotechnology Information (BioProject accession numbers PRJNA386115 and PRJNA248164). Datasets related to the pangenome of white lupin are available at the National Center for Biotechnology Information (BioProject accession number PRJNA608889) as well as at https://www.whitelupin.fr/. Accession numbers of NLL genes are given in table S1. The exact coding sequences of all the genes characterized in this study are provided in data S1. Accession numbers and source for the named varieties of lupin that were analyzed in this study are provided in the supplementary materials. Seeds of the NLL mutants are available from F.G.-F. pending scientific review and a completed material transfer agreement with the University of Copenhagen. Raw NMR data have been deposited to Zenodo (*60*). All other data needed to evaluate the conclusions of the paper are presented in the main text or supplementary materials. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

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Supplementary Text; Figs. S1 to S35; Tables S1 to S3; References (*63*–*83*); MDAR Reproducibility Checklist; Data S1

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RESEARCH ARTICLE SUMMARY

BACTERIAL IMMUNITY

The human gut microbiome primes fever after vaccination

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Full article and list of author affiliations: https://doi.org/10.1126/science.aea7733

INTRODUCTION: Vaccination is a critical tool for global public health, but wide variability in human vaccine responses influences vaccine effectiveness. Vaccine adverse reactions, including common side effects such as fever and flu-like symptoms, are a factor in vaccine hesitancy, and these reactions are currently challenging to predict. Vaccine hesitancy is considered a top- 10 global health threat by the World Health Organization as a result of its impact on vaccine-preventable diseases such as measles, cervical cancer, and COVID-19.

RATIONALE: People differ markedly in their adverse responses to vaccination, but this variation is not well understood. The gut microbiome has been implicated in adaptive immune responses to vaccination; we hypothesized that interactions between commensal gut microbes and the immune system may influence the risk of adverse inflammatory vaccine reactions.

RESULTS: The μHEAT (Microbial-Human Ecology and Temperature) Study longitudinally tracked 171 healthy young adults receiving severe acute respiratory syndrome coronavirus 2 mRNA vaccines. Thrice-daily oral temperature records were used to quantify fever, and fecal and serum samples were collected for microbiome and immune profiling. We found that fever had very little effect on the microbiome; by contrast, fever-positive participants displayed chronic low-grade intestinal inflammation and a microbiome characterized by high expression of bacterial flagellins, both pre- and postvaccination. Microbiomes from fever-positive participants induced heightened inflammatory responses in human colonic organoids and exacerbated acute vaccine responses when transplanted into gnotobiotic mice, and this was dependent on sensing of bacterial flagellin through host innate immune receptors. Moreover, microbial flagellin production was modifiable by diet: Dietary additives increased bacterial flagellin production in vitro, and a Western-style diet exacerbated vaccine reactions in mice. Consistent with this, fever was associated with dietary and metabolic markers in humans. Although interventional human studies are needed to establish causality, this suggests that diet modification could be a viable strategy to influence microbiome-dependent vaccine responses.

CONCLUSION: These data suggest that chronic production of flagellin within the human gut microbiome sets the immune tone to drive inflammatory vaccine responses. The ability to predict and control the risk of vaccine-induced adverse reactions is a crucial factor in vaccine hesitancy and global public health; our findings establish the possibility that microbiome-targeted strategies could mitigate vaccine side effect risk in humans. The identified microbiome-immune axis may also have implications for other inflammatory processes beyond vaccination. □

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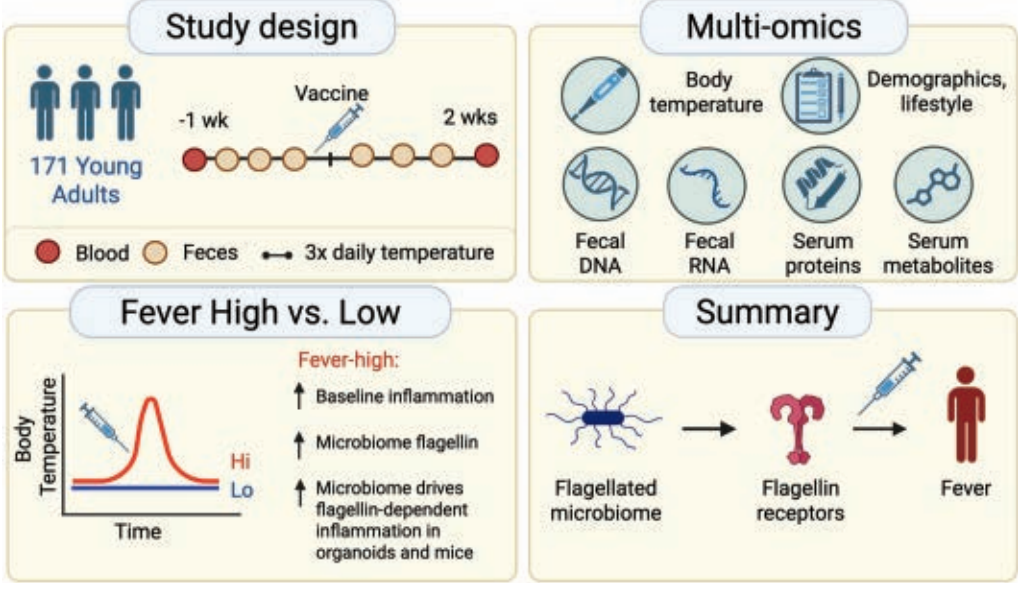

The μHEAT (Microbial-Human Ecology and Temperature) Study indicates that flagellated human gut microbes prime fever after vaccination. [Figure created with Biorender.com]

VACCINATION

The human gut microbiome primes fever after vaccination

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Fever is a common adverse reaction to vaccination, contributing to vaccine hesitancy and reduced uptake. To understand variation in fever risk, we longitudinally profiled fecal microbiota, oral temperature, and serological markers in 171 healthy adults receiving Severe acute respiratory syndrome coronavirus 2 mRNA vaccines. Fever risk correlated with low-grade intestinal inflammation, increased abundance of flagellated *Lachnospiraceae* bacteria, and increased flagellin expression prevaccine. Microbiomes from fever-high donors triggered stronger inflammation in human intestinal organoids and drove flagellin-dependent vaccine reactions in gnotobiotic mice. Moreover, microbiome flagellin phenotypes and murine vaccine reactions were modifiable by diet. Consistent with this, human fever risk was associated with self-reported diet and metabolic markers. These findings identify the gut microbiome as a driver of vaccine-induced fever, suggesting that microbiome-targeting strategies could modulate immune tone and improve vaccine side effects.

Vaccination is a crucial public health tool for disease prevention (*1*). However, vaccine responses vary widely between people and across populations, leading to limited immunity in some recipients and excessive side effects in others (*2*, *3*). Common systemic adverse events, including fever and flu-like symptoms, account for considerable discomfort and lost productivity (*4*) and limit the immunogenic dose that can be safely administered. Adverse events contribute to societal vaccine hesitancy and therefore to unnecessary, vaccine-preventable morbidity and mortality (*5*, *6*). Vaccine hesitancy has been named one of the top 10 risks to global health by the World Health Organization as a result of its serious and widespread impact on vaccine-preventable diseases such as measles, cervical cancer, and COVID-19 (*6*).

Fever is one major adverse effect of vaccination and is relatively common with newly developed mRNA- and lipid nanoparticle–based vaccine platforms (*7*). During the COVID-19 pandemic, fear of adverse events induced by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) vaccination contributed to widespread vaccine hesitancy and mistrust, and therefore to significant excess morbidity and mortality (*6*, *8*–*11*). Notably, experiencing common adverse events such as fever was directly associated with decreased intention to receive future vaccines (*9*). New mRNA-based vaccines in development show similar rates of fever and adverse reactions (*12*). A better understanding of vaccine-induced fever is critical both to manage public expectations around vaccine safety and to prevent excessive vaccine reactogenicity.

Fever risk after vaccination is variable, and this variation in risk is largely unexplained. Fevers induced by SARS-CoV-2 vaccines were individualized and consistent across doses (*13*–*16*), pointing to the importance of individual risk factors for fever. Recently, the commensal microbiome has gained considerable interest as a source of human immune variation: The microbiota tonically stimulates the host immune system and is furthermore susceptible to dietary and xenobiotic modification, creating a potentially targetable modulator of human immunity (*17*, *18*). Indeed, the presence of an intact microbiome promotes vaccine-induced antibodies in mice and humans (*19*–*23*). One hypothesis for this phenomenon is that gut microbes produce "adjuvanting" ligands that stimulate innate immune receptors (*17*); however, it is unknown whether microbiota influences vaccine-induced, innate-immune-driven adverse reactions such as fever. Furthermore, although febrile temperatures amplify immune function and could plausibly affect microbial growth, it is unclear to what extent the commensal microbiome is itself affected by human fever (*24*).

To address these fundamental knowledge gaps, we established the µHEAT (Microbial-Human Ecology and Temperature) Study. We aimed to (i) characterize the effect of an acute febrile response on the human intestinal microbiome and (ii) assess how the microbiome affects fever risk after vaccination.

Results

Febrile responses vary widely in µHEAT participants

The µHEAT (Microbial-Human Ecology and Temperature) study followed 171 individuals before and after mRNA vaccination for SARS-CoV-2, most of whom received third-dose BNT162b2 booster vaccinations (Fig. 1A and tables S1 and S2). Participants provided three oral temperature records daily over a two-week period (total $n = 7079$ temperature records; Fig. 1, B to E). The median body temperature was 36.5°C, consistent with reports that 37°C is no longer a valid benchmark in industrialized human populations (*25*). Baseline body temperature showed a strong circadian rhythm (Fig. 1D) and was higher on average in women than in men (Fig. 1E), as well as in women who took oral contraceptives compared with those who did not (fig. S1A). These data reflect well-known patterns in healthy human body temperature (*25*, *26*), validating the utility of the participant temperature records.

Vaccination induced a pronounced increase in participant body temperature, indicating febrile responses (Fig. 1, B and C). We defined ΔT as the increase in body temperature postvaccine, relative to each participant's median temperature prevaccine for that time of day (see methods), which allowed us to account for variation in personal baseline temperature. We further divided participants into "fever-high" ($n = 72$) and "fever-low" ($n = 99$) categories based on the ΔT group mean. This resulted in participants being classified as fever-high if their temperature rose 0.8°C or more above personal baseline (Fig. 1C). Unlike baseline body temperature, ΔT was unaffected by gender, oral contraceptive use, or time of day (fig. S1, B to D). Further, ΔT did not differ by type or dose of vaccine received during the study, nor by prior vaccine type or SARS-CoV-2 infection history (table S3). Instead, ΔT correlated positively with the number of flu-like symptoms reported

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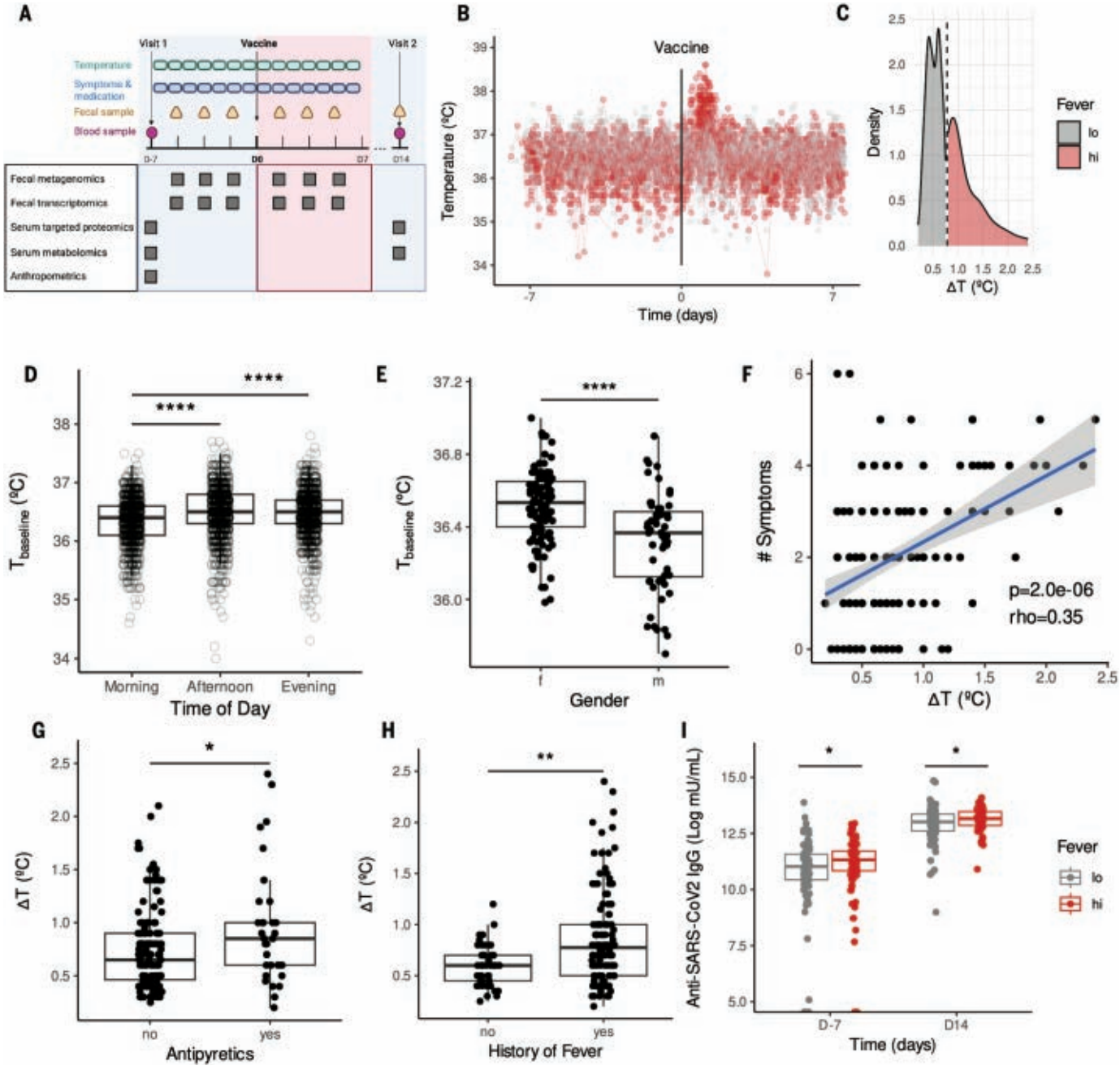


Fig. 1. Febrile responses vary widely in the μHEAT cohort. (**A**) Schematic of the μHEAT study protocol and multiomics sampling. (**B**) Oral body temperature recorded three times per day per participant ($n = 7079$ records from 171 people). (**C**) Definition of fever category based on distribution of ΔT, the degree of postvaccine temperature increase compared with personal baseline. (**D**) Baseline (prevaccine) temperature records according to time of day at measurement. (**E**) Median baseline temperature per person by gender. (**F** to **H**) Degree of postvaccine temperature increase (ΔT) versus (F) total number of flu-like symptoms reported by the participant postvaccine; (G) reported consumption of antipyretic medication postvaccine; and (H) history of fever, i.e., whether the participant reported experiencing any fever in response to prior vaccines or illnesses in the past two years. (**I**) Serum anti–SARS-CoV-2 IgG titres prevaccine (D-7) and postvaccine (D14), according to fever category. Statistical significance was determined through analysis of variance (ANOVA) with post hoc Tukey's test (D), Wilcoxon Rank-Sum test (E) and (G to I), or Spearman's correlation (F). Box plots are drawn from quartile 1 to 3 with a line at the median, and the whiskers are drawn at ± 1.5*interquartile range. *$P < 0.05$; **$P < 0.01$; ****$P < 0.0001$. Lo, fever-low; hi, fever-high. IgG mU/mL, standardized binding antibody units/mL.

by the participant after vaccination (Fig. 1F) and with subsequent consumption of antipyretic medication (Fig. 1G), supporting the ability of ΔT to reflect inflammatory vaccine responses.

Prior studies have found that the risk of vaccine-induced fever is strongly correlated across doses (*13*–*16*). Indeed, participants who reported having experienced a fever in response to previous vaccine doses measured significantly higher ΔT during the μHEAT study, suggesting an individualized risk of fever after vaccination (Fig. 1H). Fever-high participants also had higher total anti-SARS-CoV-2 antibody titres both before and after study vaccination (Fig. 1I), consistent with an established correlation between fever and antibody responses (*27*–*30*); however, the relative increase in antibody titres after booster vaccination was

unaffected by fever (fig. S1E). The relationship between prior and current fever remained significant when accounting for baseline antibodies, the time since last vaccination, and vaccine dose, and type in a general linear model ($P = 0.006$ and table S3). Together, these data support the finding that some individuals are consistently prone to stronger febrile responses following SARS-CoV-2 vaccination (*13*, *16*, *27*–*33*), making this a useful system in which to explore human fever-microbiome dynamics.

Fever is associated with self-reported diet and markers of cardiometabolic health

Dietary patterns shape immune function, with a large body of recent work showing the impact of industrialized diets on microbial ecology and chronic inflammatory processes (*34*, *35*). Notably, we observed that participants who habitually consumed a plant-based diet (self-reported vegans and vegetarians, $n = 46$) had a significantly lower ΔT compared with omnivores (self-reported omnivores, pescatarians, and other specialty diets, $n = 125$, Fig. 2A). Accordingly, significantly fewer fever-high participants were vegetarian (Fig. 2B and fig. S2A; $P = 0.03$ by Fisher's exact test).

Self-reported diet may be confounded by other variables. However, ΔT was lower in vegans compared with vegetarians, as well as in vegetarians compared with pescatarians and omnivores (fig. S2B). The maximum absolute temperature attained after vaccination was lower in vegans and vegetarians within both male and female participants (fig. S2C); furthermore, the association between fever and diet remained significant after controlling for the effects of gender, age, BMI, vaccine dose and type, antipyretic usage, prior fever, and baseline anti–SARS-CoV2 antibodies ($P = 0.004$, table S3). By contrast, antibody levels were similar by self-reported diet, both before and after vaccination (Fig. 2C) and by fold-change increase (fig. S2D), suggesting that diet-associated fever did not necessarily translate into altered antibody responses. Together, these data indicate an association between habitual self-reported diet and vaccine-induced fever.

To better understand the metabolic and immune signatures associated with self-reported dietary patterns and fever risk, we performed metabolomic and proteomic profiling of baseline sera. As expected, there were significant and substantial differences between self-reported vegetarians and omnivores (fig. S3, A and B, and tables S4 and S5). Omnivores displayed higher levels of serum triglycerides and diglycerides (fig. S3, C to E); higher LDL:HDL cholesterol ratios (fig. S3F); and higher abundance of peptides, amino acids, and ferritin in serum (fig. S3. A and B), compared with those who followed a plant-based diet. These patterns are consistent with increased consumption of animal-derived protein, fat, and iron in omnivores, and with improved cardiometabolic health in vegetarians (*36*).

Consistent with the reduced adherence to plant-based diets among fever-high participants, we found that fever-high individuals also showed less healthy cholesterol and lipid profiles at baseline (Fig. 2, D to F, and tables S6 and S7), and that this effect was largely attributable to diet (table S8). However, there was a stepwise trend for several markers of cardiometabolic dysregulation to be increased in fever-high participants within each dietary category, although these did not reach statistical significance (fig. S3, G to H). Nevertheless, apolipoprotein E (ApoE), a major cholesterol transporter, was distinctly elevated in fever-high participants (Fig. 2D), as were two metabolic features annotated as ethyl linolenate, a precursor of arachidonic acid which is used in the biosynthesis of inflammatory prostaglandins (*37*) (Fig. 2E), and these features remained significant after correction for habitual diet (tables S6 and S7). This suggests that both habitual consumption of animal products, as well as other factors, affect metabolic profiles in fever-high participants. In summary, the risk of vaccine-induced fever is associated with broad self-reported dietary patterns and metabolic health.

Fever-high individuals have chronic low-grade inflammation

Fever is an innate immune response derived from inflammatory processes; moreover, poor cardiometabolic health is frequently associated with chronic inflammation (*34*, *38*). Indeed, we found that fever-high individuals displayed a distinct innate immune signature at baseline. Pro-inflammatory signaling molecules such as RANKL (TNFSF11) and the interferon receptor IFNAR1 were elevated in fever-high participant sera, alongside anti-inflammatory cytokines such as IL-10, IL-37, and IL-36RN (Fig. 2D); these last cytokines might reflect an attempt by the host to control prior or ongoing inflammation. Several proteins with possible roles in microbial control, including IL-17RE (important for intestinal barrier integrity) (*39*), and ITLN1 (intelectin-1, an intestinal lectin which binds bacterial glycans) (*40*), were also significantly elevated compared with the low-fever group.

We thus further tested whether intestinal inflammation differed by fever category, using fecal calprotectin as a clinically relevant inflammatory marker. Fever-high participants had significantly higher levels of fecal calprotectin compared with the fever-low group, two weeks after vaccination (Fig. 2G). To determine whether intestinal inflammation was stable over time, we measured transcripts of the calprotectin subunit S100A9 through qPCR at two additional time points in RNA-buffered stool samples. These data indicated a positive, although nonsignificant, trend towards elevated calprotectin in fever-high participants both before and after vaccination, with no change over time associated with either vaccination or fever (fig. S4, A to C). The mean S100A9 expression per person was significantly higher in fever-high individuals (Fig. 2H). Moreover, calprotectin was significantly increased in fever-high participants when accounting for all time points per person, and for sex, age, and starting antibody levels, in a mixed linear model (table S3). By contrast, there was no association between intestinal inflammatory markers and habitual vegetarianism (fig. S4, D to F). Together these results suggest that intestinal inflammation is chronically elevated in fever-high participants.

Given the higher antibody titres seen in fever-high individuals, we also asked whether any serum markers predicted postvaccine antibodies. To follow an analysis analogous to our fever approach, we divided participants into "IgG-high" ($n = 71$) and "IgG-low" ($n = 100$) based on the group mean anti–SARS-CoV2 IgG titres two weeks after vaccination (fig. S4G). Very few metabolic or immune features were significantly associated with postvaccine antibody levels in this cohort (fig. S4, H and I, and tables S9 and S10), even when controlling for baseline anti–SARS-Cov2 IgG levels or restricting the analysis to third-dose BNT162b2 recipients without infection history (most participants; $n = 117$) to standardize the immune exposure. Most of the serum proteins and metabolites that were elevated in fever-high participants, including ApoE, IFNAR1, IL37, IL-36RN, and ethyl linolenate, also remained significantly associated with fever when controlling for prevaccine anti–SARS-CoV2 IgG levels (tables S6 and S7). There was no positive association between fecal calprotectin and antibody titres (fig. S4, J to L). In conclusion, although serum and fecal markers of inflammation associate with vaccine-induced fever, we do not find corresponding associations with vaccine-induced antibodies.

Taken together, these data reveal a baseline immunometabolic signature associated with the risk of vaccine-induced fever. Importantly, fever-high individuals also show elevated levels of intestinal inflammation, implicating a possible role for innate immune signaling by means of the gut microbiome.

The gut microbiome is resilient to acute fever

Febrile temperatures can influence microbial replication and invasion, and have an important role in amplifying immune function (*41*, *42*), leading us to speculate that acute fever might affect the resident gut microbiome. To determine the impact of fever on the human gut microbiome, we generated shotgun metagenomic sequencing from six

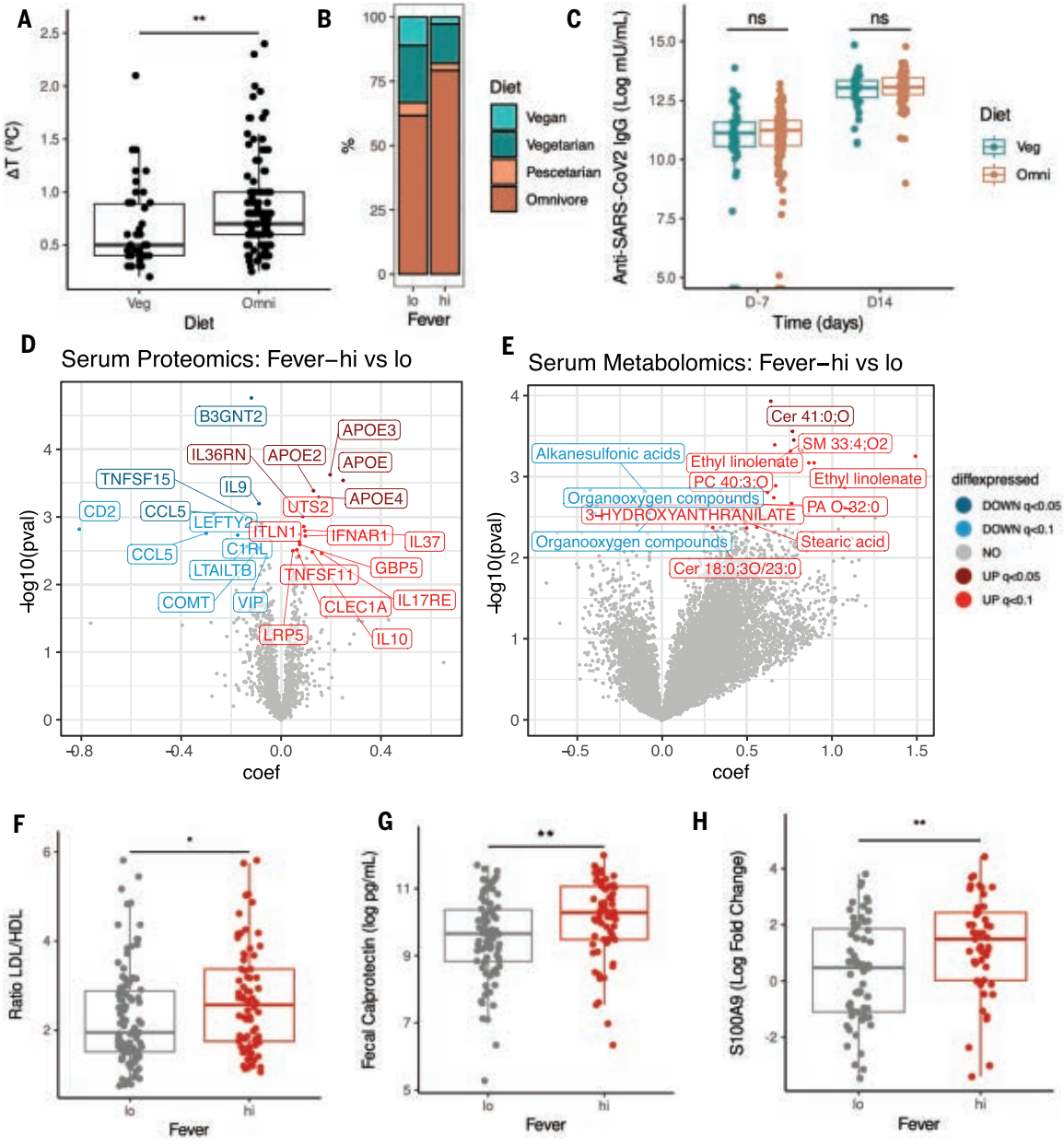


Fig. 2. Vaccine-induced fever is associated with habitual diet, poor cardiometabolic health, and chronic inflammation. (**A**) ΔT according to self-reported habitual diet. (**B**) Distribution of diet within fever-low and fever-high participant categories. (**C**) Serum anti–SARS-CoV-2 IgG titres prevaccine (D-7) and postvaccine (D14), according to dietary category. (**D** to **E**) Volcano plot of serum proteins (D) and metabolites (E) significantly enriched (red) or depleted (blue) in fever-high participants compared with fever-low participants at day D-7 prevaccine. (**F**) Ratio of LDL to HDL cholesterol in baseline serum of fever-high and -low participants. (**G**) Fecal calprotectin at day D14 postvaccine in fever-high and -low participants. (**H**) Expression of the S100A9 calprotectin subunit in stools averaged across two additional time points, D-1 prevaccine and D+7 postvaccine. Statistical significance was determined through MaAsLin2 linear models, taking age and sex as covariates, and adjusting *P*-values through false discovery rate (FDR) (D) to (E); or through Wilcoxon Rank-Sum test (A), (C), and (F) to (H). For additional covariates in linear models, see tables S6 and S7. Box plots are drawn from quartile 1 to 3 with a line at the median, and the whiskers are drawn at ± 1.5*interquartile range. *$P < 0.05$; **$P < 0.01$; n.s., not significant. $P > 0.1$. Lo, fever-low; hi, fever-high. Veg, vegans and vegetarians; omni, pescatarians and omnivores. LDL, low-density lipoprotein; HDL, high-density lipoprotein.

longitudinal fecal samples per participant over the 2-week period before and after vaccination ($n = 1022$ total samples, fig. S5A). We also generated longitudinal fecal transcriptomics from a subset of $n = 38$ participants to assess the functional activity of the microbiome ($n = 230$ samples, fig. S5B; see methods for subset selection). As expected, people had individualized and relatively stable microbiomes, with longitudinal samples from the same individual more similar to one another than to samples from unrelated individuals, at both the overall community level (fig. S5, C to E) and the level of individual strains (fig. S5, F and G).

We first tested whether vaccination or vaccine-induced fever might affect the microbiome. However, we did not observe any changes in microbiome alpha or beta diversity following vaccination, regardless of fever category (fig. S6, A to D, and table S3). Predicted bacterial

growth rates were also stable (fig. S6E), although the average growth rate was modestly elevated in fever-high participants across all time points. Measures of strain-level persistence and evolution were stable, with the exception of a slight overall decrease in strain persistence over time independently of fever (fig. S6, F and G). There was limited evidence of any individual genera- or gene-level transcription changing in abundance in fever-high individuals after vaccination (fig. S6, H and I, and table S11). Overall, we observed very minimal changes in the microbiome over time following either vaccination or fever.

This was initially surprising to us, as we expected fever to limit microbial growth through either direct (temperature) or indirect (immune) mechanisms (*24*, *42*). However, in vitro growth assays of 24 prevalent human gut bacteria indicated that although growth is significantly limited at 32°C compared with 37°C, it increases on average up to 42°C, a physiologically unlikely extreme (fig. S6J). Even the most temperature-sensitive bacterium in our screen, *Segatella copri*, showed only limited transcriptional changes in vitro when shifted from 37°C to 42°C, compared with a more robust response to below-host temperatures (fig. S6K). Moreover, as previously noted, we found no change in intestinal inflammation after vaccination or fever (fig. S4, A to C). Reported gastrointestinal side effects of vaccination such as appetite suppression, diarrhea, or nausea were rare (table S12). We therefore concluded that fever-like temperatures do not meaningfully affect gut commensals in vitro, and that short-term increases in body temperature do not affect intestinal inflammation or the human gut microbiome in vivo.

Fever-high microbiome is characterized by high flagellin expression and abundance of flagellated *Lachnospiraceae* spp.

Having observed that mild fever does not affect the temporal dynamics of diversity, growth, or strain persistence within the microbiome, we asked whether any stable microbiome characteristics were associated with fever risk. We proceeded with an analysis of the gut microbiome by fever category, using all longitudinal samples per person for greater statistical power, and leveraging mixed linear models through MaAsLin2 (*43*).

Transcriptomic data indicated that multiple flagellins and motility-associated proteins were up-regulated in fever-high participants across all time points (fig. S7, A and B, and table S13). This signature became even more pronounced when RNA abundances were normalized to the respective DNA abundances for that gene and sample, indicating a real transcriptional up-regulation (Fig. 3A and fig. S7C). Elevated flagellin expression in fever-high compared with fever-low participants occurred prevaccination and remained stable over time (Fig. 3B). Fever-high microbiomes also expressed a higher diversity of flagellins at baseline (Fig. 3C).

Flagellin is a canonical stimulator of innate immunity, raising the hypothesis that flagellin-rich microbiomes might elevate the immune tone, resulting in increased fever after vaccination. However, many commensal flagellins also evade inflammatory signaling (*44*). We applied a machine-learning classifier to annotate the flagellins based on protein identity, categorizing them as "silent" or "active" based on their predicted ability to signal through toll-like receptor (TLR5) (*45*). This prediction suggested that both active and silent flagellins were overexpressed in fever-high microbiomes compared with fever-low microbiomes (Fig. 3D and table S14).

We experimentally confirmed that the specific flagellins enriched in fever-high participants could activate TLR5. We expressed recombinant flagellins in vitro and used them to stimulate TLR5 reporter cells. These results confirmed that the flagellin sequences enriched in fever-high microbiomes included TLR5-stimulatory phenotypes (fig. S8, A and B). Thus, the microbiomes of fever-high individuals are enriched for diverse flagellins, including those with potent inflammatory activity.

To explore which microbiota species were responsible for the enriched flagellins, we performed a taxon-stratified analysis. This revealed that a large portion (67.2%) of the up-regulated flagellins were expressed by the genus *Waltera* (formerly *Acetatifactor*, family *Lachnospiraceae*), with most of the flagellins produced by the *Lachnospiraceae* family overall (76.3%) (Fig. 4E and table S13). *Waltera* was also the only genus that was significantly more abundant in fever-high participants across the study period according to metagenomic data (Fig. 3F, fig. S8C, and table S15). Moreover, two of the most strongly TLR5-stimulatory flagellins tested above were annotated as being produced by *Waltera* species. To verify the inflammatory phenotype of *Waltera* flagellin, we sheared flagella from pure cultures of *Waltera* intestinalis. Flagellin-containing bacterial supernatants activated a strong TLR5 response in TLR5 reporter cell lines (Fig. 3G) and were sufficient to induce robust IL-8 production in human organoids (fig. S8D). Thus, the microbiomes of fever-high participants are characterized by a high relative abundance of *Waltera*, which produces proinflammatory flagellin.

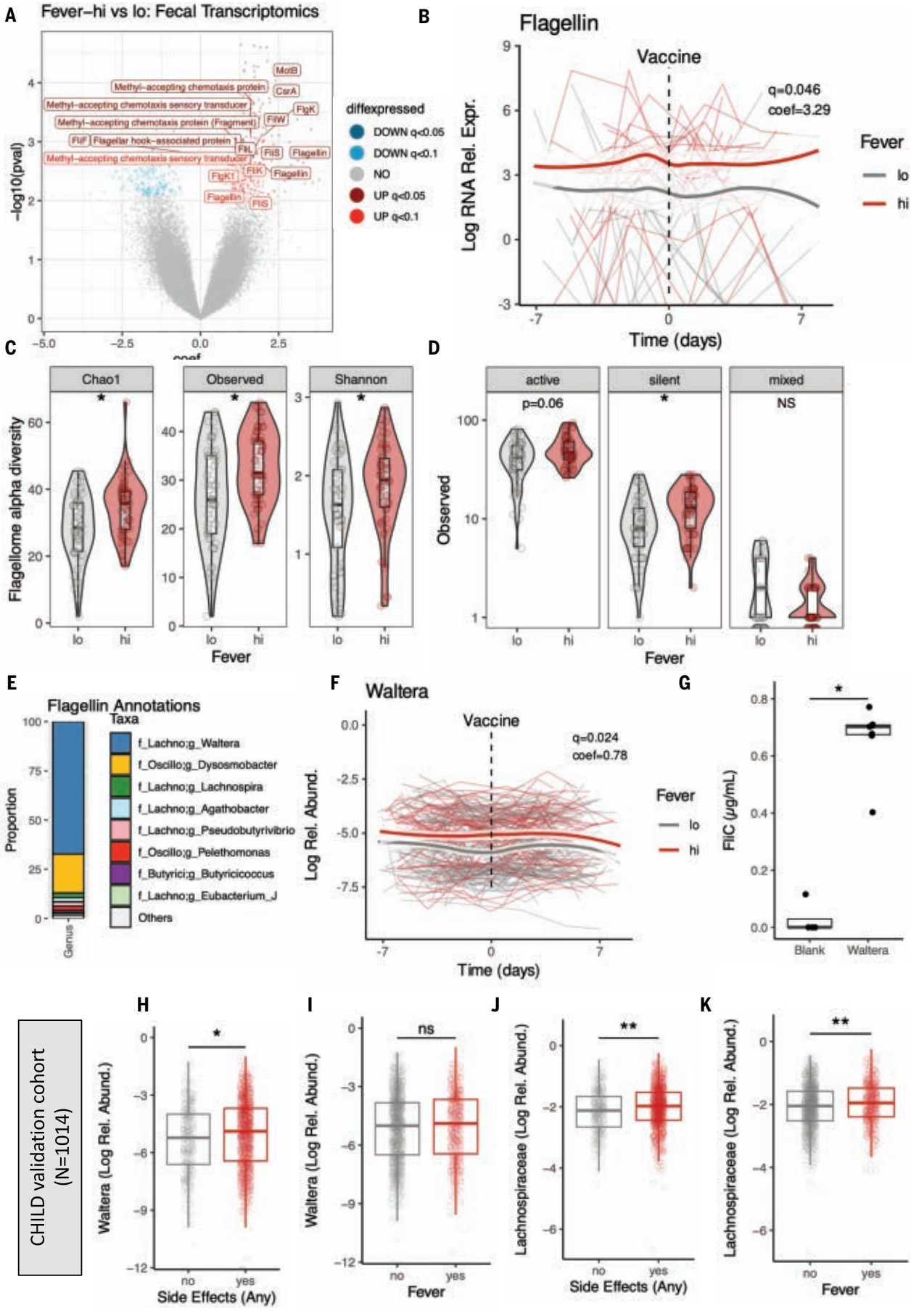
A popular hypothesis suggests that microbiome-driven innate immunity acts as an adjuvant for adaptive immune responses (*17*). We therefore looked at associations between the microbiome and antibody responses to vaccination. A smaller number of genes were changed at a transcriptional level in high- versus low-antibody participant microbiomes compared with high- versus-low fever microbiomes, and only one of the antibody-associated genes was related to flagellar motility (fig. S9A and table S16). These results were similar whether or not we corrected for baseline antibody titres (table S16). The association between flagellin transcriptional alpha diversity and fever also remained significant when correcting for baseline antibody levels (table S14), as did several individual motility genes (table S13). Although *Waltera* abundance also trended higher in high-antibody individuals over time, this association did not reach statistical significance (fig. S9B), nor was *Waltera* correlated with the fold-change increases in antibodies with vaccination ($P = 0.6$). *Waltera* abundance was significantly higher in fever-high individuals even after correcting for baseline immunity through prevaccine antibody titres, vaccine dose and type, and prior fever ($P = 0.008$). Therefore, although fever-high participants had modestly higher antibody levels (Fig. 1I), the *Waltera* flagellin signature was more strongly associated with fever at booster vaccination than with postvaccine antibodies in this cohort.

Finally, we asked whether this microbiome signature could be replicated in other populations beyond our study dataset. We analyzed data from 1014 Canadian adults who self-reported side effects after SARS-CoV-2 vaccines (*46*) by answering yes/no questions about postvaccine fever, headache, fatigue, myalgia, and injection site pain ($n = 1549$ fecal metagenomic samples; 90.0% of participants received a 2nd-dose mRNA vaccine). In this cohort, participants who reported any side effects after vaccination had significantly greater fecal abundance of *Waltera* (Fig. 3H and fig. S9, C to G). *Waltera* was significantly associated with self-reported headache (fig. S9C) and trended higher with each of the other individual self-reported symptoms (Fig. 3I and fig. S9, D to F). Moreover, the entire *Lachnospiraceae* family was significantly associated with total vaccine side effects (Fig. 3J) and with self-reported fever, fatigue, and myalgia (Fig. 3K and fig. S9, H to K). Neither *Waltera* nor *Lachnospiraceae* associated with postvaccine antibody titres (fig. S9, G and L). These data support an association between human gut microbiome and vaccine reactogenicity and suggest that *Waltera* and its relatives are broadly more abundant in those who experience acute, systemic adverse vaccine reactions.

In summary, the microbiomes of fever-high participants are characterized by chronic high expression of flagella and a chronic high abundance of flagellated bacteria.

Industrialized dietary components increase microbial flagellin production and exacerbate vaccine responses

Diet shapes microbial ecology and immune health (*34*) and was associated with fever risk in the μHEAT cohort (Fig. 2); we therefore assessed whether self-reported dietary categories would explain the flagellin-rich microbiome patterns. Although many taxa and genes were associated with vegetarianism or omnivory (fig. S10, A and B), baseline flagellin


A
Fever–hi vs lo: Fecal Transcriptomics
−log10(pval)
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diffexpressed
DOWN q<0.05
DOWN q<0.1
NO
UP q<0.05
UP q<0.1
Methyl−accepting chemotaxis protein
Methyl−accepting chemotaxis sensory transducer
Methyl−accepting chemotaxis protein (Fragment)
Flagellar hook−associated protein
Flagellin
MotB
CsrA
FlgK
FliW
FliF
FliL
FliS
FlgK1
FliK
B
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Vaccine
q=0.046
coef=3.29
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Time (days)
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Flagellome alpha diversity
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silent
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p=0.06
NS
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Fever
E
Flagellin Annotations
Taxa
Proportion
Genus
f_Lachno;g_Waltera
f_Oscillo;g_Dysosmobacter
f_Lachno;g_Lachnospira
f_Lachno;g_Agathobacter
f_Lachno;g_Pseudobutyrivibrio
f_Oscillo;g_Pelethomonas
f_Butyrici;g_Butyricicoccus
f_Lachno;g_Eubacterium_J
Others
F
Waltera
Vaccine
q=0.024
coef=0.78
Log Rel. Abund.
Time (days)
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G
FliC (μg/mL)
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CHILD validation cohort (N=1014)
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Side Effects (Any)
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Lachnospiraceae (Log Rel. Abund.)
Fever

Fig. 3. The fever-high microbiome is characterized by high flagellin expression and abundance of flagellated bacteria. (**A**) Volcano plot of genes (UniRef50) significantly enriched and depleted at the transcriptional level, relative to DNA abundance for that same gene, in fever-high compared with fever-low individuals. Genes involved in motility are labeled. (**B**) Temporal dynamics of the expression of flagellin with UniRef50 ID R6ZZ21, one of the most highly differentially expressed flagellins, by fever category. Log RNA Rel. Expr. is the log-normalized ratio of RNA abundance relative to DNA abundance for that gene. (**C**) Alpha diversity of all expressed flagellins in prevaccination samples by fever category, based on RNA detection. Each data point is an individual sample. (**D**) Machine-learning prediction of expressed flagellin phenotypes, defined as either stimulatory to TLR5 (active), silent to TLR5, or mixed phenotype, and quantified by observed alpha diversity per sample. (**E**) Taxonomic annotations for all flagellins that were significantly differentially expressed by fever category according to taxa-stratified analysis. (**F**) Relative abundance of the genus *Waltera* at the metagenomic level, by fever category over time. (**G**) Bioactive flagellin detected in *Waltera intestinalis* cultures according to TLR5 reporter assay. (**H** to **K**) Relative abundance of *Waltera* or *Lachnospiraceae* by self-reported vaccine side effects (H) and (J) and self-reported fever (I) and (K) in an independent dataset of $n = 1014$ Canadian adults receiving a SARS-CoV-2 vaccine ($n = 1549$ total samples). Statistical significance was determined through MaAsLin2 linear models taking sequencing depth, Bristol Stool Score, and time at room temperature as technical covariates, treating participant ID as a random effect, and adjusting *P*-values through FDR, using all samples per participants (A) and (F) or by means of mixed linear models that adjusted for age and sex and used only baseline prevaccination samples (C) and (D). Additional covariates can be found in table S13. Statistical significance in (G) was determined by Wilcoxon and in (H) to (K) by mixed linear models that control for sequencing depth and use participant ID as a random effect. Box plots are drawn from quartile 1 to 3 with a line at the median, and the whiskers are drawn at ± 1.5*interquartile range. $*P < 0.05$; $**P < 0.01$; n.s., not significant. $P > 0.1$. Lo, fever-low; hi, fever-high.

diversity remained higher in fever-high participants when controlling for age, sex, and diet (table S14). Similarly, although *Waltera* trended higher in omnivores compared with vegetarians (fig. S10C), its abundance remained higher in fever-high participants even when accounting for habitual diet (fig. S10D).

The dietary data collected in μHEAT was limited to broad self-reported categories, and may therefore miss important variations in real dietary intake. Indeed, the baseline inflammatory signature associated with fever was independent of dietary category (table S7 and fig. S4, D to F). To gain more detailed insight into dietary patterns, we therefore leveraged MEDI (Metagenomic Estimation of Dietary Intake), a recently developed tool that identifies food-derived DNA present in stools (*47*). MEDI confirmed differences by self-reported diet, indicating that vegetarians consumed a greater proportion of plant-based foods (fig. S11A), no pork, and more flax (fig. S11, B and C, and table S17), compared with omnivores.

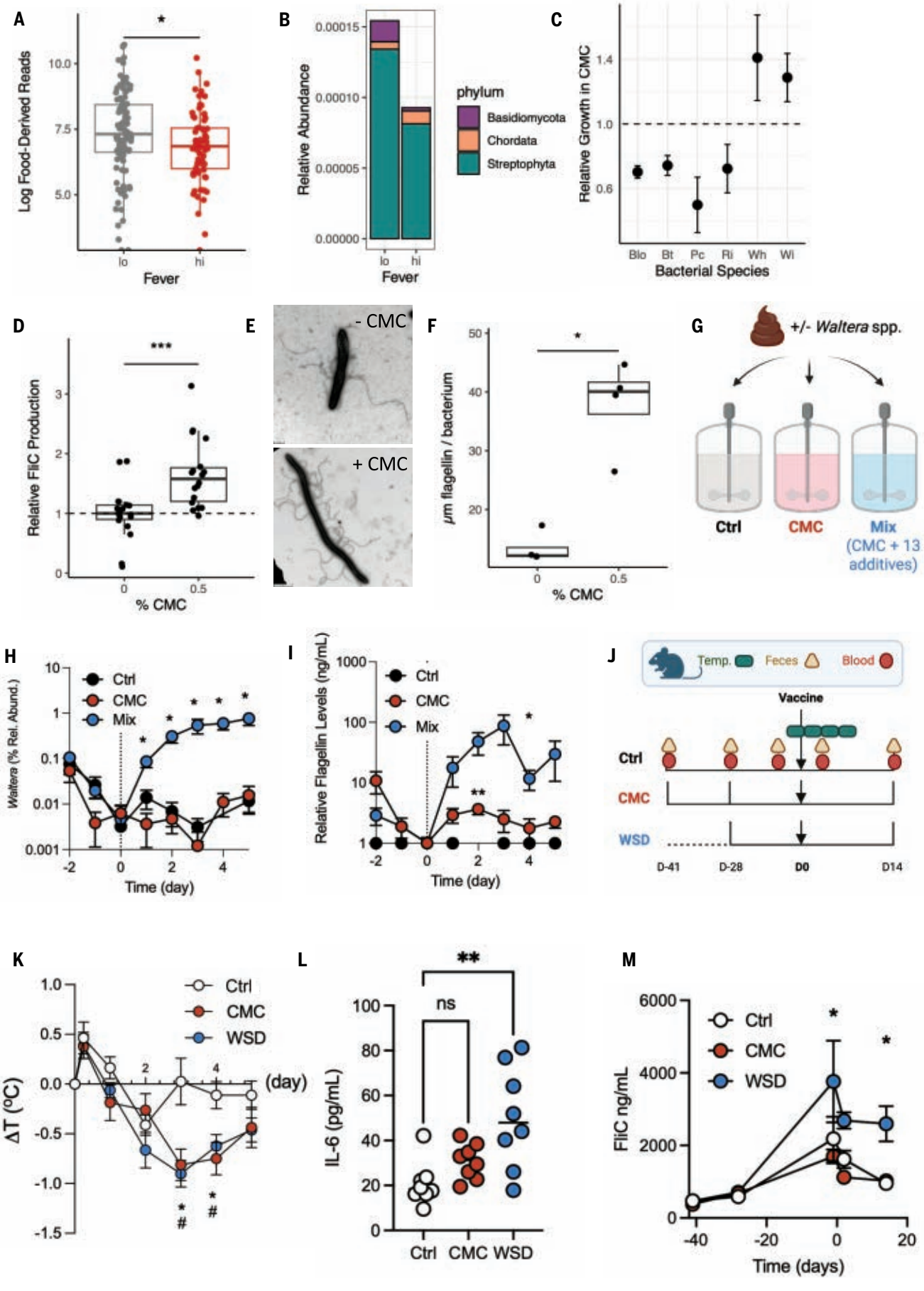
Notably, although total food DNA detected in stool did not differ by habitual diet (fig. S11D), significantly less food DNA was detected in the stool of fever-high compared with fever-low participants (Fig. 4, A and B), particularly amongst omnivores (fig. S11E). MEDI is biased towards the detection of whole, unprocessed foods and has been shown to detect less food DNA in participants consuming a "Western" diet compared with a high-fiber one (*47*). Reduced food DNA in fever-high participant stool may therefore indicate higher consumption of processed foods. The association between fecal food reads and fever was robust in mixed linear models which used only baseline (prevaccine) samples and which accounted for sequencing depth, Bristol Stool Score, age, sex, habitual diet, and prior immunity through vaccine dose and baseline antibody levels ($P = 0.032$). Moreover, two immune proteins that were elevated in fever-high individuals (CLEC1A, a C-type lectin; and TNFSF11, encoding the cytokine RANKL) displayed a strong negative correlation with fecal food reads (fig. S11, F to H, and table S18), indicating an association between food quality and immune profiles. These findings suggested the idea that low dietary quality—rather than meat consumption per se—might drive inflammatory phenotypes.

Ultraprocessed foods are consistently linked to low-grade inflammation in both humans and mice (*35*, *48*). In particular, emulsifiers such as carboxymethylcellulose (CMC) are common food additives that have been reported to up-regulate microbiome expression of flagellin and drive low-grade intestinal inflammation, consistent with the patterns observed in our cohort (*49*, *50*). During in vitro culture, we found that CMC enhanced the growth of both *W. intestinalis* and *Waltera hominis*, while suppressing the growth of other common gut commensals from the Bacillota and Bacteroidota phyla (Fig. 4C). Moreover, growth in CMC significantly increased the production of TLR5-stimulatory flagellin by *W. intestinalis*, after normalizing for bacterial density (Fig. 4, D to F).

We next tested whether ultraprocessed food additives would impact flagellin in complex human microbiomes, using bioreactors inoculated with fecal slurry from two different human donors and supplemented or not with pure *W. intestinalis* and *W. hominis* (Fig. 4G). In these human-derived microbiomes, CMC alone had no impact on *Waltera* abundance, but caused a modest increase in total flagellin levels (Fig. 4, H and I, and fig. S12). By contrast, a cocktail of food additives (CMC plus 13 other common food additives; see methods) substantially enhanced the relative abundance of *Waltera* (Fig. 4H and fig. S12, A to D), the abundance of *Waltera* flagellin genes (fig. S12, E to H), and the total production of TLR5-stimulatory flagellin (Fig. 4I and fig. S12, I to L). The additives cocktail also increased the abundance of other flagellated *Lachnospiraceae* members, such as *Agathobacter* (fig. S12M). These data suggest that there may be additive effects of multiple ultraprocessed food components on community structure. Together, they provide proof-of-principle that components of an industrialized human diet can support the growth of flagellated human gut bacteria and stimulate total flagellin production.

To test whether diet influences vaccine adverse reactions, we turned to a murine model. In conventionally housed mice, systemic inflammation manifests as hypothermia instead of fever (*51*). The hypothermic response in rodents is immunologically similar to a febrile response; it is dependent on cytokine release and prostaglandin signaling, and responsive to antipyretic medication (*52*–*54*). Specific pathogen-free (SPF) mice were either fed CMC via the drinking water or were fed a western-style diet (WSD; high fat, low fiber), compared with a control group fed polysaccharide-rich normal chow (ctrl); we then measured systemic inflammatory responses to an mRNA SARS-CoV-2 vaccine (Fig. 4J). Vaccination induced an acute drop in mouse body temperature (Fig. 4K and fig. S13A) and body weight (fig. S13B), accompanied by increases in systemic inflammatory cytokines (Fig. 4L and fig. S13, C and D). We observed that both the CMC and WSD diets exacerbated the inflammatory response to vaccination in mice compared with regular chow, with WSD having a more marked impact than CMC (Fig. 4, K and L). Consumption of a WSD led to a significantly stronger temperature response to vaccination (Fig. 4K) and increased levels of IL-6 24 hours after vaccination (Fig. 4L). Intestinal inflammation was not significantly different between the groups, although it trended higher in both WSD and CMC mice (fig. S13E). Furthermore, the WSD-fed mice not only experienced increased short-term inflammation, but additionally displayed increased antibody titres at the endpoint (fig. S13F).

Consumption of a WSD led to a disrupted microbiome structure in mice, characterized by a loss of Bacteroidota spp. and a bloom of Bacillota spp. (fig. S13G), consistent with many prior reports. Notably, there was a trend for native mouse *Waltera* to bloom in the WSD group (fig. S13H), and this taxon correlated significantly with postvaccine serum IL-6 (fig. S13I). Moreover, WSD-fed mice had higher fecal flagellin


A
Log Food-Derived Reads
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Chordata
Streptophyta
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Bacterial Species
Blo
Bt
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Ri
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Wi
D
Relative FliC Production
% CMC
E
- CMC
+ CMC
F
μm flagellin / bacterium
% CMC
G
+/- Waltera spp.
Ctrl
CMC
Mix
(CMC + 13 additives)
H
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Time (day)
I
Relative Flagellin Levels (ng/mL)
Time (day)
J
Temp.
Feces
Blood
Vaccine
Ctrl
CMC
WSD
D-41
D-28
D0
D14
K
ΔT (°C)
(day)
L
IL-6 (pg/mL)
ns
M
FliC ng/mL
Time (days)

Fig. 4. Industrialized diet controls microbial flagellin production. (**A**) Total food reads detected in stools, summed value per person, according to fever category. (**B**) Relative abundance of food-derived DNA at the phylum level, as a proportion of all stool DNA, in fever-low and fever-high participants. (**C**) Relative growth of intestinal commensal bacteria at 24 hours in media containing 0.5% CMC compared with 0% CMC. Blo, *Blautia obeum*; Bt, *Bacteroides thetaiotamicron*; Pc, *Prevotella (Segatella) copri*; Ri, *Roseburia intestinalis*; Wh, *Waltera hominis*; Wi, *W. intestinalis*. (**D**) Units of FliC, detected through the TLR5 reporter assay and normalized to bacterial density, in flagellin sheared from *W. intestinalis* cultures after growth in 0 or 0.5% CMC. (**E**) Negative stain microscopy of *W. intestinalis* grown in the absence (0%) or presence (0.5%) of CMC. (**F**) Quantification of flagellin length per cell in negative stain microscopy. (**G**) Schematic for bioreactor experiment. Reactors were inoculated with two different human fecal donors, with or without supplementation of pure *W. intestinalis* culture (total of four inocula with three replicates each, for $n = 12$ per dietary group). (**H**) Relative abundance of *Waltera* over time (pooled data, $n = 12$). (**I**) Bioactive flagellin levels over time, normalized to control ($n = 12$). (**J**) Schematic for a dietary intervention in specific pathogen-free mice. WSD, Western-Style Diet. (**K**) Mouse hypothermic response (ΔT) postvaccine. Asterisk indicates CMC versus ctrl; pound sign indicates WSD versus ctrl. (**L**) Serum IL-6 at 24 hours postvaccine. (**M**) Fecal bioactive flagellin levels over time. Statistical significance was determined through Wilcoxon Rank Sum test (A), (D), and (F), two-way ANOVA with Dunnett's post hoc test (H) to (I), (K), and (M), and Kruskal-Wallis (L). In (D) and (F), each data point represents an individual bacterial culture. Data points in (C), (H), (I), (K), and (M) represent mean ± SEM. Box plots are drawn from quartile 1 to 3 with a line at the median, and the whiskers are drawn at ± 1.5*interquartile range. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$. Lo, fever-low; hi, fever-high.

levels as measured by TLR5 reporter assay (Fig. 4M), and transiently higher fecal LPS levels around the time of vaccination (fig. S13J). These data are consistent with a role for the diet-responsive microbiome in modulating inflammatory responses to vaccination.

In summary, industrialized dietary interventions boost *Waltera* abundance and microbiome flagellin production both in human microbiomes and in mice, and exacerbate the innate immune response to vaccination.

Human microbiome drives flagellin-dependent inflammation and vaccine responses

Having established that proinflammatory flagellins were responsive to diet and were enriched in the microbiomes of fever-high individuals, we next tested whether entire microbiomes would exacerbate inflammatory responses in human cell culture or in mice.

We cultured human colonic organoids in vitro and confirmed that stimulation with bacterial flagellin elicited a proinflammatory transcriptional response and production of IL-8 (fig. S14, A and B), consistent with the expected TLR5-dependent stimulation of NF-kB (*44*). Human colonic organoids stimulated with fever-high fecal supernatants produced significantly more IL-8 compared with stimulation with fever-low fecal samples (Fig. 5A), and up-regulated genes related to TLR signaling cascades (Fig. 5B, fig. S14C, and table S19). Increased inflammation from fever-high samples was stronger when data were stratified to male participants and to women who took oral contraceptives, perhaps because a constant baseline temperature in these two groups allows for more accurate fever categorization (fig. S14, D and E). IL-8 production correlated strongly with the concentration of bioactive flagellin in the fecal sample, as measured through the TLR5 reporter cell line (Fig. 5C). In total, these data indicate that fecal flagellin from fever-high microbiomes drives inflammation in human intestinal cells.

To directly test whether microbiome-driven inflammation could alter vaccine responses, we conducted a fecal microbiota transplant experiment in which germ-free C57BL/6 mice were colonized with fever-high or fever-low human microbiomes and vaccinated against SARS-CoV-2 (Fig. 5D and fig. S15A). Vaccination of gnotobiotic mice induced an acute drop in mouse body temperature (Fig. 5E and fig. S15B), accompanied by increases in systemic inflammatory markers (fig. S15, C and D), recapitulating a systemic adverse reaction to the vaccine as previously observed in SPF mice.

After one week of colonization, prior to vaccination, mouse recipient microbiomes successfully recapitulated many of dominant genera of the human microbiome, although *Waltera* itself colonized poorly (fig. S15A). *Waltera* was not different by group, and mice that were colonized with fever-high or fever-low microbiomes displayed no differences in baseline inflammatory markers prior to vaccination (fig. S15, C to E). However, mice colonized with fever-high fecal samples showed a significantly greater drop in body temperature following vaccination (Fig. 5E and fig. S15B). Furthermore, there was a strong correlation between the fever of the human donor and the hypothermic response of the recipient gnotobiotic mouse (Fig. 5F). Fever-high-colonized mice also had higher levels of inflammatory serum IL-6 post-vaccine (Fig. 5G and fig. S15C), and this response correlated significantly with the flagellin content of the donor stool sample (Fig. 5H). Moreover, despite the lack of *Waltera* colonization, total bioactive flagellin levels were higher in the microbiomes of fever-high–colonized mice, and showed a tendency to increase over time, highlighting the contribution of other flagellated bacteria (Fig. 5I). Fecal LPS levels were not significantly different by group (Fig. 5J), nor was fecal inflammation (Fig. 5K). Consistent with the organoid data, inflammatory responses in mice tended to be more robust in males than females (fig. S16, A to C; donors and recipients were sex-matched). Together, these data indicate that the human intestinal microbiome affects systemic inflammatory responses to vaccination.

To directly test for flagellin dependency of this immune response, we repeated the experiment in germ-free mice lacking TLR5 and, additionally, NLRC4, the intracellular inflammasome receptor for flagellin [i.e., TLR5/NLRC4 double knock-out mice (DKO)]. An exacerbated inflammatory response to vaccination was once again observed in fever-high microbiome recipients, indicating a role for flagellin-independent immune signaling; nevertheless, the effect size was dampened in DKO compared with WT mice (Fig. 5E and fig. S15B) and inflammatory markers were overall decreased in DKO mice (Fig. 5, G and K, and fig. S15, C to E). These data suggest that flagellin-dependent inflammatory signaling from the microbiome contributes to the innate immune response to vaccination.

In contrast to the previous experiment in SPF mice, there was no difference in antibody levels by microbiome group at experimental endpoint (Fig. 5L). Vaccine antibody titres were moreover sufficiently induced in DKO mice (Fig. 5L). Our data indicate that the human microbiome can influence innate immune responses to vaccination without influencing the generation of vaccine-induced specific antibodies.

In total, our data suggest that innate immune detection of a flagellated human gut microbiome primes vaccine-induced adverse reactions, and that this can be modulated through diet.

Discussion

The fear of vaccine-induced adverse reactions is a major underlying cause of vaccine hesitancy and thus poses an important risk to global public health (*9*, *55*). The risk of fever and other side effects after vaccination is individualized; here, we find that the gut microbiome sets inflammatory immune tone to prime vaccine-induced fevers. Our data suggest a mechanism whereby overgrowth and motility of flagellated gut bacteria trigger flagellin-dependent inflammation and exacerbate the risk of vaccine-induced inflammatory responses. We moreover demonstrate that industrialized dietary patterns drive microbiome flagellin production in vitro, exacerbate vaccine reactions in vivo, and associate with fever responses in healthy adults. Together, our results establish the possibility of microbiome-targeted interventions to attenuate vaccine-induced adverse events.

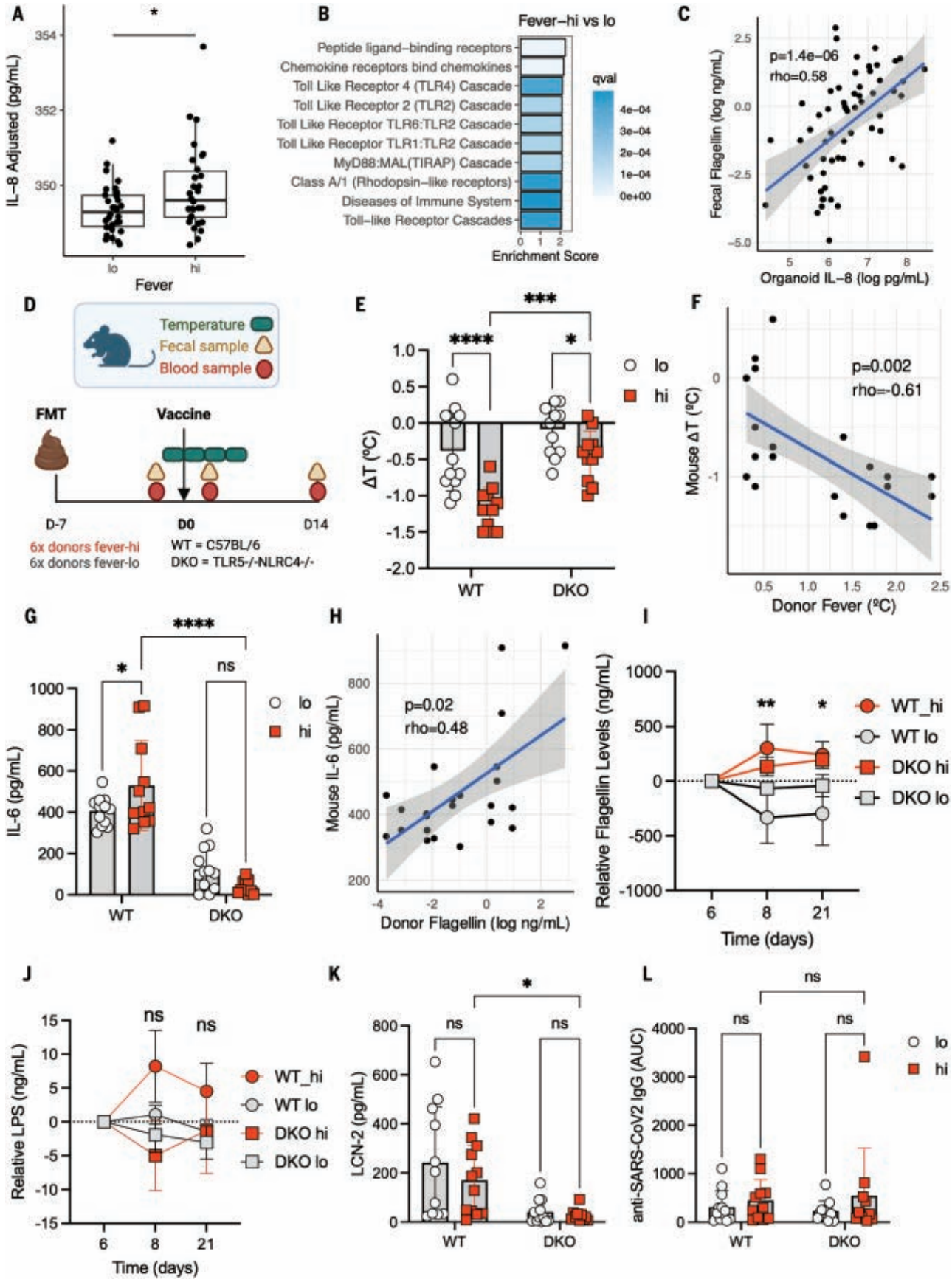

Fig. 5. The fever-high human microbiome drives flagellin-dependent inflammation. (**A**) IL-8 measured in the supernatants of human colonic organoids after stimulation with fecal bacterial products derived from fever-low and fever-high participant samples ($n = 30$ per group; each data point represents cells stimulated with a sample from a

different person, averaged across two technical replicates). (**B**) Gene set enrichment analysis (GSEA) of pathways significantly overrepresented ($q < 0.05$) at the transcriptional level in organoids stimulated with fever-high samples compared with fever-low samples. The label "MyD88:MAL (TIRAP) Cascade" was shortened from "MyD88:MAL (TIRAP) cascade initiated on plasma membrane" for space constraints. (**C**) Correlation between organoid IL-8 production and bioactive flagellin levels of the fecal sample. (**D**) Schematic for the germ-free mouse experiment. FMT, fecal microbiota transplant. WT, wild type; DKO, double knock-out mice lacking TLR5 and NLRC4 and thus unable to sense flagellin. (**E**) Change in mouse body temperature (ΔT) at 24 hours postvaccine. (**F**) Correlation between mouse ΔT at 24 hours postvaccine and postvaccine fever of the corresponding human donor. (**G**) Serum IL-6 at 24 hours postvaccine. (**H**) Correlation between mouse serum IL-6 at 24 hours postvaccine and flagellin content of the donor fecal sample. (**I**) Bioactiveflagellin levels in recipient mouse stool over time. (**J**) Bioactive LPS levels in recipient mouse stool over time. (**K**) Fecal LCN-2 levels at 24 hours postvaccine. (**L**) Anti–SARS-CoV-2 IgG antibody titres at experimental endpoint, D14 postvaccine. Statistical significance was determined through mixed linear model adjusting for donor Bristol stool score (as a proxy for bacterial biomass) and taking batch as a random effect (A); Spearman's correlation (C), (F), and (H); two-way ANOVA with posthoc Sidak's test (E), (G), (K), (L), and two-way ANOVA with posthoc Dunnett's test (I) and (J). Data points in (I) and (J) represent mean ± SEM. Box plots are drawn from quartile 1 to 3 with a line at the median, and the whiskers are drawn at ± 1.5*interquartile range. $*P < 0.05$; $***P < 0.001$; $****P < 0.0001$; n.s., not significant. $P > 0.1$. Lo, fever-low; hi, fever-high.

Our data provide a clear causal link between flagellin-stimulated TLR5 signaling and vaccine innate immune responses in humans. Notably, two recent papers have also suggested a role for *Lachnospiraceae* and microbiome-produced flagellins in the fever responses of critically ill patients (*56*, *57*). Moreover, elevated commensal flagellin production has been shown to drive low-grade inflammation and metabolic syndrome in mice (*58*–*60*), and antibodies against commensal flagellins are associated with chronic inflammatory diseases in humans (*61*–*63*). Of note, the implicated taxa in these studies, and our own, are flagellated commensal Bacillota species, rather than the Proteobacteria bloom more commonly implicated in dysbiosis. Thus, *Lachnospiraceae* flagellins may broadly influence the inflammatory immune tone, including during acute febrile responses.

We also found evidence of microbiome-dependent but flagellin-independent pathways, as TLR5/NLRC4 DKO mice still exhibited moderately increased inflammation when colonized with fever-high microbiomes. One alternative candidate pathway is type-1 interferon (IFN-1) signaling, being important for the immune response to SARS-CoV-2 and influenza vaccines (*64*–*66*), and tonically stimulated by the gut microbiome (*67*–*71*). A prior study found that IFN-1 signaling was dampened in germ-free mice compared with SPF mice after single-dose mRNA SARS-CoV-2 vaccination (*72*). Moreover, fever-high participants had higher IFNAR (IFN-1 receptor) protein levels in the blood at baseline (Fig. 2D). This additional potential mechanism could be a potential avenue for future research.

Our data also highlight diet as a possible route to manipulate pro-inflammatory, flagellin-rich microbiomes. Dietary additives directly led to flagellin overproduction in vitro, and a Western-style diet exacerbated the inflammatory vaccine response in mice. In humans, fever risk was associated with diet and metabolic markers: fever-low participants were more likely to self-report as vegetarian, had healthier cholesterol profiles, and had higher detection of food-derived DNA in stool, suggesting higher dietary quality. Detailed dietary questionnaire data would have been valuable to further support these findings. However, it is well established that industrialized diets increase the risk of chronic diseases (*36*, *73*–*78*); moreover, a recent meta analysis found that vaccine adverse events were more common in high-income countries (*79*). Our data raise the possibility that diet could affect acute vaccine reactions, although interventional human studies are needed to test this hypothesis.

The finding that flagellin-TLR5 sensing exacerbates vaccine-induced fever is mechanistically consistent with prior work demonstrating that flagellins are important for vaccine-induced antibody responses in mice, and with prior associations between TLR signaling and antibody responses in humans (*19*, *20*, *80*–*84*). However, despite the strong fever-microbiome signal in the μHEAT cohort, antibody associations in this study were mixed. This could be due in part to the fact that mRNA vaccines are already highly immunogenic, and baseline antibody responses were high, thus masking the signal. In prior research, a causal role for the gut microbiome in vaccine adaptive immunity has been found mainly in immunologically naive participants, at early vaccine doses (*20*–*23*, *72*, *85*), and in response to unadjuvanted vaccines (*19*). Moreover, it is unclear whether fever itself is useful for antibody responses after mRNA booster vaccines; although fever associates modestly with anti–SARS-CoV-2 antibody titres (*13*, *16*, *27*–*33*), antipyretic medication does not appear to interfere with antibody responses (*86*). Importantly, these data do not exclude the possibility that microbiome-driven innate immune responses stimulate adaptive immunity in other contexts.

Fever does have context-dependent utility: Moderate fevers are associated with improved clearance of microbial infections (*87*). Although temperature influences biochemical reactions, including bacterial growth rates (*88*) and pathogen virulence (*89*, *90*), the main effect of fever on infection seems to be through enhanced immune activity (*42*). We hypothesized that a febrile environment might affect intestinal microbes through either of these two mechanisms (*24*). However, we found that gut microbes were largely resilient to fever and febrile temperatures. Many intestinal microbes have codiversified with humans over evolutionary timescales, which we previously found influences their sensitivity to below-host temperatures (*91*); the μHEAT study suggests that gut microbes are nevertheless well-adapted to periods of host fever, and may even stimulate the febrile response.

The flagellated genus *Waltera*, a *Lachnospiraceae* member, was strongly associated with fever in our dataset, and with self-reported vaccine side effects in an independent cohort of more than 1000 Canadian adults. *Waltera* has only recently been described and isolated, and thus little is known about its role in human health (*92*, *93*). However, the mouse relative *Acetatifactor muris* was first isolated from mice fed a high-fat diet, and has been shown to exacerbate inflammation in murine colitis models (*94*, *95*). Our findings highlight *Waltera* as an understudied genus in the human gut microbiome; nevertheless, in both mice and bioreactors, other members of the *Lachnospiraceae* family contributed to flagellin production. The *Lachnospiraceae* family was significantly associated with self-reported vaccine side effects in Canadians, supporting an association between *Lachnospiraceae* and vaccine reactogenicity across populations. Flagellin expression in the human gut microbiome is largely attributable to *Lachnospiraceae*, but many different species within the family can contribute, providing functional redundancy (*44*). Future work to validate the association between bacterial flagellin production and vaccine-induced fever in diverse populations would be valuable, particularly given the possible discrepancy between direct temperature measurements and self-reported symptoms.

Although our work also raises the possibility that commensal motility can be manipulated through diet, the dietary observations in humans are intended primarily as hypothesis-generating. Further work is needed to test therapeutic strategies that reduce the overall motility of the commensal microbiome, which could include assessment of plant-based, minimally processed dietary interventions or other strategies to harness anti-flagellin antibodies (*59*, *96*) or flagellin-sequestering intestinal proteins such as Lypd8 and alpha-defensin 6 (*97*, *98*).

In conclusion, flagellin production by the human gut microbiome drives host inflammation and promotes vaccine-induced fever. This work has implications for optimizing vaccine reactogenicity and immunogenicity

in human populations, and furthers our fundamental understanding of the gut microbiome in human fever.

Materials and methods

Ethics statement

The μHEAT (Microbial-Human Ecology And Temperature) Study was approved by the Ethics Committee of the Universitätsklinikum Tübingen under project number 398/2021BO2. All participants provided written informed consent. Ethical approval for the CHILD Cohort Study (validation cohort), including the oversight of the CHILD biological samples and the CHILD database (CHILDdb), was obtained from the local Research Ethics Board of each study site: the University of British Columbia, the University of Alberta, the University of Manitoba, the Hospital for Sick Children and McMaster University. Animal experiments were performed at Georgia State University according to protocol A24001.

μHEAT inclusion and exclusion criteria and sample size

This study targeted healthy young adults in Germany, aged 18-40, who intended to be vaccinated with a licensed vaccine against COVID-19. Detailed inclusion criteria were as follows: adults 18-40 years of age, with no underlying health conditions, belonging to the Robert Koch Institute priority group 4 (de-prioritized for vaccination). Participants needed to want to be vaccinated against COVID-19 and be able to receive their vaccination during the study timeline. Any of the COVID-19 vaccinations approved in Germany at the time of study participation (e.g., Pfizer-BioNTech, Moderna, Johnson & Johnson or other commercially available vaccines) were permitted for inclusion in the study, including booster shots. Participants needed to be able to read and understand German or English to provide informed consent for themselves. Detailed exclusion criteria were as follows: known gastrointestinal disorders (e.g., Inflammatory Bowel Disease or Celiac Disease), recent antibiotics use (any antibiotics taken in the 3 months prior to study participation), and pregnancy, as well as any health factor which placed someone at higher risk of COVID-19 disease, and therefore in a higher vaccine priority group (1-3) according to the Robert Koch Institute. Immunocompromised and immunosuppressed people were also explicitly excluded.

To estimate the ideal sample size at the outset of the study, we considered two hypotheses: first, that microbiome might influence fever; and second, that fever might affect the microbiome. For the first hypothesis, we used a power calculation based on a general linear model approach, where a continuous immune outcome (ΔT) was taken as the dependent variable and microbiota parameters (alpha diversity) were taken as the independent variable. We estimated that a biologically relevant effect size (f2) would be 5% of the variation in immune responsiveness explained by baseline microbiota diversity. We used a significance level (α) of 0.05 and a power of 0.90 and implemented the calculation in R using the pwr.f2.test function. This yielded a sample size of 256 participants. To further confirm that this sample size was sufficient for the second hypothesis, we estimated that at least 20% of participants (18-40 years old) in our study would experience ΔT, or 51 out of 256 participants. We used a power calculation based on a two-sample t-test to compare microbiome parameters of participants with and without ΔT. Based on data previously collected by our group on the microbiomes of healthy adults in Tübingen, we estimated an effect size of 4.6 units Shannon Diversity and a standard deviation of 0.68. Taking uneven sample sizes of n1=51 (with ΔT) and n2=205 (without ΔT), and α=0.05, a power calculation with pwr.t2n.test suggested we would be highly powered (99%) to also detect differences in the microbiome induced by categorical ΔT in this study design.

A total of 182 people completed participation between November 2021 and May 2022, at which point the majority of people in Tübingen who wished to be were already fully vaccinated and therefore recruitment was ended.

We further excluded participants from downstream analysis who (a) experienced a COVID-19 or other infection during the study; (b) provided unreliable thermometer data, such that estimating ΔT was not possible; or (c) received a non-mRNA-based vaccine (n=1 who received Novavax, an adjuvanted vaccine). This yielded a final sample size for analysis of 171 people. Final metadata for participants can be found in table S1. The median age was 25, and 69.6% of the cohort were born in Germany.

Within this population, n=2 (1.7%) received dose 1, n=20 (11.7%) received dose 2, n=143 (83.6%) received dose 3 and n=6 (3.5%) received dose 4 during the study. $n = 157$ (91.8%) of doses were BioNTech and the remaining $n = 14$ (8.9%) were Moderna.

Gender categories in this study were self-reported and were defined as male, female and diverse. Due to the limited number of gender-diverse participants, analyses which directly report gender associations (e.g. with body temperature) are shown for male and female participants only, and oral contraceptive use with female participants only.

μHEAT study design

Study participation for each subject began 1 week (6-9 days) prior to their vaccine (Fig. 1A). At an enrollment visit, participants provided a blood sample and filled out a demographic questionnaire which included information on prior vaccination, prior COVID-19 infection, as well as lifestyle factors such as diet, smoking and alcohol consumption. Height and weight were measured and oral temperature was taken under the supervision of a study physician. Participants were provided with a study thermometer to take home and instructed by a study physician how to properly take oral temperature. Participants were also provided with a commercial stool collection kit (Zymo DNA/RNA Shield) to use for at-home collection, as well as an extra empty 10 mL tube (no stabilization solution) to be filled with fresh stool within 24h of their last study visit for immediate freezing at the clinic.

Participants took their oral temperature 3 times per day and recorded it in their study diary, along with the time of measurement. They were instructed to take their temperature at approximately the same times every day: once each morning, afternoon and evening. The study diary also provided spaces to record other possible systemic symptoms of vaccination every day (before and after the vaccine), namely headache, fatigue, chills, loss of appetite, diarrhea, and fever (subjective/self-reported) (symptoms data summarized in table S12). Participants were also asked to record any instance of pain- or fever-relieving medication they took throughout the study.

For each stool sample collected at home, participants also recorded in their diary the date and time, and estimated the Bristol Stool Score according to a provided Bristol Stool Score chart. They were instructed to provide 3 samples in the week before vaccination and 3 samples in the week after vaccination.

At a final study visit two weeks after vaccination, participants returned their study diaries and fecal kits and provided a final blood sample for antibody titres. Freshly provided fecal samples (sample #7, no buffer, collected within prior 24h) and sera were immediately frozen at -80°C. Buffered fecal samples (samples #1-6) were transferred to the laboratory at MPI Tübingen where they were aliquoted for DNA and RNA extraction prior to freezing at -70°C.

Temperature

To define ΔT, we first defined a personal baseline temperature as the median body temperature for a given time of day (morning, afternoon, or evening recording) for that person, in the week prior to vaccination. ΔT was then considered as the maximum observed increase in body temperature relative to baseline for the same time of day, in the week following vaccination. This variable ranges between 0.2°C and 2.4°C with a median of 0.7°C and a mean of 0.8°C. We categorized people as fever-high or fever-low based on the group mean. In practice, participants were therefore considered as fever-high if they had an increase

in body temperature following vaccination of 0.8°C or more above their own personal baseline. For comparison, the average standard deviation of personal body temperature at baseline was 0.3°C.

For self-reported prior fevers, participants were asked if they had experienced a fever (body temperature ≥38.0°C and/or fever symptoms such as aches and chills) to prior doses of the SARS-CoV-2 vaccine or at any point during the prior two years. In the CHILD Canadian validation cohort, participants were asked to indicate whether they had experienced "fever or chills" as a side effect after vaccination.

We would like to note that the definitions of human fever used in this paper, either relative to personal baseline or self-reported, are for research purposes and may not meet the criteria for clinical fever diagnosis

Diet

Self-reported vegans (consuming no animal-derived foods) and vegetarians (consuming animal products such as eggs and dairy, but no meat) were designated as following a plant-based diet (Veg). All other self-reported dietary categories, including flexitarians, pescatarians, ketogenic and fish-free diets, were considered omnivores (Omni) unless otherwise specified.

Serum antibodies

Anti-SARS-CoV-2 IgG was quantified in serum via ELISA using the Invitrogen Human SARS-CoV-2 Spike (trimer) IgG ELISA kit (Catolog #: BMS2325), according to the manufacturer's instructions. Serum samples were diluted 1:25 000 in kit assay buffer and were run in a randomized order. Units from the kit were converted to a universal standard reference scale, using the conversion 150 000 units/mL of kit = 364.5 mU binding antibody units/mL NIBSC Standard (National Institute for Biological Standards and Control, NIBSC code: 20/136).

Serum metabolomics

For metabolomics, 45 µL of serum was extracted using 244 µL of a butanol-methanol solvent (1:1 butanol:methanol containing 1.13 µg/mL EquiSPLASH and 0.32 µg/mL Trp13C11). Samples with solvent were vortexed for 15 minutes at room temperature using a multi-sample vortex, centrifuged 13000g for 10 min, and the supernatants were transferred to clean Eppendorf tubes, aliquoted and stored at -20°C until mass spectrometry analysis. Serum was processed and run in a randomized order. A QC pool was created by mixing 5 µL of each sample; an aliquot of this QC pool was then extracted every 10 samples to account for temporal variation in mass spec data.

Samples were analyzed in a Dionex UltiMate 3000 HPLC (Thermo Scientific, USA) coupled with a high-resolution mass spectrometer (Impact II mass spectrometer, Bruker, Germany). HPLC separation and mass spectrometric conditions were as reported previously (*100*). Acquired data were then converted to mzML format using MSConvert tool from ProteoWizard (*101*) prior to data processing by Mzmine3 (*102*). Features were annotated using MoNA database and/or manually checked.

Preprocessed metabolomics data was further blank-filtered at a cutoff of 0.3, such that metabolites more than one-third as abundant in blank samples compared with real samples were excluded. Analysis was then performed separately on negative- and positive-mode data using MaAsLin2 linear models. Data was normalized via total sum scaling and log transformation and a prevalence cutoff of 0.25 was applied. Age and sex, and optionally habitual diet or baseline antibody titres (table S6) were taken as covariates and p-values were corrected via FDR.

Cholesterol assay

Serum cholesterol was quantified using the HDL and LDL/VLDL Cholesterol Assay Kit (Antibodies Online ABIN2345055) according to the manufacturer's protocol. Serum samples were run at a 1:4 total dilution in a randomized order.

Serum proteomics

Serum samples were profiled by Somalogic aptamer technology for a semi-targeted panel of 1500 immune and metabolic proteins. The custom panel can be found in table S20. Serum order was randomized for profiling and per-plate control normalization steps were performed by Somalogic. Analysis was then performed using MaAsLin2 linear models correcting for age and sex, and optionally for habitual diet or baseline antibody levels. All p-values were corrected via FDR.

Generation of fecal supernatants

Freshly frozen (i.e. unbuffered) fecal samples collected within 24 hours of the last study visit were used for protein-based assays and experimental work. Fecal supernatants were prepared by resuspending stool in PBS containing an EDTA-free protease inhibitor at a concentration of 200 mg/mL. Approximately 150 mg (+/- 10%) was aliquoted from frozen stool using a biopsy punch and weighed into an empty 2 mL FastPrep tube; the exact weight was recorded, and the PBS volume was adjusted to achieve a final concentration of 200 mg/mL (approximately 750 µL). Samples were then homogenized using a 2x2 min FastPrep cycles (without the addition of beads) at 6 m/s, which should also achieve shearing of bacterial flagellins. Bacteria and debris were pelleted via two spin cycles at 13700g 10 min 4°C. For cell culture experiments, the supernatant was further sterilized and flagella were depolymerized via a 15 min boil at 98°C in a heating block. For other ELISA assays unboiled supernatant was used. Supernatants were stored at -70°C until used.

Fecal calprotectin (ELISA)

Fecal calprotectin was measured in fresh-frozen fecal samples via the human Calprotectin L1/S100-A8/A9 Complex ELISA Kit (Invitrogen EH62RB) according to manufacturer's instructions. Fecal supernatants were first generated for all samples according to the Methods section "generation of fecal supernatants". Samples were run at 1:20 dilution in a randomized order using an overnight sample incubation step.

Fecal S100A9 (qPCR)

For detection of the calprotectin subunit S100A9 via RT-qPCR in ZYMO Shield-buffered samples, RNA was extracted using the ZymoBIOMICS RNA Miniprep Kit, including the DNAse step and eluting in 85 µL. cDNA was generated using RevertAid First Strand cDNA Synthesis Kit (ThermoFisher), with oligo dT primers and an incubation of 60 min at 42°C plus 5 min at 70° in a thermocycler, from RNA normalized to 20 ng/µL. qPCR was then performed using Platinum SYBR Green Master Mix for a total reaction volume of 10 µL with 2 µL of cDNA and 0.4 µL of primers. Primers were as follows: S100A9_F, GCACCCAGACACCCTGAACCA; S100A9_R, TGTGTCCAGGTCCTCCATGATG; GAPDH_F, GGAGCGAGATCCCTCCAAAAT; GAPDH_R, GGCTGTTGTCATACTTCTCATGG (Harvard Primer Bank). Cycling conditions were: 50°C for 2 min, followed by 40 cycles of 95°C for 15 sec, 55°C for 1 min. Undetectable CT values (no amplification) were imputed as 40 cycles. Fold changes were calculated using the formula 2-ΔΔCT, in which ΔCT compared S100A9 to GAPDH per sample and ΔΔCT was relative to the average ΔCT for all samples in that cDNA batch (plate). Samples in which GAPDH was completely undetectable (both technical replicates CT value ≥40) or in which the calculated fold change exceeded the higher and lower quartiles of the total fold-change distribution by more than 10x (i.e., fold changes >100 or <0.000001, which were suspected to reflect technical errors) were excluded from further analysis.

Metagenomic sequencing

A total of 1145 longitudinal samples were processed for DNA extraction and metagenomics sequencing. DNA was extracted in a randomized order using ZymoBIOMICS 96 MagBead DNA kit. Samples were first bead-beat using Zymo "bashing" bead tubes with 8 cycles of 40 seconds,

6 m/s and 5 minutes rest between cycles on a BioSpec Bead Beater. Samples were then transferred to plate format and processed on a Hamilton robot. Each plate contained at least one extraction blank and one mock microbial community (Zymo D6300) for quality control.

Library preparation on the extracted DNA was performed as previously described (*91*), using a Nextera (Illumina, San Diego, USA) Tn5 tagmentation reaction to fragment and ligate adaptors in a single reaction. Following a cleanup step, DNA was amplified via Q5 polymerase in a total volume of 25 µL over 8 PCR cycles, using multiplexed barcoded primers. Consistent amplification was confirmed via qPCR. Samples were then pooled at equal volumes. A final PCR cleanup followed by size selection (300-800 bp) with BluePippin (Sage Science, Beverly, USA) on a 1.5% gel was performed, and final library size and distribution was confirmed on a Bioanalyzer using a High-Sensitivity DNA Chip. Sequencing was performed by Novogene on a NovaSeq6000 using a P4/300 cycle kit in three randomized batches.

Sequencing data was processed using in-house standardized Snakemake-based pipelines. A quality-control pipeline used bbduk to remove Illumina adapters and bbmap to filter out reads mapping to the human genome (https://jgi.doe.gov/data-and-tools/bbtools/). Seqkit (*103*) and fastqc (https://github.com/s-andrews/FastQC) were applied before and after QC for inspection for read depth and quality. Post-QC samples had an average read depth of 12.7 million reads (M). Four samples with a depth of less than 0.5 M, comparable to sequenced blank controls, were excluded from further analysis. In total, after technical and biological exclusions, 1022 samples with sequencing information from 171 participants were retained for analysis.

Diversity and abundance profiling was performed using kraken2 (*104*) and bracken (*105*) for abundance estimates and for alpha and beta diversity, leveraging a custom database built via Struo2 (*106*) on GTDB release 207 (*107*). Diversity was performed on data rarefied to 5 M reads, which retained 942 samples. Humann3 (*108*) was used for gene-level abundance estimates. We note that in the GTDB database used for annotation (release 207), *Waltera* is still referred to as Acetatifactor; however, since the human strains have been reclassified more recently (*93*), we have chosen to refer to it as *Waltera* throughout the manuscript.

CoPTR (https://coptr.readthedocs.io/en/latest/index.html) was used for estimation of bacterial growth rates. inStrain (*109*) was used to detect strain-level persistence and evolution.

Abundance differences were tested using MaAsLin2 (*43*) linear models on log-scaled longitudinal relative abundance data, with an abundance cut-off of 0.0001 and a prevalence cut-off of 0.25. Sequencing depth, Bristol Stool Score and the amount of time the sample spent at room temperature before freezing were taken as technical covariates and participant ID was treated as a random effect. Other biological covariates are reported in the text and tables where appropriate. P-values were corrected via FDR.

MEDI (metagenomic estimation of dietary intake)

MEDI (*47*) was applied for classification of food-derived DNA according to the author's instructions (https://github.com/Gibbons-Lab/medi). Briefly, this tool maps metagenomic reads against a database of the genomes of commonly consumed human foods. Decoys were used when building the database in order to remove background mapping to microbial genomes, and the tool has been tested against ground-truth datasets. Complete methodological details can be found in (*47*). All reads from the genus Hibiscus were manually removed, after MEDI processing and prior to analysis, due to a mapping error discovered by the authors of the tool in which the Hibiscus reference genome was contaminated with non-decoyed gut bacterial reads. For total food-derived reads, data are represented as the sum across all fecal samples provided per person, unless otherwise specified; alternatively, where specified, mixed linear models were used that accounted for all prevaccination samples. For food relative abundance metrics, data are the mean per person of all provided samples.

Transcriptomic sequencing

Samples were manually selected for RNAseq based on the presence of complete and clear ΔT profiles, attempting to maximize fever differences by category (i.e. highest and lowest fevers in the study population) while matching as closely as possible between high- and fever-low samples for age, sex and vaccine history. Samples used for RNA-seq are flagged in table S1. This subpopulation was overall similar to the total population, with basic demographics as follows: age 24.5 yrs, 60.5% female, 84.2% dose 3, and 89.5% BioNTech.

Similar to metagenomic library preparation (above), samples were first bead-beat with Zymo bead bashing tubes as described, but with reduced program of 4 bead-beating cycles, 40 sec, 6 m/sec with 1 min cooling on ice between samples. RNA extraction then proceeded with either the ZYMO RNA miniprep kit (pool 1) or with the NEB Monarch tRNA Miniprep Kit (pools 2-3), which have similar extraction chemistries. In both cases, an optional reduced ethanol volume was used at the initial ethanol step to exclude small RNAs ≤200 bp. RNA quality was spot-check via Bioanalyzer and quantified via Qubit. Library preparation was then performed using an Illumina Stranded Total RNA Prep, Ligation with Ribo-Zero Plus Microbiome. Library quality was checked on the Bioanalyzer and quantified via Qubit, normalized and pooled equimolar. A final 0.9x bead clean was performed on the final pool prior to sequencing.

Sequencing was performed by Novogene on a NovaSeq6000 with a P4/300 cycle kit in three randomized batches.

Microbial transcriptomic sequencing data was preprocessed and profiled using the same pipelines as for metagenomic sequencing, above. Post-QC samples had an average read depth of 49.2 M. Diversity analyses were performed on data rarefied to 5 M reads, which retained 226 samples.

Gene-level abundance differences in transcriptomic data were tested using MaAsLin2 linear models on log-scaled longitudinal relative abundance tables, with an abundance cut-off of 0.00001 and a prevalence cut-off of 0.25. Sequencing depth, Bristol Stool Score and the amount of time the sample spent at room temperature before freezing were taken as covariates and participant ID was treated as a random effect. Biological covariates were also included as reported in the text and tables. P-values were corrected via FDR.

In vitro bacterial temperature phenotypes

High-throughput bacterial growth curves were performed as previously described (*91*, *110*). Briefly, human gut commensals (table S21) were grown anaerobically in mGAM media. Strains were inoculated from glycerol at 37°C, subcultured overnight 1:100 at 37°C, and finally inoculated into 100 µL cultures in 96 well plates at a normalized O.D. of 0.02 under the desired experimental temperature. Due to the necessity of changing the incubator temperature, each temperature condition was run separately. Half of a plate (48 technical replicates) per strain per condition were used, using an automated stacker and plate reader to read O.D. over a 24 hour period, and each strain was run over 2-3 biological replicates at that temperature. The growth rate of each species was then calculated in R by manually separating the linear part of each growth curve (exponential phase) and then obtaining the slope via linear fit.

For RNA-seq of Segatella copri, the bacterium was cultured and subcultured as described above in 10 mL volumes. Temperature shock was then performed for 1 hour via water baths at 32°C and 42°C, versus a control condition maintained at 37°C. RNA from each full culture was extracted in QIAZOL using the miRNEAsy kit (Qiagen) and quantified via Qubit. RNAseq library preparation was performed using the Illumina Stranded Total RNA Prep, Ligation with Ribo-Zero-Plus.

Sequencing was performed in-house on a NextSeq2000. Analysis of the RNA-seq data followed an in-house custom pipeline which used BBDuk to trim adaptors (https://jgi.doe.gov/data-and-tools/bbtools/), FastQC to assess read quality (https://github.com/s-andrews/FastQC), and Bowtie2 to align reads to the S. copri reference genome (*111*).

Flagellin in vitro expression

Flagellins were expressed and purified as previously described (*44*). Briefly, codon-optimized flagellin sequences were cloned into pETM11 and transformed into BL21-DE3 E. coli competent cells. E. coli was grown at 37°C with shaking until mid-log phase. Protein expression was then induced via 0.1 mM IPTG overnight at 18°C with shaking. Cultures were then pelleted at 5000 rpm for 20 min at 4°C and resuspended in lysis buffer (20 mM Tris-HCL, 300 mM NaCl, 5 mM NaCl, 1% Triton X-100, pH 7.5). Samples were sonicated using a Bandelin Sonopuls GM3200 with 40% amplitude for a total of 4 minutes with a cycle of 0.5seconds off, 0.5seconds on. His-tagged proteins were then bound using HisPur Ni-NTA resin (Thermo Scientific) for 1 hour rocking. Finally, the beads were washed and purified proteins were eluted in an elution buffer (20 mM Tris-HCl, 300 mM NaCl, 300 mM imidazole, pH 7.5) and dialysed into 1XPBS overnight at 4°C. Protein concentration was quantified via Nanodrop.

Waltera growth and sheared flagella

Waltera intestinalis and *W. hominis* were a generous gift from the Clavel lab. *Waltera* was routinely grown in anaerobic YCFA broth without the addition of vitamins, using stoppered Balch tubes with an N2/CO2 atmosphere. Overnight cultures were inoculated using 100 µL glycerol stock aliquots into 10 mL of YCFA for 24 hours at 37°C without shaking. Subcultures were then prepared using 100 µL of 24h culture into 10 mL of fresh YCFA overnight at 37°C without shaking.

To shear flagellins from *Waltera* cultures, bacterial cells were pelleted and resuspended in PBS containing protease inhibitor, and normalized (concentrated) to an O.D. of 0.5. Cells were then Fastprepped, spun and boiled as described previously for the fecal supernatant preparation. The end product was a sterile supernatant preparation, containing neither bacterial cells nor media nor secreted metabolites, only heat-resistant bacterial products including flagellin. *Waltera* sheared flagellins were then added to organoids at 20 µL/well as described above.

TLR5 reporter assay

The HEK TLR5 (Invivogen hkb-htlr5) and TLR4 reporter cell lines (Invivogen hkb-htlr4) were grown routinely in DMEM containing 10% FBS. For reporter assays, a single confluent T75 was used to seed a 96 well plate, using 180 µL of resuspended cells per well and directly adding 20 µL of stimulus. FliC (expressed and purified in-house as previously described, according to the genetic sequence for Salmonella typhimurium (*44*) was used as a standard curve for TLR5 at a top concentration of 10 µg/mL, and purified LPS was used for TLR4 reporter standards. Other commensal flagellins were also expressed and purified as previously described (*44*). Plates were incubated at 37°C 5% CO_2 for 24h. Per well, 180 µL of QuantiDetect dye was then mixed with 20 µL of cell supernatants and O.D. 635 nm was read at 10 minutes and at 30 minutes.

Waltera growth curves

Growth curves from subcultures were inoculated in an anaerobic chamber to a normalized starting O.D.600 of 0.02 in YCFA with or without 0.5% CMC, in a 96 well plate sealed with a BreatheEasy permeable membrane. O.D. was measured over a 24–48h period with double orbital shaking at 37°C. Comparison species *Bacteroides thetaiotamicron*, *Segatella copri*, *Roseburia intestinalis* and *Blautia obeum* (table S21) were also grown in YCFA following the same protocol as for *Waltera*. Growth curves were repeated in triplicate.

Organoid experiments and RNAseq

Human colonic enteroids line xcol4 were thawed, passaged and seeded according to the protocol "Human Enteroid Protocol Handbook" by Julia Y. Co, Stanford University School of Medicine, as previously described (*44*). Briefly, organoids were grown in 24-well-plates with 400 µL/well of IntestiCult Human Basal Medium (StemCell Technologies) supplemented with Organoid Supplement (StemCell) ("Growth Media"). For passaging, 1 mL of TrypLE Express/well was used to digest organoids, which were then aspirated and pooled into a falcon tube and incubated for several minutes at 37°C, before inactivating with FBS and spinning at 500g for 3 min to pellet cells. Cells were then resuspended in cold Matrigel on ice and seeded into a fresh 24 well plate at 40 µL/well. After passaging, media contained additional inhibitors (10 µM of Y-27632 and 250 nM of CHIR99021, BioTechne) ("Passage Media") for 2-3 days until replaced with fresh growth medium. Cells were seeded approximately one week before the experiment until mature organoids were observed.

Fecal donors for cell culture experiments are flagged in table S1 and were chosen as paired fever-high and fever-low samples, by matching as evenly as possible according to age, sex and vaccination history, while maximizing differences in fever. Donor temperature data were also manually inspected to ensure robust presence or absence of fever (participants were prioritized if the data showed (a) low noise; (b) multiple >37°C fever-high reads if a fever-high participant, indicating a robust fever; and (c) no antipyretic consumption if a fever-low participant, indicating an absence of fever). Frozen fecal supernatants were thawed on ice and then boiled for 15 minutes to sterilize and to depolymerize any flagellins. Any remaining debris was settled with a short spin. In addition, flagellin and empty vector controls (extracted as described above) were thawed on ice and diluted in PBS to a working stock concentration of 100 nM. The organoid media was exchanged for 360 µL of fresh Growth Media. 40 µL of each fecal supernatant or control was added per well, using two wells (technical replicates) per sample. Moreover, three replicates or batches of organoid cells were used, with different participant fecal samples but identical flagellin controls per batch. From each experiment, cell supernatants were collected at 4h (with replacement of 200 µL fresh media) and at the endpoint of 18h. At 18h, RNA was also saved for extraction according to the ReliaPrep RNA Cell Miniprep System with modifications for organoid cells (Promega protocol Application Note #AN296). Briefly, after removal of media, 200 µL of cold LBA buffer containing TG was added per well and cells were resuspended via pipetting. This suspension was then frozen at -70°C until extraction.

IL-8 was measured in undiluted or 1:1.3 diluted cell supernatants using the R&D DuoSet IL-8 ELISA according to kit instructions. Cytokines reported in the paper correspond to the 4h time point.

RNA extraction was continued in a randomized order across both experimental replicates, according to kit instructions. RNA libraries were generated using the NEBNext Ultra II Directional RNA Library Prep Kit for Illumina in randomized batches of 24 samples. Sequencing was performed in-house on a NovaSeq2000.

Waltera microscopy

Fresh subcultures of *W. Intestinalis* in YCFA with or without 0.5% CMC were prepared for microscopy. For negative stain, glow-discharged formvar- and carbon-coated copper grids were incubated with concentrated cells, and stained with 1% uranyl acetate. Images were acquired with a CMOS camera (TemCam-F416, TVIPS, Gilching, Germany) mounted on a Tecnai Spirit transmission electron microscope (Thermo Fisher Scientific, Eindhoven, The Netherlands) operated at 120 kV. Flagellin length from negative stain images were quantified using ImageJ's line function by tracing each flagellin present in the image and calculating the resultant line length in µm. Flagellin length was then normalized to the number of bacterial cells

visible in the image to obtain μm flagellin/cell. Partially visible bacteria in the image were counted as 0.5 of a bacterium. Flagellin measurements were performed blinded to the experimental condition (YCFA vs CMC).

Bioreactor experiment

The MiniBioReactor Array (MBRA) system was assembled and used as described previously (*112*). Briefly, the MBRA system is housed in an anaerobic chamber and consists, per system, of 24 individual chambers. MBRA chambers were held on a magnetic stand for continual homogenization and were connected to two 24-channel peristaltic pumps with low flow rate capabilities (205S peristaltic pump with 24-channel drive, Watson-Marlow). After autoclaving, MBRA system was placed in an anaerobic chamber to out-gass for at least 72 hours. MBRA chambers were then filled with 15 mL of Bioreactor Medium (BRM) and subsequently inoculated with freshly prepared faecal slurry. For this, faecal samples from 2 donors were resuspended at 3% w/v in anaerobic phosphate buffered saline (D-PBS) (Gibco–Life Technologies) within an anaerobic environment, vortexed for 5 min and centrifuged at 800 rpm for 5 min at 20°C. Supernatants were transferred to a fresh tube, filtered through a 100 μm filter to eliminate residual particles and kept anaerobically. Finally, 1 mL of this faecal slurry was used to inoculate each MBRA chambers, with eighteen independent chambers inoculated per donor. In half of the chambers (nine per donor) a culture containing 10^7 colony-forming units of *W. intestinalis* and *W. hominis* (grown in brain-heart-infusion broth supplemented with 0.0005 % w/v hemin and 0.5 % w/v yeast extract) was added. Twenty-four hours after inoculation, constant flow was initiated at 1.875 mL/hour, corresponding to an 8-hour retention time. Microbiota were allowed to stabilize for 72 h under these flow conditions before any experimental treatment.

At the end of the stabilization phase the reactors were subjected to three different media in triplicate per donor: i) control BRM without any additive, ii) BRM containing 0.1 % w/v CMC, and iii) BRM containing 0.1 % w/v CMC together with a mixture of thirteen common food additives(table S22). All additives were incorporated into the medium before autoclaving. Treatments were applied continuously for 120 h. Every 24 h(0, 24, 48, 72, 96 and 120 h) a 400 μL aliquot was withdrawn from each chamber and stored at −80°C until analysis.

Bioreactor experiment: bacterial dna extraction, qPCR and 16S rRNA sequencing

As described previously (*113*), DNA was extracted from frozen MBRA suspension (50 μL) using a QIAamp 96 PowerFecal QIAcube HT kit(Qiagen Laboratories) following instruction of manufacturing with mechanical disruption (Qiagen TissueLyser II) and QIAcube high-throughput robot. In order to follow the number of copies of the FliC2 *Waltera* flagellin gene over time, extracted DNAs were diluted 1/10 with sterile DNA-free water and subjected to quantitative PCR using the *Waltera* Flic2 specific primers Flic F 5”- GAACTTGAAGCTGACATTGC-3” and Flic R 5”- CTCTGTTGATCTTGTAACCG-3” on a CFX Opus 96(Biorad) using Luna qPCR Kit(New England Biolab). Data were analyzed by Biorad CFX Maestro software(Biorad), and results were expressed as relative values.

The V4 region of 16S rRNA gene was amplified from each sample using a composite forward primer and a reverse primer containing a unique 12-base barcode, designed using the Golay error-correcting scheme, which was used to tag PCR products from respective samples. The full primer sequences are:

515F: 5” AATGATACGGCGACCACCGAGATCTACACGCTXXXXX XXXXXXXTATGGTAA TTGTGTGYCAGCMGCCGCGGTAA-3” and 806R used was 5”-CAAGCAGAAGACGGCATACGAGATAGTCAGCCAGCCG GACTACNVGGGTWTCTAAT-3” with the italicized sequence is the Illumina adapter, the 12 X sequence is the Golay barcode, the bold sequence is the primer pad, the italicized and bold sequence is the primer linker and the underlined sequence is the conserved bacterial primer 515F or 806R. PCR reactions consisted of 5PRIME HotMasterMix(Quantabio, Beverly, Massachusetts, USA), 0.2 μM of each primer, 10–100 ng template and reaction conditions were 3 min at 95°C, followed by 30 cycles of 45 s at 95°C, 60 s at 50°C and 90 s at 72°C on a Biorad thermocycler. Amplicons were visualized on agarose gels, quantified with the Quant-iT PicoGreen dsDNA assay, pooled in equimolar ratios and purified with AMPure XP magnetic beads(Agencourt, Brea, California, USA). Pooled library was sequenced using an Illumina NextSeq Gen 200 sequencer(paired-end reads, 301 cycles) at the Biomic sequencing platform from Institut Pasteur, Paris, France.

Raw reads were demultiplexed and denoised with the DADA2 plugin in QIIME 2 v2025.7, generating an ASV table and representative sequences. A phylogenetic tree was built with the align-to-tree-mafft-fasttree workflow; alpha- and beta-diversity metrics were obtained using the core-metrics-phylogenetic command. Taxonomic classification of ASVs was performed against the Greengenes 2022.10 reference database at a 99 % identity threshold.

Germ-free animal experiments

6-week-old germ-free C57BL/6 mice (housed at Georgia State University) were inoculated by oral gavage with fecal slurries prepared from fever-high and fever-low donors. Six fever-high and six fever-low donors were selected from the samples used for the organoid experiment. Donor samples are flagged in table S1 and were chosen on the basis of extreme fever values using the same criteria as for the organoid experiment (matching as evenly as possible according to age, sex and vaccination history), with additional priority given to samples with high available biomass for fecal oral gavage. Two cohoused mice were used per donor (total $n = 24$). Donors were split evenly by sex (3 female and 3 male per ΔT category), and sex-matched to recipient mice to ensure balanced groups. Mice were colonized for 1 week prior to intramuscular vaccination with 5 μg of SARS-CoV-2 mRNA vaccine (Bivalent Omicron BA.1/BA.5 strain spike-encoding mRNA). Body temperature at 0h, 5h and 24h post-vaccine were tracked via rectal thermometer. Blood was collected intraorbitally pre and post-vaccine to track serum cytokine responses using ELISA kits (IL-6; DY406, Cxcl1; DY453) and LCN-2 was detected in fecal supernatants (DY1857). Serum anti-IgG antibodies were measured using a homemade ELISA coated with recombinant SARS-CoV-2 spike protein (20 ng/50 μL/well, 40589-V08H32), using anti-mouse IgG HRP (Invitrogen 31430, 1:5000) as the secondary, with sera serially diluted from 1:100 to 1:12500. The experiment was repeated once in wild-type C57BL/6N mice (Taconic) and once in TLR5/NLRC4 DKO mice bred in-house.

Conventional animal experiment (dietary intervention)

6-week old female specific pathogen free(SPF) C57BL/6J mice (The Jackson Laboratory, JAX: 000664; housed at Georgia State University) were divided into three groups of 8 mice each (4 mice per cage) as visualized in Fig. 4J. The control (ctrl) group received generic mouse chow and normal drinking water throughout the experiment. The carboxymethylcellulose (CMC) (Sigma, 419311-1KG) group received generic mouse chow and 1% CMC in drinking water starting at 6 weeks prevaccine. The western-style diet (WSD) group received normal drinking water and a high-fat, low-fibre diet (Research diet, D12492) starting at 4 weeks prevaccine. All mice were then vaccinated, in parallel, with an intramuscular SARS-CoV-2 mRNA vaccine (5 μg, Bivalent Omicron BA.1/BA.5 strain spike-encoding mRNA). Body temperature, blood and fecal measurements were performed as described above. This experiment was performed once.

CHILD validation cohort

Recruitment and sample collection for the CHILD validation cohort are described in detail elsewhere (*46*). Briefly, parents and children enrolled in the CHILD Cohort Study (*114*, *115*) were invited to participate

in the COVID Add-On Study, which aimed to assess COVID-19 exposure, vaccination status and serology during the pandemic (*46*). Pandemic experience questionnaires were collected from participants quarterly. Participants also provided ethanol-buffered stool samples using an at-home collection kit (1-2 samples per person collected between March 2021 and Jan 2022). Blood samples were collected over the same time period using the dried blood spot method, and SARS-CoV-2 serology was performed using automated chemiluminescent ELISAs to detect anti-spike and anti-nucleoprotein antibodies, as described previously (*116*).

Stool samples from the CHILD Cohort COVID-19 sub-study were subjected to metagenomic sequencing on the Illumina NovaSeq platform using a previously described library normalization protocol (*117*). Metagenomic data processing was performed as described above for the μHEAT cohort, using the same pipelines and tools. For analysis, we used data from the first adult questionnaire provided that included vaccine side effects and symptoms (collected August-September 2021), and we excluded any participant who(a) lacked sequencing data,(b) did not receive a vaccine or(c) did not provide data on vaccine side effects. This led to a final sample size of $n = 1014$ Canadian adults providing $n = 1549$ fecal metagenomic samples. The majority of participants (90.0%) reported a 2nd-dose mRNA vaccine as their most recent vaccine. For serology analysis, we divided participants into antibody-high and antibody-low groups based on the group mean value for scaled luminescence to anti-spike IgG antibodies, using only blood samples collected at least 2 weeks following vaccine dose 2 ($n = 759$ individuals).

Statistical analysis

Statistical analysis was performed in R 4.1.2 and in GraphPad Prism 10. The details of each statistical test are reported in the figure legends or text and in tables S3 and S4-17.

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Data from the CHILD cohort study (validation cohort) are available from the CHILD Cohort Study (child@mcmaster.ca) under a material agreement with McMaster University. More information about data access for the CHILD Cohort Study can be found at https://childstudy.ca/for-researchers/data-access/. Researchers interested in accessing CHILD Cohort Study data should contact child@mcmaster.ca. Sequencing data from the CHILD study are available under PRJEB108450. 16S data from the bioreactor experiments are available in the ENA under accession number PRJEB107428. No materials were generated for this study.
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Recent strengthening of eastern Pacific ENSO in the last millennium paleorecord

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The Pacific El Niño–Southern Oscillation (ENSO) generates climate extremes that endanger ecosystems, infrastructure, and human well-being worldwide. The response of this system to climate warming is poorly constrained, owing to both data scarcity and uncertainties in climate models. The geochemistry of Galápagos coral skeletons across the past millennium reveals a large recent increase in interannual variability of sea surface temperature in the eastern equatorial Pacific compared with that in existing paleorecords that exceeds simulated natural variability. This increase parallels rising global temperature and results from stronger El Niño events. Central Pacific coral data also show increased variability, although less distinctly than in the Galápagos. Our results provide long-term context for understanding ENSO variability trends, with troubling implications for climate extremes.

El Niño–Southern Oscillation (ENSO) is the leading source of interannual extremes in modern climate, linked to droughts, floods, infrastructural damages, coral bleaching, wildfires, and agricultural and health impacts (*1*–*3*). ENSO is defined by coupled variability of the ocean and atmosphere in the equatorial Pacific, emerging from multiple processes as a coherent pattern of changes in tropical Pacific sea surface temperature (SST) and associated wind, hydrologic, and oceanographic features (*4*, *5*). The system's complexity and remote influences (*6*) create large internal variability that masks a clear response to external climate forcing (*7*–*9*) and hinders model projections, making the attribution of ENSO behavior to specific drivers and the projection of future extremes difficult. Thus, there is a pressing need for more long-term observations from the equatorial Pacific to help constrain models, establish a better record of past ENSO variability, and provide context for its recent behavior. We report paleoclimate records from Galápagos corals (Fig. 1) that allowed us to compare recent ENSO variability with that of the previous millennium.

Spatial and temporal ENSO variability

Several large and impactful El Niño events have occurred in recent decades, each exhibiting warming that extends from South America to the international dateline. This increase in strong, basin-wide El Niño events results in increasing skewness of eastern Pacific (EP) SST anomalies (fig. S1) (*10*, *11*). However, the large internal variability of ENSO and the sparse instrumental record (fig. S1) (*12*) complicate attribution of these trends to anthropogenic warming (*13*, *14*). The first systematic SST observations of the remote region defining ENSO variability (for example, Niño 3.4: 5°S–5°N; 120°–180°W) began in the 1980s with satellite observations, later augmented by the in situ Tropical Ocean Global Atmosphere (TOGA) Tropical Atmosphere Ocean (TAO) Array (*1*).

Spatial variability complicates this picture: Warm SST anomalies in the EP are typically stronger than those in the central Pacific (CP) (*15*, *16*) or along the South American coastline (*17*). Spatial modes of ENSO may respond differently to changing climate forcing—for example, over recent millennia (*18*, *19*) and under anthropogenic influences (*4*, *11*, *20*). Paleorecords from CP corals are beginning to reveal long-term CP variability (*21*–*26*), but the EP lacks comparable data to place current trends in long-term context.

Model projections of future climate change tend to disagree on how ENSO-related SST variability will respond to climate warming. Using the latest coupled climate models (phases 5 and 6 of the Coupled Model Intercomparison Project, hereafter CMIP5/6), the 6th Assessment Report of the Intergovernmental Panel on Climate Change found no consensus on changing variability in the CP (Niño 3.4) SST, even under high emissions scenarios (*7*). Later studies that used a subset of models, and large single-model ensembles, identified increased ENSO SST variability under anthropogenic forcing (*27*, *28*). In studies that distinguished EP and CP modes, most CMIP5/6 models show a forced increase in ENSO-SST variability that is stronger, and emerges sooner, in the EP relative to the CP (*11*, *29*).

Despite the emerging view that ENSO SST variability will increase under anthropogenic forcing, underlying mechanisms and trajectories differ among models, highlighting that we lack unambiguous physical understanding of simulated changes (*30*). Improving the observational record will help to identify models that best capture observed patterns and may be more likely to accurately assess future variability. High-resolution paleodata from the tropical Pacific fill a critical gap in defining long-term baselines and patterns of ENSO variability.

ENSO paleo-records from Galápagos corals

We used modern and subfossil Galápagos coral samples to quantify changes in interannual SST variance across the past millennium in the eastern equatorial Pacific (table S1). The Galápagos archipelago lies in the heart of the EP ENSO mode (Fig. 1), where the strongest basin-wide ENSO extremes with the greatest global impacts create maximum SST anomalies. Interannual variability in Galápagos SST reflects ENSO dynamics through changes in thermocline depth, upwelling, and advection. Coral skeletal strontium-to-calcium ratio (Sr/Ca) and $\delta^{18}O$ (a measure of the ratio of stable isotopes ^{18}O:^{16}O) closely track past Galápagos SST (*31*–*34*). In the main archipelago, seawater $\delta^{18}O$ ($\delta^{18}O_{sw}$) variations are subtle and reflect circulation changes, not freshwater balance (*35*). Coral Sr/Ca and $\delta^{18}O$ are reproducible and correlate significantly with local and large-scale SST patterns, supporting their interpretation as EP-ENSO proxies (figs. S2 and S3).

We used coral Sr/Ca and $\delta^{18}O$ to characterize SST variations over the past millennium. Our results derive from analysis of 28 time series at five islands across the Galápagos archipelago (Fig. 1, tables S1 to S3, and figs. S4 and S5), using local SST-geochemistry calibrations (tables S4 and S5 and fig. S6). We empirically accounted for SST mean and variability gradients across the region (Fig. 1, fig. S7, and table S3) and calculated interannual SST variability on the basis of filtered bimonthly SST anomalies reconstructed from coral Sr/Ca and $\delta^{18}O$ [supplementary materials (SM), materials and methods]. We quantified the change in variability from a modern benchmark period (Table 1 and table S6) and compared our results with CP coral data (table S7) and other ENSO reconstructions using this "percent change" metric.

To assess whether our results exceed the range expected from internal variability or natural forcing, we compared coral-reconstructed changes with simulations of the preindustrial last millennium (850 to

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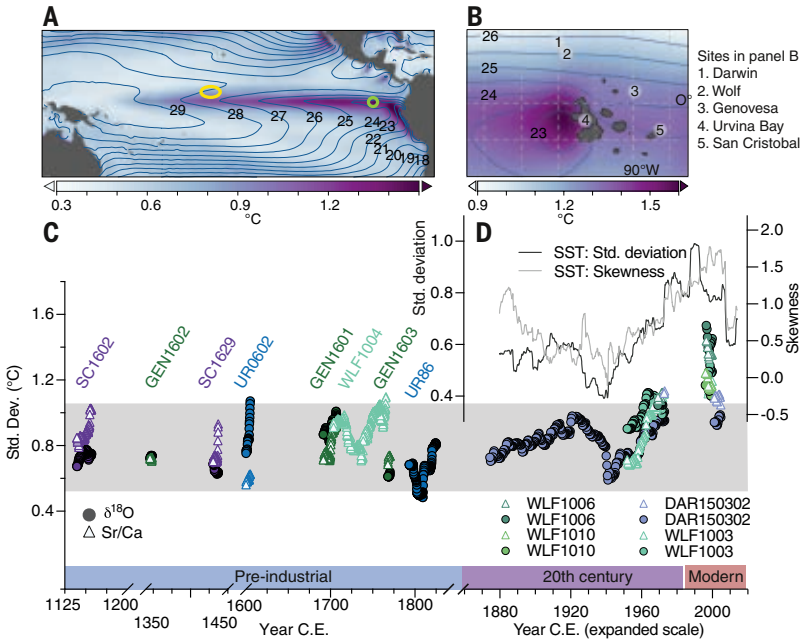

Fig. 1. EP SST variability, modern period and last millennium. (**A**) Standard deviation of observed monthly SST anomalies (1982 to 2024; colors), superimposed on mean SST (labeled contours in degrees Celsius). The Galápagos and Line Island archipelagos are indicated with green and yellow symbols, respectively. (**B**) As in (A), but for Galápagos archipelago (different scaling). Coral sites are numbered. Data were accessed from OISSTv2.1 product (*58*) by using Climate Explorer (https://climexp.knmi.nl). (**C**) Colored symbols indicate the 20-year running standard deviation of interannual SST anomalies reconstructed from Galápagos corals, corrected for interisland offsets. Circles indicate coral $\delta^{18}O$–derived SST variability, and triangles indicate SST variability derived from coral Sr/Ca. Time series lose 10 years off their start and end dates because of the 20-year running calculation, and the gap centered in the 1980s results from the death horizon in all modern Galápagos corals. Preindustrial corals are labeled with sample IDs; post-1860 samples are identified in the legend. The shaded gray range indicates $\pm 2\sigma$ range (95% distribution) for preindustrial variability. The *x* axis is discontinuous. (**D**) (Inset) Plot showing the 20-year running standard deviation for instrumental Galápagos SST in dark gray and the running 20-year skewness of interannual anomalies in light gray (HadISST: 2°S–2°N, 88°–92°W) (*59*). The shaded bars at bottom indicate preindustrial, 20th-century, and modern epochs. SST and coral data are processed as in the SM materials and methods.

Table 1. Percent change in ENSO variability compared with modern epoch. Percent change was calculated for each coral record as the difference from the modern benchmark interval, using 20-year standard deviation (SD20) values. Percent change = [(modern benchmark SD20 – older SD20)/modern benchmark SD20]. Benchmark intervals are given in table S6, and statistical tests are described in the SM materials and methods, with results provided in table S9. We compare the values in this table for Galápagos with model results in Fig. 3 and fig. S17.

	Preindustrial last millennium	**Postindustrial ~20th century**
Years	1000 to 1850 CE	1851 to 1982 CE
Galápagos	$36.5 \pm 8\%$; $P < 0.0001$	$16.2 \pm 10\%$; $P < 0.0001$
Line Islands	$33.8 \pm 15\%$; $P < 0.0001$	$18.5 \pm 15\%$; $P < 0.05$

Results: Increasing variability and asymmetry in EP SST

Over the past millennium, EP-ENSO variability recorded in Galápagos corals parallels global temperature, with notable amplification in the past ~40 years (Figs. 1 and 2). Relatively low and consistent variance characterizes the 1000–1850 CE interval (hereafter "preindustrial"). In the following period (1850 to 1981; "20th century"), variability began to rise in the late 19th century, emerging above the preindustrial range by the late 20th century. SST variability in the "modern" period (since 1984) is consistently above that reconstructed during the preindustrial period. Modern SST variability in the eastern equatorial Pacific is exceptional compared with the background state of this naturally dynamic system.

The coral reconstructions also track rising skewness (Fig. 1 and fig. S14) of EP-SST that reflects more frequent strong El Niño events. The skewness data are noisier than the variability results, but the shift from mostly neutral skewness in the preindustrial period to mostly positive skewness in the modern interval indicates that stronger warm events underlie increased variance (Fig. 2). In the instrumental era, changes in coral and instrumental SST skewness tracked one another across the 20th century, although a recent decrease in coral skewness preceded a similar change in the instrumental record.

Each of the three intervals (preindustrial, 20th century, and modern) (Fig. 1) exhibits variability that is statistically distinct from the others, increasing through time ($P \leq 0.0001$) (SM materials and methods and table S9). Using the percent change in standard deviation, normalized to a recent 20-year interval, we found that modern EP ENSO is $36.5 \pm 8\%$ stronger than in the preindustrial, and $16.2 \pm 10\%$ stronger than in the 20th century (Table 1). Values in parentheses indicate the standard deviation among individual cores. Modern and 20th-century samples derive exclusively from the northernmost islands (Wolf and Darwin), and preindustrial samples derive mostly from the central archipelago (Fig. 1). Geographic normalization (SM materials and methods) corrects for variability differences that reflect distance from upwelling centers (fig. S7), but even without this correction, the modern epoch exhibits significantly higher variability ($P < 0.0001$) than that in the earlier intervals (fig. S15 and table S9). At Isla Wolf, data from four colonies spanning preindustrial, 20th-century, and modern epochs confirm that the dramatic increase in SST variability is independent of site corrections.

Our analysis of last-millennium $\delta^{18}O$ data from Line Islands corals (table S7) extends and largely agrees with earlier work (*24*). Before 1850 CE, the range of variability among records is larger in the Line Islands than in the Galápagos, and preindustrial variability sometimes overlaps with modern ranges (Fig. 2). Nevertheless, modern Line Islands variability (which includes substantial seawater $\delta^{18}O$ influence)(*39*) is greater than preindustrial by $33.8 \pm 15\%$, which is consistent with Galápagos results, and mean values of percent change are statistically distinct among epochs ($P < 0.0001$) (Table 1 and table S10).

Fossil coral records from the eastern Indian Ocean and western Pacific Ocean (Sumatra and Vanuatu) show increasing interannual variance over the past millennium that has been attributed to ENSO (*26*, *40*). The reconstructed changes in variability span the same range as Galápagos and Line Islands data but do not show the clear distinction between preindustrial and modern seen at Galápagos (Fig. 2). Continuous ENSO

1850 CE) driven by natural forcings (*36*–*38*), and their corresponding preindustrial control simulations. Our analysis incorporates 12 different climate models (table S8 and figs. S8 to S13), 11 of which are CMIP5/PMIP3 or CMIP6/PMIP4 members with one last-millennium simulation. The 12th is the CESM Last Millennium Ensemble (LME), with 13 fully forced simulations of the last millennium (SM materials and methods).

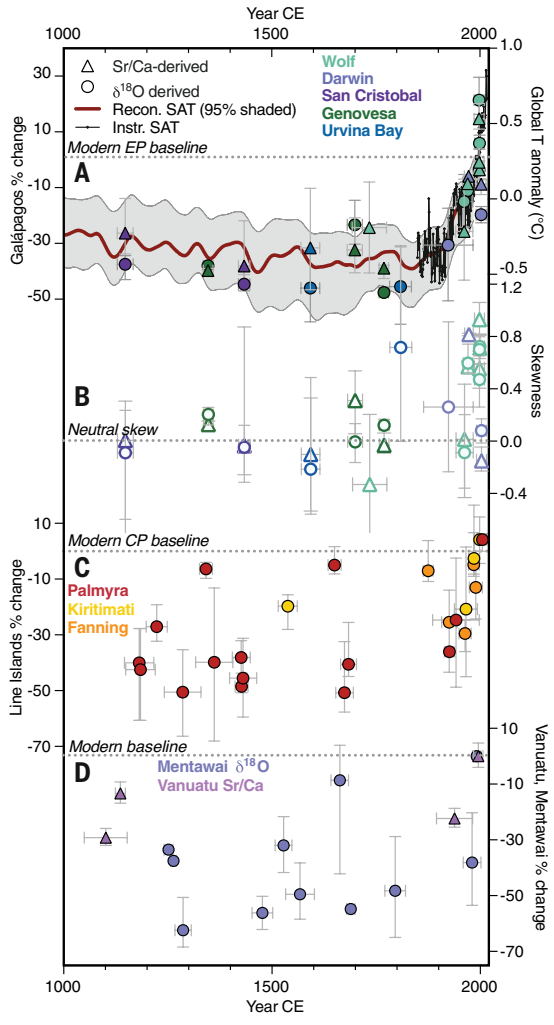

Fig. 2. Comparison of coral-reconstructed interannual variability in the Indo-Pacific and global surface air temperature (SAT). (**A**) Change in interannual SST variability from Galápagos corals, expressed as percentage change from the modern period. Each symbol indicates a single record: Colors indicate different islands, and shapes indicate proxy type (circles, $\delta^{18}O$; triangles, Sr/Ca). Vertical and horizontal bars indicate ranges of percent change and ages, respectively. SAT change is indicated with the black line (instrumental) and bold red line (reconstructed, with 95% confidence indicated with light gray shading) (*60*, *61*). (**B**) Skewness of Galápagos coral SST anomalies, shown with open symbols and using the same color scheme as in (A), calculated on the same filtered SST anomaly data used for the running standard deviation metric. Vertical bars indicate minimum and maximum 20-year running values, and horizontal bars reflect the total length of each coral record. (**C**) Change in interannual variability of coral $\delta^{18}O$ data from the northern Line Islands, expressed as percent change from modern as for (A). Colors indicate different islands as in legend. (**D**) Change in interannual variability of coral $\delta^{18}O$ and Sr/Ca data from Mentawai and Vanuatu, expressed as percent change from modern as for (A). Colors indicate different islands as in legend. Data sources are provided in tables S1 and S7.

reconstructions that merge a wider spatial network of coral, tree-ring, and other paleodata (*20*, *41*, *42*) show increasing variance in the mid-19th century (fig. S16), preceding the Galápagos shift, and preindustrial variability that spans a wider range than in the EP. By contrast, individual foraminifera analysis (IFA) from the EP shows modest variability, lowest early in the record, and peaking in the late 1700s (*43*).

Analysis of preindustrial EP ENSO variability in last-millennium simulations and their associated preindustrial control simulations (table S8) reveals that the Galápagos coral-reconstructed change is significantly greater than simulated ENSO changes resulting from natural forcing or internal variability ($P < 0.05$) (Fig. 3). The variability change reconstructed from Galápagos corals (36.5%) exceeds the 95% confidence interval (CI) defined by EP-ENSO SST in (i) a multimodel ensemble of 11 fully forced simulations of the preindustrial last millennium; (ii) a single-model ensemble of 13 fully forced simulations of the preindustrial last millennium; and (iii) a multimodel ensemble of preindustrial (1850) control simulations corresponding to the forced experiments.

Spatial diversity in the evolution of ENSO variance

Over the past millennium, EP and CP coral reconstructions roughly agree on the mean amplitude of ENSO variability change. Neither indicates coherent shifts concurrent with the extratropical Little Ice Age (~1500 to 1850 CE) or Medieval (~800 to 1350 CE) (*44*) eras, although coral coverage is sparse before 1500 CE. However, the EP and CP coral reconstructions differ in other ways. The Galápagos coral analysis reveals a tight range of preindustrial variability across cores and sites and a pronounced distinction between modern/20th century and preindustrial variability. By contrast, the Line Islands records show a greater range in reconstructed ENSO variability throughout the preindustrial period and greater overlap of modern and preindustrial variability.

This difference likely reflects the mix of factors that determine coral $\delta^{18}O$ in the EP and CP. In the Line Islands, interannual SST anomalies are relatively small, and coral $\delta^{18}O$ records a substantial $\delta^{18}O_{sw}$ component that varies among events and islands (*39*, *45*). By contrast, in Galápagos, larger interannual SST changes dominate the coral $\delta^{18}O$ signal, as supported by long-term $\delta^{18}O_{sw}$ monitoring and agreement between paired $\delta^{18}O$ and Sr/Ca records. Galápagos $\delta^{18}O_{sw}$ reflects circulation-driven changes that do not correlate with local SST or ENSO indices (*35*). The wider range of variability in Line Islands coral records likely results from substantial influence of $\delta^{18}O_{sw}$ (*45*), estimated at 30% (Kiritimati) to >50% (Palmyra) of the coral $\delta^{18}O$ signal during 2015, caused by circulation and rainfall variability (*39*, *46*). The $\delta^{18}O_{sw}$ signal cannot be separated from SST in Line Islands coral records with available data. Conceivably, gaps in datasets could also create differences between Galapagos and Line Island results; the Galápagos data sample 48% of the years between 1100 and 2015 CE, and the Line Islands sample 59% of this interval.

To the extent that these records reflect variability in EP (Galápagos) and CP (Line Islands) ENSO SST, they describe a pattern that is consistent with model inferences of EP-ENSO and CP-ENSO responses to external forcing. CMIP5/6 models tend to project forced increases in ENSO SST variability that are more consistent and emerge more rapidly in the EP than in the CP (*27*, *29*). This pattern can result from several interrelated features, among them a reduced zonal SST gradient, increased stratification due to warming and intensified rainfall, and the nonlinear Bjerknes feedback associated with EP warming and convection (*27*–*29*, *47*). The coral reconstructions do not distinguish among mechanisms; they provide an independent set of observational constraints that contextualize model simulations.

Comparison with continuous and sedimentary ENSO reconstructions

Continuous ENSO reconstructions based on multiple overlapping records (*20*, *41*, *42*) suggest that ENSO variability increased notably before 1900 (fig. S16), well before substantial global warming and the EP-ENSO variability increase at Galápagos. This may reflect biases that would create this particular timing. First, age model errors of even a year cause misalignment among records of years with extreme SST. Combining such records will dampen interannual variance as age model errors accumulate downcore (*48*, *49*). The start of instrumental SST records in the mid to late 1800s provides a reference for coral data that reduce such errors from that time forward. Second, almost all coral

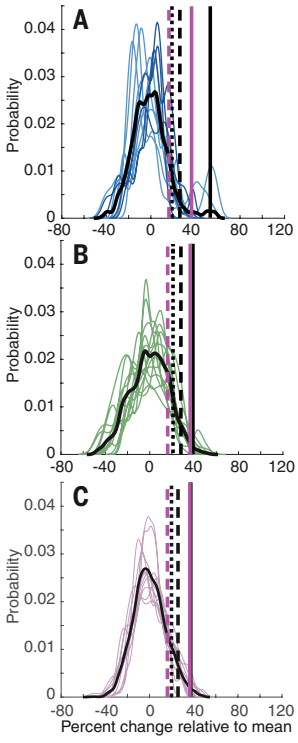

Fig. 3. Distribution of percent-change values for climate model simulations, compared with Galápagos coral. (**A** to **C**) The distribution of percent-change values calculated for all 20-year standard deviations in a simulation compared with the mean 20-year standard deviation. Vertical lines indicate the Galápagos coral changes in magenta (solid lines, modern relative to preindustrial; dashed lines, modern relative to 20th century), and the model 90, 95, and 99% significance levels values in black dotted, dashed, and solid lines, respectively. (A) CMIP5 (light blue) and CMIP6 (dark blue) models. Black is the mean of all. These were calculated for coral years only. (B) LME ensemble (individual members are in green, and mean is in black). These were calculated for coral years only. (C) Preindustrial control simulations for each of the CMIP5/6 models (individual models are in pink; mean are in black; and 99% and coral modern–preindustrial lines overlap). Values for control simulations were calculated across the full 500-year simulations. The distribution of percent change values over the full lengths of CMIP5/6 and LME simulations are provided in fig. S17.

records from within 5° to 10° of the equator, where ENSO variance dominates, begin between 1850 and 1900 CE; reconstructed interannual variance increases as these records join the reconstructions (*49*). The age-model bias is minimized in tree-ring-only ENSO reconstructions (*41*), but the scarcity of old tropical records remains a source of inhomogeneity. A third bias comes from using remote records that may be nonlinearly or inconsistently related to equatorial Pacific SST (*50*–*52*). Before the late 1800s, the continuous reconstructions do not consistently agree on the timing of changes in variability.

Individual foraminifera analysis (*43*) provides a different picture of past ENSO variability (fig. S16). Percent-change values range over ±20% compared with that of the modern period, with highest variability during the 1700s. Compared with other records, IFA suggests relatively quiescent behavior and no modern strengthening. However, in the EP, this method (by using surface-dwelling species) is more likely to reflect seasonal than interannual changes (*53*). In the data from the last millennium, the topmost sample has fewer individuals than does the rest of the core, making identification of change from the modern interval even more challenging (SM materials and methods).

The continuous records (*20*, *41*, *42*) show scales of variability comparable with coral data and concur that variability was greater in the most recent period relative to the preindustrial period. These powerful points of agreement collectively support the recent strengthening of ENSO. However, the continuous records suggest that this strengthening started in the 19th century, before substantial anthropogenic forcing. The 19th-century rise in variability may reflect the age model misalignment and sample coverage biases inherent to those reconstructions—biases that are absent in the Galápagos coral data. Although sparse in the 19th century, the Galápagos data indicate that the largest increase in EP-ENSO SST variability did not begin until the mid-late 20th century, as global warming accelerated (Fig. 2), and that variability was low with little century-scale structure before that time. Fossil coral reconstructions lack the continuity of multisite paleoclimate syntheses and sediment cores, but they provide an alternate and compelling view of last-millennium ENSO variability that avoids biases described above.

Comparison with naturally forced and internal ENSO variability

To compare simulated and reconstructed EP-ENSO variability, we used the percent-change value at the 95th percentile of the cumulative density function across all models, using the EP SST index defined for each simulation (table S8 and SM materials and methods). The coral-reconstructed SST variability is highly unusual ($P < 0.05$) compared with naturally forced and internal variability. This was true whether we used the full millennium-long simulations or just the subset of years covered by the corals to define model variability (Fig. 3 and fig. S17). The reconstructed increase in EP-ENSO exceeds the variability in the unforced control simulations at a level approaching 99% significance (Fig. 3).

EP-ENSO variability differs among models, particularly in two of the CMIP-5 models (95% CI of 58 and 54% for GISS-ER2-R and MPI-ESM-P, respectively). In these simulations, spikes in standard deviation coincide with volcanic forcing (fig. S9). This behavior is not seen in other forced simulations, nor in the preindustrial controls. In control simulations, EP-ENSO variability is more consistent, with little difference between CMIP5 and CMIP6 generations (table S8), suggesting that forced responses, not internal variability, drive intermodel differences in EP-ENSO variability. As in earlier work (*18*, *23*), forcing does not generally lead to greater ENSO variability; the only models that show substantial differences in EP-ENSO SST variability between forced and control simulations are the two in which volcanic eruptions produce outsize responses. Eight of the 11 CMIP5/6 models show similar levels of variance between forced and control simulations (95% CI differences <10%) (table S8).

The individual ensemble members of the LME yield similar percent-change values (table S8), comparable with the multimodel ensemble and control simulations. Our multimodel approach provides a robust background of naturally forced and internal ENSO variability against which reconstructed changes in EP-ENSO stand out distinctly. A formal attribution study is beyond the scope of this paper, but a recent synthesis of simulated changes in ENSO variability concluded that anthropogenic forcing is likely responsible for post-1960 increases in EP variability (*27*). Regarding detectability, in a study of 68 CMIP5/6 models (*29*), 63% show EP-ENSO variability rising above background levels between 1940 and 2100, with a mean emergence date of ~2030. Our coral data, although discontinuous, are consistent with these results, particularly with the models showing early emergence. Additional coral data from the eastern equatorial Pacific to fill in temporal gaps would provide a more robust basis for comparison.

Conclusions

We documented an unambiguous increase in EP-ENSO SST variability since the 19th century, particularly in recent decades, that reflects the observed intensification of El Niño warm extremes and exceeds simulated internal and naturally forced variability. This amplification is consistent with Earth system model projections that ENSO-related SST variance will likely increase and emerge first in the EP (*27*, *29*). We found that SST variance in the EP is rising above background variability more rapidly than expected from most model simulations, whereas in the CP, increasing variance is statistically distinct but has not yet unambiguously exceeded that of the past millennium. These data describe a preindustrial range of EP-ENSO SST variability across a time span far longer than the instrumental record, highlighting the extraordinary nature of recent EP SST variability compared with that of the past millennium. We found no evidence for changes in EP-ENSO variability associated with the Little Ice Age and Medieval climate eras.

These results have important implications for climate hazards. ENSO extremes impose tremendous economic costs, ecological damages,

and human suffering globally (*2*, *3*). The impact of more frequent and extreme El Niño events, superimposed on background ocean warming, has been particularly devastating to coral reef ecosystems and to marine ecosystems more broadly (*54*–*56*). As the climate warms, the hydroclimatic impacts of ENSO are expected to amplify regardless of changes in SST variability (*57*). Our results highlight that the underlying SST variability is also strengthening; if this trend continues, it will further intensify the impact of ENSO within and beyond the tropical Pacific. By demonstrating a clear increase in EP ENSO variability under unprecedented anthropogenic forcing, our results provide a point of comparison for identifying climate model simulations that successfully capture this change, which may improve our ability to anticipate and prepare for future risks.

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SUPPLEMENTARY MATERIALS

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QUANTUM COMPUTING

High-fidelity entangling gates and nonlocal circuits with neutral atoms

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The generation and manipulation of entanglement with low error are essential in quantum information systems. In practice, two-qubit entangling gates constitute a dominant error source, limiting circuit depths and performance in fault-tolerant architectures. Using a neutral-atom quantum processor, we realized entangling controlled-*Z* gates with a high-Rabi-frequency smooth-amplitude pulse, employing state-selective readout and qubit reuse for fast calibration, and achieved a fidelity of 99.854(4)%, which improved to 99.941(3)% upon loss postselection, with stable performance for 10 hours. We then used these low-error gates in quantum circuits with coherent atom rearrangement. Performance was benchmarked by creating and disentangling cluster states, and subsequently, we studied nonlocally entangled states with scrambling circuits featuring longer-range connectivity. Our approach provides a route toward deep-circuit, efficient fault-tolerant quantum computation.

High-fidelity entanglement is at the heart of quantum science and technology, enabling wide-ranging applications in computing, simulation, metrology, and sensing (*1*). For instance, error rates around 0.1% are generally believed to be necessary to enable efficient performance of quantum error correction (QEC), which is central to realizing utility-scale fault-tolerant quantum computation (*2*–*5*). In particular, achieving such high-fidelity performance for nonlocal quantum circuits over large numbers of qubits could enable the implementation of efficient, high-rate QEC codes (*6*, *7*) and appreciable algorithmic speedups with transversal gate architectures (*8*, *9*).

Neutral atom processors are a leading quantum computing approach, combining nonlocal qubit connectivity (*10*), scalable control of large qubit numbers (*11*, *12*), continuous operation (*13*–*15*), and the ability to efficiently implement QEC (*16*–*19*). In this approach, atomic qubits are entangled using controlled-phase gates based on Rydberg blockade (*20*, *21*). The fidelity of these entangling operations has rapidly advanced (*22*–*28*), with recent benchmarking demonstrations reaching ~99.7% fidelity (*29*). Moreover, methods for loss detection and erasure conversion have been shown in a range of settings, including loss-postselected fidelities reaching ~99.8 to 99.85% (*30*–*32*) and improved QEC with erasure information (*33*, *34*), including below-threshold performance (*35*).

Here, we demonstrate an entangling controlled-*Z* (CZ) gate fidelity of 99.854(4)% for neutral atom qubits, which improves to 99.941(3)% with postselection on neither atom being lost. Our approach combines high-Rabi-frequency gates with a smooth-amplitude pulse profile based on optimal control (*24*, *36*, *37*) with fast calibration based on loss-resolved qubit readout (*35*), and we demonstrate how this fidelity can be maintained for >10 hours without recalibration.

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Neutral atom entangling gates

Our approach made use of a reconfigurable neutral atom processor architecture (*16*, *35*), with several key upgrades to achieve and explore low-error regimes. Quantum information was encoded in long-lived hyperfine clock states of ^{87}Rb atoms, and entangling gates were performed by exciting the pair of atomic qubits to $n = 53$ Rydberg states (*24*). We used a gate profile based on optimal control (*36*, *37*) with a smooth amplitude and phase profile, as shown in Fig. 1A (*24*), and additional amplitude corrections to compensate for pulse imperfections (*25*, *38*). This gate was optimized to suppress scattering from the intermediate excited state $|e\rangle$ by maximizing population in the dark state not containing $|e\rangle$ (*24*). To suppress errors from the Rydberg-state lifetime, we used high Rabi frequencies reaching a peak of roughly $2\pi \times 17$ MHz. The required high laser intensities induced substantial light shifts on the Rydberg transition and necessitated careful stabilization of Rydberg laser positions and powers to minimize the detrimental effect on ground-Rydberg coherence T_2^* (*38*). Additionally, the coupling to the other Rydberg level $|r'\rangle$ with $m_J = +1/2$ grew as the Rabi frequency was increased, which we addressed by applying a stronger magnetic field (larger Zeeman splitting) to suppress off-resonant excitations (fig. S1B).

Our quantum processor used up to 442 atomic qubits arranged into separated zones (Fig. 1B), with 17 gate sites used for high-fidelity benchmarking and quantum circuits and a large reservoir for refilling lost atoms. We used state-selective readout based on spin-to-position conversion (*35*), which enabled the direct detection of atom loss errors at the end of the circuit. By reusing qubits and refilling lost atoms from the reservoir, we were able to perform fast calibration of the CZ gate, for up to 80 repetition cycles in a single round of atom loading, at a cycle rate of 20 to 30 Hz (Fig. 1C). We found that calibrating the gate from scratch, which consisted of scanning each gate parameter in series, typically took <40 min, with calibrated gate parameters remaining stable such that subsequent recalibration was faster.

Benchmarking low-error CZ gates

To characterize the resulting CZ gate performance, we implemented randomized benchmarking (RB) based on repeated application of CZ gates interspersed with single-qubit rotations (Fig. 2A). First, we used a global echo RB sequence using a single-qubit π pulse in between pairs of CZ gates (*24*) and measured a raw fidelity of 99.854(4)% and a fidelity of 99.941(3)% when postselecting on neither atom being lost at the end of the sequence (Fig. 2B). The detection of these loss errors can be used to improve QEC thresholds (*35*, *39*) and quantum simulation results (*40*). In particular, we found that the CZ error resulting from atom loss is 0.087(5)%, comprising ~60% of the total error. We measured a loss probability of 0.054(2)% per atom per CZ gate, which exceeds half of this total loss error and indicates correlated loss of both atoms during the gates; this information can be used in QEC decoding (*41*). Additionally, we explored the effects of leakage to other hyperfine levels (*29*) in the ground state $5S_{1/2}$, $F = 1, 2$ manifolds (from Rydberg T_1 and intermediate-state scattering) through measurements of $F = 1$ populations at the end of the RB sequence, which increased by 0.008(1)% per atom per CZ gate (fig. S4) (*38*).

To test the robustness of the gate performance, we measured the CZ fidelity across several different benchmarking and readout methods (Fig. 2C). First, we compared echo RB with the symmetric stabilizer benchmarking (SSB) protocol developed in (*29*), which is theoretically insensitive to single-qubit gate errors and requires calibration of the single-qubit phase from the CZ gate (fig. S3). We found that both the raw and loss-postselected fidelities are equivalent for the two RB methods, consistent with no measurable effect from single-qubit gate errors

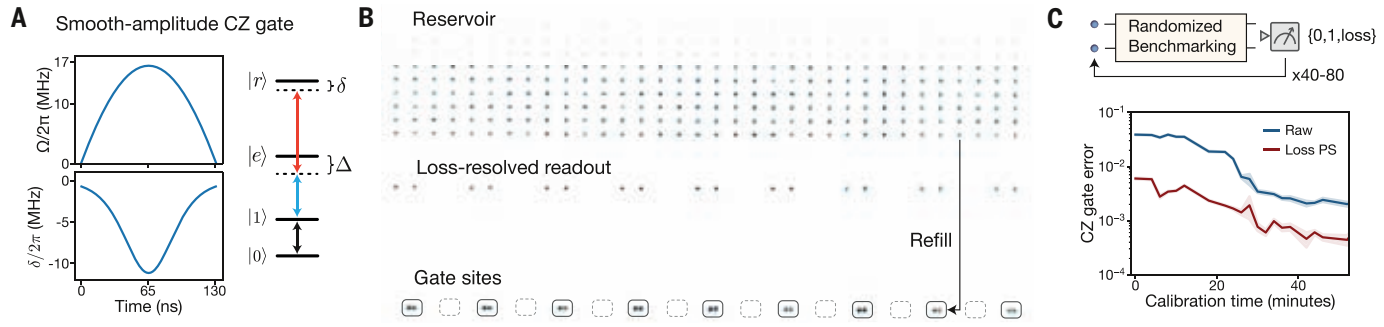

Fig. 1. CZ entangling gates in a neutral atom processor with fast cycle rates. (**A**) Rydberg gate realized through a smooth-amplitude gate pulse at a peak Rabi frequency of roughly $\Omega = 2\pi \times 17$ MHz, with a detuning profile $\delta(t)$. The two-photon excitation scheme consists of transitions at 420 and 1015 nm, exciting from the qubit state $|1\rangle$ to the Rydberg state $|r\rangle = 53S_{1/2}, m_J = -1/2$ through the intermediate state manifold $|e\rangle = 6P_{3/2}$, with intermediate state detuning $\Delta = 2\pi \times 7.8$ GHz (see fig. S1B for a more detailed excitation diagram). (**B**) Images of the atoms separated into processor zones, with gate sites used for benchmarking and high-fidelity quantum circuits, and state-selective readout to perform loss detection. Fast calibration of CZ gates is enabled by qubit reuse and refilling lost atoms from a large atom reservoir. (**C**) RB was performed over 40 to 80 repetitions, with a cycle rate of 20 to 30 Hz. Example calibration of the CZ gate from starting parameters near the theoretical prediction, showing both raw and loss-postselected (PS) CZ errors. Each point corresponds to the fitted maximum of a parabola scan over one of the gate parameters, with error bars indicated by the shaded region.

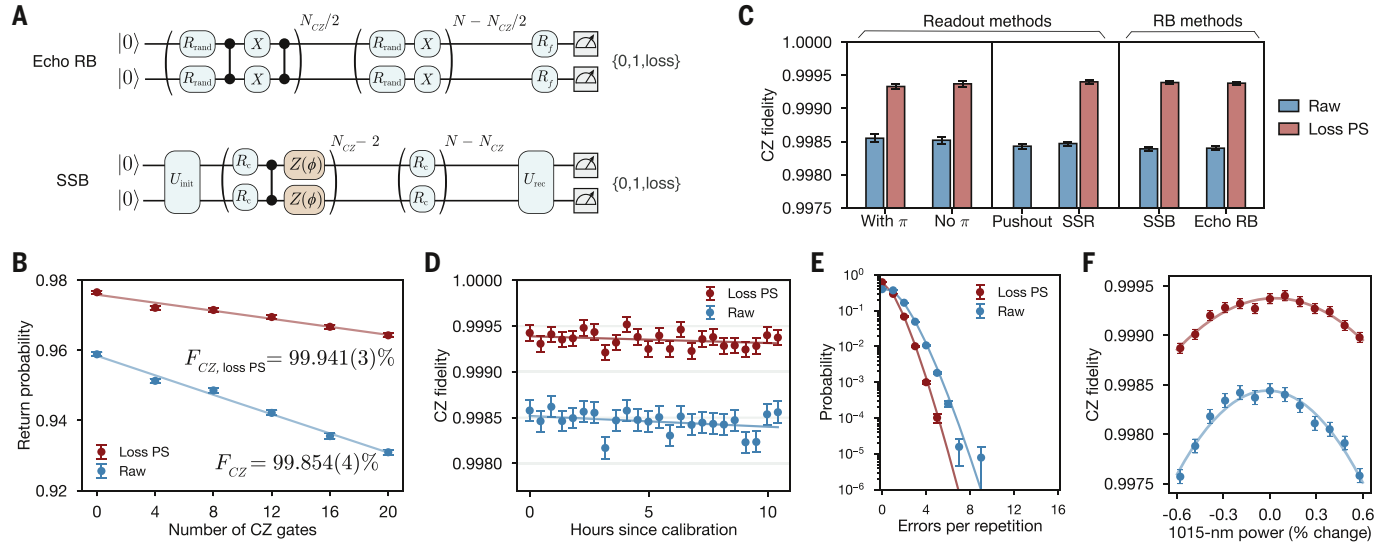

Fig. 2. Stable low-error CZ gate performance. (**A**) Benchmarking methods based on repeated CZ gate applications: global echo randomized benchmarking ("echo RB") (*24*) and symmetric stabilizer benchmarking ("SSB") (*29*). (**B**) Return probability as a function of the number of CZ gates, fitted to an exponential decay, yielding a raw fidelity of 99.854(4)% and a loss-PS fidelity of 99.941(3)%. In this work, "raw" refers to no postselection. (**C**) Comparison of CZ gate fidelity measured with different benchmarking protocols: echo RB with and without an extra single-qubit π pulse before readout, resonant pushout of $|1\rangle$ versus state-selective readout (SSR), and SSB versus echo RB. (**D**) Stability of the CZ fidelity measured for >10 hours with echo RB, with no recalibration of the gate. Each data point consists of a fit to the decay of return probabilities for 0, 8, 12, and 20 CZ gates. (**E**) Number of errors per repetition (raw and loss-PS), where a repetition consists of benchmarking on two rows of 16 atoms brought into the entangling zone, for the data shown in (D), consisting of 127,680 repetitions. Solid curves are sums of individual Poissonian fits for each number of CZ gates. (**F**) Sensitivity of the raw and loss-PS CZ gate fidelities under perturbation in the 1015-nm Rydberg beam power, keeping the gate pulse parameters unchanged.

or miscalibration of the single-qubit phase of the gate. We additionally compared state-selective readout with destructive spin-to-loss readout using resonant blowout of one of the qubit states and again found comparable results for echo RB (fig. S2B). Finally, we measured similar fidelities with and without an extra single-qubit π pulse before readout, which suggests that any residual effects of leakage on the RB results are relatively balanced between the $F = 1$ and $F = 2$ levels (fig. S2A).

To test the stability of our gates, we tracked the fidelity over long durations (Fig. 2D). We observed stable performance at ~99.85% fidelity for >10 hours without any recalibration of the gate. For this large dataset, consisting of roughly 2 million runs of the RB sequence averaged over the array, we further observed that both the raw and loss-postselected return probabilities follow Poisson distributions with no observed large-scale correlated error events spanning the system (Fig. 2E). We found that the residual drift in the gate fidelity typically originates from drifts in the Rydberg laser powers and positions, such that peak performance can generally be recovered by small adjustments to beam powers and positions, with the calibrated gate pulse parameters remaining unchanged. To understand the magnitude of the effect of drifts, we intentionally adjusted the power of the 1015-nm Rydberg beam away from optimal and observed that the gate is sensitive to power changes on the scale of ~0.1% (Fig. 2F). We further found that the sensitivity of the CZ gate to perturbations in the Rydberg beam

powers and positions is well captured by numerical error simulations that include the effects on gate detuning and T_2^* (fig. S5). These observations suggest that neutral-atom devices with sufficiently stable Rydberg beams can achieve high-fidelity performance over long durations passively. Over even longer timescales, this performance could be maintained with automated stabilization of beam positions and powers, for example, through real-time steering of QEC performance (*42*).

The remaining CZ gate errors can be largely explained by a numerical error model, with the dominant contributions being decay from the Rydberg state resulting from finite lifetime and from the intermediate excited state in the form of off-resonant scattering (Fig. 3). Using our numerical error model, we projected that raw CZ gate fidelities at the level of 99.9 to 99.95% could be enabled by improved T_2^*, suppressed coupling to $|r'\rangle$, and higher laser intensities (fig. S6).

High-fidelity circuits with atom motion

We extended our benchmarking of CZ gates to quantum circuits based on coherent rearrangement of atomic qubits. Several additional challenges can arise when incorporating high-fidelity entangling gates into quantum circuits. In our RB circuits, we placed the two atoms in two traps generated by the same spatial light modulator (SLM), which allowed us to position the atoms uniformly close together (≈1.7-μm spacing). By contrast, in quantum circuits, we performed gates between atoms in stationary SLM traps and atoms in moving traps generated by acousto-optic deflectors (AODs) (*16*). We found that to perform these gates with high fidelity, the spacing between traps must be uniform across the array and the focus of the two trap arrays must be in the same plane (*38*). When properly calibrated, it was possible to realize similarly high CZ gate performance (Fig. 4A).

To understand CZ performance in quantum circuits with atom motion, we performed a characterization circuit that repeatedly creates and disentangles 20-atom cluster states (Fig. 4B). We measured the probability that all atoms in the array return to the initial state $|0\rangle^{\otimes 20}$ after 0, 4, 8, 12, and 16 nearest-neighbor CZ gate layers, defined as the global return probability P_{return}. By fitting P_{return} as a function of the total number of CZ gates applied across the entire qubit array, we inferred the corresponding CZ gate fidelity (Fig. 4C). We found a raw fidelity of 99.843(6)% and a loss-postselected value of 99.956(5)%, similar to the fidelity measured in gate-benchmarking sequences. Although this characterization circuit is not a rigorous benchmarking method, we observed in numerical simulations that the extracted fidelity from the circuit closely matches the CZ gate fidelity (fig. S7). To test the stability of the CZ gate performance within this circuit, we ran the experiment continuously for 15 hours and found that the extracted CZ fidelity remains high without any recalibration (Fig. 4D).

Chaotic nonlocal circuits

Finally, we explored high-fidelity entanglement in more-complex circuits with longer-range qubit connectivity. Given their high degree of connectivity, such circuits are known to be effective in scrambling quantum information and can be difficult to simulate classically (*43*, *44*). As an example, we designed and studied a class of nonlocal circuits with increasingly long-range gates as a function of depth, interspersed with $\pi/4$ single-qubit gates between layers (Fig. 5). This circuit structure led to superballistic scrambling of quantum information, in which entanglement entropy saturates at the Page entropy, the near-maximal value expected for a random pure state, at a depth that scales as $\sqrt{N}$ for system size N (Fig. 5B and fig. S9). Our circuits can also be interpreted as nonlocal kicked Ising models, where the corresponding nearest-neighbor circuit does not scramble (fig. S9A) (*38*). The output random states are related to the physics of black holes (*45*) and have been used for benchmarking quantum devices (*46*, *47*) and quantum advantage experiments (*48*).

We first assessed the fidelity of the CZ gate in these nonlocal quantum circuits applied on 20 qubits through measurement of atom loss, which we found to be a useful proxy for the in situ performance of the gate in circuits. We measured the loss per qubit per CZ gate to be consistent across the RB, cluster state, and nonlocal circuits throughout this work (Fig. 5C). Additionally, we observed that the loss per shot for both circuits follows a Poisson distribution, with no evidence of large-scale loss events (fig. S8C).

Next, we sampled from the output of this circuit by measuring all qubits in the Pauli Z basis and plotting the probability of each outcome ordered by theoretical expectation. For random states, the ideal probability distribution $p(x)$ is an exponentially distributed Porter-Thomas distribution PT(x) (*49*). Such probability distributions can be studied through cross-entropy benchmarking (XEB), where $\text{XEB} = 2^n \sum_x q(x)p(x) - 1$ for outcomes x and system size n. The ideal value in the absence of noise is given by the second moment

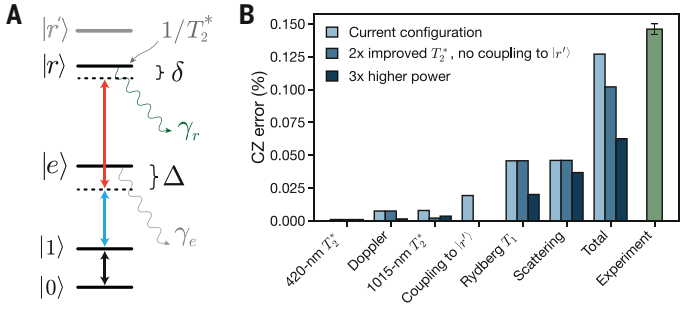

Fig. 3. Error breakdown and path to further improvement. (**A**) Illustration of error sources for Rydberg entangling gates, including scattering from the intermediate state γ_e, Rydberg state lifetime γ_r, ground-Rydberg coherence T_2^*, and coupling to the other Rydberg state $|r'\rangle = 53S_{1/2}, m_J = +1/2$. (**B**) Numerical error budget showing contributions of individual errors, total theoretical errors, and experimental data. Error breakdowns are shown for the present configuration, as well as projected values attainable with technical improvements and higher laser intensities (*38*).

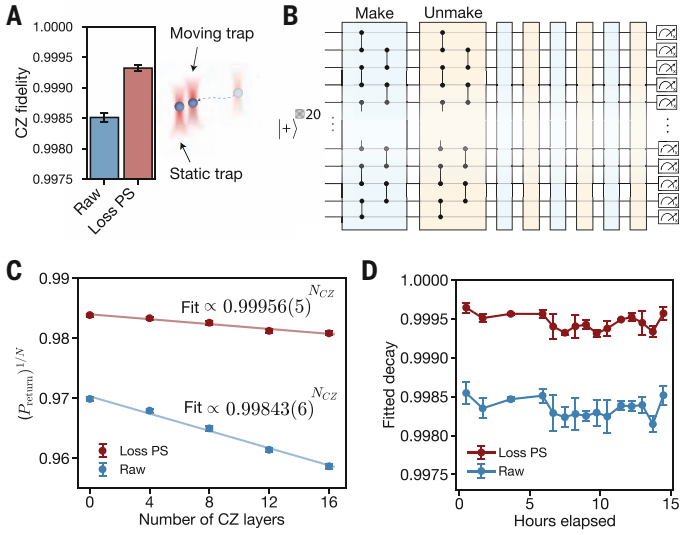

Fig. 4. High-fidelity quantum circuits with atom motion. (**A**) Raw and loss-PS CZ fidelities measured with echo RB, with one atom in a static trap generated by an SLM and one atom in a moving trap generated by an AOD. (**B**) Protocol for creating and disentangling one-dimensional cluster states to study CZ gate performance in a quantum circuit. (**C**) Decay of global return probability P_{return} as a function of the number of total CZ gates, fit to an exponential decay, for $N = 20$ qubits. (**D**) Stability of the fitted decay over time, without recalibration.

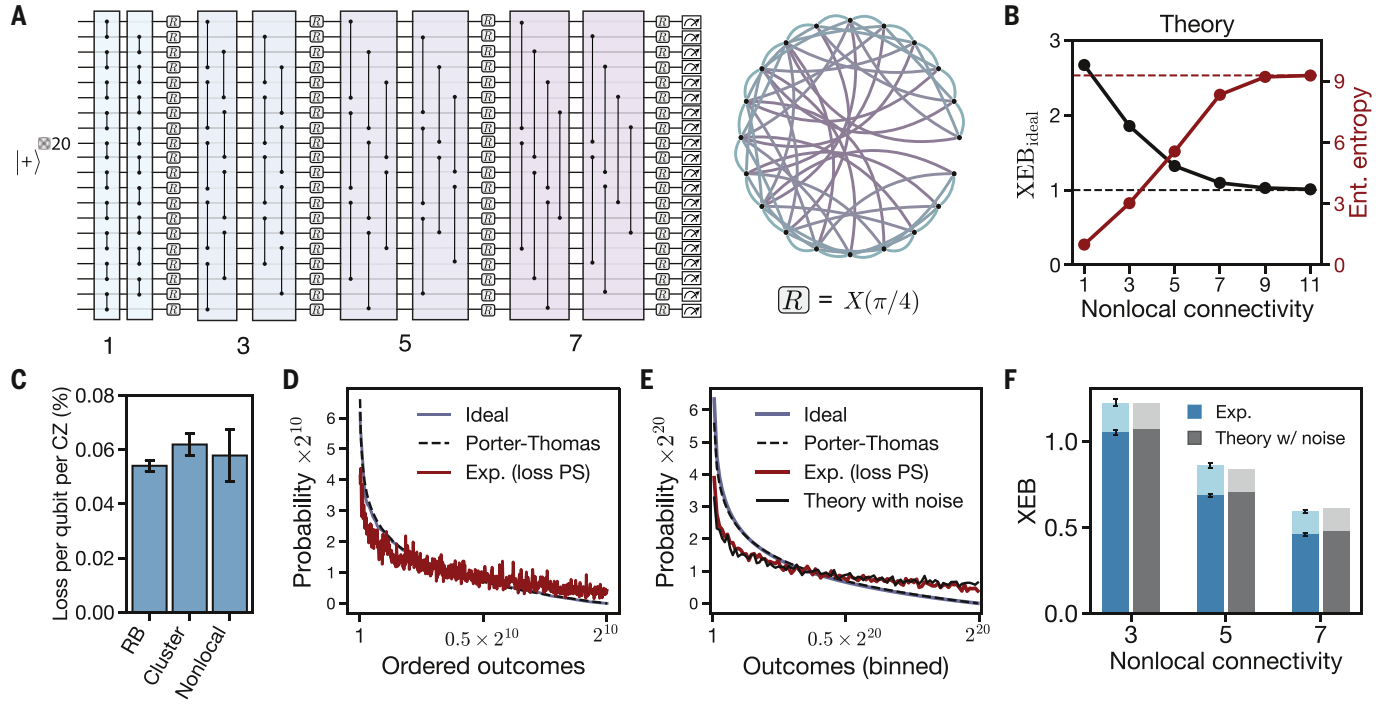

Fig. 5. High-fidelity scrambling circuits with nonlocal connectivity. (**A**) Nonlocal quantum circuit implementation. CZ gates of increasing nonlocal connectivity (with labels given below the circuit) are interspersed with single-qubit gates, $R = X(\pi/4)$. The resulting connectivity for the 20-qubit circuit is illustrated on the right. (**B**) Theoretical values for the half-system-size von Neumann entanglement entropy and XEB_{ideal} as a function of the degree of nonlocal CZ gates added to the circuit ("nonlocal connectivity"). Dashed lines indicate the Page entropy value, and $XEB_{ideal} = 1$. (**C**) Loss per qubit per CZ gate compared for different circuits in this work: RB, the cluster state circuit in Fig. 4B, and the nonlocal quantum circuit in (A). (**D**) Sampled outcome distribution for the 10-qubit nonlocal quantum circuit, run with the depth shown in (A), comparing the experimental data with the ideal theory expectation and a Porter-Thomas distribution. (**E**) Sampled outcome distribution for the 20-qubit nonlocal circuit, with the outcomes binned in 100 groups by their probability. Comparison to numerical simulations with noise is shown. (**F**) Experimentally measured XEB for the 20-qubit circuit for different degrees of nonlocal connectivity, compared with numerical simulations with noise. For both theory and experiment, darker bars are raw data, and lighter bars correspond to postselection on no atom loss.

of the probability distribution, $XEB_{ideal} = 2^n \sum_x p^2(x) - 1$, often called the collision probability, for which $PT(x)$ has $XEB_{ideal} = 1$. We find that the theoretical output distributions of our circuits converge to $PT(x)$ at sufficient depth (Fig. 5B and fig. S9).

Experimentally, we started by studying the distribution in a 10-qubit circuit, where we could measure the probability of all 2^N outcomes (fig. S8B). We found that the resulting distribution is similar to $PT(x)$ for outcomes x (Fig. 5D). Next, we extended to sampling from a 20-qubit circuit by binning the outcomes into groups (Fig. 5E). We found that our results are well described by numerical simulations with noise, with decoherence making the distribution closer to the uniform distribution. To characterize the 2^{20} sampled outcomes more quantitatively, we compared XEB for the 20-qubit circuit sampling distribution with numerical simulations (Fig. 5F) and found good agreement for the same error model across three different circuit depths (*38*). Although we did not directly probe the CZ gate fidelity, as in Fig. 4, the combination of low atom loss per gate and agreement with numerical simulations suggests high-fidelity performance in these complex nonlocal circuits.

Discussion and outlook

Our results illustrate a route for practical fault-tolerant quantum computation with atomic processors. Corresponding to a more than threefold improvement compared with our previous CZ gates used in below-threshold performance of a surface code logical qubit (*35*), these higher fidelities, in combination with improving single-qubit operations, should enable >10× below-threshold QEC performance while leveraging the ability to detect atom loss errors (*39*). These improved fidelities were achieved with ≈2.3× higher Rabi frequency, requiring suppression of coupling to $|r'\rangle$, stronger blockade with closer spacing, improved Rydberg coherence (T_2^*), and fast calibration of the gate pulse (*38*). We found that the remaining CZ gate errors are dominated by Rydberg-state lifetime and intermediate-state scattering. Through technical improvements to further suppress coupling to $|r'\rangle$ and improve the Rydberg T_2^*, we expect that reaching a raw CZ fidelity of 99.9 to 99.95% will be achievable by scaling up the available laser power (fig. S6).

These techniques can be extended to larger arrays, suitable for operations on large atomic code blocks, using efficient beam shaping (*50*, *51*), fast coherent atom rearrangement (*38*), fast beam scanning (*34*, *52*, *53*), higher-power and pulsed lasers (*54*), and different Rydberg excitation schemes (*25*, *29*). Furthermore, our techniques can be extended to other gate schemes, including entangling gates at constant atomic velocity (*32*) and multiqubit gates such as controlled-controlled-Z (CCZ) gates (*24*, *55*).

The high-fidelity quantum circuits demonstrated in this work form the basis for wide-ranging explorations of highly entangled quantum systems and fault-tolerant architectures. Extensions of our nonlocal circuits could be used to simulate nonlocal quantum models, including efficient encoding of fermions with qubits (*56*) and models for gravity (*57*). Additionally, such circuits can be used to probe complex properties of quantum many-body systems, providing optimal system size–independent sample complexity for measuring linear observables (*58*). Notably, implementing complex nonlocal circuits with high fidelity should enable the realization of high-rate quantum error–correcting codes to achieve low error rates on large numbers of logical qubits (*6*, *7*).

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S9; Tables S1 and S2; References (*60*–*72*)

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OCEAN CARBON

Marine snow viscosity regulates microbial degradation and the ocean carbon sink

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Bacterial degradation of marine snow aggregates is a major component of global carbon cycling. Whether aggregate carbon is released in the upper ocean or is sequestered at depth depends on its sinking speed and degradation rate. However, little is known about the physical constraints of bacterial colonization and degradation of individual aggregates. Using molecular rotors to measure the nanoscale viscosity field of natural aggregates, we show that aggregates are highly structured microhabitats with less-viscous regions that are accessible for colonization and more-viscous regions that correlate to a barrier to bacterial infiltration. By quantifying the viscosity degradation rate, we demonstrate that more viscous aggregates take longer to degrade, sink farther, and contribute more to carbon sequestration, revealing a microscale physical constraint of the global ocean carbon pump.

Marine snow, a flurry of heterogeneous organic particles sinking to the deep ocean, accounts for 70% of ocean carbon export (*1*). Marine snow originates in the upper water column from phytoplankton exudates and blooms, zooplankton fecal matter and carcasses, viral cell lysis, and other debris (*2*–*4*). The combined spatial and temporal variability of marine snow sources and dynamic ocean conditions makes predicting the efficiency of the carbon pump challenging, a topic that has been the focus of decades of research (*5*–*7*). Fundamentally, marine snow sinking speeds and bacterial degradation rates determine whether the organic matter is recycled in the upper ocean or contributes to the ocean carbon sink: Rapid decomposition by bacteria releases carbon dioxide back into the sunlit shallows, where it was originally fixed by photosynthesis; faster sinking transports particulate organic carbon toward the seafloor, where it is sequestered for hundreds to thousands of years (*8*–*14*). Despite the fundamental importance of these two rates, little is known about the physical structure of marine snow and how it impacts bacterial colonization and degradation over time (*15*–*17*).

Observations of the distribution of marine snow in the water column using sedimentation traps, marine snow catchers, and imaging profilers have repeatedly shown a substantial decline in the flux of particulate organic matter as a function of depth (*18*–*24*), a relationship known as the Martin curve (*25*). This empirical description of the vertical attenuation of the oceanic carbon flux is frequently used in carbon budgets and modern climate models (*26*). The collective microscale interactions between microbes and marine aggregates that influence the Martin curve are not well understood and may explain variability or deviations from the canonical attenuation curve in different ocean regions (*26*–*29*). Improving our understanding of bacterial degradation of the physical structure of marine snow is critical for predicting the efficiency of ocean carbon fluxes in an Earth climate system that is undergoing rapid changes (*30*).

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Marine snow typically contains transparent mucus material that comprises the majority of the particle or acts as a glue for dead cells, ballast minerals, and other solid detritus (*31*). The sources of marine snow differ in their chemical composition and density (*32*, *33*), and so the material properties of marine snow from different ecosystems may display considerable variability. The material properties of marine snow have not been measured directly, but indirect evidence suggests that the mucus is viscous. Phytoplankton increase the microscale viscosity of the phycosphere near their cells by exuding extracellular polymeric substances, a precursor to marine snow (*34*, *35*). An "invisible comet tail" of viscous, low-density mucus can slow the sinking speed of marine snow particles (*36*). Such studies rely on tracking seeded microspheres to estimate viscosities and flow; these microspheres do not readily penetrate into mucus and preclude concurrent observations of bacteria, which are similar in size to the microspheres. Bacteria rapidly colonize marine snow and collectively produce hydrolytic enzymes to digest the descending particles (*37*). To date, methods used to simultaneously visualize bacteria and marine snow (e.g., alcian blue, catalyzed reporter deposition–fluorescence in situ hybridization) require fixation, filtration, washing, or drying, which damage the fragile structures of the mucus and distort three-dimensional (3D) spatial relationships (*31*, *38*, *39*).

To overcome these limitations, we developed a method to measure the 3D nanoscale viscous structure of individual marine snow particles ("mucus aggregates") using molecular rotors and confocal microscopy (*40*). A molecular rotor fluoresces when twisting of a functional group is slowed by crowding of the local environment by nonsolvent molecules. Microviscosities measured with molecular rotors (referred to as "viscosity" herein) correlate with but are not the same as hydrodynamic viscosities (fig. S1). These viscosity-sensing dyes readily penetrate into mucus to reveal the interior physical structure of undisturbed marine aggregates. Combined with a nucleic acid stain to locate bacteria, we can quantify bacterial colonization and the associated transformation of viscosity within individual aggregates over time. Notably, this method is ratiometric and normalized to background fluorescence and viscosity standards to account for variability in dye concentration, temperature, and other environmental and optical artifacts (see materials and methods). The molecular rotor viscosity measurements showed excellent agreement with independent microrheological measurements of aggregates from the same species, verifying the reliability of the method in a complex biofluid (see supplementary materials). In this work, we present the first 3D quantitative view of the viscous structure—the "mucoscape"—of mucus aggregates from laboratory diatom cultures and environmental samples. We found that viscosity (relative to seawater) within aggregates correlates to a physical constraint on colonization and degradation by bacteria. Bacteria can readily colonize lower-viscosity "surface" regions but are not found in higher-viscosity inner "barrier" regions until they are broken down below a viscous threshold. By quantifying the rate of viscosity degradation, we show that marine snow viscosity is a notable, previously unrecognized rate determinant of the ocean carbon sink.

Bacteria and the viscous microhabitat of marine aggregates

The molecular rotors revealed a high degree of spatial variability in the relative viscosity of the exterior of a mucus aggregate sampled from a laboratory culture of the centric diatom, *Chaetoceros affinis* (Fig. 1A, blue color scale, and movie S1). Nanoscale heterogeneity composed of bulges and crevasses, smaller than the bacteria, was visible across the exteriors and inside of all laboratory aggregates analyzed ($N = 30$ aggregates; Fig. 1B and movie S2). Despite the heterogeneity, viscosity tended to increase toward the interior of aggregates (Fig. 1C). The median relative viscosity of all aggregates was 2.1, and the 95th percentile viscosity was 12.4, compared with 1.0 and 1.2, respectively, for the background volume outside of the aggregates (seawater). Bacteria (~1 μm) were located near the "surface" regions of the aggregates

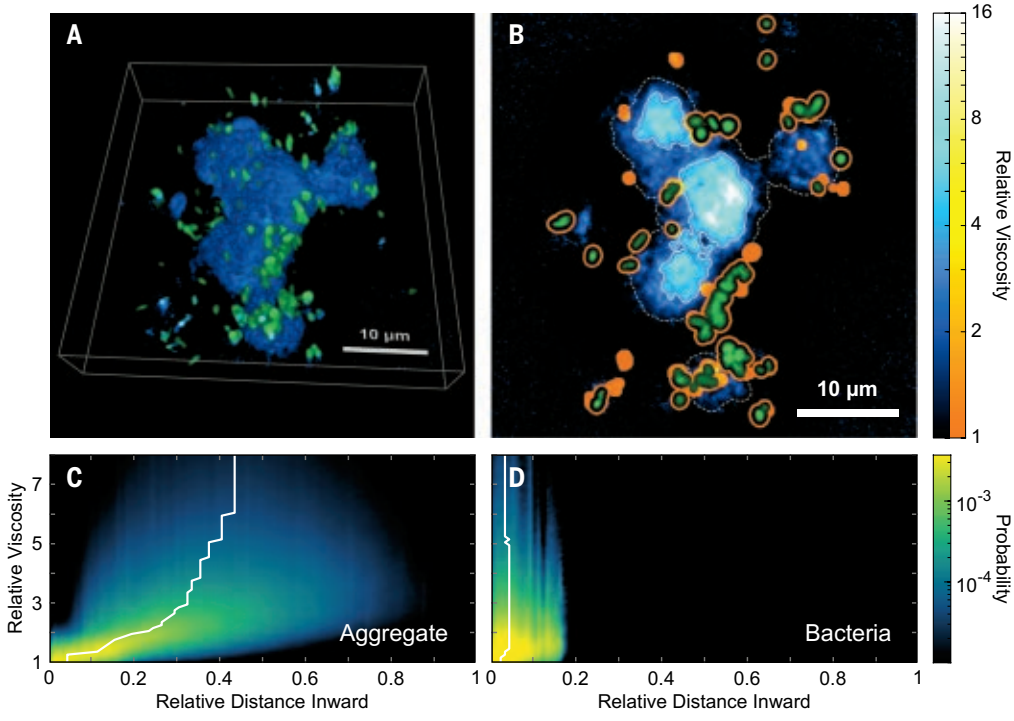

Fig. 1. The viscous "mucoscape" of an aggregate colonized by bacteria. (**A**) A composite 3D reconstruction of mucus viscosity relative to seawater (blue color scale) and bacteria identified with a nucleic acid stain (green). Bacteria can colonize the surface of the aggregate, which displays substantial nanoscale viscous heterogeneity. (**B**) A thin slice through the middle of the same aggregate shows internal complexity and viscosity increasing toward the center. Viscosities within 0.3 µm of bacterial cells are highlighted with the orange color scale. Bacteria colonize lower-viscosity edge regions (between the dotted and solid white lines) but are not found in regions with viscosities >4 (inside the solid white line). (**C**) The mean probability of a relative viscosity occurring at a given distance inward from the aggregate edges, normalized by the maximum distance ($N = 30$ aggregates). The white line indicates the most probable distance for each viscosity. (**D**) The same as (C) but for viscosities in close proximity to bacteria [orange color scale in (B)]. Bacteria are rarely located more than 20% inward from the edge of the aggregate ($N = 30$ aggregates).

as individuals or clusters (Fig. 1, B and D). Aggregate volumes ranged from 548 to 10,100 µm^3, with a mean of 3300 ± 478 µm^3. Larger millimeter-scale aggregates were abundant in the cultures, but smaller aggregates were chosen for analysis to maximize spatial resolution of both the viscosity and the bacteria.

To better understand bacterial associations with aggregate viscosities, we quantified properties within 0.3 µm of the bacterial cells (orange color scale, Fig. 1B). The probability of a given viscosity occurring in the proximity of bacteria in contact with the aggregate differed from that of bacteria not in contact with the aggregate (Fig. 2A). The most probable viscosity for aggregate-colonizing bacteria was 1.25 ($N = 2843$ bacteria) compared with 1.00 for bacteria outside the aggregates ($N = 3607$ bacteria). Hence, a viscosity of ~1.25 correlates to a "surface" to bacteria and can be used to determine the portions of mucus aggregates that are physically accessible for colonization. Bacteria likely remain on these surface regions by binding to aggregate polymers using lectins embedded in their cell membranes (*41*, *42*). The ratio of the probability of a viscosity occurring in the proximity of bacteria versus the rest of the aggregate volume is shown in Fig. 2B. A probability ratio >1 indicates a preferential association of the bacteria with an aggregate viscosity, whereas a probability ratio <1 indicates that bacteria are less associated with those viscosities within the aggregate. The probabilities are equal at a viscosity of ~2, signifying that viscosities >2 are increasingly inaccessible to bacteria. Bacteria are half as likely to encounter an aggregate viscosity of 4, which correlates to a physical "barrier" to bacteria. Elastic properties of the mucus may also contribute to bacterial colonization patterns but are not quantified with the molecular rotor assay. Analysis of viscosity gradients within the aggregates supports similar definitions of accessible surfaces and inaccessible barriers for bacteria (fig. S9).

Bacteria were rarely found inside regions with relative viscosities >4 (Fig. 1B, solid white contour). Even in areas where bacteria appear to have infiltrated inside the aggregate, local viscosities were generally <4. This suggests that bacteria must reduce the viscosity of inner aggregate material through enzymatic hydrolysis in order to break through and proliferate. The percentage of aggregate volume inaccessible to bacteria (contiguous internal regions of viscosity >4) ranged from 0.43 to 71%, with a mean of 27 ± 3.3%. Thus, there was substantial variability among aggregates in the volume that is directly accessible to bacteria, even though these aggregates were sampled from the same laboratory diatom culture.

Aggregates from coastal environmental samples collected off of the Scripps Institution of Oceanography pier exhibited similar patterns of bacterial associations (Fig. 2, C and D; see sampling conditions in table S1). These aggregates were somewhat less viscous than laboratory samples, with a median and 95th percentile viscosity of 1.6 and 10.5, respectively ($N = 11$ aggregates). Bacteria are mostly associated with relative viscosities of ~1.25 (Fig. 2C). Relative viscosities >2 appear to represent a barrier to bacterial infiltration of these aggregates (Fig. 2D), which is lower than the barrier viscosity found in laboratory aggregates owing to the relatively lower aggregate viscosities in coastal samples in the denominator of the ratio. Although the barrier viscosity threshold differed between laboratory and environmental samples, both support our definition of a physical niche space for bacteria in marine snow: Lower-viscosity regions are accessible for colonization, whereas higher-viscosity regions may be impenetrable until the viscous material is broken down. The accessible surface and inaccessible barrier viscosity thresholds may vary depending on the source and chemical composition of the marine snow and the enzymatic capabilities of the colonizing bacteria in other parts of the ocean. Nonetheless, this niche space definition represents a useful framework for interpreting those differences and their significance for the biological carbon pump.

The viscosity degradation rate of individual aggregates

Marine bacteria, particularly those associated with marine snow, produce a suite of hydrolytic enzymes that break down high–molecular weight polymers, such as polysaccharides, proteins, lipids, and nucleic acids (*37*, *43*–*45*). Some polymers (e.g., laminarin) are easily hydrolyzed by bacteria, wheras others (e.g., fucose-containing sulphated polysaccharides) are more resistant to enzymatic degradation (*46*, *47*). The ultimate remineralization of aggregates into dissolved organic matter is attributed to a combination of bacterial enzyme activity and

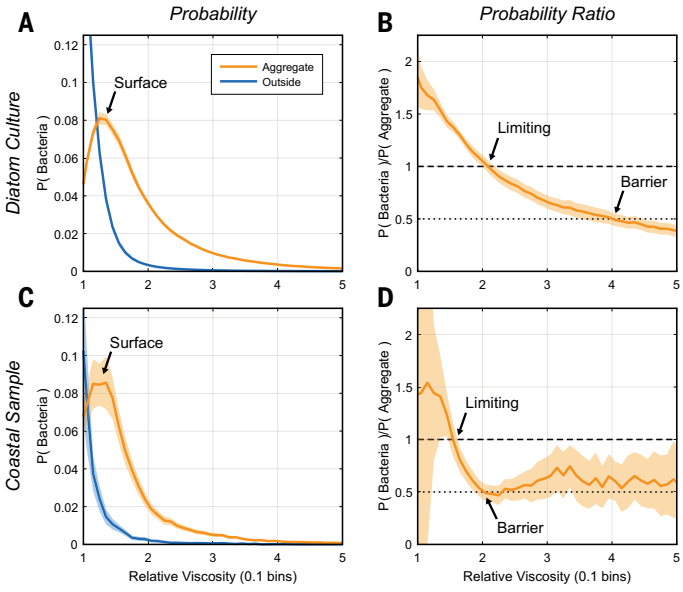

Fig. 2. Bacterial associations with mucus aggregate viscosities. (**A**) The probability (P) of a viscosity near a bacterial cell (orange, Fig. 1B) in 0.1 relative viscosity bins for aggregates from diatom cultures. Bacteria colonizing the aggregate (orange line) are most associated with a viscosity of 1.25, which represents a colonizable surface for bacterial activity. The most probable viscosity for free-living bacteria outside the aggregate (blue line) is 1, i.e., seawater. (**B**) The ratio of the probability of a viscosity occurring in the proximity of bacteria versus the rest of the aggregate volume. Bacteria are less associated with aggregate viscosities >2 (dashed line crossing) and half as likely to encounter viscosities >4 (dotted line crossing), indicative of a physical mucus barrier. More than 6450 bacteria were analyzed across $N = 30$ aggregates. (**C** and **D**) The same analyses as (A) and (B), respectively, but for aggregates sampled from the coastal environment. More than 1060 bacteria were analyzed across $N = 11$ environmental marine snow samples. Error bars represent the standard error.

physical dissolution, but this breakdown (and associated reductions in viscosity) has never been directly observed in individual aggregates (*48*, *49*). Unlike bulk assays of specific enzyme activities (e.g., glucosidase and protease), our molecular rotor method tracks the physical degradation of mucus by various enzymes through decreases in viscosity. We observed reductions in viscosity over time within single aggregates colonized by bacteria (Fig. 3A). Over the course of 7 hours, the probabilities of higher viscosities decreased, and the probabilities of lower viscosities increased (Fig. 3B). This is consistent with a conversion of higher to lower viscosities throughout the aggregate. The average change in the relative viscosity of laboratory-generated diatom aggregates was -0.017 ± 0.0033 hours^{-1} ($N = 7$ aggregates at three time points). At the same time, the aggregate volumes shrank at a rate of -0.024 ± 0.011 hours^{-1}, corresponding to 34% in one day (Fig. 3C). Based on simulated particle fluxes fit to ocean observations, the volume shrinkage rate of marine snow ranged from −0.0017 to −0.045 hours^{-1} (*50*–*53*). In these studies, the magnitude of the shrinkage rate increased primarily with the bacterial remineralization rate. Here, both the high degree of bacterial colonization (mean of 95 ± 16 per aggregate) and high shrinkage rate of aggregates suggest that the observed degradation was driven by enzymatic activity rather than physical dissolution.

Viscosity degradation occurs throughout the aggregate rather than just near surface-embedded bacteria, which suggests that exoenzyme activity is responsible for the observed breakdown of mucus structure. At an advanced stage of degradation, aggregates appear to break apart into low-viscosity fragments that are highly infiltrated by bacteria (fig. S11). Because bacteria are not found in higher-viscosity regions, viscosity degradation may allow bacteria to infiltrate aggregates at an average rate of -0.012 ± 0.0053 hours^{-1} (normalized by the maximum aggregate radial distance) and expose their membrane-bound enzymes to new, nondiffusing aggregate material (Fig. 3C). Subsequent breakdown of polymers near the bacterial membrane, transport of smaller substrates into the cells, and internal metabolism result in the ultimate decomposition of organic matter into inorganic constituents (*54*–*56*). Thus, microscale degradation of mucus viscosity—breakdown of the physical structure of mucus and subsequent exposure of new material to bacterial surfaces—is a rate-limiting step in oceanic cycling of particulate organic matter. Our measurement of the viscosity degradation rate provides a first glimpse into the microscale decomposition of aggregates that may vary with different strains of bacteria and different marine snow sources across the world's oceans. Quantifying the variability of this rate in other ocean regimes is an important next step for understanding particle breakdown and its influence on the ocean carbon pump.

Marine snow viscosity and the ocean carbon sink

Marine snow forms from a diversity of biological sources in the upper ocean, including zooplankton fecal pellets and carcasses, phytoplankton exudates and aggregates, viral cell lysis, discarded appendicularian mucus houses, ballast biominerals, and other detritus (*4*, *57*). These sources vary substantially in their molecular composition and density (*32*, *33*) and may display a high degree of viscosity variability across different sources and oceanic ecosystems. Our data shows that even mucus particles from the same source can exhibit appreciable differences in viscosity: the median relative viscosity per aggregate sampled from laboratory diatom cultures ranged from 1.6 to 7.9 ($N = 30$ aggregates). Because the viscous structure of mucus degrades over time according to the measured rate of −0.017 hours^{-1}, aggregates with higher median viscosities take longer to degrade than lower viscosity aggregates. At equivalent sinking speeds, higher-viscosity aggregates will sink deeper than lower-viscosity aggregates before they fully decompose and remineralize.

Particle sinking is also influenced by morphological properties, such as material density or drag, and a variety of external environmental factors, such as density stratification, turbulence, or zooplankton grazing (*10*, *18*, *36*, *58*–*63*). To explore the effect of viscosity degradation on marine snow sinking, we formulated an exponential decay model that depends primarily on the viscosity degradation rate, volume shrinkage rate, and degree of bacterial colonization (Fig. 4; see materials and methods). Converting particle viscosity to a density difference above seawater, diminishing density and size slowed the descent of the particle until it stopped sinking (Fig. 4A, white line, and eq. S2). A particle with an initial median viscosity of 1.25, representing the lower end of coastal environmental samples studied, would descend 530 m. A marine snow particle with an initial median viscosity of 2, typical of the relatively fresh diatom-derived aggregates from laboratory cultures, would reach 1015 m, roughly twice the distance. These sinking speeds and maximum depth values are consistent with mucilaginous marine snow attenuation observed in various ocean regions (*20*, *24*, *64*–*66*). Note that varying the initial particle size, drag coefficient, and reference density in eq. S2 changes the absolute values of the maximum depth but not the factor increase in sinking distance in the example above. This means that higher-viscosity particles always sink farther than lower-viscosity particles, all else being equal.

The depth at which sinking particles may contribute to carbon sequestration can vary depending on the physics and ecology of the water column (*67*). When defined as "the depth below which carbon is stored for 100 years or more," the sequestration depth can be located between 100 and 1000 m (*68*). Based on the functional shape of the maximum depth curve (Fig. 4, white line), small variations in

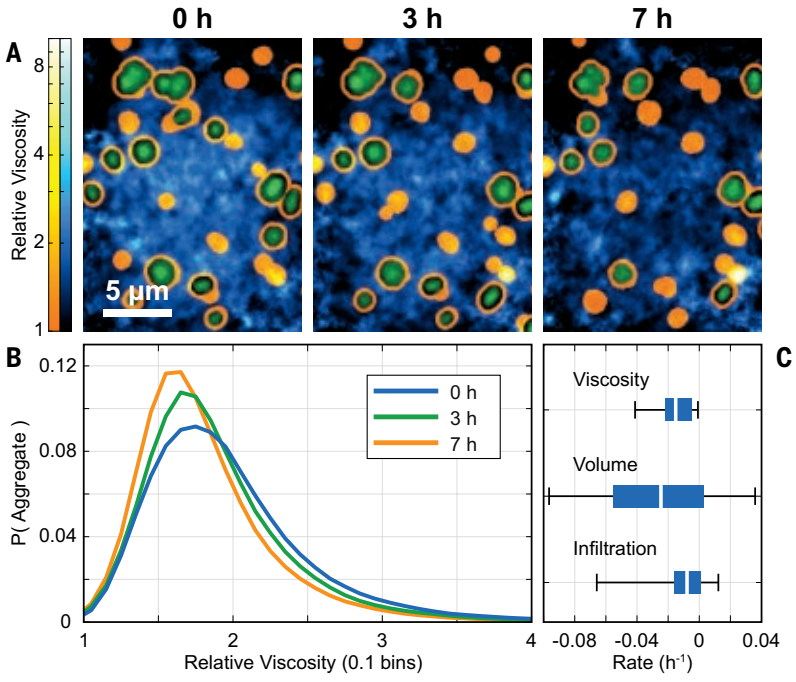

Fig. 3. Degradation and transformation of the viscous structure within a mucus aggregate. (**A**) A representative thin slice of the relative viscosity near the edge of an aggregate at 0, 3, and 7 hours. Internal aggregate viscosities (blue color scale) reduce over time, holes appear, and the aggregate ultimately fragments. Viscosities also decrease near the walls (orange color scale) of bacterial cells (green nucleic acid stain). (**B**) The probability distribution of viscosities in a single aggregate over time, corresponding to the time points in (A). The leftward shift of the probability distributions indicates that higher viscosities are converted to lower viscosities throughout the aggregate. (**C**) The viscosity degradation, volume shrinkage, and bacterial infiltration rates for $N = 7$ aggregates at three time points. Viscosity degradation may allow bacteria to infiltrate inside the aggregates, reducing their distance to the aggregate centroid over time. The white line in between the blue bars indicates that the median should be the same width in all three cases.

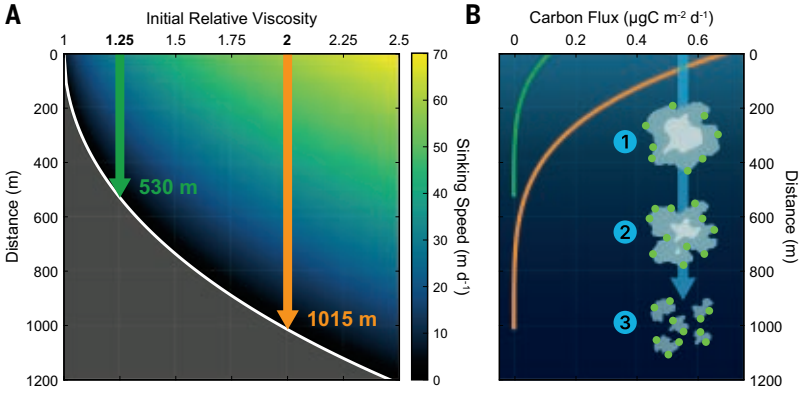

Fig. 4. Higher-viscosity aggregates take longer to degrade, sink deeper, and contribute more to the ocean carbon sink. (**A**) Modeled decay of relative viscosity and sinking speed with depth for marine snow given an initial median viscosity (top axis) and an equivalent spherical radius of 0.25 mm. The viscosity degradation rate reduces the density of the particle, and the volume shrinks, both slowing the descent until the particle stops sinking (white line). An aggregate with an initial viscosity of 2 (orange arrow) sinks roughly twice as far as an aggregate with an initial viscosity of 1.25 (green arrow). (**B**) (Left) The single-particle carbon flux attenuates with depth for aggregates with an initial viscosity of 1.25 (green line) and 2 (orange line). (Right) A conceptual illustration of the physical degradation of marine snow: (1) Bacteria (green) colonize the surface of an aggregate as it sinks in the upper ocean. (2) As the internal viscous structure degrades, bacteria gain access to inner regions. (3) The infiltrated aggregate shrinks and may eventually fragment.

particle viscosity at the lower end of initial viscosities lead to relatively large changes in sinking distance. Because higher-viscosity particles contain a greater concentration of polymers and thus more carbon (*69*, *70*), particles with an initial viscosity of 2 contribute more than a factor of six increase in carbon flux at any given depth compared with particles with an initial viscosity of 1.25 (Fig. 4B). This suggests that the ultimate fate of a particle (recycling in the upper ocean or sinking to depth) is strongly influenced by the viscosity degradation rate. As such, variability in the starting viscosity of particles and the degradation rate may mechanistically account for some of the observed variation in carbon export from the upper ocean described by the classic Martin curve (*71*). This analysis demonstrates that higher-viscosity aggregates take longer to degrade, tend to sink farther, and are more likely to contribute to carbon sequestration than low-viscosity aggregates.

Conclusions

Bacteria-mucus interactions are important drivers of carbon cycling and food web structure in most aquatic and terrestrial ecosystems (*72*–*74*). Using molecular rotors and confocal microscopy to reveal the viscous structure of undisturbed mucus aggregates, we show that viscosity correlates to a physical barrier, separating aggregates into regions that are accessible or impenetrable to bacteria. This represents the first quantitative description of mucus as a complex spatial habitat and physical niche space occupied by bacteria. Our methodology and conceptual framework highlight previously unrecognized controls of marine snow remineralization and present new opportunities for quantifying bacterial colonization and infiltration of mucus in marine snow and other biological systems. These include phytoplankton exudates (phycospheres), fish skin, kelp forests, coral reefs, land plant root systems (rhizospheres), and animal mucus membranes. This approach can also provide new insights into the physical regulation of bacterial colonization of the human gut and infection of nasal, throat, and lung mucosa.

Imaging of qualitative, visual features of marine snow morphology is considered an important determinant of sinking and carbon sequestration (*24*). Indeed, "hidden," transparent mucus trails associated with visible aggregates impact sinking rates, yet it is unclear how common this is in marine snow from different oceanic regimes (*36*). Our method provides a means to visually and physically quantify both the external and internal structure of transparent marine snow and associated bacteria, which have previously evaded simultaneous imaging. It is important to note that these relationships between bacteria and aggregate viscosities are correlative. Other factors, including binding of surface-expressed lectins, bacterial motility, and the elastic and porous properties of aggregates, may also affect bacterial colonization, and additional research is required to establish a causal relationship. Further, microviscosities inferred from molecular rotors reflect the local free volume and crowding of the local environment by nonsolvent molecules. Although molecular rotor measurements correlate with macroviscosities in our calibrations (fig. S1), they should not be interpreted as reporting resistance to flow at the macroscale (hydrodynamic viscosity) or elastic properties of the medium. We quantified the viscosity degradation rate of mucus aggregates and demonstrated that higher-viscosity

aggregates take longer to degrade, potentially sinking farther and contributing more to the ocean carbon sink than lower-viscosity aggregates. Viscosity degradation provides a mechanistic link between bacterial remineralization and physical breakdown of particles, contributing to our understanding of the attenuation of carbon export given by the Martin curve. Marine snow viscosity is thus a physical constraint—a "particulate recalcitrance"—of deep-sea carbon export in the world's oceans.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adx4170
Materials and Methods; Supplementary Text; Figs. S1 to S11; Tables S1 and S2; References (*77*–*94*); Movies S1 and S2

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NITROGEN CYCLE

Natural $^{15}N^{15}N$ abundances constrain fixed nitrogen loss

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Nitrogen regulates primary productivity across much of the biosphere, yet fixed nitrogen loss remains poorly constrained because existing methods rely on indirect proxies or ex situ experiments. In this study, we show that natural abundances of the rare $^{15}N^{15}N$ isotopologue of dinitrogen (N_2) provide a direct tracer of biological N_2 production across diverse aquatic environments. N_2 produced by denitrification and anammox has a near-stochastic $^{15}N^{15}N$ distribution [0 per mil (‰)], whereas atmospheric N_2 carries a distinct $^{15}N^{15}N$ excess (19‰), allowing the two sources to be quantitatively distinguished. Across aquifers, stratified lakes, coastal basins, oxygen minimum zones, and marine sediments, $^{15}N^{15}N$ measurements reveal widespread nitrogen loss previously obscured by physical gas accumulation and nitrogen fixation. Natural $^{15}N^{15}N$ abundances therefore provide a general framework for directly constraining fixed nitrogen loss across the aquatic nitrogen cycle.

Nitrogen (N) is an essential component of life, and the balance between its sources and sinks in aquatic environments regulates ecosystem productivity, water quality, and the long-term stability of the global fixed N inventory. Microbial conversion of fixed N to inert dinitrogen (N_2) through denitrification and anaerobic ammonium oxidation (anammox) returns biologically available nitrogen to the atmosphere, closing the global nitrogen cycle (*1*, *2*). Yet the magnitude of fixed N loss remains one of the greatest uncertainties in Earth's major biogeochemical cycles (*3*–*7*).

Despite decades of study, quantifying fixed N loss in natural environments remains difficult (*8*, *9*). Global budgets are typically inferred indirectly from stoichiometric and mass balance approaches, excess N_2 estimates based on N_2/Ar ratios, and extrapolated rates from incubation experiments (*10*–*13*). These methods either integrate signals over large spatial and temporal scales, requiring assumptions about background conditions and transport, or depend on ex situ rate measurements that represent only ephemeral conditions. Stoichiometric approaches such as N^* depend on assumed N:P relationships that vary in space and time and remain poorly constrained (*10*, *14*), whereas mass balance and inverse models infer fixed N loss indirectly from budget residuals or nutrient distributions and are therefore sensitive to flux constraints and model assumptions, introducing substantial uncertainty into globally integrated rates (*9*, *13*, *15*). N_2/Ar ratios are sensitive to air-water gas exchange, dissolution of entrained air bubbles, and temperature-dependent solubility; because Ar is approximately twice as soluble as N_2, these processes can shift N_2/Ar independently of biological N_2 production (*16*–*18*). Incubation experiments using ^{15}N-labeled tracer or inhibitors may perturb microbial activity, alter in situ rates, and provide only snapshots that may not be representative at broader spatiotemporal scales (*8*, *12*). Together, these limitations underscore the need for a direct tracer of biological N_2 production independent of physical gas exchange and experimental manipulation.

We use the natural abundance of the rare $^{15}N^{15}N$ isotopologue of N_2 as a direct and sensitive tracer of fixed N loss. The parameter Δ_{30}, which quantifies the deviation in $^{15}N^{15}N$ abundance from that expected for a random distribution of nitrogen isotopes among N_2 molecules, is defined as $\Delta_{30} = {}^{30}R/({}^{15}R)^2 - 1$ (‰), where ${}^{30}R = {}^{15}N^{15}N/{}^{14}N^{14}N$ and ${}^{15}R = {}^{15}N/{}^{14}N$ for the gas of interest. Atmospheric N_2 exhibits a well-defined Δ_{30} of 19.1 per mil (‰) ($\Delta_{30,atm}$), an enrichment interpreted to originate from upper-atmospheric nitrogen chemistry, where energetic solar radiation and charged particles dissociate and reform N_2 (*19*). Isotopologue-specific effects during these bond-breaking and bond-forming reactions are thought to enrich the rare $^{15}N^{15}N$ isotopologue, generating the high-Δ_{30} anomaly observed in atmospheric N_2. Conversely, N_2 produced by microbial denitrification and anammox ($\Delta_{30,bio}$) exhibits a near-stochastic isotopologue distribution, in which ^{14}N and ^{15}N atoms are paired randomly during N_2 formation (*19*, *20*). In the stochastic endmember, the relative abundance of $^{15}N^{15}N$ scales with the square of the bulk $^{15}N/^{14}N$ ratio, and Δ_{30} is therefore 0‰. Nitrogen fixation does not measurably alter Δ_{30} (*19*). Mixing between atmospheric and biogenic N_2 therefore produces a predictable decrease in Δ_{30} that directly records the fraction of biogenic N_2 (f_{bio}) in a sample (Eq. 1) (*20*, *21*).

$$\Delta_{30,sample} = (1 - f_{bio})\Delta_{30,atm} + f_{bio}\Delta_{30,bio} \quad (1)$$

Unlike tracers such as N_2/Ar, Δ_{30} reflects molecular origin rather than N_2 accumulation, making it insensitive to the direct effects of gas exchange, temperature change, or addition of air bubbles. Although air-water gas exchange can shift the measured Δ_{30} value through mixing with atmospheric N_2, it does not fractionate Δ_{30} and instead transports the existing biological signal. Thus, natural $^{15}N^{15}N$ abundances provide a source-specific tracer of biologically produced N_2. Below, we apply high-precision Δ_{30} measurements to a diverse range of aquatic environments—spanning nitrate-rich groundwater, stratified lakes, coastal hypoxic basins, open-ocean oxygen minimum zones, and marine sediments—to demonstrate that natural $^{15}N^{15}N$ abundances can constrain fixed N loss without relying on the corrections and assumptions required by conventional gas- and nutrient-based approaches.

Resolving groundwater denitrification independent of excess air

Groundwater denitrification is commonly evaluated by using dissolved N_2/Ar ratios (*22*, *23*), in which Ar serves as a proxy for the atmospheric component of dissolved N_2 inherited from air-water equilibrium. However, fluctuations in the water table can trap air bubbles that dissolve into groundwater, producing excess air and gas supersaturation that are ubiquitous in aquifers worldwide (*24*). Because N_2 and Ar differ in solubility, this process alters N_2/Ar ratios and complicates identification of biogenic N_2. In principle, excess air can be constrained using additional noble gases (e.g., Ne, Ar, Kr, Xe) through inverse modeling (*25*). In practice,

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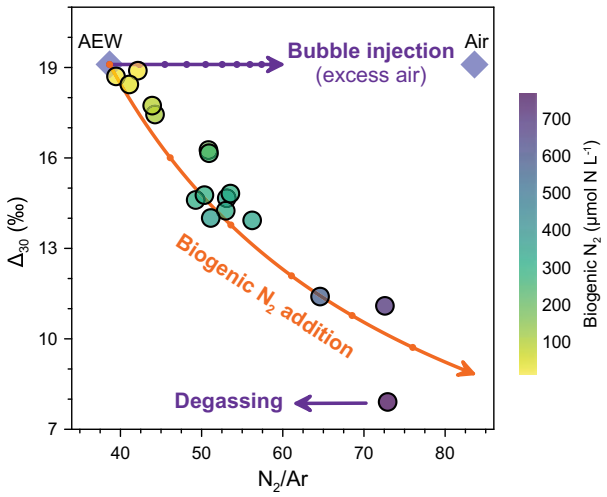

Fig. 1. Constraints on biogenic N_2 in groundwater using Δ_{30} and dissolved gas ratios. Relationship between Δ_{30} and N_2/Ar ratios for groundwater samples from an industrial facility in central Texas. Air-equilibrated water (AEW) defines the initial composition. Dissolution of trapped air bubbles (excess air) increases N_2/Ar with no change in Δ_{30} (purple), whereas addition of biogenic N_2 decreases Δ_{30} while increasing N_2/Ar (orange). Biogenic N_2 concentrations are calculated from Δ_{30} by mass balance between atmospheric and biologically produced N_2. Dots along the modeled trajectories indicate addition of N_2 in increments of 200 μmol N $liter^{-1}$ to the initial dissolved gas pool, supplied either as biogenic N_2 or through atmospheric-air addition. Error bars are smaller than symbols.

however, very few studies use the multiple noble-gas measurements required for modeling (*26*), and nearly all estimates of groundwater biogenic N_2 rely solely on N_2/Ar ratios. Consequently, interpretation of groundwater denitrification hinges on assumptions about the mechanism and magnitude of excess-air dissolution, leading to large systematic uncertainty.

Natural $^{15}N^{15}N$ abundances can resolve these uncertainties. We analyzed groundwater samples from monitoring wells in a nitrate-rich aquifer in central Texas (fig. S1) and found Δ_{30} values ranging from 7.9 to 18.9‰, corresponding to biogenic N_2 concentrations of 12 to 768 μmol N $liter^{-1}$ (Fig. 1). The lowest Δ_{30} value (7.9‰) indicates that 59% of dissolved N_2 was derived from microbial fixed N loss (Eq. 1), providing direct evidence of substantial biological N_2 production within the aquifer.

Biogenic N_2 concentrations derived from Δ_{30} and those inferred from N_2/Ar ratios agree to first order (Fig. 1). In the absence of gas-exchange processes that alter N_2/Ar, all data points in a Δ_{30} versus N_2/Ar plot should lie along the biogenic addition trajectory (orange curve). However, despite this overall agreement, several samples deviate substantially from this trend. Dissolution of trapped air bubbles increases N_2/Ar by introducing air with a higher N_2/Ar ratio than that of dissolved gases at solubility equilibrium (*24*), whereas degassing decreases N_2/Ar through preferential loss of N_2 relative to Ar (*27*). Because Δ_{30} tracks the molecular origin of N_2, scatter away from the orange curve reflects physical overprinting of N_2/Ar rather than ambiguity in the Δ_{30}-based estimate of biogenic N_2.

Natural $^{15}N^{15}N$ abundances therefore provide a source-specific tracer of groundwater fixed N loss without assumptions about excess air or gas equilibration. Replicate analyses yielded a Δ_{30} reproducibility of <0.1‰, equivalent to <0.5% of dissolved N_2 derived from microbial fixed N loss. This high sensitivity allows detection of relatively small biogenic N_2 signals that are difficult to resolve with conventional N_2/Ar-based approaches. This capability has direct implications for groundwater management and remediation monitoring, in which distinguishing biological nitrogen removal from physical gas processes remains a central challenge.

Constraining fixed nitrogen loss in stratified lakes

Stratified lakes provide a natural setting for testing Δ_{30} to quantify biological nitrogen loss under conditions where dissolved-gas inventories are strongly modified by physical processes. In permanently ice-covered Lake Untersee in Antarctica, dissolved gases are supersaturated because freezing at the base of the ice cover rejects gases into the underlying water, while the perennial ice cover suppresses gas exchange and prevents degassing (*28*). Under these conditions, N_2/Ar ratios exceed equilibrium values even at oxic, shallower depths where fixed N loss is absent (figs. S2 and S3).

Despite this gas enrichment, Δ_{30} decreases systematically from the atmospheric value of 19.1 to ~18.7‰ across the permanent oxycline in the southern basin of the lake (Fig. 2A). This decline indicates the accumulation of biologically produced N_2 in the anoxic region of the basin. By deriving the concentration of biogenic N_2 from Δ_{30}, converting its vertical gradient into a turbulent diffusive flux using a vertical eddy diffusivity (K_z) of ~0.004 cm^2 s^{-1} (see supplementary materials), and equating this flux to an areal rate of fixed N loss under a steady-state vertical balance between in situ N_2 production and upward diffusive transport, we estimate a fixed N loss rate of ~0.12 mmol m^{-2} day^{-1}. Although uncertainty in K_z remains difficult to quantify rigorously, the combined Δ_{30} and transport analysis provides a first-order basin-scale constraint on fixed N loss based on a biogenic N_2 gradient that conventional N_2/Ar measurements cannot readily resolve.

Measurements in temperate lakes in Minnesota further illustrate how Δ_{30} isolates gross biological N_2 production despite physical overprinting and concurrent nitrogen fixation. A recent study reported widespread N_2 supersaturation in Upper Midwest lakes on the basis of N_2/Ar measurements and attributed it to denitrification and/or anammox (*29*). However, supersaturation was also observed in oxygenated surface waters, where microbial fixed N loss should be suppressed (*29*). In such settings, elevated N_2/Ar ratios may arise from temperature-driven changes in gas solubility or air-bubble injection during wind-driven turbulence (*30*).

By contrast, Δ_{30} is not affected by these physical processes. Brownie Lake, a lake influenced by anthropogenic nutrient inputs, exhibits a pronounced decrease in Δ_{30} to ~18.4‰ at the oxycline, corresponding to substantial accumulation of biogenic N_2 (Fig. 2B). By contrast, Δ_{30} decreases only slightly to ~18.9‰ in Deming Lake, a forested lake less influenced by nutrient inputs, indicating lower biological N_2 production and fixed N loss (Fig. 2C). The stronger Δ_{30} shift in Brownie Lake reflects elevated anthropogenic nitrogen inputs that support primary productivity above the oxycline and enhance denitrification and/or anammox, with rates estimated at ~2.0 mmol m^{-2} day^{-1} from reactive transport modeling (fig. S4; see supplementary materials). By comparison, primary productivity in Deming Lake is likely nitrogen and phosphorus colimited (*31*) and concentrated near the oxycline, as indicated by chlorophyll a distributions (fig. S5). There, productivity is primarily sustained by ammonium from deeper waters, most of which is recycled within the system rather than converted to N_2, with modeled fixed N loss rates of ~0.2 mmol m^{-2} day^{-1}.

Notably, N_2/Ar ratios in Deming Lake fall below atmospheric equilibrium values, consistent with a net N_2 sink driven by active N_2 fixation (fig. S5) (*31*). This interpretation is supported by previous work showing that Deming Lake is dominated by cyanobacteria and that fixed N availability is low in the oxygenated upper water column, where active N_2 fixation has been demonstrated by $^{15}N_2$-labeling experiments in both the epilimnion and the subsurface oxygen-maximum layer (*31*). Because N_2 fixation does not measurably alter Δ_{30} (*19*), Δ_{30} records the accumulation of biogenic N_2 from gross fixed N loss, independent of concurrent fractional consumption of the N_2 pool by fixation, whereas N_2/Ar reflects the net balance between these opposing fluxes. Together, our measurements in these lakes demonstrate that Δ_{30} provides a robust tracer of biological N_2 production despite N_2 fixation and confounding physical gas processes that modify dissolved-gas inventories.

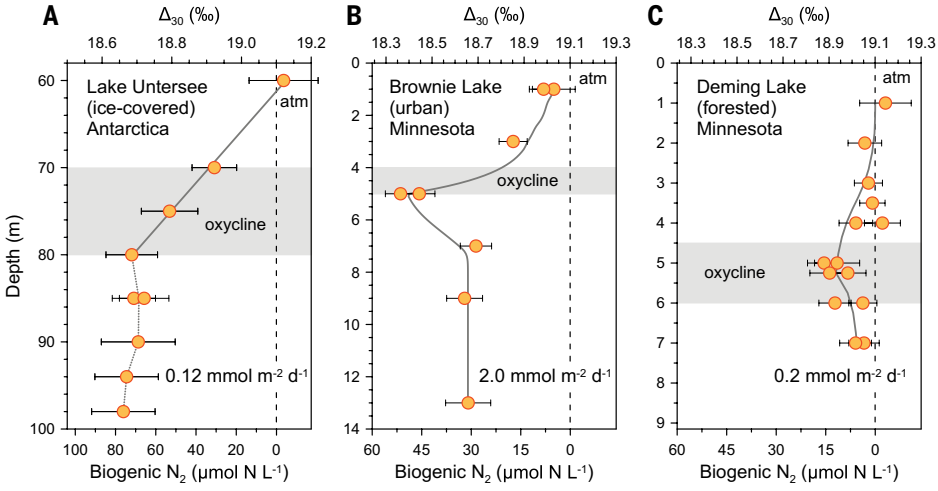

Fig. 2. Vertical distributions of Δ_{30} and biogenic N_2 in stratified lakes. Depth profiles of Δ_{30} and biogenic N_2 concentrations derived from Δ_{30} in three stratified lakes: (**A**) Lake Untersee, (**B**) Brownie Lake, and (**C**) Deming Lake. The dashed vertical line labeled "atm" indicates the atmospheric Δ_{30} value (19.1‰), and the shaded band marks the oxycline. Error bars indicate 1 standard error. The gray line represents a linear regression fit to the biogenic N_2 profile in Lake Untersee and the best-fit reactive transport model results for Brownie and Deming Lakes. Corresponding areal fixed N loss rates are shown in each panel.

The three lakes also show a gradient in how efficiently upward nutrient supply is converted to N_2. Upward ammonium fluxes are ~0.13, 4.8, and 5.5 mmol m^{-2} day^{-1} in Lake Untersee, Brownie Lake, and Deming Lake, respectively (figs. S3 to S5), whereas corresponding Δ_{30}-derived fixed N loss rates are ~0.12, 2.0, and 0.2 mmol m^{-2} day^{-1}. This comparison suggests that most of the upward ammonium supply is converted to N_2 in Lake Untersee, a substantial fraction in Brownie Lake, and only a small fraction in Deming Lake. The offset between ammonium supply and Δ_{30}-derived N_2 production therefore constrains how much regenerated nitrogen is retained within the water column, highlighting the utility of Δ_{30} for assessing nitrogen removal during eutrophication.

Reconciling nutrient and gas proxies of marine nitrogen loss

Using our Δ_{30} measurements in the hypoxic Santa Barbara Basin, we compared Δ_{30}-derived estimates of fixed N loss against N^* and excess N_2 on the basis of N_2/Ar ratios. During spring ventilation events, oxygenated water enters the basin and then becomes progressively depleted in oxygen through microbial respiration (*32*). As oxygen declines through summer and fall, Δ_{30} decreases and biogenic N_2 accumulates (Fig. 3).

In spring 2025, biogenic N_2 concentrations inferred from Δ_{30}, N_2/Ar, and N^* were similar to first order (Fig. 3A). By summer and fall 2025, however, N^* predicts larger fixed N deficits than Δ_{30}-derived biogenic N_2 at greater depths by roughly 20 to 30% (Fig. 3, B and C). This mismatch likely reflects deviations from the assumed Redfield N:P stoichiometry, benthic phosphate release during iron-oxyhydroxide reduction, and/or nitrate storage by sulfide-oxidizing bacteria, which alter nutrient ratios without directly producing N_2 (*33*, *34*). Such bias in N^* would imply corresponding overestimates of fixed N loss, with important implications for marine productivity and nutrient cycling. Interpretation of N_2/Ar is also complicated by corrections for background oceanic N_2 supersaturation, which introduce uncertainties that are difficult to quantify (*17*, *35*). By contrast, Δ_{30} directly records the fraction of biologically produced N_2 in the dissolved N_2 pool without requiring physical corrections.

Vertical gradients in Δ_{30}-derived biogenic N_2 were converted to upward diffusive fluxes by using vertical eddy diffusivity estimates (3.4 ± 1.0 cm^2 s^{-1}) (*36*), yielding fixed N loss rates of 4.9 to 6.1 mmol m^{-2} day^{-1} across individual sampling periods in the Santa Barbara Basin. Accounting for both diffusive export and temporal accumulation of biogenic N_2 within the basin produces integrated rates of 8.7 ± 3.0 mmol m^{-2} day^{-1} from April to August and 13.6 ± 2.8 mmol m^{-2} day^{-1} from August to November (see supplementary materials). These integrated Δ_{30}-based rates represent total N_2 production in the bottom-water layer and thus include both sedimentary and water column denitrification and anammox. Independent constraints from $\delta^{15}N$ of nitrate in bottom waters indicate that 11 to 49% of nitrate loss occurs in the water column (figs. S6 to S9), suggesting that although sedimentary denitrification dominates, water column denitrification is also significant in the Santa Barbara Basin, particularly under anoxic conditions (*37*). In the nearby San Pedro Basin, $\delta^{15}N$ of water column nitrate indicates that denitrification occurs almost exclusively in the sediments, yet the resulting biogenic N_2 accumulates in bottom waters and is captured by Δ_{30} (figs. S10 and S11), demonstrating that bottom-water Δ_{30} integrates sedimentary fixed N loss even where water column denitrification is minimal.

Overall, when interpreted within appropriate transport constraints, Δ_{30} quantifies the absolute magnitude of fixed N loss at the water mass–to-ecosystem scale by integrating N_2 production from both denitrification and anammox, whereas incubation experiments resolve short-term rates and pathway partitioning between denitrification and anammox (*38*), and $\delta^{15}N$ of nitrate constrains the relative contributions of water column versus sedimentary denitrification rather than total N_2 production (*37*). Thus, Δ_{30} offers a long-sought tracer capable of directly quantifying integrated fixed N loss in natural systems.

Oxygen minimum zones (OMZs) are major sites of oceanic water column nitrogen loss (*39*). However, fixed N loss can be difficult to quantify in weak OMZs such as the Bay of Bengal OMZ, where O_2 declines to submicromolar levels but rarely reaches complete anoxia (*40*). Earlier studies found no clear evidence for fixed N loss on the basis of stoichiometric indicators (*41*), whereas more recent work suggested modest loss in the Bay of Bengal (*40*). Uncertainty remains because at these low levels, nutrient-deficit and N_2/Ar-based approaches have limited sensitivity and depend on assumed stoichiometric and background conditions.

Across the core of the Bay of Bengal OMZ, Δ_{30} indicates relatively uniform biogenic N_2 concentrations of ~9.2 ± 2.8 μmol N $liter^{-1}$ (Fig. 3D and fig. S12), which are lower than values reported for major OMZs in the eastern Pacific and the Arabian Sea (~11 to 20 μmol N $liter^{-1}$) (*17*, *35*). Combined with water mass residence time, this inventory corresponds to a fixed N loss rate of ~0.4 ± 0.1 mmol m^{-2} day^{-1} (see supplementary materials). These values exceed previous Bay of Bengal estimates based on N^* (2 to 3 μmol N $liter^{-1}$) and N_2/Ar excess (4 to 6 μmol N $liter^{-1}$; Fig. 3D) (*40*), indicating that Δ_{30} resolves a larger biogenic N_2 inventory than conventional nutrient and dissolved-gas proxies. Because Δ_{30} constrains biogenic N_2 accumulation independent of assumed nutrient stoichiometry or background gas corrections, these results suggest that fixed N loss in the sampled Bay of Bengal OMZ waters may be more extensive than previously inferred, while recognizing that temporal variability may also contribute to the discrepancy with earlier N^* and N_2/Ar estimates. Although individual Δ_{30} measurements currently have relatively large analytical uncertainties owing to counting statistics, these can be substantially reduced through longer measurement times and replicate analyses. Future high-precision Δ_{30} transects across major OMZs, ideally

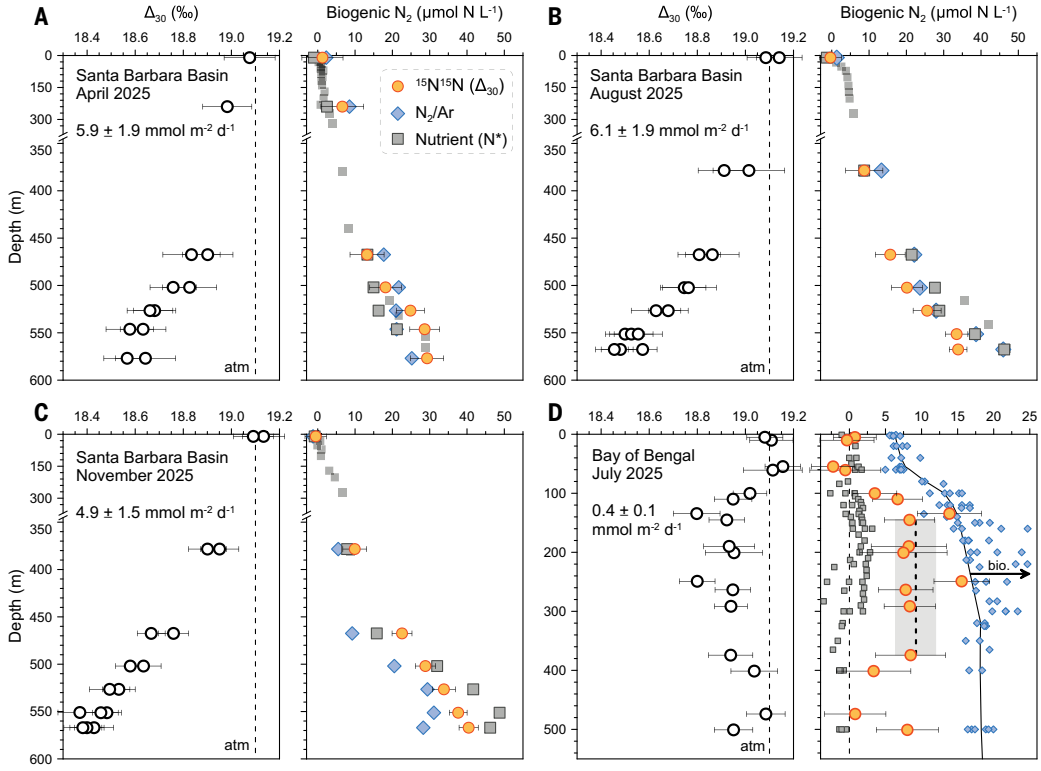

Fig. 3. Comparison of Δ_{30}-, gas-, and nutrient-based proxies of marine nitrogen loss. Depth profiles of Δ_{30} and biogenic N_2 concentrations in Santa Barbara Basin during (**A**) April 2025, (**B**) August 2025, and (**C**) November 2025, and in the Bay of Bengal during (**D**) July 2025. The dashed vertical line marks the atmospheric Δ_{30} value (19.1‰). Biogenic N_2 concentrations derived from Δ_{30} are compared with those inferred from N_2/Ar ratios and nutrient deficits (N^*). Error bars indicate 1 standard error. In (D), the thick dashed line indicates the mean Δ_{30}-derived biogenic N_2 concentration within the OMZ core, and the gray band shows ±1 SD. The areal fixed N loss rates shown in (A) to (C) are snapshot diffusive-flux estimates derived from the vertical gradients in Δ_{30}-derived biogenic N_2, whereas the rate shown in (D) is an inventory-based estimate derived from the biogenic N_2 inventory within the OMZ core and its estimated water mass turnover time. Bay of Bengal N_2/Ar and N^* data are from Bristow *et al*. (*40*). Despite correction for background N_2/Ar, a residual excess N_2 signal remains, indicating incomplete removal of physically enriched N_2 in (D). The solid black line represents a background profile estimated with locally weighted scatterplot smoothing (LOWESS) regression, and deviations from this curve are interpreted as biogenic N_2.

paired with synoptic nutrient and N_2/Ar measurements, could provide direct observational constraints on the magnitude of oceanic fixed N loss.

Identifying sedimentary hotspots of nitrogen loss

Marine sediments associated with methane seeps are localized hotspots of nitrogen loss (*42*). Sulfide-oxidizing bacteria in microbial mats can accumulate nitrate within intracellular vacuoles and couple sulfide oxidation to nitrate reduction, producing either N_2 or ammonium (*43*, *44*). In sediments from methane seeps along the Aleutian Trench, Δ_{30} decreases from 19.1‰ in bottom water to ~16‰ in the porewater of surface sediments (Fig. 4 and fig. S13), indicating substantial accumulation of biogenic N_2 in mat-covered sediments. Sediments without visible microbial mats show much smaller Δ_{30} shifts and lower biogenic N_2 concentrations, confirming that microbial mats drive enhanced fixed N loss at seep sites. These results demonstrate that natural $^{15}N^{15}N$ abundances can identify spatially localized hotspots of microbial N_2 production in sediments, providing a new tool for constraining heterogeneous sedimentary contributions to fixed N loss.

A universal tracer of nitrogen loss across aquatic ecosystems

Across aquifers, lakes, coastal basins, OMZs, and marine sediments, natural $^{15}N^{15}N$ abundances provide a direct tracer of biological N_2 production associated with fixed N loss. Unlike existing proxies, Δ_{30} distinguishes biological N_2 production from physical gas-exchange processes and opposing biological fluxes such as nitrogen fixation. By integrating Δ_{30}-derived biogenic N_2 concentrations with transport modeling, N_2 isotopologue measurements can be converted into areal fixed N loss rates that are directly comparable across environments. Together, these results establish Δ_{30} as a general framework for quantifying fixed N loss across aquatic ecosystems.

The ability to quantify biogenic N_2 without background correction or experimental perturbation is immediately relevant to groundwater remediation, lake eutrophication, and expanding ocean deoxygenation (*45*–*47*). Δ_{30} is particularly powerful in redox-sensitive boundary environments, where fixed N budgets remain poorly constrained (*15*, *29*, *39*, *48*–*50*). Application of Δ_{30} to isolated systems could also be informative. For example, measurements in subglacial Lake Vostok could test whether biogenic N_2 accumulates in a long-isolated, ice-sealed reservoir, with possible relevance to astrobiological exploration of other ice-covered ocean worlds. Because Δ_{30} directly records the molecular origin of N_2, it provides an independent constraint on fixed N loss, complementing and, in key respects, surpassing nutrient-, tracer-, and noble gas–based approaches used in Earth system models.

Improved fixed N loss estimates may have important global implications, as revisions to biogenic N_2 inventories translate into changes in total fixed N loss rates and inferred budget imbalances. As shown above, stoichiometric approaches based on nutrient deficits may either over- or underestimate fixed N loss, depending on how background nutrient stoichiometry is represented. More recent ocean circulation inverse models have improved on simple N^* methods by jointly interpreting nutrient distributions, transport, and variable stoichiometry.

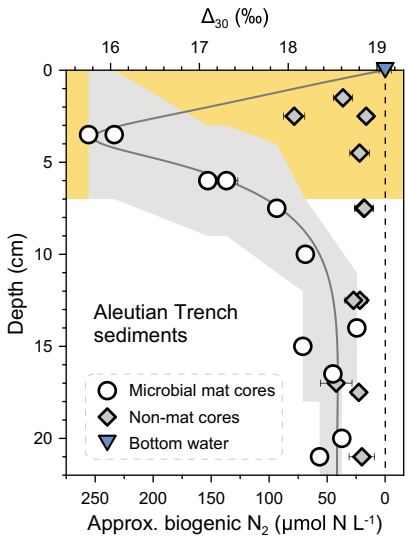

Fig. 4. Nitrogen loss in methane-seep sediments of the Aleutian Trench. Profiles show Δ_{30} with sediment depth. Error bars indicate 1 standard error. The dashed vertical line indicates the bottom-water Δ_{30} value. Yellow shading indicates the depth interval of microbial mat occurrence. Gray shading denotes the vertical range represented by each measurement, reflecting integration of the signal over several centimeters of sediment. The lower axis provides an approximate linearized conversion from Δ_{30} to biogenic N_2; the exact relationship is slightly nonlinear because total dissolved N_2 varies with biogenic N_2 accumulation. The gray curve shows the best-fit reactive transport model to the microbial mat-core Δ_{30} profile. The model is an illustration of how Δ_{30} could constrain rates of sedimentary fixed N loss, given greater depth resolution than that shown here (see supplementary materials).

For example, Wang *et al.* (*13*) inferred ~160 ± 50 Tg N $year^{-1}$ of microbial N_2 fixation, plus ~30 to 40 Tg N $year^{-1}$ of external fixed N inputs, balanced by a total marine fixed N loss of ~200 ± 60 Tg N $year^{-1}$. Yet these global rates remain weakly bounded, particularly for benthic denitrification. Independent constraints such as Δ_{30} could therefore help tighten estimates of total fixed N loss, reveal transient imbalances that are otherwise difficult to detect, and ultimately address one of the central questions in biogeochemistry: whether the global nitrogen cycle is balanced (*3*, *6*, *7*, *13*, *14*). Realizing this potential will require expanded Δ_{30} measurements across the major marine, freshwater, and terrestrial settings of fixed N loss to determine how local estimates of biological N_2 production can be integrated into regional and global nitrogen budgets.

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SUPPLEMENTARY MATERIALS

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High-capacity associative memory in a quantum-optical spin glass

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The Hopfield neural network stores memories using all-to-all-coupled spins and recalls those memories through equilibrium dynamics. Storing too many hampers recall because frustration causes an exponential number of spurious patterns to arise as the network becomes a spin glass. Despite this, we show that memory recall can be restored, and even enhanced, under quantum-optical nonequilibrium dynamics that convert spurious patterns into reliable memories. We experimentally observe associative memory with high storage capacity in a driven-dissipative spin glass made of atoms and photons. The capacity surpasses that of the Hopfield model under Hebbian learning by up to sevenfold in a 16-spin network. Atomic motion boosts capacity by dynamically modifying connectivity akin to short-term synaptic plasticity in neural networks, realizing a precursor to learning in a quantum-optical system.

Content-addressable associative memories can convert corrupted or incomplete data into clean, stored memory patterns through the process of pattern completion. Ordered states of matter can store such memories: For example, an Ising ferromagnet encodes one bit of information—namely, whether the spins are mostly up or down. This encoding is highly redundant, and therefore robust against errors, but stores only one pattern as a memory, the spins-all-aligned state. Simultaneously maximizing both memory storage capacity and robustness to recall error has wide-ranging implications for fields such as artificial intelligence (*1*) and neuroscience (*2*).

Associative memory in the Hopfield model

The Hopfield model is an example of a physically motivated associative memory that can store multiple patterns (*3*–*5*). It uses a recurrent neural network based on Ising spins that can (i) learn and store new memories through a biologically plausible mechanism [Hebbian learning (*6*)] and (ii) effect pattern completion via equilibrium dynamics such as Metropolis-Hastings (MH) dynamics (*7*). These dynamics consist of a stochastic update that corresponds to equilibrium energy exchange with a heat bath. The network consists of n binary variables (called "spins" or "neurons") $s_i = \pm 1$. The energy of a spin configuration is $E = -\sum_{ij} J_{ij} s_i s_j$ where the Hebbian coupling weights (or "synapses") are $J_{ij} = \sum_{p=1}^{P} \xi_i^p \xi_j^p$ and each n-dimensional binary vector ξ_i^p is a stored pattern, i.e., one of the intended stored memories.

By analogy with the ferromagnet, one might expect that memories are retrievable in the Hopfield model so long as the system is in a (generalized) ferromagnetic phase, where the P patterns ξ_i^p generalize the all up or down "pattern" in the conventional ferromagnet. Indeed, such ferromagnetic retrieval states exist for P less than the Hebbian learning bound $P_0 \approx 0.14n$ (*8*). In this regime, the patterns ξ_i^p form local minima of E, and pattern completion occurs via stochastic energy descent to those minima. However, when $P \geq P_0$, the retrieval states become thermodynamically unstable, and an equilibrium spin glass forms (*9*, *10*). This hampers recall through the formation of a rugged energy landscape with exponentially many local minima residing in deep, nested valleys separated by high energy barriers. Most minima do not coincide with any intended pattern ξ_i^p and are thus "spurious." Moreover, the spurious patterns cannot serve as reliable memories under MH dynamics: An error can lead to other patterns in nearby energy valleys. For this reason, spin glasses have long been considered incompatible with associative memory.

However, a recent theoretical work suggested that switching to a particular form of nonequilibrium dynamics would open access to the exponentially larger storage capacity of glassy systems (*11*). Driven-dissipative dynamics in a system of atoms and photons were theoretically found to induce energy-lowering spin flips at a rate proportional to the energy lost upon flipping. This realizes deterministic, "steepest descent" dynamics that reduces the multiplicity of possible relaxation pathways, thereby enlarging the basin size of any given spurious pattern. (By contrast, under MH dynamics, any spin flip that lowers the energy is equally likely to happen.) Decreased entropy generation due to steepest descent dynamics enhances recall fidelity, thereby turning what would have been spurious patterns under equilibrium recall dynamics into high-fidelity memories under steepest descent dynamics: What were vanishing basins under equilibrium dynamics become finite and possibly extensive in system size (*11*). The spurious pattern, residing at the minimum of a basin of attraction, can then be considered an effective memory of the network defined by the spin-connectivity weights J_{ij}. By doing so, one might be able to use an exponential number of spin glass minima for memory storage without requiring the engineering of multispin interactions (*12*), thereby greatly exceeding previously known storage limits (*13*). The attractors of the associative memory are not preprogrammed but are the natural minima of the glassy landscape under nonequilibrium dynamics. However, it is possible to retain the Hopfield network's ability to programmatically store patterns by implementing a map that associates each pattern with a minimum of the spin glass (*11*) via a learning rule (*14*).

Enhancing memory capacity

We experimentally demonstrate the enhanced memory capacity of a spin glass using an all-to-all Ising spin network created with multimode cavity quantum electrodynamics (QED) (*15*), as theoretically proposed (*11*, *16*–*18*). Driven-dissipative dynamics are natural to cavity QED-based quantum-optical systems. Not discussed, however, is a mechanism we report here by which the J_{ij} evolve as dynamical quantities that depend on spin position and contribute to capacity enhancement. Although many optics and nonlinear optics experiments have realized associative memory (*19*—*22*), none have observed memory enhancement caused by spin glass ordering. "Dense" associative memory networks can also possess enhanced capacity, though at the expense of engineering greater-than-two-body spin interactions (*12*, *23*). Photonic neural networks at the few-quanta level are under development (*24*).

We perform associative memory recall in artificial neural networks of size up to $n = 20$ and extensively characterize networks with $n \leq 16$. Bose-condensed gases of atoms play the role of the spins (neurons) while photons resonating in an optical cavity mediate the synaptic connections (J_{ij} weights). The memory capacity can exceed the Hebbian bound of the Hopfield model P_0 by up to sevenfold at $n = 16$, using a 50% recall probability threshold (*14*). This also exceeds the capacity limit equal to n attained with a pseudoinverse learning rule (*25*).

Origins of capacity enhancement

The high memory capacity is attributed to two experimental features that are absent in the Hopfield model. First, a cavity-cooling mechanism (*26*) intrinsic to the experiment (*11*) likely contributes to deterministic spin relaxation, rather than a solely stochastic energy descent. Second, the J_{ij} depend on the spin positions $\mathbf{r}_i$ that are not frozen in space, but can respond elastically to a "synaptic" stimulus. This renders

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J a dynamical quantity through a spin-motion coupling $J_{ij} = J(\mathbf{r}_i, \mathbf{r}_j)$. In response to optical forces generated by the spin-dependent cavity field, each atomic gas shifts its position to perturb its connectivity $J(\mathbf{r}_i, \mathbf{r}_j)$ with all others. The modified J_{ij} can then flip spins to drive the system deeper into a basin of attraction. The cavity field is modified with each spin flip, inducing more movement, resulting in a self-reinforcing interplay between spin and motion until the system evolves to a configuration deep within a basin of attraction. Consequently, the energy landscape elastically deforms to assist the spin evolution toward the memory. Simulations presented in (*14*) replicate this elastic response and indicate that this enhances memory capacity while inhibiting the effects of "*J*-chaos" (*27*, *28*). Thus, we realize a form of "polaronic spin glass," one whose connectivity dynamically changes to provide self-reinforced memory recall. This is akin to the creation of polarons in crystal lattices where the position of a charge leads to a deformation of the lattice, which in turn acts to trap the charge (*29*).

Our elastic enhancement of memory is reminiscent of "short-term synaptic plasticity" in neurobiology (*30*), where transient facilitation of synapses can enhance memory capacity (*31*). Altered synaptic connectivity deepens and reinforces the energy basin in which neural activity currently resides, analogous to the trampoline-like mechanism of enhanced recall in (*32*). Although the training of physical neural networks has been considered (*33*), synaptic self-reinforcement through natural dynamics in magnetic and photonic systems remains in early-stage development (*34*, *35*).

Experimental protocol

The experimental system has been discussed in (*15*, *36*); see (*14*) for parameters and procedures specific to this work. Figure 1A sketches the apparatus, which has been augmented with bias fields for each spin. Briefly, each spin is represented by the motional excitation of a rubidium Bose-Einstein condensate (BEC) trapped by an optical tweezer. These motional states are one of two possible checkerboard density waves representing collective spin up or down states. This yields an effective Ising spin degree of freedom $\hat{S}_i^z$ (*15*). The density wave states are created by the interference of a transverse pump laser and the field of the multimode cavity within which the BECs are confined (*37*–*39*).

BECs in spin up (down) density-wave states scatter the pump light into the cavity with phase 0 (π) with respect to this transverse pump. Recording the phase and amplitude of the cavity emission from each of these BECs allows us to measure all $\langle \hat{S}_i^x \rangle$ of a microscopic spin configuration (*15*, *40*, *41*); e.g., see Fig. 1D. Longitudinal pump beams provide local fields f_i that bias these effective spins toward a stimulus pattern $\langle \hat{S}_i^x \rangle \propto f_i$; see Fig. 1B (*42*).

The multimode cavity light—composed of thousands of near-degenerate modes (*38*, *41*)—also induces all-to-all, sign-changing interactions among the effective spins (*16*, *38*, *40*). The interaction yields a frustrated J_{ij} connectivity matrix sufficient to create a spin glass (*11*, *14*, *15*, *34*).

Ising spin glass model

The multimode cavity-mediated interactions yield a spin-glass Hamiltonian of Ising form when the cavity is tuned to the so-called "4/7" configuration of a Fabry-Pérot resonator (*15*). The 4/7 cavity geometry is realized when the ratio of mirror spacing L to radius of curvature R is ~1.22; here, $R = 1$ cm. The spin-glass Hamiltonian arises in the dispersive pumping limit, wherein the pumps are red-detuned $\Delta_C \approx -2\pi \times 20$ MHz from the cavity modes. The multimode Dicke model description of the system then simplifies to a frustrated, transverse-field Ising model with longitudinal fields (*14*):

$$\hat{H} = \omega_z \sum_{i=1}^{n} \hat{S}_i^z - g \sum_{ij=1}^{n} J(\mathbf{r}_i, \mathbf{r}_j) \hat{S}_i^x \hat{S}_j^x - \sum_{i=1}^{n} f_i \hat{S}_i^x \quad (1)$$

Each spin is represented by a collective spin operator $\hat{S}_i^{x/y/z}$ of size $S = M/2$, where $M \approx 4 \times 10^4$ is the average number of atoms per ensemble. The form of $J(\mathbf{r}_i, \mathbf{r}_j)$ is in (*14*). We approximate the experimental dynamics by unitary evolution through Eq. 1 combined with dissipation using a Lindbladian treatment (*11*, *14*). The term proportional to the atomic recoil frequency $\omega_z \approx 2\pi \times 7.5$ kHz plays the role of a transverse field. The strength of the interaction term is $g = -g_0^2 \Omega^2 / (\Delta_A^2 \Delta_C)$, where the Rabi rate squared Ω^2 is proportional to the transverse pump power, $g_0 = 2\pi \times 1.35$ MHz is the Rb atom-cavity coupling strength, the pump-atomic transition detuning is $\Delta_A \approx -2\pi \times 97.2$ GHz, and the cavity linewidth is $\kappa = 2\pi \times 140$ kHz. The system is in the single-atom, large cooperativity limit of cavity QED, even without multimode field enhancement (*14*, *41*).

Method for associative memory recall

The transverse pump power is exponentially increased until the spin interactions are as large as $2\pi \times 2$ kHz at 8 ms. Simultaneously, a memory stimulus is input into the network by ramping up the longitudinal fields over 3 ms. This induces $|f_i|$ whose strengths are approximately ω_z. Figure 1B shows an example stimulus pattern for the connectivity matrix we call $\mathbf{J}_1$; see (*14*) for all J_{ij} elements. The transverse and longitudinal pumps drive spin evolution from the stimulus configuration down toward lower-energy configurations within a basin of attraction (Fig. 1C). The biasing fields are held constant for another ~3 ms to allow spin organization to continue at fixed $|f_i|$. We then ramp f_i down and turn it off 1 ms before imaging to allow unbiased spin evolution toward a memory configuration. An example of successful pattern recall is shown in Fig. 1D for what we designate as memory 1 of $\mathbf{J}_1$. Recall fidelity is qualitatively insensitive to small changes in this procedure (*14*). To operate the neural network again, fresh BECs are created and positioned to approximately realize the same **J** (*14*, *15*).

Different coupling matrices **J** realize different neural networks, each with their own memories. We study five disorder realizations of ***J*** for each system size $n = 4$, 8, 12, and 16. To obtain different **J**'s at fixed n,

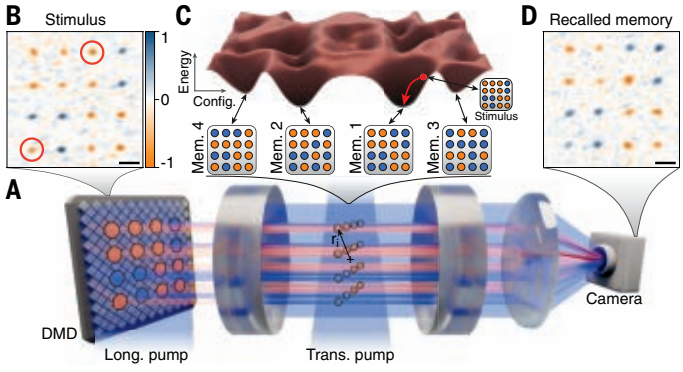

Fig. 1. Sketch of the apparatus and memory recall process for an $n = 16$ network. (**A**) Transverse and longitudinal fields (light blue) pump the cavity. A digital micromirror device (DMD) scatters the longitudinal pump into 16 beams of phase 0 (π) colored blue (orange). Each imposes a longitudinal field f_i onto a single atomic ensemble at position $\mathbf{r}_i$ within the cavity midplane, thereby inputting into the network a stimulus field. (Tweezer traps not shown.) The intracavity field contains both local (blue and orange) and nonlocal (light blue) components. Atomic ensembles are represented as spheres. They are colored blue (orange) to denote their effective spin up (down) states, which scatter into local fields with 0 (π) phase. A camera holographically images the emitted cavity field (*15*,*40*). (**B**) Example stimulus field pattern. Spin-flip errors compared to memory 1 are circled in red. Image taken by recording the transmission of the longitudinal pump beams through an empty cavity. Images are processed to show only local fields and normalized to their maximum value. Remaining background is residual noise; see (*15*) for image analysis. Black bars are of length $w_0 = 34.8$ μm. (**C**) Memory recall dynamics in a low-dimensional representation of a spin-glassy energy landscape versus spin configuration. Red arrow indicates gradient descent from the corrupted stimulus pattern (red dot) in (B) to the memory pattern 1 at the bottom of its basin of attraction. Three additional representative memory patterns are indicated at their respective basin minima. (**D**) Cavity output image showing the local fields from the atomic ensembles after successful recall of memory 1, starting from the stimulus field in (B).

we simply move the ensembles within the cavity midplane. This causes J_{ij} to change sign, resulting in distinct sets of memory patterns (*11*, *15*, *36*). In practice, it suffices to trap the ensembles in a rectilinear array of spacing roughly 50 μm; few-micrometer-scale position adjustments yield different **J**'s (*14*). We report memories recalled from a single **J** at $n = 20$ in fig. S10. The rapid proliferation of memories renders capacity measurements prohibitively time consuming for $n > 16$. The system can still be used as an associative memory at larger n because pattern storage and recall do not require exhaustively identifying every possible memory pattern (*14*).

Memory capacity measurements

Memory capacities are determined by cataloging all memories with finite basins of attraction that are naturally stored by each **J**. That is, the **J** are not trained to realize specific patterns in this work. Rather, we sample the patterns found by performing up to 400 experimental recall cycles, each with the same **J** but a randomly selected stimulus pattern $\{f_i\}$. A hierarchical clustering algorithm (*44*) organizes the observed patterns into groups of similar patterns to allow for small fluctuations in $\left\langle \hat{S}_i^x \right\rangle$ between experimental shots (*14*). Each group serves as a "candidate memory."

We define the basin size of a candidate memory as the average number of randomized spin flip errors that may be tolerated in the stimulus while exceeding a 50% recall probability threshold. We find an average of 51 candidates for the five **J**'s studied at $n = 16$. However, most candidates exhibit basin sizes smaller than one spin flip and are therefore not useful as associative memories. This is expected: Reference (*11*) predicted a typical basin size of the spin glass to be $\sim 0.013n$, which is less than one for $n = 16$. Thus, for small n, it is reasonable to restrict "memories" to mean only those candidates with basin sizes greater than or equal to one spin flip; this defines the memory capacity. We estimate that we find > 95% of all such memories for $n \leq 16$ (*14*).

Figure 2 shows recall fidelity curves for representative memories of $\mathbf{J}_1$. Each point is the average of 30 trials (*14*). Memory 1 in Fig. 2A exhibits a particularly large basin size of nearly four spin flips. Figure 2, B to D, presents typical recall curves for three additional memories of $\mathbf{J}_1$, with basin sizes between one and three spin flips. We find that $\mathbf{J}_1$ has a memory capacity of 14(1) memories (*14*). See fig. S8 for a gallery of the 10 other memories.

Scaling of memory capacity

We plot the capacity versus n in Fig. 3A. The average capacity 11.9(6) of the $n = 16$ networks is much greater than the 3.6 average number of memories storable by the Hopfield model under Hebbian learning (*14*). As expected, however, the average basin size of the memories of $\mathbf{J}_1$ is 2.1, which is lower than that of the Hopfield model, 3.9. This highlights the trade-off between a spin glass's larger capacity and its smaller average basin size (*11*); ultimately, application requirements may prefer one over the other.

Plotting the number of effective memories in the Sherrington-Kirkpatrick (SK) model (*9*, *45*) allows us to compare the capacities to a model known to possess an exponential number of minima in n (*46*). [The SK model describes an all-to-all spin glass similar to what we realize (*11*, *15*, *36*).] The mean experimental capacity is consistent with the SK model under steepest descent (SD) dynamics (up to $n = 16$). Moreover, the spreads in capacity versus disorder realization **J** are consistent. By contrast, the SK model simulated under MH dynamics falls below the data. We cannot yet directly prove that purely steepest descent dynamics are at play, but recording the spins' temporal evolution would provide direct evidence. The match with SD dynamics may provide only a lower bound to the possible enhancement to memory capacity for two reasons: (i) simulations in section IV.C (*14*) show that eliminating digital micromirror device (DMD) phase noise on the stimulus field improves capacity by two to three times, and (ii) the maximum possible enhancement provided by spin-motion coupling is not yet known.

Polaronic enhancement of capacity

Experimental observations additionally reveal an aspect of the dynamics not considered in (*11*): The atomic ensembles shift in position depending on which memory pattern is recalled. This polaronic effect occurs because the tweezer potentials are not infinitely stiff, and thus optical forces from the emergent cavity field shift the atomic ensembles to positions away from their trap minima. Figure 4A shows a typical example of this effect: After memory recall, several ensembles are observed at locations up to 4 μm away from the trap minima, which is 20% of the tweezer waist. Moreover, the data indicate a spin-configuration dependence to

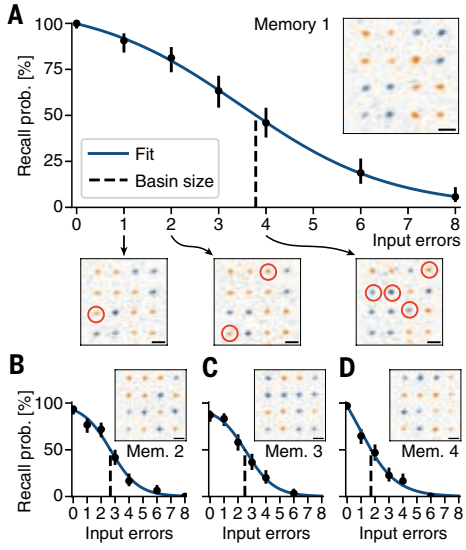

Fig. 2. Memory recall fidelity. (**A**) Recall probability versus number of spin-flip errors in memory pattern 1 of connectivity realization $\mathbf{J}_1$, shown as the inset (same image as in Fig. 1D). Below are examples of three defective stimulus images including one, two, and four spin-flip errors, circled in red. The dashed line shows the basin size, which is chosen where the fitted curve (blue line) intersects the 50% probability threshold; a hyperbolic tangent is empirically found to be a suitable fitting function (*14*). (**B** to **D**) Recall curves for three additional memories of $\mathbf{J}_1$. Insets are emission images showing the memory patterns. Black scale bars are of length w_0.

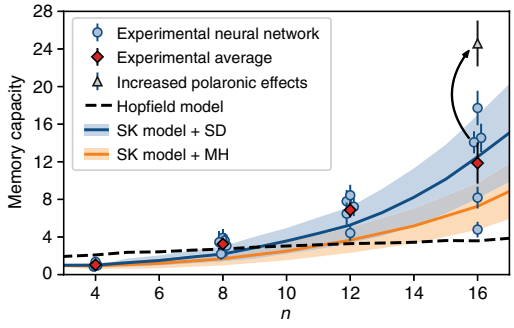

Fig. 3. Memory capacity versus n. Memory capacity of five different **J**'s for each of four different system sizes n. Their average is shown as red diamonds. Error bars are standard error and stem from basin size estimation (*14*). Arrow indicates the polaronic enhancement of memory capacity for the $\mathbf{J}_1$ network, where capacity is measured at both the nominal trap elasticity condition (blue circles) and at a higher elasticity (gray triangle). The solid curves show the simulated memory capacity in the SK spin glass under steepest descent (SD, blue) or Metropolis-Hastings (MH, orange) dynamics. The dashed black curve is for the Hopfield model under Hebbian learning. The SK (Hopfield) simulations include 1000 (10,000) independent disorder realizations of **J** per n. The light blue (orange) band shows the standard deviation in SD (MH) capacities due to random variations in the disordered energy landscape versus **J**. This band is ~1.3 memories for the Hopfield model (*14*). See fig. S11 for a study of basin volumes.

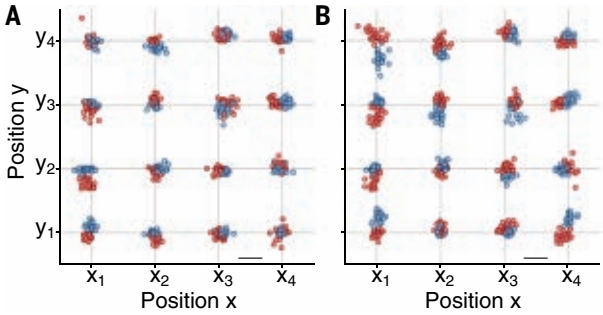

Fig. 4. Polaronic response of atomic ensemble positions. (**A**) Each dot is the center-of-mass (CoM) position of an ensemble within the 4×4 array of the $n = 16$ network. Grid lines at x_i and y_j indicate the tweezer trap locations. The scale bar is 5 μm. Positions of the ensembles at the time of imaging are shown in blue (red) after successful recall of memory pattern 1 (2) of $\mathbf{J}_1$. These are determined by fits to recall images like that shown in Fig. 1D. The CoM data from a total of 24 experimental shots are displayed per memory pattern. The average deviation of the CoM data from the trap locations is 1.32(2) μm. (**B**) Same as (A), but with trap power reduced fourfold to enhance polaronic elasticity, corresponding to the triangle data point in Fig. 3. The average deviation increases to 1.82(3) μm. The spacings between adjacent x_i (and y_j) are decreased by ~60% to visually magnify the spread in positions.

the movement, because certain ensembles move differently when recalling one memory pattern versus another. Reducing the transverse pump power down toward zero reverts the ensembles back to the tweezer trap centers, rendering this phenomenon a form of elastic rather than plastic response. This acts to change the positions, and therefore the connectivity. Simulations in section IV.C (*14*) indicate that the motional shift deepens the energy well. This can lead to the generation of new memory patterns that are self-consistently stable under both spin and positional dynamics, increasing memory capacity.

We experimentally verify that the $\mathbf{J}_1$ memory capacity increases when the polaronic deformations are made stronger. The laser power of each tweezer is lowered by a factor of 4 to yield a weaker, and therefore more elastic, trap. Figure 3A shows that this yields an almost twofold increase in memory capacity for $\mathbf{J}_1$, resulting in 25(2) memories in a $n = 16$ network, seven times the Hopfield capacity under Hebbian learning. We also measure an increased elastic response: The positions shown in Fig. 4B are on average ~38% farther away from the initial tweezer locations.

Discussion and outlook

Future work could engineer a long-term form of plasticity that persists even without the pump field. This could be achieved, e.g., by shaping optical potentials to have additional minima in learned locations. Plastic self-reinforcement of memories in this quantum-optical system would realize a natural form of learning while enjoying capacity enhancement and robustness to coupling errors. Photonic network denoisers and amplifiers are two possible applications once intracavity networks are increased in size using larger tweezer arrays (*47*). Capacity-enhanced quantum associative memory might be possible using intracavity ensembles acting as effective spin-1/2 degrees of freedom (*48*) or color centers in crystals.

This work experimentally reveals a distinct paradigm of associative memory with enhanced memory capacity that goes far beyond the Hopfield limit under Hebbian learning. It does so by exploiting naturally occurring energy minima in a glassy energy landscape that the original Hopfield model must avoid. Such glassy minima yield robust memories because they are stabilized by a form of polaronic elasticity as well as by steepest, rather than stochastic, energy descent dynamics.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aec3917
Supplementary Text; Figs. S1 to S13; Table S1; References (*50*–*56*)

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POPULATION GENETICS

Message in a bottleneck: Nested founder effects from French Polynesia to Rapa Nui and Hawai'i

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Founder effects can lead to strong, population-specific allele frequency differentiation, including autosomal recessive conditions. Their characterization in European-descent groups, such as the Finnish and Amish, has resulted in actionable genetic tests for each. We show that Polynesians experienced order-of-magnitude stronger cumulative founder effects during their settlement of Eastern Polynesia. Using whole-genome sequences spanning Polynesia, we show that two of the most bottlenecked populations are Hawai'i and Easter Island (Rapa Nui) and that these two most closely genetically related yet geographically distant islands were likely settled by the same ultravoyagers with earlier ancestral roots in Mangareva. We identify clinical genetic variation generated by this sequence of successive oceanic founder events, including a loss-of-function *FAN1* mutation common in Eastern Polynesia (>1%) but entirely absent from gnomAD.

The founder effect describes a genetic bottleneck that occurs when a population is established by a small number of individuals, leading to an overall reduction in genetic variation and an increase in frequency in retained rare variants. The clinical importance of founder effects has been extensively documented in populations such as the Finns, whose genetic history has resulted in the well-characterized Finnish disease heritage (*1*). Similarly, the Ashkenazi Jewish population experienced a strong founder effect and thus exhibits higher frequencies of particular pathogenic variants, including BRCA1/2 and certain recessive disorders (*2*, *3*). In Polynesia, more intense successive founder events compounded during the island-to-island settlement process (*4*), increasing the frequency of retained rare variants along the path of settlement. In this work, we focus on a settlement sequence that began in Samoa and ended one-fifth of the planet away in a common final founder population out of the Mangareva Islands (hereafter referred to as Mangareva). We show that this ultimate voyager group settled both Hawai'i, the only Polynesian population in the Northern Hemisphere, and Easter Island (Rapa Nui), 7000 km to the south in the cool southern Pacific, at a longitude directly below Salt Lake City, Utah. This makes Native Hawaiians at once one of the largest and one of the most intense founder populations in both Polynesia and the world.

Despite Native Hawaiians and other Pacific Islanders (NHPIs) being recognized as founder populations, there remain few characterizations of their clinically actionable genetic variants. The underrepresentation in genomic databases of NHPIs is concerning, given that the American College of Medical Genetics and Genomics (ACMG) calls for the use of such databases, for instance the Genome Aggregation Database (gnomAD), to inform founder group screening panels. For inclusion on these population-specific screening panels, the ACMG recommends a carrier frequency of >1/200 (*5*, *6*). The ACMG considers this carrier screening to be medically necessary in many founder populations, and additions in gnomAD v3.1.1 incorporating groups such as the Amish, Finnish, and Middle Easterners have identified 71 additional genes exceeding this carrier frequency threshold (*7*). Many major insurers adhere to these ACMG guidelines; for instance, Aetna's medical policy includes a genetic screening panel for 14 conditions for Ashkenazi Jewish individuals (*8*). Similarly, the Amish are well-studied for their population-specific genetic conditions, revealing previously unknown monogenic diseases (*9*). Although the Native Hawaiian population in the US is nearly twice the size of the Amish population, and the broader NHPI population is much larger still, the frequencies and variants for specific monogenic diseases enriched in these Pacific founder populations are largely unknown and thus cannot be represented on gene panels nor be reimbursed by insurance. We characterize some of these frequencies in Eastern Polynesians, who sit at the terminus of the Polynesian migration and therefore experienced the strongest founder effects in Polynesia—more pronounced, we show, than those of the well-studied founder groups mentioned above.

We describe the genetic founder effects across Polynesia using whole-genome sequencing data from 1050 individuals across French Polynesia, including the Austral Islands, the Society Islands, the Marquesas Islands, the Tuamotu Archipelago, and Mangareva, nearly all published in previous studies (*10*, *11*). We include 12 additional, previously unpublished genomes from remote Rapa Nui (Easter Island) and complete coverage of far Eastern Polynesia by integrating genomic data from Native Hawaiians in All of Us (*12*); finally, we also include Samoans and Tongans from the same biobank to provide a Western Polynesian reference (table S1). In investigating these strong genetic bottlenecks, we find that the most remote islands in Polynesia, Hawai'i, and Rapa Nui, each more than 3000 km from the nearest inhabited Polynesian island and ~7000 km from one another, are the most genetically related, suggesting the development of an ultralong-distance voyaging culture, which we refer to as *Holomoana Nui* (great ocean voyaging), that settled them both. Finally, we report frequencies of previously identified pathogenic founder mutations in the South Pacific, showing them to have rates that would qualify affected Polynesian islands for carrier testing according to ACMG guidelines.

Population structure of Polynesia

To isolate Polynesian ancestry segments in these genomes, we performed local ancestry inference with Gnomix, using a reference panel of European-, African-, East Asian–, Philippines-, and Polynesian-like individuals (*13*). Determining Polynesian-specific genome segments allowed us to condition our analyses on Polynesian ancestry, focusing on the demography of this ancestry, which has a different history from the others. We use snputils (*14*) to plot a downsampled ancestry-specific principal components analysis (PCA) (Fig. 1A), revealing the population structure of Polynesia (see materials and methods). We note that there are many single-nucleotide polymorphisms (SNPs) common (>5%) in these Polynesian sequences that are entirely absent from gnomAD (Fig. 1B) (*6*).

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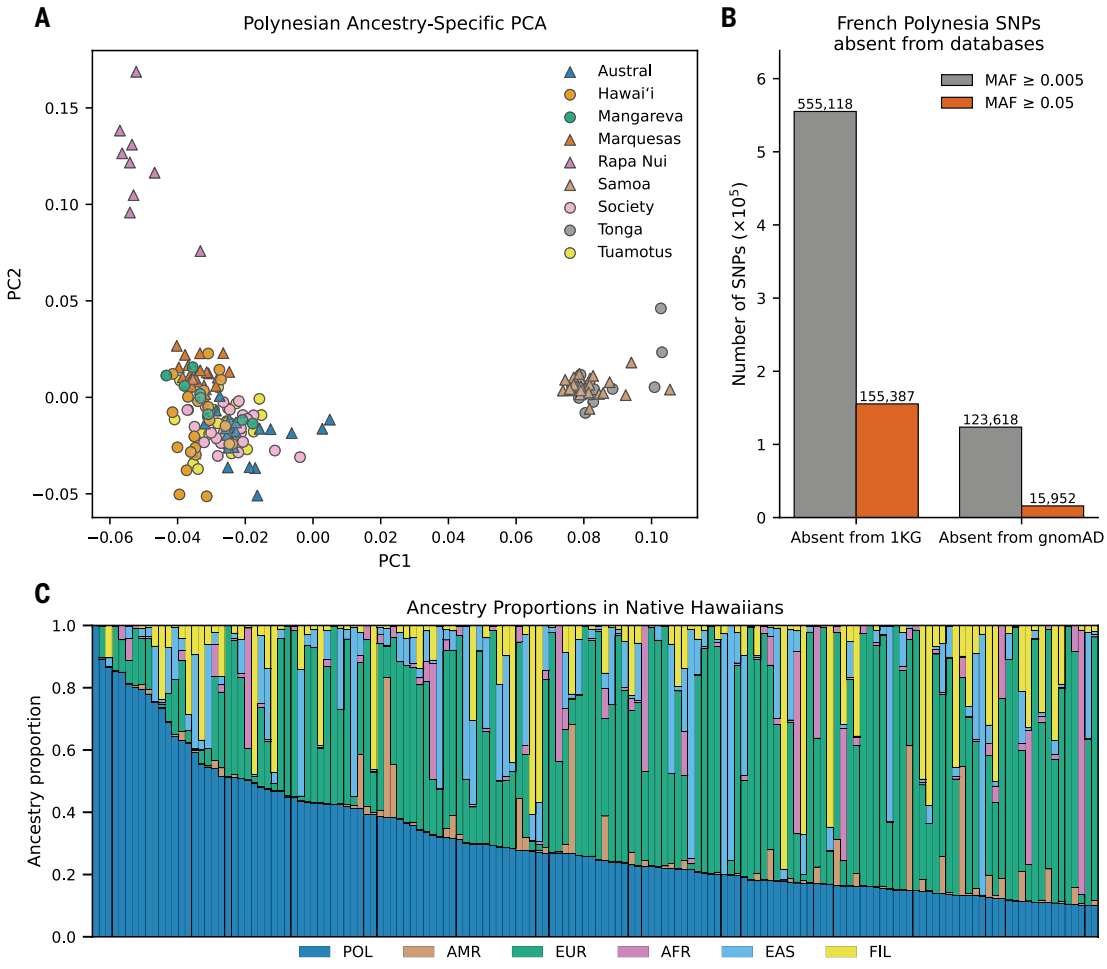

Fig. 1. Polynesian population structure. (**A**) Polynesian ancestry–specific PCA of islanders and related populations. Polynesian genome segments were inferred using local ancestry inference. Before running PCA, the dataset was downsampled to a maximum of 25 individuals per population to prevent visual overcrowding. Population point shapes are chosen only to maximize visual clarity. (**B**) Count of SNPs identified in French Polynesia that are entirely absent from existing sequencing databases. Variants are stratified by minor allele frequency (MAF). (**C**) Ancestry proportion plot of Native Hawaiians based on local ancestry results. Each vertical bar represents one individual with color height proportional to the corresponding genetic ancestry fraction: Polynesian-like (POL) in dark blue, Indigenous American–like (AMR) in green, European-like (EUR) in green, African-like (AFR) in violet, East Asian–like (EAS) in light blue, and Philippines-like (FIL) in yellow. Only individuals with greater than 10% inferred Polynesian-like ancestry are included.

We find high amounts of inferred non-Polynesian ancestry in Native Hawaiians, highlighting the history of recent admixture on the archipelago (Fig. 1C). There was a substantial amount of inferred Philippines-like ancestry in the unsupervised genetic clustering analysis (figs. S1 and S2), which motivated the use of local ancestry to identify Philippines-like haplotypes (fig. S3). The inferred Philippines-like ancestry proportion in Native Hawaiians, shown in fig. S3, has a high variance, which indicates that it originates from very recent historical admixture without time to reach equilibrium across the population (*15*). This recent admixture likely corresponds to the 20th-century recruitment of workers known as "Sakadas" from the Philippines by the Hawaiian Sugar Planters' Association (*16*).

Strong founder effects in Polynesia

We applied ancestry-specific IBDNe to 581 Society Islanders in Tahiti to reconstruct the effective population size of Tahiti's Polynesian population through time (*17*). This method requires substantial sample counts; in particular, the effective number of complete genomes in the ancestry of interest should exceed 100 (*18*), so we apply it to the island with the most samples in the dataset to illustrate an example founder effect in an Eastern Polynesian island. The method is based on identity-by-descent (IBD) segments and has previously been applied to Samoa, where it suggested that European contact reduced population size (*19*). We apply IBDNe to be Polynesian specific (considering only Polynesian ancestry segments) to remove the confounding effect of admixture (*18*) because European genetic ancestry on the island reflects French colonization and France's population history. We find that the minimum effective Tahitian single-generation population size (Ne) of 837 [95% confidence interval (CI), 639 to 1004] occurred 34 generations before the present (Fig. 2A), providing an estimate for Tahiti's founding bottleneck's size and date. Using the standard 28 years per generation (an average across paternal and maternal age for each child born over a lifetime) yields a founding date of 952 years from the sampling date, or 1069 AD (*20*–*22*). This matches previous radiocarbon and genetic timing estimates of the settlement of the Society Islands in ~1025 to 1120 AD and the 11th century, respectively (*4*, *23*). From this small founder population, we find rapid, exponential population growth postbottleneck until 9 to 10 generations ago, when there is evidence of a substantial decline in population size with approximately one in every two islanders dying (fig. S4). This likely reflects the spread of disease after European colonial contact in the 1700s (*24*, *25*). (Note that census population size is generally larger than the per-generation

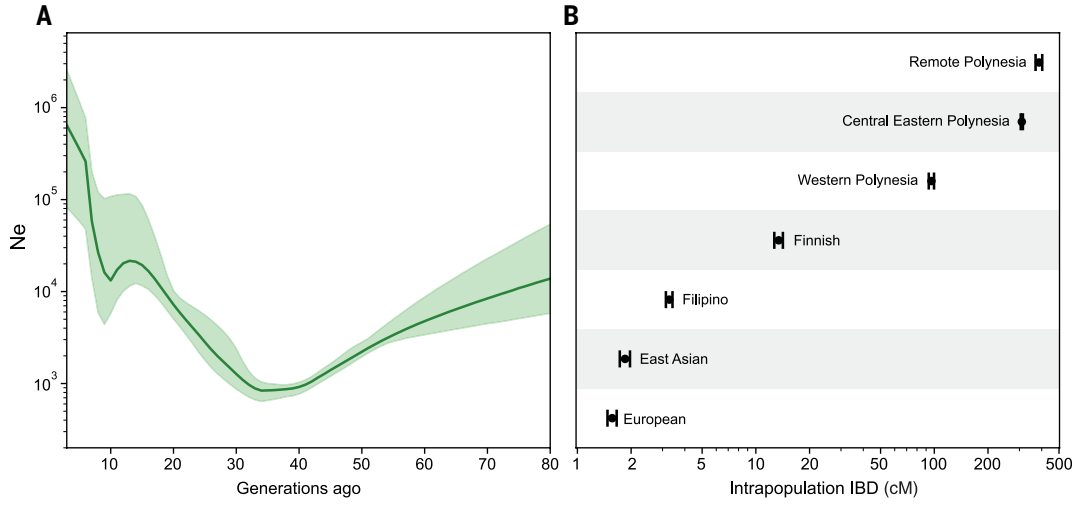

Fig. 2. Polynesian island populations exhibit strong founder effects. (**A**) Polynesian ancestry–specific effective population size per generation (Ne) through time on Tahiti, inferred by IBDNe, indicates a strong founder bottleneck 34 generations ago of ~800 individuals followed by exponential population growth (linear on the logscale). A Polynesian population collapse of nearly 40% at European contact (9 to 10 generations ago) is also visible. Census population size is larger because multiple generations (often approximated to be three) are alive at the same time. The plot begins three generations ago, and the 95% CI is displayed in lighter green. (**B**) For Polynesian populations, the sum of IBD (in centimorgans) in Polynesian ancestry tracts between individuals within a population, normalized by the effective pairs of Polynesian ancestry, is shown. We grouped together the larger regions by geography. Western Polynesia consists of Samoa and Tonga. Central Eastern Polynesia consists of the Austral Islands, the Society Islands, and the Tuamotu Archipelago. Remote Polynesia consists of the Mangareva Islands, the Marquesas Islands, Rapa Nui, and Hawai'i. Standard error bars are indicated from jackknife resampling.

effective population size plotted, as approximately three generations will generally be alive together in a population at any given time—i.e., adults, children, and elders.) Similar population reductions resulting from European colonial contact have been demonstrated in the Americas (*26*).

To further quantify the strength of compounding bottlenecks, we compute the intrapopulation IBD on each island (*27*). This is the sum of shared inherited segments between pairs of individuals, normalized by the number of pairwise comparisons. These scores are expected to be proportional to the risk for autosomal recessive disease and are more intense the further along the sequence of serial settlement bottlenecks an island lies and the smaller the population and shared bottleneck. We made this ancestry specific and normalized by the effective number of Polynesian individual pairs (materials and methods). For reference, we also computed this IBD score for other worldwide populations. We find that remote Eastern Polynesia has the highest intrapopulation IBD (Fig. 2B) and that Polynesians have scores at least an order of magnitude higher than those found in the Finnish, a widely studied founder population.

Hawai'i is most genetically related to Rapa Nui

We calculated the outgroup-F3 statistic (*28*) to quantify the pairwise genetic drift shared between Hawai'i and the other Polynesian populations, using Samoa in Western Polynesia as the outgroup (materials and methods). When calculating these statistics, we used local ancestry (materials and methods) to consider only Polynesian genome segments and to avoid confounding from European ancestry segments. Conventionally, outgroup-F3 is used to capture the shared drift between populations, typically resulting from tens to hundreds of generations of shared population history (*29*). In this context, however, where the settlement of islands may be separated only by a few generations or less, outgroup-F3 would not be expected to show much signal because the time for significant shared genetic drift to accumulate is too short. However, we do see a strong sequential signal of shared drift between islands, given that we are capturing the other source of genetic drift—namely, small population size during shared bottlenecks. In this situation, outgroup-F3 is not predominantly measuring time since divergence of two island populations, which is only a few generations, but rather the number of founding bottlenecks shared by two islands in their settlement sequence. We find that Mangareva and even more so Rapa Nui are the populations most closely related to Hawai'i by this metric, which implies that Rapa Nui and Hawai'i share the longest path through the Polynesian island settlement sequence with one another and the second longest path with Mangareva, from whence their shared deeper ancestors likely originated (Fig. 3A). We confirm these results with an ancestry-specific ancestral recombination graph (ARG)–based outgroup-F3 approach (fig. S5) (*30*, *31*).

To further disentangle relatedness between Hawai'i and other islands, we used the IBD segments shared between populations (Fig. 3, B and C). This approach relies on shared chromosomal segments, inherited from a common ancestor and decreasing in size down a pedigree owing to recombination, rather than using allele frequencies as in the outgroup-F3 analysis. We compute the interpopulation IBD, looking at these IBD segments between pairs of individuals on different islands. This is similar to the intrapopulation IBD used earlier but is now computed between islands rather than on the same island. We restrict to IBD in Polynesian ancestry genomic segments and normalize accordingly because overall IBD sharing would otherwise be lower in island populations that today have substantial non-Polynesian portions of the genome (materials and methods). When considering Hawai'i's sharing with other islands, we see that the most IBD in Hawaiians' Polynesian chromosomal segments is shared with Rapa Nui followed by Mangareva (Fig. 3C and fig. S6). In addition to the total sum of IBD in Polynesian ancestry, we calculated the mean length of IBD of segments between 5 and 15 centimorgans (cM) shared between islands (fig. S7). We again find Hawai'i most closely clustering with Rapa Nui. We generated bootstrapped replicates by resampling within each island population with replacement and then sampling these distributions of mean-length IBD to build 2500 neighbor-joining trees (Fig. 3B). In most trees, Hawai'i and Rapa Nui form their own immediate clade (61.00% indicated with yellow lines). Hawai'i forms an immediate clade with Mangareva in 10.45% of bootstrapped trees (indicated with blue lines) and with Marquesas in only 2.75% (indicated with red lines). The long mean IBD segment length sharing of individuals on Rapa Nui with those on Hawai'i indicates greater

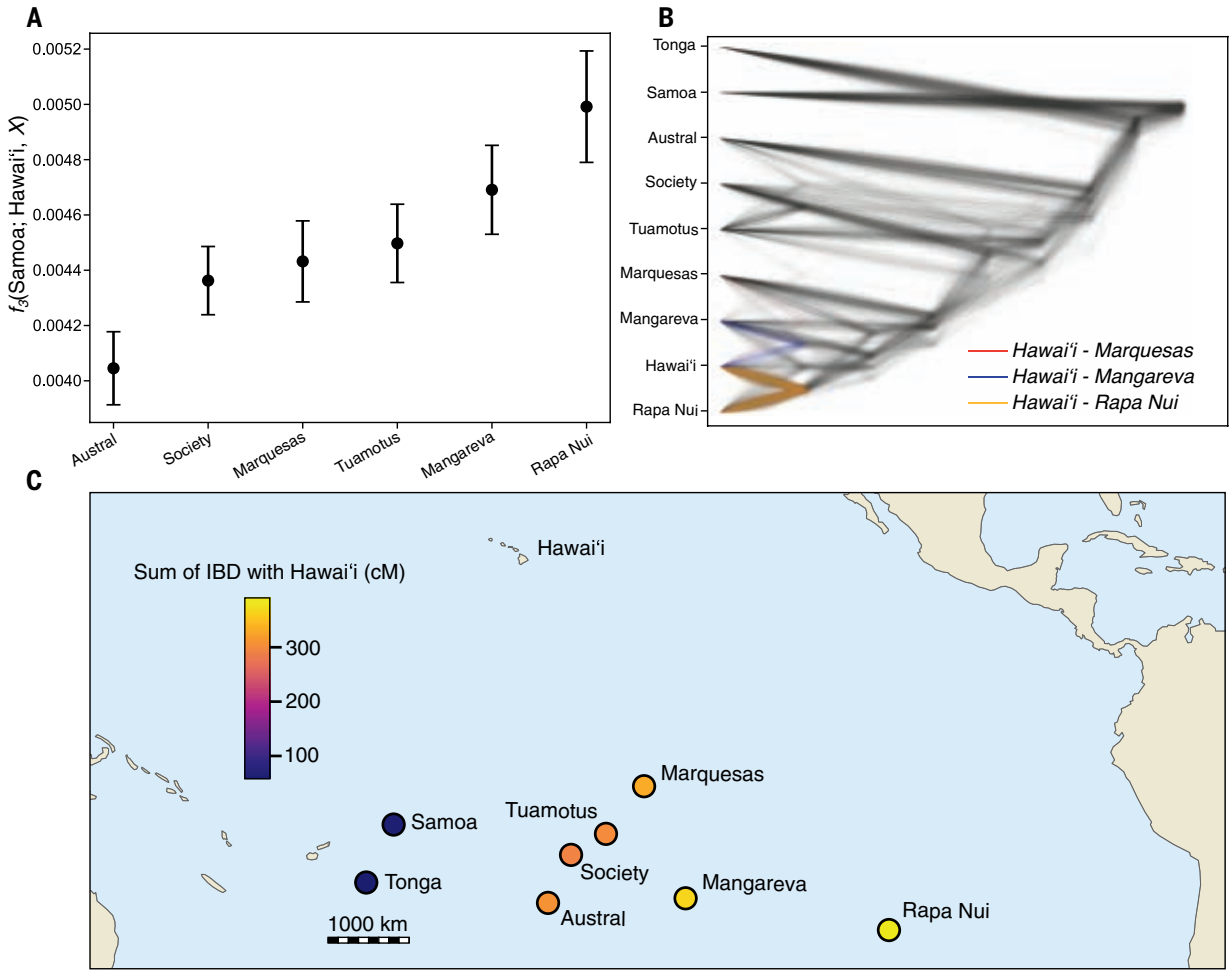

Fig. 3. Genetic relatedness of islands to Hawai'i. (**A**) Polynesian ancestry–specific F3 statistic with 95% CIs measuring shared genetic drift with Hawai'i since divergence from Samoa as a Western Polynesian outgroup. (**B**) Density tree plot created using mean IBD length between islands. We generated standard errors by resampling with replacement from each population. We sampled these distributions to produce 2500 neighbor-joining trees that are overlaid to visualize robustness of clades. Immediate clades formed between Hawai'i and the Marquesas Islands, Mangareva, and Rapa Nui are highlighted. (**C**) Map of Polynesian-specific total interpopulation IBD between Hawai'i and each Polynesian island.

recent genetic relatedness between those two island populations versus the others.

Because Hawai'i and Rapa Nui are among the most remote islands in Polynesia and are extremely far from each other, the simplest interpretation of these results is that Hawai'i and Rapa Nui were both peopled by the same ultralong-distance voyaging group whose deeper origins appear to trace back to Mangareva. Prior studies have suggested that both Hawai'i and Rapa Nui were settled around the same time (*32*–*34*) and that long-distance Polynesian voyagers from this far Eastern Polynesian region also made contact with Native Americans around this time (*35*, *36*). This further strengthens the idea that a particular ultravoyaging group (*Holomoana Nui*) with the capabilities of exploring the most far-flung edges of the Pacific Ocean existed in this region during this period.

Our IBD tree in Fig. 3B shows the hypothesized ultravoyaging *Holomoana Nui* clade (Mangareva followed by the more closely linked Rapa Nui and Hawai'i) as being rooted by Marquesas. This matches findings on the human-mediated dispersal of the Polynesian rat, which also show a Hawai'i-Marquesas clade, although lacking any data from Mangareva or Rapa Nui (*37*). This genetic structure also resembles the linguistic model advanced by Robert Green in 1966 that postulated a proto-Marquesic linguistic clade encompassing Hawai'i, Mangareva, and Marquesas, with Rapa Nui as a closely associated outlier (*38*). Green also postulated a separate proto-Tahitic clade encompassing the Austral Islands, Society Islands, and Tuamotu Archipelago that is also reflected in our IBD tree (Fig. 3B). Archaeological, anthropological, and linguistic evidence has often implicated Marquesas as a major dispersal center for remote Polynesia and as a settlement origin for Hawai'i (*39*). Our findings support the former and qualify the latter, insofar as we find that the founding sequence of Hawai'i, although rooted by Marquesas, more proximally passed through Mangareva and was associated most closely with the founders of Rapa Nui (see the supplementary materials section "On the founding population of Hawai'i").

Because of the rapid initial island population growth (demonstrated as exponential in Fig. 2A and fig. S4) of the founding Polynesian settlers, the genetic contribution of these settlers to the present-day gene pool was amplified and consolidated (*40*). To have a sizeable genetic contribution to the postgrowth population, later admixture with newcomers from other distant islands would require orders-of-magnitude more newcomers—voyaging across thousands of kilometers of open ocean—than the dozens of passengers that the Polynesian double-hulled sailing catamarans could carry (*41*). Although post-settlement interisland trade and voyaging existed (*42*, *43*), we still observe a pronounced single founder effect from these initial settlers

Table 1. Pathogenic founder mutations in French Polynesia. Variants are based on GRCh38. gnomAD frequencies across all exomes and genomes available on gnomAD v4.1. Allele frequencies were computed over all individuals in the French Polynesian cohort, excluding second-degree relatives ($n = 896$), regardless of their self-reported ethnicities. Polynesian-specific allele frequency was computed by only considering individuals with Polynesian ancestry on both haplotypes over the variant coordinates (511, 535, and 537 individuals for *ADAMTSL4*, *PDE6B*, and *FAN1*, respectively). All individuals in French Polynesia were heterozygous for these variants; the carrier frequency is estimated as double the allele frequency.

Gene	Variant	Amino acid change	Frequency (%)	Polynesian-specific frequency (%)	gnomAD frequency (%)	Carrier frequency (%)	Disease	Reference
ADAMTSL4	chr1:150558004-G-A	p.Arg746His	1.23	1.27	1.18×10^{-3}	2.46	Ectopia lentis	(*44*)
PDE6B	chr4:665258-G-C	p.Ala733Pro	1.84	2.62	2.48×10^{-4}	3.68	Rod-cone retinal dystrophy	(*45*)
FAN1	chr15:30922302-G-A	p.Trp707*	1.11	1.49	0	2.22	KMIN	(*46*)

as the dominant demographic signal in the genomes of present-day islanders (Fig. 2A).

Frequency of clinically actionable founder mutations in French Polynesia

Despite the strong founder events in Polynesia, few pathogenic founder mutations have been characterized to date. We use large sample counts from French Polynesia to report the allele frequencies for three variants recently discovered in Polynesians that cause autosomal recessive diseases (Table 1), showing them to have high population prevalence. These variants have only recently been described in clinical settings, so their population-wide frequency has yet to be understood. To better reflect the current risk of disease for islanders, we do not condition allele frequencies on only Polynesian ancestry segments in this frequency analysis.

In Table 1, we note three variants, beginning with *ADAMTSL4* p.Arg746His. Individuals homozygous for this variant develop early-onset ectopia lentis, also referred to as dislocated lens (*44*). Next, *PDE6B* p.Ala733Pro, which causes rod-cone retinal dystrophy leading to progressive vision loss (*45*). Finally, *FAN1* p.Trp707*, which exceeds 1% frequency in French Polynesia yet cannot be found in a single individual in the genome aggregation database (gnomAD), even though that dataset exceeds 800,000 combined whole genomes and exomes (*46*). This variant causes karyomegalic interstitial nephritis (KMIN), a globally ultrarare disease typically resulting in renal failure. This demonstrates that clinically important variants can be entirely absent from large-scale genomic databases but common in a founder population. Because *FAN1* is in the Fanconi anemia DNA repair pathway, the heterogeneity in progression of KMIN patients is possibly modified by environment exposures that damage DNA, making this French Polynesian founder mutation particularly important in population health given the radiation from nuclear tests in French Polynesia still present in the environment of the Tuamotu islands (*47*, *48*).

Sampling of French Polynesia was conducted to be representative (materials and methods); accordingly, the reported allele frequencies should accurately reflect the population-level disease risk for these variants in French Polynesia (*10*). Owing to insufficient sampling, we do not report frequencies of alleles in other archipelagos or islands. We also do not address any variants of uncertain significance because they are difficult to evaluate, especially considering the strong drift and serial founder effects in the region. We find that many rare and common SNPs in this French Polynesian cohort are completely absent from prominent sequencing datasets, including 1000 Genomes and gnomAD (Fig. 1B), which suggests that existing databases do not adequately capture the genetic diversity present in Polynesian founder populations.

These results provide a genetic basis for the importance of disaggregating NHPIs from other Asian ethnicities for clinical purposes. Often, disaggregating NHPIs from Asian ethnicities reveals otherwise hidden disparities in health outcomes (*49*). Given the demographic history of Polynesia traced in this work, consisting of a series of strong separate founder effects terminating in the settlement of Hawai‘i and Rapa Nui, Hawaiians likely carry alleles of clinical importance at substantially different frequencies compared with Samoans and other American Pacific Islanders from further west, which highlights the need for more sequencing data from that group. This also reinforces the importance of distinguishing NHPI populations across different islands when conducting studies that identify founder mutations and when considering ACMG carrier screening guidelines. The broad NHPI label is lacking in actionable resolution for carrier screening recommendations, just as the analogous label of German ancestry would be for Amish or Ashkenazi Jewish patients in such founder population testing guidelines (*50*).

Because NHPI populations are understudied, it is still unclear which Mendelian diseases may be enriched. In a retrospective study on end-stage kidney disease (ESKD) in younger individuals (below 45 years of age) in French Polynesia, 32% of occurrences had an identifiable genetic cause (*51*). Roughly 75% of the cohort had ESKD unrelated to diabetes mellitus, hypertension, or obesity. We believe that these results highlight the importance of globally inclusive sequencing in advancing equitable precision medicine. In particular, sequencing founder populations has already enabled the identification of certain Mendelian diseases (*52*–*54*). As such, future efforts in diverse sequencing will improve the detection of clinically relevant recessive variants, reducing inequities in carrier screening, diagnosis, treatment, and prevention.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S9; Tables S1 and S2; References (*57–77*); MDAR Reproducibility Checklist

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QUANTUM MATERIALS

Wafer-scale low-symmetry graphene moiré superlattice for integrated quantum rectifiers

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Low-symmetry materials unlock rich Berry curvature physics and anomalous transport phenomena that are forbidden in high-symmetry quantum systems through rigorous crystalline symmetry constraints. Current approaches rely on external fields or complex heterogeneous stacking to break crystalline symmetries, which hinders scalable device integration. In this work, we demonstrate surface premelting engineering to create wafer-scale low-symmetry graphene on germanium-110 [Ge(110)]. Controlled premelting forms striped germanium surface patterns that reduce graphene symmetry from D_{6h} to C_{1v}, producing a room-temperature nonlinear Hall conductivity of ~11 micrometers per volt per ohm. First-principles calculations attribute this to originating from overtilted massive Dirac cones of hybridized germanium bands and the graphene Dirac cone. Integrated nonlinear Hall rectifiers generate >20 millivolts of output from radio frequency input and drive commercial voltage boosters and light-emitting diodes, establishing a complementary metal-oxide semiconductor–compatible route toward wafer-scale nonlinear quantum devices.

The electronic properties of quantum materials are governed by crystalline symmetries through rigorous selection rules that dictate the allowed physical phenomena. In highly symmetric crystals, these constraints eliminate intrinsic anomalous transport through exact cancellation mechanisms, whereas low-symmetry materials systematically break these constraints to unlock rich Berry curvature physics and enable previously forbidden quantum geometric responses. For instance, twisted bilayer graphene aligned with hexagonal boron nitride reduces the point group from D_{6h} to C_{1v}, breaking inversion symmetry and activating Berry curvature–driven anomalous transport phenomena (*1*–*3*) that are strictly forbidden in pristine graphene.

Current approaches to low-symmetry systems primarily rely on external field manipulation and complex heterogeneous stacking to break crystalline symmetries. The twisted graphene (*3*) and two-dimensional (2D) transition metal dichalcogenide (TMD) moiré superlattices (*4*, *5*), for example, require strain engineering (*1*, *2*) or perpendicular electric fields (*3*, *4*) to break both inversion and C_3 rotational symmetries to observe intrinsic nonlinear Hall effects (NLHEs). The fabrication of such heterostructures typically requires complex transfer processes that introduce structural defects and interface contamination, substantially reducing device reproducibility. Moreover, the requirement for additional external fields complicates on-chip system integration and increases the complexity of practical device architectures. Alternatively, intrinsic structural approaches exploit the inherent crystal structure symmetry of materials to achieve permanent symmetry reduction without external stimuli. For instance, quasi-1D moiré superlattices, such as nanotubes or nanoscrolls (*6*, *7*), have been reported in which the C_3 rotational symmetry is broken, thus generating strong anisotropic responses with short-circuit current densities that are orders of magnitude larger than those in bulk materials. However, it is rather challenging to assemble these 1D materials into uniform arrays across macroscopic scales for large-area applications. Therefore, the synthesis of wafer-scale uniform low-symmetry moiré superlattices becomes essential for the realization of NLHEs in practical applications.

In this work, we develop surface premelting engineering (SPE) to create wafer-scale, low-symmetry graphene moiré superlattices with intrinsic Berry curvature–driven anomalous transport. By precisely controlling Ge(110) surface premelting and subsequent cooling during the growth of a single-crystalline graphene overlayer, we sculpt striped patterns on the top surface of Ge(110) that reduce the symmetry of the graphene overlayer from D_{6h} to C_{1v}. This symmetry reduction enables an unprecedented nonlinear Hall conductivity of χ_{yxx} ~ 11 μm V^{-1} ohm^{-1} at room temperature, which exceeds those of all state-of-the-art moiré or bulk materials systems. Scaling analysis and first-principles calculations reveal that this response originates from overtilted band anticrossings, where Ge-derived parabolic bands intersect the graphene Dirac cone 23.2 meV above the Fermi level. Strong interlayer hybridization opens tiny gaps at these crossings, creating overtilted massive Dirac cones—quasiparticles whose group velocity exceeds the Dirac velocity—that host divergent Berry curvature hotspots. Exploiting this exceptional χ_{yxx}, we demonstrate nonlinear Hall rectifiers (NRs) that generate stable dc voltages from 2.4-GHz Wi-Fi signals without external power. Because NR arrays can be fabricated on wafer-scale low-symmetry graphene/Ge(110) moiré superlattices, with the back-end-of-line (BEOL) processes, self-contained rectenna dies consisting of 3×3, 4×4, or 5×5 NR arrays can be built as well. For instance, a 4×4 NR array is able to generate more than 20 mV of dc output at –16 dBm of input power, which is sufficient to drive commercial voltage boosters and power light-emitting diodes (LEDs) under ambient conditions. Our study establishes a general and versatile platform for engineering diverse symmetry-broken moiré materials and opens an avenue toward design and scalable manufacturing of quantum devices.

Low-symmetry graphene moiré superlattices wafer

Premelting is known as the localized loss of crystalline order on surfaces of crystal bulks at temperatures slightly below the bulk melting transition (*8*). In fact, the premelting of a Ge surface has been observed, where only the first bilayer became disordered, whereas deeper layers remained ordered (*9*). To approach the desired premelting status, we heat the Ge(110) substrate to $T = 914$°C, ~23°C below the bulk melting point (937°C), where the first bilayer surface atoms become mobile while bulk crystallinity is preserved. This quasiliquid surface layer simultaneously reduces defect formation while providing orientational guidance for the subsequent single-crystal graphene growth. After the premelting, controlled cooling rates enable atomic-scale tailoring of the quasiliquid surface as it transitions back to the solid state,

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producing diverse morphologies that template distinct graphene symmetries: Disordered or amorphous surfaces yield D_{6h}-graphene (Fig. 1A) with preserved higher-order rotational symmetries, whereas highly ordered striped patterns (0.56-nm periodicity) template C_{1v}-graphene moiré superlattices (Fig. 1B) with fundamentally broken symmetry. We refer to this strategy—combining surface premelting with controlled cooling to precisely tailored substrate morphology—as SPE. This SPE approach is potentially applicable to other reconstructed surfaces, offering a general pathway for tailoring substrate templates through controlled thermal processing. The detailed growth process (see the materials and methods) of the C_{1v}-graphene moiré superlattices wafer is shown in fig. S1. The optical second-harmonic generation (SHG) intensity patterns (fig. S2) provide direct experimental evidence of this symmetry reduction, given that the SHG intensity obtained from C_{1v}-graphene is much higher than that of D_{6h}-graphene. Meanwhile, C_{1v}-graphene displays distinctive twofold patterns, whereas D_{6h}-graphene exhibits characteristic circular patterns.

To understand the SPE mechanism, we performed both density functional theory (DFT) calculations and molecular dynamics (MD) simulations. Figure S3 illustrates that the stripe phase is actually energetically preferred. Notably, previous experimental studies have rarely observed the stripe phase on Ge surfaces, predominantly reporting (6×2) reconstructions (*10*) or amorphous structures (*11*). MD simulations of the cooling process from 1000 K to 300 K (Fig. 1C and movie S1) suggest that the realization of a thermodynamically favorable stripe phase requires precise kinetic control. It is shown that the low cooling rate (70 K/ns) allows the system to reach the energetically preferred stripe-like Ge surfaces, whereas the high cooling rate (700 K/ns) kinetically restricts the system in amorphous structures.

Scanning tunneling microscopy (STM) characterizations reveal the spatial-hierarchical atomic configurations of the epitaxial C_{1v}-graphene moiré superlattice (Fig. 1, D to F). At bias (V_{bias}) of −1 V, the STM image resolves 1D Ge atomic chains along the [−110] direction with well-defined 0.56-nm stripe periodicity confirmed by fast Fourier transform (FFT) filtering (Fig. 1D, inset). At bias V_{bias} of −0.1 V, the graphene layer exhibits a uniform honeycomb lattice characteristic of high-quality monolayer graphene (Fig. 1E), with the armchair direction precisely aligned to the underlying stripe phases. At bias V_{bias} of −0.06 V, hexagonal moiré superlattices emerge with characteristic periodicity defined by basis vectors $|a_1| = |a_2| = 1.4$ nm intersecting at 60° (Fig. 1F). To understand the symmetry of this C_{1v}-graphene moiré superlattice, atomic structural modeling was subsequently performed. As shown in Fig. 1G, the striped-phase Ge(110) surface inherits an anisotropic twofold symmetry (*p2mg* plane group) characterized by a twofold rotational axis normal to the surface

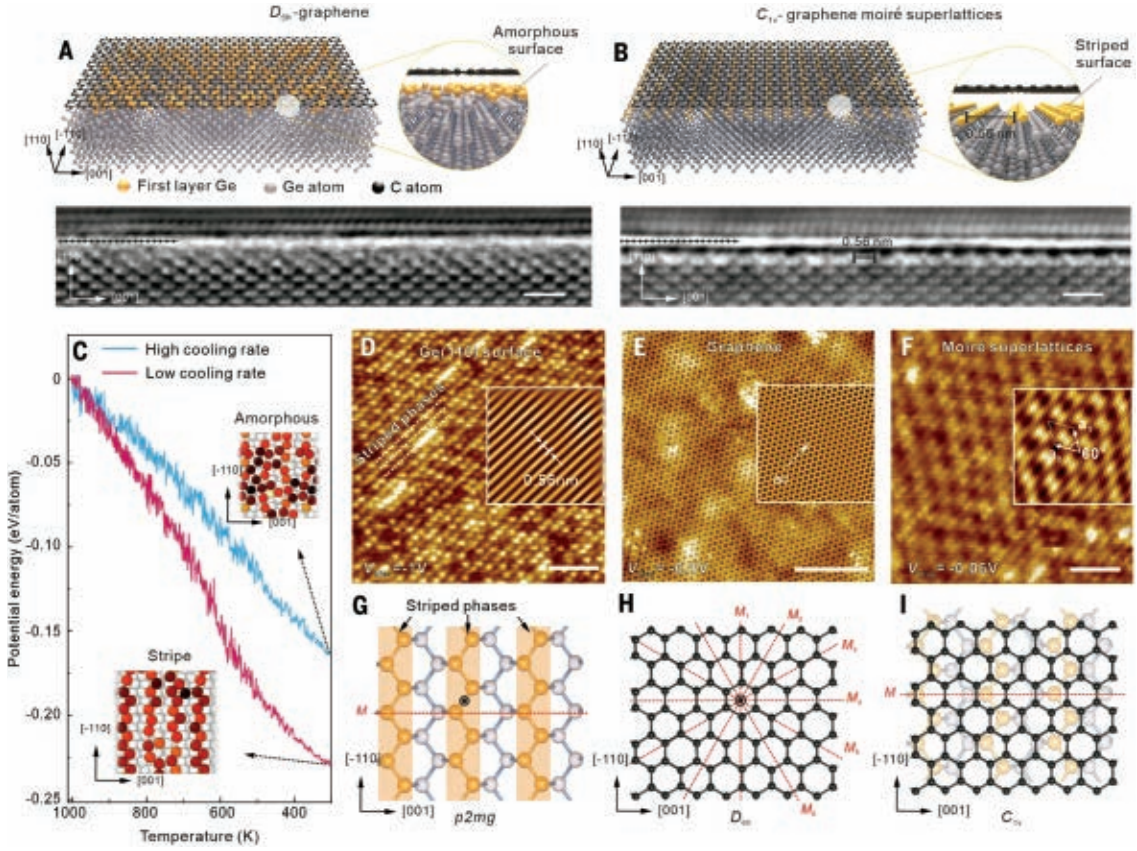

Fig. 1. Low-symmetry C_{1v}-graphene moiré superlattices wafer. (**A**) Schematic illustration (top) and cross-sectional transmission electron microscopy (TEM) image (bottom) of D_{6h}-graphene deposited on Ge(110) substrates with amorphous subsurface reconstruction. The 3D representation shows the layered architecture with graphene (black carbon atoms) overlaying the Ge crystal substrate (gray atoms), where the first Ge layer (yellow spheres) exhibits random subsurface disorder. (**B**) Schematic illustration (top) and cross-sectional TEM image (bottom) of C_{1v}-graphene on stripe-phase reconstructed Ge(110) surfaces. Stripe periodicity of the Ge(110) surface is 0.56 nm. [(A) and (B)] Scale bars, 1 nm. (**C**) MD simulation of the cooling process of the graphene/Ge(110) heterostructure from 1000 K to 300 K. (Insets) Top views of final snapshots from the simulations at high cooling rate (700 K/ns) and slow cooling rate (70 K/ns). (**D**) Morphological STM image of 1D Ge atomic chains (striped phases) along the [−110] direction of the Ge(110) surface with a 0.56-nm periodicity. $V_{bias} = -1$ V. (**E**) Morphological STM image of graphene covered on Ge(110). $V_{bias} = -0.1$ V. (**F**) Morphological STM image of moiré superlattices with hexagonal topology. $V_{bias} = -0.06$ V. [(D) to (F)] Scale bars, 3 nm. (**G**) Schematic of striped phases on Ge(110) with *p2mg* symmetry composed of first layer (yellow globules). (**H**) Schematic of freestanding monolayer graphene with D_{6h} symmetry. (**I**) Schematic of C_{1v}-graphene moiré superlattices with C_{1v} symmetry. Red dashed lines in (G) to (I) represent vertical mirror planes. Black circled dots in (G) and (H) represent rotational axes.

and a mirror plane (*M*) parallel to the [001] direction (*12*). By contrast, the pristine monolayer graphene has a hexagonal symmetry with sixfold rotational symmetry and six mirror planes (D_{6h} point group) (Fig. 1H). Owing to the inherent symmetry incompatibility between D_{6h}-graphene and *p2mg*-Ge(110), the interaction between these two materials at the interface results in the formation of a C_{1v} symmetric heterostructure with only a single mirror plane parallel to the [001] direction of Ge (*13*), as shown in Fig. 1I. Additional structural characterizations of the C_{1v}-graphene moiré superlattice through Raman spectroscopy, selected-area electron diffraction (SAED), FFT, and polarization-dependent SHG intensity are shown in figs. S4 and S5, further confirming the high-quality C_{1v}-graphene moiré superlattice at wafer scale.

Giant NLHEs in a low-symmetry Gr-Ge moiré superlattice

To experimentally demonstrate the effects of reduced symmetry, the planar Hall effect was systematically measured in both low-symmetry and high-symmetry graphene. Figure 2A shows the six-probe Hall bar geometry for C_{1v}-graphene moiré superlattices, where longitudinal current $\left(I_x^{\omega}\right)$ flows along the stripe phase of Ge(110) and transverse Hall resistance (R_y) is measured (materials and methods). The key parameters for various devices are summarized in table S2. With the rotation of the in-plane magnetic field (8 T, 2 K), R_y exhibits marked symmetry-dependent behavior (Fig. 2B), where φ is the angle between the magnetic field and mirror plane (Fig. 2B, inset). Notably, the C_{1v}-graphene displays 2π-periodic angular dependence [Fig. 2B, red line; proportional to cos(φ)], whereas D_{6h}-graphene shows negligible signal. This anomalous planar Hall effect (APHE) has been reported in low-symmetry materials with a single mirror plane (*14*), where the planar magnetic field orientation determines whether crystal symmetry constrains or permits anomalous magnetotransport. APHE measurements from two devices with stripe phases oriented at 45.3° (Fig. 2C) and 25.5° (fig. S6) relative to each other produce ~47° and 30° angular shifts, respectively, which confirms that APHE stems from symmetry-induced modulation rather than out-of-plane field components. Size-dependent APHE measurements were performed (fig. S7) on devices with different dimensions to assess possible influence of coherent stripe-phase size.

Whereas APHE requires magnetic fields to break time reversal symmetry, the low crystal symmetry permits nonlinear electronic transport through intrinsic inversion symmetry breaking under time-reversal-symmetric conditions (*15*–*17*). The harmonic electronic transport measurements

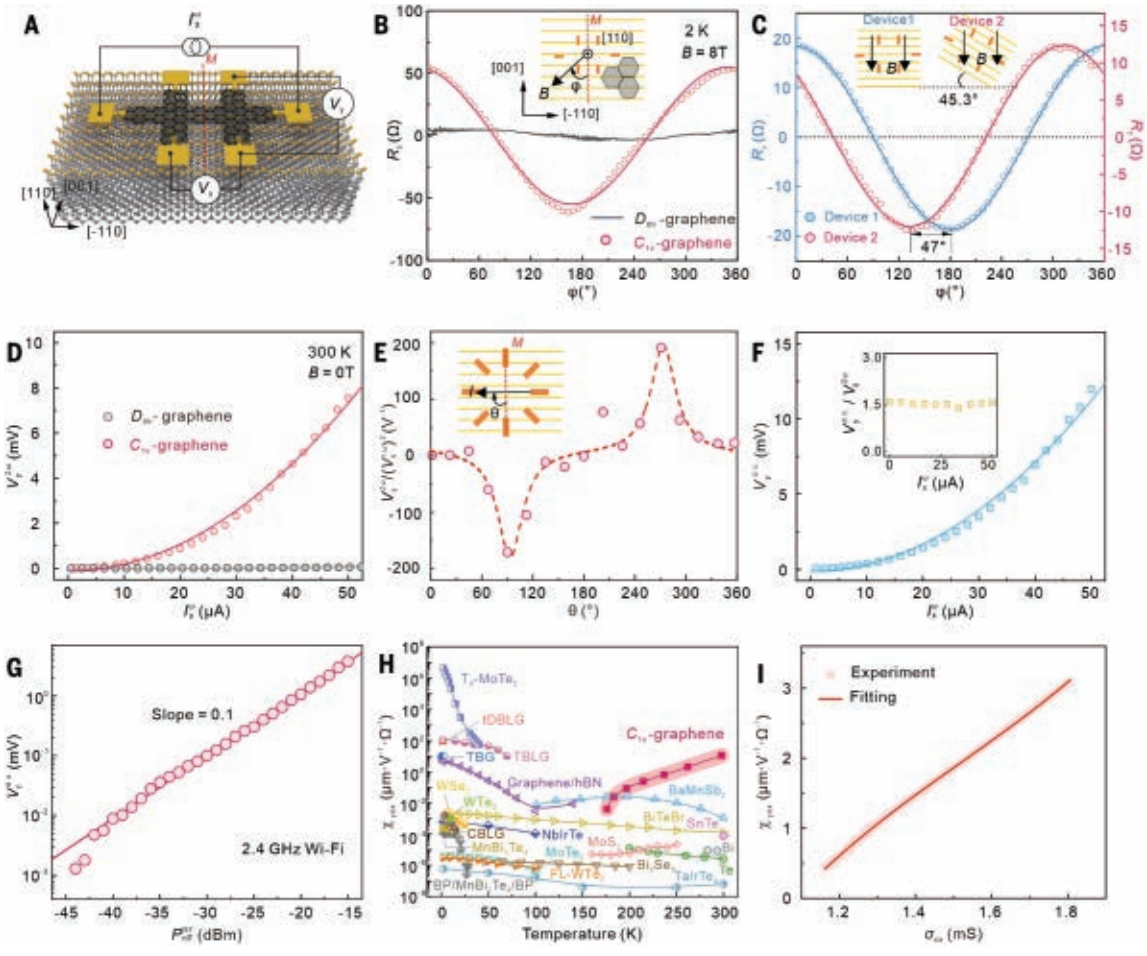

Fig. 2. Magnetotransport and electrical transport properties of low-symmetry C_{1v}-graphene moiré superlattices. (**A**) Schematic illustration of the Hall bar device geometry. (**B**) Angular-dependent APHE measurements at 2 K under rotating in-plane magnetic field $B = 8$ T, comparing C_{1v}-graphene (red circles) with D_{6h}-graphene (gray line). The 2π-periodic cos(φ) behavior reflects broken inversion symmetry, where φ is the angle between the field and mirror plane (inset). (**C**) APHE measurements from two devices with stripe phases oriented at 45.3° relative to each other. (**D**) NLHE measurements at 300 K without a magnetic field, comparing C_{1v}-graphene (red circles) with D_{6h}-graphene (gray circles). The second-harmonic voltage $\left(V_y^{2\omega}\right)$ is measured as a function of the input ac current $\left(I_x^{\omega}\right)$. (**E**) Angular-dependent NLHE measurements. (Inset) Current *I* was injected through one of eight electrodes that form an angle θ with the mirror plane. (**F**) The rectified nonlinear Hall voltage $\left(V_y^{dc}\right)$ and $V_y^{2\omega}$ versus I_x^{ω}. The inset shows that the ratio between V_y^{dc} and $V_y^{2\omega}$ keeps constant with various I_x^{ω}. (**G**) The V_y^{dc} as a function of effective incident rf power $\left(P_{eff}^{rf}\right)$ at a frequency of 2.4-GHz Wi-Fi. (**H**) Comparison of the temperature dependence of the second-order nonlinear conductivity (χ_{yxx}) for various material systems. (**I**) The scaling behavior of the χ_{yxx} and linear conductivity (σ_{xx}).

were performed, and second-harmonic responses for C_{1v}- and D_{6h}-symmetry graphene at room temperature are summarized in Fig. 2D. For C_{1v}-graphene, the second-harmonic transverse voltages ($V_y^{2\omega}$) display a quadratic dependence on the input I_x^{ω}. By contrast, D_{6h}-graphene has nearly no second-harmonic transverse voltages. Additionally, second-harmonic transverse voltages have not been obtained on graphene/non-premelted Ge(110), as shown in fig. S8. To further validate the symmetry origin, $V_y^{2\omega}$ as a function of current direction was measured using disc devices (Fig. 2E, inset). Current I was injected through one of eight electrodes that form an angle θ with the mirror plane, and voltage drops across electrodes along the diameter and perpendicular to the diameter were measured. The nonlinear response shows a 2π period with onefold angular dependence, reaching maximum when the current is perpendicular to the mirror plane and vanishing when the current is parallel to the mirror plane. In Fig. 2F, the measured transverse rectified dc Hall voltage (V_y^{dc}) at zero magnetic field shows a quadratic dependence on I_x^{ω} that mirrors the behavior of the $V_y^{2\omega}$, with its frequency-dependent response shown in fig. S9. It is worth mentioning that the amplitude of V_y^{dc} exceeds that of $V_y^{2\omega}$ by a factor of ≈1.4 (Fig. 2F, inset), in good agreement with the theoretical ratio of $V_y^{dc} / V_y^{2\omega} = \sqrt{2}$ (supplementary text). It should be noted that $V_y^{2\omega}$ is stable under the ambient environment and is uniform across the wafer, as shown in figs. S10 and S11, respectively.

Fig. 2G shows the radio frequency (rf)–to-dc rectification effect of our Hall bar device in reference to a 2.4-GHz Wi-Fi band rf source, where the V_y^{dc} in log-scale is linearly proportional to the effective incident rf power (P_{eff}^{rf}). Figure 2H presents a comprehensive comparison of NLHE nonlinear Hall conductivity (χ_{yxx}) across diverse material platforms. The C_{1v}-graphene moiré superlattice maintains a robust nonlinear Hall conductivity of χ_{yxx} ~ 11 μm V^{-1} ohm^{-1}, which is among the highest values reported to date under room-temperature conditions (table S3).

To understand the physical mechanism underlying the giant NLHE in the C_{1v}-graphene moiré superlattice, a temperature-dependent scaling analysis based on the polynomial relation between the nonlinear Hall conductivity χ_{yxx} and linear longitudinal conductivity σ_{xx} was performed (Fig. 2I). In the presence of time reversal symmetry, three mechanisms may coexist, including the geometric Berry curvature dipole mechanisms (*18*) as well as two extrinsic mechanisms—i.e., the side jump and skew scattering (*19*, *20*). The scaling law of the three mechanisms can be written as

$$\chi_{yxx} = C_{SK}\sigma_{xx}^{3}\left(\sigma_{xx}^{-1} - \sigma_{xx0}^{-1}\right)^{2} + C_{SJ}\sigma_{xx}^{2}\left(\sigma_{xx}^{-1} - \sigma_{xx0}^{-1}\right) + C_{BCD}\sigma_{xx} + C_0$$

where C_{SK}, C_{SJ}, and C_{BCD} are the fitting parameters for the skew scattering (SK), side jump (SJ), and Berry curvature dipole (BCD), respectively. Because the skew scattering and side jump also have the cubic and quadratic terms of σ_{xx}, the nearly linear scaling relationship between χ_{yxx} and σ_{xx} in Fig. 2I indicates that only BCD makes a dominant contribution to the giant NLHE in the C_{1v}-graphene moiré superlattice (more details about the scaling analysis are provided in figs. S12 and S13 and the supplementary text). The gate-dependent scaling analyses also reveal that BCD mainly contributes to NLHEs (figs. S14 and S15).

Giant NLHEs from overtilted Dirac cones in a Gr-Ge moiré superlattice

The Berry curvature acts like a "pseudomagnetic field" in momentum space that describes the curvature of the Hilbert space of quantum states in materials (*21*). The Berry curvature dipole describes a dipole-like, uneven distribution of Berry curvature. The existence of a giant Berry curvature dipole is revealed during first-principles calculations of the energy band structure of the C_{1v}-graphene moiré superlattice (fig. S16), as shown in Fig. 3A. The Fermi level is found at ~77 meV

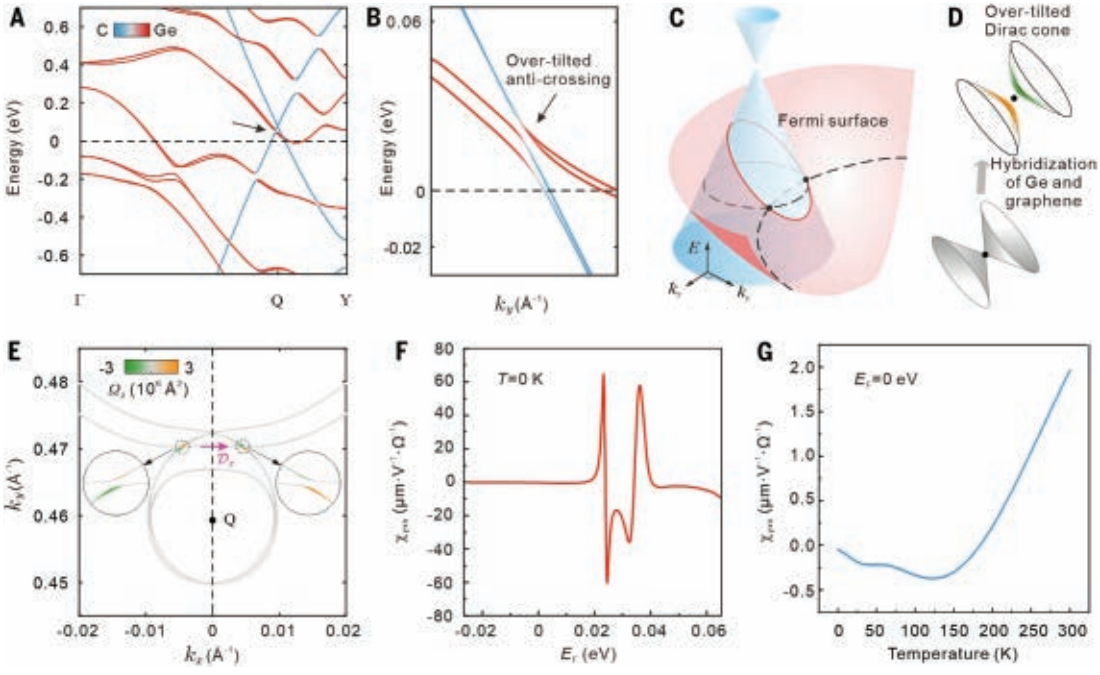

Fig. 3. Calculated giant nonlinear Hall conductivity originated from overtilted massive Dirac cones of hybridized Ge bands and massless graphene cones. (**A**) Orbital-projected band structure along the Γ-Q-Y direction. The color bar indicates the weight of orbitals from graphene (blue) and Ge (red). The arrow marks the hybridization-induced band anticrossing near the Fermi level (dashed line). (**B**) Zoom-in of the band anticrossing, revealing a hybridization-induced tiny band splitting. (**C**) Schematic of Ge bands and graphene Dirac cone hybridized to form the overtilted anticrossing. E is the energy, and k_x and k_y are the wave vectors. (**D**) The description of overtilted anticrossing as a 2D tilted massive Dirac cone. (**E**) Berry curvature $\Omega_z(k)$ (color scale) as a function of the wavevectors k_x and k_y at the chemical potential $E_F = 23.2$ meV, showing divergent hotspots near the overtilted massive Dirac cones. The magenta arrow indicates schematically the Berry curvature dipole of the uneven distribution of the Berry curvature. (**F**) Calculated Berry curvature dipole–induced second-order nonlinear Hall conductivity χ_{yxx} as a function of chemical potential (E_F) at $T = 0$ K. (**G**) Calculated χ_{yxx} as a function of temperature (T) at $E_F = 0$ meV.

below the graphene Dirac point according to the scanning tunneling spectroscopy measurements (fig. S17). Crucially, along the Γ-Q-Y line, where Q marks the projected Dirac point, a Ge-derived parabolic band (red bands) intersects the graphene Dirac cone (blue bands) 23.2 meV above the Fermi level, forming a band anticrossing. A zoom-in of the anticrossing in the [−0.02, 0.06] eV energy window measured from the Fermi level in Fig. 3B demonstrates a tiny splitting due to orbital hybridization and overtilted band dispersions, at which the group velocities of the energy bands have the same sign. Furthermore, the energy E as a function of k_x and k_y in Fig. 3C illustrates that the intersection of the graphene Dirac cone with the Ge parabolic band forms a closed tilted red ring. The Fermi surface (black dashed line) intersects this ring at discrete points (black dots), where hybridization can give the overtilted anticrossings or a 2D tilted massive Dirac cone. The tilted massive Dirac cone with a tiny energy splitting can host a divergent distribution of Berry curvature as well as a giant nonlinear Hall signal (*22*). The Berry curvature and Berry curvature dipole–induced nonlinear Hall conductivity near the overtilted massive Dirac cone in Fig. 3D can be calculated with the help of a first-principles Wannier tight-binding Hamiltonian (supplementary text). Figure 3E shows the distribution of the Berry curvature on the k_x-k_y plane at an energy of 23.2 meV above the Fermi level in the vicinity of the anticrossing, which reveals the divergent hotspots of the Berry curvature near the Dirac anticrossings, leading to a giant Berry curvature dipole, as indicated by the magenta arrow. The second-order nonlinear Hall conductivity χ_{yxx} is defined as $j_y = \chi_{yxx}E_xE_x$, where E_x is the electric field of the driving current along the x direction and j_y is the second-order current density along the y direction driven by the longitudinal current. Figure 3, F and G, shows the second-order nonlinear Hall conductivity as a function of the chemical potential (E_F) and temperature (T), respectively. As shown in Fig. 3F, the zero-temperature nonlinear Hall conductivity $\chi_{yxx} = -0.1855\ \mu m\ V^{-1}\ ohm^{-1}$ at the charge neutrality ($E_F = 0$) and shows a strong peak at $E_F \simeq 0.0232$ eV near the tilted anticrossing, reaching $\chi_{yxx} = 64.45\ \mu m\ V^{-1}\ ohm^{-1}$ with a neighboring negative peak around $E_F \simeq 0.0245$ eV. Figure 3G shows the calculated temperature dependence of χ_{yxx} at $E_F = 0$. At low temperatures, χ_{yxx} remains small because only states near the charge-neutrality point contribute to the BCD response. As temperatures increase, thermally activated carriers may distribute a broader energy range around the hybridized anticrossing, leading to the marked change of χ_{yxx}. This temperature-dependent trend is consistent with the experimental observation of a weak NLHE signal at low temperature and a pronounced response at room temperature (Fig. 2H).

NLHE-based rectennas for low-powered rf energy harvesting

In Fig. 4A, a NR array is fabricated through standard BEOL processing. The fabrication processes are presented schematically in fig. S18 and are described in the materials and methods. Notably, the second-layer Au interconnects serve a dual function as both signal routing paths and integrated receiving antenna, enabling the NR array to operate as a self-contained rectenna system without requiring external antenna components. The NR array is integrated on a printed circuit board

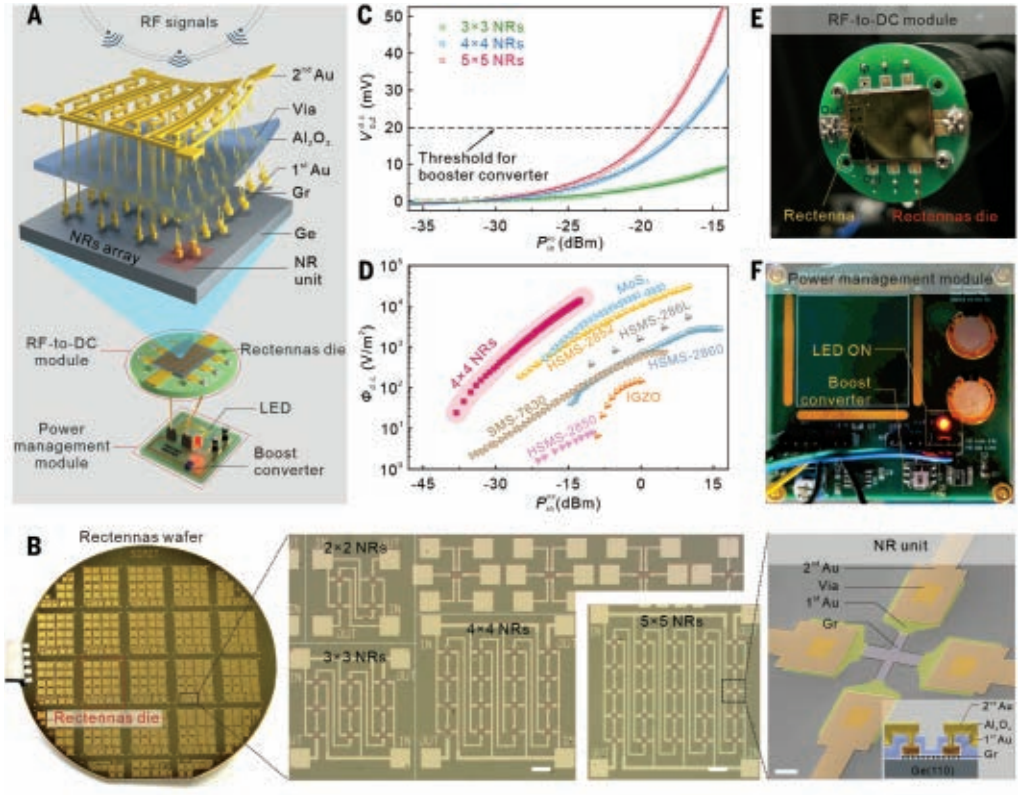

Fig. 4. Rectennas for low-powered rf energy harvesting. (**A**) Schematic diagram of an integrated rf energy harvesting module comprising a 4×4 NR array with series-connected Hall-probe configuration coupled with a commercial power management module for ambient Wi-Fi energy harvesting, which serves as a rectenna. (**B**) Rectenna wafer and architecture. From left to right: photography of a 4-inch wafer containing rectenna dies, optical image of multiscale rectennas (2×2, 3×3, 4×4, and 5×5 NRs with series-connected Hall-probe configuration), and false-colored SEM of a single NR. The 5×5 NR is spatially isolated from other arrays on the die. (Inset) Cross-sectional artistic view of the via contact. Scale bars (left to right), 300 μm, 300 μm, and 30 μm. (**C**) The rectified voltage (V_{out}^{dc}) versus input power (P_{in}^{rf}) for 3×3, 4×4, and 5×5 NR arrays. (**D**) The rectified voltage density (Φ_{dc}) versus P_{in}^{rf} of the 4×4 NRs with various rectification schemes. (**E**) Photograph of the rf-to-dc module with one rectenna die, where a NR rectenna is wire bonded. (**F**) Demonstration of lighting an LED with the NR rectenna under a rf radiation for low-powered rf energy harvesting.

(PCB) as an rf-to-dc module, where the rectenna front end efficiently collects and rectifies ambient rf energy to dc voltage, and the cascaded power management module provides regulated output power, as demonstrated by powering an LED. NR arrays can be mass-produced with high integration density, as demonstrated by the rectenna dies (each 2 cm by 1.5 cm) housing multiscale rectennas (2×2, 3×3, 4×4, and 5×5 NRs) on a 4-inch graphene/Ge(110) rectennas wafer in which C_{1v}-graphene moiré superlattice exists (Fig. 4B). At the fundamental level, the false-colored angled scanning electron microscopy (SEM) image (Fig. 4B, rightmost) details the stack structure of a single NR—the minimal functional unit within the rectenna—highlighting the dielectric isolation layer and via contacts (Fig. 4B, inset, cross-sectional schematic) critical for reliable operation of these nanoscale components. To optimize the rectenna performance, we systematically investigate both series- and parallel-connected configurations of four NRs (fig. S19). The accumulated $V_{out}^{2\omega}$ from the series-connected NRs shows an amplitude approximately four times that of a single discrete NR unit, whereas parallel-connected NRs demonstrate a relatively constant $V_{out}^{2\omega}$ but a decreased output resistance with the number of devices. These results can be interpreted by the proposed theoretical circuit model predictions. To maximize the output voltage for practical applications, we therefore adopt series-connected Hall-probe configurations in subsequent rectenna arrays.

The rectified voltages $\left(V_{out}^{dc}\right)$ of the rectenna versus input power $\left(P_{in}^{rf}\right)$ for 3×3, 4×4, and 5×5 NRs with series-connected Hall probes are summarized in Fig. 4C. Additional NR array results are presented in fig. S20. Under a comparable input voltage range, a substantial enhancement in V_{out}^{dc} is observed as the number of series-connected Hall-probe NRs increases. Among them, when the 4×4 NR array is operating at the input power of −16 dBm, it is able to generate a $V_{out}^{dc} = 24$ mV, which exceeds the threshold voltage (20 mV) required for driving a commercial booster converter. In terms of the rectified voltage density $\Phi_{dc}=V_{out}^{dc}/S$ (where S is the occupied area of rectenna), a key performance metric, the integrated 4×4 NR array rectenna with series-connected Hall probes outperforms other reported semiconductor junction or diode-type devices, including the MoS_2 phase junction (*23*), commercial silicon Schottky diodes (*24*–*28*), and indium gallium zinc oxide (IGZO) diodes (*29*) (Fig. 4D). It should be noted that our study demonstrates the footprint-normalized voltage-generation capability of the NLHE rectifier, and further improvement of its power efficiency needs to consider rf coupling, impedance matching, and more. To evaluate the practicability of series-connected Hall-probe NRs for rf energy harvesting, we further demonstrate the wireless powering of a red LED by using the 4×4 NR array–based rectenna die packaged on a PCB board as the rf-to-dc conversion module (Fig. 4E) under ambient conditions. As shown in Fig. 4F, when the module is positioned in proximity to the rf signal source, the rectenna captures and converts the wireless rf energy into a dc voltage signal. This rectified voltage is subsequently channeled through conducting wires to a power management module, where a boost converter manages to upconvert the initial threshold voltage of 20 mV to light the red LED (movie S2).

Conclusions

We have demonstrated SPE as a scalable platform for wafer-scale synthesis of low-symmetry C_{1v}-graphene moiré superlattices with giant NLHEs. First-principles calculations reveal that this exceptional response originates from overtilted massive Dirac cones at band anticrossing, where Ge-graphene hybridization creates divergent Berry curvature hotspots. The NLHE-based rectifiers fabricated using a complementary metal-oxide semiconductor (CMOS)–compatible process are capable of generating stable dc voltages by converting the 2.4-GHz Wi-Fi energy. The optimized rectenna composed of a 4×4 NR array with series-connected Hall-probe configuration is able to generate voltages sufficient to drive commercial boosters, which then light an LED. Our findings may provide a practical solution for future self-powered microelectronic devices that collect the ambient wireless energy. Furthermore, this low-symmetry moiré platform provides a pathway to explore rich nonlinear phenomena, paving the way for new nonlinear electronic, photonic, and optoelectronic applications.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aeh3915
Materials and Methods; Supplementary Text; Figs. S1 to S20; Tables S1 to S3; References (*30*–*47*); Movies S1 and S2

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SOLAR CELLS

Volatile eutectics to tailor crystallization for perovskite optoelectronics

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Improvements to the radiative efficiency of halide perovskite thin films are necessary to approach fundamental solar cell performance limits. We tailored perovskite crystallization by leveraging the eutectic interaction between zinc bromide and methylammonium chloride additives. The resulting films had high photoluminescence quantum yield and charge carrier decay times and low surface recombination velocity. Nanoprobe x-ray microscopy revealed local melting of the eutectic at grain boundaries during annealing that reduced intragranular structural defect density. Annealing also homogenized the halide distribution and caused zinc segregation to grain boundaries. Solar cells fabricated with the volatile eutectic reached 26.5% power conversion efficiency (PCE, certified 25.9%), and perovskite minimodules (areas more than 15 square centimeters) had 23.9% PCE. Small-area cells with a poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] hole transporter retained 85.8% of their initial 24.1% PCE after 1400 hours of continuous operation at 85°C. Furthermore, this approach enables 26.1% external quantum efficiency as light-emitting diodes.

Metal halide perovskites show exceptional promise for photovoltaics (PV), but single-junction perovskite power conversion efficiency (PCE) remains lower than that of both silicon and gallium arsenide (GaAs) records (*1*–*3*). Perovskite PCE is currently limited by losses in open circuit voltage (V_{OC}) and fill factor (FF), which are tied to excess nonradiative carrier recombination due to poor perovskite quality and resistive losses at interfaces (*4*, *5*). Perovskite PV faces additional challenges at scale, suffering from large cell-to-module efficiency losses (*5*, *6*). To approach the detailed balance limit of efficiency, excellent radiative efficiency is required (*7*). Despite considerable advances, state-of-the-art perovskite active layers typically still exhibit photoluminescence quantum yields (PLQYs) of <15%, which limits PV efficiency (*8*, *9*).

Perovskite thin films are polycrystalline and have a high density of defects at interfaces and grain boundaries that increase surface recombination, shorten carrier lifetimes, and reduce PLQY (*10*). The control of perovskite crystallization with additives and the manipulation of interface chemistry have played a central role (*2*, *11*–*18*) in improving optoelectronic outcomes. Additives of several distinct classes are commonly used, including alkylammonium chloride (R-NH_3Cl), metal halides, and ionic liquid additives. Volatile R-NH_3Cl salts are used to favor the formation of the black, cubic phase of formamidinium lead iodide ($FAPbI_3$) during crystallization before evaporating from the film during extended annealing (*2*, *11*). Among metal salts, Zn additives have been reported to increase apparent grain domain sizes, improve crystallinity, and suppress molecular I_2 formation (*18*–*20*). Ionic liquid additives (*21*–*24*) modulate grain growth, an effect attributed to both limiting the nucleation of the perovskite because of their low volatility and to Ostwald ripening during annealing (*21*, *24*). Although promising, this strategy lacks precise control and understanding of the distribution of the liquid additive in the final film and its effects on crystallographic orientation (*25*).

Ideally, new mechanisms and their understanding could offer pathways to deterministic control over crystallization, termination, and orientation in the final film and device. Recently, eutectics, which are homogeneous mixtures with a depressed melting point relative to the individual components at a self-regulating equilibrium, were suggested as a possible path to improving perovskite phase purity (*26*). However, beyond a theoretically calculated enhanced interaction with corner-sharing lead hexaiodide (PbI_6) octahedra, very little is understood about how they might affect film formation and defect passivation in the final film.

Here, we designed and leveraged the dynamic formation of a volatile eutectic composed of zinc halide and methylammonium chloride (MACl) salts to control both crystallization and passivation. Local melting of the eutectic mixture at grain boundaries during annealing enhanced grain ripening through a solid-liquid-solid growth mechanism in which grain growth propagates through a locally melted eutectic at grain boundaries. This redistributes Zn to grain boundaries and interfaces, where theoretical calculations suggested that a stable surface-bound zinc iodide (ZnI_2) interstitial efficiently passivated nonradiative defects. Thin films fabricated with the eutectic additives exhibited exceptional optoelectronic properties, including >65% external PLQY, >200-μs photoluminescence (PL) lifetime, and <0.5 cm/s surface recombination velocity (SRV). The improved radiative efficiency facilitated by eutectic-tailored crystallization resulted in small-area devices (0.08-cm^2 aperture area) with 26.5% (25.9% certified) PCE as PV devices and 26.1% external quantum efficiency (EQE) as light-emitting diodes (LEDs). This process translated well to large-area fabrication, enabling 15.58-cm^2 aperture area minimodules fabricated with the eutectic complex that exhibit certified 23.9% PCE.

Eutectic formation and impacts on optoelectronic performance

We began by investigating the interaction between MACl and zinc halide salts. Zinc halides form eutectics when complexed with hydrogen bond donors, including quaternary ammonium salts (*27*). Because of the ability of MA to donate hydrogen bonds, we hypothesized that a strong interaction between zinc halide salts and MACl might enable the formation of a eutectic that could manipulate perovskite crystallization. Mixtures of MACl and zinc bromide ($ZnBr_2$) melted at substantially lower temperatures than either pure MACl (230°C) or pure

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$ZnBr_2$ (394°C), with a eutectic temperature of 87°C at a 1:1 molar ratio (Fig. 1A and fig. S1). We observed similar eutectics in mixtures of MACl and zinc chloride ($ZnCl_2$), demonstrating generalizability across different zinc halides (fig. S2). Visually, we observed a complete melting and the formation of a gel-like compound of equimolar MACl and $ZnBr_2$ at 100°C (fig. S3). We observed true congruent melting in the eutectic rather than a volatilization of MACl, which occurs at higher temperatures (~200°C), as verified by thermogravimetric analysis (fig. S4). We found that the addition of $FAPbI_3$ to the MACl:$ZnCl_2$ mixture formed an even deeper eutectic (73.5°C at a 1:4:4 molar ratio; Fig. 1B).

Perovskite films fabricated with both MACl and $ZnBr_2$ additives, hereafter labeled "eutectic," demonstrated improved optoelectronic properties compared with previously state-of-the-art "reference" samples with MACl but no $ZnBr_2$. Figure 1C shows steady-state PL spectra of the reference and eutectic perovskite thin films measured at 1-sun equivalent absorbed photon flux (*28*). The reference perovskite, which has previously been used to fabricate efficient perovskite solar cells (*29*), presented a high PLQY of 34.2%. We measured a PLQY of 65.8% for the eutectic perovskite, one of the highest values reported (*14*, *28*, *30*).

Consistent with the enhancement in radiative efficiency, the eutectic perovskite demonstrated a longer time-resolved photoluminescence (TRPL) lifetime than the reference (Fig. 1D). At low injection levels (i.e., initial carrier density, $N_0 \sim 5 \times 10^{13}$ cm^{-3}), a typical average PL lifetime in the reference sample was 20 μs, whereas the eutectic perovskite exhibited a PL lifetime of >200 μs when fit with a stretched exponential. These fitted values could be sensitive to the noise floor of the instrument, especially toward the tail of the decays (see supplementary discussion 1 and fig. S5). In the most conservative fitting scenario, we observed a >500-μs differential decay time, which, to the best of our knowledge, is the longest decay time reported for perovskite thin films and any direct bandgap semiconductor to date (fig. S6 and table S1) (*31*, *32*).

To better understand the dominant recombination mechanisms under device-relevant carrier densities (10^{14} to 10^{16} cm^{-3}), we performed intensity-dependent TRPL measurements. The TRPL data at different initial carrier densities were fit globally to the standard kinetic rate equation (fig. S7) (*31*, *33*, *34*). Figure 1E shows a log-log scatter plot of the nonradiative recombination rate (k_1) and the effective external bimolecular recombination rate ($k_{2,eff}^{ext}$) for the reference and eutectic perovskite. Most notably, k_1 was reduced for the eutectic perovskite by a factor of ~10, consistent with the low injection limit data shown in Fig. 1D ($1/k_1$ = 281.3 μs). In addition, the small difference between the $k_{2,eff}^{ext}$ of the reference and the eutectic perovskite likely indicated that remnant Zn^{2+} ions did not introduce substantial surface field effects that slowed recombination by spatially separating electron and holes, as was previously observed for two-dimensional (2D)/3D heterostructures (*31*). The slightly lower $k_{2,eff}^{ext}$ observed in the eutectic sample was likely indicative of the larger contribution from photon recycling (self-absorption) at higher PLQY values (*35*).

We used the TRPL data and previously described methods (*36*, *37*) to extract an average SRV of 0.3 ± 0.1 cm/s for the eutectic perovskite, compared with 4.3 ± 2.1 cm/s for the reference samples (Fig. 1F). The range in SRV accounted for the varying k_1 and $k_{2,eff}^{ext}$ values (Fig. 1E), as well as differences in the background doping densities and mobilities determined from terahertz spectroscopy and supported by intensity-dependent PLQY measurements (see supplementary discussion 2 and fig. S8). We also used an alternative approach to determine SRV based on thickness-dependent PLQY of eutectic perovskite films with thicknesses of 250 to 700 nm. This method yielded a corroborating SRV of 0.3 ± 0.02 cm/s (fig. S9). To contextualize these values, even the best perovskite thin films and single crystals typically demonstrate SRVs of 1 to 10 cm/s (fig. S10 and table S2). To the best of our knowledge, the eutectic SRV value is the lowest reported for lead halide perovskites and rivals the lowest reported for mature semiconductors such as passivated Si measured at 0.15 cm/s (*38*, *39*).

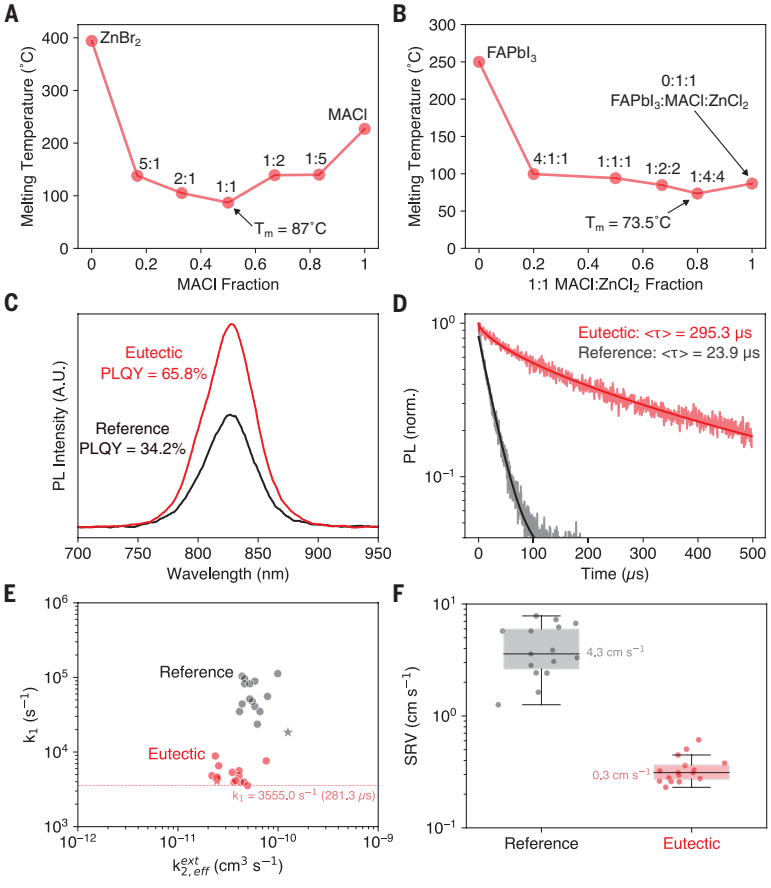

Fig. 1. Eutectic formation and optoelectronic properties of thin films. (**A**) Melting onset temperatures as a function of the $ZnBr_2$:MACl ratio. (**B**) Melting onset temperatures for different ratios of $FAPbI_3$ to equimolar MACl:$ZnCl_2$. (**C**) PL spectra of the reference and eutectic perovskite thin films with absolute PLQY values measured at 1-sun equivalent absorbed photon flux (excitation at 485 nm). (**D**) TRPL traces of the reference and eutectic perovskite thin films with stretched exponential fits and average lifetime values (<τ>). (**E**) Scatterplot of the nonradiative recombination rate constant k_1 versus the effective external radiative recombination rate constant $k_{2,eff}^{ext}$ for reference and eutectic perovskite thin films (15 samples each). Starred points represent the films shown in the TRPL in (D). (**F**) Boxplots of the SRV for the reference and eutectic perovskite thin films, with mean values shown.

Film structural and morphological properties

Microscopic analysis revealed that the eutectic additives facilitated improved morphological and structural film quality. Scanning electron microscopy (SEM) images of reference and eutectic perovskite thin films (Fig. 2A) showed that the eutectic perovskite had larger apparent domains of several micrometers. Liquid additives are known to increase apparent perovskite domain size (*21*, *22*, *24*), approaching the size of those seen in the reference films here of ~1 μm. The eutectic film apparent domains were larger, typically ~2 to 4 μm. Films fabricated with no additives or $ZnBr_2$ alone showed submicrometer apparent domain sizes, suggesting that the domain growth is facilitated by the eutectic (fig. S11). To further validate that this effect is specific to a synergistic interaction between $ZnBr_2$ and MACl, we fabricated antisolvent-based films with different

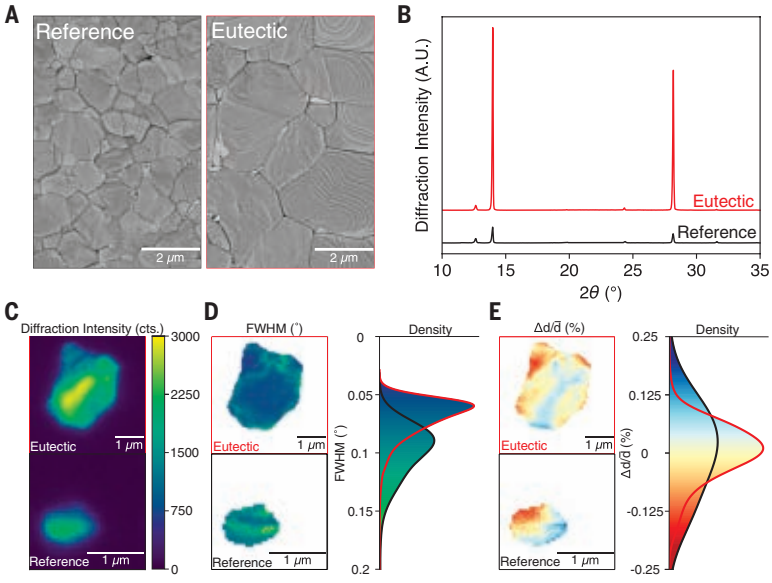

Fig. 2. Structural and morphological properties. (**A**) SEM images of reference and eutectic films. (**B**) Benchtop XRD patterns of reference and eutectic films. (**C**) Nanoprobe XRD mapping of diffraction intensity from a single grain in the reference and eutectic films. (**D**) Maps of FWHM of diffraction within a grain in each film (left) and corresponding kernel density profiles (right). (**E**) Maps of the change in local *d*-spacing relative to the mean *d*-spacing within a grain in each film (left) and corresponding kernel density profiles (right).

amounts of excess MACl with and without $ZnBr_2$ (fig. S12). Films fabricated with lower MACl loadings (<21% molar excess) showed comparable apparent domain sizes with or without $ZnBr_2$. At higher MACl loadings (≥21% molar excess), only films with $ZnBr_2$ had larger apparent domain sizes.

Benchtop x-ray diffraction (XRD) patterns of optimized films containing 35% molar excess MACl with (eutectic) and without (reference) 0.46% molar excess $ZnBr_2$ showed that the eutectic films exhibited a >10× increase in diffraction intensity relative to reference films and a lower $(001)_c$ full width at half maximum (FWHM) of 0.075° 2θ relative to 0.107° 2θ in the reference (where θ is the Bragg angle), indicating improved crystal quality in the films with a eutectic interaction (Fig. 2B). Crystallographic terracing was also visible on the surface of the eutectic perovskite (Fig. 2A). These improvements were not specific to antisolvent-based crystallization, because we observed similar effects on PL and morphology using antisolvent-free crystallization with the eutectic additives (fig. S13). We did not observe appreciable differences in the intermediate phases between films fabricated with and without antisolvent in in situ wide-angle x-ray scattering, and in both antisolvent and antisolvent-free cases, the eutectic films exhibited a large increase in cubic diffraction intensity (fig. S14).

To investigate the mechanism behind the structural improvements induced by the eutectic additives, we measured synchrotron-based nanoprobe XRD (nano-XRD). With an ~25-nm probe FWHM, nano-XRD enabled examination of crystallographic grain size and intragranular structural defects in reference and eutectic films. Both samples showed sufficient x-ray stability for high-fidelity repeated nano-XRD maps with a 0.1-s dwell per point (fig. S15). Diffraction intensity maps of a representative grain from each sample (Fig. 2C) showed that eutectic films exhibited larger grain size and higher intragranular diffraction intensity than reference films, corroborating benchtop results. Nano-XRD provided a true measure of crystallographic grain size, as opposed to the apparent domains shown by SEM. Larger-scale nano-XRD maps and statistical grain size distributions of both the eutectic and reference perovskite highlighted the increase in crystallographic grain size by introducing the eutectic (fig. S16). We attributed the increase in diffraction intensity to improved structural order in eutectic films, not merely an increase in preferential orientation.

Nanoprobe rocking curves revealed that the FWHM of diffraction at the nanoscale was substantially lower in the eutectic film than in the reference, affirming the removal of structural defects (decrease in mosaicity; Fig. 2D and fig. S17). We ruled out the contribution of grain size (Scherrer broadening) to FWHM because this effect was only observed in samples with grain sizes of <200 nm (*40*). The reference already has a grain size >750 nm. The eutectic sample also exhibited a homogenized local *d*-spacing relative to the reference, indicating that the crystallites in the eutectic film contain less microstrain (Fig. 2E and fig. S17), which can lead to improved PL properties (*41*). This nanoscopic analysis confirmed a reduction in intragranular structural defect density in the film annealed with eutectic additives.

Microscopic evolution of the film during annealing

To understand how the eutectic enabled these improvements, we turned to microcharacterization of films at different stages of annealing. We performed synchrotron-based x-ray fluorescence (XRF) and x-ray excited optical luminescence (XEOL) microscopy to track the evolution of the film composition and optical luminescence during annealing, with spatial resolution beyond the visible-light diffraction limit (*42*) (Fig. 3, A and B). XEOL is the x-ray analog of PL and determines band-to-band luminescence after excitation by a hard x-ray (*43*, *44*). Samples showed sufficient x-ray stability for nano-XRF and XEOL mapping (fig. S18).

The snapshots of elemental distribution taken during ex situ annealing of the eutectic film for 1 hour at 100°C revealed that the distribution of Zn and halides changed substantially, including a large change coincident with a sharp increase in XEOL intensity at the end of the process (Fig. 3, A and B, and fig. S19). Before annealing, Zn was distributed in submicrometer clusters, and the XEOL intensity was low. At between 5 and 20 min of annealing, Zn pooled into clusters several micrometers in size that gradually elongated, and the XEOL intensity increased slightly. Zn increasingly segregated from Pb during annealing, as shown by a monotonically decreasing Kendall rank correlation coefficient of Zn and Pb distributions as a function of annealing time (fig. S20), which are consistent with solid-to-liquid segregation of impurities during crystallization (*45*).

In the fully annealed film (60 min), Zn segregated to the crystallographic grain boundaries, as verified by correlative nano-XRD and nano-XRF (fig. S21). Depth-dependent x-ray photoelectron spectroscopy and time-of-flight secondary ion mass spectrometry measurements revealed segregation of Zn to the top and bottom interfaces of the perovskite as well, corroborating the segregation to boundaries observed through XRF and suggesting that Zn forms a full shell on the outside of each grain (figs. S22 and S23). XEOL intensity in the fully annealed eutectic film increased by about a factor of 3 relative to the earlier annealing time points. Notably, the sharp increase in XEOL intensity at longer annealing times as Zn segregated fully to grain boundaries was not present in reference samples without $ZnBr_2$ (figs. S24 and S25).

XRF maps of Br and I over the course of annealing indicated that the eutectic additives had substantial impacts on halide redistribution throughout annealing (fig. S19). Kendall rank correlations of the distributions of Br and I revealed that the halide stoichiometry became more homogeneous in eutectic films relative to the reference (fig. S26). Halide homogeneity is critical to perovskite PV efficiency and operando stability (*46*, *47*).

Given the segregation of Zn to the boundaries of the film, we hypothesized that the eutectic interaction between zinc halides and MACl in the presence of $FAPbI_3$ (Fig. 1B) could play an active role during film annealing and lead to local melting at the boundaries (Fig. 3C). In a model experiment using bulk $FAPbI_3$ single crystals and

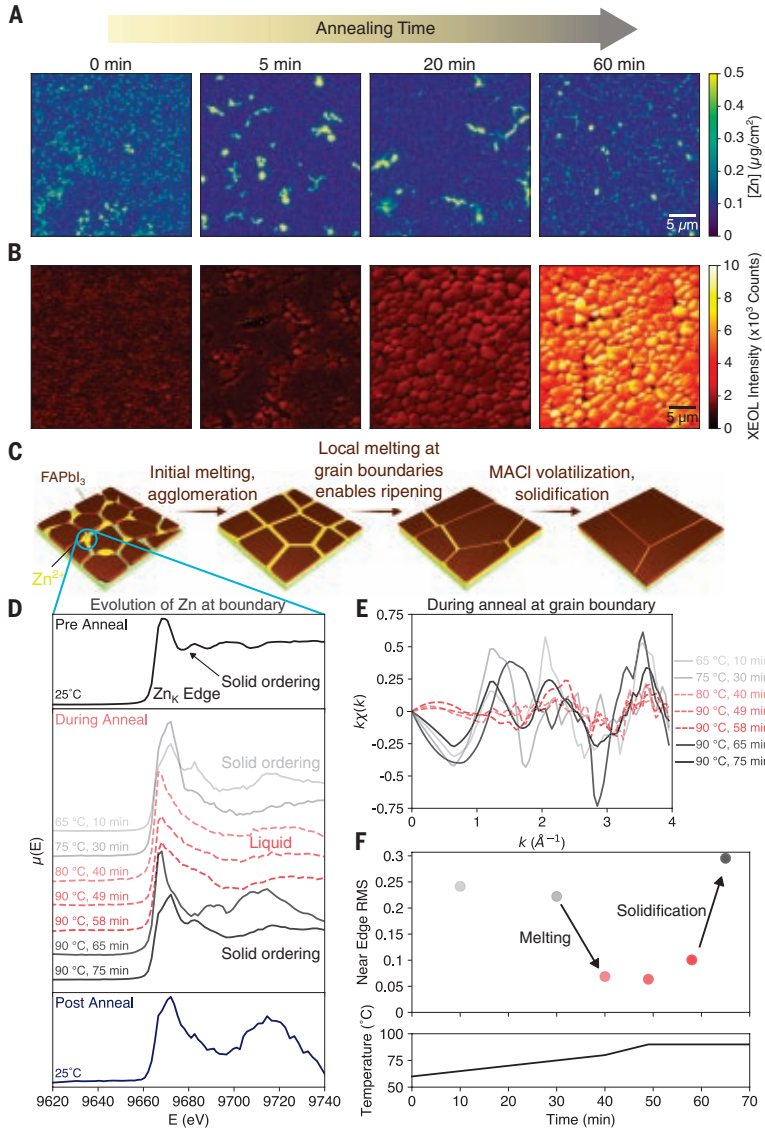

Fig. 3. Film formation mechanism. (**A**) Microprobe XRF maps of Zn distribution in the film as a function of annealing time. (**B**) Microprobe XEOL maps for the same areas plotted in (A). (**C**) Schematic of the mechanism for eutectic-assisted crystallization. (**D**) Nanoprobe XAS of the Zn K absorption edge capturing a grain boundary before, during, and after annealing. (**E**) *k*-weighted $\chi(k)$ transformed from the absorption data taken during annealing shown in the middle panel of (D). (**F**) Top: RMS values of the *k*-weighted $\chi(k)$ data plotted in (E) as a function of annealing time. Bottom: annealing profile for the corresponding time points.

Figure 3D shows the normalized Zn K-edge absorption spectra of Zn at a grain boundary before, during, and after annealing. Before annealing, there were characteristic oscillations in the spectrum that indicated local structural ordering around the Zn atoms (*48*). During annealing, most of these oscillations disappeared once the temperature reached ~80°C, suggesting that the local coordination environment around the Zn atom lost its crystalline short-range order characteristics (*49*). Dampening of fine structure oscillations indicating the loss of structural order was seen during heating and is indicated by red dashed lines in Fig. 3D. Given that the loss of structural ordering was isothermal or within minimal absolute temperature changes (~1%) and that the near-edge region was dominated by multiple scattering paths, the role of the Debye-Waller factor in reducing scattering oscillations was small. Instead, the data were consistent with a transition from a structured crystalline phase into a locally disordered, liquid-like state at the grain boundaries.

To inspect the structural transition more closely, we examined a *k*-weighted $\chi(k)$ plot (Fig. 3E) where k is the wave vector and $\chi(k)$ tracks deviation in absorption from the smooth atomic absorption background. The *k*-weighting reduced the dominance of the edge step and emphasized oscillatory deviations from the background. To provide a measure of the changes in oscillation magnitude over the course of annealing, we show the root-mean-square (RMS) amplitude of $\chi(k)$ (Fig. 3F). A loss of structural ordering around the Zn occurred between 40 and 60 min of annealing at a temperature of 80° to 90°C, when there was a decrease in the RMS oscillation. As isothermal 90°C annealing continued for 65 min, structural oscillations reappeared. Finally, when the sample returned to room temperature, near-edge structure was again seen.

We discuss this evolution in local order about Zn at the grain boundary in consideration of a local melting mechanism. The initial molar ratio in the film of R-NH_3Cl to zinc halide was high. However, annealing volatilized the R-NH_3Cl (fig. S32) and concentrated the Zn at the boundaries (Fig. 3A). As the local composition moved toward the 1:1 eutectic composition, per the phase diagram of Fig. 1, the local melting temperature decreased until the annealing temperature was above the local melting temperature. In this case, local short-range order was lost, as shown by the XAS (Fig. 3D). Continued annealing would drive off more of the R-NH_3Cl. The local composition at the grain boundary would become R-NH_3Cl poor (<1:1), and the melting temperature would increase. Well past the eutectic composition, the local melting temperature would rise above the annealing temperature, enabling a reordering of the Zn into a solid at the final stages of the isothermal annealing. This mechanism of a time-delayed, isothermal loss of structural order followed by its return at later times is consistent with the in situ XAS microscopy of Zn shown in Fig. 3D.

$MACl + ZnBr_2$ powder, we observed interfacial melting and dissolution of the crystalline interfaces into the eutectic mixture at 150°C, followed by a resolidification of the molten interfaces upon cooldown and fusion of the single crystals (see supplementary discussion 3 and fig. S27).

To investigate whether such a mechanism was active in the thin films with eutectic additives, we examined the local structural environment of Zn around a grain boundary during annealing using in situ synchrotron-based nanoprobe x-ray absorption spectroscopy (XAS) (Fig. 3, D to F, and fig. S28). These studies were performed on films fabricated with isopropylammonium chloride (iPACl) in place of MACl to slow the volatilization of the R-NH_3Cl and extend the crystallization window (fig. S29), thus facilitating the in situ spectroscopy. We emphasize that iPACl and MACl formed similar eutectics with Zn, and films fabricated with iPACl and MACl exhibited similar properties when combined with zinc halides (figs. S30 and S31).

SEM images of apparent domains throughout film annealing also revealed time-delayed domain growth (fig. S33). The similar domain size and density early in the annealing relative to reference indicated that enhanced domain growth in eutectic films was enabled by the additive interaction during annealing rather than by a substantive alteration of nucleation. Experiments with $ZnBr_2$ alone showed a limited increase in domain size (fig. S11), indicating that the interaction of MACl and $ZnBr_2$ led to grain growth.

The observation of the loss of structural order at Zn-enriched grain boundary regions (Fig. 3, D to F), combined with knowledge of the eutectic formed by the additives (Fig. 1, A and B), suggested that local melting at boundaries was key to grain growth and reduction in structural defect density observed in eutectic films. Diffusion coefficients

were much higher in liquid phases relative to solids [~10^4 times (*50*)] and enabled increased mass transport across these liquid boundaries during annealing, thus accelerating grain ripening (Fig. 2A and fig. S33). The loss of structural order at the boundaries during crystallization was also consistent with the decreased strain observed in the final eutectic perovskite films (Fig. 2, D and E), and grain growth with minimal interfacial strains was expected to result in reduced structural defect density (*51*), as was observed. The eutectic-mediated grain growth emerged from enhanced ripening due to an interfacial liquid layer, similar to the liquid droplet present in the vapor-liquid-solid (VLS) growth technique commonly used in nanowire fabrication (*52*, *53*) and also applied to indium phosphide (InP) polycrystalline thin films (*54*). However, unlike in traditional VLS growth, the mass transfer in our approach specifically occurred between grains in a ripening process rather than from an externally introduced gaseous precursor.

Because the XRF and XPS results showed that Zn decorated grain boundaries of the perovskite in the fully annealed film, we examined the driving force for and consequences of Zn surface decoration with density functional theory computations of surface slabs. First, to investigate the electronic role of Zn impurities in the bulk, we simulated native point defects and Zn substitutional and interstitial defects in bulk $FAPbI_3$ (figs. S34 and S35). Zn interstitial defects created a low-formation-energy donor-type defect. Along with the low-formation-energy native acceptor defect (Pb vacancy, V_{Pb}), the Zn interstitial donor pinned the equilibrium Fermi level as intrinsic under the Pb-rich conditions, as expected based on the synthesis procedure (see the materials and methods).

Figures S36 to S39 show optimized structures of $FAPbI_3$ surface slabs, with Zn and ZnI_2 units inserted on Pb and I and coordinated at surface interstitial sites on the PbI_2-terminated (010) surface. We calculated the surface energies for the surface in its pristine form and in the presence of various native defects and Zn or ZnI_2; these values are shown for Pb-rich conditions in fig. S40 and table S3. The I_2-terminated (010) pristine surface of $FAPbI_3$ had a surface energy $\sigma = 0.42$ J/m^2, which was lower than that of the FAI-terminated surface, and it slightly lowered to $\sigma = 0.41$ J/m^2 in the presence of an iodine vacancy (V_I). The lowest surface energy was found in the presence of a ZnI_2 interstitial defect: $\sigma = 0.40$ J/m^2. All other native defects and alternative Zn and ZnI_2 defect sites increased σ and thus made the surface less stable, with the modest exception of Zn at the Pb site, which showed a comparable $\sigma = 0.46$ J/m^2. ZnI_2 was stabilized at the surface by the formation of a ZnI_4 tetrahedron, where Zn bonded with two bridging I atoms in neighboring Pb–I–Pb units, as shown in fig. S36.

The presence of low-energy Zn_{Pb} and ZnI_2 interstitials at the surface reduced the density of midgap states compared with the native vacancy-rich surface (V_I). Figure S41 shows the computed electronic density of states for the pristine, defect-free $FAPbI_3$ surface compared with that in the presence of the reasonably stable surface defects V_I, Zn_{Pb}, and ZnI_2 interstitial. The low-energy Zn defects led to a reduction of electronic defect states compared with the V_I surface, as shown in the yellow highlighted region in fig. S41, effectively passivating defects at boundaries and interfaces.

Optoelectronic device properties

We fabricated solar cells in a n-i-p configuration using both the reference and eutectic perovskites (cross-section shown in fig. S42). The improved crystalline quality and passivation in the eutectic perovskite resulted in an increased average PCE of 25.5%, compared with an average PCE of 24.8% for the reference devices (Fig. 4A). This improvement was primarily driven by enhanced V_{OC} and FF resulting from the suppression of nonradiative recombination in the eutectic perovskite. Figure 4B shows the current density-voltage (*J-V*) curve of a champion eutectic device with an antireflection coating applied, exhibiting PCEs of 26.5 and 26.4% for forward and reverse scans with V_{OC} and FF exceeding 1.21 V and 84.5% (table S4), respectively. The device PCE was certified by Newport, an accredited testing center, achieving efficiencies of 25.9 and 25.8% for reverse and forward scans, respectively (fig. S43). The certified EQE spectrum is shown in fig. S44. To the best of our knowledge, a certified V_{OC} of 1.208 V stands as the highest among all certified V_{OC} values reported to date for n-i-p structured perovskite solar cells (table S5). As the nonradiative voltage loss was reduced, the V_{OC} measured without an aperture mask reached 1.227 V, corresponding to >97% of the detailed balance limit (fig. S45).

The optoelectronic enhancements seen in the eutectic perovskite also enabled efficient near-infrared LEDs. Figure 4C shows the electroluminescence EQE as a function of voltage for the eutectic solar cell under forward bias, showing a peak EQE of 24.2% (the full electroluminescence spectrum is shown in fig. S46). Additionally, it is well established that decreasing the thickness of the perovskite active layer can enhance LED EQE due to diminished light reabsorption. We fabricated LEDs with the same device architecture and a thinner (~350-nm) perovskite layer, which resulted in an LED EQE of 26.1% (Fig. 4C),

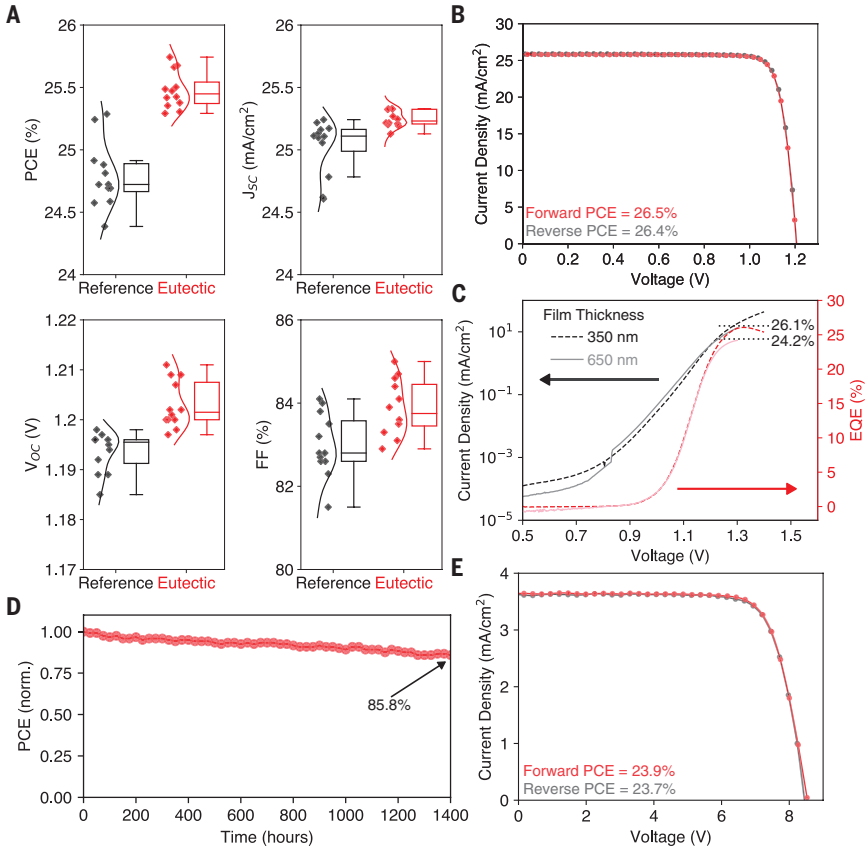

Fig. 4. Photovoltaic and LED device performance. (**A**) PCE, short-circuit current density (J_{SC}), V_{OC}, and FF distributions of the reference and eutectic devices (12 devices each). (**B**) *J-V* curves for the champion eutectic device with an antireflection coating applied. (**C**) LED EQE of eutectic devices with 350- and 650-nm perovskite thicknesses. (**D**) Maximum power point tracking of a eutectic device at 85°C in an N_2 atmosphere. (**E**) *J-V* curves for the best-performing eutectic minimodule (15.58 cm^2) with an antireflection coating applied.

surpassing the highest EQEs reported for near-infrared $FAPbI_3$ LEDs based on an n-i-p solar cell structure to date (*55*). The eutectic LED exhibited an operational half-lifetime of 1182 min (fig. S47).

To evaluate the long-term operational stability of the eutectic PV device, we tested it under ISOS-L-2I conditions: maximum power point tracking under continuous 1-sun equivalent illumination at 85°C (Fig. 4D). For the long-term stability test, 2,2′,7,7′-tetrakis(*N*,*N*-di-*p*-methoxyphenylamine)-9,9′-spirobifluorene (spiro-OMeTAD) was replaced with a thermally stable hole-transporting layer (HTL), poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine], and the PCE of this device was found to be 24.1%. The eutectic device exhibited exceptional long-term stability, retaining 85.8% of its initial PCE after 1400 hours of continuous testing.

We also fabricated perovskite solar modules to evaluate the scalability of the eutectic mechanism with spiro-OMeTAD as the HTL. The module consisted of seven subcells connected in series. As shown in Fig. 4E, the best-performing eutectic module showed a certified aperture PCE of 23.7 and 23.9% for reverse and forward scans, respectively (fig. S48 and table S6). This high PCE could again be attributed to the lowered defect density, where the V_{OC} per subcell was calculated to be 1.219 V, which approaches the best unmasked V_{OC} of the small-area devices. The previously reported high-efficiency module with a PCE ≥23% showed hysteresis (*55*), whereas that of the present eutectic module was negligible.

Discussion

The volatile eutectic mechanism presented in this work represents a new strategy for additive design to simultaneously modulate crystallization and control passivation for perovskites and more broadly across optoelectronics. With a small set of precursors, we have been able to set several optoelectronic records, with substantial room for the discovery of even more effective designer volatile eutectics. These results demonstrate that the improvements seen in small-area devices based on the eutectic interaction translate to improved large-area fabrication of perovskite optoelectronic devices.

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SUPPLEMENTARY MATERIALS

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MACHINE LEARNING

Advancing mathematics research with AI-driven formal proof search

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Large language models (LLMs) increasingly excel at mathematics tasks, but their unreliability limits their utility in mathematics research. A mitigation is to use LLMs to generate formal proofs in languages such as Lean, in which the compiler verifies every proof step. We present the first demonstration of this method's value in solving open problems at scale. We built an artificial intelligence agent for formal proof search that autonomously resolved nine of 353 open Erdős problems, proved 44/492 On-Line Encyclopedia of Integer Sequences conjectures, and is being deployed in combinatorics, optimization, graph theory, algebraic geometry, and quantum optics research. Even a basic agent alternating LLM-based generation with Lean-based verification replicated the Erdős successes. These findings demonstrate the power of formal proof search as an enabler of autonomous mathematical discovery.

Large language models (LLMs) have recently shown promise in solving complex mathematics problems (*1*, *2*), but unreliability remains a primary barrier to their integration into mathematics research. Because LLM-generated natural language proofs can contain subtle logical errors or "hallucinations," they require expensive expert review. Mistakes in unreviewed intermediate steps can cascade through an informal proof, limiting the complexity of tasks that can be delegated to artificial intelligence (AI).

Recent efforts (*3*, *4*) mitigate these issues by using AI to generate proofs in formal languages such as Lean (*5*), in which a compiler automatically verifies every logical step. So far, successes of this paradigm have been concentrated in competition mathematics and the human-aided formalization of natural language arguments (*6*). In this paper, we demonstrate its broader potential through a large-scale evaluation on open research-level problems.

To this end, we developed a framework, AlphaProof Nexus, for LLM-aided proof generation, and used it to build a basic agent in which a set of subagents independently searches for formal proofs with feedback from the Lean compiler. We also developed a "full-featured" agent in which subagents are coordinated using an evolutionary algorithm (*7*) and can use AlphaProof (*3*), a system for olympiad-level Lean theorem-proving based on reinforcement learning, as a focused proof tool.

Our full-featured agent autonomously solved 9 Erdős problems out of 353 attempted, including two questions that had been open for 56 years (*8*–*10*). It also proved 44 out of 492 open conjectures from the On-Line Encyclopedia of Integer Sequences (OEIS), resolved an open question on Hilbert functions in algebraic geometry, improved an open bound in convex optimization by discovering an algorithmic parameter schedule, identified several misformalizations in the literature, helped resolve an open problem from Green's well-known list (*11*), and is aiding ongoing research efforts on quantum optics and graph theory. We provide expert-vetted, natural-language translations of the key formal proofs in the supplementary materials (SM).

To understand the impact of the agent design on these results, we did a post hoc analysis of the performance of the full-featured and basic agents, as well as two agents with intermediate capabilities, on the nine Erdős problems solved by the full-featured agent. Notably, the basic agent solved all nine problems.

Overall, our results demonstrate the utility of an LLM-powered formal proof search as an enabler of reliable, automated mathematical discovery. They also point to an ongoing shift from specialized trained systems toward simple agentic loops as LLMs become more capable. Agent-generated Lean proofs for all problems we report solving are available in the accompanying open-source repository (*12*).

AlphaProof Nexus

Lean

Lean (*5*) is a proof assistant in which definitions, theorems, and proofs are all mechanically verified code. Proofs are constructed through a sequence of applications of tactics, or elementary proof steps. Lean's compiler "executes" a proof tactic-by-tactic, tracking the proof goals pending after every tactic. A proof is correct if it leads the compiler to a state with no pending goals.

Lean includes a special "sorry" tactic, which immediately closes pending goals while passing the type checker. Proving a theorem thus amounts to the generation of type-safe code without sorry tactics.

Input and output

AlphaProof Nexus is a new framework for agents that query frontier LLMs and the Lean compiler. The agents take as input a Lean file consisting of a target theorem with sorry in place of a proof, along with definitions and library imports on which the theorem depends (we refer to such a file as a proof sketch). Optionally, the user may input additional natural-language context and domain knowledge encoded in Lean.

The input sketch is annotated with user-provided markers that delineate which code segments the agent may modify (see Fig. 1). Within EVOLVE-BLOCK markers, the agent can introduce helper lemmas, definitions, and proof steps. EVOLVE-VALUE markers are used to enclose expressions (e.g., parameters) whose values the agent can change. On successful termination, the agent outputs a sorry-free proof of the target theorem.

Agent architecture

The basic agent in AlphaProof Nexus [agent (A)] consists of a set of prover subagents that execute independently with no shared state. Each subagent is a "Ralph loop" (*13*) consisting of a sequence of episodes: multiturn LLM inference loops (based on Gemini 3.1 Pro) in which the subagent can reason through chain-of-thought and refine the sketch using a search-and-replace tool. After each turn, the subagent uses Lean to check that the current proof sketch compiles; if it does not, Lean's error message is used to direct the next turn. If the sketch contains sorry when the episode terminates, the subagent adds a comment to the sketch summarizing the lessons learned from the episode. The resulting sketch is the input to the next episode.

We extended the basic agent into one (B) that can query AlphaProof (*3*) to fill out missing parts of sketches. Queries to AlphaProof can return a proof, a disproof (a proof that the submitted subgoal is false), or a failure message; proofs are directly substituted into the sketch, and disproofs and failure messages are fed into the prover's prompt.

Separately, we developed an evolutionary agent (C), inspired by AlphaEvolve (*7*), in which prover subagents sample from and contribute to a shared population database of sketches. A challenge here is

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Fig. 1. Example inputs/outputs for an AlphaProof-equipped agent (applied to Erdős #125). The user provides a Lean file with a specification of the problem, and an empty proof body replaced with the sorry placeholder. (**A**) Modifications are permitted only within EVOLVE-BLOCK and EVOLVE-VALUE markers. (**B**) During sketch refinement, the prover subagent is shown an assembled prompt template with the current proof, and optionally prior attempts/sketches, their Elo ratings, and feedback from AlphaProof's attempts on unsolved goals. (**C**) The prover reasons about the problem informally and invokes tools. In this example, the prover invoked AlphaProof which resolved all but one goal. The prover then decomposed that goal into three simpler lemmas, and called AlphaProof again, which then resolved all remaining goals. The agent also produced a natural language summary of its attempt at the end of generation.

the mismatch between evolutionary algorithms, which typically assume a graduated fitness landscape, and formal proof evaluation, which is inherently binary. To bridge this gap, we used a pool of rating agents (based on the less expensive Gemini 3.0 Flash) to construct relative rankings of sketches based on their plausibility, clarity, and novelty. We aggregated these rankings into Elo ratings for the sketches and used a sampling procedure based on the P-UCB formula (*14*, *15*) to drive the search.

Finally, we combined the AlphaProof and evolution capabilities into a "full-featured" agent (D; see Fig. 2), in which prover subagents can use AlphaProof as well as evolutionary search. We used this agent as the instrument in our exploration of open research problems.

Systematic evaluation on open problems

Erdős problems

Bloom maintains an online catalog (*16*) of more than 1200 open problems posed by Paul Erdős and his collaborators. The open-source Formal Conjectures repository (*17*) contains Lean formalizations of a subset of these problems. We ran Agent (D) on this subset (comprising 353 problems at the time of the run), terminating the search if no formal proof was found within 3000 episodes. The agent solved nine problems (Table 1 and supplementary text); after each solve, experts on our team validated that the Lean statement faithfully captured the original conjecture. We have publicly shared the proofs, and our results have been logged on Tao's wiki on AI contributions to Erdős problems (*18*).

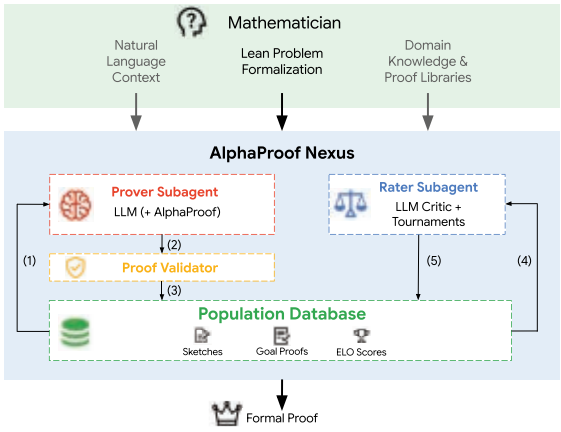

Fig. 2. Design of the full-featured AlphaProof Nexus agent. The mathematician provides as input a Lean theorem with sorry for a proof, and optionally, natural language context and additional domain knowledge encoded in Lean. The agent architecture consists of a basic generation-validation pipeline and an optional evolutionary framework. An LLM-based prover subagent attempts to solve the problem by refining proof sketches (2). The subagent may optionally call AlphaProof as a tool; each invocation of AlphaProof on a goal returns whether a formal proof or disproof was found, or whether it was unsuccessful in resolving the goal. The sketch produced in the end is checked by a validator to ensure the problem statement was not changed unsafely and the proof compiles. If all goals are successfully proved, the agent outputs the final Lean proof. This basic pipeline can be extended with an evolutionary population database and rating mechanism. In this configuration, validated sketches are admitted into the Population Database (3). Simultaneously, rater subagents sample previous attempts (4), which are then ranked by an LLM critic in matches. The match results are admitted back into the database (5) to update the Elo scores of the sketches. These scores are then used by the evolutionary algorithm to sample prior sketches from the database, constructing an overall prompt—which includes any optional inputs from the mathematician—to condition new episodes (1).

Several solutions require sophisticated constructions and the synthesis of distinct mathematical arguments. For example, problem #12-(i), posed by Erdős and Sárközy in 1970 (*19*), asks whether there is an infinite set A that satisfies a restrictive divisibility constraint—no element may divide the sum of two larger elements—and which satisfies the density condition $\liminf_{N\to\infty} \frac{|A\cap[1,N]|}{\sqrt{N}} > 0$; this problem had received attention in multiple prior works (*8*–*10*). Our agent constructed A as an infinite union of disjoint "blocks" $B_i \subseteq [P_i 1.1P_i]$ for a suitably rapidly growing sequence (P_i). To satisfy the local divisibility conditions, the proof integrates the Chinese Remainder Theorem with the properties of sets that avoid length-3 arithmetic progressions.

Problem #125 concerns the sumset $A + B$, where A is the set of nonnegative integers whose base-3 representation uses only the digits 0 and 1, and B is the analogous set in base 4. The question of whether the lower density of $(A + B)$ is positive has been open since 1996 (*20*). Our agent resolved the conjecture by synthesizing an inductive thinning argument that exploits the Diophantine proximity of the two multiplicatively independent bases ($3^m \approx 3^k$).

The agent also served as a tool for detecting and fixing misformalizations. For example, in Erdős problems #125 and #741-(i), the interpretation of "density" in the original informal statements was amended to "lower density" and "upper density" respectively, after our full-featured agent found proofs using density as "natural density." Following the correction of the ambiguity, the agent was still able to resolve the questions.

Failure analysis: We analyzed the highest-scoring sketches (measured by Elo) across a random sample of problems on which our agent failed. First, the agent frequently offloaded a problem's core difficulty into a single sorry within a helper lemma that reiterated the target statement in a slightly different form. Explicitly prompting against this behavior failed to prevent it. Second, for several problems, the top sketches relied on lemmas marked with sorry that the agent claimed were established results in the mathematical literature. Upon manual inspection, these lemmas proved to be hallucinations. These failure modes underscore the value of end-to-end formal verification.

OEIS

We also applied the agent to systematically explore open problems in the OEIS (*21*), a massive repository of integer sequences and their known and open properties. We used Gemini to autoformalize 492 open questions from the OEIS and applied the agent to the resulting Lean statements. To help guard against misformalization, the agent was required to prove "test lemmas" verifying the first few terms of each sequence against its formal definition before attempting the target conjectures. The agent found formal proofs for 44 conjectures that a manual review found to be correctly formalized and previously unproven. Two of these proofs appear in the SM.

Deployment in mathematics research

Optimization theory

Agent (D) resolved an open question in optimization: proving an exact $\mathcal{O}(1/t)$ convergence rate for the Anchored Gradient Descent-Ascent (GDA) algorithm for min-max convex-concave optimization, thereby tightening an existing slower bound (*22*). The formal proof departs from continuous-time ordinary differential equations (ODE) analysis used in previous work, instead using a discrete-time recurrence-based approach. The agent did not merely verify a fixed algorithm; we marked the learning schedule as a parameter within an EVOLVE-VALUE block in the input file, allowing the agent to simultaneously search for the schedule and the proof, ultimately discovering a novel parameter choice that yields the stronger guarantee. We previously released a deformalized version of the proof as a preprint (*23*); this informal proof is reproduced in the SM. Subsequent work (*24*) has extended this result.

Graph theory

The graph reconstruction conjecture (*25*) is one of the oldest open problems in combinatorics. It asserts that every finite simple graph with at least three vertices is determined, up to isomorphism, by the multiset of its vertex-deleted subgraphs, known as its deck. AlphaEvolve (*7*) experiments helped us formulate two bipartite variants of the graph reconstruction conjecture, along with a proposed full reconstruction algorithm. Agent (D) proved the correctness of the proposed algorithm under an additional type condition on the bipartite graph. The corresponding unrestricted bipartite reconstruction statement remains open; in that case, the agent generated proof sketches and strategies that helped clarify the structure of the problem and led to simplified reformulations of the conjectures originally suggested by AlphaEvolve.

Separately, the agent proved an open graph theory conjecture regarding a bound on the maximum number of leaves over all spanning trees of a graph G, relating it to the maximum number, over all vertices $v \in G$, of independent sets in the neighborhood of v. The problem was posed by Graffiti (*26*), an automated conjecturing system, in 1996, and points to an interesting future opportunity to close the loop between AI-based conjecturing and proof.

Algebraic geometry

We evaluated Agent (D) on eight algebraic geometry problems, ranging from textbook-style exercises to open research questions on Hilbert functions, and it solved two of the four open problems. One of these problems was Zanello's conjecture on log-concavity of pure

Table 1. Open problems from the Erdős problems repository autonomously resolved by our full-featured agent. The asterisk indicates a variant of the main problem with this number. Problem #26 (annotated with the dagger symbol) is a more general variant of a question posed by Erdős, but was not posed by Erdős himself. CRT, Chinese Remainder Theorem.

ID	Conjecture Summary	Proof Technique
12 (i)	(1970) Existence of $A \subset \mathbb{Z}^+$ s.t. $a \nmid (b+c)$ with $\lvert A \cap [1,N] \rvert / \sqrt{N} > \delta > 0$ infinitely often	Block-based construction via CRT and 3-AP avoiding sets
12 (ii)	(1970) Existence of $A \subset \mathbb{Z}^+$ s.t. $a \nmid (b+c)$ with $\lvert A \cap [1,N] \rvert \gg N^{1-\epsilon}$ for any ϵ and N sufficiently large	Block-based construction via CRT and 3-AP avoiding sets
125	(1996) $\{\sum \epsilon_k 3^k \mid \epsilon_k \in \{0,1\}\} + \{\sum \epsilon_j 4^j \mid \epsilon_j \in \{0,1\}\}$ has lower density zero	Inductive thinning via Diophantine $\sim 3^m \approx 4^k$
138*	(1981) Van der Waerden numbers satisfy $W(k+1) - W(k) \to \infty$	Greedy coloring extension with monochromatic intersection lemma
152	(1994) Sufficiently large Sidon sets A contain many isolated points in $A + A$	Elementary argument by bounds on interior points, shifted neighbors and quadruples
741 (i)	(1994) If $A + A$ has upper density, there exists a decomposition $A = A_1 \sqcup A_2$ s.t. $A_1 + A_1, A_2 + A_2$ have positive upper density	Bounding of upper density of $A_i + A_i$ via cases on upper density of A
741 (ii)	(1994) Existence of a basis A of order 2 with $A_1 + A_1, A_2 + A_2$ having bounded gaps for all A_1, A_2 such that $A = A_1 \sqcup A_2$	Explicit construction via rapidly growing sequence
846	(1992) Existence of infinite set A such that any finite subset contains many non-collinear points, but is not a finite union of non-collinear sets	Label K_∞ vertices with terms from fast-growing sequence, apply map on edges $\{i,j\} \mapsto \left(x_i + x_j, x_i^2 + x_i x_j + x_j^2\right)$ to obtain A
26*,†	(1995) Existence of A such that upper density of $A + k$ is $< 1 - \frac{1}{4}$ for all naturals k	Block-based construction using increasing sequence of primes, CRT, and bounding u. density by cases

O-sequences (*27*), which arise as Hilbert functions of monomial Artinian level algebras. For sequences with codimension 3 and type 2, the weaker unimodality statement had previously been proved (*28*). The agent's formal proof establishes log-concavity in this case, which is notable because Zanello had shown that log-concavity fails for many other families of pure O-sequences. Moreover, in the broader class of level Hilbert functions, where the defining algebra need not be monomial and hence the Hilbert function need not be a pure O-sequence, Zanello showed that non–log-concave examples exist in all remaining codimension-type pairs except the still-open case (3,2). The agent-generated argument is substantial, using a nontrivial reformulation of the Hilbert function and a detailed case analysis of the resulting second-difference inequalities.

We also deployed Agent (D) within an ongoing effort on the AI-assisted formalization of the Stacks Project algebraic geometry textbook (*29*). Here, the source material sometimes provides only brief (natural language) proof outlines or relies on results that are not widely known. The agent's ability to autonomously search for intricate formal proofs was especially valuable for these cases, such as Lemma 04KM on separable field extensions.

Additive combinatorics

Our agent helped resolve problem #57 from Green's well-known list of open conjectures (*11*). The problem asks whether two specific quadratically structured function spaces coincide. The functions of interest map elements of an Abelian group G to the complex numbers. Here, our agent autonomously solved a variant of the problem in which the functions are real-valued, but a personal communication from Green clarified that the complex-valued case was the intended challenge. Although the agent could not immediately prove the intended version of the problem, numerical heuristics using floating-point arithmetic provided a candidate counterexample (the cyclic group $\mathbb{Z}/3\mathbb{Z}$ and a specific separating functional). We formalized the problem of whether this counterexample disproves the correct conjecture, and the agent autonomously proved that it indeed does.

Quantum optics

With M. Krenn, we investigated a set of quantum optics problems concerning the existence of monochromatic quantum graphs with N vertices and d colors drawn from domains such as the reals, the complex numbers, and $\{-1, 0, 1\}$. These constructions correspond to N-particle quantum states with local Hilbert space dimension d– in particular, high-dimensional Greenberger–Horne–Zeilinger (GHZ) states realizable through linear optics (*30*). Our agent resolved multiple conjectures of this form, in particular with $N = d \in \{4, 6, 10\}$.

Impact of agent architecture and model

Because exploring a large space of open problems is computationally expensive, we deployed the full-featured agent (D) based on its strong competition benchmark performance. To determine which architectural components drive these successes, we compared its performance against agents (A), (B), and (C) on the Erdős problems listed in Table 1.

We evaluated the agents by analyzing the solve rate against the inference cost per successfully proven problem. For agents (A) and (B), we ran 100 independent, single-subagent attempts. We estimated the solve rate—the probability of finding a proof when K attempts run in parallel—by chunking these attempts. We measured inference cost in USD because it provides a natural common currency for heterogeneous compute configurations [e.g., agent (D) uses Gemini 3.0 Flash for raters and Gemini 3.1 Pro for provers], and also directly dictates the accessibility of our tools to the broader mathematics community. Detailed experimental protocols, including chunking procedures, timestamp calculations, and variance estimations, are provided in the SM.

Fig. 3 compares the agents across six Erdős problems (remaining results are in the SM). Agents (A) and (B) perform similarly—within the margin of error—on four problems, although (B) is more cost-efficient on 12-(ii) and 125. Agent (D) outperforms both on problems 138 and 125, proving to be two to five times more cost efficient, but roughly half as cost efficient on the others. Wall-clock time comparisons for agents (A), (B), and (D) broadly followed these USD cost trends, with (D) again substantially outperforming the others on problems 138 and 125. Because our agents are stochastic, per-problem inference costs exhibit high variance. Notably, the reported costs reflect only the successful runs rather than the full computational investment required for problem discovery.

Next, we evaluated AlphaProof in standalone tree-search mode and versions of agent (A) based on smaller models (Gemini 3.0 Flash,

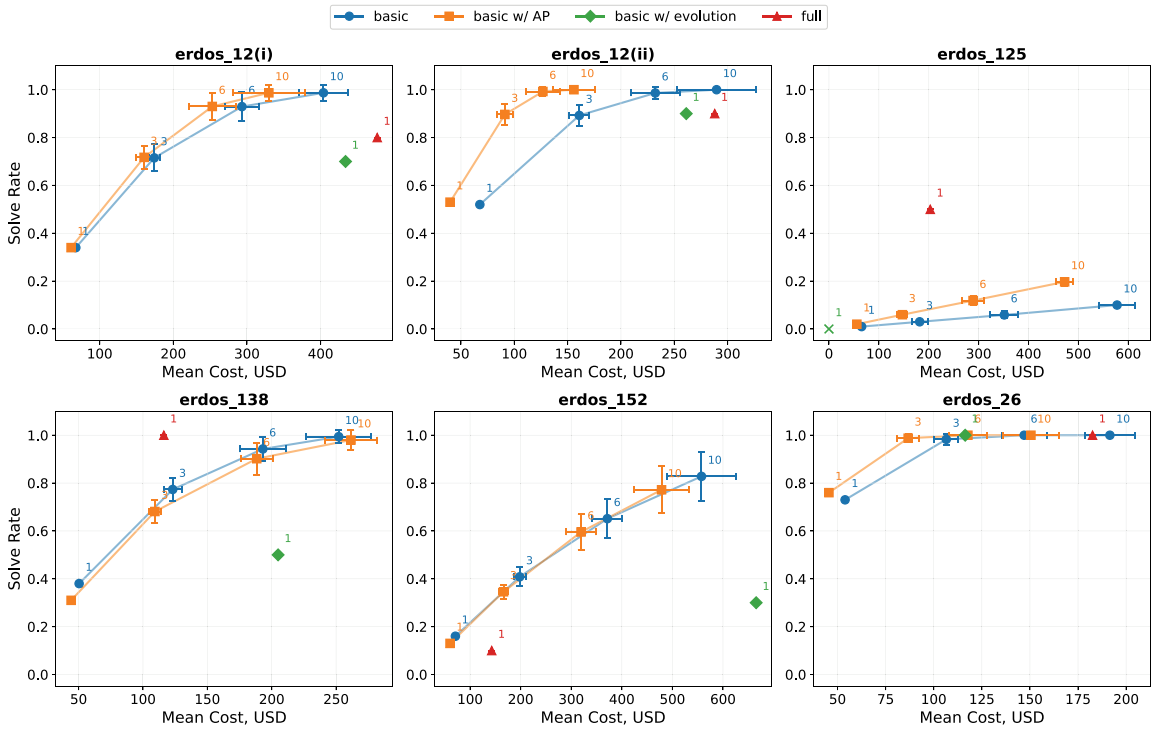

Fig. 3. Solve rate versus mean inference cost (USD) across six Erdős problem instances. The solve rates are evaluated for the four agents: (A) basic (blue circles), (B) basic with AlphaProof (orange squares), (C) basic with evolution (green diamonds), and (D) full-featured (red triangles). Numeric annotations denote the number of independent attempts $K \in \{1, 3, 6, 10\}$ grouped together; error bars indicate one standard error interval. Because of their higher costs, agents (C) and (D) lack variance estimates, as independent attempts were used to obtain a single point estimate (see SM for details). For agents (A) and (B), each curve traces the cost-performance Pareto frontier as K increases, revealing diminishing marginal returns at higher budgets. Agent (B) generally matches or exceeds the solve rate of the basic configuration at comparable cost. Costs reported for (B) and (D) do not include the inference cost of AlphaProof. Although for most problems configuration (A) or (B) is the best, for some challenging problems such as Erdős #125, the full-featured configuration (D) performs significantly better. Note that accounting for the estimated AlphaProof cost of 60 USD does not change the above outcomes beyond the margin of error.

Gemini 3.1 Flash-Lite). These systems could not solve any of the problems. Finally, we evaluated two commercial coding agents—Codex (GPT-5.5) and Claude Code (Opus 4.7)—in "goal" mode on the problems. Codex solved seven of nine problems, failing to solve problems #152 and #125 within 12 hours. We could not get Claude Code to solve any of the problems.

Discussion

We have provided a large-scale demonstration of the value of formal proof search agents on research-level mathematical problems. Recently, some natural-language reasoning systems have been shown to succeed in mathematics research tasks, including Erdős problems (*31*, *32*). However, the use of AI-generated informal proofs, either as standalone products or as inputs to a subsequent formalization stage, requires careful validation by human experts. The use of formal proofs reduces the human supervision needed, enabling large-scale sweeps through open mathematical problems.

A recurring strength of our agents was in sophisticated constructions. For Erdős Problems #12-(i) and #26, Agent (D) assembled intricate combinatorial objects using the Chinese Remainder Theorem; in the quantum optics application, it synthesized novel monochromatic quantum graphs. In the optimization setting, the agent successfully constructed a novel algorithm parameter—a learning rate schedule—and a formal proof of the convergence guarantee the parameter enables. This result can be seen as a form of formally verified algorithm discovery.

Most of the proofs our agents found follow strategies a human expert would recognize. The graph reconstruction proof, for instance, uses a conventional invariant-refinement approach. The proof of the special case of the Zanello problem is more striking: Its key step reformulates the log-concavity defect in terms of first and second differences of box-slice counts, bypassing standard techniques (real-rootedness, convolution closure, Hodge–Riemann arguments) that do not apply in this setting. To the best of our knowledge, this particular bounded-second-difference analysis of the two box-slice decomposition does not appear in the prior literature and represents a genuinely nonobvious proof strategy.

The effectiveness of our basic agent in our post hoc analysis was surprising. At the time we were planning our large-scale exploration, simpler agentic loops did not show strong performance on competition-level benchmarks, and this informed our decision to use the full-featured agent. The LLM landscape has since shifted substantially. We attribute the basic agent's success to both this shift and the power of compiler feedback in grounding LLM reasoning. The full-featured agent retains an advantage on the hardest problems for now. However, as LLM capabilities grow, this advantage may diminish.

At present, our agents' successes are concentrated in areas such as combinatorics, convex optimization, and number theory, where Lean's mathematics library (*33*) is mature and tasks often decompose into tractable subgoals. Even most Erdős problems remain out of reach, let alone problems that require extensive new theory. Additionally, our agents inherit the biases of their underlying LLMs and exhibit high search variance. Characterizing the agents' boundaries and expanding them is an important direction for future work.

Although our primary results concern autonomous discovery, our method also enriches the broader practice of mathematics. Our mathematician

collaborators found that formal proof attempts by our agents enhanced their understanding of a problem, even when an agent could not prove the claim at hand. Because the sketches were formal, experts could focus on the unresolved subgoals rather than reverifying the entire argument. Moreover, the agents were powerful tools for detecting misformalizations. These experiences suggest that AI-driven formal proof search can serve not only as an automated engine but as a means to deepen human understanding.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S8; Table S1; References (*34–92*)

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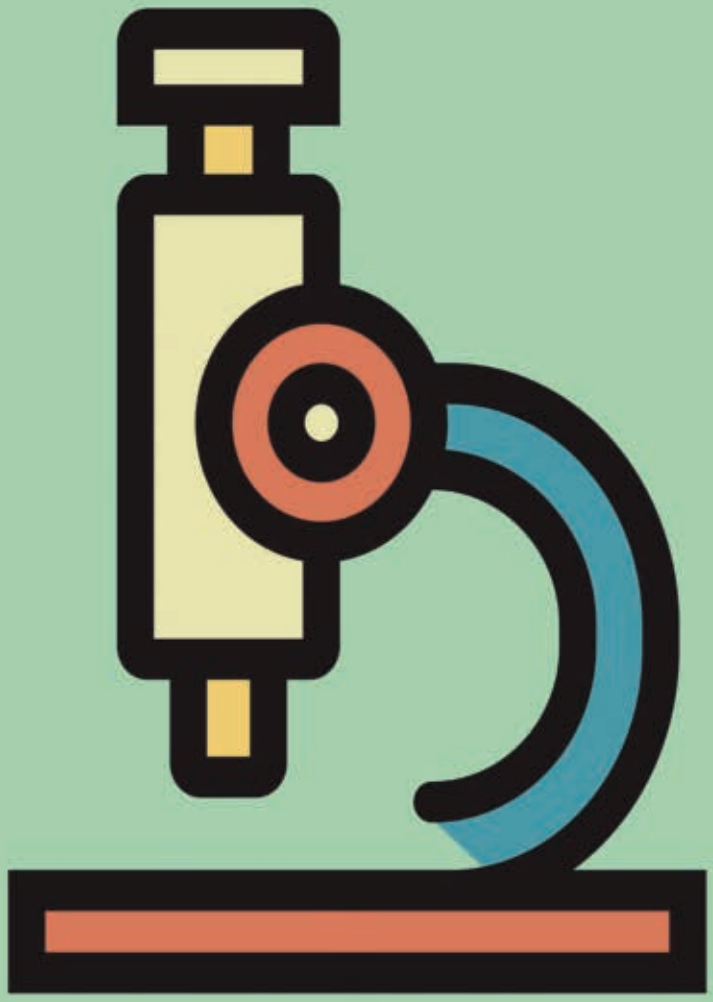

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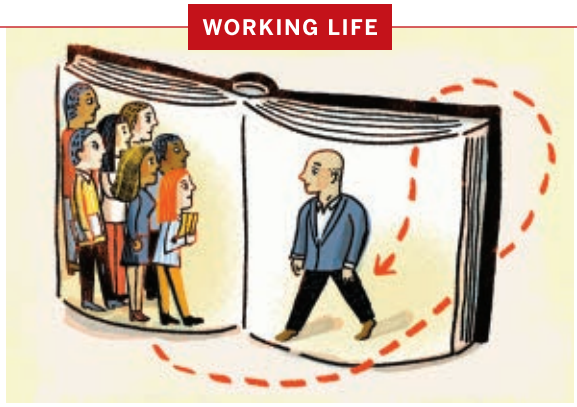

Returning home

Hamid Khazaei

Every day as I walked to my office, I felt a pang. The research institute where I worked was on a university campus, but it felt a world apart. I saw my academic colleagues building fulfilling relationships with students through teaching and mentoring—opportunities I could only engage in on a limited basis, in the time I had left after completing my research responsibilities. My decision a few years earlier to leave academia and pursue more applied research had felt completely logical. Several colleagues had made similar moves, and they seemed happy. I had packed my boxes and told myself I was closing one chapter for good. But I was starting to wonder whether there might be a way back.

For some time, I convinced myself I had made the right choice. I worked on projects I enjoyed and was proud of, building national and international collaborations, writing policy reports, and contributing to R&D that had a direct impact on farmers and farming communities.

And I still had some opportunities to interact with students. Colleagues from the university sometimes invited me to cosupervise Ph.D. students with relevant research interests. I jumped at the chance, even though I had to do it on my own time as it was not an official part of my job responsibilities. I gave and attended occasional lectures at the university, and the resulting interactions with students were always invigorating. I loved that every year new students arrived with fresh perspectives, different cultural backgrounds, and questions that sometimes forced me to rethink assumptions I did not even realize I was making. After a lecture a few years ago, one student asked why a particular crop was not grown in a specific region—a simple question that stopped me midsentence. That conversation eventually grew into a research proposal. Every time I supervised or mentored a student or gave a lecture, I left feeling energized. I couldn't deny that I wanted more.

I was also growing frustrated with some of the challenges and limitations of working at a research institute, where bureaucracy and multiple levels of management can slow even seemingly simple actions such as bringing visiting researchers on for collaborations. Universities, of course, have their own challenges, limitations, and bureaucracies. Competition for funding is intense. Teaching takes time. I was under no illusion that returning to academia would solve every frustration.

Still, whenever I came across a faculty job posting, I felt a spark of excitement and often found myself wondering whether I should give it a try. I even bookmarked a couple of faculty job ads, returning to them for days before finally closing the tabs.

A committee meeting for a Ph.D. student I co-supervise was the final push I needed. The student mentioned how overwhelming it felt to start a Ph.D. in a new country, and we spent a few minutes talking about how he could build a support network. When the meeting ended, I realized the most enjoyable part of my week had been that hour with a student. I no longer wanted to relegate the most rewarding parts of my professional life to essentially a side project. Having made what I thought was a career-defining pivot away from academia, I decided to find a way back.

It's often said that once you leave the established path to the professoriate it is hard to get back on, and returning hasn't been easy. My CV looks quite different from those of many others going up for university professor jobs through the traditional tenure-track route. But I talked to my professional network, explored unconventional on-ramps back to academia, and applied for openings, highlighting my unique background and the skills I've developed as a result. I was offered a part-time university appointment, and I now have two students to help get my research program off the ground.

I have come to realize teaching, mentoring, and academic life are part of who I am. No matter where my career or destiny takes me, I do not think I can completely walk away from them. □

Hamid Khazaei is an associate professor at the Natural Resources Institute of Finland and a docent at the University of Helsinki. Do you have an interesting career story to share? See our author guidelines at https://scim.ag/WorkingLife.

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